Supplementary Information

Title

Xiaochan Cai[1] and Gang Li[1,2,*]

[1]*School of Physical Science and Technology, ShanghaiTech University, Shanghai 200031, China*
[2]*ShanghaiTech Laboratory for Topological Physics, ShanghaiTech University, Shanghai 200031, China*

No	$\mathcal{G}_{\text{nspg}}$	No	$\mathcal{G}_{\text{spg}}$	collinear		coplanar	noncoplanar	degeneracy
				FM	AFM			
1	${}^{1}1$	1.1.1	${}^{1}1$			✓	✓	1
		1.2.2	${}^{1}1^{m_z}1$			✓		1
		1.3.3	${}^{1}1^{\infty_z}1 \to {}^{1}1$	✓	✓			1
		1.4.4	${}^{1}1^{\infty_z m_x}1 \to {}^{1}1^{m_x}1$	✓	✓			1
2	${}^{1}\bar{1}$	2.1.5	${}^{1}\bar{1}$			✓	✓	1
		2.2.6	${}^{1}\bar{1}^{m_z}1$			✓		1
		2.3.7	${}^{1}\bar{1}^{\infty_z}1 \to {}^{1}\bar{1}$	✓	✓			1
		2.4.8	${}^{1}\bar{1}^{\infty_z m_x}1 \to {}^{1}\bar{1}^{m_x}1$	✓	✓			1
3	${}^{2}\bar{1}$	3.1.9	${}^{2}\bar{1}$			✓	✓	1
		3.2.10	${}^{2_z}\bar{1}^{m_z}1$			✓		1
		3.3.11	${}^{2_z}\bar{1}^{m_x}1$			✓		1
		3.4.12	${}^{2_z}\bar{1}^{\infty_x}1 \to {}^{2_z}\bar{1}^{\Sigma_x}1$		✓			2
4	${}^{m}\bar{1}$	4.1.13	${}^{m}\bar{1}$			✓	✓	1
		4.2.14	${}^{m_z}\bar{1}^{\infty_x}1 \to {}^{m}\bar{1}$	✓	✓			1
5	${}^{\bar{1}}\bar{1}$	5.1.15	${}^{\bar{1}}\bar{1}$			✓	✓	1
		5.2.16	${}^{\bar{1}}\bar{1}^{\infty_z}1 \to {}^{\bar{1}}\bar{1}^{\Sigma_z}1$		✓			2
		5.3.17	${}^{\bar{1}}\bar{1}^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{1}^{\Sigma_z m_x}1$		✓			2
6	${}^{1}2$	6.1.18	${}^{1}2$			✓	✓	1
		6.2.19	${}^{1}2^{m_z}1$			✓		1
		6.3.20	${}^{1}2^{\infty_z}1 \to {}^{1}2$	✓	✓			1
		6.4.21	${}^{1}2^{\infty_z m_x}1 \to {}^{1}2^{m_x}1$	✓	✓			1
7	${}^{2}2$	7.1.22	${}^{2}2$			✓	✓	1

* ligang@shanghaitech.edu.cn

		7.2.23	${}^{2_y}2^{m_y}1$			✓		1
		7.3.24	${}^{2_y}2^{m_z}1$			✓		1
		7.4.25	${}^{2_y}2^{\infty_x}1 \to {}^{2_y}2^{\Sigma_x}1$		✓			2
8	${}^{m}2$	8.1.26	${}^{m}2$			✓	✓	1
		8.2.27	${}^{m_y}2^{\infty_x}1 \to {}^{m}2$	✓	✓			1
9	${}^{\bar{1}}2$	9.1.28	${}^{\bar{1}}2$			✓	✓	1
		9.2.29	${}^{\bar{1}}2^{\infty_z}1 \to {}^{\bar{1}}2^{\Sigma_z}1$		✓			2
		9.3.30	${}^{\bar{1}}2^{\infty_z m_x}1 \to {}^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2
10	${}^{1}m$	10.1.31	${}^{1}m$			✓	✓	1
		10.2.32	${}^{1}m^{m_z}1$			✓		1
		10.3.33	${}^{1}m^{\infty_z}1 \to {}^{1}m$	✓	✓			1
		10.4.34	${}^{1}m^{\infty_z m_x}1 \to {}^{1}m^{m_x}1$	✓	✓			1
11	${}^{2}m$	11.1.35	${}^{2}m$			✓	✓	1
		11.2.36	${}^{2_y}m^{m_y}1$			✓		1
		11.3.37	${}^{2_y}m^{m_z}1$			✓		1
		11.4.38	${}^{2_y}m^{\infty_x}1 \to {}^{2_y}m^{\Sigma_x}1$		✓			2
12	${}^{m}m$	12.1.39	${}^{m}m$			✓	✓	1
		12.2.40	${}^{m_y}m^{\infty_x}1 \to {}^{m}m$	✓	✓			1
13	${}^{\bar{1}}m$	13.1.41	${}^{\bar{1}}m$			✓	✓	1
		13.2.42	${}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		13.3.43	${}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
14	${}^{1}2/{}^{1}m$	14.1.44	${}^{1}2/{}^{1}m$			✓	✓	1
		14.2.45	${}^{1}2/{}^{1}m^{m_z}1$			✓		1
		14.3.46	${}^{1}2/{}^{1}m^{\infty_z}1 \to {}^{1}2/{}^{1}m$	✓	✓			1
		14.4.47	${}^{1}2/{}^{1}m^{\infty_z m_x}1 \to {}^{1}2/{}^{1}m^{m_x}1$	✓	✓			1
15	${}^{1}2/{}^{2}m$	15.1.48	${}^{1}2/{}^{2}m$			✓	✓	1
		15.2.49	${}^{1}2/{}^{2_y}m^{m_y}1$			✓		1
		15.3.50	${}^{1}2/{}^{2_y}m^{m_z}1$			✓		1
		15.4.51	${}^{1}2/{}^{2_y}m^{\infty_x}1 \to {}^{1}2/{}^{2_y}m^{\Sigma_x}1$		✓			2
16	${}^{1}2/{}^{m}m$	16.1.52	${}^{1}2/{}^{m}m$			✓	✓	1
		16.2.53	${}^{1}2/{}^{m_y}m^{\infty_x}1 \to {}^{1}2/{}^{m}m$	✓	✓			1
17	${}^{1}2/{}^{\bar{1}}m$	17.1.54	${}^{1}2/{}^{\bar{1}}m$			✓	✓	1

		17.2.55	${}^{1}2/{}^{\bar{1}}m^{\infty_z}1 \to {}^{1}2/{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		17.3.56	${}^{1}2/{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}2/{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
18	${}^{2}2/{}^{1}m$	18.1.57	${}^{2}2/{}^{1}m$			✓	✓	1
		18.2.58	${}^{2_y}2/{}^{1}m^{m_y}1$			✓		1
		18.3.59	${}^{2_y}2/{}^{1}m^{m_z}1$			✓		1
		18.4.60	${}^{2_y}2/{}^{1}m^{\infty_x}1 \to {}^{2_y}2/{}^{1}m^{\Sigma_x}1$		✓			2
19	${}^{m}2/{}^{1}m$	19.1.61	${}^{m}2/{}^{1}m$			✓	✓	1
		19.2.62	${}^{m_y}2/{}^{1}m^{\infty_x}1 \to {}^{m}2/{}^{1}m$	✓	✓			1
20	${}^{\bar{1}}2/{}^{1}m$	20.1.63	${}^{\bar{1}}2/{}^{1}m$			✓	✓	1
		20.2.64	${}^{\bar{1}}2/{}^{1}m^{\infty_z}1 \to {}^{\bar{1}}2/{}^{1}m^{\Sigma_z}1$		✓			2
		20.3.65	${}^{\bar{1}}2/{}^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}2/{}^{1}m^{\Sigma_z m_x}1$		✓			2
21	${}^{2}2/{}^{2}m$	21.1.66	${}^{2}2/{}^{2}m$			✓	✓	1
		21.2.67	${}^{2_y}2/{}^{2_y}m^{m_y}1$			✓		1
		21.3.68	${}^{2_y}2/{}^{2_y}m^{m_z}1$			✓		1
		21.4.69	${}^{2_y}2/{}^{2_y}m^{\infty_x}1 \to {}^{2_y}2/{}^{2_y}m^{\Sigma_x}1$		✓			2
22	${}^{m}2/{}^{m}m$	22.1.70	${}^{m}2/{}^{m}m$			✓	✓	1
		22.2.71	${}^{m_y}2/{}^{m_y}m^{\infty_x}1 \to {}^{m}2/{}^{m}m$	✓	✓			1
23	${}^{\bar{1}}2/{}^{\bar{1}}m$	23.1.72	${}^{\bar{1}}2/{}^{\bar{1}}m$			✓	✓	1
		23.2.73	${}^{\bar{1}}2/{}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}2/{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		23.3.74	${}^{\bar{1}}2/{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}2/{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
24	${}^{2_z}2/{}^{2_x}m$	24.1.75	${}^{2_z}2/{}^{2_x}m$			✓	✓	1
		24.2.76	${}^{2_z}2/{}^{2_x}m^{m_z}1$			✓		1
		24.3.77	${}^{2_z}2/{}^{2_x}m^{m_x}1$			✓		1
		24.4.78	${}^{2_z}2/{}^{2_x}m^{m_y}1$			✓		1
25	${}^{2_z}2/{}^{m_x}m$	25.1.79	${}^{2_z}2/{}^{m_x}m$			✓	✓	1
26	${}^{m_x}2/{}^{m_y}m$	26.1.80	${}^{m_x}2/{}^{m_y}m$			✓	✓	1
27	${}^{m_x}2/{}^{2_z}m$	27.1.81	${}^{m_x}2/{}^{2_z}m$			✓	✓	1
28	${}^{2}2/{}^{m}m$	28.1.82	${}^{2}2/{}^{m}m$			✓	✓	1
		28.2.83	${}^{2_y}2/{}^{m_y}m^{\infty_x}1 \to {}^{2_y}2/{}^{m_y}m^{\Sigma_x}1$		✓			2
29	${}^{2}2/{}^{\bar{1}}m$	29.1.84	${}^{2}2/{}^{\bar{1}}m$			✓	✓	1
		29.2.85	${}^{2_y}2/{}^{\bar{1}}m^{\infty_x}1 \to {}^{2_y}2/{}^{\bar{1}}m^{\Sigma_x}1$		✓			2
30	${}^{\bar{1}}2/{}^{m}m$	30.1.86	${}^{\bar{1}}2/{}^{m}m$			✓	✓	1

		30.2.87	${}^{\bar{1}}2/^{m_y}m^{\infty_x}1 \to {}^{\bar{1}}2/^{m_y}m^{\Sigma_x}1$		✓			2
31	${}^{\bar{1}}2/^{2}m$	31.1.88	${}^{\bar{1}}2/^{2}m$			✓	✓	1
		31.2.89	${}^{\bar{1}}2/^{2_y}m^{\infty_x}1 \to {}^{\bar{1}}2/^{2_y}m^{\Sigma_x}1$		✓			2
32	${}^{m}2/^{\bar{1}}m$	32.1.90	${}^{m}2/^{\bar{1}}m$			✓	✓	1
		32.2.91	${}^{m_y}2/^{\bar{1}}m^{\infty_x}1 \to {}^{m_y}2/^{\bar{1}}m^{\Sigma_x}1$		✓			2
33	${}^{m}2/^{2}m$	33.1.92	${}^{m}2/^{2}m$			✓	✓	1
		33.2.93	${}^{m_y}2/^{2_y}m^{\infty_x}1 \to {}^{m_y}2/^{2_y}m^{\Sigma_x}1$		✓			2
34	${}^{1}2^{1}2^{1}2$	34.1.94	${}^{1}2^{1}2^{1}2$			✓	✓	1
		34.2.95	${}^{1}2^{1}2^{1}2^{m_z}1$			✓		1
		34.3.96	${}^{1}2^{1}2^{1}2^{\infty_z}1 \to {}^{1}2^{1}2^{1}2$	✓	✓			1
		34.4.97	${}^{1}2^{1}2^{1}2^{\infty_z m_x}1 \to {}^{1}2^{1}2^{1}2^{m_x}1$	✓	✓			1
35	${}^{1}2^{2}2^{2}2$	35.1.98	${}^{1}2^{2}2^{2}2$			✓	✓	1
		35.2.99	${}^{1}2^{2_z}2^{2_z}2^{m_z}1$			✓		1
		35.3.100	${}^{1}2^{2_z}2^{2_z}2^{m_x}1$			✓		1
		35.4.101	${}^{1}2^{2_z}2^{2_z}2^{\infty_x}1 \to {}^{1}2^{2_z}2^{2_z}2^{\Sigma_x}1$		✓			2
36	${}^{1}2^{m}2^{m}2$	36.1.102	${}^{1}2^{m}2^{m}2$			✓	✓	1
		36.2.103	${}^{1}2^{m_z}2^{m_z}2^{\infty_x}1 \to {}^{1}2^{m}2^{m}2$	✓	✓			1
37	${}^{1}2^{\bar{1}}2^{\bar{1}}2$	37.1.104	${}^{1}2^{\bar{1}}2^{\bar{1}}2$			✓	✓	1
		37.2.105	${}^{1}2^{\bar{1}}2^{\bar{1}}2^{\infty_z}1 \to {}^{1}2^{\bar{1}}2^{\bar{1}}2^{\Sigma_z}1$		✓			2
		37.3.106	${}^{1}2^{\bar{1}}2^{\bar{1}}2^{\infty_z m_x}1 \to {}^{1}2^{\bar{1}}2^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2
38	${}^{2_x}2^{2_y}2^{2_z}2$	38.1.107	${}^{2_x}2^{2_y}2^{2_z}2$			✓	✓	1
		38.2.108	${}^{2_x}2^{2_y}2^{2_z}2^{m_x}1$			✓		1
39	${}^{m_x}2^{m_y}2^{2_z}2$	39.1.109	${}^{m_x}2^{m_y}2^{2_z}2$			✓	✓	1
40	${}^{2}2^{\bar{1}}2^{m}2$	40.1.110	${}^{2}2^{\bar{1}}2^{m}2$			✓	✓	1
		40.2.111	${}^{2_z}2^{\bar{1}}2^{m_z}2^{\infty_x}1 \to {}^{2_z}2^{\bar{1}}2^{m_z}2^{\Sigma_x}1$		✓			2
41	${}^{1}m^{1}m^{1}2$	41.1.112	${}^{1}m^{1}m^{1}2$			✓	✓	1
		41.2.113	${}^{1}m^{1}m^{1}2^{m_z}1$			✓		1
		41.3.114	${}^{1}m^{1}m^{1}2^{\infty_z}1 \to {}^{1}m^{1}m^{1}2$	✓	✓			1
		41.4.115	${}^{1}m^{1}m^{1}2^{\infty_z m_x}1 \to {}^{1}m^{1}m^{1}2^{m_x}1$	✓	✓			1
42	${}^{2}m^{2}m^{1}2$	42.1.116	${}^{2}m^{2}m^{1}2$			✓	✓	1
		42.2.117	${}^{2_z}m^{2_z}m^{1}2^{m_z}1$			✓		1
		42.3.118	${}^{2_z}m^{2_z}m^{1}2^{m_x}1$			✓		1

		42.4.119	${}^{2_z}m^{2_z}m^{1}2^{\infty_x}1 \to {}^{2_z}m^{2_z}m^{1}2^{\Sigma_x}1$		✓			2
43	${}^{m}m^{m}m^{1}2$	43.1.120	${}^{m}m^{m}m^{1}2$			✓	✓	1
		43.2.121	${}^{m_z}m^{m_z}m^{1}2^{\infty_x}1 \to {}^{m}m^{m}m^{1}2$	✓	✓			1
44	${}^{\bar{1}}m^{\bar{1}}m^{1}2$	44.1.122	${}^{\bar{1}}m^{\bar{1}}m^{1}2$			✓	✓	1
		44.2.123	${}^{\bar{1}}m^{\bar{1}}m^{1}2^{\infty_z}1 \to {}^{\bar{1}}m^{\bar{1}}m^{1}2^{\Sigma_z}1$		✓			2
		44.3.124	${}^{\bar{1}}m^{\bar{1}}m^{1}2^{\infty_z m_x}1 \to {}^{\bar{1}}m^{\bar{1}}m^{1}2^{\Sigma_z m_x}1$		✓			2
45	${}^{1}m^{2}m^{2}2$	45.1.125	${}^{1}m^{2}m^{2}2$			✓	✓	1
		45.2.126	${}^{1}m^{2_z}m^{2_z}2^{m_z}1$			✓		1
		45.3.127	${}^{1}m^{2_z}m^{2_z}2^{m_x}1$			✓		1
		45.4.128	${}^{1}m^{2_z}m^{2_z}2^{\infty_x}1 \to {}^{1}m^{2_z}m^{2_z}2^{\Sigma_x}1$		✓			2
46	${}^{1}m^{m}m^{m}2$	46.1.129	${}^{1}m^{m}m^{m}2$			✓	✓	1
		46.2.130	${}^{1}m^{m_z}m^{m_z}2^{\infty_x}1 \to {}^{1}m^{m}m^{m}2$	✓	✓			1
47	${}^{1}m^{\bar{1}}m^{\bar{1}}2$	47.1.131	${}^{1}m^{\bar{1}}m^{\bar{1}}2$			✓	✓	1
		47.2.132	${}^{1}m^{\bar{1}}m^{\bar{1}}2^{\infty_z}1 \to {}^{1}m^{\bar{1}}m^{\bar{1}}2^{\Sigma_z}1$		✓			2
		47.3.133	${}^{1}m^{\bar{1}}m^{\bar{1}}2^{\infty_z m_x}1 \to {}^{1}m^{\bar{1}}m^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2
48	${}^{2_x}m^{2_y}m^{2_z}2$	48.1.134	${}^{2_x}m^{2_y}m^{2_z}2$			✓	✓	1
		48.2.135	${}^{2_x}m^{2_y}m^{2_z}2^{m_x}1$			✓		1
		48.3.136	${}^{2_x}m^{2_y}m^{2_z}2^{m_z}1$			✓		1
49	${}^{m_x}m^{m_y}m^{2_z}2$	49.1.137	${}^{m_x}m^{m_y}m^{2_z}2$			✓	✓	1
50	${}^{m_x}m^{2_z}m^{m_y}2$	50.1.138	${}^{m_x}m^{2_z}m^{m_y}2$			✓	✓	1
51	${}^{m}m^{\bar{1}}m^{2}2$	51.1.139	${}^{m}m^{\bar{1}}m^{2}2$			✓	✓	1
		51.2.140	${}^{m_z}m^{\bar{1}}m^{2_z}2^{\infty_x}1 \to {}^{m_z}m^{\bar{1}}m^{2_z}2^{\Sigma_x}1$		✓			2
52	${}^{m}m^{2}m^{\bar{1}}2$	52.1.141	${}^{m}m^{2}m^{\bar{1}}2$			✓	✓	1
		52.2.142	${}^{m_z}m^{2_z}m^{\bar{1}}2^{\infty_x}1 \to {}^{m_z}m^{2_z}m^{\bar{1}}2^{\Sigma_x}1$		✓			2
53	${}^{\bar{1}}m^{2}m^{m}2$	53.1.143	${}^{\bar{1}}m^{2}m^{m}2$			✓	✓	1
		53.2.144	${}^{\bar{1}}m^{2_z}m^{m_z}2^{\infty_x}1 \to {}^{\bar{1}}m^{2_z}m^{m_z}2^{\Sigma_x}1$		✓			2
54	${}^{1}m^{1}m^{1}m$	54.1.145	${}^{1}m^{1}m^{1}m$			✓	✓	1
		54.2.146	${}^{1}m^{1}m^{1}m^{m_z}1$			✓		1
		54.3.147	${}^{1}m^{1}m^{1}m^{\infty_z}1 \to {}^{1}m^{1}m^{1}m$	✓	✓			1
		54.4.148	${}^{1}m^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}m^{1}m^{1}m^{m_x}1$	✓	✓			1
55	${}^{1}m^{2}m^{2}m$	55.1.149	${}^{1}m^{2}m^{2}m$			✓	✓	1
		55.2.150	${}^{1}m^{2_z}m^{2_z}m^{m_z}1$			✓		1

		55.3.151	${}^{1}m^{2_z}m^{2_z}m^{m_x}1$			✓		1
		55.4.152	${}^{1}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
56	${}^{1}m^{m}m^{m}m$	56.1.153	${}^{1}m^{m}m^{m}m$			✓	✓	1
		56.2.154	${}^{1}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}m^{m}m^{m}m$	✓	✓			1
57	${}^{1}m^{\bar{1}}m^{\bar{1}}m$	57.1.155	${}^{1}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		57.2.156	${}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		57.3.157	${}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
58	${}^{1}m^{1}m^{2}m$	58.1.158	${}^{1}m^{1}m^{2}m$			✓	✓	1
		58.2.159	${}^{1}m^{1}m^{2_z}m^{m_z}1$			✓		1
		58.3.160	${}^{1}m^{1}m^{2_z}m^{m_x}1$			✓		1
		58.4.161	${}^{1}m^{1}m^{2_z}m^{\infty_x}1 \to {}^{1}m^{1}m^{2_z}m^{\Sigma_x}1$		✓			2
59	${}^{1}m^{1}m^{m}m$	59.1.162	${}^{1}m^{1}m^{m}m$			✓	✓	1
		59.2.163	${}^{1}m^{1}m^{m_z}m^{\infty_x}1 \to {}^{1}m^{1}m^{m}m$	✓	✓			1
60	${}^{1}m^{1}m^{\bar{1}}m$	60.1.164	${}^{1}m^{1}m^{\bar{1}}m$			✓	✓	1
		60.2.165	${}^{1}m^{1}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}m^{1}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		60.3.166	${}^{1}m^{1}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}m^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
61	${}^{2}m^{2}m^{2}m$	61.1.167	${}^{2}m^{2}m^{2}m$			✓	✓	1
		61.2.168	${}^{2_z}m^{2_z}m^{2_z}m^{m_z}1$			✓		1
		61.3.169	${}^{2_z}m^{2_z}m^{2_z}m^{m_x}1$			✓		1
		61.4.170	${}^{2_z}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{2_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
62	${}^{m}m^{m}m^{m}m$	62.1.171	${}^{m}m^{m}m^{m}m$			✓	✓	1
		62.2.172	${}^{m_z}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{m}m^{m}m^{m}m$	✓	✓			1
63	${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$	63.1.173	${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		63.2.174	${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		63.3.175	${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
64	${}^{2_z}m^{2_z}m^{2_x}m$	64.1.176	${}^{2_z}m^{2_z}m^{2_x}m$			✓	✓	1
		64.2.177	${}^{2_z}m^{2_z}m^{2_x}m^{m_z}1$			✓		1
		64.3.178	${}^{2_z}m^{2_z}m^{2_x}m^{m_x}1$			✓		1
		64.4.179	${}^{2_z}m^{2_z}m^{2_x}m^{m_y}1$			✓		1
65	${}^{m_y}m^{m_y}m^{m_x}m$	65.1.180	${}^{m_y}m^{m_y}m^{m_x}m$			✓	✓	1
66	${}^{m_y}m^{m_y}m^{2_z}m$	66.1.181	${}^{m_y}m^{m_y}m^{2_z}m$			✓	✓	1
67	${}^{2_z}m^{2_z}m^{m_y}m$	67.1.182	${}^{2_z}m^{2_z}m^{m_y}m$			✓	✓	1

68	${}^{m}m^{m}m^{\bar{1}}m$	68.1.183	${}^{m}m^{m}m^{\bar{1}}m$			✓	✓	1
		68.2.184	${}^{m_z}m^{m_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}m^{m_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2
69	${}^{m}m^{m}m^{2}m$	69.1.185	${}^{m}m^{m}m^{2}m$			✓	✓	1
		69.2.186	${}^{m_z}m^{m_z}m^{2_z}m^{\infty_x}1 \to {}^{m_z}m^{m_z}m^{2_z}m^{\Sigma_x}1$		✓			2
70	${}^{2}m^{2}m^{\bar{1}}m$	70.1.187	${}^{2}m^{2}m^{\bar{1}}m$			✓	✓	1
		70.2.188	${}^{2_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2
71	${}^{2}m^{2}m^{m}m$	71.1.189	${}^{2}m^{2}m^{m}m$			✓	✓	1
		71.2.190	${}^{2_z}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{2_z}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
72	${}^{\bar{1}}m^{\bar{1}}m^{2}m$	72.1.191	${}^{\bar{1}}m^{\bar{1}}m^{2}m$			✓	✓	1
		72.2.192	${}^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\Sigma_x}1$		✓			2
73	${}^{\bar{1}}m^{\bar{1}}m^{m}m$	73.1.193	${}^{\bar{1}}m^{\bar{1}}m^{m}m$			✓	✓	1
		73.2.194	${}^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
74	${}^{2_x}m^{2_y}m^{1}m$	74.1.195	${}^{2_x}m^{2_y}m^{1}m$			✓	✓	1
		74.2.196	${}^{2_x}m^{2_y}m^{1}m^{m_x}1$			✓		1
		74.3.197	${}^{2_x}m^{2_y}m^{1}m^{m_z}1$			✓		1
75	${}^{m_x}m^{m_y}m^{1}m$	75.1.198	${}^{m_x}m^{m_y}m^{1}m$			✓	✓	1
76	${}^{m_y}m^{2_z}m^{1}m$	76.1.199	${}^{m_y}m^{2_z}m^{1}m$			✓	✓	1
77	${}^{m}m^{\bar{1}}m^{1}m$	77.1.200	${}^{m}m^{\bar{1}}m^{1}m$			✓	✓	1
		77.2.201	${}^{m_z}m^{\bar{1}}m^{1}m^{\infty_x}1 \to {}^{m_z}m^{\bar{1}}m^{1}m^{\Sigma_x}1$		✓			2
78	${}^{m}m^{2}m^{1}m$	78.1.202	${}^{m}m^{2}m^{1}m$			✓	✓	1
		78.2.203	${}^{m_z}m^{2_z}m^{1}m^{\infty_x}1 \to {}^{m_z}m^{2_z}m^{1}m^{\Sigma_x}1$		✓			2
79	${}^{\bar{1}}m^{2}m^{1}m$	79.1.204	${}^{\bar{1}}m^{2}m^{1}m$			✓	✓	1
		79.2.205	${}^{\bar{1}}m^{2_z}m^{1}m^{\infty_x}1 \to {}^{\bar{1}}m^{2_z}m^{1}m^{\Sigma_x}1$		✓			2
80	${}^{2_x}m^{2_y}m^{2_z}m$	80.1.206	${}^{2_x}m^{2_y}m^{2_z}m$			✓	✓	1
		80.2.207	${}^{2_x}m^{2_y}m^{2_z}m^{m_x}1$			✓		1
81	${}^{m_x}m^{m_y}m^{2_z}m$	81.1.208	${}^{m_x}m^{m_y}m^{2_z}m$			✓	✓	1
82	${}^{2}m^{\bar{1}}m^{m}m$	82.1.209	${}^{2}m^{\bar{1}}m^{m}m$			✓	✓	1
		82.2.210	${}^{2_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
83	${}^{m_x}m^{m_y}m^{m_z}m$	83.1.211	${}^{m_x}m^{m_y}m^{m_z}m$			✓	✓	1
84	${}^{m_x}m^{2_y}m^{2_z}m$	84.1.212	${}^{m_x}m^{2_y}m^{2_z}m$			✓	✓	1
85	${}^{m_z}m^{2_x}m^{2_z}m$	85.1.213	${}^{m_z}m^{2_x}m^{2_z}m$			✓	✓	1
86	${}^{2_z}m^{m_x}m^{m_z}m$	86.1.214	${}^{2_z}m^{m_x}m^{m_z}m$			✓	✓	1

87	${}^{2_z}m^{\bar{1}}m^{m_x}m$	87.1.215	${}^{2_z}m^{\bar{1}}m^{m_x}m$			✓	✓	1
88	${}^{2_z}m^{\bar{1}}m^{2_x}m$	88.1.216	${}^{2_z}m^{\bar{1}}m^{2_x}m$			✓	✓	1
89	${}^{\bar{1}}m^{m_z}m^{m_x}m$	89.1.217	${}^{\bar{1}}m^{m_z}m^{m_x}m$			✓	✓	1
90	${}^{1}4$	90.1.218	${}^{1}4$			✓	✓	1
		90.2.219	${}^{1}4^{m_z}1$			✓		1, 2
		90.3.220	${}^{1}4^{\infty_z}1 \to {}^{1}4$	✓	✓			1
		90.4.221	${}^{1}4^{\infty_z m_x}1 \to {}^{1}4^{m_x}1$	✓	✓			1, 2
91	${}^{2}4$	91.1.222	${}^{2}4$			✓	✓	1
		91.2.223	${}^{2_z}4^{m_z}1$			✓		1, 2
		91.3.224	${}^{2_z}4^{m_x}1$			✓		1, 2
		91.4.225	${}^{2_z}4^{\infty_x}1 \to {}^{2_z}4^{\Sigma_x}1$		✓			2
92	${}^{m}4$	92.1.226	${}^{m}4$			✓	✓	1, 2
		92.2.227	${}^{m_z}4^{\infty_x}1 \to {}^{m}4$	✓	✓			1, 2
93	${}^{\bar{1}}4$	93.1.228	${}^{\bar{1}}4$			✓	✓	1, 2
		93.2.229	${}^{\bar{1}}4^{\infty_z}1 \to {}^{\bar{1}}4^{\Sigma_z}1$		✓			2
		93.3.230	${}^{\bar{1}}4^{\infty_z m_x}1 \to {}^{\bar{1}}4^{\Sigma_z m_x}1$		✓			2
94	${}^{4}4$	94.1.231	${}^{4}4$			✓	✓	1
		94.2.232	${}^{4_z}4^{m_z}1$			✓		1, 2
95	${}^{\bar{4}}4$	95.1.233	${}^{\bar{4}}4$			✓	✓	1, 2
96	${}^{1}\bar{4}$	96.1.234	${}^{1}\bar{4}$			✓	✓	1
		96.2.235	${}^{1}\bar{4}^{m_z}1$			✓		1, 2
		96.3.236	${}^{1}\bar{4}^{\infty_z}1 \to {}^{1}\bar{4}$	✓	✓			1
		96.4.237	${}^{1}\bar{4}^{\infty_z m_x}1 \to {}^{1}\bar{4}^{m_x}1$	✓	✓			1, 2
97	${}^{2}\bar{4}$	97.1.238	${}^{2}\bar{4}$			✓	✓	1
		97.2.239	${}^{2_z}\bar{4}^{m_z}1$			✓		1, 2
		97.3.240	${}^{2_z}\bar{4}^{m_x}1$			✓		1, 2
		97.4.241	${}^{2_z}\bar{4}^{\infty_x}1 \to {}^{2_z}\bar{4}^{\Sigma_x}1$		✓			2
98	${}^{m}\bar{4}$	98.1.242	${}^{m}\bar{4}$			✓	✓	1, 2
		98.2.243	${}^{m_z}\bar{4}^{\infty_x}1 \to {}^{m}\bar{4}$	✓	✓			1, 2
99	${}^{\bar{1}}\bar{4}$	99.1.244	${}^{\bar{1}}\bar{4}$			✓	✓	1, 2
		99.2.245	${}^{\bar{1}}\bar{4}^{\infty_z}1 \to {}^{\bar{1}}\bar{4}^{\Sigma_z}1$		✓			2
		99.3.246	${}^{\bar{1}}\bar{4}^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{4}^{\Sigma_z m_x}1$		✓			2

100	${}^{4}\bar{4}$	100.1.247	${}^{4}\bar{4}$			✓	✓	1
		100.2.248	${}^{4_z}\bar{4}^{m_z}1$			✓		1, 2
101	${}^{\bar{4}}\bar{4}$	101.1.249	${}^{\bar{4}}\bar{4}$			✓	✓	1, 2
102	${}^{1}4/{}^{1}m$	102.1.250	${}^{1}4/{}^{1}m$			✓	✓	1
		102.2.251	${}^{1}4/{}^{1}m^{m_z}1$			✓		1, 2
		102.3.252	${}^{1}4/{}^{1}m^{\infty_z}1 \to {}^{1}4/{}^{1}m$	✓	✓			1
		102.4.253	${}^{1}4/{}^{1}m^{\infty_z m_x}1 \to {}^{1}4/{}^{1}m^{m_x}1$	✓	✓			1, 2
103	${}^{2}4/{}^{1}m$	103.1.254	${}^{2}4/{}^{1}m$			✓	✓	1
		103.2.255	${}^{2_z}4/{}^{1}m^{m_z}1$			✓		1, 2
		103.3.256	${}^{2_z}4/{}^{1}m^{m_x}1$			✓		1, 2
		103.4.257	${}^{2_z}4/{}^{1}m^{\infty_x}1 \to {}^{2_z}4/{}^{1}m^{\Sigma_x}1$		✓			2
104	${}^{m}4/{}^{1}m$	104.1.258	${}^{m}4/{}^{1}m$			✓	✓	1, 2
		104.2.259	${}^{m_z}4/{}^{1}m^{\infty_x}1 \to {}^{m}4/{}^{1}m$	✓	✓			1, 2
105	${}^{\bar{1}}4/{}^{1}m$	105.1.260	${}^{\bar{1}}4/{}^{1}m$			✓	✓	1, 2
		105.2.261	${}^{\bar{1}}4/{}^{1}m^{\infty_z}1 \to {}^{\bar{1}}4/{}^{1}m^{\Sigma_z}1$		✓			2
		105.3.262	${}^{\bar{1}}4/{}^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}4/{}^{1}m^{\Sigma_z m_x}1$		✓			2
106	${}^{2}4/{}^{2}m$	106.1.263	${}^{2}4/{}^{2}m$			✓	✓	1
		106.2.264	${}^{2_z}4/{}^{2_z}m^{m_z}1$			✓		1, 2
		106.3.265	${}^{2_z}4/{}^{2_z}m^{m_x}1$			✓		1, 2
		106.4.266	${}^{2_z}4/{}^{2_z}m^{\infty_x}1 \to {}^{2_z}4/{}^{2_z}m^{\Sigma_x}1$		✓			2
107	${}^{m}4/{}^{m}m$	107.1.267	${}^{m}4/{}^{m}m$			✓	✓	1, 2
		107.2.268	${}^{m_z}4/{}^{m_z}m^{\infty_x}1 \to {}^{m}4/{}^{m}m$	✓	✓			1, 2
108	${}^{\bar{1}}4/{}^{\bar{1}}m$	108.1.269	${}^{\bar{1}}4/{}^{\bar{1}}m$			✓	✓	1, 2
		108.2.270	${}^{\bar{1}}4/{}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}4/{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		108.3.271	${}^{\bar{1}}4/{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}4/{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
109	${}^{1}4/{}^{2}m$	109.1.272	${}^{1}4/{}^{2}m$			✓	✓	1
		109.2.273	${}^{1}4/{}^{2_z}m^{m_z}1$			✓		1, 2
		109.3.274	${}^{1}4/{}^{2_z}m^{m_x}1$			✓		1, 2
		109.4.275	${}^{1}4/{}^{2_z}m^{\infty_x}1 \to {}^{1}4/{}^{2_z}m^{\Sigma_x}1$		✓			2
110	${}^{1}4/{}^{m}m$	110.1.276	${}^{1}4/{}^{m}m$			✓	✓	1, 2
		110.2.277	${}^{1}4/{}^{m_z}m^{\infty_x}1 \to {}^{1}4/{}^{m}m$	✓	✓			1, 2
111	${}^{1}4/{}^{\bar{1}}m$	111.1.278	${}^{1}4/{}^{\bar{1}}m$			✓	✓	1, 2

		111.2.279	${}^{1}4/{}^{\bar{1}}m^{\infty_z}1 \to {}^{1}4/{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		111.3.280	${}^{1}4/{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}4/{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
112	${}^{4}4/{}^{2}m$	112.1.281	${}^{4}4/{}^{2}m$			✓	✓	1
		112.2.282	${}^{4_z}4/{}^{2_z}m^{m_z}1$			✓		1, 2
113	${}^{\bar{4}}4/{}^{2}m$	113.1.283	${}^{\bar{4}}4/{}^{2}m$			✓	✓	1, 2
114	${}^{4}4/{}^{1}m$	114.1.284	${}^{4}4/{}^{1}m$			✓	✓	1
		114.2.285	${}^{4_z}4/{}^{1}m^{m_z}1$			✓		1, 2
115	${}^{\bar{4}}4/{}^{1}m$	115.1.286	${}^{\bar{4}}4/{}^{1}m$			✓	✓	1, 2
116	${}^{2_z}4/{}^{2_x}m$	116.1.287	${}^{2_z}4/{}^{2_x}m$			✓	✓	1
		116.2.288	${}^{2_z}4/{}^{2_x}m^{m_z}1$			✓		1, 2
		116.3.289	${}^{2_z}4/{}^{2_x}m^{m_x}1$			✓		1, 2
		116.4.290	${}^{2_z}4/{}^{2_x}m^{m_y}1$			✓		1, 2
117	${}^{2_z}4/{}^{m_x}m$	117.1.291	${}^{2_z}4/{}^{m_x}m$			✓	✓	1, 2
118	${}^{m_x}4/{}^{2_z}m$	118.1.292	${}^{m_x}4/{}^{2_z}m$			✓	✓	1, 2
119	${}^{m_x}4/{}^{m_y}m$	119.1.293	${}^{m_x}4/{}^{m_y}m$			✓	✓	1, 2
120	${}^{2}4/{}^{\bar{1}}m$	120.1.294	${}^{2}4/{}^{\bar{1}}m$			✓	✓	1, 2
		120.2.295	${}^{2_z}4/{}^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}4/{}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
121	${}^{2}4/{}^{m}m$	121.1.296	${}^{2}4/{}^{m}m$			✓	✓	1, 2
		121.2.297	${}^{2_z}4/{}^{m_z}m^{\infty_x}1 \to {}^{2_z}4/{}^{m_z}m^{\Sigma_x}1$		✓			2
122	${}^{\bar{1}}4/{}^{2}m$	122.1.298	${}^{\bar{1}}4/{}^{2}m$			✓	✓	1, 2
		122.2.299	${}^{\bar{1}}4/{}^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}4/{}^{2_z}m^{\Sigma_x}1$		✓			2, 4
123	${}^{\bar{1}}4/{}^{m}m$	123.1.300	${}^{\bar{1}}4/{}^{m}m$			✓	✓	1, 2
		123.2.301	${}^{\bar{1}}4/{}^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}4/{}^{m_z}m^{\Sigma_x}1$		✓			2
124	${}^{m}4/{}^{2}m$	124.1.302	${}^{m}4/{}^{2}m$			✓	✓	1, 2
		124.2.303	${}^{m_z}4/{}^{2_z}m^{\infty_x}1 \to {}^{m_z}4/{}^{2_z}m^{\Sigma_x}1$		✓			2, 4
125	${}^{m}4/{}^{\bar{1}}m$	125.1.304	${}^{m}4/{}^{\bar{1}}m$			✓	✓	1, 2
		125.2.305	${}^{m_z}4/{}^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}4/{}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
126	${}^{4}4/{}^{m}m$	126.1.306	${}^{4}4/{}^{m}m$			✓	✓	1, 2
127	${}^{4}4/{}^{\bar{1}}m$	127.1.307	${}^{4}4/{}^{\bar{1}}m$			✓	✓	1, 2
128	${}^{\bar{4}}4/{}^{m}m$	128.1.308	${}^{\bar{4}}4/{}^{m}m$			✓	✓	1, 2
129	${}^{\bar{4}}4/{}^{\bar{1}}m$	129.1.309	${}^{\bar{4}}4/{}^{\bar{1}}m$			✓	✓	1, 2
130	${}^{1}4{}^{1}2{}^{1}2$	130.1.310	${}^{1}4{}^{1}2{}^{1}2$			✓	✓	1, 2

		130.2.311	${}^{1}4^{1}2^{1}2^{m_z}1$			✓		1, 2
		130.3.312	${}^{1}4^{1}2^{1}2^{\infty_z}1 \to {}^{1}4^{1}2^{1}2$	✓	✓			1, 2
		130.4.313	${}^{1}4^{1}2^{1}2^{\infty_z m_x}1 \to {}^{1}4^{1}2^{1}2^{m_x}1$	✓	✓			1, 2
131	${}^{1}4^{2}2^{2}2$	131.1.314	${}^{1}4^{2}2^{2}2$			✓	✓	1, 2
		131.2.315	${}^{1}4^{2_z}2^{2_z}2^{m_z}1$			✓		1, 2
		131.3.316	${}^{1}4^{2_z}2^{2_z}2^{m_x}1$			✓		1, 2
		131.4.317	${}^{1}4^{2_z}2^{2_z}2^{\infty_x}1 \to {}^{1}4^{2_z}2^{2_z}2^{\Sigma_x}1$		✓			2
132	${}^{1}4^{m}2^{m}2$	132.1.318	${}^{1}4^{m}2^{m}2$			✓	✓	1
		132.2.319	${}^{1}4^{m_z}2^{m_z}2^{\infty_x}1 \to {}^{1}4^{m}2^{m}2$	✓	✓			1
133	${}^{1}4^{\bar{1}}2^{\bar{1}}2$	133.1.320	${}^{1}4^{\bar{1}}2^{\bar{1}}2$			✓	✓	1
		133.2.321	${}^{1}4^{\bar{1}}2^{\bar{1}}2^{\infty_z}1 \to {}^{1}4^{\bar{1}}2^{\bar{1}}2^{\Sigma_z}1$		✓			2
		133.3.322	${}^{1}4^{\bar{1}}2^{\bar{1}}2^{\infty_z m_x}1 \to {}^{1}4^{\bar{1}}2^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4
134	${}^{2}4^{1}2^{2}2$	134.1.323	${}^{2}4^{1}2^{2}2$			✓	✓	1, 2
		134.2.324	${}^{2_z}4^{1}2^{2_z}2^{m_z}1$			✓		1, 2
		134.3.325	${}^{2_z}4^{1}2^{2_z}2^{m_x}1$			✓		1, 2
		134.4.326	${}^{2_z}4^{1}2^{2_z}2^{\infty_x}1 \to {}^{2_z}4^{1}2^{2_z}2^{\Sigma_x}1$		✓			2
135	${}^{m}4^{1}2^{m}2$	135.1.327	${}^{m}4^{1}2^{m}2$			✓	✓	1, 2
		135.2.328	${}^{m_z}4^{1}2^{m_z}2^{\infty_x}1 \to {}^{m}4^{1}2^{m}2$	✓	✓			1, 2
136	${}^{\bar{1}}4^{1}2^{\bar{1}}2$	136.1.329	${}^{\bar{1}}4^{1}2^{\bar{1}}2$			✓	✓	1, 2
		136.2.330	${}^{\bar{1}}4^{1}2^{\bar{1}}2^{\infty_z}1 \to {}^{\bar{1}}4^{1}2^{\bar{1}}2^{\Sigma_z}1$		✓			2
		136.3.331	${}^{\bar{1}}4^{1}2^{\bar{1}}2^{\infty_z m_x}1 \to {}^{\bar{1}}4^{1}2^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2
137	${}^{2_x}4^{2_y}2^{2_z}2$	137.1.332	${}^{2_x}4^{2_y}2^{2_z}2$			✓	✓	1, 2
		137.2.333	${}^{2_x}4^{2_y}2^{2_z}2^{m_x}1$			✓		1, 2
		137.3.334	${}^{2_x}4^{2_y}2^{2_z}2^{m_y}1$			✓		1, 2
138	${}^{2_z}4^{m_x}2^{m_y}2$	138.1.335	${}^{2_z}4^{m_x}2^{m_y}2$			✓	✓	1
139	${}^{m_x}4^{2_z}2^{m_y}2$	139.1.336	${}^{m_x}4^{2_z}2^{m_y}2$			✓	✓	1, 2
140	${}^{2}4^{\bar{1}}2^{m}2$	140.1.337	${}^{2}4^{\bar{1}}2^{m}2$			✓	✓	1
		140.2.338	${}^{2_z}4^{\bar{1}}2^{m_z}2^{\infty_x}1 \to {}^{2_z}4^{\bar{1}}2^{m_z}2^{\Sigma_x}1$		✓			2
141	${}^{\bar{1}}4^{2}2^{m}2$	141.1.339	${}^{\bar{1}}4^{2}2^{m}2$			✓	✓	1, 2
		141.2.340	${}^{\bar{1}}4^{2_z}2^{m_z}2^{\infty_x}1 \to {}^{\bar{1}}4^{2_z}2^{m_z}2^{\Sigma_x}1$		✓			2
142	${}^{m}4^{\bar{1}}2^{2}2$	142.1.341	${}^{m}4^{\bar{1}}2^{2}2$			✓	✓	1, 2
		142.2.342	${}^{m_z}4^{\bar{1}}2^{2_z}2^{\infty_x}1 \to {}^{m_z}4^{\bar{1}}2^{2_z}2^{\Sigma_x}1$		✓			2, 4

143	${}^{4_z}4^{2_x}2^{2_{xy}}2$	143.1.343	${}^{4_z}4^{2_x}2^{2_{xy}}2$			✓	✓	1, 2
		143.2.344	${}^{4_z}4^{2_x}2^{2_{xy}}2^{m_z}1$			✓		1, 2
144	${}^{4_z}4^{m_x}2^{m_{xy}}2$	144.1.345	${}^{4_z}4^{m_x}2^{m_{xy}}2$			✓	✓	1
145	${}^{\bar{4}_z}4^{2_x}2^{m_{xy}}2$	145.1.346	${}^{\bar{4}_z}4^{2_x}2^{m_{xy}}2$			✓	✓	1, 2
146	${}^{1}4^{1}m^{1}m$	146.1.347	${}^{1}4^{1}m^{1}m$			✓	✓	1, 2
		146.2.348	${}^{1}4^{1}m^{1}m^{m_z}1$			✓		1, 2
		146.3.349	${}^{1}4^{1}m^{1}m^{\infty_z}1 \to {}^{1}4^{1}m^{1}m$	✓	✓			1, 2
		146.4.350	${}^{1}4^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}4^{1}m^{1}m^{m_x}1$	✓	✓			1, 2
147	${}^{1}4^{2}m^{2}m$	147.1.351	${}^{1}4^{2}m^{2}m$			✓	✓	1, 2
		147.2.352	${}^{1}4^{2_z}m^{2_z}m^{m_z}1$			✓		1, 2
		147.3.353	${}^{1}4^{2_z}m^{2_z}m^{m_x}1$			✓		1, 2
		147.4.354	${}^{1}4^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}4^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
148	${}^{1}4^{m}m^{m}m$	148.1.355	${}^{1}4^{m}m^{m}m$			✓	✓	1
		148.2.356	${}^{1}4^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}4^{m}m^{m}m$	✓	✓			1
149	${}^{1}4^{\bar{1}}m^{\bar{1}}m$	149.1.357	${}^{1}4^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		149.2.358	${}^{1}4^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}4^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		149.3.359	${}^{1}4^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}4^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
150	${}^{2}4^{1}m^{2}m$	150.1.360	${}^{2}4^{1}m^{2}m$			✓	✓	1, 2
		150.2.361	${}^{2_z}4^{1}m^{2_z}m^{m_z}1$			✓		1, 2
		150.3.362	${}^{2_z}4^{1}m^{2_z}m^{m_x}1$			✓		1, 2
		150.4.363	${}^{2_z}4^{1}m^{2_z}m^{\infty_x}1 \to {}^{2_z}4^{1}m^{2_z}m^{\Sigma_x}1$		✓			2
151	${}^{m}4^{1}m^{m}m$	151.1.364	${}^{m}4^{1}m^{m}m$			✓	✓	1, 2
		151.2.365	${}^{m_z}4^{1}m^{m_z}m^{\infty_x}1 \to {}^{m}4^{1}m^{m}m$	✓	✓			1, 2
152	${}^{\bar{1}}4^{1}m^{\bar{1}}m$	152.1.366	${}^{\bar{1}}4^{1}m^{\bar{1}}m$			✓	✓	1, 2
		152.2.367	${}^{\bar{1}}4^{1}m^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}4^{1}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		152.3.368	${}^{\bar{1}}4^{1}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}4^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
153	${}^{2_x}4^{2_y}m^{2_z}m$	153.1.369	${}^{2_x}4^{2_y}m^{2_z}m$			✓	✓	1, 2
		153.2.370	${}^{2_x}4^{2_y}m^{2_z}m^{m_x}1$			✓		1, 2
		153.3.371	${}^{2_x}4^{2_y}m^{2_z}m^{m_y}1$			✓		1, 2
154	${}^{2_z}4^{m_x}m^{m_y}m$	154.1.372	${}^{2_z}4^{m_x}m^{m_y}m$			✓	✓	1
155	${}^{m_x}4^{2_z}m^{m_y}m$	155.1.373	${}^{m_x}4^{2_z}m^{m_y}m$			✓	✓	1, 2
156	${}^{2}4^{\bar{1}}m^{m}m$	156.1.374	${}^{2}4^{\bar{1}}m^{m}m$			✓	✓	1

		156.2.375	${}^{2_z}4^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}4^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
157	${}^{\bar{1}}4^{2}m^{m}m$	157.1.376	${}^{\bar{1}}4^{2}m^{m}m$			✓	✓	1, 2
		157.2.377	${}^{\bar{1}}4^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}4^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
158	${}^{m}4^{2}m^{\bar{1}}m$	158.1.378	${}^{m}4^{2}m^{\bar{1}}m$			✓	✓	1, 2
		158.2.379	${}^{m_z}4^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}4^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
159	${}^{4_z}4^{2_x}m^{2_{xy}}m$	159.1.380	${}^{4_z}4^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
		159.2.381	${}^{4_z}4^{2_x}m^{2_{xy}}m^{m_z}1$			✓		1, 2
160	${}^{4_z}4^{m_x}m^{m_{xy}}m$	160.1.382	${}^{4_z}4^{m_x}m^{m_{xy}}m$			✓	✓	1
161	${}^{\bar{4}_z}4^{m_x}m^{2_{xy}}m$	161.1.383	${}^{\bar{4}_z}4^{m_x}m^{2_{xy}}m$			✓	✓	1, 2
162	${}^{1}\bar{4}^{1}2^{1}m$	162.1.384	${}^{1}\bar{4}^{1}2^{1}m$			✓	✓	1, 2
		162.2.385	${}^{1}\bar{4}^{1}2^{1}m^{m_z}1$			✓		1, 2
		162.3.386	${}^{1}\bar{4}^{1}2^{1}m^{\infty_z}1 \to {}^{1}\bar{4}^{1}2^{1}m$	✓	✓			1, 2
		162.4.387	${}^{1}\bar{4}^{1}2^{1}m^{\infty_z m_x}1 \to {}^{1}\bar{4}^{1}2^{1}m^{m_x}1$	✓	✓			1, 2
163	${}^{1}\bar{4}^{2}2^{2}m$	163.1.388	${}^{1}\bar{4}^{2}2^{2}m$			✓	✓	1, 2
		163.2.389	${}^{1}\bar{4}^{2_z}2^{2_z}m^{m_z}1$			✓		1, 2
		163.3.390	${}^{1}\bar{4}^{2_z}2^{2_z}m^{m_x}1$			✓		1, 2
		163.4.391	${}^{1}\bar{4}^{2_z}2^{2_z}m^{\infty_x}1 \to {}^{1}\bar{4}^{2_z}2^{2_z}m^{\Sigma_x}1$		✓			2
164	${}^{1}\bar{4}^{m}2^{m}m$	164.1.392	${}^{1}\bar{4}^{m}2^{m}m$			✓	✓	1
		164.2.393	${}^{1}\bar{4}^{m_z}2^{m_z}m^{\infty_x}1 \to {}^{1}\bar{4}^{m}2^{m}m$	✓	✓			1
165	${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m$	165.1.394	${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m$			✓	✓	1
		165.2.395	${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\infty_z}1 \to {}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\Sigma_z}1$		✓			2
		165.3.396	${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
166	${}^{2}\bar{4}^{2}2^{1}m$	166.1.397	${}^{2}\bar{4}^{2}2^{1}m$			✓	✓	1, 2
		166.2.398	${}^{2_z}\bar{4}^{2_z}2^{1}m^{m_z}1$			✓		1, 2
		166.3.399	${}^{2_z}\bar{4}^{2_z}2^{1}m^{m_x}1$			✓		1, 2
		166.4.400	${}^{2_z}\bar{4}^{2_z}2^{1}m^{\infty_x}1 \to {}^{2_z}\bar{4}^{2_z}2^{1}m^{\Sigma_x}1$		✓			2
167	${}^{m}\bar{4}^{m}2^{1}m$	167.1.401	${}^{m}\bar{4}^{m}2^{1}m$			✓	✓	1, 2
		167.2.402	${}^{m_z}\bar{4}^{m_z}2^{1}m^{\infty_x}1 \to {}^{m}\bar{4}^{m}2^{1}m$	✓	✓			1, 2
168	${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m$	168.1.403	${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m$			✓	✓	1, 2
		168.2.404	${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\infty_z}1 \to {}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\Sigma_z}1$		✓			2
		168.3.405	${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\Sigma_z m_x}1$		✓			2
169	${}^{2}\bar{4}^{1}2^{2}m$	169.1.406	${}^{2}\bar{4}^{1}2^{2}m$			✓	✓	1, 2

		169.2.407	${}^{2_z}\bar{4}^{1}2^{2_z}m^{m_z}1$			✓		1, 2
		169.3.408	${}^{2_z}\bar{4}^{1}2^{2_z}m^{m_x}1$			✓		1, 2
		169.4.409	${}^{2_z}\bar{4}^{1}2^{2_z}m^{\infty_x}1 \to {}^{2_z}\bar{4}^{1}2^{2_z}m^{\Sigma_x}1$		✓			2
170	${}^{m}\bar{4}^{1}2^{m}m$	170.1.410	${}^{m}\bar{4}^{1}2^{m}m$			✓	✓	1, 2
		170.2.411	${}^{m_z}\bar{4}^{1}2^{m_z}m^{\infty_x}1 \to {}^{m}\bar{4}^{1}2^{m}m$	✓	✓			1, 2
171	${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m$	171.1.412	${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m$			✓	✓	1, 2
		171.2.413	${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\Sigma_z}1$		✓			2
		171.3.414	${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
172	${}^{2_x}\bar{4}^{2_y}2^{2_z}m$	172.1.415	${}^{2_x}\bar{4}^{2_y}2^{2_z}m$			✓	✓	1, 2
		172.2.416	${}^{2_x}\bar{4}^{2_y}2^{2_z}m^{m_x}1$			✓		1, 2
		172.3.417	${}^{2_x}\bar{4}^{2_y}2^{2_z}m^{m_y}1$			✓		1, 2
		172.4.418	${}^{2_x}\bar{4}^{2_y}2^{2_z}m^{m_z}1$			✓		1, 2
173	${}^{2_z}\bar{4}^{m_x}2^{m_y}m$	173.1.419	${}^{2_z}\bar{4}^{m_x}2^{m_y}m$			✓	✓	1
174	${}^{m_x}\bar{4}^{2_z}2^{m_y}m$	174.1.420	${}^{m_x}\bar{4}^{2_z}2^{m_y}m$			✓	✓	1, 2
175	${}^{m_x}\bar{4}^{m_y}2^{2_z}m$	175.1.421	${}^{m_x}\bar{4}^{m_y}2^{2_z}m$			✓	✓	1, 2
176	${}^{2}\bar{4}^{\bar{1}}2^{m}m$	176.1.422	${}^{2}\bar{4}^{\bar{1}}2^{m}m$			✓	✓	1
		176.2.423	${}^{2_z}\bar{4}^{\bar{1}}2^{m_z}m^{\infty_x}1 \to {}^{2_z}\bar{4}^{\bar{1}}2^{m_z}m^{\Sigma_x}1$		✓			2
177	${}^{2}\bar{4}^{m}2^{\bar{1}}m$	177.1.424	${}^{2}\bar{4}^{m}2^{\bar{1}}m$			✓	✓	1
		177.2.425	${}^{2_z}\bar{4}^{m_z}2^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}\bar{4}^{m_z}2^{\bar{1}}m^{\Sigma_x}1$		✓			2
178	${}^{\bar{1}}\bar{4}^{2}2^{m}m$	178.1.426	${}^{\bar{1}}\bar{4}^{2}2^{m}m$			✓	✓	1, 2
		178.2.427	${}^{\bar{1}}\bar{4}^{2_z}2^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}\bar{4}^{2_z}2^{m_z}m^{\Sigma_x}1$		✓			2
179	${}^{\bar{1}}\bar{4}^{m}2^{2}m$	179.1.428	${}^{\bar{1}}\bar{4}^{m}2^{2}m$			✓	✓	1, 2
		179.2.429	${}^{\bar{1}}\bar{4}^{m_z}2^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}\bar{4}^{m_z}2^{2_z}m^{\Sigma_x}1$		✓			2
180	${}^{m}\bar{4}^{2}2^{\bar{1}}m$	180.1.430	${}^{m}\bar{4}^{2}2^{\bar{1}}m$			✓	✓	1, 2
		180.2.431	${}^{m_z}\bar{4}^{2_z}2^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}\bar{4}^{2_z}2^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
181	${}^{m}\bar{4}^{\bar{1}}2^{2}m$	181.1.432	${}^{m}\bar{4}^{\bar{1}}2^{2}m$			✓	✓	1, 2
		181.2.433	${}^{m_z}\bar{4}^{\bar{1}}2^{2_z}m^{\infty_x}1 \to {}^{m_z}\bar{4}^{\bar{1}}2^{2_z}m^{\Sigma_x}1$		✓			2, 4
182	${}^{4_z}\bar{4}^{2_x}2^{2_{xy}}m$	182.1.434	${}^{4_z}\bar{4}^{2_x}2^{2_{xy}}m$			✓	✓	1, 2
		182.2.435	${}^{4_z}\bar{4}^{2_x}2^{2_{xy}}m^{m_z}1$			✓		1, 2
183	${}^{4_z}\bar{4}^{m_x}2^{m_{xy}}m$	183.1.436	${}^{4_z}\bar{4}^{m_x}2^{m_{xy}}m$			✓	✓	1
184	${}^{\bar{4}_z}\bar{4}^{2_x}2^{m_{xy}}m$	184.1.437	${}^{\bar{4}_z}\bar{4}^{2_x}2^{m_{xy}}m$			✓	✓	1, 2
185	${}^{\bar{4}_z}\bar{4}^{m_x}2^{2_{xy}}m$	185.1.438	${}^{\bar{4}_z}\bar{4}^{m_x}2^{2_{xy}}m$			✓	✓	1, 2

186	${}^{1}4/{}^{1}m{}^{1}m{}^{1}m$	186.1.439	${}^{1}4/{}^{1}m{}^{1}m{}^{1}m$			✓	✓	1, 2
		186.2.440	${}^{1}4/{}^{1}m{}^{1}m{}^{1}m^{m_z}1$			✓		1, 2
		186.3.441	${}^{1}4/{}^{1}m{}^{1}m{}^{1}m^{\infty_z}1 \to {}^{1}4/{}^{1}m{}^{1}m{}^{1}m$	✓	✓			1, 2
		186.4.442	${}^{1}4/{}^{1}m{}^{1}m{}^{1}m^{\infty_z m_x}1 \to {}^{1}4/{}^{1}m{}^{1}m{}^{1}m^{m_x}1$	✓	✓			1, 2
187	${}^{2}4/{}^{2}m{}^{2}m{}^{1}m$	187.1.443	${}^{2}4/{}^{2}m{}^{2}m{}^{1}m$			✓	✓	1, 2
		187.2.444	${}^{2_z}4/{}^{2_z}m{}^{2_z}m{}^{1}m^{m_z}1$			✓		1, 2
		187.3.445	${}^{2_z}4/{}^{2_z}m{}^{2_z}m{}^{1}m^{m_x}1$			✓		1, 2
		187.4.446	${}^{2_z}4/{}^{2_z}m{}^{2_z}m{}^{1}m^{\infty_x}1 \to {}^{2_z}4/{}^{2_z}m{}^{2_z}m{}^{1}m^{\Sigma_x}1$		✓			2, 4
188	${}^{m}4/{}^{m}m{}^{m}m{}^{1}m$	188.1.447	${}^{m}4/{}^{m}m{}^{m}m{}^{1}m$			✓	✓	1, 2
		188.2.448	${}^{m_z}4/{}^{m_z}m{}^{m_z}m{}^{1}m^{\infty_x}1 \to {}^{m}4/{}^{m}m{}^{m}m{}^{1}m$	✓	✓			1, 2
189	${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m$	189.1.449	${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m$			✓	✓	1, 2
		189.2.450	${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m^{\infty_z}1 \to {}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m^{\Sigma_z}1$		✓			2, 4
		189.3.451	${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m^{\Sigma_z m_x}1$		✓			2, 4
190	${}^{1}4/{}^{2}m{}^{1}m{}^{1}m$	190.1.452	${}^{1}4/{}^{2}m{}^{1}m{}^{1}m$			✓	✓	1, 2
		190.2.453	${}^{1}4/{}^{2_z}m{}^{1}m{}^{1}m^{m_z}1$			✓		1, 2
		190.3.454	${}^{1}4/{}^{2_z}m{}^{1}m{}^{1}m^{m_x}1$			✓		1, 2
		190.4.455	${}^{1}4/{}^{2_z}m{}^{1}m{}^{1}m^{\infty_x}1 \to {}^{1}4/{}^{2_z}m{}^{1}m{}^{1}m^{\Sigma_x}1$		✓			2, 4
191	${}^{1}4/{}^{m}m{}^{1}m{}^{1}m$	191.1.456	${}^{1}4/{}^{m}m{}^{1}m{}^{1}m$			✓	✓	1, 2
		191.2.457	${}^{1}4/{}^{m_z}m{}^{1}m{}^{1}m^{\infty_x}1 \to {}^{1}4/{}^{m}m{}^{1}m{}^{1}m$	✓	✓			1, 2
192	${}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m$	192.1.458	${}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m$			✓	✓	1, 2
		192.2.459	${}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m^{\infty_z}1 \to {}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m^{\Sigma_z}1$		✓			2, 4
		192.3.460	${}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m^{\infty_z m_x}1 \to {}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m^{\Sigma_z m_x}1$		✓			2, 4
193	${}^{2}4/{}^{1}m{}^{1}m{}^{2}m$	193.1.461	${}^{2}4/{}^{1}m{}^{1}m{}^{2}m$			✓	✓	1, 2
		193.2.462	${}^{2_z}4/{}^{1}m{}^{1}m{}^{2_z}m^{m_z}1$			✓		1, 2
		193.3.463	${}^{2_z}4/{}^{1}m{}^{1}m{}^{2_z}m^{m_x}1$			✓		1, 2
		193.4.464	${}^{2_z}4/{}^{1}m{}^{1}m{}^{2_z}m^{\infty_x}1 \to {}^{2_z}4/{}^{1}m{}^{1}m{}^{2_z}m^{\Sigma_x}1$		✓			2
194	${}^{m}4/{}^{1}m{}^{1}m{}^{m}m$	194.1.465	${}^{m}4/{}^{1}m{}^{1}m{}^{m}m$			✓	✓	1, 2
		194.2.466	${}^{m_z}4/{}^{1}m{}^{1}m{}^{m_z}m^{\infty_x}1 \to {}^{m}4/{}^{1}m{}^{1}m{}^{m}m$	✓	✓			1, 2
195	${}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m$	195.1.467	${}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m$			✓	✓	1, 2
		195.2.468	${}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		195.3.469	${}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2
196	${}^{1}4/{}^{1}m{}^{2}m{}^{2}m$	196.1.470	${}^{1}4/{}^{1}m{}^{2}m{}^{2}m$			✓	✓	1, 2

		196.2.471	${}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{m_z}1$			✓		1, 2
		196.3.472	${}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{m_x}1$			✓		1, 2
		196.4.473	${}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
197	${}^{1}4/{}^{1}m^{m}m^{m}m^{m}$	197.1.474	${}^{1}4/{}^{1}m^{m}m^{m}m^{m}$			✓	✓	1
		197.2.475	${}^{1}4/{}^{1}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}4/{}^{1}m^{m}m^{m}m^{m}$	✓	✓			1
198	${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m$	198.1.476	${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		198.2.477	${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		198.3.478	${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
199	${}^{1}4/{}^{2}m^{2}m^{2}m$	199.1.479	${}^{1}4/{}^{2}m^{2}m^{2}m$			✓	✓	1, 2
		199.2.480	${}^{1}4/{}^{2_z}m^{2_z}m^{2_z}m^{m_z}1$			✓		1, 2
		199.3.481	${}^{1}4/{}^{2_z}m^{2_z}m^{2_z}m^{m_x}1$			✓		1, 2
		199.4.482	${}^{1}4/{}^{2_z}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}4/{}^{2_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
200	${}^{1}4/{}^{m}m^{m}m^{m}m$	200.1.483	${}^{1}4/{}^{m}m^{m}m^{m}m$			✓	✓	1, 2
		200.2.484	${}^{1}4/{}^{m_z}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}4/{}^{m}m^{m}m^{m}m$	✓	✓			1, 2
201	${}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$	201.1.485	${}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1, 2
		201.2.486	${}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2, 4
		201.3.487	${}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
202	${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m$	202.1.488	${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m$			✓	✓	1, 2
		202.2.489	${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m^{m_z}1$			✓		1, 2
		202.3.490	${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m^{m_x}1$			✓		1, 2
203	${}^{2_z}4/{}^{2_z}m^{m_x}m^{m_y}m$	203.1.491	${}^{2_z}4/{}^{2_z}m^{m_x}m^{m_y}m$			✓	✓	1
204	${}^{m_x}4/{}^{m_x}m^{2_z}m^{m_y}m$	204.1.492	${}^{m_x}4/{}^{m_x}m^{2_z}m^{m_y}m$			✓	✓	1, 2
205	${}^{2}4/{}^{2}m^{\bar{1}}m^{m}m$	205.1.493	${}^{2}4/{}^{2}m^{\bar{1}}m^{m}m$			✓	✓	1
		205.2.494	${}^{2_z}4/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}4/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
206	${}^{\bar{1}}4/{}^{\bar{1}}m^{2}m^{m}m$	206.1.495	${}^{\bar{1}}4/{}^{\bar{1}}m^{2}m^{m}m$			✓	✓	1, 2
		206.2.496	${}^{\bar{1}}4/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}4/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
207	${}^{m}4/{}^{m}m^{2}m^{\bar{1}}m$	207.1.497	${}^{m}4/{}^{m}m^{2}m^{\bar{1}}m$			✓	✓	1, 2
		207.2.498	${}^{m_z}4/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}4/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
208	${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m$	208.1.499	${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m$			✓	✓	1, 2
		208.2.500	${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m^{m_x}1$			✓		1, 2
		208.3.501	${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m^{m_y}1$			✓		1, 2
		208.4.502	${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m^{m_z}1$			✓		1, 2

209	${}^{1}4/^{2_z}m^{m_x}m^{m_x}m$	209.1.503	${}^{1}4/^{2_z}m^{m_x}m^{m_x}m$			✓	✓	1
210	${}^{1}4/^{m_x}m^{2_z}m^{2_z}m$	210.1.504	${}^{1}4/^{m_x}m^{2_z}m^{2_z}m$			✓	✓	1, 2
211	${}^{1}4/^{m_x}m^{m_y}m^{m_y}m$	211.1.505	${}^{1}4/^{m_x}m^{m_y}m^{m_y}m$			✓	✓	1, 2
212	${}^{1}4/^{2}m^{\bar{1}}m^{\bar{1}}m$	212.1.506	${}^{1}4/^{2}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		212.2.507	${}^{1}4/^{2_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \to {}^{1}4/^{2_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$		✓			2
213	${}^{1}4/^{2}m^{m}m^{m}m$	213.1.508	${}^{1}4/^{2}m^{m}m^{m}m$			✓	✓	1
		213.2.509	${}^{1}4/^{2_z}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}4/^{2_z}m^{m_z}m^{m_z}m^{\Sigma_x}1$		✓			2
214	${}^{1}4/^{\bar{1}}m^{2}m^{2}m$	214.1.510	${}^{1}4/^{\bar{1}}m^{2}m^{2}m$			✓	✓	1, 2
		214.2.511	${}^{1}4/^{\bar{1}}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}4/^{\bar{1}}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
215	${}^{1}4/^{\bar{1}}m^{m}m^{m}m$	215.1.512	${}^{1}4/^{\bar{1}}m^{m}m^{m}m$			✓	✓	1, 2
		215.2.513	${}^{1}4/^{\bar{1}}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}4/^{\bar{1}}m^{m_z}m^{m_z}m^{\Sigma_x}1$		✓			2
216	${}^{1}4/^{m}m^{2}m^{2}m$	216.1.514	${}^{1}4/^{m}m^{2}m^{2}m$			✓	✓	1, 2
		216.2.515	${}^{1}4/^{m_z}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}4/^{m_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
217	${}^{1}4/^{m}m^{\bar{1}}m^{\bar{1}}m$	217.1.516	${}^{1}4/^{m}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1, 2
		217.2.517	${}^{1}4/^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \to {}^{1}4/^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
218	${}^{2_z}4/^{1}m^{2_x}m^{2_y}m$	218.1.518	${}^{2_z}4/^{1}m^{2_x}m^{2_y}m$			✓	✓	1, 2
		218.2.519	${}^{2_z}4/^{1}m^{2_x}m^{2_y}m^{m_z}1$			✓		1, 2
		218.3.520	${}^{2_z}4/^{1}m^{2_x}m^{2_y}m^{m_x}1$			✓		1, 2
219	${}^{2_z}4/^{1}m^{m_x}m^{m_y}m$	219.1.521	${}^{2_z}4/^{1}m^{m_x}m^{m_y}m$			✓	✓	1
220	${}^{m_x}4/^{1}m^{2_z}m^{m_y}m$	220.1.522	${}^{m_x}4/^{1}m^{2_z}m^{m_y}m$			✓	✓	1, 2
221	${}^{2}4/^{1}m^{\bar{1}}m^{m}m$	221.1.523	${}^{2}4/^{1}m^{\bar{1}}m^{m}m$			✓	✓	1
		221.2.524	${}^{2_z}4/^{1}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}4/^{1}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
222	${}^{\bar{1}}4/^{1}m^{2}m^{m}m$	222.1.525	${}^{\bar{1}}4/^{1}m^{2}m^{m}m$			✓	✓	1, 2
		222.2.526	${}^{\bar{1}}4/^{1}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}4/^{1}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
223	${}^{m}4/^{1}m^{2}m^{\bar{1}}m$	223.1.527	${}^{m}4/^{1}m^{2}m^{\bar{1}}m$			✓	✓	1, 2
		223.2.528	${}^{m_z}4/^{1}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}4/^{1}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
224	${}^{2_z}4/^{2_x}m^{1}m^{2_z}m$	224.1.529	${}^{2_z}4/^{2_x}m^{1}m^{2_z}m$			✓	✓	1, 2
		224.2.530	${}^{2_z}4/^{2_x}m^{1}m^{2_z}m^{m_z}1$			✓		1, 2
		224.3.531	${}^{2_z}4/^{2_x}m^{1}m^{2_z}m^{m_x}1$			✓		1, 2
		224.4.532	${}^{2_z}4/^{2_x}m^{1}m^{2_z}m^{m_y}1$			✓		1, 2
225	${}^{2_z}4/^{m_x}m^{1}m^{2_z}m$	225.1.533	${}^{2_z}4/^{m_x}m^{1}m^{2_z}m$			✓	✓	1, 2
226	${}^{m_x}4/^{2_z}m^{1}m^{m_x}m$	226.1.534	${}^{m_x}4/^{2_z}m^{1}m^{m_x}m$			✓	✓	1, 2

227	${}^{m_x}4/^{m_y}m^1m^{m_x}m$	227.1.535	${}^{m_x}4/^{m_y}m^1m^{m_x}m$			✓	✓	1, 2
228	${}^{2}4/^{\bar{1}}m^1m^2m$	228.1.536	${}^{2}4/^{\bar{1}}m^1m^2m$			✓	✓	1, 2
		228.2.537	${}^{2_z}4/^{\bar{1}}m^1m^{2_z}m^{\infty_x}1 \to {}^{2_z}4/^{\bar{1}}m^1m^{2_z}m^{\Sigma_x}1$		✓			2, 4
229	${}^{2}4/^{m}m^1m^2m$	229.1.538	${}^{2}4/^{m}m^1m^2m$			✓	✓	1, 2
		229.2.539	${}^{2_z}4/^{m_z}m^1m^{2_z}m^{\infty_x}1 \to {}^{2_z}4/^{m_z}m^1m^{2_z}m^{\Sigma_x}1$		✓			2
230	${}^{\bar{1}}4/^{2}m^1m^{\bar{1}}m$	230.1.540	${}^{\bar{1}}4/^{2}m^1m^{\bar{1}}m$			✓	✓	1, 2
		230.2.541	${}^{\bar{1}}4/^{2_z}m^1m^{\bar{1}}m^{\infty_x}1 \to {}^{\bar{1}}4/^{2_z}m^1m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
231	${}^{\bar{1}}4/^{m}m^1m^{\bar{1}}m$	231.1.542	${}^{\bar{1}}4/^{m}m^1m^{\bar{1}}m$			✓	✓	1, 2
		231.2.543	${}^{\bar{1}}4/^{m_z}m^1m^{\bar{1}}m^{\infty_x}1 \to {}^{\bar{1}}4/^{m_z}m^1m^{\bar{1}}m^{\Sigma_x}1$		✓			2
232	${}^{m}4/^{2}m^1m^{m}m$	232.1.544	${}^{m}4/^{2}m^1m^{m}m$			✓	✓	1, 2
		232.2.545	${}^{m_z}4/^{2_z}m^1m^{m_z}m^{\infty_x}1 \to {}^{m_z}4/^{2_z}m^1m^{m_z}m^{\Sigma_x}1$		✓			2, 4
233	${}^{m}4/^{\bar{1}}m^1m^{m}m$	233.1.546	${}^{m}4/^{\bar{1}}m^1m^{m}m$			✓	✓	1, 2
		233.2.547	${}^{m_z}4/^{\bar{1}}m^1m^{m_z}m^{\infty_x}1 \to {}^{m_z}4/^{\bar{1}}m^1m^{m_z}m^{\Sigma_x}1$		✓			2, 4
234	${}^{2_z}4/^{2_x}m^{2_x}m^{2_y}m$	234.1.548	${}^{2_z}4/^{2_x}m^{2_x}m^{2_y}m$			✓	✓	1, 2
		234.2.549	${}^{2_z}4/^{2_x}m^{2_x}m^{2_y}m^{m_z}1$			✓		1, 2
		234.3.550	${}^{2_z}4/^{2_x}m^{2_x}m^{2_y}m^{m_x}1$			✓		1, 2
		234.4.551	${}^{2_z}4/^{2_x}m^{2_x}m^{2_y}m^{m_y}1$			✓		1, 2
235	${}^{2_z}4/^{m_x}m^{m_x}m^{m_y}m$	235.1.552	${}^{2_z}4/^{m_x}m^{m_x}m^{m_y}m$			✓	✓	1, 2
236	${}^{m_x}4/^{2_z}m^{2_z}m^{m_y}m$	236.1.553	${}^{m_x}4/^{2_z}m^{2_z}m^{m_y}m$			✓	✓	1, 2
237	${}^{m_x}4/^{m_y}m^{m_y}m^{2_z}m$	237.1.554	${}^{m_x}4/^{m_y}m^{m_y}m^{2_z}m$			✓	✓	1, 2
238	${}^{2}4/^{\bar{1}}m^{\bar{1}}m^{m}m$	238.1.555	${}^{2}4/^{\bar{1}}m^{\bar{1}}m^{m}m$			✓	✓	1, 2
		238.2.556	${}^{2_z}4/^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}4/^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2, 4
239	${}^{2}4/^{m}m^{m}m^{\bar{1}}m$	239.1.557	${}^{2}4/^{m}m^{m}m^{\bar{1}}m$			✓	✓	1, 2
		239.2.558	${}^{2_z}4/^{m_z}m^{m_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}4/^{m_z}m^{m_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2
240	${}^{\bar{1}}4/^{2}m^2m^{m}m$	240.1.559	${}^{\bar{1}}4/^{2}m^2m^{m}m$			✓	✓	1, 2
		240.2.560	${}^{\bar{1}}4/^{2_z}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}4/^{2_z}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2, 4
241	${}^{\bar{1}}4/^{m}m^{m}m^2m$	241.1.561	${}^{\bar{1}}4/^{m}m^{m}m^2m$			✓	✓	1, 2
		241.2.562	${}^{\bar{1}}4/^{m_z}m^{m_z}m^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}4/^{m_z}m^{m_z}m^{2_z}m^{\Sigma_x}1$		✓			2
242	${}^{m}4/^{2}m^2m^{\bar{1}}m$	242.1.563	${}^{m}4/^{2}m^2m^{\bar{1}}m$			✓	✓	1, 2
		242.2.564	${}^{m_z}4/^{2_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}4/^{2_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
243	${}^{m}4/^{\bar{1}}m^{\bar{1}}m^2m$	243.1.565	${}^{m}4/^{\bar{1}}m^{\bar{1}}m^2m$			✓	✓	1, 2
		243.2.566	${}^{m_z}4/^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\infty_x}1 \to {}^{m_z}4/^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\Sigma_x}1$		✓			2, 4

244	${}^{4_z}4/^{1}m^{2_x}m^{2_{xy}}m$	244.1.567	${}^{4_z}4/^{1}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
		244.2.568	${}^{4_z}4/^{1}m^{2_x}m^{2_{xy}}m^{m_z}1$			✓		1, 2
245	${}^{4_z}4/^{1}m^{m_x}m^{m_{xy}}m$	245.1.569	${}^{4_z}4/^{1}m^{m_x}m^{m_{xy}}m$			✓	✓	1
246	${}^{\bar{4}_z}4/^{1}m^{2_x}m^{m_{xy}}m$	246.1.570	${}^{\bar{4}_z}4/^{1}m^{2_x}m^{m_{xy}}m$			✓	✓	1, 2
247	${}^{4_z}4/^{2_z}m^{2_x}m^{2_{xy}}m$	247.1.571	${}^{4_z}4/^{2_z}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
		247.2.572	${}^{4_z}4/^{2_z}m^{2_x}m^{2_{xy}}m^{m_z}1$			✓		1, 2
248	${}^{4_z}4/^{2_z}m^{m_x}m^{m_{xy}}m$	248.1.573	${}^{4_z}4/^{2_z}m^{m_x}m^{m_{xy}}m$			✓	✓	1
249	${}^{\bar{4}_z}4/^{2_z}m^{m_x}m^{2_{xy}}m$	249.1.574	${}^{\bar{4}_z}4/^{2_z}m^{m_x}m^{2_{xy}}m$			✓	✓	1, 2
250	${}^{m_z}4/^{2_x}m^{m_x}m^{2_y}m$	250.1.575	${}^{m_z}4/^{2_x}m^{m_x}m^{2_y}m$			✓	✓	1, 2
251	${}^{m_z}4/^{m_x}m^{2_x}m^{m_y}m$	251.1.576	${}^{m_z}4/^{m_x}m^{2_x}m^{m_y}m$			✓	✓	1, 2
252	${}^{2_z}4/^{m_x}m^{2_x}m^{2_y}m$	252.1.577	${}^{2_z}4/^{m_x}m^{2_x}m^{2_y}m$			✓	✓	1, 2
253	${}^{m_z}4/^{2_z}m^{2_x}m^{m_y}m$	253.1.578	${}^{m_z}4/^{2_z}m^{2_x}m^{m_y}m$			✓	✓	1, 2
254	${}^{2_z}4/^{m_z}m^{2_x}m^{2_y}m$	254.1.579	${}^{2_z}4/^{m_z}m^{2_x}m^{2_y}m$			✓	✓	1, 2
255	${}^{2_z}4/^{m_z}m^{m_x}m^{m_y}m$	255.1.580	${}^{2_z}4/^{m_z}m^{m_x}m^{m_y}m$			✓	✓	1, 2
256	${}^{m_z}4/^{2_x}m^{2_z}m^{\bar{1}}m$	256.1.581	${}^{m_z}4/^{2_x}m^{2_z}m^{\bar{1}}m$			✓	✓	1, 2
257	${}^{m_z}4/^{m_x}m^{\bar{1}}m^{2_z}m$	257.1.582	${}^{m_z}4/^{m_x}m^{\bar{1}}m^{2_z}m$			✓	✓	1, 2
258	${}^{m_z}4/^{\bar{1}}m^{m_x}m^{2_y}m$	258.1.583	${}^{m_z}4/^{\bar{1}}m^{m_x}m^{2_y}m$			✓	✓	1, 2
259	${}^{2_z}4/^{2_x}m^{m_x}m^{m_y}m$	259.1.584	${}^{2_z}4/^{2_x}m^{m_x}m^{m_y}m$			✓	✓	1
260	${}^{2_z}4/^{2_x}m^{\bar{1}}m^{m_z}m$	260.1.585	${}^{2_z}4/^{2_x}m^{\bar{1}}m^{m_z}m$			✓	✓	1
261	${}^{2_z}4/^{\bar{1}}m^{m_x}m^{m_y}m$	261.1.586	${}^{2_z}4/^{\bar{1}}m^{m_x}m^{m_y}m$			✓	✓	1, 2
262	${}^{2_z}4/^{m_x}m^{\bar{1}}m^{m_z}m$	262.1.587	${}^{2_z}4/^{m_x}m^{\bar{1}}m^{m_z}m$			✓	✓	1, 2
263	${}^{2_z}4/^{\bar{1}}m^{2_x}m^{2_y}m$	263.1.588	${}^{2_z}4/^{\bar{1}}m^{2_x}m^{2_y}m$			✓	✓	1, 2
264	${}^{\bar{1}}4/^{2_x}m^{2_z}m^{m_z}m$	264.1.589	${}^{\bar{1}}4/^{2_x}m^{2_z}m^{m_z}m$			✓	✓	1, 2
265	${}^{\bar{1}}4/^{m_x}m^{m_z}m^{2_z}m$	265.1.590	${}^{\bar{1}}4/^{m_x}m^{m_z}m^{2_z}m$			✓	✓	1, 2
266	${}^{4_z}4/^{m_z}m^{m_x}m^{m_{xy}}m$	266.1.591	${}^{4_z}4/^{m_z}m^{m_x}m^{m_{xy}}m$			✓	✓	1, 2
267	${}^{4_z}4/^{\bar{1}}m^{m_x}m^{m_{xy}}m$	267.1.592	${}^{4_z}4/^{\bar{1}}m^{m_x}m^{m_{xy}}m$			✓	✓	1, 2
268	${}^{4_z}4/^{m_z}m^{2_x}m^{2_{xy}}m$	268.1.593	${}^{4_z}4/^{m_z}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
269	${}^{4_z}4/^{\bar{1}}m^{2_x}m^{2_{xy}}m$	269.1.594	${}^{4_z}4/^{\bar{1}}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
270	${}^{\bar{4}_z}4/^{m_z}m^{m_x}m^{2_{xy}}m$	270.1.595	${}^{\bar{4}_z}4/^{m_z}m^{m_x}m^{2_{xy}}m$			✓	✓	1, 2
271	${}^{\bar{4}_z}4/^{\bar{1}}m^{m_x}m^{2_{xy}}m$	271.1.596	${}^{\bar{4}_z}4/^{\bar{1}}m^{m_x}m^{2_{xy}}m$			✓	✓	1, 2
272	${}^{1}3$	272.1.597	${}^{1}3$			✓	✓	1
		272.2.598	${}^{1}3^{m_z}1$			✓		1, 2

		272.3.599	${}^{1}3^{\infty_z}1 \to {}^{1}3$	✓	✓			1
		272.4.600	${}^{1}3^{\infty_z m_x}1 \to {}^{1}3^{m_x}1$	✓	✓			1, 2
273	${}^{3}3$	273.1.601	${}^{3}3$			✓	✓	1
		273.2.602	${}^{3_z}3^{m_z}1$			✓		1, 2
274	${}^{1}\bar{3}$	274.1.603	${}^{1}\bar{3}$			✓	✓	1
		274.2.604	${}^{1}\bar{3}^{m_z}1$			✓		1, 2
		274.3.605	${}^{1}\bar{3}^{\infty_z}1 \to {}^{1}\bar{3}$	✓	✓			1
		274.4.606	${}^{1}\bar{3}^{\infty_z m_x}1 \to {}^{1}\bar{3}^{m_x}1$	✓	✓			1, 2
275	${}^{2}\bar{3}$	275.1.607	${}^{2}\bar{3}$			✓	✓	1
		275.2.608	${}^{2_z}\bar{3}^{m_z}1$			✓		1, 2
		275.3.609	${}^{2_z}\bar{3}^{m_x}1$			✓		1, 2
		275.4.610	${}^{2_z}\bar{3}^{\infty_x}1 \to {}^{2_z}\bar{3}^{\Sigma_x}1$		✓			2
276	${}^{m}\bar{3}$	276.1.611	${}^{m}\bar{3}$			✓	✓	1, 2
		276.2.612	${}^{m_z}\bar{3}^{\infty_x}1 \to {}^{m}\bar{3}$	✓	✓			1, 2
277	${}^{\bar{1}}\bar{3}$	277.1.613	${}^{\bar{1}}\bar{3}$			✓	✓	1, 2
		277.2.614	${}^{\bar{1}}\bar{3}^{\infty_z}1 \to {}^{\bar{1}}\bar{3}^{\Sigma_z}1$		✓			2
		277.3.615	${}^{\bar{1}}\bar{3}^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{3}^{\Sigma_z m_x}1$		✓			2, 4
278	${}^{3}\bar{3}$	278.1.616	${}^{3}\bar{3}$			✓	✓	1
		278.2.617	${}^{3_z}\bar{3}^{m_z}1$			✓		1, 2
279	${}^{6}\bar{3}$	279.1.618	${}^{6}\bar{3}$			✓	✓	1
		279.2.619	${}^{6_z}\bar{3}^{m_z}1$			✓		1, 2
280	${}^{\bar{3}}\bar{3}$	280.1.620	${}^{\bar{3}}\bar{3}$			✓	✓	1, 2
281	${}^{\bar{6}}\bar{3}$	281.1.621	${}^{\bar{6}}\bar{3}$			✓	✓	1, 2
282	${}^{1}3_1{}^{1}2$	282.1.622	${}^{1}3_1{}^{1}2$			✓	✓	1, 2
		282.2.623	${}^{1}3_1{}^{1}2^{m_z}1$			✓		1, 2
		282.3.624	${}^{1}3_1{}^{1}2^{\infty_z}1 \to {}^{1}3_1{}^{1}2$	✓	✓			1, 2
		282.4.625	${}^{1}3_1{}^{1}2^{\infty_z m_x}1 \to {}^{1}3_1{}^{1}2^{m_x}1$	✓	✓			1, 2
283	${}^{1}3_1{}^{2}2$	283.1.626	${}^{1}3_1{}^{2}2$			✓	✓	1, 2
		283.2.627	${}^{1}3_1{}^{2_z}2^{m_z}1$			✓		1, 2
		283.3.628	${}^{1}3_1{}^{2_z}2^{m_x}1$			✓		1, 2
		283.4.629	${}^{1}3_1{}^{2_z}2^{\infty_x}1 \to {}^{1}3_1{}^{2_z}2^{\Sigma_x}1$		✓			2
284	${}^{1}3_1{}^{m}2$	284.1.630	${}^{1}3_1{}^{m}2$			✓	✓	1

		284.2.631	${}^{1}3_{1}{}^{m_z}2^{\infty_x}1 \to {}^{1}3_{1}{}^{m}2$	✓	✓			1
285	${}^{1}3_{1}{}^{\bar{1}}2$	285.1.632	${}^{1}3_{1}{}^{\bar{1}}2$			✓	✓	1
		285.2.633	${}^{1}3_{1}{}^{\bar{1}}2^{\infty_z}1 \to {}^{1}3_{1}{}^{\bar{1}}2^{\Sigma_z}1$		✓			2
		285.3.634	${}^{1}3_{1}{}^{\bar{1}}2^{\infty_z m_x}1 \to {}^{1}3_{1}{}^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4
286	${}^{3_z}3_{1}{}^{2_x}2$	286.1.635	${}^{3_z}3_{1}{}^{2_x}2$			✓	✓	1, 2
		286.2.636	${}^{3_z}3_{1}{}^{2_x}2^{m_z}1$			✓		1, 2
287	${}^{3_z}3_{1}{}^{m_x}2$	287.1.637	${}^{3_z}3_{1}{}^{m_x}2$			✓	✓	1
288	${}^{1}3^{1}m_1$	288.1.638	${}^{1}3^{1}m_1$			✓	✓	1, 2
		288.2.639	${}^{1}3^{1}m_1{}^{m_z}1$			✓		1, 2
		288.3.640	${}^{1}3^{1}m_1{}^{\infty_z}1 \to {}^{1}3^{1}m_1$	✓	✓			1, 2
		288.4.641	${}^{1}3^{1}m_1{}^{\infty_z m_x}1 \to {}^{1}3^{1}m_1{}^{m_x}1$	✓	✓			1, 2
289	${}^{1}3^{2}m_1$	289.1.642	${}^{1}3^{2}m_1$			✓	✓	1, 2
		289.2.643	${}^{1}3^{2_z}m_1{}^{m_z}1$			✓		1, 2
		289.3.644	${}^{1}3^{2_z}m_1{}^{m_x}1$			✓		1, 2
		289.4.645	${}^{1}3^{2_z}m_1{}^{\infty_x}1 \to {}^{1}3^{2_z}m_1{}^{\Sigma_x}1$		✓			2
290	${}^{1}3^{m}m_1$	290.1.646	${}^{1}3^{m}m_1$			✓	✓	1
		290.2.647	${}^{1}3^{m_z}m_1{}^{\infty_x}1 \to {}^{1}3^{m}m_1$	✓	✓			1
291	${}^{1}3^{\bar{1}}m_1$	291.1.648	${}^{1}3^{\bar{1}}m_1$			✓	✓	1
		291.2.649	${}^{1}3^{\bar{1}}m_1{}^{\infty_z}1 \to {}^{1}3^{\bar{1}}m_1{}^{\Sigma_z}1$		✓			2
		291.3.650	${}^{1}3^{\bar{1}}m_1{}^{\infty_z m_x}1 \to {}^{1}3^{\bar{1}}m_1{}^{\Sigma_z m_x}1$		✓			2, 4
292	${}^{3_z}3^{2_x}m_1$	292.1.651	${}^{3_z}3^{2_x}m_1$			✓	✓	1, 2
		292.2.652	${}^{3_z}3^{2_x}m_1{}^{m_z}1$			✓		1, 2
293	${}^{3_z}3^{m_x}m_1$	293.1.653	${}^{3_z}3^{m_x}m_1$			✓	✓	1
294	${}^{1}\bar{3}_{1}{}^{1}m$	294.1.654	${}^{1}\bar{3}_{1}{}^{1}m$			✓	✓	1, 2
		294.2.655	${}^{1}\bar{3}_{1}{}^{1}m^{m_z}1$			✓		1, 2
		294.3.656	${}^{1}\bar{3}_{1}{}^{1}m^{\infty_z}1 \to {}^{1}\bar{3}_{1}{}^{1}m$	✓	✓			1, 2
		294.4.657	${}^{1}\bar{3}_{1}{}^{1}m^{\infty_z m_x}1 \to {}^{1}\bar{3}_{1}{}^{1}m^{m_x}1$	✓	✓			1, 2
295	${}^{1}\bar{3}_{1}{}^{2}m$	295.1.658	${}^{1}\bar{3}_{1}{}^{2}m$			✓	✓	1, 2
		295.2.659	${}^{1}\bar{3}_{1}{}^{2_z}m^{m_z}1$			✓		1, 2
		295.3.660	${}^{1}\bar{3}_{1}{}^{2_z}m^{m_x}1$			✓		1, 2
		295.4.661	${}^{1}\bar{3}_{1}{}^{2_z}m^{\infty_x}1 \to {}^{1}\bar{3}_{1}{}^{2_z}m^{\Sigma_x}1$		✓			2
296	${}^{1}\bar{3}_{1}{}^{m}m$	296.1.662	${}^{1}\bar{3}_{1}{}^{m}m$			✓	✓	1

		296.2.663	${}^{1}\bar{3}_{1}{}^{m_z}m^{\infty_x}1 \to {}^{1}\bar{3}_{1}{}^{m}m$	✓	✓			1
297	${}^{1}\bar{3}_{1}{}^{\bar{1}}m$	297.1.664	${}^{1}\bar{3}_{1}{}^{\bar{1}}m$			✓	✓	1
		297.2.665	${}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\infty_z}1 \to {}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		297.3.666	${}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
298	${}^{2}\bar{3}_{1}{}^{1}m$	298.1.667	${}^{2}\bar{3}_{1}{}^{1}m$			✓	✓	1, 2
		298.2.668	${}^{2_z}\bar{3}_{1}{}^{1}m^{m_z}1$			✓		1, 2
		298.3.669	${}^{2_z}\bar{3}_{1}{}^{1}m^{m_x}1$			✓		1, 2
		298.4.670	${}^{2_z}\bar{3}_{1}{}^{1}m^{\infty_x}1 \to {}^{2_z}\bar{3}_{1}{}^{1}m^{\Sigma_x}1$		✓			2, 4
299	${}^{m}\bar{3}_{1}{}^{1}m$	299.1.671	${}^{m}\bar{3}_{1}{}^{1}m$			✓	✓	1, 2
		299.2.672	${}^{m_z}\bar{3}_{1}{}^{1}m^{\infty_x}1 \to {}^{m}\bar{3}_{1}{}^{1}m$	✓	✓			1, 2
300	${}^{\bar{1}}\bar{3}_{1}{}^{1}m$	300.1.673	${}^{\bar{1}}\bar{3}_{1}{}^{1}m$			✓	✓	1, 2
		300.2.674	${}^{\bar{1}}\bar{3}_{1}{}^{1}m^{\infty_z}1 \to {}^{\bar{1}}\bar{3}_{1}{}^{1}m^{\Sigma_z}1$		✓			2, 4
		300.3.675	${}^{\bar{1}}\bar{3}_{1}{}^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{3}_{1}{}^{1}m^{\Sigma_z m_x}1$		✓			2, 4
301	${}^{2}\bar{3}_{1}{}^{2}m$	301.1.676	${}^{2}\bar{3}_{1}{}^{2}m$			✓	✓	1, 2
		301.2.677	${}^{2_z}\bar{3}_{1}{}^{2_z}m^{m_z}1$			✓		1, 2
		301.3.678	${}^{2_z}\bar{3}_{1}{}^{2_z}m^{m_x}1$			✓		1, 2
		301.4.679	${}^{2_z}\bar{3}_{1}{}^{2_z}m^{\infty_x}1 \to {}^{2_z}\bar{3}_{1}{}^{2_z}m^{\Sigma_x}1$		✓			2, 4
302	${}^{m}\bar{3}_{1}{}^{m}m$	302.1.680	${}^{m}\bar{3}_{1}{}^{m}m$			✓	✓	1, 2
		302.2.681	${}^{m_z}\bar{3}_{1}{}^{m_z}m^{\infty_x}1 \to {}^{m}\bar{3}_{1}{}^{m}m$	✓	✓			1, 2
303	${}^{\bar{1}}\bar{3}_{1}{}^{\bar{1}}m$	303.1.682	${}^{\bar{1}}\bar{3}_{1}{}^{\bar{1}}m$			✓	✓	1, 2
		303.2.683	${}^{\bar{1}}\bar{3}_{1}{}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}\bar{3}_{1}{}^{\bar{1}}m^{\Sigma_z}1$		✓			2, 4
		303.3.684	${}^{\bar{1}}\bar{3}_{1}{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{3}_{1}{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
304	${}^{2_x}\bar{3}_{1}{}^{2_z}m$	304.1.685	${}^{2_x}\bar{3}_{1}{}^{2_z}m$			✓	✓	1, 2
		304.2.686	${}^{2_x}\bar{3}_{1}{}^{2_z}m^{m_x}1$			✓		1, 2
		304.3.687	${}^{2_x}\bar{3}_{1}{}^{2_z}m^{m_z}1$			✓		1, 2
		304.4.688	${}^{2_x}\bar{3}_{1}{}^{2_z}m^{m_y}1$			✓		1, 2
305	${}^{2_z}\bar{3}_{1}{}^{m_x}m$	305.1.689	${}^{2_z}\bar{3}_{1}{}^{m_x}m$			✓	✓	1
306	${}^{m_x}\bar{3}_{1}{}^{m_y}m$	306.1.690	${}^{m_x}\bar{3}_{1}{}^{m_y}m$			✓	✓	1, 2
307	${}^{m_x}\bar{3}_{1}{}^{2_z}m$	307.1.691	${}^{m_x}\bar{3}_{1}{}^{2_z}m$			✓	✓	1, 2
308	${}^{2}\bar{3}_{1}{}^{m}m$	308.1.692	${}^{2}\bar{3}_{1}{}^{m}m$			✓	✓	1
		308.2.693	${}^{2_z}\bar{3}_{1}{}^{m_z}m^{\infty_x}1 \to {}^{2_z}\bar{3}_{1}{}^{m_z}m^{\Sigma_x}1$		✓			2
309	${}^{2}\bar{3}_{1}{}^{\bar{1}}m$	309.1.694	${}^{2}\bar{3}_{1}{}^{\bar{1}}m$			✓	✓	1

		309.2.695	${}^{2_z}\bar{3}_1{}^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}\bar{3}_1{}^{\bar{1}}m^{\Sigma_x}1$		✓			2
310	${}^{\bar{1}}\bar{3}_1{}^{m}m$	310.1.696	${}^{\bar{1}}\bar{3}_1{}^{m}m$			✓	✓	1, 2
		310.2.697	${}^{\bar{1}}\bar{3}_1{}^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}\bar{3}_1{}^{m_z}m^{\Sigma_x}1$		✓			2
311	${}^{\bar{1}}\bar{3}_1{}^{2}m$	311.1.698	${}^{\bar{1}}\bar{3}_1{}^{2}m$			✓	✓	1, 2
		311.2.699	${}^{\bar{1}}\bar{3}_1{}^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}\bar{3}_1{}^{2_z}m^{\Sigma_x}1$		✓			2
312	${}^{m}\bar{3}_1{}^{\bar{1}}m$	312.1.700	${}^{m}\bar{3}_1{}^{\bar{1}}m$			✓	✓	1, 2
		312.2.701	${}^{m_z}\bar{3}_1{}^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}\bar{3}_1{}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
313	${}^{m}\bar{3}_1{}^{2}m$	313.1.702	${}^{m}\bar{3}_1{}^{2}m$			✓	✓	1, 2
		313.2.703	${}^{m_z}\bar{3}_1{}^{2_z}m^{\infty_x}1 \to {}^{m_z}\bar{3}_1{}^{2_z}m^{\Sigma_x}1$		✓			2, 4
314	${}^{3_z}\bar{3}_1{}^{2_x}m$	314.1.704	${}^{3_z}\bar{3}_1{}^{2_x}m$			✓	✓	1, 2
		314.2.705	${}^{3_z}\bar{3}_1{}^{2_x}m^{m_z}1$			✓		1, 2
315	${}^{3_z}\bar{3}_1{}^{m_x}m$	315.1.706	${}^{3_z}\bar{3}_1{}^{m_x}m$			✓	✓	1
316	${}^{6_z}\bar{3}_1{}^{2_x}m$	316.1.707	${}^{6_z}\bar{3}_1{}^{2_x}m$			✓	✓	1, 2
		316.2.708	${}^{6_z}\bar{3}_1{}^{2_x}m^{m_z}1$			✓		1, 2
317	${}^{\bar{3}_z}\bar{3}_1{}^{m_x}m$	317.1.709	${}^{\bar{3}_z}\bar{3}_1{}^{m_x}m$			✓	✓	1, 2
318	${}^{\bar{3}_z}\bar{3}_1{}^{2_x}m$	318.1.710	${}^{\bar{3}_z}\bar{3}_1{}^{2_x}m$			✓	✓	1, 2
319	${}^{6_z}\bar{3}_1{}^{m_x}m$	319.1.711	${}^{6_z}\bar{3}_1{}^{m_x}m$			✓	✓	1
320	${}^{\bar{6}_z}\bar{3}_1{}^{m_x}m$	320.1.712	${}^{\bar{6}_z}\bar{3}_1{}^{m_x}m$			✓	✓	1, 2
321	${}^{\bar{6}_z}\bar{3}_1{}^{2_x}m$	321.1.713	${}^{\bar{6}_z}\bar{3}_1{}^{2_x}m$			✓	✓	1, 2
322	${}^{1}6$	322.1.714	${}^{1}6$			✓	✓	1
		322.2.715	${}^{1}6^{m_z}1$			✓		1, 2
		322.3.716	${}^{1}6^{\infty_z}1 \to {}^{1}6$	✓	✓			1
		322.4.717	${}^{1}6^{\infty_z m_x}1 \to {}^{1}6^{m_x}1$	✓	✓			1, 2
323	${}^{2}6$	323.1.718	${}^{2}6$			✓	✓	1
		323.2.719	${}^{2_z}6^{m_z}1$			✓		1, 2
		323.3.720	${}^{2_z}6^{m_x}1$			✓		1, 2
		323.4.721	${}^{2_z}6^{\infty_x}1 \to {}^{2_z}6^{\Sigma_x}1$		✓			2
324	${}^{m}6$	324.1.722	${}^{m}6$			✓	✓	1, 2
		324.2.723	${}^{m_z}6^{\infty_x}1 \to {}^{m}6$	✓	✓			1, 2
325	${}^{\bar{1}}6$	325.1.724	${}^{\bar{1}}6$			✓	✓	1, 2
		325.2.725	${}^{\bar{1}}6^{\infty_z}1 \to {}^{\bar{1}}6^{\Sigma_z}1$		✓			2
		325.3.726	${}^{\bar{1}}6^{\infty_z m_x}1 \to {}^{\bar{1}}6^{\Sigma_z m_x}1$		✓			2, 4

326	${}^{3}6$	326.1.727	${}^{3}6$			✓	✓	1
		326.2.728	${}^{3_z}6^{m_z}1$			✓		1, 2
327	${}^{6}6$	327.1.729	${}^{6}6$			✓	✓	1
		327.2.730	${}^{6_z}6^{m_z}1$			✓		1, 2
328	${}^{\bar{3}}6$	328.1.731	${}^{\bar{3}}6$			✓	✓	1, 2
329	${}^{\bar{6}}6$	329.1.732	${}^{\bar{6}}6$			✓	✓	1, 2
330	${}^{1}\bar{6}$	330.1.733	${}^{1}\bar{6}$			✓	✓	1
		330.2.734	${}^{1}\bar{6}^{m_z}1$			✓		1, 2
		330.3.735	${}^{1}\bar{6}^{\infty_z}1 \to {}^{1}\bar{6}$	✓	✓			1
		330.4.736	${}^{1}\bar{6}^{\infty_z m_x}1 \to {}^{1}\bar{6}^{m_x}1$	✓	✓			1, 2
331	${}^{2}\bar{6}$	331.1.737	${}^{2}\bar{6}$			✓	✓	1
		331.2.738	${}^{2_z}\bar{6}^{m_z}1$			✓		1, 2
		331.3.739	${}^{2_z}\bar{6}^{m_x}1$			✓		1, 2
		331.4.740	${}^{2_z}\bar{6}^{\infty_x}1 \to {}^{2_z}\bar{6}^{\Sigma_x}1$		✓			2
332	${}^{m}\bar{6}$	332.1.741	${}^{m}\bar{6}$			✓	✓	1, 2
		332.2.742	${}^{m_z}\bar{6}^{\infty_x}1 \to {}^{m}\bar{6}$	✓	✓			1, 2
333	${}^{\bar{1}}\bar{6}$	333.1.743	${}^{\bar{1}}\bar{6}$			✓	✓	1, 2
		333.2.744	${}^{\bar{1}}\bar{6}^{\infty_z}1 \to {}^{\bar{1}}\bar{6}^{\Sigma_z}1$		✓			2
		333.3.745	${}^{\bar{1}}\bar{6}^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{6}^{\Sigma_z m_x}1$		✓			2, 4
334	${}^{3}\bar{6}$	334.1.746	${}^{3}\bar{6}$			✓	✓	1
		334.2.747	${}^{3_z}\bar{6}^{m_z}1$			✓		1, 2
335	${}^{6}\bar{6}$	335.1.748	${}^{6}\bar{6}$			✓	✓	1
		335.2.749	${}^{6_z}\bar{6}^{m_z}1$			✓		1, 2
336	${}^{\bar{3}}\bar{6}$	336.1.750	${}^{\bar{3}}\bar{6}$			✓	✓	1, 2
337	${}^{\bar{6}}\bar{6}$	337.1.751	${}^{\bar{6}}\bar{6}$			✓	✓	1, 2
338	${}^{1}6/{}^{1}m$	338.1.752	${}^{1}6/{}^{1}m$			✓	✓	1
		338.2.753	${}^{1}6/{}^{1}m^{m_z}1$			✓		1, 2
		338.3.754	${}^{1}6/{}^{1}m^{\infty_z}1 \to {}^{1}6/{}^{1}m$	✓	✓			1
		338.4.755	${}^{1}6/{}^{1}m^{\infty_z m_x}1 \to {}^{1}6/{}^{1}m^{m_x}1$	✓	✓			1, 2
339	${}^{2}6/{}^{2}m$	339.1.756	${}^{2}6/{}^{2}m$			✓	✓	1
		339.2.757	${}^{2_z}6/{}^{2_z}m^{m_z}1$			✓		1, 2
		339.3.758	${}^{2_z}6/{}^{2_z}m^{m_x}1$			✓		1, 2

		339.4.759	${}^{2_z}6/{}^{2_z}m^{\infty_x}1 \to {}^{2_z}6/{}^{2_z}m^{\Sigma_x}1$		✓			2
340	${}^{m}6/{}^{m}m$	340.1.760	${}^{m}6/{}^{m}m$			✓	✓	1, 2
		340.2.761	${}^{m_z}6/{}^{m_z}m^{\infty_x}1 \to {}^{m}6/{}^{m}m$	✓	✓			1, 2
341	${}^{\bar{1}}6/{}^{\bar{1}}m$	341.1.762	${}^{\bar{1}}6/{}^{\bar{1}}m$			✓	✓	1, 2
		341.2.763	${}^{\bar{1}}6/{}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}6/{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		341.3.764	${}^{\bar{1}}6/{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}6/{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
342	${}^{2}6/{}^{1}m$	342.1.765	${}^{2}6/{}^{1}m$			✓	✓	1
		342.2.766	${}^{2_z}6/{}^{1}m^{m_z}1$			✓		1, 2
		342.3.767	${}^{2_z}6/{}^{1}m^{m_x}1$			✓		1, 2
		342.4.768	${}^{2_z}6/{}^{1}m^{\infty_x}1 \to {}^{2_z}6/{}^{1}m^{\Sigma_x}1$		✓			2
343	${}^{m}6/{}^{1}m$	343.1.769	${}^{m}6/{}^{1}m$			✓	✓	1, 2
		343.2.770	${}^{m_z}6/{}^{1}m^{\infty_x}1 \to {}^{m}6/{}^{1}m$	✓	✓			1, 2
344	${}^{\bar{1}}6/{}^{1}m$	344.1.771	${}^{\bar{1}}6/{}^{1}m$			✓	✓	1, 2
		344.2.772	${}^{\bar{1}}6/{}^{1}m^{\infty_z}1 \to {}^{\bar{1}}6/{}^{1}m^{\Sigma_z}1$		✓			2
		344.3.773	${}^{\bar{1}}6/{}^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}6/{}^{1}m^{\Sigma_z m_x}1$		✓			2, 4
345	${}^{1}6/{}^{2}m$	345.1.774	${}^{1}6/{}^{2}m$			✓	✓	1
		345.2.775	${}^{1}6/{}^{2_z}m^{m_z}1$			✓		1, 2
		345.3.776	${}^{1}6/{}^{2_z}m^{m_x}1$			✓		1, 2
		345.4.777	${}^{1}6/{}^{2_z}m^{\infty_x}1 \to {}^{1}6/{}^{2_z}m^{\Sigma_x}1$		✓			2
346	${}^{1}6/{}^{m}m$	346.1.778	${}^{1}6/{}^{m}m$			✓	✓	1, 2
		346.2.779	${}^{1}6/{}^{m_z}m^{\infty_x}1 \to {}^{1}6/{}^{m}m$	✓	✓			1, 2
347	${}^{1}6/{}^{\bar{1}}m$	347.1.780	${}^{1}6/{}^{\bar{1}}m$			✓	✓	1, 2
		347.2.781	${}^{1}6/{}^{\bar{1}}m^{\infty_z}1 \to {}^{1}6/{}^{\bar{1}}m^{\Sigma_z}1$		✓			2
		347.3.782	${}^{1}6/{}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}6/{}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
348	${}^{3}6/{}^{1}m$	348.1.783	${}^{3}6/{}^{1}m$			✓	✓	1
		348.2.784	${}^{3_z}6/{}^{1}m^{m_z}1$			✓		1, 2
349	${}^{2_z}6/{}^{2_x}m$	349.1.785	${}^{2_z}6/{}^{2_x}m$			✓	✓	1
		349.2.786	${}^{2_z}6/{}^{2_x}m^{m_z}1$			✓		1, 2
		349.3.787	${}^{2_z}6/{}^{2_x}m^{m_x}1$			✓		1, 2
		349.4.788	${}^{2_z}6/{}^{2_x}m^{m_y}1$			✓		1, 2
350	${}^{2_z}6/{}^{m_x}m$	350.1.789	${}^{2_z}6/{}^{m_x}m$			✓	✓	1, 2
351	${}^{m_x}6/{}^{2_z}m$	351.1.790	${}^{m_x}6/{}^{2_z}m$			✓	✓	1, 2

352	${}^{m_x}6/{}^{m_y}m$	352.1.791	${}^{m_x}6/{}^{m_y}m$			✓	✓	1, 2
353	${}^{2}6/{}^{\bar{1}}m$	353.1.792	${}^{2}6/{}^{\bar{1}}m$			✓	✓	1, 2
		353.2.793	${}^{2_z}6/{}^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}6/{}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
354	${}^{2}6/{}^{m}m$	354.1.794	${}^{2}6/{}^{m}m$			✓	✓	1, 2
		354.2.795	${}^{2_z}6/{}^{m_z}m^{\infty_x}1 \to {}^{2_z}6/{}^{m_z}m^{\Sigma_x}1$		✓			2, 4
355	${}^{\bar{1}}6/{}^{2}m$	355.1.796	${}^{\bar{1}}6/{}^{2}m$			✓	✓	1, 2
		355.2.797	${}^{\bar{1}}6/{}^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}6/{}^{2_z}m^{\Sigma_x}1$		✓			2, 4
356	${}^{\bar{1}}6/{}^{m}m$	356.1.798	${}^{\bar{1}}6/{}^{m}m$			✓	✓	1, 2
		356.2.799	${}^{\bar{1}}6/{}^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}6/{}^{m_z}m^{\Sigma_x}1$		✓			2, 4
357	${}^{m}6/{}^{2}m$	357.1.800	${}^{m}6/{}^{2}m$			✓	✓	1, 2
		357.2.801	${}^{m_z}6/{}^{2_z}m^{\infty_x}1 \to {}^{m_z}6/{}^{2_z}m^{\Sigma_x}1$		✓			2, 4
358	${}^{m}6/{}^{\bar{1}}m$	358.1.802	${}^{m}6/{}^{\bar{1}}m$			✓	✓	1, 2
		358.2.803	${}^{m_z}6/{}^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}6/{}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
359	${}^{3}6/{}^{2}m$	359.1.804	${}^{3}6/{}^{2}m$			✓	✓	1
		359.2.805	${}^{3_z}6/{}^{2_z}m^{m_z}1$			✓		1, 2
360	${}^{3}6/{}^{\bar{1}}m$	360.1.806	${}^{3}6/{}^{\bar{1}}m$			✓	✓	1, 2
361	${}^{3}6/{}^{m}m$	361.1.807	${}^{3}6/{}^{m}m$			✓	✓	1, 2
362	${}^{6}6/{}^{1}m$	362.1.808	${}^{6}6/{}^{1}m$			✓	✓	1
		362.2.809	${}^{6_z}6/{}^{1}m^{m_z}1$			✓		1, 2
363	${}^{\bar{3}}6/{}^{1}m$	363.1.810	${}^{\bar{3}}6/{}^{1}m$			✓	✓	1, 2
364	${}^{\bar{6}}6/{}^{1}m$	364.1.811	${}^{\bar{6}}6/{}^{1}m$			✓	✓	1, 2
365	${}^{6}6/{}^{2}m$	365.1.812	${}^{6}6/{}^{2}m$			✓	✓	1
		365.2.813	${}^{6_z}6/{}^{2_z}m^{m_z}1$			✓		1, 2
366	${}^{\bar{3}}6/{}^{\bar{1}}m$	366.1.814	${}^{\bar{3}}6/{}^{\bar{1}}m$			✓	✓	1, 2
367	${}^{\bar{6}}6/{}^{m}m$	367.1.815	${}^{\bar{6}}6/{}^{m}m$			✓	✓	1, 2
368	${}^{6}6/{}^{m}m$	368.1.816	${}^{6}6/{}^{m}m$			✓	✓	1, 2
369	${}^{6}6/{}^{\bar{1}}m$	369.1.817	${}^{6}6/{}^{\bar{1}}m$			✓	✓	1, 2
370	${}^{\bar{3}}6/{}^{m}m$	370.1.818	${}^{\bar{3}}6/{}^{m}m$			✓	✓	1, 2
371	${}^{\bar{3}}6/{}^{2}m$	371.1.819	${}^{\bar{3}}6/{}^{2}m$			✓	✓	1, 2
372	${}^{\bar{6}}6/{}^{\bar{1}}m$	372.1.820	${}^{\bar{6}}6/{}^{\bar{1}}m$			✓	✓	1, 2
373	${}^{\bar{6}}6/{}^{2}m$	373.1.821	${}^{\bar{6}}6/{}^{2}m$			✓	✓	1, 2
374	${}^{1}6{}^{1}2{}^{1}2$	374.1.822	${}^{1}6{}^{1}2{}^{1}2$			✓	✓	1, 2

		374.2.823	${}^{1}6^{1}2^{1}2^{m_z}1$			✓		1, 2
		374.3.824	${}^{1}6^{1}2^{1}2^{\infty_z}1 \to {}^{1}6^{1}2^{1}2$	✓	✓			1, 2
		374.4.825	${}^{1}6^{1}2^{1}2^{\infty_z m_x}1 \to {}^{1}6^{1}2^{1}2^{m_x}1$	✓	✓			1, 2
375	${}^{1}6^{2}2^{2}2$	375.1.826	${}^{1}6^{2}2^{2}2$			✓	✓	1, 2
		375.2.827	${}^{1}6^{2_z}2^{2_z}2^{m_z}1$			✓		1, 2
		375.3.828	${}^{1}6^{2_z}2^{2_z}2^{m_x}1$			✓		1, 2
		375.4.829	${}^{1}6^{2_z}2^{2_z}2^{\infty_x}1 \to {}^{1}6^{2_z}2^{2_z}2^{\Sigma_x}1$		✓			2
376	${}^{1}6^{m}2^{m}2$	376.1.830	${}^{1}6^{m}2^{m}2$			✓	✓	1
		376.2.831	${}^{1}6^{m_z}2^{m_z}2^{\infty_x}1 \to {}^{1}6^{m}2^{m}2$	✓	✓			1
377	${}^{1}6^{\bar{1}}2^{\bar{1}}2$	377.1.832	${}^{1}6^{\bar{1}}2^{\bar{1}}2$			✓	✓	1
		377.2.833	${}^{1}6^{\bar{1}}2^{\bar{1}}2^{\infty_z}1 \to {}^{1}6^{\bar{1}}2^{\bar{1}}2^{\Sigma_z}1$		✓			2
		377.3.834	${}^{1}6^{\bar{1}}2^{\bar{1}}2^{\infty_z m_x}1 \to {}^{1}6^{\bar{1}}2^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4
378	${}^{2}6^{1}2^{2}2$	378.1.835	${}^{2}6^{1}2^{2}2$			✓	✓	1, 2
		378.2.836	${}^{2_z}6^{1}2^{2_z}2^{m_z}1$			✓		1, 2
		378.3.837	${}^{2_z}6^{1}2^{2_z}2^{m_x}1$			✓		1, 2
		378.4.838	${}^{2_z}6^{1}2^{2_z}2^{\infty_x}1 \to {}^{2_z}6^{1}2^{2_z}2^{\Sigma_x}1$		✓			2, 4
379	${}^{m}6^{1}2^{m}2$	379.1.839	${}^{m}6^{1}2^{m}2$			✓	✓	1, 2
		379.2.840	${}^{m_z}6^{1}2^{m_z}2^{\infty_x}1 \to {}^{m}6^{1}2^{m}2$	✓	✓			1, 2
380	${}^{\bar{1}}6^{1}2^{\bar{1}}2$	380.1.841	${}^{\bar{1}}6^{1}2^{\bar{1}}2$			✓	✓	1, 2
		380.2.842	${}^{\bar{1}}6^{1}2^{\bar{1}}2^{\infty_z}1 \to {}^{\bar{1}}6^{1}2^{\bar{1}}2^{\Sigma_z}1$		✓			2, 4
		380.3.843	${}^{\bar{1}}6^{1}2^{\bar{1}}2^{\infty_z m_x}1 \to {}^{\bar{1}}6^{1}2^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4
381	${}^{2_x}6^{2_y}2^{2_z}2$	381.1.844	${}^{2_x}6^{2_y}2^{2_z}2$			✓	✓	1, 2
		381.2.845	${}^{2_x}6^{2_y}2^{2_z}2^{m_x}1$			✓		1, 2
		381.3.846	${}^{2_x}6^{2_y}2^{2_z}2^{m_y}1$			✓		1, 2
382	${}^{2_z}6^{m_x}2^{m_y}2$	382.1.847	${}^{2_z}6^{m_x}2^{m_y}2$			✓	✓	1
383	${}^{m_x}6^{2_z}2^{m_y}2$	383.1.848	${}^{m_x}6^{2_z}2^{m_y}2$			✓	✓	1, 2
384	${}^{2}6^{\bar{1}}2^{m}2$	384.1.849	${}^{2}6^{\bar{1}}2^{m}2$			✓	✓	1
		384.2.850	${}^{2_z}6^{\bar{1}}2^{m_z}2^{\infty_x}1 \to {}^{2_z}6^{\bar{1}}2^{m_z}2^{\Sigma_x}1$		✓			2
385	${}^{\bar{1}}6^{2}2^{m}2$	385.1.851	${}^{\bar{1}}6^{2}2^{m}2$			✓	✓	1, 2
		385.2.852	${}^{\bar{1}}6^{2_z}2^{m_z}2^{\infty_x}1 \to {}^{\bar{1}}6^{2_z}2^{m_z}2^{\Sigma_x}1$		✓			2
386	${}^{m}6^{2}2^{\bar{1}}2$	386.1.853	${}^{m}6^{2}2^{\bar{1}}2$			✓	✓	1, 2
		386.2.854	${}^{m_z}6^{2_z}2^{\bar{1}}2^{\infty_x}1 \to {}^{m_z}6^{2_z}2^{\bar{1}}2^{\Sigma_x}1$		✓			2, 4

387	${}^{3_z}6^{2_x}2^{2_{xy}}2$	387.1.855	${}^{3_z}6^{2_x}2^{2_{xy}}2$			✓	✓	1, 2
		387.2.856	${}^{3_z}6^{2_x}2^{2_{xy}}2^{m_z}1$			✓		1, 2
388	${}^{3_z}6^{m_x}2^{m_{xy}}2$	388.1.857	${}^{3_z}6^{m_x}2^{m_{xy}}2$			✓	✓	1
389	${}^{6_z}6^{2_x}2^{2_1}2$	389.1.858	${}^{6_z}6^{2_x}2^{2_1}2$			✓	✓	1, 2
		389.2.859	${}^{6_z}6^{2_x}2^{2_1}2^{m_z}1$			✓		1, 2
390	${}^{\bar{3}_z}6^{2_x}2^{m_{xy}}2$	390.1.860	${}^{\bar{3}_z}6^{2_x}2^{m_{xy}}2$			✓	✓	1, 2
391	${}^{6_z}6^{m_x}2^{m_1}2$	391.1.861	${}^{6_z}6^{m_x}2^{m_1}2$			✓	✓	1
392	${}^{\bar{6}_z}6^{m_x}2^{2_1}2$	392.1.862	${}^{\bar{6}_z}6^{m_x}2^{2_1}2$			✓	✓	1, 2
393	${}^{1}6^{1}m^{1}m$	393.1.863	${}^{1}6^{1}m^{1}m$			✓	✓	1, 2
		393.2.864	${}^{1}6^{1}m^{1}m^{m_z}1$			✓		1, 2
		393.3.865	${}^{1}6^{1}m^{1}m^{\infty_z}1 \to {}^{1}6^{1}m^{1}m$	✓	✓			1, 2
		393.4.866	${}^{1}6^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}6^{1}m^{1}m^{m_x}1$	✓	✓			1, 2
394	${}^{1}6^{2}m^{2}m$	394.1.867	${}^{1}6^{2}m^{2}m$			✓	✓	1, 2
		394.2.868	${}^{1}6^{2_z}m^{2_z}m^{m_z}1$			✓		1, 2
		394.3.869	${}^{1}6^{2_z}m^{2_z}m^{m_x}1$			✓		1, 2
		394.4.870	${}^{1}6^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}6^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
395	${}^{1}6^{m}m^{m}m$	395.1.871	${}^{1}6^{m}m^{m}m$			✓	✓	1
		395.2.872	${}^{1}6^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6^{m}m^{m}m$	✓	✓			1
396	${}^{1}6^{\bar{1}}m^{\bar{1}}m$	396.1.873	${}^{1}6^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		396.2.874	${}^{1}6^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}6^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		396.3.875	${}^{1}6^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}6^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
397	${}^{2}6^{1}m^{2}m$	397.1.876	${}^{2}6^{1}m^{2}m$			✓	✓	1, 2
		397.2.877	${}^{2_z}6^{1}m^{2_z}m^{m_z}1$			✓		1, 2
		397.3.878	${}^{2_z}6^{1}m^{2_z}m^{m_x}1$			✓		1, 2
		397.4.879	${}^{2_z}6^{1}m^{2_z}m^{\infty_x}1 \to {}^{2_z}6^{1}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
398	${}^{m}6^{1}m^{m}m$	398.1.880	${}^{m}6^{1}m^{m}m$			✓	✓	1, 2
		398.2.881	${}^{m_z}6^{1}m^{m_z}m^{\infty_x}1 \to {}^{m}6^{1}m^{m}m$	✓	✓			1, 2
399	${}^{\bar{1}}6^{1}m^{\bar{1}}m$	399.1.882	${}^{\bar{1}}6^{1}m^{\bar{1}}m$			✓	✓	1, 2
		399.2.883	${}^{\bar{1}}6^{1}m^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}6^{1}m^{\bar{1}}m^{\Sigma_z}1$		✓			2, 4
		399.3.884	${}^{\bar{1}}6^{1}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}6^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
400	${}^{2_x}6^{2_y}m^{2_z}m$	400.1.885	${}^{2_x}6^{2_y}m^{2_z}m$			✓	✓	1, 2
		400.2.886	${}^{2_x}6^{2_y}m^{2_z}m^{m_x}1$			✓		1, 2

		400.3.887	${}^{2_x}6^{2_y}m^{2_z}m^{m_y}1$			✓		1, 2
401	${}^{2_z}6^{m_x}m^{m_y}m$	401.1.888	${}^{2_z}6^{m_x}m^{m_y}m$			✓	✓	1
402	${}^{m_x}6^{2_z}m^{m_y}m$	402.1.889	${}^{m_x}6^{2_z}m^{m_y}m$			✓	✓	1, 2
403	${}^{2}6^{\bar{1}}m^{m}m$	403.1.890	${}^{2}6^{\bar{1}}m^{m}m$			✓	✓	1
		403.2.891	${}^{2_z}6^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}6^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
404	${}^{\bar{1}}6^{2}m^{m}m$	404.1.892	${}^{\bar{1}}6^{2}m^{m}m$			✓	✓	1, 2
		404.2.893	${}^{\bar{1}}6^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}6^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
405	${}^{m}6^{2}m^{\bar{1}}m$	405.1.894	${}^{m}6^{2}m^{\bar{1}}m$			✓	✓	1, 2
		405.2.895	${}^{m_z}6^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}6^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
406	${}^{3_z}6^{2_x}m^{2_{xy}}m$	406.1.896	${}^{3_z}6^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
		406.2.897	${}^{3_z}6^{2_x}m^{2_{xy}}m^{m_z}1$			✓		1, 2
407	${}^{3_z}6^{m_x}m^{m_{xy}}m$	407.1.898	${}^{3_z}6^{m_x}m^{m_{xy}}m$			✓	✓	1
408	${}^{6_z}6^{2_x}m^{2_1}m$	408.1.899	${}^{6_z}6^{2_x}m^{2_1}m$			✓	✓	1, 2
		408.2.900	${}^{6_z}6^{2_x}m^{2_1}m^{m_z}1$			✓		1, 2
409	${}^{\bar{3}_z}6^{2_x}m^{m_{xy}}m$	409.1.901	${}^{\bar{3}_z}6^{2_x}m^{m_{xy}}m$			✓	✓	1, 2
410	${}^{6_z}6^{m_x}m^{m_1}m$	410.1.902	${}^{6_z}6^{m_x}m^{m_1}m$			✓	✓	1
411	${}^{\bar{6}_z}6^{m_x}m^{2_1}m$	411.1.903	${}^{\bar{6}_z}6^{m_x}m^{2_1}m$			✓	✓	1, 2
412	${}^{1}\bar{6}^{1}m^{1}2$	412.1.904	${}^{1}\bar{6}^{1}m^{1}2$			✓	✓	1, 2
		412.2.905	${}^{1}\bar{6}^{1}m^{1}2^{m_z}1$			✓		1, 2
		412.3.906	${}^{1}\bar{6}^{1}m^{1}2^{\infty_z}1 \to {}^{1}\bar{6}^{1}m^{1}2$	✓	✓			1, 2
		412.4.907	${}^{1}\bar{6}^{1}m^{1}2^{\infty_z m_x}1 \to {}^{1}\bar{6}^{1}m^{1}2^{m_x}1$	✓	✓			1, 2
413	${}^{1}\bar{6}^{2}m^{2}2$	413.1.908	${}^{1}\bar{6}^{2}m^{2}2$			✓	✓	1, 2
		413.2.909	${}^{1}\bar{6}^{2_z}m^{2_z}2^{m_z}1$			✓		1, 2
		413.3.910	${}^{1}\bar{6}^{2_z}m^{2_z}2^{m_x}1$			✓		1, 2
		413.4.911	${}^{1}\bar{6}^{2_z}m^{2_z}2^{\infty_x}1 \to {}^{1}\bar{6}^{2_z}m^{2_z}2^{\Sigma_x}1$		✓			2
414	${}^{1}\bar{6}^{m}m^{m}2$	414.1.912	${}^{1}\bar{6}^{m}m^{m}2$			✓	✓	1
		414.2.913	${}^{1}\bar{6}^{m_z}m^{m_z}2^{\infty_x}1 \to {}^{1}\bar{6}^{m}m^{m}2$	✓	✓			1
415	${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2$	415.1.914	${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2$			✓	✓	1
		415.2.915	${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\infty_z}1 \to {}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\Sigma_z}1$		✓			2
		415.3.916	${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\infty_z m_x}1 \to {}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4
416	${}^{2}\bar{6}^{1}m^{2}2$	416.1.917	${}^{2}\bar{6}^{1}m^{2}2$			✓	✓	1, 2
		416.2.918	${}^{2_z}\bar{6}^{1}m^{2_z}2^{m_z}1$			✓		1, 2

		416.3.919	${}^{2_z}\bar{6}^{1}m^{2_z}2^{m_x}1$			✓		1, 2
		416.4.920	${}^{2_z}\bar{6}^{1}m^{2_z}2^{\infty_x}1 \to {}^{2_z}\bar{6}^{1}m^{2_z}2^{\Sigma_x}1$		✓			2, 4
417	${}^{m}\bar{6}^{1}m^{m}2$	417.1.921	${}^{m}\bar{6}^{1}m^{m}2$			✓	✓	1, 2
		417.2.922	${}^{m_z}\bar{6}^{1}m^{m_z}2^{\infty_x}1 \to {}^{m}\bar{6}^{1}m^{m}2$	✓	✓			1, 2
418	${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2$	418.1.923	${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2$			✓	✓	1, 2
		418.2.924	${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\infty_z}1 \to {}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\Sigma_z}1$		✓			2, 4
		418.3.925	${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4
419	${}^{2}\bar{6}^{2}m^{1}2$	419.1.926	${}^{2}\bar{6}^{2}m^{1}2$			✓	✓	1, 2
		419.2.927	${}^{2_z}\bar{6}^{2_z}m^{1}2^{m_z}1$			✓		1, 2
		419.3.928	${}^{2_z}\bar{6}^{2_z}m^{1}2^{m_x}1$			✓		1, 2
		419.4.929	${}^{2_z}\bar{6}^{2_z}m^{1}2^{\infty_x}1 \to {}^{2_z}\bar{6}^{2_z}m^{1}2^{\Sigma_x}1$		✓			2, 4
420	${}^{m}\bar{6}^{m}m^{1}2$	420.1.930	${}^{m}\bar{6}^{m}m^{1}2$			✓	✓	1, 2
		420.2.931	${}^{m_z}\bar{6}^{m_z}m^{1}2^{\infty_x}1 \to {}^{m}\bar{6}^{m}m^{1}2$	✓	✓			1, 2
421	${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2$	421.1.932	${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2$			✓	✓	1, 2
		421.2.933	${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\infty_z}1 \to {}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\Sigma_z}1$		✓			2, 4
		421.3.934	${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\Sigma_z m_x}1$		✓			2, 4
422	${}^{2_z}\bar{6}^{2_x}m^{2_y}2$	422.1.935	${}^{2_z}\bar{6}^{2_x}m^{2_y}2$			✓	✓	1, 2
		422.2.936	${}^{2_z}\bar{6}^{2_x}m^{2_y}2^{m_z}1$			✓		1, 2
		422.3.937	${}^{2_z}\bar{6}^{2_x}m^{2_y}2^{m_x}1$			✓		1, 2
		422.4.938	${}^{2_z}\bar{6}^{2_x}m^{2_y}2^{m_y}1$			✓		1, 2
423	${}^{2_z}\bar{6}^{m_x}m^{m_y}2$	423.1.939	${}^{2_z}\bar{6}^{m_x}m^{m_y}2$			✓	✓	1
424	${}^{m_x}\bar{6}^{m_y}m^{2_z}2$	424.1.940	${}^{m_x}\bar{6}^{m_y}m^{2_z}2$			✓	✓	1, 2
425	${}^{m_x}\bar{6}^{2_z}m^{m_y}2$	425.1.941	${}^{m_x}\bar{6}^{2_z}m^{m_y}2$			✓	✓	1, 2
426	${}^{2}\bar{6}^{m}m^{\bar{1}}2$	426.1.942	${}^{2}\bar{6}^{m}m^{\bar{1}}2$			✓	✓	1
		426.2.943	${}^{2_z}\bar{6}^{m_z}m^{\bar{1}}2^{\infty_x}1 \to {}^{2_z}\bar{6}^{m_z}m^{\bar{1}}2^{\Sigma_x}1$		✓			2
427	${}^{2}\bar{6}^{\bar{1}}m^{m}2$	427.1.944	${}^{2}\bar{6}^{\bar{1}}m^{m}2$			✓	✓	1
		427.2.945	${}^{2_z}\bar{6}^{\bar{1}}m^{m_z}2^{\infty_x}1 \to {}^{2_z}\bar{6}^{\bar{1}}m^{m_z}2^{\Sigma_x}1$		✓			2
428	${}^{\bar{1}}\bar{6}^{m}m^{2}2$	428.1.946	${}^{\bar{1}}\bar{6}^{m}m^{2}2$			✓	✓	1, 2
		428.2.947	${}^{\bar{1}}\bar{6}^{m_z}m^{2_z}2^{\infty_x}1 \to {}^{\bar{1}}\bar{6}^{m_z}m^{2_z}2^{\Sigma_x}1$		✓			2
429	${}^{\bar{1}}\bar{6}^{2}m^{m}2$	429.1.948	${}^{\bar{1}}\bar{6}^{2}m^{m}2$			✓	✓	1, 2
		429.2.949	${}^{\bar{1}}\bar{6}^{2_z}m^{m_z}2^{\infty_x}1 \to {}^{\bar{1}}\bar{6}^{2_z}m^{m_z}2^{\Sigma_x}1$		✓			2
430	${}^{m}\bar{6}^{\bar{1}}m^{2}2$	430.1.950	${}^{m}\bar{6}^{\bar{1}}m^{2}2$			✓	✓	1, 2

		430.2.951	${}^{m_z}\bar{6}^{\bar{1}}m^{2_z}2^{\infty_x}1 \to {}^{m_z}\bar{6}^{\bar{1}}m^{2_z}2^{\Sigma_x}1$		✓			2, 4
431	${}^{m}\bar{6}^{2}m^{\bar{1}}2$	431.1.952	${}^{m}\bar{6}^{2}m^{\bar{1}}2$			✓	✓	1, 2
		431.2.953	${}^{m_z}\bar{6}^{2_z}m^{\bar{1}}2^{\infty_x}1 \to {}^{m_z}\bar{6}^{2_z}m^{\bar{1}}2^{\Sigma_x}1$		✓			2, 4
432	${}^{3_z}\bar{6}^{2_x}m^{2_{xy}}2$	432.1.954	${}^{3_z}\bar{6}^{2_x}m^{2_{xy}}2$			✓	✓	1, 2
		432.2.955	${}^{3_z}\bar{6}^{2_x}m^{2_{xy}}2^{m_z}1$			✓		1, 2
433	${}^{3_z}\bar{6}^{m_x}m^{m_{xy}}2$	433.1.956	${}^{3_z}\bar{6}^{m_x}m^{m_{xy}}2$			✓	✓	1
434	${}^{6_z}\bar{6}^{2_x}m^{2_1}2$	434.1.957	${}^{6_z}\bar{6}^{2_x}m^{2_1}2$			✓	✓	1, 2
		434.2.958	${}^{6_z}\bar{6}^{2_x}m^{2_1}2^{m_z}1$			✓		1, 2
435	${}^{\bar{3}_z}\bar{6}^{m_x}m^{2_{xy}}2$	435.1.959	${}^{\bar{3}_z}\bar{6}^{m_x}m^{2_{xy}}2$			✓	✓	1, 2
436	${}^{\bar{3}_z}\bar{6}^{2_x}m^{m_{xy}}2$	436.1.960	${}^{\bar{3}_z}\bar{6}^{2_x}m^{m_{xy}}2$			✓	✓	1, 2
437	${}^{6_z}\bar{6}^{m_x}m^{m_1}2$	437.1.961	${}^{6_z}\bar{6}^{m_x}m^{m_1}2$			✓	✓	1
438	${}^{\bar{6}_z}\bar{6}^{m_x}m^{2_1}2$	438.1.962	${}^{\bar{6}_z}\bar{6}^{m_x}m^{2_1}2$			✓	✓	1, 2
439	${}^{\bar{6}_z}\bar{6}^{2_x}m^{m_1}2$	439.1.963	${}^{\bar{6}_z}\bar{6}^{2_x}m^{m_1}2$			✓	✓	1, 2
440	${}^{1}6/^{1}m^{1}m^{1}m$	440.1.964	${}^{1}6/^{1}m^{1}m^{1}m$			✓	✓	1, 2
		440.2.965	${}^{1}6/^{1}m^{1}m^{1}m^{m_z}1$			✓		1, 2
		440.3.966	${}^{1}6/^{1}m^{1}m^{1}m^{\infty_z}1 \to {}^{1}6/^{1}m^{1}m^{1}m$	✓	✓			1, 2
		440.4.967	${}^{1}6/^{1}m^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}6/^{1}m^{1}m^{1}m^{m_x}1$	✓	✓			1, 2
441	${}^{2}6/^{2}m^{1}m^{2}m$	441.1.968	${}^{2}6/^{2}m^{1}m^{2}m$			✓	✓	1, 2
		441.2.969	${}^{2_z}6/^{2_z}m^{1}m^{2_z}m^{m_z}1$			✓		1, 2
		441.3.970	${}^{2_z}6/^{2_z}m^{1}m^{2_z}m^{m_x}1$			✓		1, 2
		441.4.971	${}^{2_z}6/^{2_z}m^{1}m^{2_z}m^{\infty_x}1 \to {}^{2_z}6/^{2_z}m^{1}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
442	${}^{m}6/^{m}m^{1}m^{m}m$	442.1.972	${}^{m}6/^{m}m^{1}m^{m}m$			✓	✓	1, 2
		442.2.973	${}^{m_z}6/^{m_z}m^{1}m^{m_z}m^{\infty_x}1 \to {}^{m}6/^{m}m^{1}m^{m}m$	✓	✓			1, 2
443	${}^{\bar{1}}6/^{\bar{1}}m^{1}m^{\bar{1}}m$	443.1.974	${}^{\bar{1}}6/^{\bar{1}}m^{1}m^{\bar{1}}m$			✓	✓	1, 2
		443.2.975	${}^{\bar{1}}6/^{\bar{1}}m^{1}m^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}6/^{\bar{1}}m^{1}m^{\bar{1}}m^{\Sigma_z}1$		✓			2, 4
		443.3.976	${}^{\bar{1}}6/^{\bar{1}}m^{1}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}6/^{\bar{1}}m^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
444	${}^{2}6/^{1}m^{2}m^{1}m$	444.1.977	${}^{2}6/^{1}m^{2}m^{1}m$			✓	✓	1, 2
		444.2.978	${}^{2_z}6/^{1}m^{2_z}m^{1}m^{m_z}1$			✓		1, 2
		444.3.979	${}^{2_z}6/^{1}m^{2_z}m^{1}m^{m_x}1$			✓		1, 2
		444.4.980	${}^{2_z}6/^{1}m^{2_z}m^{1}m^{\infty_x}1 \to {}^{2_z}6/^{1}m^{2_z}m^{1}m^{\Sigma_x}1$		✓			2, 4
445	${}^{m}6/^{1}m^{m}m^{1}m$	445.1.981	${}^{m}6/^{1}m^{m}m^{1}m$			✓	✓	1, 2
		445.2.982	${}^{m_z}6/^{1}m^{m_z}m^{1}m^{\infty_x}1 \to {}^{m}6/^{1}m^{m}m^{1}m$	✓	✓			1, 2

446	${}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m$	446.1.983	${}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m$			✓	✓	1, 2
		446.2.984	${}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\infty_z}1 \to {}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\Sigma_z}1$		✓			2, 4
		446.3.985	${}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\Sigma_z m_x}1$		✓			2, 4
447	${}^{1}6/{}^{1}m^{2}m^{2}m$	447.1.986	${}^{1}6/{}^{1}m^{2}m^{2}m$			✓	✓	1, 2
		447.2.987	${}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{m_z}1$			✓		1, 2
		447.3.988	${}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{m_x}1$			✓		1, 2
		447.4.989	${}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
448	${}^{1}6/{}^{1}m^{m}m^{m}m$	448.1.990	${}^{1}6/{}^{1}m^{m}m^{m}m$			✓	✓	1
		448.2.991	${}^{1}6/{}^{1}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6/{}^{1}m^{m}m^{m}m$	✓	✓			1
449	${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m$	449.1.992	${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		449.2.993	${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2
		449.3.994	${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
450	${}^{1}6/{}^{2}m^{1}m^{1}m$	450.1.995	${}^{1}6/{}^{2}m^{1}m^{1}m$			✓	✓	1, 2
		450.2.996	${}^{1}6/{}^{2_z}m^{1}m^{1}m^{m_z}1$			✓		1, 2
		450.3.997	${}^{1}6/{}^{2_z}m^{1}m^{1}m^{m_x}1$			✓		1, 2
		450.4.998	${}^{1}6/{}^{2_z}m^{1}m^{1}m^{\infty_x}1 \to {}^{1}6/{}^{2_z}m^{1}m^{1}m^{\Sigma_x}1$		✓			2, 4
451	${}^{1}6/{}^{m}m^{1}m^{1}m$	451.1.999	${}^{1}6/{}^{m}m^{1}m^{1}m$			✓	✓	1, 2
		451.2.1000	${}^{1}6/{}^{m_z}m^{1}m^{1}m^{\infty_x}1 \to {}^{1}6/{}^{m}m^{1}m^{1}m$	✓	✓			1, 2
452	${}^{1}6/{}^{\bar{1}}m^{1}m^{1}m$	452.1.1001	${}^{1}6/{}^{\bar{1}}m^{1}m^{1}m$			✓	✓	1, 2
		452.2.1002	${}^{1}6/{}^{\bar{1}}m^{1}m^{1}m^{\infty_z}1 \to {}^{1}6/{}^{\bar{1}}m^{1}m^{1}m^{\Sigma_z}1$		✓			2, 4
		452.3.1003	${}^{1}6/{}^{\bar{1}}m^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}6/{}^{\bar{1}}m^{1}m^{1}m^{\Sigma_z m_x}1$		✓			2, 4
453	${}^{1}6/{}^{2}m^{2}m^{2}m$	453.1.1004	${}^{1}6/{}^{2}m^{2}m^{2}m$			✓	✓	1, 2
		453.2.1005	${}^{1}6/{}^{2_z}m^{2_z}m^{2_z}m^{m_z}1$			✓		1, 2
		453.3.1006	${}^{1}6/{}^{2_z}m^{2_z}m^{2_z}m^{m_x}1$			✓		1, 2
		453.4.1007	${}^{1}6/{}^{2_z}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}6/{}^{2_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
454	${}^{1}6/{}^{m}m^{m}m^{m}m$	454.1.1008	${}^{1}6/{}^{m}m^{m}m^{m}m$			✓	✓	1, 2
		454.2.1009	${}^{1}6/{}^{m_z}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6/{}^{m}m^{m}m^{m}m$	✓	✓			1, 2
455	${}^{1}6/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$	455.1.1010	${}^{1}6/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1, 2
		455.2.1011	${}^{1}6/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \to {}^{1}6/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$		✓			2, 4
		455.3.1012	${}^{1}6/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}6/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4
456	${}^{2_z}6/{}^{2_z}m^{2_x}m^{2_y}m$	456.1.1013	${}^{2_z}6/{}^{2_z}m^{2_x}m^{2_y}m$			✓	✓	1, 2
		456.2.1014	${}^{2_z}6/{}^{2_z}m^{2_x}m^{2_y}m^{m_z}1$			✓		1, 2

		456.3.1015	${}^{2_z}6/{}^{2_z}m^{2_x}m^{2_y}m^{m_x}1$			✓		1, 2
457	${}^{2_z}6/{}^{2_z}m^{m_x}m^{m_y}m$	457.1.1016	${}^{2_z}6/{}^{2_z}m^{m_x}m^{m_y}m$			✓	✓	1
458	${}^{m_x}6/{}^{m_x}m^{2_z}m^{m_y}m$	458.1.1017	${}^{m_x}6/{}^{m_x}m^{2_z}m^{m_y}m$			✓	✓	1, 2
459	${}^{2}6/{}^{2}m^{\bar{1}}m^{m}m$	459.1.1018	${}^{2}6/{}^{2}m^{\bar{1}}m^{m}m$			✓	✓	1
		459.2.1019	${}^{2_z}6/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}6/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
460	${}^{\bar{1}}6/{}^{\bar{1}}m^{2}m^{m}m$	460.1.1020	${}^{\bar{1}}6/{}^{\bar{1}}m^{2}m^{m}m$			✓	✓	1, 2
		460.2.1021	${}^{\bar{1}}6/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}6/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
461	${}^{m}6/{}^{m}m^{2}m^{\bar{1}}m$	461.1.1022	${}^{m}6/{}^{m}m^{2}m^{\bar{1}}m$			✓	✓	1, 2
		461.2.1023	${}^{m_z}6/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}6/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
462	${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m$	462.1.1024	${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m$			✓	✓	1, 2
		462.2.1025	${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m^{m_z}1$			✓		1, 2
		462.3.1026	${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m^{m_x}1$			✓		1, 2
463	${}^{2_z}6/{}^{1}m^{m_x}m^{m_y}m$	463.1.1027	${}^{2_z}6/{}^{1}m^{m_x}m^{m_y}m$			✓	✓	1
464	${}^{m_x}6/{}^{1}m^{2_z}m^{m_y}m$	464.1.1028	${}^{m_x}6/{}^{1}m^{2_z}m^{m_y}m$			✓	✓	1, 2
465	${}^{2}6/{}^{1}m^{\bar{1}}m^{m}m$	465.1.1029	${}^{2}6/{}^{1}m^{\bar{1}}m^{m}m$			✓	✓	1
		465.2.1030	${}^{2_z}6/{}^{1}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}6/{}^{1}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2
466	${}^{\bar{1}}6/{}^{1}m^{2}m^{m}m$	466.1.1031	${}^{\bar{1}}6/{}^{1}m^{2}m^{m}m$			✓	✓	1, 2
		466.2.1032	${}^{\bar{1}}6/{}^{1}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}6/{}^{1}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2
467	${}^{m}6/{}^{1}m^{2}m^{\bar{1}}m$	467.1.1033	${}^{m}6/{}^{1}m^{2}m^{\bar{1}}m$			✓	✓	1, 2
		467.2.1034	${}^{m_z}6/{}^{1}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}6/{}^{1}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
468	${}^{1}6/{}^{2_z}m^{2_x}m^{2_x}m$	468.1.1035	${}^{1}6/{}^{2_z}m^{2_x}m^{2_x}m$			✓	✓	1, 2
		468.2.1036	${}^{1}6/{}^{2_z}m^{2_x}m^{2_x}m^{m_z}1$			✓		1, 2
		468.3.1037	${}^{1}6/{}^{2_z}m^{2_x}m^{2_x}m^{m_x}1$			✓		1, 2
		468.4.1038	${}^{1}6/{}^{2_z}m^{2_x}m^{2_x}m^{m_y}1$			✓		1, 2
469	${}^{1}6/{}^{2_z}m^{m_x}m^{m_x}m$	469.1.1039	${}^{1}6/{}^{2_z}m^{m_x}m^{m_x}m$			✓	✓	1
470	${}^{1}6/{}^{m_x}m^{m_y}m^{m_y}m$	470.1.1040	${}^{1}6/{}^{m_x}m^{m_y}m^{m_y}m$			✓	✓	1, 2
471	${}^{1}6/{}^{m_x}m^{2_z}m^{2_z}m$	471.1.1041	${}^{1}6/{}^{m_x}m^{2_z}m^{2_z}m$			✓	✓	1, 2
472	${}^{1}6/{}^{2}m^{m}m^{m}m$	472.1.1042	${}^{1}6/{}^{2}m^{m}m^{m}m$			✓	✓	1
		472.2.1043	${}^{1}6/{}^{2_z}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6/{}^{2_z}m^{m_z}m^{m_z}m^{\Sigma_x}1$		✓			2
473	${}^{1}6/{}^{2}m^{\bar{1}}m^{\bar{1}}m$	473.1.1044	${}^{1}6/{}^{2}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1
		473.2.1045	${}^{1}6/{}^{2_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \to {}^{1}6/{}^{2_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$		✓			2
474	${}^{1}6/{}^{\bar{1}}m^{m}m^{m}m$	474.1.1046	${}^{1}6/{}^{\bar{1}}m^{m}m^{m}m$			✓	✓	1, 2

		474.2.1047	${}^{1}6/^{\bar{1}}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6/^{\bar{1}}m^{m_z}m^{m_z}m^{\Sigma_x}1$		✓			2
475	${}^{1}6/^{\bar{1}}m^{2}m^{2}m$	475.1.1048	${}^{1}6/^{\bar{1}}m^{2}m^{2}m$			✓	✓	1, 2
		475.2.1049	${}^{1}6/^{\bar{1}}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}6/^{\bar{1}}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2
476	${}^{1}6/^{m}m^{\bar{1}}m^{\bar{1}}m$	476.1.1050	${}^{1}6/^{m}m^{\bar{1}}m^{\bar{1}}m$			✓	✓	1, 2
		476.2.1051	${}^{1}6/^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \to {}^{1}6/^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
477	${}^{1}6/^{m}m^{2}m^{2}m$	477.1.1052	${}^{1}6/^{m}m^{2}m^{2}m$			✓	✓	1, 2
		477.2.1053	${}^{1}6/^{m_z}m^{2_z}m^{2_z}m^{\infty_x}1 \to {}^{1}6/^{m_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
478	${}^{2_z}6/^{2_x}m^{1}m^{2_z}m$	478.1.1054	${}^{2_z}6/^{2_x}m^{1}m^{2_z}m$			✓	✓	1, 2
		478.2.1055	${}^{2_z}6/^{2_x}m^{1}m^{2_z}m^{m_z}1$			✓		1, 2
		478.3.1056	${}^{2_z}6/^{2_x}m^{1}m^{2_z}m^{m_x}1$			✓		1, 2
		478.4.1057	${}^{2_z}6/^{2_x}m^{1}m^{2_z}m^{m_y}1$			✓		1, 2
479	${}^{m_x}6/^{m_y}m^{1}m^{m_x}m$	479.1.1058	${}^{m_x}6/^{m_y}m^{1}m^{m_x}m$			✓	✓	1, 2
480	${}^{m_x}6/^{2_z}m^{1}m^{m_x}m$	480.1.1059	${}^{m_x}6/^{2_z}m^{1}m^{m_x}m$			✓	✓	1, 2
481	${}^{2_z}6/^{m_x}m^{1}m^{2_z}m$	481.1.1060	${}^{2_z}6/^{m_x}m^{1}m^{2_z}m$			✓	✓	1, 2
482	${}^{m}6/^{\bar{1}}m^{1}m^{m}m$	482.1.1061	${}^{m}6/^{\bar{1}}m^{1}m^{m}m$			✓	✓	1, 2
		482.2.1062	${}^{m_z}6/^{\bar{1}}m^{1}m^{m_z}m^{\infty_x}1 \to {}^{m_z}6/^{\bar{1}}m^{1}m^{m_z}m^{\Sigma_x}1$		✓			2, 4
483	${}^{\bar{1}}6/^{m}m^{1}m^{\bar{1}}m$	483.1.1063	${}^{\bar{1}}6/^{m}m^{1}m^{\bar{1}}m$			✓	✓	1, 2
		483.2.1064	${}^{\bar{1}}6/^{m_z}m^{1}m^{\bar{1}}m^{\infty_x}1 \to {}^{\bar{1}}6/^{m_z}m^{1}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
484	${}^{m}6/^{2}m^{1}m^{m}m$	484.1.1065	${}^{m}6/^{2}m^{1}m^{m}m$			✓	✓	1, 2
		484.2.1066	${}^{m_z}6/^{2_z}m^{1}m^{m_z}m^{\infty_x}1 \to {}^{m_z}6/^{2_z}m^{1}m^{m_z}m^{\Sigma_x}1$		✓			2, 4
485	${}^{2}6/^{m}m^{1}m^{2}m$	485.1.1067	${}^{2}6/^{m}m^{1}m^{2}m$			✓	✓	1, 2
		485.2.1068	${}^{2_z}6/^{m_z}m^{1}m^{2_z}m^{\infty_x}1 \to {}^{2_z}6/^{m_z}m^{1}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
486	${}^{\bar{1}}6/^{2}m^{1}m^{\bar{1}}m$	486.1.1069	${}^{\bar{1}}6/^{2}m^{1}m^{\bar{1}}m$			✓	✓	1, 2
		486.2.1070	${}^{\bar{1}}6/^{2_z}m^{1}m^{\bar{1}}m^{\infty_x}1 \to {}^{\bar{1}}6/^{2_z}m^{1}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
487	${}^{2}6/^{\bar{1}}m^{1}m^{2}m$	487.1.1071	${}^{2}6/^{\bar{1}}m^{1}m^{2}m$			✓	✓	1, 2
		487.2.1072	${}^{2_z}6/^{\bar{1}}m^{1}m^{2_z}m^{\infty_x}1 \to {}^{2_z}6/^{\bar{1}}m^{1}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
488	${}^{2_z}6/^{2_x}m^{2_y}m^{2_x}m$	488.1.1073	${}^{2_z}6/^{2_x}m^{2_y}m^{2_x}m$			✓	✓	1, 2
		488.2.1074	${}^{2_z}6/^{2_x}m^{2_y}m^{2_x}m^{m_z}1$			✓		1, 2
		488.3.1075	${}^{2_z}6/^{2_x}m^{2_y}m^{2_x}m^{m_x}1$			✓		1, 2
		488.4.1076	${}^{2_z}6/^{2_x}m^{2_y}m^{2_x}m^{m_y}1$			✓		1, 2
489	${}^{m_y}6/^{2_z}m^{m_x}m^{2_z}m$	489.1.1077	${}^{m_y}6/^{2_z}m^{m_x}m^{2_z}m$			✓	✓	1, 2
490	${}^{m_y}6/^{m_x}m^{2_z}m^{m_x}m$	490.1.1078	${}^{m_y}6/^{m_x}m^{2_z}m^{m_x}m$			✓	✓	1, 2

491	${}^{2_z}6/^{m_x}m^{m_y}m^{m_x}m$	491.1.1079	${}^{2_z}6/^{m_x}m^{m_y}m^{m_x}m$			✓	✓	1, 2
492	${}^{m}6/^{2}m^{\bar{1}}m^{2}m$	492.1.1080	${}^{m}6/^{2}m^{\bar{1}}m^{2}m$			✓	✓	1, 2
		492.2.1081	${}^{m_z}6/^{2_z}m^{\bar{1}}m^{2_z}m^{\infty_x}1 \to {}^{m_z}6/^{2_z}m^{\bar{1}}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
493	${}^{\bar{1}}6/^{2}m^{m}m^{2}m$	493.1.1082	${}^{\bar{1}}6/^{2}m^{m}m^{2}m$			✓	✓	1, 2
		493.2.1083	${}^{\bar{1}}6/^{2_z}m^{m_z}m^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}6/^{2_z}m^{m_z}m^{2_z}m^{\Sigma_x}1$		✓			2, 4
494	${}^{m}6/^{\bar{1}}m^{2}m^{\bar{1}}m$	494.1.1084	${}^{m}6/^{\bar{1}}m^{2}m^{\bar{1}}m$			✓	✓	1, 2
		494.2.1085	${}^{m_z}6/^{\bar{1}}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}6/^{\bar{1}}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
495	${}^{2}6/^{\bar{1}}m^{m}m^{\bar{1}}m$	495.1.1086	${}^{2}6/^{\bar{1}}m^{m}m^{\bar{1}}m$			✓	✓	1, 2
		495.2.1087	${}^{2_z}6/^{\bar{1}}m^{m_z}m^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}6/^{\bar{1}}m^{m_z}m^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4
496	${}^{\bar{1}}6/^{m}m^{2}m^{m}m$	496.1.1088	${}^{\bar{1}}6/^{m}m^{2}m^{m}m$			✓	✓	1, 2
		496.2.1089	${}^{\bar{1}}6/^{m_z}m^{2_z}m^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}6/^{m_z}m^{2_z}m^{m_z}m^{\Sigma_x}1$		✓			2, 4
497	${}^{2}6/^{m}m^{\bar{1}}m^{m}m$	497.1.1090	${}^{2}6/^{m}m^{\bar{1}}m^{m}m$			✓	✓	1, 2
		497.2.1091	${}^{2_z}6/^{m_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \to {}^{2_z}6/^{m_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$		✓			2, 4
498	${}^{3_z}6/^{1}m^{2_x}m^{2_{xy}}m$	498.1.1092	${}^{3_z}6/^{1}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
		498.2.1093	${}^{3_z}6/^{1}m^{2_x}m^{2_{xy}}m^{m_z}1$			✓		1, 2
499	${}^{3_z}6/^{1}m^{m_x}m^{m_{xy}}m$	499.1.1094	${}^{3_z}6/^{1}m^{m_x}m^{m_{xy}}m$			✓	✓	1
500	${}^{m_z}6/^{2_z}m^{m_x}m^{2_y}m$	500.1.1095	${}^{m_z}6/^{2_z}m^{m_x}m^{2_y}m$			✓	✓	1, 2
501	${}^{2_z}6/^{m_z}m^{m_x}m^{m_y}m$	501.1.1096	${}^{2_z}6/^{m_z}m^{m_x}m^{m_y}m$			✓	✓	1, 2
502	${}^{2_z}6/^{m_z}m^{2_x}m^{2_y}m$	502.1.1097	${}^{2_z}6/^{m_z}m^{2_x}m^{2_y}m$			✓	✓	1, 2
503	${}^{m_z}6/^{m_x}m^{2_x}m^{m_y}m$	503.1.1098	${}^{m_z}6/^{m_x}m^{2_x}m^{m_y}m$			✓	✓	1, 2
504	${}^{m_z}6/^{2_x}m^{m_x}m^{2_y}m$	504.1.1099	${}^{m_z}6/^{2_x}m^{m_x}m^{2_y}m$			✓	✓	1, 2
505	${}^{2_z}6/^{2_x}m^{m_y}m^{m_x}m$	505.1.1100	${}^{2_z}6/^{2_x}m^{m_y}m^{m_x}m$			✓	✓	1
506	${}^{2_z}6/^{m_x}m^{2_y}m^{2_x}m$	506.1.1101	${}^{2_z}6/^{m_x}m^{2_y}m^{2_x}m$			✓	✓	1, 2
507	${}^{m_z}6/^{2_x}m^{\bar{1}}m^{2_z}m$	507.1.1102	${}^{m_z}6/^{2_x}m^{\bar{1}}m^{2_z}m$			✓	✓	1, 2
508	${}^{m_z}6/^{m_x}m^{2_z}m^{\bar{1}}m$	508.1.1103	${}^{m_z}6/^{m_x}m^{2_z}m^{\bar{1}}m$			✓	✓	1, 2
509	${}^{m_z}6/^{\bar{1}}m^{2_x}m^{m_y}m$	509.1.1104	${}^{m_z}6/^{\bar{1}}m^{2_x}m^{m_y}m$			✓	✓	1, 2
510	${}^{2_z}6/^{m_x}m^{m_z}m^{\bar{1}}m$	510.1.1105	${}^{2_z}6/^{m_x}m^{m_z}m^{\bar{1}}m$			✓	✓	1, 2
511	${}^{2_z}6/^{2_x}m^{m_z}m^{\bar{1}}m$	511.1.1106	${}^{2_z}6/^{2_x}m^{m_z}m^{\bar{1}}m$			✓	✓	1
512	${}^{2_z}6/^{\bar{1}}m^{m_x}m^{m_y}m$	512.1.1107	${}^{2_z}6/^{\bar{1}}m^{m_x}m^{m_y}m$			✓	✓	1, 2
513	${}^{2_z}6/^{\bar{1}}m^{2_x}m^{2_y}m$	513.1.1108	${}^{2_z}6/^{\bar{1}}m^{2_x}m^{2_y}m$			✓	✓	1, 2
514	${}^{\bar{1}}6/^{m_x}m^{2_z}m^{m_z}m$	514.1.1109	${}^{\bar{1}}6/^{m_x}m^{2_z}m^{m_z}m$			✓	✓	1, 2
515	${}^{\bar{1}}6/^{2_x}m^{m_z}m^{2_z}m$	515.1.1110	${}^{\bar{1}}6/^{2_x}m^{m_z}m^{2_z}m$			✓	✓	1, 2

516	${}^{6_z}6/{}^{2_z}m^{2_x}m^{2_1}m$	516.1.1111	${}^{6_z}6/{}^{2_z}m^{2_x}m^{2_1}m$			✓	✓	1, 2
		516.2.1112	${}^{6_z}6/{}^{2_z}m^{2_x}m^{2_1}m^{m_z}1$			✓		1, 2
517	${}^{\bar{3}_z}6/{}^{\bar{1}}m^{2_x}m^{m_{xy}}m$	517.1.1113	${}^{\bar{3}_z}6/{}^{\bar{1}}m^{2_x}m^{m_{xy}}m$			✓	✓	1, 2
518	${}^{6_z}6/{}^{2_z}m^{m_x}m^{m_1}m$	518.1.1114	${}^{6_z}6/{}^{2_z}m^{m_x}m^{m_1}m$			✓	✓	1
519	${}^{\bar{6}_z}6/{}^{m_z}m^{m_x}m^{2_1}m$	519.1.1115	${}^{\bar{6}_z}6/{}^{m_z}m^{m_x}m^{2_1}m$			✓	✓	1, 2
520	${}^{6_z}6/{}^{1}m^{2_x}m^{2_1}m$	520.1.1116	${}^{6_z}6/{}^{1}m^{2_x}m^{2_1}m$			✓	✓	1, 2
		520.2.1117	${}^{6_z}6/{}^{1}m^{2_x}m^{2_1}m^{m_z}1$			✓		1, 2
521	${}^{\bar{3}_z}6/{}^{1}m^{m_x}m^{2_{xy}}m$	521.1.1118	${}^{\bar{3}_z}6/{}^{1}m^{m_x}m^{2_{xy}}m$			✓	✓	1, 2
522	${}^{6_z}6/{}^{1}m^{m_x}m^{m_1}m$	522.1.1119	${}^{6_z}6/{}^{1}m^{m_x}m^{m_1}m$			✓	✓	1
523	${}^{\bar{6}_z}6/{}^{1}m^{2_x}m^{m_1}m$	523.1.1120	${}^{\bar{6}_z}6/{}^{1}m^{2_x}m^{m_1}m$			✓	✓	1, 2
524	${}^{3_z}6/{}^{2_z}m^{2_x}m^{2_{xy}}m$	524.1.1121	${}^{3_z}6/{}^{2_z}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
		524.2.1122	${}^{3_z}6/{}^{2_z}m^{2_x}m^{2_{xy}}m^{m_z}1$			✓		1, 2
525	${}^{3_z}6/{}^{\bar{1}}m^{m_x}m^{m_{xy}}m$	525.1.1123	${}^{3_z}6/{}^{\bar{1}}m^{m_x}m^{m_{xy}}m$			✓	✓	1, 2
526	${}^{3_z}6/{}^{\bar{1}}m^{2_x}m^{2_{xy}}m$	526.1.1124	${}^{3_z}6/{}^{\bar{1}}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
527	${}^{3_z}6/{}^{2_z}m^{m_x}m^{m_{xy}}m$	527.1.1125	${}^{3_z}6/{}^{2_z}m^{m_x}m^{m_{xy}}m$			✓	✓	1
528	${}^{3_z}6/{}^{m_z}m^{m_x}m^{m_{xy}}m$	528.1.1126	${}^{3_z}6/{}^{m_z}m^{m_x}m^{m_{xy}}m$			✓	✓	1, 2
529	${}^{3_z}6/{}^{m_z}m^{2_x}m^{2_{xy}}m$	529.1.1127	${}^{3_z}6/{}^{m_z}m^{2_x}m^{2_{xy}}m$			✓	✓	1, 2
530	${}^{6_z}6/{}^{\bar{1}}m^{m_x}m^{m_1}m$	530.1.1128	${}^{6_z}6/{}^{\bar{1}}m^{m_x}m^{m_1}m$			✓	✓	1, 2
531	${}^{6_z}6/{}^{m_z}m^{m_x}m^{m_1}m$	531.1.1129	${}^{6_z}6/{}^{m_z}m^{m_x}m^{m_1}m$			✓	✓	1, 2
532	${}^{6_z}6/{}^{\bar{1}}m^{2_x}m^{2_1}m$	532.1.1130	${}^{6_z}6/{}^{\bar{1}}m^{2_x}m^{2_1}m$			✓	✓	1, 2
533	${}^{6_z}6/{}^{m_z}m^{2_x}m^{2_1}m$	533.1.1131	${}^{6_z}6/{}^{m_z}m^{2_x}m^{2_1}m$			✓	✓	1, 2
534	${}^{\bar{3}_z}6/{}^{m_z}m^{2_x}m^{m_{xy}}m$	534.1.1132	${}^{\bar{3}_z}6/{}^{m_z}m^{2_x}m^{m_{xy}}m$			✓	✓	1, 2
535	${}^{\bar{3}_z}6/{}^{2_z}m^{m_x}m^{2_{xy}}m$	535.1.1133	${}^{\bar{3}_z}6/{}^{2_z}m^{m_x}m^{2_{xy}}m$			✓	✓	1, 2
536	${}^{\bar{6}_z}6/{}^{\bar{1}}m^{2_x}m^{m_1}m$	536.1.1134	${}^{\bar{6}_z}6/{}^{\bar{1}}m^{2_x}m^{m_1}m$			✓	✓	1, 2
537	${}^{\bar{6}_z}6/{}^{2_z}m^{m_x}m^{2_1}m$	537.1.1135	${}^{\bar{6}_z}6/{}^{2_z}m^{m_x}m^{2_1}m$			✓	✓	1, 2
538	${}^{1}2^{1}3$	538.1.1136	${}^{1}2^{1}3$			✓	✓	1, 3
		538.2.1137	${}^{1}2^{1}3^{m_z}1$			✓		1, 2, 3
		538.3.1138	${}^{1}2^{1}3^{\infty_z}1 \to {}^{1}2^{1}3$	✓	✓			1, 3
		538.4.1139	${}^{1}2^{1}3^{\infty_z m_x}1 \to {}^{1}2^{1}3^{m_x}1$	✓	✓			1, 2, 3
539	${}^{1}2^{3}3$	539.1.1140	${}^{1}2^{3}3$			✓	✓	1, 3
		539.2.1141	${}^{1}2^{3_z}3^{m_z}1$			✓		1, 2, 3
540	${}^{2_z}2^{3_{xyz}}3$	540.1.1142	${}^{2_z}2^{3_{xyz}}3$				✓	1, 3

541	${}^{1}m^{1}\bar{3}$	541.1.1143	${}^{1}m^{1}\bar{3}$			✓	✓	1, 3
		541.2.1144	${}^{1}m^{1}\bar{3}^{m_z}1$			✓		1, 2, 3
		541.3.1145	${}^{1}m^{1}\bar{3}^{\infty_z}1 \to {}^{1}m^{1}\bar{3}$	✓	✓			1, 3
		541.4.1146	${}^{1}m^{1}\bar{3}^{\infty_z m_x}1 \to {}^{1}m^{1}\bar{3}^{m_x}1$	✓	✓			1, 2, 3
542	${}^{2}m^{2}\bar{3}$	542.1.1147	${}^{2}m^{2}\bar{3}$			✓	✓	1, 3
		542.2.1148	${}^{2_z}m^{2_z}\bar{3}^{m_z}1$			✓		1, 2, 3
		542.3.1149	${}^{2_z}m^{2_z}\bar{3}^{m_x}1$			✓		1, 2, 3
		542.4.1150	${}^{2_z}m^{2_z}\bar{3}^{\infty_x}1 \to {}^{2_z}m^{2_z}\bar{3}^{\Sigma_x}1$		✓			2, 6
543	${}^{m}m^{m}\bar{3}$	543.1.1151	${}^{m}m^{m}\bar{3}$			✓	✓	1, 2, 3
		543.2.1152	${}^{m_z}m^{m_z}\bar{3}^{\infty_x}1 \to {}^{m}m^{m}\bar{3}$	✓	✓			1, 2, 3
544	${}^{\bar{1}}m^{\bar{1}}\bar{3}$	544.1.1153	${}^{\bar{1}}m^{\bar{1}}\bar{3}$			✓	✓	1, 2, 3
		544.2.1154	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\infty_z}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\Sigma_z}1$		✓			2, 6
		544.3.1155	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\infty_z m_x}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\Sigma_z m_x}1$		✓			2, 4, 6
545	${}^{1}m^{3}\bar{3}$	545.1.1156	${}^{1}m^{3}\bar{3}$			✓	✓	1, 3
		545.2.1157	${}^{1}m^{3_z}\bar{3}^{m_z}1$			✓		1, 2, 3
546	${}^{2}m^{6}\bar{3}$	546.1.1158	${}^{2}m^{6}\bar{3}$			✓	✓	1, 3
		546.2.1159	${}^{2_z}m^{6_z}\bar{3}^{m_z}1$			✓		1, 2, 3
547	${}^{\bar{1}}m^{\bar{3}}\bar{3}$	547.1.1160	${}^{\bar{1}}m^{\bar{3}}\bar{3}$			✓	✓	1, 2, 3
548	${}^{m}m^{\bar{6}}\bar{3}$	548.1.1161	${}^{m}m^{\bar{6}}\bar{3}$			✓	✓	1, 2, 3
549	${}^{2_z}m^{3_{xyz}}\bar{3}$	549.1.1162	${}^{2_z}m^{3_{xyz}}\bar{3}$				✓	1, 3
550	${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}$	550.1.1163	${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}$				✓	1, 2, 3
551	${}^{1}4^{1}3^{1}2$	551.1.1164	${}^{1}4^{1}3^{1}2$			✓	✓	1, 2, 3
		551.2.1165	${}^{1}4^{1}3^{1}2^{m_z}1$			✓		1, 2, 3
		551.3.1166	${}^{1}4^{1}3^{1}2^{\infty_z}1 \to {}^{1}4^{1}3^{1}2$	✓	✓			1, 2, 3
		551.4.1167	${}^{1}4^{1}3^{1}2^{\infty_z m_x}1 \to {}^{1}4^{1}3^{1}2^{m_x}1$	✓	✓			1, 2, 3
552	${}^{2}4^{1}3^{2}2$	552.1.1168	${}^{2}4^{1}3^{2}2$			✓	✓	1, 2, 3
		552.2.1169	${}^{2_z}4^{1}3^{2_z}2^{m_z}1$			✓		1, 2, 3
		552.3.1170	${}^{2_z}4^{1}3^{2_z}2^{m_x}1$			✓		1, 2, 3
		552.4.1171	${}^{2_z}4^{1}3^{2_z}2^{\infty_x}1 \to {}^{2_z}4^{1}3^{2_z}2^{\Sigma_x}1$		✓			2, 6
553	${}^{m}4^{1}3^{m}2$	553.1.1172	${}^{m}4^{1}3^{m}2$			✓	✓	1, 3
		553.2.1173	${}^{m_z}4^{1}3^{m_z}2^{\infty_x}1 \to {}^{m}4^{1}3^{m}2$	✓	✓			1, 3
554	${}^{\bar{1}}4^{1}3^{\bar{1}}2$	554.1.1174	${}^{\bar{1}}4^{1}3^{\bar{1}}2$			✓	✓	1, 3

		554.2.1175	$^{\bar{1}}4^{1}3^{\bar{1}}2^{\infty_z}1 \to {}^{\bar{1}}4^{1}3^{\bar{1}}2^{\Sigma_z}1$		✓			2, 6
		554.3.1176	$^{\bar{1}}4^{1}3^{\bar{1}}2^{\infty_z m_x}1 \to {}^{\bar{1}}4^{1}3^{\bar{1}}2^{\Sigma_z m_x}1$		✓			2, 4, 6
555	$^{2_x}4^{3_z}3^{2_x}2$	555.1.1177	$^{2_x}4^{3_z}3^{2_x}2$			✓	✓	1, 2, 3
		555.2.1178	$^{2_x}4^{3_z}3^{2_x}2^{m_z}1$			✓		1, 2, 3
556	$^{m_x}4^{3_z}3^{m_x}2$	556.1.1179	$^{m_x}4^{3_z}3^{m_x}2$			✓	✓	1, 3
557	$^{4_z}4^{3_{xyz}}3^{2_{xy}}2$	557.1.1180	$^{4_z}4^{3_{xyz}}3^{2_{xy}}2$				✓	1, 2, 3
558	$^{\bar{4}_z}4^{3_{xyz}}3^{m_{xy}}2$	558.1.1181	$^{\bar{4}_z}4^{3_{xyz}}3^{m_{xy}}2$				✓	1, 3
559	$^{1}\bar{4}^{1}3^{1}m$	559.1.1182	$^{1}\bar{4}^{1}3^{1}m$			✓	✓	1, 2, 3
		559.2.1183	$^{1}\bar{4}^{1}3^{1}m^{m_z}1$			✓		1, 2, 3
		559.3.1184	$^{1}\bar{4}^{1}3^{1}m^{\infty_z}1 \to {}^{1}\bar{4}^{1}3^{1}m$	✓	✓			1, 2, 3
		559.4.1185	$^{1}\bar{4}^{1}3^{1}m^{\infty_z m_x}1 \to {}^{1}\bar{4}^{1}3^{1}m^{m_x}1$	✓	✓			1, 2, 3
560	$^{2}\bar{4}^{1}3^{2}m$	560.1.1186	$^{2}\bar{4}^{1}3^{2}m$			✓	✓	1, 2, 3
		560.2.1187	$^{2_z}\bar{4}^{1}3^{2_z}m^{m_z}1$			✓		1, 2, 3
		560.3.1188	$^{2_z}\bar{4}^{1}3^{2_z}m^{m_x}1$			✓		1, 2, 3
		560.4.1189	$^{2_z}\bar{4}^{1}3^{2_z}m^{\infty_x}1 \to {}^{2_z}\bar{4}^{1}3^{2_z}m^{\Sigma_x}1$		✓			2, 6
561	$^{m}\bar{4}^{1}3^{m}m$	561.1.1190	$^{m}\bar{4}^{1}3^{m}m$			✓	✓	1, 3
		561.2.1191	$^{m_z}\bar{4}^{1}3^{m_z}m^{\infty_x}1 \to {}^{m}\bar{4}^{1}3^{m}m$	✓	✓			1, 3
562	$^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m$	562.1.1192	$^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m$			✓	✓	1, 3
		562.2.1193	$^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\Sigma_z}1$		✓			2, 6
		562.3.1194	$^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4, 6
563	$^{2_x}\bar{4}^{3_z}3^{2_x}m$	563.1.1195	$^{2_x}\bar{4}^{3_z}3^{2_x}m$			✓	✓	1, 2, 3
		563.2.1196	$^{2_x}\bar{4}^{3_z}3^{2_x}m^{m_z}1$			✓		1, 2, 3
564	$^{m_x}\bar{4}^{3_z}3^{m_x}m$	564.1.1197	$^{m_x}\bar{4}^{3_z}3^{m_x}m$			✓	✓	1, 3
565	$^{4_z}\bar{4}^{3_{xyz}}3^{2_{xy}}m$	565.1.1198	$^{4_z}\bar{4}^{3_{xyz}}3^{2_{xy}}m$				✓	1, 2, 3
566	$^{\bar{4}_z}\bar{4}^{3_{xyz}}3^{m_{xy}}m$	566.1.1199	$^{\bar{4}_z}\bar{4}^{3_{xyz}}3^{m_{xy}}m$				✓	1, 3
567	$^{1}m^{1}\bar{3}^{1}m$	567.1.1200	$^{1}m^{1}\bar{3}^{1}m$			✓	✓	1, 2, 3
		567.2.1201	$^{1}m^{1}\bar{3}^{1}m^{m_z}1$			✓		1, 2, 3
		567.3.1202	$^{1}m^{1}\bar{3}^{1}m^{\infty_z}1 \to {}^{1}m^{1}\bar{3}^{1}m$	✓	✓			1, 2, 3
		567.4.1203	$^{1}m^{1}\bar{3}^{1}m^{\infty_z m_x}1 \to {}^{1}m^{1}\bar{3}^{1}m^{m_x}1$	✓	✓			1, 2, 3
568	$^{1}m^{1}\bar{3}^{2}m$	568.1.1204	$^{1}m^{1}\bar{3}^{2}m$			✓	✓	1, 2, 3
		568.2.1205	$^{1}m^{1}\bar{3}^{2_z}m^{m_z}1$			✓		1, 2, 3
		568.3.1206	$^{1}m^{1}\bar{3}^{2_z}m^{m_x}1$			✓		1, 2, 3

		568.4.1207	${}^{1}m^{1}\bar{3}^{2_z}m^{\infty_x}1 \to {}^{1}m^{1}\bar{3}^{2_z}m^{\Sigma_x}1$		✓			2, 6
569	${}^{1}m^{1}\bar{3}^{m}m$	569.1.1208	${}^{1}m^{1}\bar{3}^{m}m$			✓	✓	1, 3
		569.2.1209	${}^{1}m^{1}\bar{3}^{m_z}m^{\infty_x}1 \to {}^{1}m^{1}\bar{3}^{m}m$	✓	✓			1, 3
570	${}^{1}m^{1}\bar{3}^{\bar{1}}m$	570.1.1210	${}^{1}m^{1}\bar{3}^{\bar{1}}m$			✓	✓	1, 3
		570.2.1211	${}^{1}m^{1}\bar{3}^{\bar{1}}m^{\infty_z}1 \to {}^{1}m^{1}\bar{3}^{\bar{1}}m^{\Sigma_z}1$		✓			2, 6
		570.3.1212	${}^{1}m^{1}\bar{3}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{1}m^{1}\bar{3}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4, 6
571	${}^{2}m^{2}\bar{3}^{1}m$	571.1.1213	${}^{2}m^{2}\bar{3}^{1}m$			✓	✓	1, 2, 3
		571.2.1214	${}^{2_z}m^{2_z}\bar{3}^{1}m^{m_z}1$			✓		1, 2, 3
		571.3.1215	${}^{2_z}m^{2_z}\bar{3}^{1}m^{m_x}1$			✓		1, 2, 3
		571.4.1216	${}^{2_z}m^{2_z}\bar{3}^{1}m^{\infty_x}1 \to {}^{2_z}m^{2_z}\bar{3}^{1}m^{\Sigma_x}1$		✓			2, 4, 6
572	${}^{m}m^{m}\bar{3}^{1}m$	572.1.1217	${}^{m}m^{m}\bar{3}^{1}m$			✓	✓	1, 2, 3
		572.2.1218	${}^{m_z}m^{m_z}\bar{3}^{1}m^{\infty_x}1 \to {}^{m}m^{m}\bar{3}^{1}m$	✓	✓			1, 2, 3
573	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m$	573.1.1219	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m$			✓	✓	1, 2, 3
		573.2.1220	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\infty_z}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\Sigma_z}1$		✓			2, 4, 6
		573.3.1221	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\infty_z m_x}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\Sigma_z m_x}1$		✓			2, 4, 6
574	${}^{2}m^{2}\bar{3}^{2}m$	574.1.1222	${}^{2}m^{2}\bar{3}^{2}m$			✓	✓	1, 2, 3
		574.2.1223	${}^{2_z}m^{2_z}\bar{3}^{2_z}m^{m_z}1$			✓		1, 2, 3
		574.3.1224	${}^{2_z}m^{2_z}\bar{3}^{2_z}m^{m_x}1$			✓		1, 2, 3
		574.4.1225	${}^{2_z}m^{2_z}\bar{3}^{2_z}m^{\infty_x}1 \to {}^{2_z}m^{2_z}\bar{3}^{2_z}m^{\Sigma_x}1$		✓			2, 4, 6
575	${}^{m}m^{m}\bar{3}^{m}m$	575.1.1226	${}^{m}m^{m}\bar{3}^{m}m$			✓	✓	1, 2, 3
		575.2.1227	${}^{m_z}m^{m_z}\bar{3}^{m_z}m^{\infty_x}1 \to {}^{m}m^{m}\bar{3}^{m}m$	✓	✓			1, 2, 3
576	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m$	576.1.1228	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m$			✓	✓	1, 2, 3
		576.2.1229	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\infty_z}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\Sigma_z}1$		✓			2, 4, 6
		576.3.1230	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\infty_z m_x}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\Sigma_z m_x}1$		✓			2, 4, 6
577	${}^{2_y}m^{2_y}\bar{3}^{2_z}m$	577.1.1231	${}^{2_y}m^{2_y}\bar{3}^{2_z}m$			✓	✓	1, 2, 3
		577.2.1232	${}^{2_y}m^{2_y}\bar{3}^{2_z}m^{m_y}1$			✓		1, 2, 3
		577.3.1233	${}^{2_y}m^{2_y}\bar{3}^{2_z}m^{m_z}1$			✓		1, 2, 3
		577.4.1234	${}^{2_y}m^{2_y}\bar{3}^{2_z}m^{m_x}1$			✓		1, 2, 3
578	${}^{m_x}m^{m_x}\bar{3}^{m_y}m$	578.1.1235	${}^{m_x}m^{m_x}\bar{3}^{m_y}m$			✓	✓	1, 2, 3
579	${}^{2_z}m^{2_z}\bar{3}^{m_y}m$	579.1.1236	${}^{2_z}m^{2_z}\bar{3}^{m_y}m$			✓	✓	1, 3
580	${}^{m_y}m^{m_y}\bar{3}^{2_z}m$	580.1.1237	${}^{m_y}m^{m_y}\bar{3}^{2_z}m$			✓	✓	1, 2, 3
581	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{m}m$	581.1.1238	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{m}m$			✓	✓	1, 2, 3

		581.2.1239	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{m_z}m^{\infty_x}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{m_z}m^{\Sigma_x}1$		✓			2, 6
582	${}^{m}m^{m}\bar{3}^{\bar{1}}m$	582.1.1240	${}^{m}m^{m}\bar{3}^{\bar{1}}m$			✓	✓	1, 2, 3
		582.2.1241	${}^{m_z}m^{m_z}\bar{3}^{\bar{1}}m^{\infty_x}1 \to {}^{m_z}m^{m_z}\bar{3}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 4, 6
583	${}^{2}m^{2}\bar{3}^{m}m$	583.1.1242	${}^{2}m^{2}\bar{3}^{m}m$			✓	✓	1, 3
		583.2.1243	${}^{2_z}m^{2_z}\bar{3}^{m_z}m^{\infty_x}1 \to {}^{2_z}m^{2_z}\bar{3}^{m_z}m^{\Sigma_x}1$		✓			2, 6
584	${}^{m}m^{m}\bar{3}^{2}m$	584.1.1244	${}^{m}m^{m}\bar{3}^{2}m$			✓	✓	1, 2, 3
		584.2.1245	${}^{m_z}m^{m_z}\bar{3}^{2_z}m^{\infty_x}1 \to {}^{m_z}m^{m_z}\bar{3}^{2_z}m^{\Sigma_x}1$		✓			2, 4, 6
585	${}^{2}m^{2}\bar{3}^{\bar{1}}m$	585.1.1246	${}^{2}m^{2}\bar{3}^{\bar{1}}m$			✓	✓	1, 3
		585.2.1247	${}^{2_z}m^{2_z}\bar{3}^{\bar{1}}m^{\infty_x}1 \to {}^{2_z}m^{2_z}\bar{3}^{\bar{1}}m^{\Sigma_x}1$		✓			2, 6
586	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{2}m$	586.1.1248	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{2}m$			✓	✓	1, 2, 3
		586.2.1249	${}^{\bar{1}}m^{\bar{1}}\bar{3}^{2_z}m^{\infty_x}1 \to {}^{\bar{1}}m^{\bar{1}}\bar{3}^{2_z}m^{\Sigma_x}1$		✓			2, 6
587	${}^{1}m^{3_z}\bar{3}^{2_x}m$	587.1.1250	${}^{1}m^{3_z}\bar{3}^{2_x}m$			✓	✓	1, 2, 3
		587.2.1251	${}^{1}m^{3_z}\bar{3}^{2_x}m^{m_z}1$			✓		1, 2, 3
588	${}^{1}m^{3_z}\bar{3}^{m_x}m$	588.1.1252	${}^{1}m^{3_z}\bar{3}^{m_x}m$			✓	✓	1, 3
589	${}^{2_z}m^{6_z}\bar{3}^{2_x}m$	589.1.1253	${}^{2_z}m^{6_z}\bar{3}^{2_x}m$			✓	✓	1, 2, 3
		589.2.1254	${}^{2_z}m^{6_z}\bar{3}^{2_x}m^{m_z}1$			✓		1, 2, 3
590	${}^{\bar{1}}m^{\bar{3}_z}\bar{3}^{m_x}m$	590.1.1255	${}^{\bar{1}}m^{\bar{3}_z}\bar{3}^{m_x}m$			✓	✓	1, 2, 3
591	${}^{\bar{1}}m^{\bar{3}_z}\bar{3}^{2_x}m$	591.1.1256	${}^{\bar{1}}m^{\bar{3}_z}\bar{3}^{2_x}m$			✓	✓	1, 2, 3
592	${}^{2_z}m^{6_z}\bar{3}^{m_x}m$	592.1.1257	${}^{2_z}m^{6_z}\bar{3}^{m_x}m$			✓	✓	1, 3
593	${}^{m_z}m^{\bar{6}_z}\bar{3}^{2_x}m$	593.1.1258	${}^{m_z}m^{\bar{6}_z}\bar{3}^{2_x}m$			✓	✓	1, 2, 3
594	${}^{m_z}m^{\bar{6}_z}\bar{3}^{m_x}m$	594.1.1259	${}^{m_z}m^{\bar{6}_z}\bar{3}^{m_x}m$			✓	✓	1, 2, 3
595	${}^{2_z}m^{3_{xyz}}\bar{3}^{2_{xy}}m$	595.1.1260	${}^{2_z}m^{3_{xyz}}\bar{3}^{2_{xy}}m$				✓	1, 2, 3
596	${}^{2_z}m^{3_{xyz}}\bar{3}^{m_{xy}}m$	596.1.1261	${}^{2_z}m^{3_{xyz}}\bar{3}^{m_{xy}}m$				✓	1, 3
597	${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}^{m_{xy}}m$	597.1.1262	${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}^{m_{xy}}m$				✓	1, 2, 3
598	${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}^{2_{xy}}m$	598.1.1263	${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}^{2_{xy}}m$				✓	1, 2, 3

No 1.1.1 : $^{1}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$

Character table

	C_1
χ_1	1

Irrepresntations of group generators

	$\left[E \mid\mid 1\right]$
Γ_1	1

No 1.2.2 : ${}^{1}1^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	$\mathcal{K}$

No 1.3.3 : ${}^{1}1^{\infty_z}1 \to {}^{1}1$

No 1.4.4 : ${}^{1}1^{\infty_z m_x}1 \to {}^{1}1^{m_x}1$

No 2.1.5 : $^{1}\bar{1}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid -1\right]$

Character table

	C_1	C_2
χ_1	1	1
χ_2	1	-1

Irrepresntations of group generators

	$\left[E \mid\mid -1\right]$
Γ_1	1
Γ_2	-1

No 2.2.6 : ${}^{1}\bar{1}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid -1\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 2.3.7 : ${}^{1}\bar{\mathrm{I}}^{\infty_z}1 \to {}^{1}\bar{\mathrm{I}}$

No 2.4.8 : ${}^{1}\bar{1}^{\infty_z m_x}1 \rightarrow {}^{1}\bar{1}^{m_x}1$

No 3.1.9 : $^{2}\bar{1}$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{001} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid -1\right]$

Character table

	C_1	C_2
χ_1	1	1
χ_2	1	-1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid -1\right]$
Γ_1	1
Γ_2	-1

No 3.2.10 : ${}^{2_z}\bar{1}^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, -1\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 3.3.11 : ${}^{2_z}\bar{1}^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid -1\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 3.4.12 : ${}^{2_z}\bar{1}^{\infty_x}1 \rightarrow {}^{2_z}\bar{1}^{\Sigma_x}1$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[2_{100} \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\bar{E} 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Character table

	C_1	C_2	C_3
χ_1	2	0	−2

Irrepresntations of group generators

	$\left[2_{100} \mid\mid -1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$

No 4.1.13 : $^{m}\bar{1}$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[2_{001}T \mid\mid -1\right]$
Γ_1^{co}	$\mathcal{K}$

No 4.2.14 : ${}^{m_z}\bar{1}^{\infty_x}1 \to {}^{m}\bar{1}$

No 5.1.15 : $^{\bar{1}}\bar{1}$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[T \mid\mid -1\right]$
Γ_1^{co}	$\mathcal{K}$

No 5.2.16 : ${}^{\bar{1}}\bar{1}^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{1}^{\Sigma_z}1$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[T \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z T \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid -1\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\Sigma_z \mid\mid 1\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$
C_4 : $\left[\bar{\Sigma}_z \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[T \mid\mid -1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 5.3.17 : ${}^{\bar{1}}\bar{1}^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{1}^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel -1\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \parallel 1\right]$
$C_2 : \quad \left[2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$C_3 : \quad \left[\bar{E} \parallel 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \parallel -1\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 6.1.18 : $^{1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{010}\right]$

Character table

	C_1	C_2
χ_1	1	1
χ_2	1	-1

Irrepresntations of group generators

	$\left[E \mid\mid 2_{010}\right]$
Γ_1	1
Γ_2	-1

No 6.2.19 : ${}^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \parallel 2_{010}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 6.3.20 : ${}^{1}2^{\infty_z}1 \to {}^{1}2$

No 6.4.21 : ${}^{1}2^{\infty_z m_x}1 \to {}^{1}2^{m_x}1$

No 7.1.22 : $^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid 2_{010}\right]$

Character table

	C_1	C_2
χ_1	1	1
χ_2	1	-1

Irrepresntations of group generators

	$\left[2_{010} \mid\mid 2_{010}\right]$
Γ_1	1
Γ_2	-1

No 7.2.23 : ${}^{2_y}2^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 7.3.24 : ${}^{2_y}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 7.4.25 : ${}^{2_y}2^{\infty_x}1 \rightarrow {}^{2_y}2^{\Sigma_x}1$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Character table

	C_1	C_2	C_3
χ_1	2	0	−2

Irrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 8.1.26 : $^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{010}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[2_{010}T \mid\mid 2_{010}\right]$
Γ_1^{co}	$\mathcal{K}$

No 8.2.27 : ${}^{m}{}_{y}2^{\infty}{}_{x}1 \rightarrow {}^{m}2$

No 9.1.28 : $^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[T \mid\mid 2_{010}\right]$
Γ_1^{co}	$\mathcal{K}$

No 9.2.29 : ${}^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$$\left[E \parallel 1\right] \left[T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \parallel 1\right]$
$C_2 : \quad \left[\Sigma_z \parallel 1\right]$
$C_3 : \quad \left[\bar{E} \parallel 1\right]$
$C_4 : \quad \left[\bar{\Sigma}_z \parallel 1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[T \parallel 2_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 9.3.30 : ${}^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_3 : $\left[\bar{E} \parallel 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \parallel 2_{010}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 10.1.31 : ^1m

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{010}\right]$

Character table

	C_1	C_2
χ_1	1	1
χ_2	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid m_{010}\right]$
Γ_1	1
Γ_2	−1

No 10.2.32 : ${}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 10.3.33 : ${}^{1}m^{\infty_z}1 \to {}^{1}m$

No 10.4.34 : ${}^{1}m^{\infty_z m_x}1 \to {}^{1}m^{m_x}1$

No 11.1.35 : ^2m

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{010} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid m_{010}\right]$

Character table

	C_1	C_2
χ_1	1	1
χ_2	1	-1

Irrepresntations of group generators

	$\left[2_{010} \mid\mid m_{010}\right]$
Γ_1	1
Γ_2	-1

No 11.2.36 : ${}^{2_y}m^{m_y}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 1\right]$ $\left[T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[2_{010}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 11.3.37 : ${}^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 11.4.38 : ${}^{2_y}m^{\infty_x}1 \rightarrow {}^{2_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Character table

	C_1	C_2	C_3
χ_1	2	0	-2

Irrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 12.1.39 : ${}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{010}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[2_{010}T \mid\mid m_{010}\right]$
Γ_1^{co}	$\mathcal{K}$

No 12.2.40 : ${}^{m_y}m^{\infty_x}1 \to {}^{m}m$

No 13.1.41 : $^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$

Co-character table

	C_1
χ_1^{co}	1

Coirrepresntations of group generators

	$\left[T \mid\mid m_{010}\right]$
Γ_1^{co}	$\mathcal{K}$

No 13.2.42 : ${}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \mid\mid 1\right]$
$C_2 : \quad \left[\Sigma_z \mid\mid 1\right]$
$C_3 : \quad \left[\bar{E} \mid\mid 1\right]$
$C_4 : \quad \left[\bar{\Sigma}_z \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[T \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 13.3.43 : ${}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{010}\right] \left[2_{100}T \mid\mid 1\right] \left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 14.1.44 : ${}^{1}2/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{010}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	−1	−1
χ_3	1	−1	−1	1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid 2_{010}\right]$	$\left[E \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1

No 14.2.45 : ${}^{1}2/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{010}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{010}\right]$	$\left[E \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 14.3.46 : ${}^{1}2/{}^{1}m^{\infty_z}1 \rightarrow {}^{1}2/{}^{1}m$

No 14.4.47 : ${}^{1}2/{}^{1}m^{\infty_z m_x}1 \rightarrow {}^{1}2/{}^{1}m^{m_x}1$

No 15.1.48 : ${}^{1}2/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{010}\right]$
C_3 : $\left[2_{010} \mid\mid m_{010}\right]$
C_4 : $\left[2_{010} \mid\mid -1\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	−1	−1
χ_3	1	−1	−1	1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{010} \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1

No 15.2.49 : ${}^{1}2/{}^{2_y}m^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{010} \,||\, -1\right]$ $\left[2_{010}T \,||\, 1\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 2_{010}\right]$
C_3 : $\left[2_{010} \,||\, m_{010}\right]$
C_4 : $\left[2_{010} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \,||\, 2_{010}\right]$	$\left[2_{010} \,||\, m_{010}\right]$	$\left[2_{010}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 15.3.50 : ${}^{1}2/{}^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{010}\right]$
C_3 : $\left[2_{010} \mid\mid m_{010}\right]$
C_4 : $\left[2_{010} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 15.4.51 : ${}^{1}2/{}^{2_y}m^{\infty_x}1 \rightarrow {}^{1}2/{}^{2_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{010}\right]$
C_3 : $\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	2	0	−2	−2
χ_2	2	−2	0	−2	2

Irrepresntations of group generators

	$\left[E \parallel 2_{010}\right]$	$\left[\bar{E}2_{100} \parallel m_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 16.1.52 : ${}^{1}2/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{010}T \mid\mid m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 16.2.53 : ${}^{1}2/{}^{m_y}m{}^{\infty_x}1 \to {}^{1}2/{}^{m}m$

No 17.1.54 : ${}^{1}2/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 2_{010}\right] \left[T \parallel m_{010}\right] \left[T \parallel -1\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \parallel 1\right]$
$C_2 : \quad \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \parallel 2_{010}\right]$	$\left[T \parallel m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 17.2.55 : ${}^{1}2/{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}2/{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel -1\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 2_{010}\right]$ $\left[\Sigma_z T \parallel m_{010}\right]$ $\left[\Sigma_z T \parallel -1\right]$ $\left[\bar{E} \parallel 1\right]$
$\left[\bar{E} \parallel 2_{010}\right]$ $\left[\bar{E}T \parallel m_{010}\right]$ $\left[\bar{E}T \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[\bar{E} \parallel 1\right]$
C_2 : $\left[E \parallel 2_{010}\right]$ C_6 : $\left[\bar{E} \parallel 2_{010}\right]$
C_3 : $\left[\Sigma_z \parallel 1\right]$ C_7 : $\left[\bar{\Sigma}_z \parallel 1\right]$
C_4 : $\left[\Sigma_z \parallel 2_{010}\right]$ C_8 : $\left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{010}\right]$	$\left[T \parallel m_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 17.3.56 : ${}^{1}2/{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}2/{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 2_{010}\right] \left[T \parallel m_{010}\right] \left[T \parallel -1\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{010}\right]$
C_3 : $\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \parallel 2_{010}\right]$	$\left[2_{100} \parallel m_{010}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 18.1.57 : ${}^{2}2/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid 2_{010}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[2_{010} \mid\mid -1\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	−1	−1
χ_3	1	−1	−1	1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[2_{010} \mid\mid 2_{010}\right]$	$\left[E \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1

No 18.2.58 : ${}^{2_y}2/{}^{1}m^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[2_{010} \,||\, -1\right]$ $\left[2_{010}T \,||\, 1\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{010} \,||\, 2_{010}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$
C_4 : $\left[2_{010} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{010} \,||\, 2_{010}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{010}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 18.3.59 : ${}^{2_y}2/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[2_{010} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{010} \,||\, 2_{010}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$
C_4 : $\left[2_{010} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{010} \,||\, 2_{010}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 18.4.60 : ${}^{2_y}2/{}^{1}m^{\infty_x}1 \rightarrow {}^{2_y}2/{}^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$
$\left[2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid m_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	0	2	−2	−2
χ_2	2	0	−2	−2	2

Irrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$	$\left[E \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$

No 19.1.61 : ${}^{m}2/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{010}\right]$	$\left[2_{010}T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 19.2.62 : ${}^{m_y}2/{}^{1}m^{\infty_x}1 \to {}^{m}2/{}^{1}m$

No 20.1.63 : ${}^{\bar{1}}2/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{010}\right]$	$\left[T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 20.2.64 : ${}^{\bar{1}}2/{}^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}2/{}^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[\bar{E} \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{010}\right]$ C_6 : $\left[\bar{E} \mid\mid m_{010}\right]$
C_3 : $\left[\Sigma_z \mid\mid 1\right]$ C_7 : $\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_4 : $\left[\Sigma_z \mid\mid m_{010}\right]$ C_8 : $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid m_{010}\right]$	$\left[T \mid\mid 2_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 20.3.65 : ${}^{\bar{1}}2/{}^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}2/{}^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$
$\left[\bar{E} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E}2_{100}T \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{010}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid m_{010}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 21.1.66 : ${}^{2}2/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[E \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel 2_{010}\right]$
C_3 : $\left[2_{010} \parallel m_{010}\right]$
C_4 : $\left[E \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	−1	−1
χ_3	1	−1	−1	1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{010} \parallel m_{010}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1

No 21.2.67 : ${}^{2_y}2/{}^{2_y}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[E \parallel -1\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{010}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel 2_{010}\right]$
C_3 : $\left[2_{010} \parallel m_{010}\right]$
C_4 : $\left[E \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{010} \parallel m_{010}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 21.3.68 : ${}^{2_y}2/{}^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid 2_{010}\right]$
C_3 : $\left[2_{010} \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid 2_{010}\right]$	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 21.4.69 : ${}^{2_y}2/{}^{2_y}m^{\infty_x}1 \rightarrow {}^{2_y}2/{}^{2_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[E \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$
$\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid -1\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid -1\right]$
C_3 : $\left[E \mid\mid -1\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	0	2	−2	−2
χ_2	2	0	−2	−2	2

Irrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$	$\left[\bar{E}2_{100} \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 22.1.70 : ${}^{m}2/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid -1\right]$	$\left[2_{010}T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 22.2.71 : ${}^{m_y}2/{}^{m_y}m^{\infty_x}1 \to {}^{m}2/{}^{m}m$

No 23.1.72 : $^{\bar{1}}2/^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[E \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \parallel -1\right]$	$\left[T \parallel 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 23.2.73 : ${}^{\bar{1}}2/{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}2/{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[E \parallel -1\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z T \parallel 2_{010}\right]$ $\left[\Sigma_z T \parallel m_{010}\right]$ $\left[\Sigma_z \parallel -1\right]$ $\left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel 2_{010}\right]$ $\left[\bar{E}T \parallel m_{010}\right]$ $\left[\bar{E} \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[\bar{E} \parallel 1\right]$
C_2 : $\left[E \parallel -1\right]$ C_6 : $\left[\bar{E} \parallel -1\right]$
C_3 : $\left[\Sigma_z \parallel 1\right]$ C_7 : $\left[\bar{\Sigma}_z \parallel 1\right]$
C_4 : $\left[\Sigma_z \parallel -1\right]$ C_8 : $\left[\bar{\Sigma}_z \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \parallel -1\right]$	$\left[T \parallel 2_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 23.3.74 : ${}^{\bar{1}}2/{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}2/{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 2_{010}\right] \left[T \parallel m_{010}\right] \left[E \parallel -1\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel -1\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel -1\right]$
C_3 : $\left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \parallel -1\right]$	$\left[2_{100} \parallel 2_{010}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 24.1.75 : ${}^{2_z}2/{}^{2_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_3 : $\left[2_{100} \mid\mid m_{010}\right]$
C_4 : $\left[2_{010} \mid\mid -1\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	−1	−1
χ_3	1	−1	−1	1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{100} \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1

No 24.2.76 : ${}^{2_z}2/{}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[2_{010} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, 2_{010}\right]$
C_3 : $\left[2_{100} \,||\, m_{010}\right]$
C_4 : $\left[2_{010} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 2_{010}\right]$	$\left[2_{100} \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 24.3.77 : ${}^{2_z}2/{}^{2_x}m^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_3 : $\left[2_{100} \mid\mid m_{010}\right]$
C_4 : $\left[2_{010} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 24.4.78 : ${}^{2_z}2/{}^{2_x}m^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[2_{010} \,||\, -1\right]$ $\left[2_{010}T \,||\, 1\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, 2_{010}\right]$
C_3 : $\left[2_{100} \,||\, m_{010}\right]$
C_4 : $\left[2_{010} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 2_{010}\right]$	$\left[2_{100} \,||\, m_{010}\right]$	$\left[2_{010}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 25.1.79 : ${}^{2_z}2/{}^{m_x}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 2_{010}\right]$	$\left[2_{100}T \,||\, m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 26.1.80 : ${}^{m_x}2/{}^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{001} \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \parallel -1\right]$	$\left[2_{100}T \parallel 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 27.1.81 : ${}^{m_x}2/{}^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 28.1.82 : ${}^{2}2/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid 2_{010}\right]$	$\left[2_{010}T \mid\mid m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 28.2.83 : ${}^{2_y}2/{}^{m_y}m^{\infty_x}1 \rightarrow {}^{2_y}2/{}^{m_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100}T \mid\mid m_{010}\right] \left[T \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right] \left[\Sigma_z T \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right]$
$\left[2_{100} \mid\mid 2_{010}\right] \left[2_{100}T \mid\mid m_{010}\right] \left[\bar{E}T \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$	$\left[\bar{E}2_{100}T \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 29.1.84 : ${}^{2}2/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{010}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \parallel 2_{010}\right]$	$\left[T \parallel m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 29.2.85 : ${}^{2_y}2/^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_y}2/^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[T \mid\mid m_{010}\right] \left[\bar{E}2_{100}T \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right]$
$\left[2_{100} \mid\mid 2_{010}\right] \left[\bar{E}T \mid\mid m_{010}\right] \left[2_{100}T \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 30.1.86 : $^{\bar{1}}2/^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid -1\right]$	$\left[T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 30.2.87 : ${}^{\bar{1}}2/^{m_y}m^{\infty_x}1 \rightarrow {}^{\bar{1}}2/^{m_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel 2_{010}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[\bar{E}2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
C_3 : $\left[\bar{E} \parallel 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[\bar{E}2_{100} \parallel -1\right]$	$\left[T \parallel 2_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 31.1.88 : ${}^{\bar{1}}2/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 31.2.89 : ${}^{\bar{1}}2/{}^{2_y}m^{\infty_x}1 \rightarrow {}^{\bar{1}}2/{}^{2_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
C_3 : $\left[\bar{E} \parallel 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[\bar{E}2_{100} \parallel m_{010}\right]$	$\left[T \parallel 2_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 32.1.90 : ${}^{m}2/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid -1\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid -1\right]$	$\left[2_{010}T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 32.2.91 : ${}^{m_y}2/^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_y}2/^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid 2_{010}\right] \left[T \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right]$
$\left[2_{100}T \mid\mid 2_{010}\right] \left[\bar{E}T \mid\mid m_{010}\right] \left[2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[\bar{E}2_{100} \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid -1\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[\bar{E}2_{100} \mid\mid -1\right]$	$\left[\bar{E}2_{100}T \mid\mid 2_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 33.1.92 : ${}^{m}2/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{010} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[2_{010}T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 33.2.93 : ${}^{m_y}2/^{2_y}m^{\infty_x}1 \rightarrow {}^{m_y}2/^{2_y}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[T \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel -1\right] \left[\bar{E} \parallel 1\right]$
$\left[2_{100}T \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
C_3 : $\left[\bar{E} \parallel 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[\bar{E}2_{100} \parallel m_{010}\right]$	$\left[\bar{E}2_{100}T \parallel 2_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 34.1.94 : $^{1}2^{1}2^{1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[E \mid\mid 2_{010}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[E \mid\mid 2_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 34.2.95 : ${}^{1}2^{1}2^{1}2^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[E \mid\mid 2_{010}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 34.3.96 : ${}^{1}2{}^{1}2{}^{1}2{}^{\infty_z}1 \to {}^{1}2{}^{1}2{}^{1}2$

No 34.4.97 : ${}^{1}2^{1}2^{1}2^{\infty}{}_{z}m_{x}1 \rightarrow {}^{1}2^{1}2^{1}2^{m_{x}}1$

No 35.1.98 : $^{1}2^{2}2^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 35.2.99 : ${}^{1}2^{2_z}2^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 35.3.100 : ${}^{1}2^{2_z}2^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 35.4.101 : ${}^{1}2^{2_z}2^{2_z}2^{\infty_x}1 \rightarrow {}^{1}2^{2_z}2^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	2	0	−2	−2
χ_2	2	−2	0	−2	2

Irrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 36.1.102 : ${}^{1}2^{m}2^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 36.2.103 : ${}^{1}2^{m_z}2^{m_z}2^{\infty_x}1 \to {}^{1}2^{m}2^{m}2$

No 37.1.104 : ${}^{1}2^{\bar{1}}2^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 37.2.105 : ${}^{1}2^{\bar{1}}2^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{1}2^{\bar{1}}2^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 2_{100}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_5 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 2_{100}\right]$ $C_6 : \left[\bar{E} \parallel 2_{100}\right]$
$C_3 : \left[\Sigma_z \parallel 1\right]$ $C_7 : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_4 : \left[\Sigma_z \parallel 2_{100}\right]$ $C_8 : \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{100}\right]$	$\left[T \parallel 2_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 37.3.106 : ${}^{1}2^{\bar{1}}2^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{1}2^{\bar{1}}2^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 2_{100}\right]$
$\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{100}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 38.1.107 : ${}^{2}{}_{x}2^{2}{}_{y}2^{2}{}_{z}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 2_{100}\right]$
C_3 : $\left[2_{010} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[2_{010} \mid\mid 2_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 38.2.108 : ${}^{2_x}2^{2_y}2^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 2_{100}\right]$
C_3 : $\left[2_{010} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[2_{010} \mid\mid 2_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 39.1.109 : ${}^{m_x}2^{m_y}2^{2_z}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 40.1.110 : ${}^{2}2^{\bar{1}}2^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{100}\right]$	$\left[T \mid\mid 2_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 40.2.111 : ${}^{2_z}2^{\bar{1}}2^{m_z}2^{\infty_x}1 \rightarrow {}^{2_z}2^{\bar{1}}2^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100} \mid\mid 2_{100}\right] \left[T \mid\mid 2_{010}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z T \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} T \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}T \mid\mid 2_{010}\right] \left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[T \mid\mid 2_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 41.1.112 : ${}^{1}m^{1}m^{1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[E \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 41.2.113 : ${}^{1}m^{1}m^{1}2^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 41.3.114 : ${}^{1}m^{1}m^{1}2^{\infty_z}1 \to {}^{1}m^{1}m^{1}2$

No 41.4.115 : ${}^{1}m^{1}m^{1}2^{\infty}{}_{z}{}^{m_x}1 \rightarrow {}^{1}m^{1}m^{1}2^{m_x}1$

No 42.1.116 : ${}^{2}m^{2}m^{1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{100}\right]$
C_3 : $\left[2_{001} \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{001} \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 42.2.117 : ${}^{2_z}m^{2_z}m^{1}2^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 42.3.118 : ${}^{2_z}m^{2_z}m^{1}2^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 42.4.119 : ${}^{2_z}m^{2_z}m^{1}2^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}m^{1}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100} \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[E \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E} 2_{100} \mid\mid m_{100}\right] \left[\bar{E} 2_{100} \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\bar{E} 2_{100} \mid\mid m_{100}\right] \left[\bar{E} 2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	0	2	−2	−2
χ_2	2	0	−2	−2	2

Irrepresntations of group generators

	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 43.1.120 : ${}^{m}m^{m}m^{1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 43.2.121 : ${}^{m_z}m^{m_z}m^{1}2^{\infty_x}1 \to {}^{m}m^{m}m^{1}2$

No 44.1.122 : ${}^{\bar{1}}m^{\bar{1}}m^{1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 44.2.123 : ${}^{\bar{1}}m^{\bar{1}}m^{1}2^{\infty_z}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}m^{1}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_5 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 2_{001}\right]$ $C_6 : \left[\bar{E} \parallel 2_{001}\right]$
$C_3 : \left[\Sigma_z \parallel 1\right]$ $C_7 : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_4 : \left[\Sigma_z \parallel 2_{001}\right]$ $C_8 : \left[\bar{\Sigma}_z \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 44.3.124 : ${}^{\bar{1}}m^{\bar{1}}m^{1}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}m^{1}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{100}\right] \left[T \mid\mid m_{010}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid 1\right] \left[2_{100} \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid m_{100}\right]$
$\left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 45.1.125 : ${}^{1}m^{2}m^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[2_{001} \mid\mid m_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001} \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 45.2.126 : ${}^{1}m^{2_z}m^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 45.3.127 : ${}^{1}m^{2_z}m^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 45.4.128 : ${}^{1}m^{2_z}m^{2_z}2^{\infty_x}1 \rightarrow {}^{1}m^{2_z}m^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[2_{100} \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E} \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[2_{100} \mid\mid m_{010}\right] \left[2_{100} \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid m_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	2	0	−2	−2
χ_2	2	−2	0	−2	2

Irrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 46.1.129 : ${}^{1}m^{m}m^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 46.2.130 : ${}^{1}m^{m_z}m^{m_z}2^{\infty_x}1 \to {}^{1}m^{m}m^{m}2$

No 47.1.131 : ${}^{1}m^{\bar{1}}m^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{010}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 47.2.132 : ${}^{1}m^{\bar{1}}m^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{1}m^{\bar{1}}m^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel m_{100}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{010}\right]$ $\left[\Sigma_z T \parallel 2_{001}\right]$ $\left[\bar{E} \parallel 1\right]$
$\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{010}\right]$ $\left[\bar{E}T \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[\bar{E} \parallel 1\right]$
C_2 : $\left[E \parallel m_{100}\right]$ C_6 : $\left[\bar{E} \parallel m_{100}\right]$
C_3 : $\left[\Sigma_z \parallel 1\right]$ C_7 : $\left[\bar{\Sigma}_z \parallel 1\right]$
C_4 : $\left[\Sigma_z \parallel m_{100}\right]$ C_8 : $\left[\bar{\Sigma}_z \parallel m_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \parallel m_{100}\right]$	$\left[T \parallel m_{010}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 47.3.133 : ${}^{1}m^{\bar{1}}m^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{1}m^{\bar{1}}m^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, m_{100}\right] \left[T \,||\, m_{010}\right] \left[T \,||\, 2_{001}\right] \left[2_{100}T \,||\, 1\right] \left[2_{100}T \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z T \,||\, m_{010}\right] \left[\Sigma_z T \,||\, 2_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 1\right] \left[\Sigma_z 2_{100}T \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} \,||\, m_{100}\right]$
$\left[\bar{E}T \,||\, m_{010}\right] \left[\bar{E}T \,||\, 2_{001}\right] \left[\bar{E}2_{100}T \,||\, 1\right] \left[\bar{E}2_{100}T \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right]$
C_4 : $\left[\bar{E} \,||\, 1\right]$
C_5 : $\left[\bar{E} \,||\, m_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[2_{100} \,||\, m_{010}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 48.1.134 : ${}^{2_x}m^{2_y}m^{2_z}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{100}\right]$
C_3 : $\left[2_{010} \mid\mid m_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	−1	−1	1
χ_3	1	1	−1	−1
χ_4	1	−1	1	−1

Irrepresntations of group generators

	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{010} \mid\mid m_{010}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1

No 48.2.135 : ${}^{2_x}m^{2_y}m^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{100}\right]$
C_3 : $\left[2_{010} \mid\mid m_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{010} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 48.3.136 : ${}^{2_x}m^{2_y}m^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, m_{100}\right]$
C_3 : $\left[2_{010} \,||\, m_{010}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{010} \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 49.1.137 : ${}^{m_x}m^{m_y}m^{2_z}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 50.1.138 : ${}^{m_x}m^{2_z}m^{m_y}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 51.1.139 : ${}^{m}m^{\bar{1}}m^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 51.2.140 : ${}^{m_z}m^{\bar{1}}m^{2_z}2^{\infty_x}1 \rightarrow {}^{m_z}m^{\bar{1}}m^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
C_3 : $\left[\bar{E} \parallel 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \parallel 2_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 52.1.141 : ${}^{m}m^{2}m^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$

No 52.2.142 : ${}^{m_z}m^{2_z}m^{\bar{1}}2^{\infty_x}1 \rightarrow {}^{m_z}m^{2_z}m^{\bar{1}}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100}T \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[T \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100}T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z T \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E}2_{100}T \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{E}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 53.1.143 : ${}^{\bar{1}}m^{2}m^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$

No 53.2.144 : ${}^{\bar{1}}m^{2_z}m^{m_z}2^{\infty_x}1 \rightarrow {}^{\bar{1}}m^{2_z}m^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E}T \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
C_3 : $\left[\bar{E} \mid\mid 1\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	2	0	−2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 54.1.145 : ${}^{1}m^{1}m^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[E \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$ C_6 : $\left[E \mid\mid 2_{010}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$ C_7 : $\left[E \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$ C_8 : $\left[E \mid\mid -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[E \mid\mid m_{010}\right]$	$\left[E \mid\mid m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 54.2.146 : ${}^{1}m^{1}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[2_{001}T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[E \,||\, m_{001}\right]$
C_2 : $\left[E \,||\, m_{100}\right]$ C_6 : $\left[E \,||\, 2_{010}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$ C_7 : $\left[E \,||\, 2_{100}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$ C_8 : $\left[E \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 54.3.147 : ${}^{1}m^{1}m^{1}m^{\infty_z}1 \to {}^{1}m^{1}m^{1}m$

No 54.4.148 : ${}^{1}m^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}m^{1}m^{1}m^{m_x}1$

No 55.1.149 : ${}^{1}m^{2}m^{2}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[E \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[E \,||\, m_{100}\right]$	C_6 :	$\left[2_{001} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{001} \,||\, m_{010}\right]$	C_7 :	$\left[E \,||\, 2_{100}\right]$
C_4 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[E \,||\, -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 55.2.150 : ${}^{1}m^{2_z}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[2_{001}T \parallel m_{100}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[E \parallel m_{100}\right]$	C_6 :	$\left[2_{001} \parallel 2_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{010}\right]$	C_7 :	$\left[E \parallel 2_{100}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \parallel m_{100}\right]$	$\left[2_{001} \parallel m_{010}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 55.3.151 : ${}^{1}m^{2_z}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel -1\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[2_{100}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[E \parallel m_{100}\right]$	C_6 :	$\left[2_{001} \parallel 2_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{010}\right]$	C_7 :	$\left[E \parallel 2_{100}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \parallel m_{100}\right]$	$\left[2_{001} \parallel m_{010}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 55.4.152 : ${}^{1}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, 2_{001}\right] \left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, 2_{010}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, -1\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, -1\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} \,||\, m_{100}\right]$
$\left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}2_{100} \,||\, 2_{010}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, -1\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, 2_{001}\right] \left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, -1\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}2_{100} \,||\, 2_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, -1\right]$
C_4 : $\left[E \,||\, 2_{100}\right]$
C_5 : $\left[E \,||\, -1\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, m_{100}\right]$
C_8 : $\left[\bar{E} \,||\, 2_{100}\right]$
C_9 : $\left[\bar{E} \,||\, -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	2	0	2	2	−2	−2	−2	−2
χ_2	2	−2	0	2	−2	−2	2	−2	2
χ_3	2	−2	0	−2	2	−2	2	2	−2
χ_4	2	2	0	−2	−2	−2	−2	2	2

Irrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[2_{100} \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$-\sigma_0$	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 56.1.153 : ${}^{1}m^{m}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[E \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid m_{010}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$

No 56.2.154 : ${}^{1}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}m^{m}m^{m}m$

No 57.1.155 : ${}^{1}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[E \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[E \mid\mid 2_{100}\right]$	$\left[T \mid\mid m_{010}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$

No 57.2.156 : ${}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[E \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, m_{100}\right]$ $\left[\Sigma_z T \,||\, m_{010}\right]$ $\left[\Sigma_z T \,||\, 2_{001}\right]$ $\left[\Sigma_z T \,||\, m_{001}\right]$ $\left[\Sigma_z T \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\Sigma_z \,||\, -1\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, m_{100}\right]$
$\left[\bar{E}T \,||\, m_{010}\right]$ $\left[\bar{E}T \,||\, 2_{001}\right]$ $\left[\bar{E}T \,||\, m_{001}\right]$ $\left[\bar{E}T \,||\, 2_{010}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$ $\left[\bar{E} \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_9 :	$\left[\bar{E} \,||\, 1\right]$
C_2 :	$\left[E \,||\, m_{100}\right]$	C_{10} :	$\left[\bar{E} \,||\, m_{100}\right]$
C_3 :	$\left[E \,||\, 2_{100}\right]$	C_{11} :	$\left[\bar{E} \,||\, 2_{100}\right]$
C_4 :	$\left[E \,||\, -1\right]$	C_{12} :	$\left[\bar{E} \,||\, -1\right]$
C_5 :	$\left[\Sigma_z \,||\, 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \,||\, 1\right]$
C_6 :	$\left[\Sigma_z \,||\, m_{100}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \,||\, m_{100}\right]$
C_7 :	$\left[\Sigma_z \,||\, 2_{100}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$
C_8 :	$\left[\Sigma_z \,||\, -1\right]$	C_{16} :	$\left[\bar{\Sigma}_z \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0
χ_3^{co}	2	−2	−2	2	0	0	0	0	−2	2	2	−2	0	0	0	0
χ_4^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, 2_{100}\right]$	$\left[T \,||\, m_{010}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 57.3.157 : ${}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[2_{100} \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$ $\left[2_{100}T \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z \,||\, m_{100}\right]$
$\left[\Sigma_z T \,||\, m_{010}\right]$ $\left[\Sigma_z T \,||\, 2_{001}\right]$ $\left[\Sigma_z T \,||\, m_{001}\right]$ $\left[\Sigma_z T \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\Sigma_z \,||\, -1\right]$ $\left[\Sigma_z 2_{100}T \,||\, 1\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{010}\right]$
$\left[\Sigma_z 2_{100} \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$ $\left[\Sigma_z 2_{100}T \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100}T \,||\, -1\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, m_{100}\right]$ $\left[\bar{E}T \,||\, m_{010}\right]$ $\left[\bar{E}T \,||\, 2_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right]$ $\left[\bar{E}T \,||\, 2_{010}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$ $\left[\bar{E} \,||\, -1\right]$ $\left[\bar{E}2_{100}T \,||\, 1\right]$ $\left[\bar{E}2_{100}T \,||\, m_{100}\right]$ $\left[\bar{E}2_{100} \,||\, m_{010}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{001}\right]$ $\left[\bar{E}2_{100} \,||\, m_{001}\right]$
$\left[\bar{E}2_{100} \,||\, 2_{010}\right]$ $\left[\bar{E}2_{100}T \,||\, 2_{100}\right]$ $\left[\bar{E}2_{100}T \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{100}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[E \,||\, 2_{100}\right]$
C_4 : $\left[E \,||\, -1\right]$
C_5 : $\left[2_{100} \,||\, m_{010}\right]$ $\left[2_{100} \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z \,||\, m_{100}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\Sigma_z \,||\, -1\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{010}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{E}2_{100} \,||\, m_{010}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{001}\right]$ $\left[\bar{E}2_{100} \,||\, m_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, m_{100}\right]$
C_8 : $\left[\bar{E} \,||\, 2_{100}\right]$
C_9 : $\left[\bar{E} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	−2	2	−2	2
χ_3^{co}	2	−2	−2	2	0	−2	2	2	−2
χ_4^{co}	2	2	−2	−2	0	−2	−2	2	2

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, 2_{100}\right]$	$\left[2_{100} \,||\, m_{010}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 58.1.158 : ${}^{1}m^{1}m^{2}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[2_{001} \,||\, m_{001}\right]$
C_2 : $\left[E \,||\, m_{100}\right]$ C_6 : $\left[2_{001} \,||\, 2_{010}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$ C_7 : $\left[2_{001} \,||\, 2_{100}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$ C_8 : $\left[2_{001} \,||\, -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 58.2.159 : ${}^{1}m^{1}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[2_{001}T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[2_{001} \,||\, m_{001}\right]$
C_2 : $\left[E \,||\, m_{100}\right]$ C_6 : $\left[2_{001} \,||\, 2_{010}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$ C_7 : $\left[2_{001} \,||\, 2_{100}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$ C_8 : $\left[2_{001} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 58.3.160 : ${}^{1}m^{1}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$ $\left[2_{010}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[2_{001} \,||\, m_{001}\right]$
C_2 : $\left[E \,||\, m_{100}\right]$ C_6 : $\left[2_{001} \,||\, 2_{010}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$ C_7 : $\left[2_{001} \,||\, 2_{100}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$ C_8 : $\left[2_{001} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 58.4.161 : ${}^1m^1m^{2_z}m^{\infty_x}1 \rightarrow {}^1m^1m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, m_{100}\right]$ $\left[\Sigma_z \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, m_{100}\right]$
$\left[\bar{E} \,||\, m_{010}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E} 2_{100} \,||\, m_{001}\right]$ $\left[\bar{E} 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{E} 2_{100} \,||\, 2_{100}\right]$ $\left[\bar{E} 2_{100} \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$
C_5 : $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z \,||\, m_{100}\right]$ $\left[\Sigma_z \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\bar{E} 2_{100} \,||\, m_{001}\right]$ $\left[\bar{E} 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{E} 2_{100} \,||\, 2_{100}\right]$ $\left[\bar{E} 2_{100} \,||\, -1\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, m_{100}\right]$
C_8 : $\left[\bar{E} \,||\, m_{010}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	2	2	2	0	−2	−2	−2	−2
χ_2	2	−2	−2	2	0	−2	2	2	−2
χ_3	2	2	−2	−2	0	−2	−2	2	2
χ_4	2	−2	2	−2	0	−2	2	−2	2

Irrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 59.1.162 : ${}^{1}m^{1}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[E \mid\mid m_{010}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 59.2.163 : ${}^{1}m^{1}m^{m_z}m^{\infty_x}1 \to {}^{1}m^{1}m^{m}m$

No 60.1.164 : ${}^{1}m^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{100}\right]$
C_3 : $\left[E \mid\mid m_{010}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[E \mid\mid m_{010}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 60.2.165 : ${}^{1}m^{1}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}m^{1}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, m_{100}\right]$ $\left[\Sigma_z \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z T \,||\, m_{001}\right]$ $\left[\Sigma_z T \,||\, 2_{010}\right]$ $\left[\Sigma_z T \,||\, 2_{100}\right]$ $\left[\Sigma_z T \,||\, -1\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, m_{100}\right]$
$\left[\bar{E} \,||\, m_{010}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E}T \,||\, m_{001}\right]$ $\left[\bar{E}T \,||\, 2_{010}\right]$ $\left[\bar{E}T \,||\, 2_{100}\right]$ $\left[\bar{E}T \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_9 :	$\left[\bar{E} \,||\, 1\right]$
C_2 :	$\left[E \,||\, m_{100}\right]$	C_{10} :	$\left[\bar{E} \,||\, m_{100}\right]$
C_3 :	$\left[E \,||\, m_{010}\right]$	C_{11} :	$\left[\bar{E} \,||\, m_{010}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_{12} :	$\left[\bar{E} \,||\, 2_{001}\right]$
C_5 :	$\left[\Sigma_z \,||\, 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \,||\, 1\right]$
C_6 :	$\left[\Sigma_z \,||\, m_{100}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \,||\, m_{100}\right]$
C_7 :	$\left[\Sigma_z \,||\, m_{010}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \,||\, m_{010}\right]$
C_8 :	$\left[\Sigma_z \,||\, 2_{001}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	−2	−2	2	0	0	0	0	−2	2	2	−2	0	0	0	0
χ_3^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_4^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 60.3.166 : ${}^{1}m^{1}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}m^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z \,||\, m_{100}\right]$
$\left[\Sigma_z \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z T \,||\, m_{001}\right]$ $\left[\Sigma_z T \,||\, 2_{010}\right]$ $\left[\Sigma_z T \,||\, 2_{100}\right]$ $\left[\Sigma_z T \,||\, -1\right]$ $\left[\Sigma_z 2_{100}T \,||\, 1\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{100}\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{010}\right]$
$\left[\Sigma_z 2_{100}T \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, m_{100}\right]$ $\left[\bar{E} \,||\, m_{010}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right]$ $\left[\bar{E}T \,||\, 2_{010}\right]$ $\left[\bar{E}T \,||\, 2_{100}\right]$ $\left[\bar{E}T \,||\, -1\right]$ $\left[\bar{E}2_{100}T \,||\, 1\right]$ $\left[\bar{E}2_{100}T \,||\, m_{100}\right]$ $\left[\bar{E}2_{100}T \,||\, m_{010}\right]$ $\left[\bar{E}2_{100}T \,||\, 2_{001}\right]$ $\left[\bar{E}2_{100} \,||\, m_{001}\right]$
$\left[\bar{E}2_{100} \,||\, 2_{010}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{100}\right]$ $\left[\bar{E}2_{100} \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z T \,||\, -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{100}\right]$
C_3 : $\left[E \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$
C_5 : $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z \,||\, m_{100}\right]$ $\left[\Sigma_z \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\bar{E}2_{100} \,||\, m_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{010}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{100}\right]$ $\left[\bar{E}2_{100} \,||\, -1\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, m_{100}\right]$
C_8 : $\left[\bar{E} \,||\, m_{010}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	−2	−2	2	0	−2	2	2	−2
χ_3^{co}	2	2	−2	−2	0	−2	−2	2	2
χ_4^{co}	2	−2	2	−2	0	−2	2	−2	2

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[E \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 61.1.167 : ${}^{2}m^{2}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001} \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel m_{100}\right]$	C_6 :	$\left[E \parallel 2_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{010}\right]$	C_7 :	$\left[E \parallel 2_{100}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001} \parallel m_{010}\right]$	$\left[2_{001} \parallel m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 61.2.168 : ${}^{2_z}m^{2_z}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[T \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, m_{100}\right]$	C_6 :	$\left[E \,||\, 2_{010}\right]$
C_3 :	$\left[2_{001} \,||\, m_{010}\right]$	C_7 :	$\left[E \,||\, 2_{100}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 61.3.169 : ${}^{2_z}m^{2_z}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel m_{100}\right]$	C_6 :	$\left[E \parallel 2_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{010}\right]$	C_7 :	$\left[E \parallel 2_{100}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001} \parallel m_{010}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 61.4.170 : ${}^{2_z}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} 2_{100} \,||\, m_{100}\right]$
$\left[\bar{E} 2_{100} \,||\, m_{010}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E} 2_{100} \,||\, m_{001}\right]$ $\left[\bar{E} \,||\, 2_{010}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$ $\left[\bar{E} 2_{100} \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\bar{E} 2_{100} \,||\, m_{100}\right]$ $\left[\bar{E} 2_{100} \,||\, m_{010}\right]$ $\left[\bar{E} 2_{100} \,||\, m_{001}\right]$ $\left[\bar{E} 2_{100} \,||\, -1\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[E \,||\, 2_{010}\right]$
C_5 : $\left[E \,||\, 2_{100}\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_8 : $\left[\bar{E} \,||\, 2_{010}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	−2	2	2
χ_3	2	0	−2	−2	2	−2	2	2	−2
χ_4	2	0	−2	2	−2	−2	2	−2	2

Irrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{100} \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 62.1.171 : ${}^{m}m^{m}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{010}\right]$	$\left[2_{001}T \parallel m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 62.2.172 : ${}^{m_z}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{m}m^{m}m^{m}m$

No 63.1.173 : ${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{010}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{010}\right]$	$\left[T \parallel m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 63.2.174 : ${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right]$ $\left[\Sigma_z T \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid m_{100}\right]$
$\left[\bar{E}T \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E} \mid\mid 2_{010}\right]$ $\left[\bar{E} \mid\mid 2_{100}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_9 :	$\left[\bar{E} \mid\mid 1\right]$
C_2 :	$\left[E \mid\mid 2_{001}\right]$	C_{10} :	$\left[\bar{E} \mid\mid 2_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{010}\right]$	C_{11} :	$\left[\bar{E} \mid\mid 2_{010}\right]$
C_4 :	$\left[E \mid\mid 2_{100}\right]$	C_{12} :	$\left[\bar{E} \mid\mid 2_{100}\right]$
C_5 :	$\left[\Sigma_z \mid\mid 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_6 :	$\left[\Sigma_z \mid\mid 2_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
C_7 :	$\left[\Sigma_z \mid\mid 2_{010}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \mid\mid 2_{010}\right]$
C_8 :	$\left[\Sigma_z \mid\mid 2_{100}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	−2	2	0	0	0	0	−2	2	2	−2	0	0	0	0
χ_4^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 63.3.175 : ${}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[T \,||\, m_{100}\right] \left[T \,||\, m_{010}\right] \left[E \,||\, 2_{001}\right] \left[T \,||\, m_{001}\right] \left[E \,||\, 2_{010}\right] \left[E \,||\, 2_{100}\right] \left[T \,||\, -1\right] \left[2_{100}T \,||\, 1\right]$
$\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100}T \,||\, 2_{001}\right] \left[2_{100} \,||\, m_{001}\right] \left[2_{100}T \,||\, 2_{010}\right] \left[2_{100}T \,||\, 2_{100}\right] \left[2_{100} \,||\, -1\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z T \,||\, m_{100}\right]$
$\left[\Sigma_z T \,||\, m_{010}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z T \,||\, m_{001}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z T \,||\, -1\right] \left[\Sigma_z 2_{100}T \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right]$
$\left[\Sigma_z 2_{100}T \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{010}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{100}\right] \left[\Sigma_z 2_{100} \,||\, -1\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}T \,||\, m_{100}\right] \left[\bar{E}T \,||\, m_{010}\right] \left[\bar{E} \,||\, 2_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right] \left[\bar{E} \,||\, 2_{010}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E}T \,||\, -1\right] \left[\bar{E}2_{100}T \,||\, 1\right] \left[\bar{E}2_{100} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100}T \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right]$
$\left[\bar{E}2_{100}T \,||\, 2_{010}\right] \left[\bar{E}2_{100}T \,||\, 2_{100}\right] \left[\bar{E}2_{100} \,||\, -1\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z T \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z T \,||\, m_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z T \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{100}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 2_{001}\right]$
C_3 : $\left[E \,||\, 2_{010}\right]$
C_4 : $\left[E \,||\, 2_{100}\right]$
C_5 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, -1\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 2_{100}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, -1\right] \left[\bar{E}2_{100} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}2_{100} \,||\, -1\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_8 : $\left[\bar{E} \,||\, 2_{010}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	−2	−2	2	2
χ_3^{co}	2	−2	−2	2	0	−2	2	2	−2
χ_4^{co}	2	−2	2	−2	0	−2	2	−2	2

Coirrepresntations of group generators

	$\left[E \,||\, 2_{001}\right]$	$\left[E \,||\, 2_{010}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$

No 64.1.176 : ${}^{2_z}m^{2_z}m^{2_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{100} \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel m_{100}\right]$	C_6 :	$\left[2_{010} \parallel 2_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{010}\right]$	C_7 :	$\left[2_{010} \parallel 2_{100}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{100} \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001} \parallel m_{010}\right]$	$\left[2_{100} \parallel m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 64.2.177 : ${}^{2_z}m^{2_z}m^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[T \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$ $\left[2_{010}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{100} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, m_{100}\right]$	C_6 :	$\left[2_{010} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{001} \,||\, m_{010}\right]$	C_7 :	$\left[2_{010} \,||\, 2_{100}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{100} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 64.3.178 : ${}^{2_z}m^{2_z}m^{2_x}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{100} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, m_{100}\right]$	C_6 :	$\left[2_{010} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{001} \,||\, m_{010}\right]$	C_7 :	$\left[2_{010} \,||\, 2_{100}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{100} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 64.4.179 : ${}^{2_z}m^{2_z}m^{2_x}m^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010}T \,||\, 1\right]$
$\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{100} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, m_{100}\right]$	C_6 :	$\left[2_{010} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{001} \,||\, m_{010}\right]$	C_7 :	$\left[2_{010} \,||\, 2_{100}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{100} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{010}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 65.1.180 : ${}^{m_y}m^{m_y}m^{m_x}m$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[2_{010}T \mid\mid m_{100}\right] \left[2_{010}T \mid\mid m_{010}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid m_{001}\right] \left[2_{001} \mid\mid 2_{010}\right] \left[2_{001} \mid\mid 2_{100}\right] \left[2_{100}T \mid\mid -1\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{010}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 66.1.181 : ${}^{m_y}m{}^{m_y}m{}^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{010}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 67.1.182 : ${}^{2_z}m^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$ $\left[2_{010}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{010}T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 68.1.183 : ${}^{m}m^{m}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 68.2.184 : ${}^{m_z}m^{m_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}m^{m_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100}T \mid\mid m_{100}\right] \left[2_{100}T \mid\mid m_{010}\right] \left[E \mid\mid 2_{001}\right] \left[T \mid\mid m_{001}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[T \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z T \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z T \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid m_{100}\right]$
$\left[\bar{E}2_{100}T \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}T \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}T \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 69.1.185 : ${}^{m}m^{m}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 69.2.186 : ${}^{m_z}m^{m_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}m^{m_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{100}\right]$
$\left[\bar{E}2_{100}T \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 70.1.187 : ${}^{2}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 70.2.188 : ${}^{2_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 2_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{100}\right] \left[T \parallel -1\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z T \parallel -1\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100} \parallel m_{010}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 71.1.189 : ${}^{2}m^{2}m^{m}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{001} \,||\, m_{010}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001} \,||\, m_{010}\right]$	$\left[2_{001}T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 71.2.190 : ${}^{2_z}m^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 72.1.191 : ${}^{\bar{1}}m^{\bar{1}}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 72.2.192 : ${}^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{100}\right] \left[T \mid\mid m_{010}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100} \mid\mid m_{001}\right] \left[2_{100}T \mid\mid 2_{010}\right] \left[2_{100}T \mid\mid 2_{100}\right] \left[2_{100} \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid m_{100}\right]$
$\left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100}T \mid\mid 2_{010}\right] \left[\bar{E}2_{100}T \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 73.1.193 : ${}^{\bar{1}}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 73.2.194 : ${}^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid m_{100}\right] \left[T \mid\mid m_{010}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid m_{001}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[2_{100}T \mid\mid -1\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100}T \mid\mid -1\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid m_{100}\right]$
$\left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100}T \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 74.1.195 : ${}^{2_x}m^{2_y}m^{1}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[E \,||\, m_{001}\right]$
C_2 :	$\left[2_{100} \,||\, m_{100}\right]$	C_6 :	$\left[2_{100} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{010} \,||\, m_{010}\right]$	C_7 :	$\left[2_{010} \,||\, 2_{100}\right]$
C_4 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{010} \,||\, m_{010}\right]$	$\left[E \,||\, m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 74.2.196 : ${}^{2_x}m^{2_y}m^{1}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$ $\left[2_{010}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[E \,||\, m_{001}\right]$
C_2 : $\left[2_{100} \,||\, m_{100}\right]$ C_6 : $\left[2_{100} \,||\, 2_{010}\right]$
C_3 : $\left[2_{010} \,||\, m_{010}\right]$ C_7 : $\left[2_{010} \,||\, 2_{100}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$ C_8 : $\left[2_{001} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{010} \,||\, m_{010}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 74.3.197 : ${}^{2_x}m^{2_y}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[E \,||\, m_{001}\right]$
C_2 :	$\left[2_{100} \,||\, m_{100}\right]$	C_6 :	$\left[2_{100} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{010} \,||\, m_{010}\right]$	C_7 :	$\left[2_{010} \,||\, 2_{100}\right]$
C_4 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{010} \,||\, m_{010}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 75.1.198 : ${}^{m_x}m^{m_y}m^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 76.1.199 : ${}^{m_y}m^{2_z}m^1m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{010}T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{100}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{010}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 77.1.200 : ${}^{m}m^{\bar{1}}m^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 77.2.201 : ${}^{m_z}m^{\bar{1}}m^{1}m^{\infty_x}1 \rightarrow {}^{m_z}m^{\bar{1}}m^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\Sigma_z T \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{100}\right]$
$\left[\bar{E}T \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid m_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$

No 78.1.202 : ${}^{m}m^{2}m^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 78.2.203 : ${}^{m_z}m^{2_z}m^1m^{\infty_x}1 \rightarrow {}^{m_z}m^{2_z}m^1m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{100}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid m_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$

No 79.1.204 : ${}^{\bar{1}}m^{2}m^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{010}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{010}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 79.2.205 : ${}^{\bar{1}}m^{2_z}m^{1}m^{\infty_x}1 \rightarrow {}^{\bar{1}}m^{2_z}m^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid m_{100}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid m_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid m_{010}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 80.1.206 : ${}^{2_x}m^{2_y}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[E \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[2_{100} \parallel m_{100}\right]$	C_6 :	$\left[2_{010} \parallel 2_{010}\right]$
C_3 :	$\left[2_{010} \parallel m_{010}\right]$	C_7 :	$\left[2_{100} \parallel 2_{100}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	1	−1
χ_3	1	−1	−1	1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1	−1	−1
χ_5	1	1	−1	−1	−1	−1	1	1
χ_6	1	−1	1	−1	1	−1	1	−1
χ_7	1	−1	1	−1	−1	1	−1	1
χ_8	1	1	−1	−1	1	1	−1	−1

Irrepresntations of group generators

	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{010} \parallel m_{010}\right]$	$\left[2_{001} \parallel m_{001}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	−1
Γ_3	−1	−1	1
Γ_4	1	1	−1
Γ_5	1	−1	−1
Γ_6	−1	1	1
Γ_7	−1	1	−1
Γ_8	1	−1	1

No 80.2.207 : ${}^{2_x}m^{2_y}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[2_{100}T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[2_{100} \,||\, m_{100}\right]$	C_6 :	$\left[2_{010} \,||\, 2_{010}\right]$
C_3 :	$\left[2_{010} \,||\, m_{010}\right]$	C_7 :	$\left[2_{100} \,||\, 2_{100}\right]$
C_4 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[E \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	1	−1
χ_3^{co}	1	−1	−1	1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1
χ_5^{co}	1	1	−1	−1	−1	−1	1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1
χ_8^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{010} \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	1	$\mathcal{K}$
Γ_4^{co}	1	1	−1	$\mathcal{K}$
Γ_5^{co}	1	−1	−1	$\mathcal{K}$
Γ_6^{co}	−1	1	1	$\mathcal{K}$

Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	1	-1	1	$\mathcal{K}$

No 81.1.208 : ${}^{m_x}m^{m_y}m^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	-1	-1
χ_3^{co}	1	-1	-1	1
χ_4^{co}	1	-1	1	-1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$

No 82.1.209 : ${}^{2}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[E \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{100}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$	$\left[T \mid\mid m_{010}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$

No 82.2.210 : ${}^{2_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100} \,||\, m_{100}\right] \left[T \,||\, m_{010}\right] \left[2_{100}T \,||\, 2_{001}\right] \left[2_{100}T \,||\, m_{001}\right] \left[T \,||\, 2_{010}\right] \left[2_{100} \,||\, 2_{100}\right] \left[E \,||\, -1\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z T \,||\, m_{010}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{001}\right] \left[\Sigma_z 2_{100}T \,||\, m_{001}\right] \left[\Sigma_z T \,||\, 2_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\Sigma_z \,||\, -1\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}2_{100} \,||\, m_{100}\right]$
$\left[\bar{E}T \,||\, m_{010}\right] \left[\bar{E}2_{100}T \,||\, 2_{001}\right] \left[\bar{E}2_{100}T \,||\, m_{001}\right] \left[\bar{E}T \,||\, 2_{010}\right] \left[\bar{E}2_{100} \,||\, 2_{100}\right] \left[\bar{E} \,||\, -1\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, m_{001}\right] \left[\bar{\Sigma}_z T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\Sigma_z \,||\, -1\right] \left[\bar{E}2_{100} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, 2_{100}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, -1\right]$
C_3 : $\left[E \,||\, -1\right]$
C_4 : $\left[\bar{E} \,||\, 1\right]$
C_5 : $\left[\bar{E} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{100} \,||\, 2_{100}\right]$	$\left[T \,||\, m_{010}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 83.1.211 : ${}^{m_x}m^{m_y}m^{m_z}m$

Full symmetry operations

$$\left[E \parallel 1\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{001} \parallel 2_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{100}\right] \left[T \parallel -1\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right]$
C_3 : $\left[2_{010} \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{100}T \parallel m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 84.1.212 : ${}^{m_x}m^{2_y}m^{2_z}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[T \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{010} \,||\, m_{010}\right]$
C_3 : $\left[2_{001} \,||\, m_{001}\right]$
C_4 : $\left[2_{100} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{010} \,||\, m_{010}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 85.1.213 : ${}^{m_z}m^{2_x}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{100}T \parallel -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{100} \parallel m_{010}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 86.1.214 : ${}^{2_z}m^{m_x}m^{m_z}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{100} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{010} \,||\, 2_{100}\right]$
C_4 : $\left[2_{100} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{010} \,||\, 2_{100}\right]$	$\left[2_{100}T \,||\, m_{010}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$

No 87.1.215 : ${}^{2_z}m^{\bar{1}}m^{m_x}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, m_{100}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, m_{001}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{010} \,||\, -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{100}\right]$
C_3 : $\left[2_{100} \,||\, 2_{100}\right]$
C_4 : $\left[2_{010} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{100} \,||\, 2_{100}\right]$	$\left[T \,||\, m_{010}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$

No 88.1.216 : ${}^{2_z}m^{\bar{1}}m^{2_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid m_{100}\right]$
C_3 : $\left[2_{100} \mid\mid m_{001}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	−1	1
χ_3^{co}	1	1	−1	−1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{010}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 89.1.217 : ${}^{\bar{1}}m^{m_z}m^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid -1\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{010}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	1	−1	−1	1
χ_4^{co}	1	−1	1	−1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$

No 90.1.218 : $^{1}4$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	i	-1	$-i$
χ_4	1	$-i$	-1	i

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	i
Γ_4	$-i$

No 90.2.219 : ${}^{1}4^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 90.3.220 : ${}^{1}4^{\infty_z}1 \to {}^{1}4$

No 90.4.221 : ${}^{1}4^{\infty_z m_x}1 \to {}^{1}4^{m_x}1$

No 91.1.222 : $^{2}4$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	i	-1	$-i$
χ_4	1	$-i$	-1	i

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	i
Γ_4	$-i$

No 91.2.223 : ${}^{2_z}4^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 91.3.224 : ${}^{2_z}4^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{010}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[2_{010}T \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 91.4.225 : ${}^{2_z}4^{\infty_x}1 \rightarrow {}^{2_z}4^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	0	2	−2	−2
χ_2	2	0	−2	−2	2

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_2	$-i\sigma_y \mathcal{K}$	$\sigma_z \mathcal{K}$

No 92.1.226 : $^{m}4$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	2	-2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$

No 92.2.227 : ${}^{m_z}4^{\infty_x}1 \to {}^{m}4$

No 93.1.228 : $^{\bar{1}}4$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	2	-2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$

No 93.2.229 : ${}^{\bar{1}}4^{\infty_z}1 \rightarrow {}^{\bar{1}}4^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_5 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 2_{001}\right]$ $C_6 : \left[\bar{E} \parallel 2_{001}\right]$
$C_3 : \left[\Sigma_z \parallel 1\right]$ $C_7 : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_4 : \left[\Sigma_z \parallel 2_{001}\right]$ $C_8 : \left[\bar{\Sigma}_z \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 93.3.230 : ${}^{\bar{1}}4^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[T \mid\mid 4^{-}_{001}\right] \left[2_{100}T \mid\mid 1\right] \left[2_{100} \mid\mid 4^{+}_{001}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[2_{100} \mid\mid 4^{-}_{001}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z T \mid\mid 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid 4^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}T \mid\mid 4^{-}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 4^{+}_{001}\right] \left[2_{100} \mid\mid 4^{-}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 94.1.231 : $^{4}4$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001} \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001} \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^{+}_{001} \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^{-}_{001} \mid\mid 4^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	i	-1	$-i$
χ_4	1	$-i$	-1	i

Irrepresntations of group generators

	$\left[4^{+}_{001} \mid\mid 4^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	i
Γ_4	$-i$

No 94.2.232 : ${}^{4_z}4^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[4^-_{001}T \mid\mid 4^+_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[4^+_{001}T \mid\mid 4^-_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid 4^+_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^-_{001} \mid\mid 4^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 95.1.233 : ${}^{\bar{4}}4$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid 4^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	2	-2

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$

No 96.1.234 : ${}^{1}\bar{4}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	i	-1	$-i$
χ_4	1	$-i$	-1	i

Irrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	i
Γ_4	$-i$

No 96.2.235 : ${}^{1}\bar{4}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 96.3.236 : ${}^{1}\bar{4}^{\infty_z}1 \to {}^{1}\bar{4}$

No 96.4.237 : ${}^{1}\bar{4}^{\infty_z m_x}1 \to {}^{1}\bar{4}^{m_x}1$

No 97.1.238 : ${}^{2}\bar{4}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	i	-1	$-i$
χ_4	1	$-i$	-1	i

Irrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	i
Γ_4	$-i$

No 97.2.239 : ${}^{2_z}\bar{4}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 97.3.240 : ${}^{2_z}\bar{4}^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{010}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 97.4.241 : ${}^{2_z}\bar{4}^{\infty_x}1 \rightarrow {}^{2_z}\bar{4}^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel 1\right]$
$\left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	2	0	2	−2	−2
χ_2	2	0	−2	−2	2

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_2	$-i\sigma_y \mathcal{K}$	$\sigma_z \mathcal{K}$

No 98.1.242 : ${}^{m}\bar{4}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	2	-2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$

No 98.2.243 : ${}^{m_z}\bar{4}^{\infty_x}1 \to {}^{m}\bar{4}$

No 99.1.244 : ${}^{\bar{1}}\bar{4}$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	2	-2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$

No 99.2.245 : ${}^{\bar{1}}\bar{4}^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{4}^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$
$\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[\bar{E} \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$ C_6 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_3 : $\left[\Sigma_z \mid\mid 1\right]$ C_7 : $\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_4 : $\left[\Sigma_z \mid\mid 2_{001}\right]$ C_8 : $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	2	2	0	0	−2	−2	0	0
χ_2^{co}	2	−2	0	0	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 99.3.246 : ${}^{\bar{1}}\bar{4}^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{4}^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[T \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100}T \mid\mid 1\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 100.1.247 : $^{4}\bar{4}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	-1	1	-1
χ_3	1	i	-1	$-i$
χ_4	1	$-i$	-1	i

Irrepresntations of group generators

	$\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	i
Γ_4	$-i$

No 100.2.248 : ${}^{4_z}\bar{4}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[4^{-}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[4^{+}_{001}T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 101.1.249 : ${}^{\bar{4}}\bar{4}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid \bar{4}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2
χ_1^{co}	1	1
χ_2^{co}	2	-2

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[4^{+}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$

No 102.1.250 : ${}^{1}4/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[E \mid\mid m_{001}\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$	C_6 :	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_7 :	$\left[E \mid\mid -1\right]$
C_4 :	$\left[E \mid\mid 4^{-}_{001}\right]$	C_8 :	$\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 102.2.251 : ${}^{1}4/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$
$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[E \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[E \mid\mid -1\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 102.3.252 : ${}^{1}4/{}^{1}m^{\infty_z}1 \rightarrow {}^{1}4/{}^{1}m$

No 102.4.253 : ${}^{1}4/{}^{1}m^{\infty_z m_x}1 \rightarrow {}^{1}4/{}^{1}m^{m_x}1$

No 103.1.254 : ${}^{2}4/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[E \mid\mid m_{001}\right]$
C_2 :	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	C_6 :	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_7 :	$\left[E \mid\mid -1\right]$
C_4 :	$\left[2_{001} \mid\mid 4^{-}_{001}\right]$	C_8 :	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 103.2.255 : ${}^{2_z}4/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[T \parallel 4^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[E \parallel m_{001}\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_7 : $\left[E \parallel -1\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 103.3.256 : ${}^{2_z}4/{}^{1}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel 4^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[E \parallel m_{001}\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_7 : $\left[E \parallel -1\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 103.4.257 : ${}^{2_z}4/{}^{1}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[E \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 2_{001}\right]$
C_8 : $\left[\bar{E} \parallel m_{001}\right]$
C_9 : $\left[\bar{E} \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	−2	2	2
χ_3	2	0	−2	−2	2	−2	2	2	−2
χ_4	2	0	−2	2	−2	−2	2	−2	2

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$-i\sigma_y \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_4	$-i\sigma_y \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$

No 104.1.258 : ${}^{m}4/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 104.2.259 : ${}^{m_z}4/{}^{1}m^{\infty_x}1 \to {}^{m}4/{}^{1}m$

No 105.1.260 : ${}^{\bar{1}}4/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 105.2.261 : ${}^{\bar{1}}4/{}^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}4/{}^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z \parallel m_{001}\right]$ $\left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \parallel -1\right]$ $\left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^{-}_{001}\right]$ $\left[\bar{E} \parallel m_{001}\right]$ $\left[\bar{E}T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \parallel -1\right]$ $\left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel -1\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_3 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[\bar{E} \parallel m_{001}\right]$
C_4 :	$\left[E \parallel -1\right]$	C_{12} :	$\left[\bar{E} \parallel -1\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel m_{001}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel m_{001}\right]$
C_8 :	$\left[\Sigma_z \parallel -1\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	−2	2	0	0	0	0	−2	2	2	−2	0	0	0	0
χ_4^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 105.3.262 : ${}^{\bar{1}}4/{}^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4/{}^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel -1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right]$
$\left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 2_{001}\right]$
C_8 : $\left[\bar{E} \parallel m_{001}\right]$
C_9 : $\left[\bar{E} \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	−2	−2	2	2
χ_3^{co}	2	−2	−2	2	0	−2	2	2	−2
χ_4^{co}	2	−2	2	−2	0	−2	2	−2	2

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 106.1.263 : ${}^{2}4/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[2_{001} \mid\mid m_{001}\right]$
C_2 :	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	C_6 :	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_7 :	$\left[2_{001} \mid\mid -1\right]$
C_4 :	$\left[2_{001} \mid\mid 4^{-}_{001}\right]$	C_8 :	$\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 106.2.264 : ${}^{2_z}4/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[T \parallel 4^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$	C_6 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_7 :	$\left[2_{001} \parallel -1\right]$
C_4 :	$\left[2_{001} \parallel 4^{-}_{001}\right]$	C_8 :	$\left[E \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 106.3.265 : ${}^{2_z}4/{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[2_{010}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[2_{010}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{010}T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[2_{001} \mid\mid m_{001}\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[2_{001} \mid\mid -1\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 106.4.266 : ${}^{2_z}4/{}^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_5 : $\left[E \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 2_{001}\right]$
C_8 : $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	−2	2	2
χ_3	2	0	−2	$-i$	i	−2	2	i	$-i$
χ_4	2	0	−2	i	$-i$	−2	2	$-i$	i

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$-i\sigma_y\mathcal{K}$	$\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-i\sigma_y\mathcal{K}$	$-\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 107.1.267 : ${}^{m}4/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	-1	-1
χ_3^{co}	2	-2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 107.2.268 : ${}^{m_z}4/{}^{m_z}m^{\infty_x}1 \to {}^{m}4/{}^{m}m$

No 108.1.269 : ${}^{\bar{1}}4/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_4 : $\left[E \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 108.2.270 : ${}^{\bar{1}}4/{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}4/{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z T \parallel m_{001}\right]$ $\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \parallel -1\right]$ $\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^{-}_{001}\right]$ $\left[\bar{E}T \parallel m_{001}\right]$ $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \parallel -1\right]$ $\left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_3 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$	C_{11} :	$\left[\bar{E} \parallel \bar{4}^{-}_{001}\right]$
C_4 :	$\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{12} :	$\left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
C_8 :	$\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	0	0	0	0	$-i$	i	−2	2	0	0	0	0	i	$-i$
χ_4^{co}	2	−2	0	0	0	0	i	$-i$	−2	2	0	0	0	0	$-i$	i

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$i\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 108.3.271 : ${}^{\bar{1}}4/{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4/{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$
$\left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$
C_5 : $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid \bar{4}^{-}_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	−2	−2	2	2
χ_3^{co}	4	−4	0	0	0	−4	4	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -i\sigma_0 & 0 \\ 0 & i\sigma_0 \end{pmatrix}$	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -i\sigma_y \\ i\sigma_y & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 109.1.272 : $^{1}4/^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[2_{001} \mid\mid m_{001}\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$	C_6 :	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_7 :	$\left[2_{001} \mid\mid -1\right]$
C_4 :	$\left[E \mid\mid 4^{-}_{001}\right]$	C_8 :	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 109.2.273 : ${}^{1}4/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_7 : $\left[2_{001} \parallel -1\right]$
C_4 : $\left[E \parallel 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 109.3.274 : ${}^{1}4/{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{010}T \mid\mid m_{001}\right]$ $\left[2_{010}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \mid\mid -1\right]$ $\left[2_{010}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[2_{001} \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[2_{001} \mid\mid -1\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 109.4.275 : ${}^{1}4/{}^{2_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 4^{-}_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 4^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	2	2	2	0	−2	−2	−2	−2
χ_2	2	−2	2	−2	0	−2	2	−2	2
χ_3	2	i	−2	$-i$	0	−2	$-i$	2	i
χ_4	2	$-i$	−2	i	0	−2	i	2	$-i$

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 110.1.276 : ${}^{1}4/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 110.2.277 : ${}^{1}4/{}^{m_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{m}m$

No 111.1.278 : ${}^{1}4/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 111.2.279 : ${}^{1}4/{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z \parallel 4^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z T \parallel m_{001}\right]$ $\left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \parallel -1\right]$ $\left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E} \parallel 4^{-}_{001}\right]$ $\left[\bar{E}T \parallel m_{001}\right]$ $\left[\bar{E}T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \parallel -1\right]$ $\left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel -1\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_{11} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_4 :	$\left[E \parallel 4^{-}_{001}\right]$	C_{12} :	$\left[\bar{E} \parallel 4^{-}_{001}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 4^{+}_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_8 :	$\left[\Sigma_z \parallel 4^{-}_{001}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0
χ_3^{co}	2	0	−2	0	0	i	0	$-i$	−2	0	2	0	0	$-i$	0	i
χ_4^{co}	2	0	−2	0	0	$-i$	0	i	−2	0	2	0	0	i	0	$-i$

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 111.3.280 : ${}^{1}4/{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$
C_5 : $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	−2	2	−2	2
χ_3^{co}	4	0	−4	0	0	−4	0	4	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} i\sigma_0 & 0 \\ 0 & -i\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 112.1.281 : ${}^{4}4/{}^{2}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^{-}_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[4^{-}_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[4^{+}_{001} \,||\, \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$	C_6 :	$\left[4^{-}_{001} \,||\, \bar{4}^{-}_{001}\right]$
C_3 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_7 :	$\left[E \,||\, -1\right]$
C_4 :	$\left[4^{-}_{001} \,||\, 4^{-}_{001}\right]$	C_8 :	$\left[4^{+}_{001} \,||\, \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 112.2.282 : ${}^{4_z}4/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[4^-_{001} \mid\mid \bar{4}^-_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$
$\left[4^-_{001}T \mid\mid 4^+_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[4^+_{001}T \mid\mid 4^-_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[4^+_{001}T \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[4^-_{001}T \mid\mid \bar{4}^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[2_{001} \mid\mid m_{001}\right]$
C_2 :	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	C_6 :	$\left[4^-_{001} \mid\mid \bar{4}^-_{001}\right]$
C_3 :	$\left[2_{001} \mid\mid 2_{001}\right]$	C_7 :	$\left[E \mid\mid -1\right]$
C_4 :	$\left[4^-_{001} \mid\mid 4^-_{001}\right]$	C_8 :	$\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 113.1.283 : ${}^{\bar{4}}4/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[4^{-}_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[4^{+}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 114.1.284 : ${}^{4}4/{}^{1}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^{-}_{001} \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[4^{+}_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[4^{-}_{001} \,||\, \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[E \,||\, m_{001}\right]$
C_2 :	$\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$	C_6 :	$\left[4^{+}_{001} \,||\, \bar{4}^{-}_{001}\right]$
C_3 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_7 :	$\left[2_{001} \,||\, -1\right]$
C_4 :	$\left[4^{-}_{001} \,||\, 4^{-}_{001}\right]$	C_8 :	$\left[4^{-}_{001} \,||\, \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 114.2.285 : ${}^{4_z}4/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[4^{+}_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[4^{-}_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[4^{-}_{001}T \parallel 4^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[4^{+}_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[4^{-}_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[4^{+}_{001}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$	C_6 :	$\left[4^{+}_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 :	$\left[2_{001} \parallel 2_{001}\right]$	C_7 :	$\left[2_{001} \parallel -1\right]$
C_4 :	$\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$	C_8 :	$\left[4^{-}_{001} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 115.1.286 : ${}^{\bar{4}}4/{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[4^{+}_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[4^{-}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 116.1.287 : ${}^{2_z}4/{}^{2_x}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{100} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	C_6 :	$\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_7 :	$\left[2_{100} \,||\, -1\right]$
C_4 :	$\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_8 :	$\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1
χ_3	1	−1	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1	−1	1
χ_5	1	i	−1	$-i$	−1	$-i$	1	i
χ_6	1	i	−1	$-i$	1	i	−1	$-i$
χ_7	1	$-i$	−1	i	−1	i	1	$-i$
χ_8	1	$-i$	−1	i	1	$-i$	−1	i

Irrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	i	−1
Γ_6	i	1
Γ_7	$-i$	−1

Γ_8	$-i$	1

No 116.2.288 : ${}^{2_z}4/{}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[T \,||\, 4^{+}_{001}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, 4^{-}_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \,||\, -1\right]$ $\left[2_{100}T \,||\, \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[2_{100} \,||\, m_{001}\right]$
C_2 : $\left[2_{001} \,||\, 4^{+}_{001}\right]$ C_6 : $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$ C_7 : $\left[2_{100} \,||\, -1\right]$
C_4 : $\left[2_{001} \,||\, 4^{-}_{001}\right]$ C_8 : $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 116.3.289 : ${}^{2_z}4/{}^{2_x}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel 4^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$	C_6 :	$\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_7 :	$\left[2_{100} \parallel -1\right]$
C_4 :	$\left[2_{001} \parallel 4^{-}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 116.4.290 : ${}^{2_z}4/{}^{2_x}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel 1\right]$
$\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[2_{100} \parallel m_{001}\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ C_6 : $\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_7 : $\left[2_{100} \parallel -1\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$ C_8 : $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	−2	0	2	0
χ_6^{co}	2	0	−2	0	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_6^{co}	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$

No 117.1.291 : ${}^{2_z}4/{}^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 118.1.292 : ${}^{m_x}4/{}^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{010}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{010}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 119.1.293 : ${}^{m_x}4/{}^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{010}T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 120.1.294 : ${}^{2}4/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 120.2.295 : ${}^{2_z}4/^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}4/^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{4}^-_{001}\right] \left[T \parallel -1\right] \left[2_{100}T \parallel \bar{4}^+_{001}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^-_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^+_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 4^+_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^-_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^-_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{E}2_{100} \parallel 4^+_{001}\right] \left[\bar{E}2_{100} \parallel 4^-_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	4	0	−4	−4	4

Coirrepresntations of group generators

	$\left[2_{100} \parallel 4^+_{001}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 121.1.296 : ${}^{2}4/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	2	0	-2	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 121.2.297 : ${}^{2_z}4/{}^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 122.1.298 : ${}^{\bar{1}}4/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 122.2.299 : ${}^{\bar{1}}4/^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel -1\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 123.1.300 : $^{\bar{1}}4/^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 123.2.301 : ${}^{\bar{1}}4/^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 124.1.302 : ${}^{m}4/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	−2	2
χ_4^{co}	2	−2	2	−2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$

No 124.2.303 : ${}^{m_z}4/{}^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}4/{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 125.1.304 : ${}^{m}4/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 125.2.305 : ${}^{m_z}4/^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}4/^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100}T \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid 4^{-}_{001}\right] \left[T \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[T \mid\mid -1\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 4^{-}_{001}\right] \left[\Sigma_z T \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z T \mid\mid -1\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid 4^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid 4^{-}_{001}\right] \left[\bar{E}T \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}T \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 126.1.306 : ${}^{4}4/{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[4^{-}_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[4^{+}_{001}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 127.1.307 : ${}^{4}4/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[4^{+}_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[4^{-}_{001}T \parallel \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	−1	1	−1
χ_3^{co}	2	0	−2	0

Coirrepresntations of group generators

	$\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$

No 128.1.308 : ${}^{\bar{4}}4/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	-1	-1
χ_3^{co}	2	-2	0	0

Coirrepresntations of group generators

	$\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 129.1.309 : ${}^{\bar{4}}4/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[4^{+}_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[4^{-}_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[4^{+}_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_4 : $\left[4^{-}_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	-1	-1
χ_3^{co}	2	-2	0	0

Coirrepresntations of group generators

	$\left[4^{+}_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[4^{+}_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$-i\sigma_y\mathcal{K}$

No 130.1.310 : ${}^{1}4^{1}2^{1}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_5 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	1	1	−1	−1
χ_4	1	−1	1	−1	1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	1	−1
Γ_4	−1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 130.2.311 : ${}^{1}4^{1}2^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_5 : $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 130.3.312 : ${}^{1}4^{1}2^{1}2^{\infty_z}1 \to {}^{1}4^{1}2^{1}2$

No 130.4.313 : ${}^{1}4^{1}2^{1}2^{\infty}{}_{z}{}^{m_x}1 \rightarrow {}^{1}4^{1}2^{1}2^{m_x}1$

No 131.1.314 : ${}^{1}4^{2}2^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	1	1	−1	−1
χ_4	1	−1	1	−1	1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	1	−1
Γ_4	−1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 131.2.315 : ${}^{1}4^{2_z}2^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$
$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 131.3.316 : ${}^{1}4^{2_z}2^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 131.4.317 : ${}^{1}4^{2_z}2^{2_z}2^{\infty_x}1 \rightarrow {}^{1}4^{2_z}2^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_5 : $\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	2	2	0	0	0	−2	−2	−2
χ_2	2	−2	2	0	0	0	−2	2	−2
χ_3	2	0	−2	0	i	$-i$	−2	0	2
χ_4	2	0	−2	0	$-i$	i	−2	0	2

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 132.1.318 : ${}^{1}4^{m}2^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 132.2.319 : ${}^{1}4^{m_z}2^{m_z}2^{\infty_x}1 \rightarrow {}^{1}4^{m}2^{m}2$

No 133.1.320 : ${}^{1}4^{\bar{1}}2^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 133.2.321 : ${}^{1}4^{\bar{1}}2^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{1}4^{\bar{1}}2^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z \parallel 4^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z T \parallel 2_{100}\right]$ $\left[\Sigma_z T \parallel 2_{110}\right]$ $\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \parallel 2_{010}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E} \parallel 4^{-}_{001}\right]$ $\left[\bar{E}T \parallel 2_{100}\right]$ $\left[\bar{E}T \parallel 2_{110}\right]$ $\left[\bar{E}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_{11} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_4 :	$\left[E \parallel 4^{-}_{001}\right]$	C_{12} :	$\left[\bar{E} \parallel 4^{-}_{001}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 4^{+}_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_8 :	$\left[\Sigma_z \parallel 4^{-}_{001}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0
χ_3^{co}	2	i	−2	$-i$	0	0	0	0	−2	$-i$	2	i	0	0	0	0
χ_4^{co}	2	$-i$	−2	i	0	0	0	0	−2	i	2	$-i$	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 133.3.322 : ${}^{1}4^{\bar{1}}2^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{1}4^{\bar{1}}2^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{100}T \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_5 : $\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	4	0	−4	0	0	0	−4	0	4

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 134.1.323 : ${}^{2}4^{1}2^{2}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	1	1	−1	−1
χ_4	1	−1	1	−1	1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	1	−1
Γ_4	−1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 134.2.324 : ${}^{2_z}4^1 2^{2_z} 2^{m_z} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[E \parallel 2_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right] \left[2_{001}T \parallel 1\right]$
$\left[T \parallel 4^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[T \parallel 4^-_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 134.3.325 : ${}^{2_z}4^1 2^{2_z} 2^{m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[E \parallel 2_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel 4^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{010}T \parallel 4^-_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 134.4.326 : ${}^{2_z}4^1 2^{2_z} 2^{\infty_x} 1 \rightarrow {}^{2_z}4^1 2^{2_z} 2^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[E \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_5 : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_6 : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	0	0	−2	−2	−2
χ_2	2	0	2	−2	0	0	−2	−2	2
χ_3	2	0	−2	0	2	−2	−2	2	0
χ_4	2	0	−2	0	−2	2	−2	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^+_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$-i\sigma_y \mathcal{K}$	σ_z	$\sigma_z \mathcal{K}$
Γ_4	$-i\sigma_y \mathcal{K}$	$-\sigma_z$	$\sigma_z \mathcal{K}$

No 135.1.327 : ${}^{m}4^{1}2^{m}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 135.2.328 : ${}^{m_z}4^1 2^{m_z} 2^{\infty_x} 1 \rightarrow {}^{m}4^1 2^{m} 2$

No 136.1.329 : ${}^{\bar{1}}4^{1}2^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 136.2.330 : ${}^{\bar{1}}4^{1}2^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{\bar{1}}4^{1}2^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z \parallel 2_{100}\right]$ $\left[\Sigma_z T \parallel 2_{110}\right]$ $\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z \parallel 2_{010}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^{-}_{001}\right]$ $\left[\bar{E} \parallel 2_{100}\right]$ $\left[\bar{E}T \parallel 2_{110}\right]$ $\left[\bar{E}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_3 :	$\left[E \parallel 2_{100}\right]$	C_{11} :	$\left[\bar{E} \parallel 2_{100}\right]$
C_4 :	$\left[E \parallel 2_{010}\right]$	C_{12} :	$\left[\bar{E} \parallel 2_{010}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 2_{100}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_8 :	$\left[\Sigma_z \parallel 2_{010}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	0	0	0	0	2	−2	−2	2	0	0	0	0	−2	2
χ_4^{co}	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0	2	−2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 136.3.331 : ${}^{\bar{1}}4^{1}2^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4^{1}2^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
C_5 : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_6 : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	2	−2	−2	2	0
χ_4^{co}	2	−2	0	0	−2	2	−2	2	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{100}\right]$	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_z	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 137.1.332 : ${}^{2_x}4^{2_y}2^{2_z}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100} \parallel 4^{-}_{001}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right]$ $\left[2_{100} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	1	1	−1	−1
χ_4	1	−1	1	−1	1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	1	−1
Γ_4	−1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 137.2.333 : ${}^{2_x}4^{2_y}2^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[T \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{010} \mid\mid 2_{100}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 137.3.334 : ${}^{2_x}4^{2_y}2^{2_z}2^{m_y}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 1\right]$
$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{010}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{100}T \mid\mid 2_{110}\right]$ $\left[2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{010} \mid\mid 2_{100}\right]$	$\left[2_{010}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 138.1.335 : ${}^{2_z}4^{m_x}2^{m_y}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100}T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 139.1.336 : ${}^{m_x}4^{2_z}2^{m_y}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right]$
C_4 : $\left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 140.1.337 : ${}^{2}4^{\bar{1}}2^{m}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 140.2.338 : ${}^{2_z}4^{\bar{1}}2^{m_z}2^{\infty_x}1 \rightarrow {}^{2_z}4^{\bar{1}}2^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 141.1.339 : ${}^{\bar{1}}4^{2}2^{m}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right]$
C_4 : $\left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 141.2.340 : ${}^{\bar{1}}4^{2_z}2^{m_z}2^{\infty_x}1 \rightarrow {}^{\bar{1}}4^{2_z}2^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 142.1.341 : ${}^{m}4^{\bar{1}}2^{2}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{110}\right]$
C_4 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	−2	0	0
χ_3^{co}	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel 2_{110}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$

No 142.2.342 : ${}^{m_z}4^{\bar{1}}2^{2_z}2^{\infty_x}1 \rightarrow {}^{m_z}4^{\bar{1}}2^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel 2_{110}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 143.1.343 : ${}^{4_z}4^{2_x}2^{2_{xy}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	1	1	−1	−1
χ_4	1	−1	1	−1	1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	1	−1
Γ_4	−1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 143.2.344 : ${}^{4_z}4^{2_x}2^{2_{xy}}2^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$
$\left[4^-_{001}T \mid\mid 4^+_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[4^+_{001}T \mid\mid 4^-_{001}\right]$ $\left[2_{010}T \mid\mid 2_{100}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{110}\right]$ $\left[2_{110}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 144.1.345 : ${}^{4_z}4^{m_x}2^{m_{xy}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid 4^+_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^-_{001} \mid\mid 4^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 145.1.346 : ${}^{\bar{4}_z}4^{2_x}2^{m_{xy}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001}T \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001}T \mid\mid 4^-_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{100}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[4^+_{001}T \mid\mid 4^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 146.1.347 : ${}^{1}4^{1}m^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 146.2.348 : ${}^{1}4^{1}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$
$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 146.3.349 : ${}^{1}4^{1}m^{1}m^{\infty_z}1 \to {}^{1}4^{1}m^{1}m$

No 146.4.350 : ${}^{1}4^{1}m^{1}m^{\infty_z m_x}1 \to {}^{1}4^{1}m^{1}m^{m_x}1$

No 147.1.351 : ${}^{1}4^{2}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 147.2.352 : ${}^{1}4^{2_z}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 147.3.353 : ${}^{1}4^{2_z}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{010}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 147.4.354 : ${}^{1}4^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}4^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, 4^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 4^{-}_{001}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, m_{010}\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, 4^{+}_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} \,||\, 4^{+}_{001}\right]$
$\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 4^{-}_{001}\right] \left[\bar{E} 2_{100} \,||\, m_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 4^{+}_{001}\right] \left[E \,||\, 4^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, m_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\bar{E} 2_{100} \,||\, m_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$
C_5 : $\left[\Sigma_z \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, 4^{+}_{001}\right] \left[\bar{E} \,||\, 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	2	2	0	0	0	−2	−2	−2
χ_2	2	−2	2	0	0	0	−2	2	−2
χ_3	2	0	−2	0	i	$-i$	−2	0	2
χ_4	2	0	−2	0	$-i$	i	−2	0	2

Irrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 148.1.355 : ${}^{1}4^{m}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 148.2.356 : ${}^{1}4^{m_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{1}4^{m}m^{m}m$

No 149.1.357 : ${}^{1}4^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 149.2.358 : ${}^{1}4^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}4^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_{11} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_4 :	$\left[E \parallel 4^{-}_{001}\right]$	C_{12} :	$\left[\bar{E} \parallel 4^{-}_{001}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 4^{+}_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_8 :	$\left[\Sigma_z \parallel 4^{-}_{001}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0
χ_3^{co}	2	i	−2	$-i$	0	0	0	0	−2	$-i$	2	i	0	0	0	0
χ_4^{co}	2	$-i$	−2	i	0	0	0	0	−2	i	2	$-i$	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 149.3.359 : ${}^{1}4^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}4^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, 4^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 4^{-}_{001}\right] \left[T \,||\, m_{100}\right] \left[T \,||\, m_{110}\right] \left[T \,||\, m_{1\bar{1}0}\right] \left[T \,||\, m_{010}\right] \left[2_{100}T \,||\, 1\right]$
$\left[2_{100}T \,||\, 4^{+}_{001}\right] \left[2_{100}T \,||\, 2_{001}\right] \left[2_{100}T \,||\, 4^{-}_{001}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, m_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 4^{+}_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 4^{-}_{001}\right] \left[\Sigma_z T \,||\, m_{100}\right] \left[\Sigma_z T \,||\, m_{110}\right] \left[\Sigma_z T \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z T \,||\, m_{010}\right] \left[\Sigma_z 2_{100}T \,||\, 1\right] \left[\Sigma_z 2_{100}T \,||\, 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{001}\right]$
$\left[\Sigma_z 2_{100}T \,||\, 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} \,||\, 4^{+}_{001}\right] \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 4^{-}_{001}\right]$
$\left[\bar{E}T \,||\, m_{100}\right] \left[\bar{E}T \,||\, m_{110}\right] \left[\bar{E}T \,||\, m_{1\bar{1}0}\right] \left[\bar{E}T \,||\, m_{010}\right] \left[\bar{E}2_{100}T \,||\, 1\right] \left[\bar{E}2_{100}T \,||\, 4^{+}_{001}\right] \left[\bar{E}2_{100}T \,||\, 2_{001}\right] \left[\bar{E}2_{100}T \,||\, 4^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{100}\right]$
$\left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{110}\right]$
$\left[\bar{\Sigma}_z T \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 4^{+}_{001}\right] \left[E \,||\, 4^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, m_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$
C_5 : $\left[\Sigma_z \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, 4^{+}_{001}\right] \left[\bar{E} \,||\, 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	4	0	−4	0	0	0	−4	0	4

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 150.1.360 : ${}^{2}4^{1}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right] \left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 150.2.361 : ${}^{2_z}4^1 m^{2_z} m^{m_z} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right] \left[2_{001}T \parallel 1\right]$
$\left[T \parallel 4^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[T \parallel 4^-_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 150.3.362 : ${}^{2_z}4^1 m^{2_z} m^{m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel 4^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{010}T \parallel 4^-_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 150.4.363 : ${}^{2_z}4^1 m^{2_z} m^{\infty_x} 1 \rightarrow {}^{2_z}4^1 m^{2_z} m^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\Sigma_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_6 : $\left[\Sigma_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	0	0	−2	−2	−2
χ_2	2	0	2	−2	0	0	−2	−2	2
χ_3	2	0	−2	0	2	−2	−2	2	0
χ_4	2	0	−2	0	−2	2	−2	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$-i\sigma_y \mathcal{K}$	σ_z	$\sigma_z \mathcal{K}$
Γ_4	$-i\sigma_y \mathcal{K}$	$-\sigma_z$	$\sigma_z \mathcal{K}$

No 151.1.364 : ${}^{m}4^{1}m^{m}m$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[2_{001}T \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[2_{001}T \mid\mid 4^{-}_{001}\right] \left[E \mid\mid m_{100}\right] \left[2_{001}T \mid\mid m_{110}\right] \left[2_{001}T \mid\mid m_{1\bar{1}0}\right] \left[E \mid\mid m_{010}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{100}\right]$
C_4 : $\left[E \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 151.2.365 : ${}^{m_z}4^1m^{m_z}m^{\infty_x}1 \to {}^{m}4^1m^{m}m$

No 152.1.366 : ${}^{\bar{1}}4^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right]$
C_4 : $\left[E \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 152.2.367 : ${}^{\bar{1}}4^{1}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}4^{1}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel 4^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$ $\left[\Sigma_z \parallel m_{010}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^{-}_{001}\right]$ $\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_3 :	$\left[E \parallel m_{100}\right]$	C_{11} :	$\left[\bar{E} \parallel m_{100}\right]$
C_4 :	$\left[E \parallel m_{010}\right]$	C_{12} :	$\left[\bar{E} \parallel m_{010}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel m_{100}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel m_{100}\right]$
C_8 :	$\left[\Sigma_z \parallel m_{010}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	0	0	0	0	2	−2	−2	2	0	0	0	0	−2	2
χ_4^{co}	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0	2	−2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 152.3.368 : ${}^{\bar{1}}4^{1}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[T \,||\, 4^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[T \,||\, 4^{-}_{001}\right] \left[E \,||\, m_{100}\right] \left[T \,||\, m_{110}\right] \left[T \,||\, m_{1\bar{1}0}\right] \left[E \,||\, m_{010}\right] \left[2_{100}T \,||\, 1\right]$
$\left[2_{100} \,||\, 4^{+}_{001}\right] \left[2_{100}T \,||\, 2_{001}\right] \left[2_{100} \,||\, 4^{-}_{001}\right] \left[2_{100}T \,||\, m_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100}T \,||\, m_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z T \,||\, 4^{+}_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z T \,||\, 4^{-}_{001}\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z T \,||\, m_{110}\right] \left[\Sigma_z T \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, m_{010}\right] \left[\Sigma_z 2_{100}T \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \,||\, m_{010}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}T \,||\, 4^{+}_{001}\right] \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E}T \,||\, 4^{-}_{001}\right]$
$\left[\bar{E} \,||\, m_{100}\right] \left[\bar{E}T \,||\, m_{110}\right] \left[\bar{E}T \,||\, m_{1\bar{1}0}\right] \left[\bar{E} \,||\, m_{010}\right] \left[\bar{E}2_{100}T \,||\, 1\right] \left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{E}2_{100}T \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{E}2_{100}T \,||\, m_{100}\right]$
$\left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z T \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z T \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{110}\right]$
$\left[\bar{\Sigma}_z T \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 2_{001}\right]$
C_3 : $\left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right]$
C_4 : $\left[2_{100} \,||\, 4^{+}_{001}\right] \left[2_{100} \,||\, 4^{-}_{001}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
C_5 : $\left[\Sigma_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right]$
C_6 : $\left[\Sigma_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_9 : $\left[\bar{E} \,||\, m_{100}\right] \left[\bar{E} \,||\, m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	2	−2	−2	2	0
χ_4^{co}	2	−2	0	0	−2	2	−2	2	0

Coirrepresntations of group generators

	$\left[E \,||\, m_{100}\right]$	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_z	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 153.1.369 : ${}^{2_x}4^{2_y}m^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{010} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{010} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{010} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 153.2.370 : ${}^{2_x}4^{2_y}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$ $\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 1\right]$
$\left[T \,||\, 4^{+}_{001}\right]$ $\left[2_{100}T \,||\, 2_{001}\right]$ $\left[T \,||\, 4^{-}_{001}\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[2_{010}T \,||\, m_{110}\right]$ $\left[2_{010}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{001}T \,||\, m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_5 : $\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[2_{010} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 153.3.371 : ${}^{2_x}4^{2_y}m^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$ $\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 1\right]$
$\left[2_{001}T \,||\, 4^{+}_{001}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{110}\right]$ $\left[2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[T \,||\, m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_5 : $\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[2_{010} \,||\, m_{100}\right]$	$\left[2_{010}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 154.1.372 : ${}^{2_z}4^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 155.1.373 : ${}^{m_x}4^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right]$
C_4 : $\left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 156.1.374 : ${}^{2}4^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 156.2.375 : ${}^{2_z}4^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}4^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100} \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100} \mid\mid 4^{-}_{001}\right] \left[T \mid\mid m_{100}\right] \left[2_{100}T \mid\mid m_{110}\right] \left[2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[T \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{110}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{E}T \mid\mid m_{100}\right] \left[\bar{E}2_{100}T \mid\mid m_{110}\right] \left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 4^{+}_{001}\right] \left[2_{100} \mid\mid 4^{-}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 157.1.376 : ${}^{\bar{1}}4^{2}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right]$
C_4 : $\left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 157.2.377 : ${}^{\bar{1}}4^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[T \mid\mid 4^{-}_{001}\right] \left[2_{100} \mid\mid m_{100}\right] \left[2_{100}T \mid\mid m_{110}\right] \left[2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z T \mid\mid 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{110}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid 4^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}T \mid\mid 4^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{100}\right] \left[\bar{E}2_{100}T \mid\mid m_{110}\right] \left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 158.1.378 : ${}^{m}4^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right]$
C_4 : $\left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 158.2.379 : ${}^{m_z}4^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}4^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 159.1.380 : ${}^{4_z}4^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$
C_5 : $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 159.2.381 : ${}^{4_z}4^{2_x}m^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[4^-_{001}T \parallel 4^+_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[4^+_{001}T \parallel 4^-_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{110}\right]$ $\left[2_{110}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_5 : $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 160.1.382 : ${}^{4_z}4^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid 4^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid 4^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid 4^+_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^-_{001} \mid\mid 4^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid 4^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 161.1.383 : ${}^{\bar{4}_z}4^{m_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001}T \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001}T \parallel 4^-_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right]$
C_3 : $\left[2_{110} \parallel m_{110}\right]$
C_4 : $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[2_{110} \parallel m_{110}\right]$	$\left[4^+_{001}T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 162.1.384 : ${}^{1}\bar{4}^{1}2^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_5 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 162.2.385 : ${}^{1}\bar{4}^{1}2^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_5 : $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 162.3.386 : ${}^{1}\bar{4}^{1}2^{1}m^{\infty_z}1 \to {}^{1}\bar{4}^{1}2^{1}m$

No 162.4.387 : ${}^{1}\bar{4}^{1}2^{1}m^{\infty_z m_x}1 \to {}^{1}\bar{4}^{1}2^{1}m^{m_x}1$

No 163.1.388 : ${}^{1}\bar{4}^{2}2^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 163.2.389 : ${}^{1}\bar{4}^{2_z}2^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 163.3.390 : ${}^{1}\bar{4}^{2_z}2^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 163.4.391 : ${}^{1}\bar{4}^{2_z}2^{2_z}m^{\infty_x}1 \rightarrow {}^{1}\bar{4}^{2_z}2^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, \bar{4}^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, \bar{4}^{-}_{001}\right] \left[2_{100} \,||\, 2_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, \bar{4}^{+}_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, 2_{010}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} \,||\, \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \,||\, 2_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, \bar{4}^{+}_{001}\right] \left[E \,||\, \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{100} \,||\, 2_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, 2_{010}\right] \left[\bar{E} 2_{100} \,||\, 2_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right]$
C_5 : $\left[\Sigma_z \,||\, \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \,||\, \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^{+}_{001}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, \bar{4}^{+}_{001}\right] \left[\bar{E} \,||\, \bar{4}^{-}_{001}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	2	2	0	0	0	−2	−2	−2
χ_2	2	−2	2	0	0	0	−2	2	−2
χ_3	2	0	−2	0	i	$-i$	−2	0	2
χ_4	2	0	−2	0	$-i$	i	−2	0	2

Irrepresntations of group generators

	$\left[E \,\|\, \bar{4}^{+}_{001}\right]$	$\left[2_{100} \,\|\, 2_{100}\right]$	$\left[\Sigma_z \,\|\, 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 164.1.392 : ${}^{1}\bar{4}^{m}2^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 164.2.393 : ${}^{1}\bar{4}^{m_z}2^{m_z}m^{\infty_x}1 \to {}^{1}\bar{4}^{m}2^{m}m$

No 165.1.394 : ${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 165.2.395 : ${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, 2_{100}\right]$ $\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z \,||\, \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z \,||\, \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \,||\, 2_{100}\right]$ $\left[\Sigma_z T \,||\, m_{110}\right]$ $\left[\Sigma_z T \,||\, m_{1\bar{1}0}\right]$ $\left[\Sigma_z T \,||\, 2_{010}\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E} \,||\, \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \,||\, 2_{100}\right]$ $\left[\bar{E}T \,||\, m_{110}\right]$ $\left[\bar{E}T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{E}T \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{110}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_9 :	$\left[\bar{E} \,||\, 1\right]$
C_2 :	$\left[E \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[\bar{E} \,||\, \bar{4}^{+}_{001}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_{11} :	$\left[\bar{E} \,||\, 2_{001}\right]$
C_4 :	$\left[E \,||\, \bar{4}^{-}_{001}\right]$	C_{12} :	$\left[\bar{E} \,||\, \bar{4}^{-}_{001}\right]$
C_5 :	$\left[\Sigma_z \,||\, 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \,||\, 1\right]$
C_6 :	$\left[\Sigma_z \,||\, \bar{4}^{+}_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \,||\, \bar{4}^{+}_{001}\right]$
C_7 :	$\left[\Sigma_z \,||\, 2_{001}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
C_8 :	$\left[\Sigma_z \,||\, \bar{4}^{-}_{001}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \,||\, \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	−2	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0
χ_3^{co}	2	i	−2	$-i$	0	0	0	0	−2	$-i$	2	i	0	0	0	0
χ_4^{co}	2	$-i$	−2	i	0	0	0	0	−2	i	2	$-i$	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, \bar{4}^{+}_{001}\right]$	$\left[T \,||\, 2_{100}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 165.3.396 : ${}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}\bar{4}^{\bar{1}}2^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid \bar{4}^{-}_{001}\right] \left[T \mid\mid 2_{100}\right] \left[T \mid\mid m_{110}\right] \left[T \mid\mid m_{1\bar{1}0}\right] \left[T \mid\mid 2_{010}\right] \left[2_{100}T \mid\mid 1\right]$
$\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[2_{100} \mid\mid m_{110}\right] \left[2_{100} \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z T \mid\mid 2_{100}\right] \left[\Sigma_z T \mid\mid m_{110}\right] \left[\Sigma_z T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z T \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{110}\right] \left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid 2_{100}\right] \left[\bar{E}T \mid\mid m_{110}\right] \left[\bar{E}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}T \mid\mid 2_{010}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{110}\right] \left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{110}\right]$
$\left[\bar{\Sigma}_z T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid 2_{100}\right] \left[2_{100} \mid\mid m_{110}\right] \left[2_{100} \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{110}\right] \left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$
C_5 : $\left[\Sigma_z \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid \bar{4}^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid \bar{4}^{+}_{001}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E} \mid\mid \bar{4}^{-}_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	4	0	−4	0	0	0	−4	0	4

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 166.1.397 : ${}^{2}\bar{4}^{2}2^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{001} \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[2_{001} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{001} \mid\mid 2_{100}\right] \left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{1\bar{1}0}\right] \left[2_{001} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{100}\right] \left[2_{001} \mid\mid 2_{010}\right]$
C_5 : $\left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 166.2.398 : ${}^{2_z}\bar{4}^{2_z}2^1m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_5 : $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 166.3.399 : ${}^{2_z}\bar{4}^{2_z}2^1m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_5 : $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 166.4.400 : ${}^{2_z}\bar{4}^{2_z}2^1m^{\infty_x}1 \rightarrow {}^{2_z}\bar{4}^{2_z}2^1m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{4}^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel \bar{4}^-_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{4}^+_{001}\right] \left[2_{100} \parallel \bar{4}^-_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[\Sigma_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$
C_6 : $\left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	0	0	−2	−2	−2
χ_2	2	0	2	−2	0	0	−2	−2	2
χ_3	2	0	−2	0	2	−2	−2	2	0
χ_4	2	0	−2	0	−2	2	−2	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^+_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$-i\sigma_y\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-i\sigma_y\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 167.1.401 : ${}^{m}\bar{4}^{m}2^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel m_{110}\right]$
C_4 : $\left[E \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{110}\right]$	$\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 167.2.402 : ${}^{m_z}\bar{4}^{m_z}2^1m^{\infty_x}1 \to {}^{m}\bar{4}^{m}2^1m$

No 168.1.403 : ${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{110}\right]$
C_4 : $\left[E \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{110}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 168.2.404 : ${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_9 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 2_{001}\right]$ $C_{10} : \left[\bar{E} \parallel 2_{001}\right]$
$C_3 : \left[E \parallel m_{110}\right]$ $C_{11} : \left[\bar{E} \parallel m_{110}\right]$
$C_4 : \left[E \parallel m_{1\bar{1}0}\right]$ $C_{12} : \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
$C_5 : \left[\Sigma_z \parallel 1\right]$ $C_{13} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_6 : \left[\Sigma_z \parallel 2_{001}\right]$ $C_{14} : \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$C_7 : \left[\Sigma_z \parallel m_{110}\right]$ $C_{15} : \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$C_8 : \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$ $C_{16} : \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	0	0	0	0	2	−2	−2	2	0	0	0	0	−2	2
χ_4^{co}	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0	2	−2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{110}\right]$	$\left[T \parallel \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 168.3.405 : ${}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{4}^{\bar{1}}2^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_4 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_5 : $\left[\Sigma_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$
C_6 : $\left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	2	−2	−2	2	0
χ_4^{co}	2	−2	0	0	−2	2	−2	2	0

Coirrepresntations of group generators

	$\left[E \parallel m_{110}\right]$	$\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_z	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 169.1.406 : ${}^{2}\bar{4}^{1}2^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[E \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 169.2.407 : ${}^{2_z}\bar{4}^{1}2^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$
$\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 169.3.408 : ${}^{2_z}\bar{4}^{1}2^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 1\right]$
$\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 169.4.409 : ${}^{2_z}\bar{4}^1 2^{2_z} m^{\infty_x} 1 \rightarrow {}^{2_z}\bar{4}^1 2^{2_z} m^{\Sigma_x} 1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100} \,||\, \bar{4}^+_{001}\right] \left[E \,||\, 2_{001}\right] \left[2_{100} \,||\, \bar{4}^-_{001}\right] \left[E \,||\, 2_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[E \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{001}\right]$
$\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, \bar{4}^+_{001}\right] \left[2_{100} \,||\, \bar{4}^-_{001}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_5 : $\left[\Sigma_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right]$
C_6 : $\left[\Sigma_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1	2	0	2	2	0	0	−2	−2	−2
χ_2	2	0	2	−2	0	0	−2	−2	2
χ_3	2	0	−2	0	2	−2	−2	2	0
χ_4	2	0	−2	0	−2	2	−2	2	0

Irrepresntations of group generators

	$\left[2_{100} \,||\, \bar{4}^+_{001}\right]$	$\left[E \,||\, 2_{100}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$-i\sigma_y \mathcal{K}$	σ_z	$\sigma_z \mathcal{K}$
Γ_4	$-i\sigma_y \mathcal{K}$	$-\sigma_z$	$\sigma_z \mathcal{K}$

No 170.1.410 : ${}^{m}\bar{4}^{1}2^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 170.2.411 : ${}^{m_z}\bar{4}^{1}2^{m_z}m^{\infty_x}1 \to {}^{m}\bar{4}^{1}2^{m}m$

No 171.1.412 : ${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 171.2.413 : ${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \parallel 2_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$ $\left[\Sigma_z \parallel 2_{010}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \parallel 2_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_3 :	$\left[E \parallel 2_{100}\right]$	C_{11} :	$\left[\bar{E} \parallel 2_{100}\right]$
C_4 :	$\left[E \parallel 2_{010}\right]$	C_{12} :	$\left[\bar{E} \parallel 2_{010}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 2_{100}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_8 :	$\left[\Sigma_z \parallel 2_{010}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	−2	−2	0	0	0	0	−2	−2	2	2	0	0	0	0
χ_3^{co}	2	−2	0	0	0	0	2	−2	−2	2	0	0	0	0	−2	2
χ_4^{co}	2	−2	0	0	0	0	−2	2	−2	2	0	0	0	0	2	−2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[T \parallel \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 171.3.414 : ${}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{4}^{1}2^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right]$
$\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_6 : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	2	−2	−2	2	0
χ_4^{co}	2	−2	0	0	−2	2	−2	2	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{100}\right]$	$\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_z	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 172.1.415 : ${}^{2_x}\bar{4}^{2_y}2^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{010} \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 172.2.416 : ${}^{2_x}\bar{4}^{2_y}2^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 1\right]$
$\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{010} \mid\mid 2_{100}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 172.3.417 : ${}^{2_x}\bar{4}^{2_y}2^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 1\right]$
$\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 172.4.418 : ${}^{2_x}\bar{4}^{2_y}2^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 1\right]$
$\left[2_{010}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$ $\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$
C_5 : $\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$	$\left[2_{010} \,||\, 2_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 173.1.419 : ${}^{2_z}\bar{4}^{m_x}2^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 174.1.420 : ${}^{m_x}\bar{4}^{2_z}2^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right]$
C_4 : $\left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 175.1.421 : ${}^{m_x}\bar{4}^{m_y}2^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{110}\right]$
C_4 : $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{110}\right]$	$\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 176.1.422 : ${}^{2}\bar{4}^{\bar{1}}2^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 176.2.423 : ${}^{2_z}\bar{4}^{\bar{1}}2^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}\bar{4}^{\bar{1}}2^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[T \mid\mid 2_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 177.1.424 : ${}^{2}\bar{4}^{m}2^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 177.2.425 : ${}^{2_z}\bar{4}^{m_z}2^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}\bar{4}^{m_z}2^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[\Sigma_z \parallel 1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel 2_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100}T \parallel 2_{010}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \parallel 2_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[\bar{E} \parallel 1\right]$
C_5 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	0	2	−2	−2
χ_2^{co}	2	0	−2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$	$\left[2_{100}T \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-i\sigma_y\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 178.1.426 : ${}^{\bar{1}}\bar{4}^{2}2^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{100}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 178.2.427 : ${}^{\bar{1}}\bar{4}^{2_z}2^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}\bar{4}^{2_z}2^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[T \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[2_{100}T \mid\mid m_{110}\right] \left[2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{110}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100}T \mid\mid m_{110}\right] \left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{100}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 179.1.428 : ${}^{\bar{1}}\bar{4}^{m}2^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{110}\right]$
C_4 : $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{110}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 179.2.429 : ${}^{\bar{1}}\bar{4}^{m_z}2^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}\bar{4}^{m_z}2^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{100} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	2	−2	0	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{110}\right]$	$\left[T \mid\mid \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 180.1.430 : ${}^{m}\bar{4}^{2}2^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{100}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 180.2.431 : ${}^{m_z}\bar{4}^{2_z}2^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}\bar{4}^{2_z}2^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[T \mid\mid m_{110}\right] \left[T \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right]$
$\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z T \mid\mid m_{110}\right] \left[\Sigma_z T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}T \mid\mid m_{110}\right] \left[\bar{E}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{100}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$
C_4 : $\left[\bar{E} \mid\mid 1\right]$
C_5 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 181.1.432 : ${}^{m}\bar{4}^{\bar{1}}2^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{110}\right]$
C_4 : $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{110}\right]$	$\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 181.2.433 : ${}^{m_z}\bar{4}^{\bar{1}}2^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}\bar{4}^{\bar{1}}2^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100}T \,||\, \bar{4}^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[T \,||\, 2_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[T \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 1\right]$
$\left[\Sigma_z 2_{100}T \,||\, \bar{4}^{+}_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[\Sigma_z T \,||\, 2_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z T \,||\, 2_{010}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}2_{100}T \,||\, \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E}2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[\bar{E}T \,||\, 2_{100}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 2_{001}\right]$
C_3 : $\left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
C_4 : $\left[\bar{E} \,||\, 1\right]$
C_5 : $\left[\bar{E} \,||\, 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	2	2	0	−2	−2
χ_2^{co}	4	−4	0	−4	4

Coirrepresntations of group generators

	$\left[E \,||\, 2_{001}\right]$	$\left[2_{100} \,||\, m_{110}\right]$	$\left[2_{100}T \,||\, \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 182.1.434 : ${}^{4_z}\bar{4}^{2_x}2^{2_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[4^{-}_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	1	−1	1	−1	1
χ_4	1	1	1	−1	−1
χ_5	2	0	−2	0	0

Irrepresntations of group generators

	$\left[4^{+}_{001} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	$i\sigma_y$	$-\sigma_x$

No 182.2.435 : ${}^{4_z}\bar{4}^{2_x}2^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$
$\left[4^-_{001}T \mid\mid \bar{4}^+_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[4^+_{001}T \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{010}T \mid\mid 2_{100}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{110}\right]$ $\left[2_{110}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$ $\left[4^-_{001} \mid\mid \bar{4}^-_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
C_5 : $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 183.1.436 : ${}^{4_z}\bar{4}^{m_x}2^{m_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001} \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_4 : $\left[4^-_{001} \mid\mid \bar{4}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	-1	1	-1
χ_3^{co}	1	i	-1	$-i$
χ_4^{co}	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[4^+_{001} \mid\mid \bar{4}^+_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	i	$\mathcal{K}$
Γ_4^{co}	$-i$	$\mathcal{K}$

No 184.1.437 : ${}^{\bar{4}_z}\bar{4}^{2_x}2^{m_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^+_{001}T \mid\mid \bar{4}^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^-_{001}T \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{100}\right]$
C_4 : $\left[2_{010} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[4^+_{001}T \mid\mid \bar{4}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 185.1.438 : ${}^{\bar{4}_z}\bar{4}^{m_x}2^{2_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[4^{+}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[4^{-}_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_3 : $\left[2_{110} \mid\mid m_{110}\right]$
C_4 : $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	−1	−1
χ_3^{co}	2	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 2_{001}\right]$	$\left[2_{110} \mid\mid m_{110}\right]$	$\left[4^{+}_{001}T \mid\mid \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$

No 186.1.439 : ${}^{1}4/{}^{1}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_6 :	$\left[E \mid\mid -1\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$	C_7 :	$\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_8 :	$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$
C_4 :	$\left[E \mid\mid m_{001}\right]$	C_9 :	$\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 :	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[E \mid\mid m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 186.2.440 : ${}^{1}4/{}^{1}m{}^{1}m{}^{1}m{}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
$\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$
$\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_6 :	$\left[E \mid\mid -1\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$	C_7 :	$\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_8 :	$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$
C_4 :	$\left[E \mid\mid m_{001}\right]$	C_9 :	$\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 :	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 186.3.441 : ${}^{1}4/{}^{1}m{}^{1}m{}^{1}m{}^{\infty_z}1 \rightarrow {}^{1}4/{}^{1}m{}^{1}m{}^{1}m$

No 186.4.442 : ${}^{1}4/{}^{1}m{}^{1}m{}^{1}m{}^{\infty_z m_x}1 \rightarrow {}^{1}4/{}^{1}m{}^{1}m{}^{1}m{}^{m_x}1$

No 187.1.443 : ${}^{2}4/{}^{2}m{}^{2}m{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_6 :	$\left[2_{001} \mid\mid -1\right]$
C_2 :	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_8 :	$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \mid\mid m_{001}\right]$	C_9 :	$\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 :	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 187.2.444 : ${}^{2_z}4/{}^{2_z}m^{2_z}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[E \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[E \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001} \,||\, m_{100}\right]$
$\left[E \,||\, m_{110}\right]$ $\left[E \,||\, m_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, 4^{+}_{001}\right]$
$\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[2_{001}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{110}\right]$ $\left[2_{001}T \,||\, m_{1\bar{1}0}\right]$
$\left[T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{110}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{001} \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[E \,||\, m_{110}\right]$ $\left[E \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \,||\, m_{001}\right]$	C_9 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 :	$\left[E \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 187.3.445 : ${}^{2_z}4/{}^{2_z}m{}^{2_z}m{}^{1}m{}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^{-}_{001}\right]$ $\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \parallel m_{001}\right]$	C_9 :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$
C_5 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 187.4.446 : ${}^{2_z}4/{}^{2_z}m^{2_z}m^1m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{2_z}m^{2_z}m^1m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel \bar{4}^-_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel \bar{4}^+_{001}\right] \left[2_{100} \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^+_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel \bar{4}^+_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^+_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\bar{E} 2_{100} \parallel 4^+_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^+_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel \bar{4}^-_{001}\right] \left[E \parallel \bar{4}^+_{001}\right]$
C_5 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel \bar{4}^-_{001}\right] \left[\bar{E} \parallel \bar{4}^+_{001}\right]$
C_{10} : $\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
C_{11} : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1	2	0	2	2	2	2	−2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	2	−2	−2	2	2	−2
χ_3	2	0	2	−2	2	−2	−2	−2	2	−2	2

χ_4	2	0	2	2	−2	−2	−2	−2	−2	2	2
χ_5	4	0	−4	0	0	0	−4	4	0	0	0

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -i\sigma_y \\ i\sigma_y & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -\sigma_z \\ -\sigma_z & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 188.1.447 : ${}^{m}4/{}^{m}m^{m}m^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[E \parallel m_{110}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 188.2.448 : ${}^{m_z}4/{}^{m_z}m^{m_z}m^1m^{\infty_x}1 \to {}^{m}4/{}^{m}m^{m}m^1m$

No 189.1.449 : ${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$
C_5 : $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$	$\left[E \mid\mid m_{110}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 189.2.450 : ${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[T \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{11} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_{12} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_3 :	$\left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right]$	C_{13} :	$\left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
C_4 :	$\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$	C_{14} :	$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
C_5 :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$	C_{15} :	$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$
C_6 :	$\left[\Sigma_z \parallel 1\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_7 :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{17} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_8 :	$\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$	C_{18} :	$\left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$
C_{10} :	$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{100}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}
χ_1^{co}	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	0	0	0	0	0
χ_2^{co}	2	2	−2	−2	2	0	0	0	0	0	−2	−2	2	2	−2	0	0	0	0	0
χ_3^{co}	2	2	−2	2	−2	0	0	0	0	0	−2	−2	2	−2	2	0	0	0	0	0
χ_4^{co}	2	2	2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	0	0	0	0	0
χ_5^{co}	4	−4	0	0	0	0	0	0	0	0	−4	4	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[E \parallel m_{110}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 189.3.451 : ${}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4/{}^{\bar{1}}m{}^{\bar{1}}m^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[T \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right]$
$\left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right]$
$\left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_6 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
C_{11} : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1^{co}	2	2	2	2	2	0	−2	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	2	0	−2	−2	2	2	−2
χ_3^{co}	2	2	−2	2	−2	0	−2	−2	2	−2	2
χ_4^{co}	2	2	2	−2	−2	0	−2	−2	−2	2	2
χ_5^{co}	4	−4	0	0	0	0	−4	4	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$	$\left[E \mid\mid m_{110}\right]$	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ -\sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 190.1.452 : ${}^{1}4/{}^{2}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$	C_7 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \parallel m_{001}\right]$	C_9 :	$\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$
C_5 :	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 190.2.453 : ${}^{1}4/{}^{2_z}m^{1}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, m_{100}\right]$
$\left[E \,||\, m_{110}\right]$ $\left[E \,||\, m_{1\bar{1}0}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[2_{001}T \,||\, 4^{+}_{001}\right]$
$\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{110}\right]$ $\left[2_{001}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{001}T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{110}\right]$ $\left[T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{001} \,||\, -1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[E \,||\, m_{110}\right]$ $\left[E \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \,||\, m_{001}\right]$	C_9 :	$\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 190.3.454 : ${}^{1}4/{}^{2_z}m^{1}m^{1}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, m_{100}\right]$
$\left[E \,||\, m_{110}\right]$ $\left[E \,||\, m_{1\bar{1}0}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{100}T \,||\, 4^{+}_{001}\right]$
$\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 4^{-}_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{010}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \,||\, -1\right]$ $\left[2_{010}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{110}\right]$ $\left[2_{100}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{001} \,||\, -1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[E \,||\, m_{110}\right]$ $\left[E \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \,||\, m_{001}\right]$	C_9 :	$\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 190.4.455 : ${}^{1}4/{}^{2}{}_{z}m^{1}m^{1}m^{\infty}{}_{x}1 \rightarrow {}^{1}4/{}^{2}{}_{z}m^{1}m^{1}m^{\Sigma}{}_{x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 4^{-}_{001}\right] \left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[E \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{1\bar{1}0}\right] \left[E \mid\mid m_{010}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_{z} \mid\mid 1\right] \left[\Sigma_{z} \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_{z} \mid\mid 2_{001}\right] \left[\Sigma_{z} \mid\mid 4^{-}_{001}\right] \left[\Sigma_{z}2_{100} \mid\mid m_{001}\right] \left[\Sigma_{z}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_{z}2_{100} \mid\mid -1\right] \left[\Sigma_{z}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_{z} \mid\mid m_{100}\right] \left[\Sigma_{z} \mid\mid m_{110}\right] \left[\Sigma_{z} \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_{z} \mid\mid m_{010}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{010}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{110}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{100}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 4^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E} \mid\mid m_{100}\right] \left[\bar{E} \mid\mid m_{110}\right] \left[\bar{E} \mid\mid m_{1\bar{1}0}\right] \left[\bar{E} \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_{z} \mid\mid 1\right] \left[\bar{\Sigma}_{z} \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_{z} \mid\mid 2_{001}\right] \left[\bar{\Sigma}_{z} \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_{z}2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{100}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{110}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{010}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_{1} : $\left[E \mid\mid 1\right]$
C_{2} : $\left[E \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 4^{-}_{001}\right]$
C_{3} : $\left[E \mid\mid 2_{001}\right]$
C_{4} : $\left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{100}\right]$
$\left[\Sigma_{z} \mid\mid 1\right] \left[\Sigma_{z} \mid\mid 4^{+}_{001}\right] \left[\Sigma_{z} \mid\mid 2_{001}\right] \left[\Sigma_{z} \mid\mid 4^{-}_{001}\right] \left[\Sigma_{z}2_{100} \mid\mid m_{001}\right] \left[\Sigma_{z}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_{z}2_{100} \mid\mid -1\right] \left[\Sigma_{z}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\Sigma_{z} \mid\mid m_{100}\right] \left[\Sigma_{z} \mid\mid m_{110}\right] \left[\Sigma_{z} \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_{z} \mid\mid m_{010}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{010}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{110}\right] \left[\Sigma_{z}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_{z} \mid\mid 1\right] \left[\bar{\Sigma}_{z} \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_{z} \mid\mid 2_{001}\right] \left[\bar{\Sigma}_{z} \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_{z} \mid\mid m_{100}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{110}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_{z} \mid\mid m_{010}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_{z}2_{100} \mid\mid 2_{100}\right]$
C_{5} : $\left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{010}\right]$
C_{6} : $\left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{1\bar{1}0}\right]$
C_{7} : $\left[\bar{E} \mid\mid 1\right]$
C_{8} : $\left[\bar{E} \mid\mid 4^{+}_{001}\right] \left[\bar{E} \mid\mid 4^{-}_{001}\right]$
C_{9} : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_{10} : $\left[\bar{E} \mid\mid m_{100}\right] \left[\bar{E} \mid\mid m_{010}\right]$
C_{11} : $\left[\bar{E} \mid\mid m_{110}\right] \left[\bar{E} \mid\mid m_{1\bar{1}0}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1	2	2	2	0	2	2	−2	−2	−2	−2	−2
χ_2	2	−2	2	0	2	−2	−2	2	−2	−2	2
χ_3	2	−2	2	0	−2	2	−2	2	−2	2	−2

χ_4	2	2	2	0	−2	−2	−2	−2	−2	2	2
χ_5	4	0	−4	0	0	0	−4	0	4	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3	$-\sigma_0$	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_4	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} i\sigma_y & 0 \\ 0 & i\sigma_y \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} -\sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 191.1.456 : ${}^{1}4/{}^{m}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 191.2.457 : ${}^{1}4/{}^{m_z}m^{1}m^{1}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{m}m^{1}m^{1}m$

No 192.1.458 : ${}^{1}4/{}^{\bar{1}}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 192.2.459 : ${}^{1}4/{}^{\bar{1}}m^{1}m^{1}m^{\infty_z}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{1}m^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid m_{100}\right]$ $\left[\Sigma_z \mid\mid m_{110}\right]$ $\left[\Sigma_z \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \mid\mid 2_{110}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid m_{100}\right]$ $\left[\bar{E} \mid\mid m_{110}\right]$ $\left[\bar{E} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$
$\left[\bar{E}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \mid\mid 2_{110}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_{11} :	$\left[\bar{E} \mid\mid 1\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$	C_{12} :	$\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_{13} :	$\left[\bar{E} \mid\mid 2_{001}\right]$
C_4 :	$\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$	C_{14} :	$\left[\bar{E} \mid\mid m_{100}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$
C_5 :	$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$	C_{15} :	$\left[\bar{E} \mid\mid m_{110}\right]$ $\left[\bar{E} \mid\mid m_{1\bar{1}0}\right]$
C_6 :	$\left[\Sigma_z \mid\mid 1\right]$	C_{16} :	$\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_7 :	$\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$	C_{17} :	$\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$
C_8 :	$\left[\Sigma_z \mid\mid 2_{001}\right]$	C_{18} :	$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
C_9 :	$\left[\Sigma_z \mid\mid m_{100}\right]$ $\left[\Sigma_z \mid\mid m_{010}\right]$	C_{19} :	$\left[\bar{\Sigma}_z \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$
C_{10} :	$\left[\Sigma_z \mid\mid m_{110}\right]$ $\left[\Sigma_z \mid\mid m_{1\bar{1}0}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}
χ_1^{co}	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	0	0	0	0	0
χ_2^{co}	2	−2	2	2	−2	0	0	0	0	0	−2	2	−2	−2	2	0	0	0	0	0
χ_3^{co}	2	−2	2	−2	2	0	0	0	0	0	−2	2	−2	2	−2	0	0	0	0	0
χ_4^{co}	2	2	2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	0	0	0	0	0
χ_5^{co}	4	0	−4	0	0	0	0	0	0	0	−4	0	4	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} i\sigma_y & 0 \\ 0 & i\sigma_y \end{pmatrix}$	$\begin{pmatrix} -\sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 192.3.460 : ${}^{1}4/{}^{\bar{1}}m^{1}m^{1}m^{\infty_z m_x}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{1}m^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel -1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel m_{100}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{100}T \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{010}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right]$
$\left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right]$
$\left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_6 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{001}\right]$
C_{10} : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right]$
C_{11} : $\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1^{co}	2	2	2	2	2	0	−2	−2	−2	−2	−2
χ_2^{co}	2	−2	2	2	−2	0	−2	2	−2	−2	2
χ_3^{co}	2	−2	2	−2	2	0	−2	2	−2	2	−2
χ_4^{co}	2	2	2	−2	−2	0	−2	−2	−2	2	2
χ_5^{co}	4	0	−4	0	0	0	−4	0	4	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^+_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} i\sigma_y & 0 \\ 0 & i\sigma_y \end{pmatrix}$	$\begin{pmatrix} -\sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 193.1.461 : ${}^{2}4/{}^{1}m{}^{1}m{}^{2}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, m_{100}\right]$
$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[E \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[E \,||\, m_{001}\right]$	C_9 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 :	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[E \,||\, m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 193.2.462 : ${}^{2_z}4/{}^{1}m^{1}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, m_{100}\right]$
$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, 4^{+}_{001}\right]$
$\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, 4^{-}_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \,||\, -1\right]$ $\left[T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{110}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[E \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[E \,||\, m_{001}\right]$	C_9 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 :	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 193.3.463 : ${}^{2_z}4/{}^{1}m^{1}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[E \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_9 :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$
C_5 :	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 193.4.464 : ${}^{2_z}4/{}^{1}m^{1}m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{1}m^{1}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[E \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[E \parallel m_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_7 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_8 : $\left[\Sigma_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_9 : $\left[\Sigma_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right]$
C_{10} : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_{11} : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_{12} : $\left[\bar{E} \parallel 1\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$
C_{14} : $\left[\bar{E} \parallel m_{001}\right]$
C_{15} : $\left[\bar{E} \parallel -1\right]$
C_{16} : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right]$
C_{17} : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}
χ_1	2	0	2	2	2	2	2	0	0	0	0	−2	−2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2	0	0	0	0	−2	−2	2	2	2	−2
χ_3	2	0	2	2	2	−2	−2	0	0	0	0	−2	−2	−2	−2	2	2
χ_4	2	0	2	−2	−2	2	−2	0	0	0	0	−2	−2	2	2	−2	2
χ_5	2	0	−2	−2	2	0	0	2	−2	−2	2	−2	2	2	−2	0	0
χ_6	2	0	−2	2	−2	0	0	−2	2	−2	2	−2	2	−2	2	0	0
χ_7	2	0	−2	−2	2	0	0	−2	2	2	−2	−2	2	2	−2	0	0
χ_8	2	0	−2	2	−2	0	0	2	−2	2	−2	−2	2	−2	2	0	0

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_0$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	σ_0	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_4	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_5	$-i\sigma_y\mathcal{K}$	$-\sigma_0$	σ_z	$\sigma_z\mathcal{K}$
Γ_6	$-i\sigma_y\mathcal{K}$	σ_0	$-\sigma_z$	$\sigma_z\mathcal{K}$
Γ_7	$-i\sigma_y\mathcal{K}$	$-\sigma_0$	$-\sigma_z$	$\sigma_z\mathcal{K}$
Γ_8	$-i\sigma_y\mathcal{K}$	σ_0	σ_z	$\sigma_z\mathcal{K}$

No 194.1.465 : ${}^{m}4/{}^{1}m{}^{1}m{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[E \parallel m_{100}\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ C_6 : $\left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel m_{001}\right]$ C_7 : $\left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel -1\right]$ C_8 : $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 194.2.466 : ${}^{m_z}4/{}^{1}m^{1}m^{m_z}m^{\infty_x}1 \rightarrow {}^{m}4/{}^{1}m^{1}m^{m}m$

No 195.1.467 : ${}^{\bar{1}}4/{}^{1}m{}^{1}m{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[E \parallel m_{100}\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ C_6 : $\left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel m_{001}\right]$ C_7 : $\left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel -1\right]$ C_8 : $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 195.2.468 : ${}^{\bar{1}}4/{}^{1}m^{1}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}4/{}^{1}m^{1}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z \parallel m_{001}\right]$ $\left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \parallel -1\right]$ $\left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{010}\right]$ $\left[\Sigma_z \parallel 2_{010}\right]$ $\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \parallel 2_{110}\right]$ $\left[\Sigma_z \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^{+}_{001}\right]$ $\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right]$ $\left[\bar{E}T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \parallel -1\right]$ $\left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel m_{010}\right]$ $\left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \parallel 2_{110}\right]$ $\left[\bar{E} \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_{17} : $\left[\bar{E} \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ C_{18} : $\left[\bar{E} \parallel 2_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$ C_{19} : $\left[\bar{E} \parallel m_{001}\right]$
C_4 : $\left[E \parallel -1\right]$ C_{20} : $\left[\bar{E} \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right]$ C_{21} : $\left[\bar{E} \parallel m_{100}\right]$
C_6 : $\left[E \parallel m_{010}\right]$ C_{22} : $\left[\bar{E} \parallel m_{010}\right]$
C_7 : $\left[E \parallel 2_{010}\right]$ C_{23} : $\left[\bar{E} \parallel 2_{010}\right]$
C_8 : $\left[E \parallel 2_{100}\right]$ C_{24} : $\left[\bar{E} \parallel 2_{100}\right]$
C_9 : $\left[\Sigma_z \parallel 1\right]$ C_{25} : $\left[\bar{\Sigma}_z \parallel 1\right]$
C_{10} : $\left[\Sigma_z \parallel 2_{001}\right]$ C_{26} : $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_{11} : $\left[\Sigma_z \parallel m_{001}\right]$ C_{27} : $\left[\bar{\Sigma}_z \parallel m_{001}\right]$
C_{12} : $\left[\Sigma_z \parallel -1\right]$ C_{28} : $\left[\bar{\Sigma}_z \parallel -1\right]$
C_{13} : $\left[\Sigma_z \parallel m_{100}\right]$ C_{29} : $\left[\bar{\Sigma}_z \parallel m_{100}\right]$
C_{14} : $\left[\Sigma_z \parallel m_{010}\right]$ C_{30} : $\left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_{15} : $\left[\Sigma_z \parallel 2_{010}\right]$ C_{31} : $\left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_{16} : $\left[\Sigma_z \parallel 2_{100}\right]$ C_{32} : $\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	2	2	2	2	0	0	0	0	0	0	0	0
χ_2^{co}	2	2	−2	−2	−2	−2	2	2	0	0	0	0	0	0	0	0
χ_3^{co}	2	2	2	2	−2	−2	−2	−2	0	0	0	0	0	0	0	0

χ_4^{co}	2	2	−2	−2	2	2	−2	−2	0	0	0	0	0	0	0	0
χ_5^{co}	2	−2	−2	2	0	0	0	0	0	0	0	0	2	−2	−2	2
χ_6^{co}	2	−2	2	−2	0	0	0	0	0	0	0	0	−2	2	−2	2
χ_7^{co}	2	−2	−2	2	0	0	0	0	0	0	0	0	−2	2	2	−2
χ_8^{co}	2	−2	2	−2	0	0	0	0	0	0	0	0	2	−2	2	−2

	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}	C_{25}	C_{26}	C_{27}	C_{28}	C_{29}	C_{30}	C_{31}	C_{32}
χ_1^{co}	−2	−2	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0	0	0
χ_2^{co}	−2	−2	2	2	2	2	−2	−2	0	0	0	0	0	0	0	0
χ_3^{co}	−2	−2	−2	−2	2	2	2	2	0	0	0	0	0	0	0	0
χ_4^{co}	−2	−2	2	2	−2	−2	2	2	0	0	0	0	0	0	0	0
χ_5^{co}	−2	2	2	−2	0	0	0	0	0	0	0	0	−2	2	2	−2
χ_6^{co}	−2	2	−2	2	0	0	0	0	0	0	0	0	2	−2	2	−2
χ_7^{co}	−2	2	2	−2	0	0	0	0	0	0	0	0	2	−2	−2	2
χ_8^{co}	−2	2	−2	2	0	0	0	0	0	0	0	0	−2	2	−2	2

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_7^{co}	$-\sigma_0$	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_8^{co}	$-\sigma_0$	σ_0	σ_z	$-i\sigma_y\mathcal{K}$	$\sigma_z\mathcal{K}$

No 195.3.469 : ${}^{\bar{1}}4/{}^{1}m^{1}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4/{}^{1}m^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel -1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right]$
$\left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_7 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
C_8 : $\left[\Sigma_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_9 : $\left[\Sigma_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right]$
C_{10} : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_{11} : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_{12} : $\left[\bar{E} \parallel 1\right]$

$C_{13}: \left[\bar{E} \,||\, 2_{001}\right]$
$C_{14}: \left[\bar{E} \,||\, m_{001}\right]$
$C_{15}: \left[\bar{E} \,||\, -1\right]$
$C_{16}: \left[\bar{E} \,||\, m_{100}\right] \left[\bar{E} \,||\, m_{010}\right]$
$C_{17}: \left[\bar{E} \,||\, 2_{010}\right] \left[\bar{E} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	−2	2	0	0	0	0	0	−2	−2	2	2	2	−2
χ_3^{co}	2	2	2	2	−2	−2	0	0	0	0	0	−2	−2	−2	−2	2	2
χ_4^{co}	2	2	−2	−2	2	−2	0	0	0	0	0	−2	−2	2	2	−2	2
χ_5^{co}	2	−2	−2	2	0	0	0	2	−2	−2	2	−2	2	2	−2	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	−2	2	−2	2	−2	2	−2	2	0	0
χ_7^{co}	2	−2	−2	2	0	0	0	−2	2	2	−2	−2	2	2	−2	0	0
χ_8^{co}	2	−2	2	−2	0	0	0	2	−2	2	−2	−2	2	−2	2	0	0

Coirrepresntations of group generators

	$\left[E \,||\, m_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_6^{co}	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_7^{co}	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$
Γ_8^{co}	σ_0	σ_z	$-i\sigma_y\mathcal{K}$	$i\sigma_y$	$\sigma_z\mathcal{K}$

No 196.1.470 : ${}^{1}4/{}^{1}m^{2}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[E \parallel -1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_9 :	$\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$
C_5 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 196.2.471 : ${}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[E \parallel -1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_9 :	$\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$
C_5 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 196.3.472 : ${}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[E \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[E \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001} \,||\, m_{100}\right]$
$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{100}T \,||\, 4^{+}_{001}\right]$
$\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 4^{-}_{001}\right]$ $\left[2_{100}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \,||\, -1\right]$ $\left[2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{010}T \,||\, m_{110}\right]$ $\left[2_{010}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[E \,||\, -1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \,||\, m_{100}\right]$ $\left[2_{001} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[E \,||\, m_{001}\right]$	C_9 :	$\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$
C_5 :	$\left[E \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 196.4.473 : ${}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{1}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel -1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[E \parallel -1\right]$
C_7 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_8 : $\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_{10} : $\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$
C_{11} : $\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 1\right]$
C_{13} : $\left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
C_{14} : $\left[\bar{E} \parallel 2_{001}\right]$
C_{15} : $\left[\bar{E} \parallel m_{001}\right]$
C_{16} : $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
C_{17} : $\left[\bar{E} \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}
χ_1	2	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	−2
χ_2	2	2	2	−2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	2
χ_3	2	−2	2	2	−2	2	0	0	0	0	0	−2	2	−2	−2	2	−2
χ_4	2	−2	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2	−2	2
χ_5	2	0	−2	−2	0	2	0	i	$-i$	$-i$	i	−2	0	2	2	0	−2
χ_6	2	0	−2	2	0	−2	0	i	$-i$	i	$-i$	−2	0	2	−2	0	2
χ_7	2	0	−2	−2	0	2	0	$-i$	i	i	$-i$	−2	0	2	2	0	−2
χ_8	2	0	−2	2	0	−2	0	$-i$	i	$-i$	i	−2	0	2	−2	0	2

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	$i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	$i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_7	$-i\sigma_z$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_8	$-i\sigma_z$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 197.1.474 : ${}^{1}4/{}^{1}m^{m}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[E \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[E \mid\mid -1\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 197.2.475 : ${}^{1}4/{}^{1}m^{m_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{1}m^{m}m^{m}m$

No 198.1.476 : ${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[E \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[E \mid\mid -1\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 198.2.477 : ${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z \parallel m_{001}\right]$ $\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \parallel -1\right]$ $\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z T \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{010}\right]$ $\left[\Sigma_z T \parallel 2_{010}\right]$ $\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \parallel 2_{110}\right]$ $\left[\Sigma_z T \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E} \parallel 4^{+}_{001}\right]$ $\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right]$ $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \parallel -1\right]$ $\left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E}T \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E}T \parallel m_{010}\right]$ $\left[\bar{E}T \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \parallel 2_{110}\right]$ $\left[\bar{E}T \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_{17} : $\left[\bar{E} \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ C_{18} : $\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_{19} : $\left[\bar{E} \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 4^{-}_{001}\right]$ C_{20} : $\left[\bar{E} \parallel 4^{-}_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$ C_{21} : $\left[\bar{E} \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{4}^{-}_{001}\right]$ C_{22} : $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right]$
C_7 : $\left[E \parallel -1\right]$ C_{23} : $\left[\bar{E} \parallel -1\right]$
C_8 : $\left[E \parallel \bar{4}^{+}_{001}\right]$ C_{24} : $\left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 1\right]$ C_{25} : $\left[\bar{\Sigma}_z \parallel 1\right]$
C_{10} : $\left[\Sigma_z \parallel 4^{+}_{001}\right]$ C_{26} : $\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_{11} : $\left[\Sigma_z \parallel 2_{001}\right]$ C_{27} : $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_{12} : $\left[\Sigma_z \parallel 4^{-}_{001}\right]$ C_{28} : $\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_{13} : $\left[\Sigma_z \parallel m_{001}\right]$ C_{29} : $\left[\bar{\Sigma}_z \parallel m_{001}\right]$
C_{14} : $\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right]$ C_{30} : $\left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
C_{15} : $\left[\Sigma_z \parallel -1\right]$ C_{31} : $\left[\bar{\Sigma}_z \parallel -1\right]$
C_{16} : $\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$ C_{32} : $\left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	2	2	2	2	0	0	0	0	0	0	0	0
χ_2^{co}	2	2	2	2	−2	−2	−2	−2	0	0	0	0	0	0	0	0
χ_3^{co}	2	−2	2	−2	2	−2	2	−2	0	0	0	0	0	0	0	0

χ_4^{co}	2	−2	2	−2	−2	2	−2	2	0	0	0	0	0	0	0	0
χ_5^{co}	2	i	−2	$-i$	−2	$-i$	2	i	0	0	0	0	0	0	0	0
χ_6^{co}	2	i	−2	$-i$	2	i	−2	$-i$	0	0	0	0	0	0	0	0
χ_7^{co}	2	$-i$	−2	i	−2	i	2	$-i$	0	0	0	0	0	0	0	0
χ_8^{co}	2	$-i$	−2	i	2	$-i$	−2	i	0	0	0	0	0	0	0	0

	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}	C_{25}	C_{26}	C_{27}	C_{28}	C_{29}	C_{30}	C_{31}	C_{32}
χ_1^{co}	−2	−2	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0	0	0
χ_2^{co}	−2	−2	−2	−2	2	2	2	2	0	0	0	0	0	0	0	0
χ_3^{co}	−2	2	−2	2	−2	2	−2	2	0	0	0	0	0	0	0	0
χ_4^{co}	−2	2	−2	2	2	−2	2	−2	0	0	0	0	0	0	0	0
χ_5^{co}	−2	$-i$	2	i	2	i	−2	$-i$	0	0	0	0	0	0	0	0
χ_6^{co}	−2	$-i$	2	i	−2	$-i$	2	i	0	0	0	0	0	0	0	0
χ_7^{co}	−2	i	2	$-i$	2	$-i$	−2	i	0	0	0	0	0	0	0	0
χ_8^{co}	−2	i	2	$-i$	−2	i	2	$-i$	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4_{001}^{+}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$i\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	$i\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_7^{co}	$-i\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_8^{co}	$-i\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 198.3.478 : ${}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}4/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel -1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[T \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{010}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right]$
$\left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{11}\right.$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[E \parallel -1\right]$
C_7 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_8 : $\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_{10} : $\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$
C_{11} : $\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 1\right]$

$C_{13}: \left[\bar{E} \,||\, 4^{+}_{001}\right] \left[\bar{E} \,||\, 4^{-}_{001}\right]$
$C_{14}: \left[\bar{E} \,||\, 2_{001}\right]$
$C_{15}: \left[\bar{E} \,||\, m_{001}\right]$
$C_{16}: \left[\bar{E} \,||\, \bar{4}^{-}_{001}\right] \left[\bar{E} \,||\, \bar{4}^{+}_{001}\right]$
$C_{17}: \left[\bar{E} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	−2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	2
χ_3^{co}	2	−2	2	2	−2	2	0	0	0	0	0	−2	2	−2	−2	2	−2
χ_4^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2	−2	2
χ_5^{co}	4	0	−4	−4	0	4	0	0	0	0	0	−4	0	4	4	0	−4
χ_6^{co}	4	0	−4	4	0	−4	0	0	0	0	0	−4	0	4	−4	0	4

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & i\sigma_z \end{pmatrix}$	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_6^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 199.1.479 : ${}^{1}4/{}^{2}m{}^{2}m{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_6 :	$\left[2_{001} \mid\mid -1\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_8 :	$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \mid\mid m_{001}\right]$	C_9 :	$\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 :	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 199.2.480 : ${}^{1}4/{}^{2_z}m{}^{2_z}m{}^{2_z}m{}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
$\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$
$\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_6 :	$\left[2_{001} \mid\mid -1\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_8 :	$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \mid\mid m_{001}\right]$	C_9 :	$\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 :	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 199.3.481 : ${}^{1}4/{}^{2_z}m{}^{2_z}m{}^{2_z}m{}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \parallel m_{001}\right]$	C_9 :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$
C_5 :	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 199.4.482 : ${}^{1}4/{}^{2_z}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{2_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 4^{-}_{001}\right] \left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100} \mid\mid m_{100}\right]$
$\left[2_{100} \mid\mid m_{110}\right] \left[2_{100} \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid m_{010}\right] \left[E \mid\mid 2_{010}\right] \left[E \mid\mid 2_{1\bar{1}0}\right] \left[E \mid\mid 2_{110}\right] \left[E \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{110}\right] \left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z \mid\mid 2_{110}\right] \left[\Sigma_z \mid\mid 2_{100}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 4^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{110}\right] \left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 2_{010}\right]$
$\left[\bar{E} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E} \mid\mid 2_{110}\right] \left[\bar{E} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right] \left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100} \mid\mid m_{100}\right] \left[2_{100} \mid\mid m_{110}\right] \left[2_{100} \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid m_{010}\right]$
$\left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{110}\right] \left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z \mid\mid 2_{110}\right] \left[\Sigma_z \mid\mid 2_{100}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{110}\right] \left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$
C_5 : $\left[E \mid\mid 2_{010}\right] \left[E \mid\mid 2_{100}\right]$
C_6 : $\left[E \mid\mid 2_{1\bar{1}0}\right] \left[E \mid\mid 2_{110}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid 4^{+}_{001}\right] \left[\bar{E} \mid\mid 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_{10} : $\left[\bar{E} \mid\mid 2_{010}\right] \left[\bar{E} \mid\mid 2_{100}\right]$
C_{11} : $\left[\bar{E} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E} \mid\mid 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1	2	2	2	0	2	2	−2	−2	−2	−2	−2
χ_2	2	−2	2	0	2	−2	−2	2	−2	−2	2
χ_3	2	2	2	0	−2	−2	−2	−2	−2	2	2

χ_4	2	−2	2	0	−2	2	−2	2	−2	2	−2
χ_5	4	0	−4	0	0	0	−4	0	4	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} i\sigma_y & 0 \\ 0 & i\sigma_y \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 200.1.483 : ${}^{1}4/{}^{m}m{}^{m}m{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 : $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 200.2.484 : ${}^{1}4/{}^{m_z}m{}^{m_z}m{}^{m_z}m{}^{\infty_x}1 \rightarrow {}^{1}4/{}^{m}m{}^{m}m{}^{m}m$

No 201.1.485 : ${}^{1}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 : $\left[E \mid\mid 2_{1\bar{1}0}\right]$ $\left[E \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 201.2.486 : ${}^{1}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}4/{}^{\bar{1}}m{}^{\bar{1}}m{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[T \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$
$\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{1\bar{1}0}\right]$ $\left[E \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z \,||\, 4^{+}_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z \,||\, 4^{-}_{001}\right]$ $\left[\Sigma_z T \,||\, m_{001}\right]$ $\left[\Sigma_z T \,||\, \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \,||\, -1\right]$ $\left[\Sigma_z T \,||\, \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z T \,||\, m_{100}\right]$ $\left[\Sigma_z T \,||\, m_{110}\right]$ $\left[\Sigma_z T \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 2_{1\bar{1}0}\right]$ $\left[\Sigma_z \,||\, 2_{110}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E} \,||\, 4^{+}_{001}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E} \,||\, 4^{-}_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right]$ $\left[\bar{E}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \,||\, -1\right]$ $\left[\bar{E}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{E}T \,||\, m_{100}\right]$ $\left[\bar{E}T \,||\, m_{110}\right]$ $\left[\bar{E}T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{E}T \,||\, m_{010}\right]$ $\left[\bar{E} \,||\, 2_{010}\right]$
$\left[\bar{E} \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{E} \,||\, 2_{110}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z \,||\, 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \,||\, -1\right]$ $\left[\bar{\Sigma}_z T \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{110}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{110}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_{11} :	$\left[\bar{E} \,||\, 1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_{12} :	$\left[\bar{E} \,||\, 4^{+}_{001}\right]$ $\left[\bar{E} \,||\, 4^{-}_{001}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_{13} :	$\left[\bar{E} \,||\, 2_{001}\right]$
C_4 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$	C_{14} :	$\left[\bar{E} \,||\, 2_{010}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$
C_5 :	$\left[E \,||\, 2_{1\bar{1}0}\right]$ $\left[E \,||\, 2_{110}\right]$	C_{15} :	$\left[\bar{E} \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{E} \,||\, 2_{110}\right]$
C_6 :	$\left[\Sigma_z \,||\, 1\right]$	C_{16} :	$\left[\bar{\Sigma}_z \,||\, 1\right]$
C_7 :	$\left[\Sigma_z \,||\, 4^{+}_{001}\right]$ $\left[\Sigma_z \,||\, 4^{-}_{001}\right]$	C_{17} :	$\left[\bar{\Sigma}_z \,||\, 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 4^{-}_{001}\right]$
C_8 :	$\left[\Sigma_z \,||\, 2_{001}\right]$	C_{18} :	$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$
C_9 :	$\left[\Sigma_z \,||\, 2_{010}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$	C_{19} :	$\left[\bar{\Sigma}_z \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$
C_{10} :	$\left[\Sigma_z \,||\, 2_{1\bar{1}0}\right]$ $\left[\Sigma_z \,||\, 2_{110}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}
χ_1^{co}	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	0	0	0	0	0
χ_2^{co}	2	−2	2	2	−2	0	0	0	0	0	−2	2	−2	−2	2	0	0	0	0	0
χ_3^{co}	2	2	2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	0	0	0	0	0
χ_4^{co}	2	−2	2	−2	2	0	0	0	0	0	−2	2	−2	2	−2	0	0	0	0	0
χ_5^{co}	4	0	−4	0	0	0	0	0	0	0	−4	0	4	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, 2_{010}\right]$	$\left[T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} i\sigma_y & 0 \\ 0 & i\sigma_y \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 201.3.487 : ${}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel -1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel$
$\left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right]$
$\left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{110}\right]$
C_6 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$
C_{11} : $\left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1^{co}	2	2	2	2	2	0	−2	−2	−2	−2	−2
χ_2^{co}	2	−2	2	2	−2	0	−2	2	−2	−2	2
χ_3^{co}	2	2	2	−2	−2	0	−2	−2	−2	2	2
χ_4^{co}	2	−2	2	−2	2	0	−2	2	−2	2	−2
χ_5^{co}	4	0	−4	0	0	0	−4	0	4	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^+_{001}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} i\sigma_y & 0 \\ 0 & i\sigma_y \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 202.1.488 : ${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \parallel m_{001}\right]$	C_9 :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$
C_5 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 202.2.489 : ${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \parallel m_{001}\right]$	C_9 :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$
C_5 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 202.3.490 : ${}^{2_z}4/{}^{2_z}m^{2_x}m^{2_y}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^{-}_{001}\right]$ $\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{010}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \parallel m_{001}\right]$	C_9 :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$
C_5 :	$\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 203.1.491 : ${}^{2_z}4/{}^{2_z}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ C_6 : $\left[E \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_7 : $\left[2_{001} \parallel -1\right]$
C_4 : $\left[2_{001} \parallel 4^{-}_{001}\right]$ C_8 : $\left[E \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 204.1.492 : ${}^{m_x}4/{}^{m_x}m^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 205.1.493 : ${}^{2}4/{}^{2}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[2_{001} \mid\mid m_{001}\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[2_{001} \mid\mid -1\right]$
C_4 : $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[E \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 205.2.494 : ${}^{2_z}4/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[E \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[E \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$
$\left[2_{100}T \,||\, m_{110}\right]$ $\left[2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z \,||\, \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z T \,||\, m_{100}\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{110}\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \,||\, m_{010}\right]$ $\left[\Sigma_z 2_{100}T \,||\, 2_{010}\right]$ $\left[\Sigma_z T \,||\, 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \,||\, 2_{110}\right]$ $\left[\Sigma_z 2_{100}T \,||\, 2_{100}\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \,||\, m_{001}\right]$ $\left[\bar{E} \,||\, \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \,||\, -1\right]$ $\left[\bar{E} \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{E}T \,||\, m_{100}\right]$ $\left[\bar{E}2_{100}T \,||\, m_{110}\right]$ $\left[\bar{E}2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{E}T \,||\, m_{010}\right]$ $\left[\bar{E}2_{100}T \,||\, 2_{010}\right]$
$\left[\bar{E}T \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \,||\, 2_{110}\right]$ $\left[\bar{E}2_{100}T \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \,||\, 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right]$ $\left[\Sigma_z \,||\, \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, -1\right]$ $\left[\Sigma_z \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \,||\, m_{001}\right]$ $\left[\bar{E}2_{100} \,||\, -1\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, -1\right]$ $\left[\bar{\Sigma}_z \,||\, \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[E \,||\, \bar{4}^{-}_{001}\right]$
C_5 : $\left[E \,||\, \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_8 : $\left[\bar{E} \,||\, \bar{4}^{-}_{001}\right]$
C_9 : $\left[\bar{E} \,||\, \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	2	−2	−2	−2	−2
χ_2^{co}	2	0	2	−2	−2	−2	−2	2	2
χ_3^{co}	2	0	−2	$-i$	i	−2	2	i	$-i$
χ_4^{co}	2	0	−2	i	$-i$	−2	2	$-i$	i

Coirrepresntations of group generators

	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[T \,||\, m_{100}\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_y\mathcal{K}$	$\sigma_y\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_y\mathcal{K}$	$-\sigma_y\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 206.1.495 : ${}^{\bar{1}}4/{}^{\bar{1}}m^2m^mm$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^-_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{4}^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel \bar{4}^+_{001}\right] \left[2_{001} \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^-_{001}\right] \left[E \parallel \bar{4}^+_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^-_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 206.2.496 : ${}^{\bar{1}}4/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right]$
C_5 : $\left[\Sigma_z \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{+}_{001}\right]$
C_6 : $\left[\Sigma_z \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid \bar{4}^{-}_{001}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	$-i$	i	−2	2	0
χ_4^{co}	2	−2	0	0	i	$-i$	−2	2	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 207.1.497 : ${}^{m}4/{}^{m}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 207.2.498 : ${}^{m_z}4/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}4/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[T \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{010}\right]$
$\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
C_5 : $\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2
χ_3^{co}	4	−4	0	0	0	0	−4	4	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -i\sigma_z & 0 \\ 0 & i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 208.1.499 : ${}^{1}4/{}^{2_x}m{}^{2_y}m{}^{2_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{010} \mid\mid m_{100}\right]$
$\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_6 :	$\left[2_{100} \mid\mid -1\right]$
C_2 :	$\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$	C_7 :	$\left[2_{010} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_8 :	$\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \mid\mid m_{001}\right]$	C_9 :	$\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$
C_5 :	$\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{010} \mid\mid m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 208.2.500 : ${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{010} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{100}T \,||\, 4^{+}_{001}\right]$
$\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{110}\right]$ $\left[2_{001}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{010} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 208.3.501 : ${}^{1}4/{}^{2_x}m^{2_y}m^{2_y}m^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{010} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{010}T \,||\, 1\right]$ $\left[2_{010}T \,||\, 4^{+}_{001}\right]$
$\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, 4^{-}_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \,||\, -1\right]$ $\left[2_{001}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$ $\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$
$\left[T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{010} \,||\, m_{100}\right]$	$\left[2_{010}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 208.4.502 : ${}^{1}4/{}^{2_x}m{}^{2_y}m{}^{2_y}m{}^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{010} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[2_{001}T \,||\, 4^{+}_{001}\right]$
$\left[2_{001}T \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 4^{-}_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{010}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \,||\, -1\right]$ $\left[2_{010}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{110}\right]$ $\left[2_{100}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{100}T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{110}\right]$ $\left[T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[E \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{010} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{100} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{010} \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 209.1.503 : ${}^{1}4/{}^{2_z}m^{m_x}m^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[2_{001} \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[2_{001} \mid\mid -1\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 210.1.504 : ${}^{1}4/{}^{m_x}m{}^{2_z}m{}^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 211.1.505 : ${}^{1}4/^{m_x}m^{m_y}m^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \mid\mid m_{100}\right]$
$\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 212.1.506 : ${}^{1}4/{}^{2}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[2_{001} \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$ C_7 : $\left[2_{001} \mid\mid -1\right]$
C_4 : $\left[E \mid\mid 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 212.2.507 : ${}^{1}4/{}^{2_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{2_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[T \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 4^{-}_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	−2	2	−2	2
χ_3^{co}	2	i	−2	$-i$	0	−2	$-i$	2	i
χ_4^{co}	2	$-i$	−2	i	0	−2	i	2	$-i$

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 213.1.508 : ${}^{1}4/{}^{2}m^{m}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ C_7 : $\left[2_{001} \parallel -1\right]$
C_4 : $\left[E \parallel 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 213.2.509 : ${}^{1}4/{}^{2_z}m^{m_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{2_z}m^{m_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right]$
$\left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{010}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 4^{-}_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 4^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 4^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	−2	2	−2	2
χ_3^{co}	2	i	−2	$-i$	0	−2	$-i$	2	i
χ_4^{co}	2	$-i$	−2	i	0	−2	i	2	$-i$

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 214.1.510 : ${}^{1}4/{}^{\bar{1}}m^{2}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 214.2.511 : ${}^{1}4/{}^{\bar{1}}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel -1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_5 : $\left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	2	0	−2	0	i	$-i$	−2	0	2
χ_4^{co}	2	0	−2	0	$-i$	i	−2	0	2

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 215.1.512 : ${}^{1}4/{}^{\bar{1}}m^{m}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$
C_5 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{010}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 215.2.513 : ${}^{1}4/{}^{\bar{1}}m^{m_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{\bar{1}}m^{m_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right.$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_5 : $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	2	0	−2	0	i	$-i$	−2	0	2
χ_4^{co}	2	0	−2	0	$-i$	i	−2	0	2

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_z$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 216.1.514 : ${}^{1}4/{}^{m}m^{2}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 216.2.515 : ${}^{1}4/{}^{m_z}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{m_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$
$\left[2_{100} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid -1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \mid\mid 2_{110}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid -1\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$
$\left[\bar{E}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \mid\mid 2_{110}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$
C_5 : $\left[\Sigma_z \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	4	0	−4	0	0	0	−4	0	4

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & -i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 217.1.516 : ${}^{1}4/{}^{m}m{}^{\bar{1}}m{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 4^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{010}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 217.2.517 : ${}^{1}4/{}^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{1}4/{}^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel m_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel -1\right]$ $\left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z T \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{110}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E} \parallel 4^{+}_{001}\right]$ $\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E} \parallel 4^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right]$ $\left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \parallel -1\right]$ $\left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E}T \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E}T \parallel m_{010}\right]$ $\left[\bar{E}2_{100} \parallel 2_{010}\right]$
$\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \parallel 2_{110}\right]$ $\left[\bar{E}2_{100} \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{100}\right]$ $\left[\bar{E}2_{100} \parallel 2_{010}\right]$ $\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \parallel 2_{110}\right]$ $\left[\bar{E}2_{100} \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_5 : $\left[\Sigma_z \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right]$
C_6 : $\left[\Sigma_z \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 4^{+}_{001}\right]$ $\left[\bar{E} \parallel 4^{-}_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−2	2	0	0	0	−2	2	−2
χ_3^{co}	4	0	−4	0	0	0	−4	0	4

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{010}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} i\sigma_z & 0 \\ 0 & -i\sigma_z \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 218.1.518 : ${}^{2_z}4/{}^{1}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{110}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[E \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[E \,||\, m_{001}\right]$	C_9 :	$\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{010} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 218.2.519 : ${}^{2_z}4/{}^{1}m^{2_x}m^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[E \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_9 :	$\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 :	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 218.3.520 : ${}^{2_z}4/{}^{1}m^{2_x}m^{2_y}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 4^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[E \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_9 :	$\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 :	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 219.1.521 : ${}^{2_z}4/{}^{1}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$
$\left[2_{010}T \,||\, m_{110}\right]$ $\left[2_{010}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[E \,||\, m_{001}\right]$
C_2 : $\left[2_{001} \,||\, 4^{+}_{001}\right]$ C_6 : $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$ C_7 : $\left[E \,||\, -1\right]$
C_4 : $\left[2_{001} \,||\, 4^{-}_{001}\right]$ C_8 : $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 220.1.522 : ${}^{m_x}4/{}^{1}m^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100}T \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001} \,||\, m_{100}\right]$
$\left[2_{010}T \,||\, m_{110}\right]$ $\left[2_{010}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, m_{010}\right]$ $\left[2_{001} \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{001} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[2_{001} \,||\, m_{100}\right]$
C_2 : $\left[E \,||\, 2_{001}\right]$ C_6 : $\left[2_{001} \,||\, m_{010}\right]$
C_3 : $\left[E \,||\, m_{001}\right]$ C_7 : $\left[2_{001} \,||\, 2_{010}\right]$
C_4 : $\left[E \,||\, -1\right]$ C_8 : $\left[2_{001} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 2_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 221.1.523 : ${}^{2}4/{}^{1}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[E \mid\mid m_{001}\right]$
C_2 :	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	C_6 :	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \mid\mid 2_{001}\right]$	C_7 :	$\left[E \mid\mid -1\right]$
C_4 :	$\left[2_{001} \mid\mid 4^{-}_{001}\right]$	C_8 :	$\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 221.2.524 : ${}^{2_z}4/{}^{1}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{1}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100} \,||\, 4^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[2_{100} \,||\, 4^{-}_{001}\right] \left[E \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[E \,||\, -1\right] \left[2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[T \,||\, m_{100}\right]$
$\left[2_{100}T \,||\, m_{110}\right] \left[2_{100}T \,||\, m_{1\bar{1}0}\right] \left[T \,||\, m_{010}\right] \left[T \,||\, 2_{010}\right] \left[2_{100}T \,||\, 2_{1\bar{1}0}\right] \left[2_{100}T \,||\, 2_{110}\right] \left[T \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right] \left[\Sigma_z \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[\Sigma_z \,||\, -1\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[\Sigma_z T \,||\, m_{100}\right] \left[\Sigma_z 2_{100}T \,||\, m_{110}\right] \left[\Sigma_z 2_{100}T \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \,||\, m_{010}\right] \left[\Sigma_z T \,||\, 2_{010}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{110}\right] \left[\Sigma_z T \,||\, 2_{100}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\bar{E} \,||\, m_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[\bar{E} \,||\, -1\right] \left[\bar{E}2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[\bar{E}T \,||\, m_{100}\right] \left[\bar{E}2_{100}T \,||\, m_{110}\right] \left[\bar{E}2_{100}T \,||\, m_{1\bar{1}0}\right] \left[\bar{E}T \,||\, m_{010}\right] \left[\bar{E}T \,||\, 2_{010}\right]$
$\left[\bar{E}2_{100}T \,||\, 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \,||\, 2_{110}\right] \left[\bar{E}T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right] \left[\bar{\Sigma}_z T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{110}\right]$
$\left[\bar{\Sigma}_z T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, 4^{+}_{001}\right] \left[2_{100} \,||\, 4^{-}_{001}\right] \left[2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\Sigma_z \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[\Sigma_z \,||\, -1\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{4}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^{+}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[E \,||\, m_{001}\right]$
C_5 : $\left[E \,||\, -1\right]$
C_6 : $\left[\bar{E} \,||\, 1\right]$
C_7 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_8 : $\left[\bar{E} \,||\, m_{001}\right]$
C_9 : $\left[\bar{E} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	2	−2	−2	−2	−2
χ_2^{co}	2	0	2	−2	−2	−2	−2	2	2
χ_3^{co}	2	0	−2	−2	2	−2	2	2	−2
χ_4^{co}	2	0	−2	2	−2	−2	2	−2	2

Coirrepresntations of group generators

	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[T \,||\, m_{100}\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_y\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_y\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 222.1.525 : ${}^{\bar{1}}4/{}^{1}m^{2}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_5 : $\left[2_{001} \mid\mid m_{100}\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$ C_6 : $\left[2_{001} \mid\mid m_{010}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$ C_7 : $\left[2_{001} \mid\mid 2_{010}\right]$
C_4 : $\left[E \mid\mid -1\right]$ C_8 : $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 222.2.526 : ${}^{\bar{1}}4/{}^{1}m^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/{}^{1}m^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E} \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z \mid\mid -1\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid m_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	−2	−2	2	2
χ_3^{co}	2	−2	−2	2	0	−2	2	2	−2
χ_4^{co}	2	−2	2	−2	0	−2	2	−2	2

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 223.1.527 : ${}^{m}4/{}^{1}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_5 : $\left[2_{001} \parallel m_{100}\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ C_6 : $\left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[E \parallel m_{001}\right]$ C_7 : $\left[2_{001} \parallel 2_{010}\right]$
C_4 : $\left[E \parallel -1\right]$ C_8 : $\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 223.2.528 : ${}^{m_z}4/{}^{1}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}4/{}^{1}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid -1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z T \mid\mid m_{110}\right]$ $\left[\Sigma_z T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \mid\mid 2_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E} \mid\mid -1\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}T \mid\mid m_{110}\right]$ $\left[\bar{E}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{E}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[E \mid\mid m_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid m_{001}\right]$ $\left[\Sigma_z \mid\mid -1\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid m_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	−2	−2	2	2
χ_3^{co}	4	−4	−4	4	0	−4	4	4	−4
χ_4^{co}	4	−4	4	−4	0	−4	4	−4	4

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 224.1.529 : ${}^{2_z}4/{}^{2_x}m^1m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^-_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{100} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$	C_7 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \parallel m_{001}\right]$	C_9 :	$\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 :	$\left[2_{010} \parallel \bar{4}^-_{001}\right]$ $\left[2_{010} \parallel \bar{4}^+_{001}\right]$	C_{10} :	$\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 224.2.530 : ${}^{2_z}4/{}^{2_x}m^1m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^+_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^-_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^-_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^+_{001}\right]$ $\left[E \,||\, m_{100}\right]$
$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{110}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, 4^+_{001}\right]$
$\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, 4^-_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{4}^-_{001}\right]$ $\left[2_{010}T \,||\, -1\right]$ $\left[2_{100}T \,||\, \bar{4}^+_{001}\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^+_{001}\right]$ $\left[2_{001} \,||\, 4^-_{001}\right]$	C_7 :	$\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{001} \,||\, m_{110}\right]$ $\left[2_{001} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$
C_5 :	$\left[2_{010} \,||\, \bar{4}^-_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^+_{001}\right]$	C_{10} :	$\left[2_{010} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^+_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 224.3.531 : ${}^{2_z}4/{}^{2_x}m^1m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^-_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 4^+_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 4^-_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^-_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{100}T \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{100} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$	C_7 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \parallel m_{001}\right]$	C_9 :	$\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 :	$\left[2_{010} \parallel \bar{4}^-_{001}\right]$ $\left[2_{010} \parallel \bar{4}^+_{001}\right]$	C_{10} :	$\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 224.4.532 : ${}^{2_z}4/{}^{2_x}m^1m^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^-_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[2_{100}T \parallel 4^+_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^-_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^-_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{4}^+_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{100} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$	C_7 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \parallel m_{001}\right]$	C_9 :	$\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 :	$\left[2_{010} \parallel \bar{4}^-_{001}\right]$ $\left[2_{010} \parallel \bar{4}^+_{001}\right]$	C_{10} :	$\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{010}T \parallel 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 225.1.533 : ${}^{2_z}4/{}^{m_x}m^1m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^-_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 226.1.534 : ${}^{m_x}4/{}^{2_z}m^1m^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^-_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^-_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[E \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[E \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[2_{001} \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 227.1.535 : ${}^{m_x}4/{}^{m_y}m^1m^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^+_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^-_{001}\right]$ $\left[2_{010}T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{010}T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^+_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{4}^-_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^+_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^-_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid 4^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 228.1.536 : ${}^{2}4/{}^{\bar{1}}m^{1}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 228.2.537 : ${}^{2_z}4/^{\bar{1}}m^{1}m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/^{\bar{1}}m^{1}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[2_{100} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{110}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{110}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[\Sigma_z \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$
C_6 : $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{100}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid m_{100}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	0	0	−2	−2	−2
χ_2^{co}	2	0	2	−2	0	0	−2	−2	2
χ_3^{co}	4	0	−4	0	0	0	−4	4	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 229.1.538 : ${}^{2}4/{}^{m}m^{1}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 229.2.539 : ${}^{2_z}4/{}^{m_z}m^1m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{m_z}m^1m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100} \,||\, 4^+_{001}\right] \left[E \,||\, 2_{001}\right] \left[2_{100} \,||\, 4^-_{001}\right] \left[2_{100}T \,||\, m_{001}\right] \left[T \,||\, \bar{4}^-_{001}\right] \left[2_{100}T \,||\, -1\right] \left[T \,||\, \bar{4}^+_{001}\right] \left[E \,||\, m_{100}\right]$
$\left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[E \,||\, m_{010}\right] \left[2_{100}T \,||\, 2_{010}\right] \left[T \,||\, 2_{1\bar{1}0}\right] \left[T \,||\, 2_{110}\right] \left[2_{100}T \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{001}\right] \left[\Sigma_z 2_{100}T \,||\, m_{001}\right] \left[\Sigma_z T \,||\, \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100}T \,||\, -1\right] \left[\Sigma_z T \,||\, \bar{4}^+_{001}\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z \,||\, m_{010}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{010}\right] \left[\Sigma_z T \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z T \,||\, 2_{110}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{100}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}2_{100} \,||\, 4^+_{001}\right] \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E}2_{100} \,||\, 4^-_{001}\right]$
$\left[\bar{E}2_{100}T \,||\, m_{001}\right] \left[\bar{E}T \,||\, \bar{4}^-_{001}\right] \left[\bar{E}2_{100}T \,||\, -1\right] \left[\bar{E}T \,||\, \bar{4}^+_{001}\right] \left[\bar{E} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} \,||\, m_{010}\right] \left[\bar{E}2_{100}T \,||\, 2_{010}\right]$
$\left[\bar{E}T \,||\, 2_{1\bar{1}0}\right] \left[\bar{E}T \,||\, 2_{110}\right] \left[\bar{E}2_{100}T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, m_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, -1\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z T \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, 4^+_{001}\right] \left[2_{100} \,||\, 4^-_{001}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{001}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \,||\, 4^+_{001}\right] \left[\bar{E}2_{100} \,||\, 4^-_{001}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right]$
C_5 : $\left[\Sigma_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right]$
C_6 : $\left[\Sigma_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_9 : $\left[\bar{E} \,||\, m_{100}\right] \left[\bar{E} \,||\, m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	0	0	−2	−2	−2
χ_2^{co}	2	0	2	−2	0	0	−2	−2	2
χ_3^{co}	2	0	−2	0	2	−2	−2	2	0
χ_4^{co}	2	0	−2	0	−2	2	−2	2	0

Coirrepresntations of group generators

	$\left[2_{100} \,||\, 4^+_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_y\mathcal{K}$	σ_z	σ_0	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_y\mathcal{K}$	$-\sigma_z$	σ_0	$\sigma_z\mathcal{K}$

No 230.1.540 : ${}^{\bar{1}}4/{}^{2}m^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_5 :	$\left[E \mid\mid m_{100}\right]$
C_2 :	$\left[E \mid\mid 2_{001}\right]$	C_6 :	$\left[E \mid\mid m_{010}\right]$
C_3 :	$\left[2_{001} \mid\mid m_{001}\right]$	C_7 :	$\left[2_{001} \mid\mid 2_{010}\right]$
C_4 :	$\left[2_{001} \mid\mid -1\right]$	C_8 :	$\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 230.2.541 : ${}^{\bar{1}}4/{}^{2_z}m^1m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/{}^{2_z}m^1m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^-_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^-_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{100}T \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z T \parallel 4^+_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^-_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel m_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel \bar{4}^-_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel -1\right]$ $\left[\Sigma_z 2_{100}T \parallel \bar{4}^+_{001}\right]$ $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{010}\right]$ $\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100}T \parallel 2_{110}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^+_{001}\right]$ $\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right]$ $\left[\bar{E}2_{100}T \parallel \bar{4}^-_{001}\right]$ $\left[\bar{E}2_{100} \parallel -1\right]$ $\left[\bar{E}2_{100}T \parallel \bar{4}^+_{001}\right]$ $\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel m_{010}\right]$ $\left[\bar{E}2_{100} \parallel 2_{010}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \parallel 2_{110}\right]$ $\left[\bar{E}2_{100} \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^+_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel 4^-_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^+_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel m_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z \parallel m_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{100}\right]$ $\left[\bar{E}2_{100} \parallel m_{001}\right]$ $\left[\bar{E}2_{100} \parallel -1\right]$ $\left[\bar{E}2_{100} \parallel 2_{010}\right]$ $\left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_4 : $\left[E \parallel m_{100}\right]$
C_5 : $\left[E \parallel m_{010}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 2_{001}\right]$
C_8 : $\left[\bar{E} \parallel m_{100}\right]$
C_9 : $\left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	2	−2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	−2	2	2
χ_3^{co}	4	−4	0	0	0	−4	4	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 4^+_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 231.1.542 : ${}^{\bar{1}}4/{}^{m}m^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 231.2.543 : ${}^{\bar{1}}4/^{m_z}m^{1}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/^{m_z}m^{1}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z T \parallel 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel m_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel -1\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{010}\right]$ $\left[\Sigma_z 2_{100}T \parallel 2_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{110}\right]$ $\left[\Sigma_z 2_{100}T \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}T \parallel 4^{+}_{001}\right]$ $\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100}T \parallel -1\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel m_{010}\right]$ $\left[\bar{E}2_{100}T \parallel 2_{010}\right]$
$\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \parallel 2_{110}\right]$ $\left[\bar{E}2_{100}T \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{110}\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \parallel 2_{110}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
C_4 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$
C_5 : $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_6 : $\left[\Sigma_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	0	0	−2	−2	−2
χ_2^{co}	2	2	0	−2	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	2	−2	−2	2	0
χ_4^{co}	2	−2	0	0	−2	2	−2	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_y\mathcal{K}$	σ_z	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_y\mathcal{K}$	$-\sigma_z$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 232.1.544 : ${}^{m}4/{}^{2}m^{1}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[E \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[E \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[2_{001} \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 232.2.545 : ${}^{m_z}4/{}^{2_z}m^1m^{m_z}m^{\infty_x}1 \rightarrow {}^{m_z}4/{}^{2_z}m^1m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^-_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^-_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[T \parallel \bar{4}^+_{001}\right]$ $\left[E \parallel m_{100}\right]$
$\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z 2_{100}T \parallel 4^+_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \parallel 4^-_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel m_{001}\right]$ $\left[\Sigma_z T \parallel \bar{4}^-_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel -1\right]$ $\left[\Sigma_z T \parallel \bar{4}^+_{001}\right]$ $\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z 2_{100}T \parallel m_{110}\right]$ $\left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{010}\right]$ $\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \parallel 2_{110}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E}2_{100}T \parallel 4^+_{001}\right]$ $\left[\bar{E} \parallel 2_{001}\right]$ $\left[\bar{E}2_{100}T \parallel 4^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right]$ $\left[\bar{E}T \parallel \bar{4}^-_{001}\right]$ $\left[\bar{E}2_{100} \parallel -1\right]$ $\left[\bar{E}T \parallel \bar{4}^+_{001}\right]$ $\left[\bar{E} \parallel m_{100}\right]$ $\left[\bar{E}2_{100}T \parallel m_{110}\right]$ $\left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{E} \parallel m_{010}\right]$ $\left[\bar{E}2_{100} \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \parallel 2_{110}\right]$ $\left[\bar{E}2_{100} \parallel 2_{100}\right]$ $\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 4^+_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel 4^-_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$ $\left[\bar{\Sigma}_z T \parallel \bar{4}^+_{001}\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel 2_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel m_{001}\right]$ $\left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z \parallel m_{100}\right]$ $\left[\Sigma_z \parallel m_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{010}\right]$ $\left[\Sigma_z 2_{100} \parallel 2_{100}\right]$ $\left[\bar{E}2_{100} \parallel m_{001}\right]$ $\left[\bar{E}2_{100} \parallel -1\right]$ $\left[\bar{E}2_{100} \parallel 2_{010}\right]$ $\left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_4 : $\left[E \parallel m_{100}\right]$
C_5 : $\left[E \parallel m_{010}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 2_{001}\right]$
C_8 : $\left[\bar{E} \parallel m_{100}\right]$
C_9 : $\left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	2	−2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	−2	2	2
χ_3^{co}	4	−4	0	0	0	−4	4	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^+_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 233.1.546 : ${}^{m}4/{}^{\bar{1}}m^{1}m^{m}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001}T \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[E \,||\, m_{100}\right]$
$\left[2_{001}T \,||\, m_{110}\right]$ $\left[2_{001}T \,||\, m_{1\bar{1}0}\right]$ $\left[E \,||\, m_{010}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 2_{001}\right]$
C_3 : $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$
C_4 : $\left[E \,||\, m_{100}\right]$ $\left[E \,||\, m_{010}\right]$
C_5 : $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 233.2.547 : ${}^{m_z}4/^{\bar{1}}m^{1}m^{m_z}m^{\infty_x}1 \rightarrow {}^{m_z}4/^{\bar{1}}m^{1}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \mid\mid -1\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid 2_{110}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right]$
C_4 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$
C_5 : $\left[\Sigma_z \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{010}\right]$
C_6 : $\left[\Sigma_z \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid m_{100}\right]$
C_7 : $\left[\bar{E} \mid\mid 1\right]$
C_8 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_9 : $\left[\bar{E} \mid\mid m_{100}\right]$ $\left[\bar{E} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	0	0	−2	−2	−2
χ_2^{co}	2	2	0	−2	0	0	−2	−2	2
χ_3^{co}	4	−4	0	0	0	0	−4	4	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 234.1.548 : ${}^{2_z}4/{}^{2_x}m{}^{2_x}m{}^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{100} \parallel -1\right]$
C_2 :	$\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel 2_{001}\right]$	C_8 :	$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \parallel m_{001}\right]$	C_9 :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$
C_5 :	$\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 234.2.549 : ${}^{2_z}4/{}^{2_x}m{}^{2_x}m{}^{2_y}m{}^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[T \,||\, 4^{+}_{001}\right]$
$\left[2_{001}T \,||\, 2_{001}\right]$ $\left[T \,||\, 4^{-}_{001}\right]$ $\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \,||\, -1\right]$ $\left[2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{100}T \,||\, m_{110}\right]$ $\left[2_{100}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{001}T \,||\, 2_{010}\right]$ $\left[T \,||\, 2_{1\bar{1}0}\right]$ $\left[T \,||\, 2_{110}\right]$ $\left[2_{001}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 :	$\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 234.3.550 : ${}^{2_z}4/{}^{2_x}m{}^{2_x}m{}^{2_y}m{}^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{010}T \,||\, 4^{+}_{001}\right]$
$\left[2_{100}T \,||\, 2_{001}\right]$ $\left[2_{010}T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[2_{001}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$ $\left[2_{001}T \,||\, m_{110}\right]$ $\left[2_{001}T \,||\, m_{1\bar{1}0}\right]$
$\left[T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 :	$\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 234.4.551 : ${}^{2_z}4/{}^{2_x}m{}^{2_x}m{}^{2_y}m{}^{m_y}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[2_{010}T \,||\, 1\right]$ $\left[2_{100}T \,||\, 4^{+}_{001}\right]$
$\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 4^{-}_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[T \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \,||\, -1\right]$ $\left[T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \,||\, m_{100}\right]$ $\left[T \,||\, m_{110}\right]$ $\left[T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{001}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{100}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{100} \,||\, -1\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{100} \,||\, m_{010}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_8 :	$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \,||\, m_{001}\right]$	C_9 :	$\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 :	$\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{010}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 235.1.552 : ${}^{2_z}4/{}^{m_x}m{}^{m_x}m{}^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$
C_5 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel 2_{010}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 236.1.553 : ${}^{m_x}4/{}^{2_z}m^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[E \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 237.1.554 : ${}^{m_x}4/{}^{m_y}m^{m_y}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 4^{-}_{001}\right]$ $\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{110}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 238.1.555 : ${}^{2}4/{}^{\bar{1}}m{}^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 238.2.556 : ${}^{2_z}4/^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}4/^{\bar{1}}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$
$\left[2_{100}T \,||\, m_{110}\right]$ $\left[2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[E \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, 2_{110}\right]$ $\left[E \,||\, 2_{100}\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\Sigma_z T \,||\, m_{001}\right]$ $\left[\Sigma_z 2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z T \,||\, -1\right]$ $\left[\Sigma_z 2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z T \,||\, m_{100}\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{110}\right]$ $\left[\Sigma_z 2_{100}T \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \,||\, m_{010}\right]$ $\left[\Sigma_z \,||\, 2_{010}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{110}\right]$ $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\bar{E} \,||\, 1\right]$ $\left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{E} \,||\, 2_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right]$ $\left[\bar{E}2_{100}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[\bar{E}T \,||\, -1\right]$ $\left[\bar{E}2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{E}T \,||\, m_{100}\right]$ $\left[\bar{E}2_{100}T \,||\, m_{110}\right]$ $\left[\bar{E}2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{E}T \,||\, m_{010}\right]$ $\left[\bar{E} \,||\, 2_{010}\right]$
$\left[\bar{E}2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{110}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \,||\, -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \,||\, m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \,||\, m_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, 4^{+}_{001}\right]$ $\left[2_{100} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, 2_{110}\right]$ $\left[\Sigma_z \,||\, 1\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\Sigma_z \,||\, 2_{001}\right]$ $\left[\Sigma_z 2_{100} \,||\, 4^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100} \,||\, 2_{110}\right]$ $\left[\bar{E}2_{100} \,||\, 4^{+}_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \,||\, 2_{110}\right]$ $\left[\bar{\Sigma}_z \,||\, 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$
C_4 : $\left[E \,||\, 2_{010}\right]$ $\left[E \,||\, 2_{100}\right]$
C_5 : $\left[\Sigma_z \,||\, 2_{010}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{100}\right]$
C_6 : $\left[\Sigma_z \,||\, 2_{100}\right]$ $\left[\bar{\Sigma}_z \,||\, 2_{010}\right]$
C_7 : $\left[\bar{E} \,||\, 1\right]$
C_8 : $\left[\bar{E} \,||\, 2_{001}\right]$
C_9 : $\left[\bar{E} \,||\, 2_{010}\right]$ $\left[\bar{E} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	0	0	−2	−2	−2
χ_2^{co}	2	0	2	−2	0	0	−2	−2	2
χ_3^{co}	4	0	−4	0	0	0	−4	4	0

Coirrepresntations of group generators

	$\left[2_{100} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, 2_{010}\right]$	$\left[T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} -\sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 239.1.557 : ${}^{2}4/{}^{m}m{}^{m}m{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 4^{+}_{001}\right]$ $\left[2_{001} \mid\mid 4^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{001}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid 2_{010}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 239.2.558 : ${}^{2_z}4/{}^{m_z}m^{m_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}4/{}^{m_z}m^{m_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[T \parallel \bar{4}^-_{001}\right] \left[2_{100}T \parallel -1\right] \left[T \parallel \bar{4}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z T \parallel \bar{4}^+_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 4^+_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^-_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{4}^-_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}T \parallel \bar{4}^+_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 4^+_{001}\right] \left[\bar{E}2_{100} \parallel 4^-_{001}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	0	0	−2	−2	−2
χ_2^{co}	2	0	2	−2	0	0	−2	−2	2
χ_3^{co}	2	0	−2	0	−2	2	−2	2	0
χ_4^{co}	2	0	−2	0	2	−2	−2	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 4^+_{001}\right]$	$\left[E \parallel 2_{010}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_y\mathcal{K}$	$-\sigma_z$	σ_0	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_y\mathcal{K}$	σ_z	σ_0	$\sigma_z\mathcal{K}$

No 240.1.559 : ${}^{\bar{1}}4/{}^{2}m{}^{2}m{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[E \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 240.2.560 : ${}^{\bar{1}}4/{}^{2_z}m{}^{2_z}m{}^{m_z}m{}^{\infty_x}1 \rightarrow {}^{\bar{1}}4/{}^{2_z}m{}^{2_z}m{}^{m_z}m{}^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \mid\mid 2_{110}\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid 2_{010}\right]$
$\left[\bar{E}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \mid\mid 2_{110}\right]$ $\left[\bar{E} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$
C_5 : $\left[E \mid\mid 2_{100}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid 2_{010}\right]$
C_9 : $\left[\bar{E} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	2	−2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	−2	2	2
χ_3^{co}	4	−4	0	0	0	−4	4	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} -\sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 241.1.561 : ${}^{\bar{1}}4/{}^{m}m{}^{m}m{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{110}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 241.2.562 : ${}^{\bar{1}}4/{}^{m_z}m^{m_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}4/{}^{m_z}m^{m_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	0	0	−2	−2	−2
χ_2^{co}	2	2	0	−2	0	0	−2	−2	2
χ_3^{co}	2	−2	0	0	−2	2	−2	2	0
χ_4^{co}	2	−2	0	0	2	−2	−2	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{110}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-i\sigma_y\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-i\sigma_y\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 242.1.563 : ${}^{m}4/{}^{2}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[E \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 242.2.564 : ${}^{m_z}4/{}^{2_z}m{}^{2_z}m{}^{\bar{1}}m{}^{\infty_x}1 \rightarrow {}^{m_z}4/{}^{2_z}m{}^{2_z}m{}^{\bar{1}}m{}^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z T \mid\mid m_{110}\right]$ $\left[\Sigma_z T \mid\mid m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{010}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{110}\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E}2_{100}T \mid\mid 4^{-}_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}T \mid\mid m_{110}\right]$ $\left[\bar{E}T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{110}\right]$ $\left[\bar{E} \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 4^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{110}\right]$
$\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100} \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{010}\right]$ $\left[\Sigma_z \mid\mid 2_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100} \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{100}\right]$
C_4 : $\left[E \mid\mid 2_{010}\right]$
C_5 : $\left[E \mid\mid 2_{100}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid 2_{010}\right]$
C_9 : $\left[\bar{E} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	2	−2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	−2	2	2
χ_3^{co}	4	−4	0	0	0	−4	4	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 2_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{100}T \mid\mid 4^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$

Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} -\sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 243.1.565 : ${}^{m}4/{}^{\bar{1}}m{}^{\bar{1}}m^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$
C_5 : $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{001} \mid\mid m_{110}\right]$	$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 243.2.566 : ${}^{m_z}4/^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}4/^{\bar{1}}m^{\bar{1}}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[T \parallel -1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[T \parallel m_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{010}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\Sigma_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
C_7 : $\left[\bar{E} \parallel 1\right]$
C_8 : $\left[\bar{E} \parallel 2_{001}\right]$
C_9 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	0	2	0	0	−2	−2	−2
χ_2^{co}	2	2	0	−2	0	0	−2	−2	2
χ_3^{co}	4	−4	0	0	0	0	−4	4	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{110}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} -i\sigma_y & 0 \\ 0 & -i\sigma_y \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} -\sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & -\sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 244.1.567 : ${}^{4_z}4/{}^{1}m^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[4^{+}_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[4^{-}_{001} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{110} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_6 :	$\left[2_{001} \parallel -1\right]$
C_2 :	$\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel 2_{001}\right]$	C_8 :	$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_9 :	$\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$
C_5 :	$\left[4^{+}_{001} \parallel \bar{4}^{-}_{001}\right]$ $\left[4^{-}_{001} \parallel \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{110} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 244.2.568 : ${}^{4_z}4/{}^{1}m^{2_x}m^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^{-}_{001} \,||\, 4^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[4^{+}_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[4^{-}_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{110} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{110}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[4^{-}_{001}T \,||\, 4^{+}_{001}\right]$
$\left[T \,||\, 2_{001}\right]$ $\left[4^{+}_{001}T \,||\, 4^{-}_{001}\right]$ $\left[2_{001}T \,||\, m_{001}\right]$ $\left[4^{-}_{001}T \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[4^{+}_{001}T \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{1\bar{1}0}T \,||\, m_{110}\right]$ $\left[2_{110}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{1\bar{1}0}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{110}T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[2_{001} \,||\, -1\right]$
C_2 :	$\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$ $\left[4^{-}_{001} \,||\, 4^{-}_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_3 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[E \,||\, m_{001}\right]$	C_9 :	$\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010} \,||\, 2_{100}\right]$
C_5 :	$\left[4^{+}_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[4^{-}_{001} \,||\, \bar{4}^{+}_{001}\right]$	C_{10} :	$\left[2_{110} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[4^{+}_{001} \,||\, 4^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 245.1.569 : ${}^{4_z}4/{}^{1}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^+_{001} \,||\, 4^+_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^-_{001} \,||\, 4^-_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[4^+_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[2_{001} \,||\, -1\right]$ $\left[4^-_{001} \,||\, \bar{4}^+_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$
$\left[2_{110}T \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{110}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[E \,||\, m_{001}\right]$
C_2 : $\left[4^+_{001} \,||\, 4^+_{001}\right]$ C_6 : $\left[4^+_{001} \,||\, \bar{4}^-_{001}\right]$
C_3 : $\left[2_{001} \,||\, 2_{001}\right]$ C_7 : $\left[2_{001} \,||\, -1\right]$
C_4 : $\left[4^-_{001} \,||\, 4^-_{001}\right]$ C_8 : $\left[4^-_{001} \,||\, \bar{4}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[4^+_{001} \,||\, 4^+_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 246.1.570 : ${}^{\bar{4}_z}4/{}^{1}m^{2_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^{+}_{001}T \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^{-}_{001}T \parallel 4^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[4^{+}_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[4^{-}_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{110}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{100} \parallel m_{100}\right]$
C_2 :	$\left[2_{001} \parallel 2_{001}\right]$	C_6 :	$\left[2_{010} \parallel m_{010}\right]$
C_3 :	$\left[E \parallel m_{001}\right]$	C_7 :	$\left[2_{100} \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[4^{+}_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 247.1.571 : ${}^{4_z}4/{}^{2_z}m^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^+_{001} \,||\, 4^+_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^-_{001} \,||\, 4^-_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{110} \,||\, 2_{110}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[E \,||\, -1\right]$
C_2 :	$\left[4^+_{001} \,||\, 4^+_{001}\right]$ $\left[4^-_{001} \,||\, 4^-_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_3 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \,||\, m_{001}\right]$	C_9 :	$\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$
C_5 :	$\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$	C_{10} :	$\left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{110} \,||\, 2_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3	1	−1	1	1	−1	1	1	−1	1	−1
χ_4	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5	1	1	1	1	1	1	−1	−1	−1	−1
χ_6	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7	1	−1	1	1	−1	1	−1	1	−1	1
χ_8	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9	2	0	−2	−2	0	2	0	0	0	0
χ_{10}	2	0	−2	2	0	−2	0	0	0	0

Irrepresntations of group generators

	$\left[4^+_{001} \,||\, 4^+_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$
Γ_1	1	1	1
Γ_2	1	−1	−1
Γ_3	−1	1	1

Γ_4	-1	-1	-1
Γ_5	1	1	-1
Γ_6	1	-1	1
Γ_7	-1	1	-1
Γ_8	-1	-1	1
Γ_9	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$
Γ_{10}	$i\sigma_y$	σ_0	σ_x

No 247.2.572 : ${}^{4_z}4/{}^{2_z}m^{2_x}m^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^+_{001} \,||\, 4^+_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^-_{001} \,||\, 4^-_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$
$\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{010}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{110} \,||\, 2_{110}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[4^-_{001}T \,||\, 4^+_{001}\right]$
$\left[T \,||\, 2_{001}\right]$ $\left[4^+_{001}T \,||\, 4^-_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[4^+_{001}T \,||\, \bar{4}^-_{001}\right]$ $\left[2_{001}T \,||\, -1\right]$ $\left[4^-_{001}T \,||\, \bar{4}^+_{001}\right]$ $\left[2_{010}T \,||\, m_{100}\right]$ $\left[2_{1\bar{1}0}T \,||\, m_{110}\right]$ $\left[2_{110}T \,||\, m_{1\bar{1}0}\right]$
$\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{110}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \,||\, 2_{110}\right]$ $\left[2_{010}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_6 :	$\left[E \,||\, -1\right]$
C_2 :	$\left[4^+_{001} \,||\, 4^+_{001}\right]$ $\left[4^-_{001} \,||\, 4^-_{001}\right]$	C_7 :	$\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{010}\right]$
C_3 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_8 :	$\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$
C_4 :	$\left[2_{001} \,||\, m_{001}\right]$	C_9 :	$\left[2_{010} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$
C_5 :	$\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$	C_{10} :	$\left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{110} \,||\, 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	−1	1	1	−1	1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	1	−1	−1	1	1	−1
χ_5^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	1	−1	−1	−1	1	1	−1	−1
χ_7^{co}	1	−1	1	1	−1	1	−1	1	−1	1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	−1	1
χ_9^{co}	2	0	−2	−2	0	2	0	0	0	0
χ_{10}^{co}	2	0	−2	2	0	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \,||\, 4^+_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$

Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	-1	$\mathcal{K}$
Γ_5^{co}	1	1	-1	$\mathcal{K}$
Γ_6^{co}	1	-1	1	$\mathcal{K}$
Γ_7^{co}	-1	1	-1	$\mathcal{K}$
Γ_8^{co}	-1	-1	1	$\mathcal{K}$
Γ_9^{co}	$i\sigma_y$	$-\sigma_0$	$-\sigma_x$	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	$i\sigma_y$	σ_0	σ_x	$\sigma_0\mathcal{K}$

No 248.1.573 : ${}^{4_z}4/{}^{2_z}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^+_{001} \,||\, 4^+_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^-_{001} \,||\, 4^-_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[E \,||\, -1\right]$ $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$
$\left[2_{110}T \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{010}T \,||\, 2_{010}\right]$ $\left[2_{1\bar{1}0}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{110}T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[4^+_{001} \,||\, 4^+_{001}\right]$	C_6 :	$\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$
C_3 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_7 :	$\left[E \,||\, -1\right]$
C_4 :	$\left[4^-_{001} \,||\, 4^-_{001}\right]$	C_8 :	$\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	1	-1
χ_4^{co}	1	-1	1	-1	-1	1	-1	1
χ_5^{co}	1	i	-1	$-i$	-1	$-i$	1	i
χ_6^{co}	1	i	-1	$-i$	1	i	-1	$-i$
χ_7^{co}	1	$-i$	-1	i	-1	i	1	$-i$
χ_8^{co}	1	$-i$	-1	i	1	$-i$	-1	i

Coirrepresntations of group generators

	$\left[4^+_{001} \,||\, 4^+_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	1	$\mathcal{K}$
Γ_4^{co}	-1	-1	$\mathcal{K}$
Γ_5^{co}	i	-1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 249.1.574 : ${}^{\bar{4}_z}4/{}^{2_z}m^{m_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001}T \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001}T \parallel 4^-_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[4^-_{001}T \parallel \bar{4}^-_{001}\right]$ $\left[E \parallel -1\right]$ $\left[4^+_{001}T \parallel \bar{4}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{110} \parallel m_{110}\right]$
C_2 :	$\left[2_{001} \parallel 2_{001}\right]$	C_6 :	$\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$
C_4 :	$\left[E \parallel -1\right]$	C_8 :	$\left[2_{110} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	2	−2	−2	2	0	0	0	0
χ_4^{co}	2	−2	2	−2	0	0	0	0
χ_5^{co}	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{110} \parallel m_{110}\right]$	$\left[4^+_{001}T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	σ_z	$-i\sigma_y\mathcal{K}$
Γ_5^{co}	1	1	−1	$\mathcal{K}$
Γ_6^{co}	1	−1	1	$\mathcal{K}$

No 250.1.575 : ${}^{m_z}4/{}^{2_x}m{}^{m_x}m{}^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{010} \parallel m_{110}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{010} \parallel m_{1\bar{1}0}\right]$
C_3 :	$\left[2_{100} \parallel m_{001}\right]$	C_7 :	$\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
C_4 :	$\left[2_{100} \parallel -1\right]$	C_8 :	$\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	2	−2	−2	2	0	0	0	0
χ_4^{co}	2	−2	2	−2	0	0	0	0
χ_5^{co}	1	1	1	1	−1	−1	−1	−1
χ_6^{co}	1	1	−1	−1	1	1	−1	−1

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{010} \parallel m_{110}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$-\sigma_z$	$-i\sigma_y\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	σ_z	$-i\sigma_y\mathcal{K}$
Γ_5^{co}	1	1	−1	$\mathcal{K}$
Γ_6^{co}	1	−1	1	$\mathcal{K}$

No 251.1.576 : ${}^{m_z}4/{}^{m_x}m^{2_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$
C_3 : $\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_5 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{010} \parallel \bar{4}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 252.1.577 : ${}^{2_z}4/{}^{m_x}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_5 : $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 253.1.578 : ${}^{m_z}4/{}^{2_z}m{}^{2_x}m{}^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{100} \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{100} \parallel m_{010}\right]$
C_3 :	$\left[2_{001} \parallel m_{001}\right]$	C_7 :	$\left[2_{010} \parallel 2_{010}\right]$
C_4 :	$\left[2_{001} \parallel -1\right]$	C_8 :	$\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 254.1.579 : ${}^{2_z}4/{}^{m_z}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_5 : $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 255.1.580 : ${}^{2_z}4/{}^{m_z}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$
C_5 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 256.1.581 : ${}^{m_z}4/{}^{2_x}m{}^{2_z}m{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 4^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[2_{100} \parallel m_{001}\right]$	C_7 :	$\left[2_{010} \parallel 2_{010}\right]$
C_4 :	$\left[2_{100} \parallel -1\right]$	C_8 :	$\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 257.1.582 : ${}^{m_z}4/{}^{m_x}m^{\bar{1}}m^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{010} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[2_{010} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{010} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{010} \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$
C_5 : $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{001} \mid\mid m_{110}\right]$	$\left[2_{001}T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 258.1.583 : ${}^{m_z}4/^{\bar{1}}m^{m_x}m^{2_y}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001}T \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001}T \,||\, 4^{-}_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$
$\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 2_{001}\right]$
C_3 : $\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$
C_5 : $\left[2_{100} \,||\, 2_{010}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, \bar{4}^{-}_{001}\right]$	$\left[2_{010} \,||\, m_{110}\right]$	$\left[2_{001}T \,||\, 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 259.1.584 : ${}^{2_z}4/{}^{2_x}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$
$\left[2_{010}T \,||\, m_{110}\right]$ $\left[2_{010}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100}T \,||\, m_{010}\right]$ $\left[T \,||\, 2_{010}\right]$ $\left[2_{001}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{001}T \,||\, 2_{110}\right]$ $\left[T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$ C_5 : $\left[2_{100} \,||\, m_{001}\right]$
C_2 : $\left[2_{001} \,||\, 4^{+}_{001}\right]$ C_6 : $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 2_{001}\right]$ C_7 : $\left[2_{100} \,||\, -1\right]$
C_4 : $\left[2_{001} \,||\, 4^{-}_{001}\right]$ C_8 : $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 260.1.585 : ${}^{2_z}4/{}^{2_x}m^{\bar{1}}m^{m_z}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 4^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[2_{001} \,||\, 4^{-}_{001}\right]$ $\left[2_{100} \,||\, m_{001}\right]$ $\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$ $\left[2_{100} \,||\, -1\right]$ $\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$ $\left[T \,||\, m_{100}\right]$
$\left[2_{001}T \,||\, m_{110}\right]$ $\left[2_{001}T \,||\, m_{1\bar{1}0}\right]$ $\left[T \,||\, m_{010}\right]$ $\left[2_{100}T \,||\, 2_{010}\right]$ $\left[2_{010}T \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010}T \,||\, 2_{110}\right]$ $\left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_5 :	$\left[2_{100} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	C_6 :	$\left[2_{010} \,||\, \bar{4}^{-}_{001}\right]$
C_3 :	$\left[E \,||\, 2_{001}\right]$	C_7 :	$\left[2_{100} \,||\, -1\right]$
C_4 :	$\left[2_{001} \,||\, 4^{-}_{001}\right]$	C_8 :	$\left[2_{010} \,||\, \bar{4}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1	−1	1
χ_5^{co}	1	i	−1	$-i$	−1	$-i$	1	i
χ_6^{co}	1	i	−1	$-i$	1	i	−1	$-i$
χ_7^{co}	1	$-i$	−1	i	−1	i	1	$-i$
χ_8^{co}	1	$-i$	−1	i	1	$-i$	−1	i

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 4^{+}_{001}\right]$	$\left[2_{100} \,||\, m_{001}\right]$	$\left[T \,||\, m_{100}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	i	−1	$\mathcal{K}$

Γ_6^{co}	i	1	$\mathcal{K}$
Γ_7^{co}	$-i$	-1	$\mathcal{K}$
Γ_8^{co}	$-i$	1	$\mathcal{K}$

No 261.1.586 : ${}^{2_z}4/^{\bar{1}}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 : $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{010}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 262.1.587 : ${}^{2_z}4/{}^{m_x}m^{\bar{1}}m^{m_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 : $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{010}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 263.1.588 : ${}^{2_z}4/{}^{\bar{1}}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_5 : $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 264.1.589 : ${}^{\bar{1}}4/{}^{2_x}m{}^{2_z}m{}^{m_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[T \parallel 4^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{100}T \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$
$\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_5 :	$\left[2_{001} \parallel m_{100}\right]$
C_2 :	$\left[E \parallel 2_{001}\right]$	C_6 :	$\left[2_{001} \parallel m_{010}\right]$
C_3 :	$\left[2_{100} \parallel m_{001}\right]$	C_7 :	$\left[2_{010} \parallel 2_{010}\right]$
C_4 :	$\left[2_{100} \parallel -1\right]$	C_8 :	$\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	−1	−1	1	1	−1	−1
χ_5^{co}	2	−2	−2	2	0	0	0	0
χ_6^{co}	2	−2	2	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	$-\sigma_0$	$-\sigma_0$	σ_z	$-i\sigma_y\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	σ_0	$-\sigma_z$	$-i\sigma_y\mathcal{K}$

No 265.1.590 : ${}^{\bar{1}}4/^{m_x}m^{m_z}m^{2_z}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid -1\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 2_{001}\right]$
C_3 : $\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$ $\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{1\bar{1}0}\right]$
C_5 : $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{-}_{001}\right]$	$\left[2_{001} \mid\mid m_{110}\right]$	$\left[T \mid\mid 4^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 266.1.591 : ${}^{4_z}4/{}^{m_z}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[4^-_{001}T \parallel \bar{4}^-_{001}\right]$ $\left[T \parallel -1\right]$ $\left[4^+_{001}T \parallel \bar{4}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{100}\right]$
C_5 : $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \parallel 4^+_{001}\right]$	$\left[2_{010} \parallel 2_{010}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 267.1.592 : ${}^{4_z}4/^{\bar{1}}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[4^+_{001}T \parallel \bar{4}^-_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[4^-_{001}T \parallel \bar{4}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{110} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$
C_5 : $\left[2_{110} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel 2_{010}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	σ_x	$\sigma_0\mathcal{K}$

No 268.1.593 : ${}^{4_z}4/{}^{m_z}m{}^{2_x}m{}^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[4^+_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[4^-_{001} \parallel 4^-_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[4^-_{001}T \parallel \bar{4}^-_{001}\right] \left[T \parallel -1\right] \left[4^+_{001}T \parallel \bar{4}^+_{001}\right] \left[2_{100} \parallel m_{100}\right]$
$\left[2_{110} \parallel m_{110}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^+_{001} \parallel 4^+_{001}\right] \left[4^-_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_5 : $\left[2_{110} \parallel m_{110}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 269.1.594 : ${}^{4_z}4/{}^{\bar{1}}m^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[4^+_{001}T \parallel \bar{4}^-_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[4^-_{001}T \parallel \bar{4}^+_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{110}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[4^+_{001} \parallel 4^+_{001}\right]$ $\left[4^-_{001} \parallel 4^-_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_5 : $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	1	−1	1	−1	1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	0	−2	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \parallel 4^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$i\sigma_y$	$-\sigma_x$	$\sigma_0\mathcal{K}$

No 270.1.595 : ${}^{\bar{4}_z}4/{}^{m_z}m^{m_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[4^+_{001}T \parallel 4^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[4^-_{001}T \parallel 4^-_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[4^-_{001} \parallel \bar{4}^-_{001}\right] \left[T \parallel -1\right] \left[4^+_{001} \parallel \bar{4}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right]$
$\left[2_{110} \parallel m_{110}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right]$
C_3 : $\left[4^-_{001} \parallel \bar{4}^-_{001}\right] \left[4^+_{001} \parallel \bar{4}^+_{001}\right]$
C_4 : $\left[2_{110} \parallel m_{110}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{010} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[4^-_{001} \parallel \bar{4}^-_{001}\right]$	$\left[2_{110} \parallel m_{110}\right]$	$\left[4^+_{001}T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 271.1.596 : ${}^{\bar{4}_z}4/{}^{\bar{1}}m^{m_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[4^+_{001}T \parallel 4^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[4^-_{001}T \parallel 4^-_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[4^+_{001} \parallel \bar{4}^-_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[4^-_{001} \parallel \bar{4}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{110}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right]$
C_3 : $\left[4^+_{001} \parallel \bar{4}^-_{001}\right]$ $\left[4^-_{001} \parallel \bar{4}^+_{001}\right]$
C_4 : $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{100} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1
χ_3^{co}	1	1	−1	1	−1
χ_4^{co}	1	1	1	−1	−1
χ_5^{co}	2	−2	0	0	0

Coirrepresntations of group generators

	$\left[4^+_{001} \parallel \bar{4}^-_{001}\right]$	$\left[2_{110} \parallel m_{110}\right]$	$\left[4^+_{001}T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	$-i\sigma_y$	σ_z	$\sigma_0\mathcal{K}$

No 272.1.597 : $^{1}3$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$
Γ_1	1
Γ_2	$e^{\frac{2\pi i}{3}}$
Γ_3	$e^{\frac{-2\pi i}{3}}$

No 272.2.598 : ${}^{1}3^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 3^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\sigma_x\mathcal{K}$

No 272.3.599 : ${}^{1}3^{\infty_z}1 \to {}^{1}3$

No 272.4.600 : ${}^{1}3^{\infty_z}m_x1 \to {}^{1}3^{m_x}1$

No 273.1.601 : $^{3}3$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Irrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
Γ_1	1
Γ_2	$e^{\frac{2\pi i}{3}}$
Γ_3	$e^{\frac{-2\pi i}{3}}$

No 273.2.602 : ${}^{3_z}3^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[6^{-}_{001}T \mid\mid 3^{+}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid 3^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\sigma_x\mathcal{K}$

No 274.1.603 : ${}^{1}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel 3^{+}_{001}\right]$
C_6 : $\left[E \parallel \bar{3}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{2\pi i}{3}}$
Γ_4	$e^{\frac{-\pi i}{3}}$
Γ_5	$e^{\frac{-2\pi i}{3}}$
Γ_6	$e^{\frac{\pi i}{3}}$

No 274.2.604 : ${}^{1}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 3^{-}_{001}\right]$
$\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[E \mid\mid \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3^*	$\sigma_x\mathcal{K}$

No 274.3.605 : ${}^{1}\bar{3}^{\infty_z}1 \to {}^{1}\bar{3}$

No 274.4.606 : ${}^{1}\bar{3}^{\infty_z m_x}1 \to {}^{1}\bar{3}^{m_x}1$

No 275.1.607 : $^{2}\bar{3}$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[2_{001} \mid\mid \bar{3}^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[2_{001} \mid\mid -1\right] \left[E \mid\mid 3^{+}_{001}\right] \left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{2\pi i}{3}}$
Γ_4	$e^{\frac{-\pi i}{3}}$
Γ_5	$e^{\frac{-2\pi i}{3}}$
Γ_6	$e^{\frac{\pi i}{3}}$

No 275.2.608 : ${}^{2_z}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel 3^{+}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3^*	$\sigma_x\mathcal{K}$

No 275.3.609 : ${}^{2_z}\bar{3}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$
$\left[2_{010}T \parallel -1\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel 3^{+}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3^*	$\sigma_x\mathcal{K}$

No 275.4.610 : ${}^{2_z}\bar{3}^{\infty_x}1 \rightarrow {}^{2_z}\bar{3}^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 3^{+}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_3	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_2	$\alpha \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_3	$\alpha^* \mathcal{K}$	$\sigma_z \mathcal{K}$

No 276.1.611 : $^{m}\bar{3}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{-}_{001}\right]$	$\left[2_{001}T \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1^*	$\alpha\mathcal{K}$

No 276.2.612 : ${}^{m_z}\bar{3}^{\infty_x}1 \to {}^{m}\bar{3}$

No 277.1.613 : $^{\bar{1}}\bar{3}$

Full symmetry operations

$$\left[E \parallel 1\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1^*	$\alpha\mathcal{K}$

No 277.2.614 : ${}^{\bar{1}}\bar{3}^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{3}^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_7 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 3^{-}_{001}\right]$ $C_8 : \left[\bar{E} \parallel 3^{-}_{001}\right]$
$C_3 : \left[E \parallel 3^{+}_{001}\right]$ $C_9 : \left[\bar{E} \parallel 3^{+}_{001}\right]$
$C_4 : \left[\Sigma_z \parallel 1\right]$ $C_{10} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_5 : \left[\Sigma_z \parallel 3^{-}_{001}\right]$ $C_{11} : \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
$C_6 : \left[\Sigma_z \parallel 3^{+}_{001}\right]$ $C_{12} : \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	−1	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	0	$\sqrt{3}i$	$-\sqrt{3}i$
χ_3^{co}	2	−1	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	0	$-\sqrt{3}i$	$\sqrt{3}i$

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[T \parallel \bar{3}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1^*	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$

No 277.3.615 : ${}^{\bar{1}}\bar{3}^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{3}^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha & 0 \\ 0 & \alpha^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha \\ -\beta & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 278.1.616 : ${}^{3}\bar{3}$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[3^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right] \left[3^{-}_{001} \mid\mid 3^{-}_{001}\right] \left[E \mid\mid -1\right] \left[3^{+}_{001} \mid\mid 3^{+}_{001}\right] \left[3^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Irrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{2\pi i}{3}}$
Γ_4	$e^{\frac{-\pi i}{3}}$
Γ_5	$e^{\frac{-2\pi i}{3}}$
Γ_6	$e^{\frac{\pi i}{3}}$

No 278.2.617 : ${}^{3_z}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[6^{-}_{001}T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid 3^{-}_{001}\right]$
$\left[2_{001}T \mid\mid -1\right]$ $\left[6^{-}_{001}T \mid\mid 3^{+}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3^*	$\sigma_x\mathcal{K}$

No 279.1.618 : ${}^{6}\bar{3}$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right] \left[3^{+}_{001} \mid\mid 3^{-}_{001}\right] \left[2_{001} \mid\mid -1\right] \left[3^{-}_{001} \mid\mid 3^{+}_{001}\right] \left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Irrepresntations of group generators

	$\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{2\pi i}{3}}$
Γ_4	$e^{\frac{-\pi i}{3}}$
Γ_5	$e^{\frac{-2\pi i}{3}}$
Γ_6	$e^{\frac{\pi i}{3}}$

No 279.2.619 : ${}^{6_z}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[3^-_{001}T \mid\mid \bar{3}^+_{001}\right]$ $\left[6^-_{001}T \mid\mid 3^-_{001}\right]$
$\left[T \mid\mid -1\right]$ $\left[6^+_{001}T \mid\mid 3^+_{001}\right]$ $\left[3^+_{001}T \mid\mid \bar{3}^-_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_6 : $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3^*	$\sigma_x\mathcal{K}$

No 280.1.620 : ${}^{\bar{3}}\bar{3}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001}T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[3^{-}_{001}T \mid\mid \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$	$\left[3^{+}_{001}T \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1^*	$\alpha\mathcal{K}$

No 281.1.621 : ${}^{\bar{6}}\bar{3}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^{+}_{001}T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[6^{-}_{001}T \mid\mid \bar{3}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$	$\left[6^{+}_{001}T \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1^*	$\alpha\mathcal{K}$

No 282.1.622 : ${}^{1}3_{1}{}^{1}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

Irrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	-1
Γ_3	ω_1	ω_2^*

No 282.2.623 : ${}^{1}3_{1}{}^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$
$\left[2_{001}T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 282.3.624 : ${}^{1}3_{1}{}^{1}2^{\infty_z}1 \to {}^{1}3_{1}{}^{1}2$

No 282.4.625 : ${}^{1}3_{1}{}^{1}2^{\infty}{}_{z}{}^{m}{}_{x}1 \rightarrow {}^{1}3_{1}{}^{1}2^{m}{}_{x}1$

No 283.1.626 : ${}^{1}3_{1}{}^{2}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

Irrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	-1
Γ_3	ω_1	ω_2^*

No 283.2.627 : ${}^{1}3_{1}{}^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$
$\left[T \parallel 2_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 283.3.628 : ${}^{1}3_{1}{}^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$
$\left[2_{010}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 283.4.629 : ${}^{1}3_{1}{}^{2_z}2^{\infty_x}1 \rightarrow {}^{1}3_{1}{}^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	2	0	0	0	−2	−2
χ_2	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1
χ_3	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1

Irrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 284.1.630 : ${}^{1}3_{1}{}^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$

No 284.2.631 : ${}^{1}3_{1}{}^{m_z}2^{\infty_x}1 \rightarrow {}^{1}3_{1}{}^{m}2$

No 285.1.632 : ${}^{1}3_{1}{}^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$

No 285.2.633 : ${}^{1}3_{1}{}^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{1}3_{1}{}^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_7 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 3^{+}_{001}\right]$ $C_8 : \left[\bar{E} \parallel 3^{+}_{001}\right]$
$C_3 : \left[E \parallel 3^{-}_{001}\right]$ $C_9 : \left[\bar{E} \parallel 3^{-}_{001}\right]$
$C_4 : \left[\Sigma_z \parallel 1\right]$ $C_{10} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_5 : \left[\Sigma_z \parallel 3^{+}_{001}\right]$ $C_{11} : \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
$C_6 : \left[\Sigma_z \parallel 3^{-}_{001}\right]$ $C_{12} : \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 285.3.634 : ${}^{1}3_{1}{}^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{1}3_{1}{}^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right]$
$\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 286.1.635 : ${}^{3_z}3_1{}^{2_x}2$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[3^+_{001} \mid\mid 3^+_{001}\right] \left[3^-_{001} \mid\mid 3^-_{001}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[2_{110} \mid\mid 2_{110}\right] \left[2_{010} \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right] \left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{100}\right] \left[2_{110} \mid\mid 2_{110}\right] \left[2_{010} \mid\mid 2_{010}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

Irrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$
Γ_1	1	1
Γ_2	1	-1
Γ_3	ω_1	ω_2^*

No 286.2.636 : ${}^{3_z}3_1{}^{2_x}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[6^-_{001}T \parallel 3^+_{001}\right]$ $\left[6^+_{001}T \parallel 3^-_{001}\right]$
$\left[2_{120}T \parallel 2_{100}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{110}\right]$ $\left[2_{210}T \parallel 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 3^+_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 287.1.637 : ${}^{3_z}3_1{}^{m_x}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$

No 288.1.638 : ${}^{1}3^{1}m_{1}$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{010}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

Irrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	1	-1
Γ_3	ω_1	σ_x

No 288.2.639 : ${}^{1}3^{1}m_{1}{}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 288.3.640 : ${}^{1}3^{1}m_{1}{}^{\infty_{z}}1 \rightarrow {}^{1}3^{1}m_{1}$

No 288.4.641 : ${}^{1}3^{1}m_{1}{}^{\infty_{z}m_{x}}1 \rightarrow {}^{1}3^{1}m_{1}{}^{m_{x}}1$

No 289.1.642 : ${}^{1}3^{2}m_{1}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

Irrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	1	-1
Γ_3	ω_1	σ_x

No 289.2.643 : ${}^{1}3^{2_z}m_1{}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 3^{-}_{001}\right]$
$\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 289.3.644 : ${}^{1}3^{2_z}m_1{}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 289.4.645 : ${}^{1}3^{2_z}m_1{}^{\infty_x}1 \rightarrow {}^{1}3^{2_z}m_1{}^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right]$
$\left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	2	0	0	0	−2	−2
χ_2	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1
χ_3	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1

Irrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 290.1.646 : ${}^{1}3^{m}m_{1}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$

No 290.2.647 : ${}^{1}3^{m_z}m_1{}^{\infty_x}1 \to {}^{1}3^{m}m_1$

No 291.1.648 : ${}^{1}3^{\bar{1}}m_{1}$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \parallel 1\right]$
$C_2 : \quad \left[E \parallel 3^{+}_{001}\right]$
$C_3 : \quad \left[E \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$

No 291.2.649 : ${}^{1}3^{\bar{1}}m_1{}^{\infty_z}1 \rightarrow {}^{1}3^{\bar{1}}m_1{}^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right]$ $\left[\Sigma_z T \mid\mid m_{110}\right]$ $\left[\Sigma_z T \mid\mid m_{010}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 3^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 3^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid m_{100}\right]$ $\left[\bar{E}T \mid\mid m_{110}\right]$ $\left[\bar{E}T \mid\mid m_{010}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_7 :	$\left[\bar{E} \mid\mid 1\right]$
C_2 :	$\left[E \mid\mid 3^{+}_{001}\right]$	C_8 :	$\left[\bar{E} \mid\mid 3^{+}_{001}\right]$
C_3 :	$\left[E \mid\mid 3^{-}_{001}\right]$	C_9 :	$\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_4 :	$\left[\Sigma_z \mid\mid 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_5 :	$\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_6 :	$\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	-2	-2	-2	0	0	0
χ_2^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 291.3.650 : ${}^{1}3^{\bar{1}}m_1{}^{\infty_z m_x}1 \rightarrow {}^{1}3^{\bar{1}}m_1{}^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 292.1.651 : ${}^{3_z}3^{2_x}m_1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[3^+_{001} \mid\mid 3^+_{001}\right] \left[3^-_{001} \mid\mid 3^-_{001}\right] \left[2_{100} \mid\mid m_{100}\right] \left[2_{110} \mid\mid m_{110}\right] \left[2_{010} \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right] \left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right] \left[2_{110} \mid\mid m_{110}\right] \left[2_{010} \mid\mid m_{010}\right]$

Character table

	C_1	C_2	C_3
χ_1	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

Irrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	1	-1
Γ_3	ω_1	σ_x

No 292.2.652 : ${}^{3_z}3^{2_x}m_1{}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[6^-_{001}T \mid\mid 3^+_{001}\right]$ $\left[6^+_{001}T \mid\mid 3^-_{001}\right]$
$\left[2_{120}T \mid\mid m_{100}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{110}\right]$ $\left[2_{210}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 293.1.653 : ${}^{3_z}3^{m_x}m_1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$

No 294.1.654 : ${}^{1}\bar{3}_{1}{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_6 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 294.2.655 : ${}^{1}\bar{3}_{1}{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[E \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_6 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 294.3.656 : ${}^{1}\bar{3}_{1}{}^{1}m^{\infty_z}1 \rightarrow {}^{1}\bar{3}_{1}{}^{1}m$

No 294.4.657 : ${}^{1}\bar{3}_{1}{}^{1}m^{\infty_z m_x}1 \rightarrow {}^{1}\bar{3}_{1}{}^{1}m^{m_x}1$

No 295.1.658 : ${}^{1}\bar{3}_{1}{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid \bar{3}^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid -1\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid \bar{3}^{-}_{001}\right] \left[2_{001} \mid\mid m_{100}\right] \left[2_{001} \mid\mid 2_{110}\right] \left[2_{001} \mid\mid 2_{010}\right]$
$\left[2_{001} \mid\mid m_{010}\right] \left[2_{001} \mid\mid m_{110}\right] \left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{3}^{+}_{001}\right] \left[E \mid\mid \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[2_{001} \mid\mid m_{100}\right] \left[2_{001} \mid\mid m_{010}\right] \left[2_{001} \mid\mid m_{110}\right]$
C_6 : $\left[2_{001} \mid\mid 2_{110}\right] \left[2_{001} \mid\mid 2_{010}\right] \left[2_{001} \mid\mid 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 295.2.659 : ${}^{1}\bar{3}_{1}{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$
$\left[T \parallel m_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 295.3.660 : ${}^{1}\bar{3}_{1}{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel \bar{3}^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 295.4.661 : ${}^{1}\bar{3}_{1}{}^{2_z}m^{\infty_x}1 \rightarrow {}^{1}\bar{3}_{1}{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2
χ_4	2	1	−1	−2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	−2	−1	1	2
χ_5	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2

χ_6	2	1	-1	-2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	-2	-1	1	2

Irrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	ω_3	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 296.1.662 : ${}^{1}\bar{3}_{1}{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$
$\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[E \mid\mid \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 296.2.663 : ${}^{1}\bar{3}_{1}{}^{m_z}m^{\infty_x}1 \to {}^{1}\bar{3}_{1}{}^{m}m$

No 297.1.664 : ${}^{1}\bar{3}_{1}{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel 3^{+}_{001}\right]$
C_6 : $\left[E \parallel \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 297.2.665 : ${}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[\Sigma_z \parallel 1\right]$ $\left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$ $\left[\Sigma_z \parallel 3^{-}_{001}\right]$ $\left[\Sigma_z \parallel -1\right]$ $\left[\Sigma_z \parallel 3^{+}_{001}\right]$ $\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right]$ $\left[\Sigma_z T \parallel 2_{110}\right]$ $\left[\Sigma_z T \parallel 2_{010}\right]$ $\left[\Sigma_z T \parallel m_{010}\right]$ $\left[\Sigma_z T \parallel m_{110}\right]$ $\left[\Sigma_z T \parallel 2_{100}\right]$ $\left[\bar{E} \parallel 1\right]$ $\left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$ $\left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E} \parallel -1\right]$ $\left[\bar{E} \parallel 3^{+}_{001}\right]$ $\left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$ $\left[\bar{E}T \parallel m_{100}\right]$ $\left[\bar{E}T \parallel 2_{110}\right]$ $\left[\bar{E}T \parallel 2_{010}\right]$ $\left[\bar{E}T \parallel m_{010}\right]$ $\left[\bar{E}T \parallel m_{110}\right]$ $\left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right]$ $\left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel -1\right]$ $\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{100}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{010}\right]$ $\left[\bar{\Sigma}_z T \parallel m_{110}\right]$ $\left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel \bar{3}^{+}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
C_3 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_4 :	$\left[E \parallel -1\right]$	C_{16} :	$\left[\bar{E} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{+}_{001}\right]$	C_{17} :	$\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right]$	C_{18} :	$\left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel -1\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel -1\right]$
C_{11} :	$\left[\Sigma_z \parallel 3^{+}_{001}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_{12} :	$\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	0
χ_3^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	2	-2	2	-2	2	0	0	0	0	0	0
χ_3^{co}	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_7^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_7	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 297.3.666 : ${}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}\bar{3}_{1}{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, \bar{3}^{+}_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[E \,||\, -1\right] \left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, \bar{3}^{-}_{001}\right] \left[T \,||\, m_{100}\right] \left[T \,||\, 2_{110}\right] \left[T \,||\, 2_{010}\right]$
$\left[T \,||\, m_{010}\right] \left[T \,||\, m_{110}\right] \left[T \,||\, 2_{100}\right] \left[2_{100}T \,||\, 1\right] \left[2_{100}T \,||\, \bar{3}^{+}_{001}\right] \left[2_{100}T \,||\, 3^{-}_{001}\right] \left[2_{100}T \,||\, -1\right] \left[2_{100}T \,||\, 3^{+}_{001}\right] \left[2_{100}T \,||\, \bar{3}^{-}_{001}\right]$
$\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, 2_{010}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, \bar{3}^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right]$
$\left[\Sigma_z \,||\, -1\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z \,||\, \bar{3}^{-}_{001}\right] \left[\Sigma_z T \,||\, m_{100}\right] \left[\Sigma_z T \,||\, 2_{110}\right] \left[\Sigma_z T \,||\, 2_{010}\right] \left[\Sigma_z T \,||\, m_{010}\right] \left[\Sigma_z T \,||\, m_{110}\right] \left[\Sigma_z T \,||\, 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \,||\, 1\right] \left[\Sigma_z 2_{100}T \,||\, \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, -1\right] \left[\Sigma_z 2_{100}T \,||\, 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, 2_{110}\right] \left[\Sigma_z 2_{100} \,||\right.$
$\left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E} \,||\, \bar{3}^{+}_{001}\right] \left[\bar{E} \,||\, 3^{-}_{001}\right] \left[\bar{E} \,||\, -1\right] \left[\bar{E} \,||\, 3^{+}_{001}\right] \left[\bar{E} \,||\, \bar{3}^{-}_{001}\right]$
$\left[\bar{E}T \,||\, m_{100}\right] \left[\bar{E}T \,||\, 2_{110}\right] \left[\bar{E}T \,||\, 2_{010}\right] \left[\bar{E}T \,||\, m_{010}\right] \left[\bar{E}T \,||\, m_{110}\right] \left[\bar{E}T \,||\, 2_{100}\right] \left[\bar{E}2_{100}T \,||\, 1\right] \left[\bar{E}2_{100}T \,||\, \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \,||\, 3^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \,||\, -1\right] \left[\bar{E}2_{100}T \,||\, 3^{+}_{001}\right] \left[\bar{E}2_{100}T \,||\, \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, 2_{110}\right] \left[\bar{E}2_{100} \,||\, 2_{010}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, 2_{100}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, 2_{110}\right] \left[\bar{\Sigma}_z T \,||\, 2_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, m_{010}\right] \left[\bar{\Sigma}_z T \,||\, m_{110}\right] \left[\bar{\Sigma}_z T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, \bar{3}^{+}_{001}\right] \left[E \,||\, \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 3^{-}_{001}\right] \left[E \,||\, 3^{+}_{001}\right]$
C_4 : $\left[E \,||\, -1\right]$
C_5 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, 2_{010}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, -1\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, 2_{110}\right] \left[\Sigma_z 2_{100} \,||\, 2_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\bar{E}2_{100} \,||\, m_{100}\right] \left[\bar{E}2_{100} \,||\, 2_{110}\right]$
$\left[\bar{E}2_{100} \,||\, 2_{010}\right] \left[\bar{E}2_{100} \,||\, m_{010}\right] \left[\bar{E}2_{100} \,||\, m_{110}\right] \left[\bar{E}2_{100} \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$
C_6 : $\left[\Sigma_z \,||\, \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right]$
C_8 : $\left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right]$
C_9 : $\left[\Sigma_z \,||\, \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \,||\, 1\right]$
C_{11} : $\left[\bar{E} \,||\, \bar{3}^{+}_{001}\right] \left[\bar{E} \,||\, \bar{3}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \,||\, 3^{-}_{001}\right] \left[\bar{E} \,||\, 3^{+}_{001}\right]$
C_{13} : $\left[\bar{E} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	2	−2	−4	0	0	0	0	0	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3^* & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 298.1.667 : ${}^{2}\bar{3}_{1}{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_6 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 298.2.668 : ${}^{2_z}\bar{3}_1{}^1m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel \bar{3}^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[2_{001}T \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{110}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^+_{001}\right] \left[2_{001} \parallel \bar{3}^-_{001}\right]$
C_3 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_6 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 298.3.669 : ${}^{2_z}\bar{3}_1{}^1m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel \bar{3}^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel 3^-_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel 3^+_{001}\right] \left[2_{010}T \parallel \bar{3}^-_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^+_{001}\right] \left[2_{001} \parallel \bar{3}^-_{001}\right]$
C_3 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_6 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 298.4.670 : ${}^{2_z}\bar{3}_1{}^1m^{\infty_x}1 \rightarrow {}^{2_z}\bar{3}_1{}^1m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel m_{010}\right]$
$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_3 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{110}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2
χ_3	4	0	−2	0	−4	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 299.1.671 : ${}^{m}\bar{3}_{1}{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{001}T \mid\mid \bar{3}^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[2_{001}T \mid\mid -1\right] \left[E \mid\mid 3^{+}_{001}\right] \left[2_{001}T \mid\mid \bar{3}^{-}_{001}\right] \left[E \mid\mid m_{100}\right] \left[2_{001}T \mid\mid 2_{110}\right] \left[2_{001}T \mid\mid 2_{010}\right]$
$\left[E \mid\mid m_{010}\right] \left[E \mid\mid m_{110}\right] \left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{010}\right] \left[E \mid\mid m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{-}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 299.2.672 : ${}^{m_z}\bar{3}_1{}^1 m^{\infty_x} 1 \to {}^{m}\bar{3}_1{}^1 m$

No 300.1.673 : ${}^{\bar{1}}\bar{3}_1{}^1m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^-_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 300.2.674 : ${}^{\bar{1}}\bar{3}_1{}^1m^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{3}_1{}^1m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z T \parallel \bar{3}^-_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
$\left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}T \parallel \bar{3}^-_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$	C_8 :	$\left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
C_3 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$	C_9 :	$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{110}\right]$
C_4 :	$\left[\Sigma_z \parallel 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_5 :	$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right]$
C_6 :	$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{110}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2	0	0	0
χ_3^{co}	4	−2	0	0	0	0	−4	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^-_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel \bar{3}^+_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 300.3.675 : ${}^{\bar{1}}\bar{3}_1{}^1 m^{\infty_z m_x} 1 \rightarrow {}^{\bar{1}}\bar{3}_1{}^1 m^{\Sigma_z m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right]$
$\left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel 3^-_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel 3^+_{001}\right] \left[2_{100} \parallel \bar{3}^-_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right]$
$\left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z T \parallel \bar{3}^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{01}$
$\left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}T \parallel \bar{3}^-_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100}T \parallel 3^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 3^+_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{110}\right]$
C_4 : $\left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^-_{001}\right]$
$\left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{110}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 301.1.676 : ${}^{2}\bar{3}_{1}{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 301.2.677 : ${}^{2_z}\bar{3}_1{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$
$\left[T \parallel m_{100}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{110}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 301.3.678 : ${}^{2_z}\bar{3}_1{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 301.4.679 : ${}^{2_z}\bar{3}_1{}^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}\bar{3}_1{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2
χ_3	4	0	−2	0	−4	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 302.1.680 : ${}^{m}\bar{3}_{1}{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{110}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[E \parallel 2_{110}\right]$	$\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 302.2.681 : ${}^{m_z}\bar{3}_1{}^{m_z}m^{\infty_x}1 \to {}^{m}\bar{3}_1{}^{m}m$

No 303.1.682 : ${}^{\bar{1}}\bar{3}_1{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right] \left[T \parallel m_{110}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[E \parallel 2_{110}\right]$	$\left[T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 303.2.683 : ${}^{\bar{1}}\bar{3}_1{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{3}_1{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right] \left[T \parallel m_{110}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z T \parallel \bar{3}^-_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
$\left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}T \parallel \bar{3}^-_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$	C_8 :	$\left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
C_3 :	$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$	C_9 :	$\left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$
C_4 :	$\left[\Sigma_z \parallel 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_5 :	$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{100}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2	0	0	0
χ_3^{co}	4	−2	0	0	0	0	−4	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^-_{001}\right]$	$\left[E \parallel 2_{110}\right]$	$\left[T \parallel \bar{3}^+_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 303.3.684 : ${}^{\bar{1}}\bar{3}_1{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{3}_1{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right] \left[T \parallel m_{110}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2\right.$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{100}\right]$
C_4 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{110}\right]$	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 304.1.685 : ${}^{2_x}\bar{3}_1{}^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 304.2.686 : ${}^{2_x}\bar{3}_1{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[T \parallel -1\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 304.3.687 : ${}^{2_x}\bar{3}_1{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{001}T \parallel 3^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{010}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{110}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 304.4.688 : ${}^{2_x}\bar{3}_1{}^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{010}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[2_{010}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$
C_6 : $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 305.1.689 : ${}^{2_z}\bar{3}_1{}^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$
$\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel 3^{+}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 306.1.690 : ${}^{m_x}\bar{3}_1{}^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel m_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$
$\left[2_{010}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[2_{001} \parallel 2_{110}\right]$	$\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 307.1.691 : ${}^{m_x}\bar{3}_1{}^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 308.1.692 : ${}^{2}\bar{3}_{1}{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{010}\right]$
$\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 308.2.693 : ${}^{2_z}\bar{3}_1{}^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}\bar{3}_1{}^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right]$
$\left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{110}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right]$
C_3 : $\left[E \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 3^+_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_3^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^+_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\alpha\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha^*\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 309.1.694 : ${}^{2}\bar{3}_{1}{}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$
$\left[T \mid\mid m_{010}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 309.2.695 : ${}^{2_z}\bar{3}_1{}^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}\bar{3}_1{}^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{100} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{100}T \mid\mid 2_{110}\right]$ $\left[2_{100}T \mid\mid 2_{010}\right]$
$\left[T \mid\mid m_{010}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid m_{010}\right]$ $\left[\Sigma_z T \mid\mid m_{110}\right]$ $\left[\Sigma_z 2_{100}T \mid\mid 2_{100}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E} \mid\mid 3^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid m_{100}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{010}\right]$ $\left[\bar{E}T \mid\mid m_{010}\right]$ $\left[\bar{E}T \mid\mid m_{110}\right]$ $\left[\bar{E}2_{100}T \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{100}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right]$
$\left[\bar{\Sigma}_z T \mid\mid m_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid m_{110}\right]$ $\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[2_{100} \mid\mid -1\right]$ $\left[2_{100} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$ $\left[\Sigma_z 2_{100} \mid\mid -1\right]$ $\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[\bar{E}2_{100} \mid\mid -1\right]$ $\left[\bar{E}2_{100} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right]$
$\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$ $\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_5 : $\left[\bar{E} \mid\mid 1\right]$
C_6 : $\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \mid\mid 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_3^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{3}^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\alpha\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 310.1.696 : ${}^{\bar{1}}\bar{3}_1{}^m m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$
$\left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^-_{001}\right]$	$\left[2_{001} \parallel 2_{110}\right]$	$\left[T \parallel \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 310.2.697 : ${}^{\bar{1}}\bar{3}_1{}^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}\bar{3}_1{}^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid \bar{3}^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[T \mid\mid -1\right] \left[E \mid\mid 3^{+}_{001}\right] \left[T \mid\mid \bar{3}^{-}_{001}\right] \left[2_{100}T \mid\mid m_{100}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{010}\right]$
$\left[2_{100}T \mid\mid m_{010}\right] \left[2_{100}T \mid\mid m_{110}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z T \mid\mid \bar{3}^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z T \mid\mid -1\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\Sigma_z T \mid\mid \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid -1\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E}T \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100}T \mid\mid m_{010}\right] \left[\bar{E}2_{100}T \mid\mid m_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_4 : $\left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_5 : $\left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1
χ_3^{co}	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{-}_{001}\right]$	$\left[2_{100} \mid\mid 2_{110}\right]$	$\left[T \mid\mid \bar{3}^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 311.1.698 : ${}^{\bar{1}}\bar{3}_1{}^2m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^-_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 311.2.699 : ${}^{\bar{1}}\bar{3}_1{}^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}\bar{3}_1{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel -1\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{010}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z T \parallel \bar{3}^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
$\left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}T \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^-_{001}\right] \left[E \parallel 3^+_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
C_4 : $\left[\Sigma_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1
χ_3^{co}	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1

Coirrepresntations of group generators

	$\left[E \parallel 3^-_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel \bar{3}^+_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 312.1.700 : ${}^{m}\bar{3}_{1}{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[2_{001} \parallel 2_{110}\right]$	$\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 312.2.701 : ${}^{m_z}\bar{3}_1{}^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}\bar{3}_1{}^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right]$
$\left[T \parallel m_{010}\right] \left[T \parallel m_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[2_{100} \parallel 2_{110}\right]$	$\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 313.1.702 : ${}^{m}\bar{3}_{1}{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right]$
$\left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 313.2.703 : ${}^{m_z}\bar{3}_1{}^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}\bar{3}_1{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{-}_{001}\right] \left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 314.1.704 : ${}^{3_z}\bar{3}_1{}^{2_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$
$\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^-_{001} \mid\mid \bar{3}^-_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$
C_6 : $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 314.2.705 : ${}^{3_z}\bar{3}_1{}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[3^+_{001} \,||\, \bar{3}^+_{001}\right] \left[3^-_{001} \,||\, 3^-_{001}\right] \left[E \,||\, -1\right] \left[3^+_{001} \,||\, 3^+_{001}\right] \left[3^-_{001} \,||\, \bar{3}^-_{001}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{110} \,||\, 2_{110}\right] \left[2_{010} \,||\, 2_{010}\right]$
$\left[2_{010} \,||\, m_{010}\right] \left[2_{110} \,||\, m_{110}\right] \left[2_{100} \,||\, 2_{100}\right] \left[2_{001}T \,||\, 1\right] \left[6^-_{001}T \,||\, \bar{3}^+_{001}\right] \left[6^+_{001}T \,||\, 3^-_{001}\right] \left[2_{001}T \,||\, -1\right] \left[6^-_{001}T \,||\, 3^+_{001}\right] \left[6^+_{001}T \,||\, \bar{3}^-_{001}\right]$
$\left[2_{120}T \,||\, m_{100}\right] \left[2_{1\bar{1}0}T \,||\, 2_{110}\right] \left[2_{210}T \,||\, 2_{010}\right] \left[2_{210}T \,||\, m_{010}\right] \left[2_{1\bar{1}0}T \,||\, m_{110}\right] \left[2_{120}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[3^+_{001} \,||\, \bar{3}^+_{001}\right] \left[3^-_{001} \,||\, \bar{3}^-_{001}\right]$
C_3 : $\left[3^-_{001} \,||\, 3^-_{001}\right] \left[3^+_{001} \,||\, 3^+_{001}\right]$
C_4 : $\left[E \,||\, -1\right]$
C_5 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{010} \,||\, m_{010}\right] \left[2_{110} \,||\, m_{110}\right]$
C_6 : $\left[2_{110} \,||\, 2_{110}\right] \left[2_{010} \,||\, 2_{010}\right] \left[2_{100} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \,||\, \bar{3}^+_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 315.1.706 : ${}^{3_z}\bar{3}_1{}^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[E \mid\mid -1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_4 : $\left[E \mid\mid -1\right]$
C_5 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_6 : $\left[3^-_{001} \mid\mid \bar{3}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 316.1.707 : ${}^{6_z}\bar{3}_1{}^{2_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{210} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{010}\right]$
$\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{120} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$
C_6 : $\left[2_{210} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{010}\right]$ $\left[2_{120} \mid\mid 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	−1	1
χ_3	1	1	1	1	−1	−1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	−1
Γ_3	1	−1
Γ_4	−1	1
Γ_5	ω_1	ω_2^*
Γ_6	ω_3^*	ω_4

No 316.2.708 : ${}^{6_z}\bar{3}_1{}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{210} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{010}\right]$
$\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{120} \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[3^-_{001}T \mid\mid \bar{3}^+_{001}\right]$ $\left[6^-_{001}T \mid\mid 3^-_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[6^+_{001}T \mid\mid 3^+_{001}\right]$ $\left[3^+_{001}T \mid\mid \bar{3}^-_{001}\right]$
$\left[2_{120}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{110}T \mid\mid 2_{010}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{010}\right]$ $\left[2_{210}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$
C_6 : $\left[2_{210} \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{010}\right]$ $\left[2_{120} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 317.1.709 : ${}^{\bar{3}_z}\bar{3}_1{}^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001}T \mid\mid \bar{3}^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001}T \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_3 : $\left[2_{110} \mid\mid 2_{110}\right]$ $\left[2_{010} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^-_{001} \mid\mid 3^-_{001}\right]$	$\left[2_{110} \mid\mid 2_{110}\right]$	$\left[3^+_{001}T \mid\mid \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 318.1.710 : ${}^{\bar{3}_z}\bar{3}_1{}^{2_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001}T \mid\mid \bar{3}^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[T \mid\mid -1\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001}T \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[3^-_{001} \mid\mid 3^-_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[3^+_{001}T \mid\mid \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 319.1.711 : ${}^{6_z}\bar{3}_1{}^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{001} \mid\mid -1\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{210}T \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{010}\right]$
$\left[2_{110}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{120}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_6 : $\left[6^-_{001} \mid\mid \bar{3}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid \bar{3}^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$

No 320.1.712 : ${}^{\bar{6}_z}\bar{3}_1{}^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001}T \parallel \bar{3}^+_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[6^-_{001}T \parallel \bar{3}^-_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{210} \parallel 2_{110}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{010}\right]$ $\left[2_{110}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{120} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^-_{001}\right]$ $\left[3^-_{001} \parallel 3^+_{001}\right]$
C_3 : $\left[2_{210} \parallel 2_{110}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{010}\right]$ $\left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 3^-_{001}\right]$	$\left[2_{210} \parallel 2_{110}\right]$	$\left[6^+_{001}T \parallel \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 321.1.713 : ${}^{\bar{6}_z}\bar{3}_1{}^{2_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001}T \mid\mid \bar{3}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[2_{001}T \mid\mid -1\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[6^-_{001}T \mid\mid \bar{3}^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{210}T \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{010}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{120}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^-_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[6^+_{001}T \mid\mid \bar{3}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	σ_x	$\sigma_0\mathcal{K}$

No 322.1.714 : $^{1}6$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 322.2.715 : ${}^{1}6^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 6^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 3^{+}_{001}\right]$
$\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[2_{001}T \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 322.3.716 : ${}^{1}6^{\infty_z}1 \to {}^{1}6$

No 322.4.717 : ${}^{1}6^{\infty_z m_x}1 \to {}^{1}6^{m_x}1$

No 323.1.718 : $^{2}6$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid 6^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{001} \mid\mid 6^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 323.2.719 : ${}^{2_z}6^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 323.3.720 : ${}^{2_z}6^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{010}T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 323.4.721 : ${}^{2_z}6^{\infty_x}1 \rightarrow {}^{2_z}6^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_3	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_2	$\alpha^* \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_3	$\alpha \mathcal{K}$	$\sigma_z \mathcal{K}$

No 324.1.722 : $^{m}6$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 324.2.723 : ${}^{m_z}6^{\infty_x}1 \to {}^{m}6$

No 325.1.724 : $^{\bar{1}}6$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 325.2.725 : ${}^{\bar{1}}6^{\infty_z}1 \rightarrow {}^{\bar{1}}6^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_7 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 3^{+}_{001}\right]$ $C_8 : \left[\bar{E} \parallel 3^{+}_{001}\right]$
$C_3 : \left[E \parallel 3^{-}_{001}\right]$ $C_9 : \left[\bar{E} \parallel 3^{-}_{001}\right]$
$C_4 : \left[\Sigma_z \parallel 1\right]$ $C_{10} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_5 : \left[\Sigma_z \parallel 3^{+}_{001}\right]$ $C_{11} : \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
$C_6 : \left[\Sigma_z \parallel 3^{-}_{001}\right]$ $C_{12} : \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	−1	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_3^{co}	2	−1	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	0	$\sqrt{3}i$	$-\sqrt{3}i$

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$

No 325.3.726 : ${}^{\bar{1}}6^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha^* \\ -\beta^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 326.1.727 : $^{3}6$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 326.2.728 : ${}^{3_z}6^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[6^-_{001}T \mid\mid 6^+_{001}\right]$ $\left[6^+_{001}T \mid\mid 3^+_{001}\right]$
$\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[6^-_{001}T \mid\mid 3^-_{001}\right]$ $\left[6^+_{001}T \mid\mid 6^-_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 6^+_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[3^-_{001} \mid\mid 6^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 327.1.729 : $^{6}6$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^{+}_{001} \mid\mid 6^{+}_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[6^{-}_{001} \mid\mid 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^{+}_{001} \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[6^{-}_{001} \mid\mid 6^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[6^{+}_{001} \mid\mid 6^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 327.2.730 : ${}^{6_z}6^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[3^-_{001}T \mid\mid 6^+_{001}\right]$ $\left[6^-_{001}T \mid\mid 3^+_{001}\right]$
$\left[T \mid\mid 2_{001}\right]$ $\left[6^+_{001}T \mid\mid 3^-_{001}\right]$ $\left[3^+_{001}T \mid\mid 6^-_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid 6^+_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[6^-_{001} \mid\mid 6^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 328.1.731 : $^{\bar{3}}6$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$	$\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 329.1.732 : $^{\bar{6}}6$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^{+}_{001}T \parallel 6^{+}_{001}\right] \left[3^{+}_{001} \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[3^{-}_{001} \parallel 3^{-}_{001}\right] \left[6^{-}_{001}T \parallel 6^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$	$\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 330.1.733 : ${}^{1}\bar{6}$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$
$C_2 : \left[E \parallel \bar{6}^{+}_{001}\right]$
$C_3 : \left[E \parallel 3^{+}_{001}\right]$
$C_4 : \left[E \parallel m_{001}\right]$
$C_5 : \left[E \parallel 3^{-}_{001}\right]$
$C_6 : \left[E \parallel \bar{6}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 330.2.734 : ${}^{1}\bar{6}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 3^{+}_{001}\right]$
$\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 330.3.735 : ${}^{1}\bar{6}^{\infty_z}1 \to {}^{1}\bar{6}$

No 330.4.736 : ${}^{1}\bar{6}^{\infty_z m_x}1 \to {}^{1}\bar{6}^{m_x}1$

No 331.1.737 : ${}^{2}\bar{6}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 331.2.738 : ${}^{2_z}\bar{6}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 331.3.739 : ${}^{2_z}\bar{6}^{m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 1\right]$ $\left[2_{010}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 3^{+}_{001}\right]$
$\left[2_{010}T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid 3^{-}_{001}\right]$ $\left[2_{010}T \mid\mid \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 331.4.740 : ${}^{2_z}\bar{6}^{\infty_x}1 \rightarrow {}^{2_z}\bar{6}^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_3	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_2	$\alpha^* \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_3	$\alpha \mathcal{K}$	$\sigma_z \mathcal{K}$

No 332.1.741 : ${}^{m}\bar{6}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001}T \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 332.2.742 : ${}^{m_z}\bar{6}^{\infty_x}1 \to {}^{m}\bar{6}$

No 333.1.743 : ${}^{\bar{1}}\bar{6}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 333.2.744 : ${}^{\bar{1}}\bar{6}^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{6}^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_7 : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 3^{+}_{001}\right]$ $C_8 : \left[\bar{E} \parallel 3^{+}_{001}\right]$
$C_3 : \left[E \parallel 3^{-}_{001}\right]$ $C_9 : \left[\bar{E} \parallel 3^{-}_{001}\right]$
$C_4 : \left[\Sigma_z \parallel 1\right]$ $C_{10} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_5 : \left[\Sigma_z \parallel 3^{+}_{001}\right]$ $C_{11} : \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
$C_6 : \left[\Sigma_z \parallel 3^{-}_{001}\right]$ $C_{12} : \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	−1	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_3^{co}	2	−1	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	0	$\sqrt{3}i$	$-\sqrt{3}i$

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[T \parallel \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$

No 333.3.745 : ${}^{\bar{1}}\bar{6}^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{6}^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[T \,||\, \bar{6}^{+}_{001}\right] \left[E \,||\, 3^{+}_{001}\right] \left[T \,||\, m_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[T \,||\, \bar{6}^{-}_{001}\right] \left[2_{100}T \,||\, 1\right] \left[2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[2_{100}T \,||\, 3^{+}_{001}\right]$
$\left[2_{100} \,||\, m_{001}\right] \left[2_{100}T \,||\, 3^{-}_{001}\right] \left[2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z T \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z T \,||\, m_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\Sigma_z T \,||\, \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \,||\, 1\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}T \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E} \,||\, 3^{+}_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right] \left[\bar{E} \,||\, 3^{-}_{001}\right] \left[\bar{E}T \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \,||\, 1\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \,||\, 3^{+}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}2_{100}T \,||\, 3^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z T \,||\, \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \,||\, m_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 3^{+}_{001}\right]$
C_3 : $\left[E \,||\, 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{-}_{001}\right]$
C_5 : $\left[\bar{E} \,||\, 1\right]$
C_6 : $\left[\bar{E} \,||\, 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \,||\, 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \,||\, \bar{6}^{+}_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha^* \\ -\beta^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 334.1.746 : ${}^{3}\bar{6}$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid m_{001}\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 334.2.747 : ${}^{3_z}\bar{6}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[6^{-}_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid 3^{+}_{001}\right]$
$\left[2_{001}T \mid\mid m_{001}\right]$ $\left[6^{-}_{001}T \mid\mid 3^{-}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid m_{001}\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 335.1.748 : ${}^{6}\bar{6}$

Full symmetry operations

$$\left[E \mid\mid 1\right] \left[6^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right] \left[3^{+}_{001} \mid\mid 3^{+}_{001}\right] \left[2_{001} \mid\mid m_{001}\right] \left[3^{-}_{001} \mid\mid 3^{-}_{001}\right] \left[6^{-}_{001} \mid\mid \bar{6}^{-}_{001}\right]$$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[3^{-}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[6^{-}_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	-1	1	-1	1	-1
χ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[6^{+}_{001} \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1	1
Γ_2	-1
Γ_3	$e^{\frac{-2\pi i}{3}}$
Γ_4	$e^{\frac{\pi i}{3}}$
Γ_5	$e^{\frac{2\pi i}{3}}$
Γ_6	$e^{\frac{-\pi i}{3}}$

No 335.2.749 : ${}^{6_z}\bar{6}^{m_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid \bar{6}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001} \mid\mid \bar{6}^-_{001}\right]$ $\left[2_{001}T \mid\mid 1\right]$ $\left[3^-_{001}T \mid\mid \bar{6}^+_{001}\right]$ $\left[6^-_{001}T \mid\mid 3^+_{001}\right]$
$\left[T \mid\mid m_{001}\right]$ $\left[6^+_{001}T \mid\mid 3^-_{001}\right]$ $\left[3^+_{001}T \mid\mid \bar{6}^-_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid \bar{6}^+_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[6^-_{001} \mid\mid \bar{6}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid \bar{6}^+_{001}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 336.1.750 : ${}^{\bar{3}}\bar{6}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$	$\left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 337.1.751 : ${}^{\bar{6}}\bar{6}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	2	-1	-1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$	$\left[6^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	ω_1	$\alpha^{*}\mathcal{K}$

No 338.1.752 : ${}^{1}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_8 :	$\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[E \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 338.2.753 : ${}^{1}6/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_8 :	$\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 338.3.754 : ${}^{1}6/{}^{1}m^{\infty_z}1 \rightarrow {}^{1}6/{}^{1}m$

No 338.4.755 : ${}^{1}6/{}^{1}m^{\infty_z m_x}1 \to {}^{1}6/{}^{1}m^{m_x}1$

No 339.1.756 : ${}^{2}6/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[E \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 339.2.757 : ${}^{2_z}6/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$
$\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_7 : $\left[2_{001} \parallel m_{001}\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$ C_8 : $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ C_9 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$ C_{10} : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$ C_{11} : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$ C_{12} : $\left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 339.3.758 : ${}^{2_z}6/{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[2_{001} \,||\, 6^{+}_{001}\right]$ $\left[E \,||\, 3^{+}_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[E \,||\, 3^{-}_{001}\right]$ $\left[2_{001} \,||\, 6^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[E \,||\, \bar{3}^{-}_{001}\right]$ $\left[2_{001} \,||\, \bar{6}^{-}_{001}\right]$
$\left[E \,||\, -1\right]$ $\left[2_{001} \,||\, \bar{6}^{+}_{001}\right]$ $\left[E \,||\, \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \,||\, 1\right]$ $\left[2_{010}T \,||\, 6^{+}_{001}\right]$ $\left[2_{100}T \,||\, 3^{+}_{001}\right]$ $\left[2_{010}T \,||\, 2_{001}\right]$ $\left[2_{100}T \,||\, 3^{-}_{001}\right]$ $\left[2_{010}T \,||\, 6^{-}_{001}\right]$
$\left[2_{010}T \,||\, m_{001}\right]$ $\left[2_{100}T \,||\, \bar{3}^{-}_{001}\right]$ $\left[2_{010}T \,||\, \bar{6}^{-}_{001}\right]$ $\left[2_{100}T \,||\, -1\right]$ $\left[2_{010}T \,||\, \bar{6}^{+}_{001}\right]$ $\left[2_{100}T \,||\, \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_7 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[2_{001} \,||\, 6^{+}_{001}\right]$	C_8 :	$\left[E \,||\, \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \,||\, 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \,||\, \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_{10} :	$\left[E \,||\, -1\right]$
C_5 :	$\left[E \,||\, 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \,||\, \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \,||\, 6^{-}_{001}\right]$	C_{12} :	$\left[E \,||\, \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, 6^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 339.4.759 : ${}^{2_z}6/{}^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_6 : $\left[E \parallel -1\right]$
C_7 : $\left[E \parallel \bar{3}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel -1\right]$
C_{13} : $\left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	0	2	2	2	2	2	−2	−2	−2	−2	−2	−2
χ_2	2	0	2	2	−2	−2	−2	−2	−2	−2	2	2	2
χ_3	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$
χ_4	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$

χ_5	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$
χ_6	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\alpha^*\mathcal{K}$	$\omega_2^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$\alpha^*\mathcal{K}$	$\omega_4\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	$\alpha\mathcal{K}$	$\omega_2\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	$\alpha\mathcal{K}$	$\omega_4^*\mathcal{K}$	$\sigma_z\mathcal{K}$

No 340.1.760 : ${}^{m}6/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[E \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid \bar{3}^{-}_{001}\right]$
C_5 : $\left[E \mid\mid -1\right]$
C_6 : $\left[E \mid\mid \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{-}_{001}\right]$	$\left[2_{001}T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 340.2.761 : ${}^{m_z}6/{}^{m_z}m^{\infty_x}1 \to {}^{m}6/{}^{m}m$

No 341.1.762 : ${}^{\bar{1}}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 341.2.763 : ${}^{\bar{1}}6/{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}6/{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_{13} : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 3^{+}_{001}\right]$ $C_{14} : \left[\bar{E} \parallel 3^{+}_{001}\right]$
$C_3 : \left[E \parallel 3^{-}_{001}\right]$ $C_{15} : \left[\bar{E} \parallel 3^{-}_{001}\right]$
$C_4 : \left[E \parallel \bar{3}^{-}_{001}\right]$ $C_{16} : \left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$
$C_5 : \left[E \parallel -1\right]$ $C_{17} : \left[\bar{E} \parallel -1\right]$
$C_6 : \left[E \parallel \bar{3}^{+}_{001}\right]$ $C_{18} : \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
$C_7 : \left[\Sigma_z \parallel 1\right]$ $C_{19} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_8 : \left[\Sigma_z \parallel 3^{+}_{001}\right]$ $C_{20} : \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
$C_9 : \left[\Sigma_z \parallel 3^{-}_{001}\right]$ $C_{21} : \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
$C_{10} : \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right]$ $C_{22} : \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$
$C_{11} : \left[\Sigma_z \parallel -1\right]$ $C_{23} : \left[\bar{\Sigma}_z \parallel -1\right]$
$C_{12} : \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$ $C_{24} : \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	2	2	−2	−2	−2	0	0	0	0	0	0
χ_3^{co}	2	−1	−1	−1	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$
χ_4^{co}	2	−1	−1	1	−2	1	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$
χ_5^{co}	2	−1	−1	−1	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$
χ_6^{co}	2	−1	−1	1	−2	1	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	-2	-2	2	2	2	0	0	0	0	0	0
χ_3^{co}	-2	1	1	1	-2	1	0	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$
χ_4^{co}	-2	1	1	-1	2	-1	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$
χ_5^{co}	-2	1	1	1	-2	1	0	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$
χ_6^{co}	-2	1	1	-1	2	-1	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{-}_{001}\right]$	$\left[T \mid\mid 6^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3^*	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$

No 341.3.764 : ${}^{\bar{1}}6/{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6/{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right]$
$\left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}\right.$
$\left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel -1\right]$
C_{13} : $\left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	−2	−2	−2	0	−2	−2	−2	2	2	2
χ_3^{co}	4	−2	−2	−2	4	−2	0	−4	2	2	2	−4	2
χ_4^{co}	4	−2	−2	2	−4	2	0	−4	2	2	−2	4	−2

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{-}_{001}\right]$	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_5^* & 0 \\ 0 & \omega_5 \end{pmatrix}$	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha^* \\ -\beta^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_7 & 0 \\ 0 & \omega_7^* \end{pmatrix}$	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha^* \\ -\beta^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 342.1.765 : ${}^{2}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 342.2.766 : ${}^{2_z}6/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 342.3.767 : ${}^{2_z}6/{}^{1}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{010}T \parallel 6^{-}_{001}\right]$
$\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 342.4.768 : ${}^{2_z}6/{}^{1}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_7 : $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel m_{001}\right]$
C_{12} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	0	2	2	2	2	2	−2	−2	−2	−2	−2	−2
χ_2	2	0	2	2	−2	−2	−2	−2	−2	−2	2	2	2
χ_3	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_4	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$

χ_5	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_6	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_3	$\alpha^*\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_4	$\alpha^*\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_5	$\alpha\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_6	$\alpha\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$

No 343.1.769 : ${}^{m}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[E \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3^*	$\alpha^*\mathcal{K}$

No 343.2.770 : ${}^{m_z}6/{}^{1}m^{\infty_x}1 \to {}^{m}6/{}^{1}m$

No 344.1.771 : ${}^{\bar{1}}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[E \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^{*}\mathcal{K}$
Γ_4^{co}	ω_3^{*}	$\alpha^{*}\mathcal{K}$

No 344.2.772 : ${}^{\bar{1}}6/{}^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}6/{}^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_3 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_{16} :	$\left[\bar{E} \parallel m_{001}\right]$
C_5 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$	C_{17} :	$\left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_6 :	$\left[E \parallel \bar{6}^{+}_{001}\right]$	C_{18} :	$\left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel m_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel m_{001}\right]$
C_{11} :	$\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
C_{12} :	$\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	2	2	−2	−2	−2	0	0	0	0	0	0
χ_3^{co}	2	−1	−1	2	−1	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$	$-\sqrt{3}i$
χ_4^{co}	2	−1	−1	−2	1	1	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_5^{co}	2	−1	−1	2	−1	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_6^{co}	2	−1	−1	−2	1	1	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	$\sqrt{3}i$	$-\sqrt{3}i$

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	-2	-2	2	2	2	0	0	0	0	0	0
χ_3^{co}	-2	1	1	-2	1	1	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_4^{co}	-2	1	1	2	-1	-1	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	$\sqrt{3}i$	$-\sqrt{3}i$
χ_5^{co}	-2	1	1	-2	1	1	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$	$-\sqrt{3}i$
χ_6^{co}	-2	1	1	2	-1	-1	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	$-\sqrt{3}i$	$\sqrt{3}i$

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{6}_{001}^{-}\right]$	$\left[T \mid\mid 6_{001}^{+}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3^*	$\alpha^*\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1^*	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3	$\alpha\mathcal{K}$	$\sigma_z\mathcal{K}$

No 344.3.773 : ${}^{\bar{1}}6/{}^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6/{}^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right]$
$\left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel\right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel m_{001}\right]$
C_{12} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	−2	−2	−2	0	−2	−2	−2	2	2	2
χ_3^{co}	4	−2	−2	4	−2	−2	0	−4	2	2	−4	2	2
χ_4^{co}	4	−2	−2	−4	2	2	0	−4	2	2	4	−2	−2

Coirrepresntations of group generators

	$\left[E \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha^* \\ -\beta^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \alpha^* \\ -\beta^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 345.1.774 : ${}^{1}6/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001} \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$ C_7 : $\left[2_{001} \mid\mid m_{001}\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$ C_8 : $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$ C_9 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$ C_{10} : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$ C_{11} : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$ C_{12} : $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 345.2.775 : ${}^{1}6/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 345.3.776 : ${}^{1}6/{}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel 6^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{010}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 345.4.777 : ${}^{1}6/{}^{2_z}m^{\infty_x}1 \rightarrow {}^{1}6/{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$
C_7 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel 2_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 6^{-}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2	2	−2	2	−2	2	−2	0	−2	2	−2	2	−2	2
χ_3	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_4	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$

χ_5	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_6	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$

Irrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	ω_7	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	ω_7^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 346.1.778 : ${}^{1}6/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[2_{001}T \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 346.2.779 : ${}^{1}6/{}^{m_z}m^{\infty_x}1 \rightarrow {}^{1}6/{}^{m}m$

No 347.1.780 : ${}^{1}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 347.2.781 : ${}^{1}6/{}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}6/{}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel 6^{+}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{16} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{17} :	$\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{18} :	$\left[\bar{E} \parallel 6^{-}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel 6^{+}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_{11} :	$\left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{12} :	$\left[\Sigma_z \parallel 6^{-}_{001}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	0
χ_3^{co}	2	−1	−1	2	−1	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_4^{co}	2	1	−1	−2	−1	1	0	$\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$	$-\sqrt{3}i$
χ_5^{co}	2	−1	−1	2	−1	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$	$-\sqrt{3}i$
χ_6^{co}	2	1	−1	−2	−1	1	0	$-\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$	$\sqrt{3}i$

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	2	-2	2	-2	2	0	0	0	0	0	0
χ_3^{co}	-2	1	1	-2	1	1	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$	$-\sqrt{3}i$
χ_4^{co}	-2	-1	1	2	1	-1	0	$-\sqrt{3}i$	$-\sqrt{3}i$	0	$\sqrt{3}i$	$\sqrt{3}i$
χ_5^{co}	-2	1	1	-2	1	1	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$	$\sqrt{3}i$
χ_6^{co}	-2	-1	1	2	1	-1	0	$\sqrt{3}i$	$\sqrt{3}i$	0	$-\sqrt{3}i$	$-\sqrt{3}i$

Coirrepresntations of group generators

	$\left[E \mid\mid 6_{001}^{+}\right]$	$\left[T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 347.3.782 : ${}^{1}6/{}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}6/{}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right] \left[T \mid\mid m_{001}\right] \left[T \mid\mid \bar{3}^{-}_{001}\right] \left[T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[T \mid\mid -1\right] \left[T \mid\mid \bar{6}^{+}_{001}\right] \left[T \mid\mid \bar{3}^{+}_{001}\right] \left[2_{100}T \mid\mid 1\right] \left[2_{100}T \mid\mid 6^{+}_{001}\right] \left[2_{100}T \mid\mid 3^{+}_{001}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid 3^{-}_{001}\right] \left[2_{100}T \mid\mid 6^{-}_{001}\right]$
$\left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\Sigma_z T \mid\mid m_{001}\right] \left[\Sigma_z T \mid\mid \bar{3}^{-}_{001}\right] \left[\Sigma_z T \mid\mid \bar{6}^{-}_{001}\right] \left[\Sigma_z T \mid\mid -1\right] \left[\Sigma_z T \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z T \mid\mid \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100}T \mid\mid 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid$
$\left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\Sigma_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right] \left[\bar{E}T \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}T \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}T \mid\mid -1\right] \left[\bar{E}T \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E}T \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid 6^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid 3^{-}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 6^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right] \left[\bar{\Sigma}_z T \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$
C_7 : $\left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right]$
$\left[\Sigma_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid 1\right]$
C_9 : $\left[\bar{E} \mid\mid 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \mid\mid 3^{+}_{001}\right]$
C_{11} : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_{12} : $\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	2	−2	0	−2	2	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	−2	−2	0	−4	2	2	−4	2	2
χ_4^{co}	4	2	−2	−4	−2	2	0	−4	−2	2	4	2	−2

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_5^* & 0 \\ 0 & \omega_5 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_7 & 0 \\ 0 & \omega_7^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 348.1.783 : ${}^{3}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$	C_9 :	$\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 348.2.784 : ${}^{3_z}6/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[6^{-}_{001}T \parallel 6^{+}_{001}\right]$ $\left[6^{+}_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[6^{-}_{001}T \parallel 3^{-}_{001}\right]$ $\left[6^{+}_{001}T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[6^{-}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$	C_9 :	$\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 349.1.785 : ${}^{2_z}6/{}^{2_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 349.2.786 : ${}^{2_z}6/{}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{001}\right]$ $\left[2_{100}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{010}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100}T \parallel -1\right]$ $\left[2_{010}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 349.3.787 : ${}^{2_z}6/{}^{2_x}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{010}T \parallel 6^{-}_{001}\right]$
$\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 349.4.788 : ${}^{2_z}6/{}^{2_x}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[2_{100}T \parallel 6^{+}_{001}\right]$ $\left[2_{010}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 350.1.789 : ${}^{2_z}6/{}^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel -1\right]$ $\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 351.1.790 : ${}^{m_x}6/{}^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{010}T \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3^*	$\alpha^*\mathcal{K}$

No 352.1.791 : ${}^{m_x}6/{}^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{100}T \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 6^{-}_{001}\right]$ $\left[2_{010}T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{010}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001} \mid\mid -1\right]$ $\left[2_{010}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$
C_5 : $\left[2_{001} \mid\mid -1\right]$
C_6 : $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$	$\left[2_{100}T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 353.1.792 : ${}^{2}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 353.2.793 : ${}^{2_z}6/^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}6/^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^-_{001}\right] \left[T \parallel \bar{6}^-_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[T \parallel \bar{6}^+_{001}\right] \left[2_{100}T \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^-_{001}\right] \left[\Sigma_z T \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^+_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^-_{001}\right] \left[\bar{E}T \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^+_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^+_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right]$
C_4 : $\left[E \parallel 3^-_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	4	0	−2	−2	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^+_{001}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 354.1.794 : ${}^{2}6/{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 354.2.795 : ${}^{2_z}6/{}^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	4	0	−2	−2	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 355.1.796 : ${}^{\bar{1}}6/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^{*}\mathcal{K}$
Γ_4^{co}	ω_3^{*}	$\alpha^{*}\mathcal{K}$

No 355.2.797 : ${}^{\bar{1}}6/^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_6 & 0 \\ 0 & \omega_6^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_5 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 356.1.798 : $^{\bar{1}}6/^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 356.2.799 : ${}^{\bar{1}}6/^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_5 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 357.1.800 : ${}^{m}6/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3^*	$\alpha^*\mathcal{K}$

No 357.2.801 : ${}^{m_z}6/{}^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}6/{}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100}T \,||\, 6^{+}_{001}\right] \left[E \,||\, 3^{+}_{001}\right] \left[2_{100}T \,||\, 2_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[2_{100}T \,||\, 6^{-}_{001}\right] \left[2_{100} \,||\, m_{001}\right] \left[T \,||\, \bar{3}^{-}_{001}\right] \left[2_{100} \,||\, \bar{6}^{-}_{001}\right]$
$\left[T \,||\, -1\right] \left[2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[T \,||\, \bar{3}^{+}_{001}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z 2_{100}T \,||\, 6^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \,||\, 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z T \,||\, \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\Sigma_z T \,||\, -1\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z T \,||\, \bar{3}^{+}_{001}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}2_{100}T \,||\, 6^{+}_{001}\right] \left[\bar{E} \,||\, 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \,||\, 2_{001}\right] \left[\bar{E} \,||\, 3^{-}_{001}\right] \left[\bar{E}2_{100}T \,||\, 6^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}T \,||\, \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E}T \,||\, -1\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E}T \,||\, \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 6^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \,||\, 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 3^{+}_{001}\right]$
C_3 : $\left[E \,||\, 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \,||\, m_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{+}_{001}\right]$
C_5 : $\left[\bar{E} \,||\, 1\right]$
C_6 : $\left[\bar{E} \,||\, 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \,||\, 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \,||\, \bar{6}^{-}_{001}\right]$	$\left[2_{100}T \,||\, 6^{+}_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_6 & 0 \\ 0 & \omega_6^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_5 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 358.1.802 : ${}^{m}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 358.2.803 : ${}^{m_z}6/^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}6/^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	4	−2	−2	0	−4	2	2

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \alpha^* & 0 \\ 0 & \alpha \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_5 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 359.1.804 : ${}^{3}6/{}^{2}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001} \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001} \mid\mid -1\right]$ $\left[6^{-}_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_7 :	$\left[2_{001} \mid\mid m_{001}\right]$
C_2 :	$\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$	C_8 :	$\left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$	C_9 :	$\left[6^{+}_{001} \mid\mid \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \mid\mid 2_{001}\right]$	C_{10} :	$\left[2_{001} \mid\mid -1\right]$
C_5 :	$\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$	C_{11} :	$\left[6^{-}_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_6 :	$\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$	C_{12} :	$\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 359.2.805 : ${}^{3_z}6/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$ $\left[3^{-}_{001} \,||\, 3^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[3^{+}_{001} \,||\, 3^{-}_{001}\right]$ $\left[3^{-}_{001} \,||\, 6^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[6^{-}_{001} \,||\, \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001} \,||\, \bar{6}^{-}_{001}\right]$
$\left[2_{001} \,||\, -1\right]$ $\left[6^{-}_{001} \,||\, \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001} \,||\, \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[6^{-}_{001}T \,||\, 6^{+}_{001}\right]$ $\left[6^{+}_{001}T \,||\, 3^{+}_{001}\right]$ $\left[2_{001}T \,||\, 2_{001}\right]$ $\left[6^{-}_{001}T \,||\, 3^{-}_{001}\right]$ $\left[6^{+}_{001}T \,||\, 6^{-}_{001}\right]$
$\left[T \,||\, m_{001}\right]$ $\left[3^{+}_{001}T \,||\, \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001}T \,||\, \bar{6}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[3^{+}_{001}T \,||\, \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001}T \,||\, \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_7 :	$\left[2_{001} \,||\, m_{001}\right]$
C_2 :	$\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$	C_8 :	$\left[6^{-}_{001} \,||\, \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{-}_{001} \,||\, 3^{+}_{001}\right]$	C_9 :	$\left[6^{+}_{001} \,||\, \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, -1\right]$
C_5 :	$\left[3^{+}_{001} \,||\, 3^{-}_{001}\right]$	C_{11} :	$\left[6^{-}_{001} \,||\, \bar{6}^{+}_{001}\right]$
C_6 :	$\left[3^{-}_{001} \,||\, 6^{-}_{001}\right]$	C_{12} :	$\left[6^{+}_{001} \,||\, \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 360.1.806 : ${}^{3}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 361.1.807 : ${}^{3}6/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[6^{-}_{001}T \mid\mid \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \mid\mid -1\right]$ $\left[6^{-}_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 362.1.808 : ${}^{6}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$	C_9 :	$\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 362.2.809 : ${}^{6_z}6/{}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[6^{+}_{001} \,||\, 6^{+}_{001}\right]$ $\left[3^{+}_{001} \,||\, 3^{+}_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[3^{-}_{001} \,||\, 3^{-}_{001}\right]$ $\left[6^{-}_{001} \,||\, 6^{-}_{001}\right]$ $\left[E \,||\, m_{001}\right]$ $\left[6^{+}_{001} \,||\, \bar{3}^{-}_{001}\right]$ $\left[3^{+}_{001} \,||\, \bar{6}^{-}_{001}\right]$
$\left[2_{001} \,||\, -1\right]$ $\left[3^{-}_{001} \,||\, \bar{6}^{+}_{001}\right]$ $\left[6^{-}_{001} \,||\, \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \,||\, 1\right]$ $\left[3^{-}_{001}T \,||\, 6^{+}_{001}\right]$ $\left[6^{-}_{001}T \,||\, 3^{+}_{001}\right]$ $\left[T \,||\, 2_{001}\right]$ $\left[6^{+}_{001}T \,||\, 3^{-}_{001}\right]$ $\left[3^{+}_{001}T \,||\, 6^{-}_{001}\right]$
$\left[2_{001}T \,||\, m_{001}\right]$ $\left[3^{-}_{001}T \,||\, \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001}T \,||\, \bar{6}^{-}_{001}\right]$ $\left[T \,||\, -1\right]$ $\left[6^{+}_{001}T \,||\, \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001}T \,||\, \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_7 :	$\left[E \,||\, m_{001}\right]$
C_2 :	$\left[6^{+}_{001} \,||\, 6^{+}_{001}\right]$	C_8 :	$\left[6^{+}_{001} \,||\, \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{+}_{001} \,||\, 3^{+}_{001}\right]$	C_9 :	$\left[3^{+}_{001} \,||\, \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \,||\, 2_{001}\right]$	C_{10} :	$\left[2_{001} \,||\, -1\right]$
C_5 :	$\left[3^{-}_{001} \,||\, 3^{-}_{001}\right]$	C_{11} :	$\left[3^{-}_{001} \,||\, \bar{6}^{+}_{001}\right]$
C_6 :	$\left[6^{-}_{001} \,||\, 6^{-}_{001}\right]$	C_{12} :	$\left[6^{-}_{001} \,||\, \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^{+}_{001} \,||\, 6^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 363.1.810 : ${}^{\bar{3}}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^{*}\mathcal{K}$
Γ_4^{co}	ω_3^{*}	$\alpha^{*}\mathcal{K}$

No 364.1.811 : ${}^{\bar{6}}6/{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3^*	$\alpha^*\mathcal{K}$

No 365.1.812 : ${}^{6}6/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[6^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[3^{-}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$	C_9 :	$\left[6^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[6^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[3^{+}_{001} \parallel \bar{3}^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Irrepresntations of group generators

	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$

Γ_1	1	1
Γ_2	1	-1
Γ_3	-1	-1
Γ_4	-1	1
Γ_5	$e^{\frac{-2\pi i}{3}}$	1
Γ_6	$e^{\frac{-2\pi i}{3}}$	-1
Γ_7	$e^{\frac{\pi i}{3}}$	-1
Γ_8	$e^{\frac{\pi i}{3}}$	1
Γ_9	$e^{\frac{2\pi i}{3}}$	1
Γ_{10}	$e^{\frac{2\pi i}{3}}$	-1
Γ_{11}	$e^{\frac{-\pi i}{3}}$	-1
Γ_{12}	$e^{\frac{-\pi i}{3}}$	1

No 365.2.813 : ${}^{6_z}6/{}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[3^-_{001} \parallel \bar{3}^-_{001}\right]$ $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right]$ $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[3^+_{001} \parallel \bar{3}^+_{001}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[3^-_{001}T \parallel 6^+_{001}\right]$ $\left[6^-_{001}T \parallel 3^+_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[6^+_{001}T \parallel 3^-_{001}\right]$ $\left[3^+_{001}T \parallel 6^-_{001}\right]$
$\left[T \parallel m_{001}\right]$ $\left[6^+_{001}T \parallel \bar{3}^-_{001}\right]$ $\left[3^+_{001}T \parallel \bar{6}^-_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[3^-_{001}T \parallel \bar{6}^+_{001}\right]$ $\left[6^-_{001}T \parallel \bar{3}^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[6^+_{001} \parallel 6^+_{001}\right]$	C_8 :	$\left[3^-_{001} \parallel \bar{3}^-_{001}\right]$
C_3 :	$\left[3^+_{001} \parallel 3^+_{001}\right]$	C_9 :	$\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[3^-_{001} \parallel 3^-_{001}\right]$	C_{11} :	$\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$
C_6 :	$\left[6^-_{001} \parallel 6^-_{001}\right]$	C_{12} :	$\left[3^+_{001} \parallel \bar{3}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	−1	−1	−1	−1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1	−1	1	−1	1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	−1	−1	2	−1	−1	2	−1	−1
χ_6^{co}	2	−1	−1	2	−1	−1	−2	1	1	−2	1	1
χ_7^{co}	2	1	−1	−2	−1	1	−2	−1	1	2	1	−1
χ_8^{co}	2	1	−1	−2	−1	1	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$

Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	σ_0	$\sigma_x\mathcal{K}$
Γ_6^{co}	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_7^{co}	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$
Γ_8^{co}	ω_3	σ_0	$\sigma_x\mathcal{K}$

No 366.1.814 : ${}^{\bar{3}}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 367.1.815 : $^{\bar{6}}6/^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[6^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[3^{-}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[3^{+}_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 368.1.816 : ${}^{6}6/{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[6^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_6 : $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 369.1.817 : ${}^{6}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[3^{-}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_6 : $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	1	−1	−2	−1	1

Coirrepresntations of group generators

	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$

No 370.1.818 : ${}^{\bar{3}}6/{}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001}T \mid\mid 6^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001} \mid\mid -1\right]$ $\left[6^{-}_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$
C_5 : $\left[2_{001} \mid\mid -1\right]$
C_6 : $\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[6^{-}_{001} \mid\mid \bar{3}^{-}_{001}\right]$	$\left[3^{+}_{001}T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 371.1.819 : ${}^{\bar{3}}6/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[6^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[6^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[6^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[6^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^{*}\mathcal{K}$
Γ_4^{co}	ω_3^{*}	$\alpha^{*}\mathcal{K}$

No 372.1.820 : ${}^{\bar{6}}6/{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[3^{-}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	−1	2	−1
χ_4^{co}	2	−1	−1	1	−2	1

Coirrepresntations of group generators

	$\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1^*	$\alpha^*\mathcal{K}$
Γ_4^{co}	ω_3	$\alpha^*\mathcal{K}$

No 373.1.821 : ${}^{\bar{6}}6/{}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right]$ $\left[6^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{3}^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[6^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
C_6 : $\left[6^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1	−1
χ_3^{co}	2	−1	−1	2	−1	−1
χ_4^{co}	2	−1	−1	−2	1	1

Coirrepresntations of group generators

	$\left[6^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[6^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	$\alpha^{*}\mathcal{K}$
Γ_4^{co}	ω_3^{*}	$\alpha^{*}\mathcal{K}$

No 374.1.822 : ${}^{1}6^{1}2^{1}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{210}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 374.2.823 : ${}^{1}6^{1}2^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{120}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 374.3.824 : ${}^{1}6{}^{1}2{}^{1}2{}^{\infty_z}1 \to {}^{1}6{}^{1}2{}^{1}2$

No 374.4.825 : ${}^{1}6^{1}2^{1}2^{\infty}{}_{z}{}^{m_x}1 \rightarrow {}^{1}6^{1}2^{1}2^{m_x}1$

No 375.1.826 : ${}^{1}6^{2}2^{2}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 375.2.827 : ${}^{1}6^{2_z}2^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[T \parallel 2_{100}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 375.3.828 : ${}^{1}6^{2_z}2^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel 6^{-}_{001}\right]$
$\left[2_{010}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 375.4.829 : ${}^{1}6^{2_z}2^{2_z}2^{\infty_x}1 \rightarrow {}^{1}6^{2_z}2^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_6 : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2
χ_4	2	1	−1	−2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2
χ_5	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2

χ_6	2	1	-1	-2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	-2	-1	1	2

Irrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	ω_3	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 376.1.830 : ${}^{1}6^{m}2^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{210}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{001}T \mid\mid 2_{110}\right]$ $\left[2_{001}T \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 376.2.831 : ${}^{1}6^{m_z}2^{m_z}2^{\infty_x}1 \to {}^{1}6^{m}2^{m}2$

No 377.1.832 : ${}^{1}6^{\bar{1}}2^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 377.2.833 : ${}^{1}6^{\bar{1}}2^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{1}6^{\bar{1}}2^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[T \mid\mid 2_{100}\right]$ $\left[T \mid\mid 2_{210}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$
$\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{120}\right]$ $\left[\Sigma_z \mid\mid 1\right]$ $\left[\Sigma_z \mid\mid 6^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$ $\left[\Sigma_z \mid\mid 2_{001}\right]$ $\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$ $\left[\Sigma_z \mid\mid 6^{-}_{001}\right]$
$\left[\Sigma_z T \mid\mid 2_{100}\right]$ $\left[\Sigma_z T \mid\mid 2_{210}\right]$ $\left[\Sigma_z T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\Sigma_z T \mid\mid 2_{110}\right]$ $\left[\Sigma_z T \mid\mid 2_{010}\right]$ $\left[\Sigma_z T \mid\mid 2_{120}\right]$ $\left[\bar{E} \mid\mid 1\right]$ $\left[\bar{E} \mid\mid 6^{+}_{001}\right]$ $\left[\bar{E} \mid\mid 3^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right]$ $\left[\bar{E} \mid\mid 3^{-}_{001}\right]$ $\left[\bar{E} \mid\mid 6^{-}_{001}\right]$ $\left[\bar{E}T \mid\mid 2_{100}\right]$ $\left[\bar{E}T \mid\mid 2_{210}\right]$ $\left[\bar{E}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[\bar{E}T \mid\mid 2_{110}\right]$ $\left[\bar{E}T \mid\mid 2_{010}\right]$ $\left[\bar{E}T \mid\mid 2_{120}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right]$ $\left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$ $\left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{100}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{210}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{110}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{010}\right]$ $\left[\bar{\Sigma}_z T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_{13} :	$\left[\bar{E} \mid\mid 1\right]$
C_2 :	$\left[E \mid\mid 6^{+}_{001}\right]$	C_{14} :	$\left[\bar{E} \mid\mid 6^{+}_{001}\right]$
C_3 :	$\left[E \mid\mid 3^{+}_{001}\right]$	C_{15} :	$\left[\bar{E} \mid\mid 3^{+}_{001}\right]$
C_4 :	$\left[E \mid\mid 2_{001}\right]$	C_{16} :	$\left[\bar{E} \mid\mid 2_{001}\right]$
C_5 :	$\left[E \mid\mid 3^{-}_{001}\right]$	C_{17} :	$\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_6 :	$\left[E \mid\mid 6^{-}_{001}\right]$	C_{18} :	$\left[\bar{E} \mid\mid 6^{-}_{001}\right]$
C_7 :	$\left[\Sigma_z \mid\mid 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_8 :	$\left[\Sigma_z \mid\mid 6^{+}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_{10} :	$\left[\Sigma_z \mid\mid 2_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
C_{11} :	$\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$
C_{12} :	$\left[\Sigma_z \mid\mid 6^{-}_{001}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	-2	2	-2	2	-2	0	0	0	0	0	0
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	2	-2	2	-2	2	0	0	0	0	0	0
χ_3^{co}	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 6_{001}^{+}\right]$	$\left[T \mid\mid 2_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_7	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_7^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 377.3.834 : ${}^{1}6^{\bar{1}}2^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{1}6^{\bar{1}}2^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right] \left[T \mid\mid 2_{100}\right] \left[T \mid\mid 2_{210}\right] \left[T \mid\mid 2_{1\bar{1}0}\right]$
$\left[T \mid\mid 2_{110}\right] \left[T \mid\mid 2_{010}\right] \left[T \mid\mid 2_{120}\right] \left[2_{100}T \mid\mid 1\right] \left[2_{100}T \mid\mid 6^{+}_{001}\right] \left[2_{100}T \mid\mid 3^{+}_{001}\right] \left[2_{100}T \mid\mid 2_{001}\right] \left[2_{100}T \mid\mid 3^{-}_{001}\right] \left[2_{100}T \mid\mid 6^{-}_{001}\right]$
$\left[2_{100} \mid\mid 2_{100}\right] \left[2_{100} \mid\mid 2_{210}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{120}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\Sigma_z T \mid\mid 2_{100}\right] \left[\Sigma_z T \mid\mid 2_{210}\right] \left[\Sigma_z T \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z T \mid\mid 2_{110}\right] \left[\Sigma_z T \mid\mid 2_{010}\right] \left[\Sigma_z T \mid\mid 2_{120}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid 1\right] \left[\Sigma_z 2_{100}T \mid\mid 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \mid\mid 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100} \mid\mid$
$\left[\Sigma_z 2_{100} \mid\mid 2_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{120}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid 2_{100}\right] \left[\bar{E}T \mid\mid 2_{210}\right] \left[\bar{E}T \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}T \mid\mid 2_{110}\right] \left[\bar{E}T \mid\mid 2_{010}\right] \left[\bar{E}T \mid\mid 2_{120}\right] \left[\bar{E}2_{100}T \mid\mid 1\right] \left[\bar{E}2_{100}T \mid\mid 6^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \mid\mid 2_{001}\right] \left[\bar{E}2_{100}T \mid\mid 3^{-}_{001}\right] \left[\bar{E}2_{100}T \mid\mid 6^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{210}\right] \left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{120}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z T \mid\mid 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[2_{100} \mid\mid 2_{100}\right] \left[2_{100} \mid\mid 2_{210}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{120}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{120}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{210}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{120}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{120}\right]$
C_6 : $\left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \mid\mid 1\right]$
C_{11} : $\left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	2	−2	−4	0	0	0	0	0	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel 6_{001}^{+}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 378.1.835 : ${}^{2}6^{1}2^{2}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 378.2.836 : ${}^{2_z}6^1 2^{2_z} 2^{m_z} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel 6^-_{001}\right] \left[E \parallel 2_{100}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[2_{001} \parallel 2_{120}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[2_{001}T \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right]$
$\left[2_{001}T \parallel 2_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{010}\right] \left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^+_{001}\right] \left[2_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^+_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 378.3.837 : ${}^{2_z}6^1 2^{2_z} 2^{m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel 6^-_{001}\right] \left[E \parallel 2_{100}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[2_{001} \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^+_{001}\right] \left[2_{100}T \parallel 3^+_{001}\right] \left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^-_{001}\right] \left[2_{010}T \parallel 6^-_{001}\right]$
$\left[2_{100}T \parallel 2_{100}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^+_{001}\right] \left[2_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^+_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 378.4.838 : ${}^{2_z}6^1 2^{2_z} 2^{\infty_x} 1 \rightarrow {}^{2_z}6^1 2^{2_z} 2^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[E \parallel 2_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right]$
$\left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} 2_{100} \parallel 6^-_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right]$
$\left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel 6^+_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^-_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2
χ_3	4	0	−2	0	−4	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 6^+_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix} \mathcal{K}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} \mathcal{K}$

No 379.1.839 : ${}^{m}6^{1}2^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 6^{-}_{001}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{210}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid 2_{110}\right]$ $\left[E \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[E \mid\mid 2_{100}\right]$	$\left[2_{001}T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 379.2.840 : ${}^{m_z}6^{1}2^{m_z}2^{\infty_x}1 \to {}^{m}6^{1}2^{m}2$

No 380.1.841 : ${}^{\bar{1}}6^{1}2^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[T \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[T \mid\mid 6^{-}_{001}\right] \left[E \mid\mid 2_{100}\right] \left[T \mid\mid 2_{210}\right] \left[T \mid\mid 2_{1\bar{1}0}\right]$
$\left[E \mid\mid 2_{110}\right] \left[E \mid\mid 2_{010}\right] \left[T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 2_{100}\right] \left[E \mid\mid 2_{110}\right] \left[E \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[E \mid\mid 2_{100}\right]$	$\left[T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 380.2.842 : $\bar{1}6^{1}2^{\bar{1}}2^{\infty_z}1 \rightarrow \bar{1}6^{1}2^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_8 :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_3 :	$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$	C_9 :	$\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right]$
C_4 :	$\left[\Sigma_z \parallel 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_5 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2	0	0	0
χ_3^{co}	4	−2	0	0	0	0	−4	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel 2_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 380.3.843 : ${}^{\bar{1}}6^{1}2^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6^{1}2^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel 2_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right] \left[T \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right]$
$\left[2_{100}T \parallel 2_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1}\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{100}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{100}\right]$	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 381.1.844 : ${}^{2_x}6^{2_y}2^{2_z}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 381.2.845 : ${}^{2_x}6^{2_y}2^{2_z}2^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 381.3.846 : ${}^{2_x}6^{2_y}2^{2_z}2^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{010}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[T \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 2_{210}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 382.1.847 : ${}^{2_z}6^{m_x}2^{m_y}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 383.1.848 : ${}^{m_x}6^{2_z}2^{m_y}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 384.1.849 : ${}^{2}6^{\bar{1}}2^{m}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 384.2.850 : ${}^{2_z}6^{\bar{1}}2^{m_z}2^{\infty_x}1 \rightarrow {}^{2_z}6^{\bar{1}}2^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel 2_{110}\right] \left[T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_3^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[T \parallel 2_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\alpha^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 385.1.851 : $^{\bar{1}}6^{2}2^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[T \mid\mid 6^{-}_{001}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001}T \mid\mid 2_{210}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{100}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{100}\right]$	$\left[T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 385.2.852 : ${}^{\bar{1}}6^{2_z}2^{m_z}2^{\infty_x}1 \rightarrow {}^{\bar{1}}6^{2_z}2^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1
χ_3^{co}	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 386.1.853 : ${}^{m}6^{2}2^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel 2_{100}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{100}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 386.2.854 : ${}^{m_z}6^{2_z}2^{\bar{1}}2^{\infty_x}1 \rightarrow {}^{m_z}6^{2_z}2^{\bar{1}}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{100}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 387.1.855 : ${}^{3_z}6^{2_x}2^{2_{xy}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^+_{001} \parallel 6^+_{001}\right]$ $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$ $\left[3^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{110} \parallel 2_{010}\right]$ $\left[2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 6^+_{001}\right]$ $\left[3^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{110} \parallel 2_{010}\right]$
C_6 : $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 387.2.856 : ${}^{3_z}6^{2_x}2^{2_{xy}}2^{m_z}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[3^{+}_{001} \,||\, 6^{+}_{001}\right] \left[3^{-}_{001} \,||\, 3^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[3^{+}_{001} \,||\, 3^{-}_{001}\right] \left[3^{-}_{001} \,||\, 6^{-}_{001}\right] \left[2_{100} \,||\, 2_{100}\right] \left[2_{110} \,||\, 2_{210}\right] \left[2_{010} \,||\, 2_{1\bar{1}0}\right]$
$\left[2_{010} \,||\, 2_{110}\right] \left[2_{110} \,||\, 2_{010}\right] \left[2_{100} \,||\, 2_{120}\right] \left[2_{001}T \,||\, 1\right] \left[6^{-}_{001}T \,||\, 6^{+}_{001}\right] \left[6^{+}_{001}T \,||\, 3^{+}_{001}\right] \left[2_{001}T \,||\, 2_{001}\right] \left[6^{-}_{001}T \,||\, 3^{-}_{001}\right] \left[6^{+}_{001}T \,||\, 6^{-}_{001}\right]$
$\left[2_{120}T \,||\, 2_{100}\right] \left[2_{1\bar{1}0}T \,||\, 2_{210}\right] \left[2_{210}T \,||\, 2_{1\bar{1}0}\right] \left[2_{210}T \,||\, 2_{110}\right] \left[2_{1\bar{1}0}T \,||\, 2_{010}\right] \left[2_{120}T \,||\, 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[3^{+}_{001} \,||\, 6^{+}_{001}\right] \left[3^{-}_{001} \,||\, 6^{-}_{001}\right]$
C_3 : $\left[3^{-}_{001} \,||\, 3^{+}_{001}\right] \left[3^{+}_{001} \,||\, 3^{-}_{001}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$
C_5 : $\left[2_{100} \,||\, 2_{100}\right] \left[2_{010} \,||\, 2_{110}\right] \left[2_{110} \,||\, 2_{010}\right]$
C_6 : $\left[2_{110} \,||\, 2_{210}\right] \left[2_{010} \,||\, 2_{1\bar{1}0}\right] \left[2_{100} \,||\, 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$	$\left[2_{100} \,||\, 2_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 388.1.857 : ${}^{3_z}6^{m_x}2^{m_{xy}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid 2_{210}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{010}T \mid\mid 2_{110}\right]$ $\left[2_{110}T \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[3^{-}_{001} \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 389.1.858 : ${}^{6_z}6^{2_x}2^{2_1}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_6 : $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	1	−1
χ_4	1	−1	1	−1	−1	1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[6^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	1
Γ_4	−1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 389.2.859 : ${}^{6_z}6^{2_x}2^{2_1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{120} \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[3^-_{001}T \parallel 6^+_{001}\right]$ $\left[6^-_{001}T \parallel 3^+_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[6^+_{001}T \parallel 3^-_{001}\right]$ $\left[3^+_{001}T \parallel 6^-_{001}\right]$
$\left[2_{120}T \parallel 2_{100}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{110}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{110}\right]$ $\left[2_{210}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_6 : $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	1	−1
χ_4^{co}	1	−1	1	−1	−1	1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel 2_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 390.1.860 : ${}^{\bar{3}_z}6^{2_x}2^{m_{xy}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001}T \mid\mid 6^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001}T \mid\mid 6^-_{001}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{110}T \mid\mid 2_{210}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{110}\right]$ $\left[2_{110} \mid\mid 2_{010}\right]$ $\left[2_{100}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{100}\right]$ $\left[2_{010} \mid\mid 2_{110}\right]$ $\left[2_{110} \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^-_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100} \mid\mid 2_{100}\right]$	$\left[3^+_{001}T \mid\mid 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 391.1.861 : ${}^{6_z}6^{m_x}2^{m_1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$ $\left[2_{210}T \mid\mid 2_{210}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{120}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid 6^+_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[6^-_{001} \mid\mid 6^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{100}T \mid\mid 2_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 392.1.862 : ${}^{\bar{6}_z}6^{m_x}2^{2_1}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001}T \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001}T \parallel 6^-_{001}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$ $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 3^+_{001}\right]$	$\left[2_{210} \parallel 2_{210}\right]$	$\left[6^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 393.1.863 : ${}^{1}6^{1}m^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 393.2.864 : ${}^{1}6^{1}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel m_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{210}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[2_{001}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 393.3.865 : ${}^{1}6^{1}m^{1}m^{\infty_z}1 \to {}^{1}6^{1}m^{1}m$

No 393.4.866 : ${}^{1}6^{1}m^{1}m^{\infty_{z}m_{x}}1 \rightarrow {}^{1}6^{1}m^{1}m^{m_{x}}1$

No 394.1.867 : ${}^{1}6^{2}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 394.2.868 : ${}^{1}6^{2_z}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{210}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 394.3.869 : ${}^{1}6^{2_z}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{100}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel 6^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 394.4.870 : ${}^{1}6^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}6^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_6 : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2
χ_4	2	1	−1	−2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2
χ_5	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2

χ_6	2	1	-1	-2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	-2	-1	1	2

Irrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	ω_3	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 395.1.871 : ${}^{1}6^{m}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{210}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 395.2.872 : ${}^{1}6^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6^{m}m^{m}m$

No 396.1.873 : ${}^{1}6^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{210}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 396.2.874 : ${}^{1}6^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}6^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right] \left[T \mid\mid m_{100}\right] \left[T \mid\mid m_{210}\right] \left[T \mid\mid m_{1\bar{1}0}\right]$
$\left[T \mid\mid m_{110}\right] \left[T \mid\mid m_{010}\right] \left[T \mid\mid m_{120}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z T \mid\mid m_{210}\right] \left[\Sigma_z T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z T \mid\mid m_{110}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\Sigma_z T \mid\mid m_{120}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right]$
$\left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right] \left[\bar{E}T \mid\mid m_{100}\right] \left[\bar{E}T \mid\mid m_{210}\right] \left[\bar{E}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}T \mid\mid m_{110}\right] \left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{E}T \mid\mid m_{120}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{210}\right] \left[\bar{\Sigma}_z T \mid\mid m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z T \mid\mid m_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \mid\mid 1\right]$	C_{13} :	$\left[\bar{E} \mid\mid 1\right]$
C_2 :	$\left[E \mid\mid 6^{+}_{001}\right]$	C_{14} :	$\left[\bar{E} \mid\mid 6^{+}_{001}\right]$
C_3 :	$\left[E \mid\mid 3^{+}_{001}\right]$	C_{15} :	$\left[\bar{E} \mid\mid 3^{+}_{001}\right]$
C_4 :	$\left[E \mid\mid 2_{001}\right]$	C_{16} :	$\left[\bar{E} \mid\mid 2_{001}\right]$
C_5 :	$\left[E \mid\mid 3^{-}_{001}\right]$	C_{17} :	$\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_6 :	$\left[E \mid\mid 6^{-}_{001}\right]$	C_{18} :	$\left[\bar{E} \mid\mid 6^{-}_{001}\right]$
C_7 :	$\left[\Sigma_z \mid\mid 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \mid\mid 1\right]$
C_8 :	$\left[\Sigma_z \mid\mid 6^{+}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \mid\mid 3^{+}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_{10} :	$\left[\Sigma_z \mid\mid 2_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \mid\mid 2_{001}\right]$
C_{11} :	$\left[\Sigma_z \mid\mid 3^{-}_{001}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$
C_{12} :	$\left[\Sigma_z \mid\mid 6^{-}_{001}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	0
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	2	-2	2	-2	2	0	0	0	0	0	0
χ_3^{co}	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 6_{001}^{+}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_7	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_7^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 396.3.875 : ${}^{1}6^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}6^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_6 : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	2	−2	−4	0	0	0	0	0	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 397.1.876 : ${}^{2}6^{1}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 397.2.877 : ${}^{2_z}6^1 m^{2_z} m^{m_z} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel 6^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[2_{001}T \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^+_{001}\right] \left[2_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 397.3.878 : ${}^{2_z}6^1 m^{2_z} m^{m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel 6^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^+_{001}\right] \left[2_{100}T \parallel 3^+_{001}\right] \left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^-_{001}\right] \left[2_{010}T \parallel 6^-_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^+_{001}\right] \left[2_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 397.4.879 : ${}^{2_z}6^1 m^{2_z} m^{\infty_x} 1 \rightarrow {}^{2_z}6^1 m^{2_z} m^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} 2_{100} \parallel 6^-_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{210}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right]$
$\left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right]$
$\left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E} 2_{100} \parallel 6^+_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^-_{001}\right] \left[\bar{E} 2_{100} \parallel m_{210}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2
χ_3	4	0	−2	0	−4	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 6^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix} \mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} \mathcal{K}$

No 398.1.880 : ${}^{m}6^{1}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 398.2.881 : ${}^{m_z}6^1 m^{m_z} m^{\infty_x} 1 \to {}^{m}6^1 m^{m} m$

No 399.1.882 : ${}^{\bar{1}}6^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 399.2.883 : ${}^{\bar{1}}6^{1}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}6^{1}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_8 :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_3 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$	C_9 :	$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$
C_4 :	$\left[\Sigma_z \parallel 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_5 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_6 :	$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2	0	0	0
χ_3^{co}	4	−2	0	0	0	0	−4	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 399.3.884 : ${}^{\bar{1}}6^{1}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m\right.$
$\left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right.$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_4 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 400.1.885 : ${}^{2_x}6^{2_y}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$ $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 400.2.886 : ${}^{2_x}6^{2_y}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$ $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[T \parallel 6^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 400.3.887 : ${}^{2_x}6^{2_y}m^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$ $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{010}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{010}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$
$\left[T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{210}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right]$ $\left[2_{100} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{100} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel m_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 401.1.888 : ${}^{2_z}6^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 402.1.889 : ${}^{m_x}6^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 403.1.890 : ${}^{2}6^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{210}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{001}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 403.2.891 : ${}^{2_z}6^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}6^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
$\left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right]$
C_4 : $\left[E \parallel 3^-_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_3^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^+_{001}\right]$	$\left[T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\alpha^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 404.1.892 : ${}^{\bar{1}}6^2m^mm$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^+_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 404.2.893 : ${}^{\bar{1}}6^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1
χ_3^{co}	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 405.1.894 : ${}^{m}6^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 405.2.895 : ${}^{m_z}6^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}6^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 406.1.896 : ${}^{3_z}6^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^+_{001} \parallel 6^+_{001}\right]$ $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$ $\left[3^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{210}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{110} \parallel m_{010}\right]$ $\left[2_{100} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 6^+_{001}\right]$ $\left[3^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel m_{210}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 406.2.897 : ${}^{3_z}6^{2_x}m^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[E \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[2_{001}T \parallel 1\right] \left[6^-_{001}T \parallel 6^+_{001}\right] \left[6^+_{001}T \parallel 3^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[6^-_{001}T \parallel 3^-_{001}\right] \left[6^+_{001}T \parallel 6^-_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right] \left[2_{1\bar{1}0}T \parallel m_{210}\right] \left[2_{210}T \parallel m_{1\bar{1}0}\right] \left[2_{210}T \parallel m_{110}\right] \left[2_{1\bar{1}0}T \parallel m_{010}\right] \left[2_{120}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 407.1.898 : ${}^{3_z}6^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid m_{210}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{110}T \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 6^+_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[3^-_{001} \mid\mid 6^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 408.1.899 : ${}^{6_z}6^{2_x}m^{2_1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{210} \mid\mid m_{210}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$
$\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{120} \mid\mid m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$
C_6 : $\left[2_{210} \mid\mid m_{210}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{120} \mid\mid m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	−1	1	−1	1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[6^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	1	−1
Γ_3	−1	−1
Γ_4	−1	1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 408.2.900 : ${}^{6_z}6^{2_x}m^{2_1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{210} \parallel m_{210}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{120} \parallel m_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[3^-_{001}T \parallel 6^+_{001}\right]$ $\left[6^-_{001}T \parallel 3^+_{001}\right]$ $\left[T \parallel 2_{001}\right]$ $\left[6^+_{001}T \parallel 3^-_{001}\right]$ $\left[3^+_{001}T \parallel 6^-_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{110}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{110}\right]$ $\left[2_{210}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{210} \parallel m_{210}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{120} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 409.1.901 : ${}^{\bar{3}_z}6^{2_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001}T \mid\mid 6^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[T \mid\mid 2_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001}T \mid\mid 6^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid m_{210}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^-_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[3^+_{001}T \mid\mid 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 410.1.902 : ${}^{6_z}6^{m_x}m^{m_1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{210}T \mid\mid m_{210}\right]$ $\left[2_{1\bar{1}0}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{120}T \mid\mid m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid 6^+_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[6^-_{001} \mid\mid 6^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 411.1.903 : ${}^{\bar{6}_z}6^{m_x}m^{2_1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001}T \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001}T \parallel 6^-_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{210} \parallel m_{210}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{120} \parallel m_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[2_{210} \parallel m_{210}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{120} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 3^+_{001}\right]$	$\left[2_{210} \parallel m_{210}\right]$	$\left[6^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 412.1.904 : ${}^{1}\bar{6}^{1}m^{1}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 412.2.905 : ${}^{1}\bar{6}^{1}m^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel 2_{120}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 412.3.906 : ${}^{1}\bar{6}^{1}m^{1}2^{\infty_z}1 \to {}^{1}\bar{6}^{1}m^{1}2$

No 412.4.907 : ${}^{1}\bar{6}^{1}m^{1}2^{\infty_z m_x}1 \rightarrow {}^{1}\bar{6}^{1}m^{1}2^{m_x}1$

No 413.1.908 : ${}^{1}\bar{6}^{2}m^{2}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 413.2.909 : ${}^{1}\bar{6}^{2_z}m^{2_z}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel m_{100}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 413.3.910 : ${}^{1}\bar{6}^{2_z} m^{2_z} 2^{m_x} 1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, \bar{6}^{+}_{001}\right] \left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, m_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[E \,||\, \bar{6}^{-}_{001}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, 2_{210}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right]$
$\left[2_{001} \,||\, m_{110}\right] \left[2_{001} \,||\, m_{010}\right] \left[2_{001} \,||\, 2_{120}\right] \left[2_{100}T \,||\, 1\right] \left[2_{100}T \,||\, \bar{6}^{+}_{001}\right] \left[2_{100}T \,||\, 3^{+}_{001}\right] \left[2_{100}T \,||\, m_{001}\right] \left[2_{100}T \,||\, 3^{-}_{001}\right] \left[2_{100}T \,||\, \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \,||\, m_{100}\right] \left[2_{010}T \,||\, 2_{210}\right] \left[2_{010}T \,||\, 2_{1\bar{1}0}\right] \left[2_{010}T \,||\, m_{110}\right] \left[2_{010}T \,||\, m_{010}\right] \left[2_{010}T \,||\, 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, \bar{6}^{+}_{001}\right] \left[E \,||\, \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, 3^{-}_{001}\right]$
C_4 : $\left[E \,||\, m_{001}\right]$
C_5 : $\left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{110}\right] \left[2_{001} \,||\, m_{010}\right]$
C_6 : $\left[2_{001} \,||\, 2_{210}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right] \left[2_{001} \,||\, 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \,||\, \bar{6}^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 413.4.911 : ${}^{1}\bar{6}^{2_z} m^{2_z} 2^{\infty_x} 1 \rightarrow {}^{1}\bar{6}^{2_z} m^{2_z} 2^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_6 : $\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel m_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2
χ_4	2	1	−1	−2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2
χ_5	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2

χ_6	2	1	-1	-2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	-2	-1	1	2

Irrepresntations of group generators

	$\left[E \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	ω_3	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 414.1.912 : ${}^{1}\bar{6}^{m}m^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid 2_{210}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 414.2.913 : ${}^{1}\bar{6}^{m_z}m^{m_z}2^{\infty_x}1 \to {}^{1}\bar{6}^{m}m^{m}2$

No 415.1.914 : ${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 415.2.915 : ${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel \bar{6}^{+}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_4 :	$\left[E \parallel m_{001}\right]$	C_{16} :	$\left[\bar{E} \parallel m_{001}\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{17} :	$\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_6 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$	C_{18} :	$\left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel m_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel m_{001}\right]$
C_{11} :	$\left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{12} :	$\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	0
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}

χ_1^{co}	-2	-2	-2	-2	-2	-2	0	0	0	0	0	0
χ_2^{co}	-2	2	-2	2	-2	2	0	0	0	0	0	0
χ_3^{co}	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0
χ_4^{co}	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0
χ_5^{co}	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0
χ_6^{co}	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{6}_{001}^{+}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_7	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_7^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 415.3.916 : ${}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{1}\bar{6}^{\bar{1}}m^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_6 : $\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel m_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	2	−2	−4	0	0	0	0	0	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 416.1.917 : ${}^{2}\bar{6}^{1}m^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid 2_{210}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$
$\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$
C_6 : $\left[2_{001} \mid\mid 2_{210}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 416.2.918 : ${}^{2_z}\bar{6}^1 m^{2_z} 2^{m_z} 1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001} \,||\, \bar{6}^+_{001}\right] \left[E \,||\, 3^+_{001}\right] \left[2_{001} \,||\, m_{001}\right] \left[E \,||\, 3^-_{001}\right] \left[2_{001} \,||\, \bar{6}^-_{001}\right] \left[E \,||\, m_{100}\right] \left[2_{001} \,||\, 2_{210}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right]$
$\left[E \,||\, m_{110}\right] \left[E \,||\, m_{010}\right] \left[2_{001} \,||\, 2_{120}\right] \left[2_{001}T \,||\, 1\right] \left[T \,||\, \bar{6}^+_{001}\right] \left[2_{001}T \,||\, 3^+_{001}\right] \left[T \,||\, m_{001}\right] \left[2_{001}T \,||\, 3^-_{001}\right] \left[T \,||\, \bar{6}^-_{001}\right]$
$\left[2_{001}T \,||\, m_{100}\right] \left[T \,||\, 2_{210}\right] \left[T \,||\, 2_{1\bar{1}0}\right] \left[2_{001}T \,||\, m_{110}\right] \left[2_{001}T \,||\, m_{010}\right] \left[T \,||\, 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, \bar{6}^+_{001}\right] \left[2_{001} \,||\, \bar{6}^-_{001}\right]$
C_3 : $\left[E \,||\, 3^+_{001}\right] \left[E \,||\, 3^-_{001}\right]$
C_4 : $\left[2_{001} \,||\, m_{001}\right]$
C_5 : $\left[E \,||\, m_{100}\right] \left[E \,||\, m_{110}\right] \left[E \,||\, m_{010}\right]$
C_6 : $\left[2_{001} \,||\, 2_{210}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right] \left[2_{001} \,||\, 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \,||\, \bar{6}^+_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{001}T \,||\, 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 416.3.919 : ${}^{2_z}\bar{6}^1 m^{2_z} 2^{m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{6}^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel \bar{6}^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel \bar{6}^+_{001}\right] \left[2_{100}T \parallel 3^+_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel 3^-_{001}\right] \left[2_{010}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^+_{001}\right] \left[2_{001} \parallel \bar{6}^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 416.4.920 : ${}^{2_z}\bar{6}^1 m^{2_z} 2^{\infty_x} 1 \rightarrow {}^{2_z}\bar{6}^1 m^{2_z} 2^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right]$
$\left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right]$
$\left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2
χ_3	4	0	−2	0	−4	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	σ_0	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_0$	$\sigma_z \mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix} \mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} \mathcal{K}$

No 417.1.921 : ${}^{m}\bar{6}^{1}m^{m}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[2_{001}T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 417.2.922 : ${}^{m_z}\bar{6}^1 m^{m_z} 2^{\infty_x} 1 \to {}^{m}\bar{6}^1 m^{m} 2$

No 418.1.923 : ${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid \bar{6}^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[T \mid\mid m_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[T \mid\mid \bar{6}^{-}_{001}\right] \left[E \mid\mid m_{100}\right] \left[T \mid\mid 2_{210}\right] \left[T \mid\mid 2_{1\bar{1}0}\right]$
$\left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{010}\right] \left[T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{110}\right] \left[E \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 418.2.924 : ${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \,||\, 1\right] \left[T \,||\, \bar{6}^{+}_{001}\right] \left[E \,||\, 3^{+}_{001}\right] \left[T \,||\, m_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[T \,||\, \bar{6}^{-}_{001}\right] \left[E \,||\, m_{100}\right] \left[T \,||\, 2_{210}\right] \left[T \,||\, 2_{1\bar{1}0}\right]$
$\left[E \,||\, m_{110}\right] \left[E \,||\, m_{010}\right] \left[T \,||\, 2_{120}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z T \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z T \,||\, m_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\Sigma_z T \,||\, \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z T \,||\, 2_{210}\right] \left[\Sigma_z T \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z \,||\, m_{110}\right] \left[\Sigma_z \,||\, m_{010}\right] \left[\Sigma_z T \,||\, 2_{120}\right] \left[\bar{E} \,||\, 1\right] \left[\bar{E}T \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E} \,||\, 3^{+}_{001}\right]$
$\left[\bar{E}T \,||\, m_{001}\right] \left[\bar{E} \,||\, 3^{-}_{001}\right] \left[\bar{E}T \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E} \,||\, m_{100}\right] \left[\bar{E}T \,||\, 2_{210}\right] \left[\bar{E}T \,||\, 2_{1\bar{1}0}\right] \left[\bar{E} \,||\, m_{110}\right] \left[\bar{E} \,||\, m_{010}\right] \left[\bar{E}T \,||\, 2_{120}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z T \,||\, \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \,||\, m_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, 2_{210}\right] \left[\bar{\Sigma}_z T \,||\, 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \,||\, m_{110}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z T \,||\, 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_7 :	$\left[\bar{E} \,||\, 1\right]$
C_2 :	$\left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, 3^{-}_{001}\right]$	C_8 :	$\left[\bar{E} \,||\, 3^{+}_{001}\right] \left[\bar{E} \,||\, 3^{-}_{001}\right]$
C_3 :	$\left[E \,||\, m_{100}\right] \left[E \,||\, m_{110}\right] \left[E \,||\, m_{010}\right]$	C_9 :	$\left[\bar{E} \,||\, m_{100}\right] \left[\bar{E} \,||\, m_{110}\right] \left[\bar{E} \,||\, m_{010}\right]$
C_4 :	$\left[\Sigma_z \,||\, 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \,||\, 1\right]$
C_5 :	$\left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right]$
C_6 :	$\left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z \,||\, m_{110}\right] \left[\Sigma_z \,||\, m_{010}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{110}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2	0	0	0
χ_3^{co}	4	−2	0	0	0	0	−4	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \,||\, 3^{+}_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[T \,||\, \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 418.3.925 : ${}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{6}^{1}m^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2\right.$
$\left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_4 : $\left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[E \mid\mid m_{100}\right]$	$\left[2_{100} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{100}T \mid\mid 1\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 419.1.926 : ${}^{2}\bar{6}^{2}m^{1}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 419.2.927 : ${}^{2_z}\bar{6}^{2_z}m^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 2_{120}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 419.3.928 : ${}^{2_z}\bar{6}^{2_z}m^{1}2^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 419.4.929 : ${}^{2_z}\bar{6}^{2_z}m^1 2^{\infty_x} 1 \rightarrow {}^{2_z}\bar{6}^{2_z}m^1 2^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1	2	0	2	2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2
χ_3	4	0	−2	0	−4	2	0

Irrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	$\sigma_x \mathcal{K}$	$\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_2	$\sigma_x \mathcal{K}$	$-\sigma_x \mathcal{K}$	$\sigma_z \mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix} \mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \sigma_x & 0 \end{pmatrix} \mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} \mathcal{K}$

No 420.1.930 : ${}^{m}\bar{6}^{m}m^{1}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[E \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel 2_{210}\right]$	$\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 420.2.931 : ${}^{m_z}\bar{6}^{m_z}m^{1}2^{\infty_x}1 \to {}^{m}\bar{6}^{m}m^{1}2$

No 421.1.932 : ${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel 2_{210}\right]$	$\left[T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 421.2.933 : ${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_8 :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_3 :	$\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$	C_9 :	$\left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{120}\right]$
C_4 :	$\left[\Sigma_z \parallel 1\right]$	C_{10} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_5 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_6 :	$\left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{120}\right]$	C_{12} :	$\left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2	0	0	0
χ_2^{co}	2	2	−2	0	0	0	−2	−2	2	0	0	0
χ_3^{co}	4	−2	0	0	0	0	−4	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[E \parallel 2_{210}\right]$	$\left[T \parallel \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 421.3.934 : ${}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{6}^{\bar{1}}m^{1}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[E \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{120}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel\right.$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel 2_{210}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{120}\right]$
C_4 : $\left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	2	0	−2	−2	−2
χ_2^{co}	2	2	−2	0	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{210}\right]$	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1^* & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 422.1.935 : ${}^{2_z}\bar{6}^{2_x}m^{2_y}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 422.2.936 : ${}^{2_z}\bar{6}^{2_x}m^{2_y}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel 2_{120}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 422.3.937 : ${}^{2_z}\bar{6}^{2_x}m^{2_y}2^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel 2_{120}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 422.4.938 : ${}^{2_z}\bar{6}^{2_x}m^{2_y}2^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel 2_{120}\right]$ $\left[2_{010}T \parallel 1\right]$ $\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010}T \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[2_{010}T \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 423.1.939 : ${}^{2_z}\bar{6}^{m_x}m^{m_y}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid 2_{210}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 424.1.940 : ${}^{m_x}\bar{6}^{m_y}m^{2_z}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{210}\right]$	$\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 425.1.941 : ${}^{m_x}\bar{6}^{2_z}m^{m_y}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{010}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 426.1.942 : ${}^{2}\bar{6}^{m}m^{\bar{1}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[T \mid\mid 2_{210}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[2_{001}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 426.2.943 : ${}^{2_z}\bar{6}^{m_z}m^{\bar{1}}2^{\infty_x}1 \rightarrow {}^{2_z}\bar{6}^{m_z}m^{\bar{1}}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_3^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\alpha^*\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 427.1.944 : ${}^{2}\bar{6}^{\bar{1}}m^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[T \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid 2_{210}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid m_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$	$\left[T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 427.2.945 : ${}^{2_z}\bar{6}^{\bar{1}}m^{m_z}2^{\infty_x}1 \rightarrow {}^{2_z}\bar{6}^{\bar{1}}m^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_3^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$	$\left[T \parallel m_{100}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\alpha^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 428.1.946 : ${}^{\bar{1}}\bar{6}^{m}m^{2}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[T \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid 2_{210}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{001}T \mid\mid m_{110}\right]$ $\left[2_{001}T \mid\mid m_{010}\right]$ $\left[2_{001} \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid 2_{210}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{210}\right]$	$\left[T \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 428.2.947 : ${}^{\bar{1}}\bar{6}^{m_z}m^{2_z}2^{\infty_x}1 \rightarrow {}^{\bar{1}}\bar{6}^{m_z}m^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[T \mid\mid \bar{6}^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[T \mid\mid m_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[T \mid\mid \bar{6}^{-}_{001}\right] \left[2_{100}T \mid\mid m_{100}\right] \left[2_{100} \mid\mid 2_{210}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{100}T \mid\mid m_{110}\right] \left[2_{100}T \mid\mid m_{010}\right] \left[2_{100} \mid\mid 2_{120}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z T \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\Sigma_z T \mid\mid m_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{110}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{120}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E}T \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E}T \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \mid\mid m_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{210}\right] \left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \mid\mid m_{110}\right] \left[\bar{E}2_{100}T \mid\mid m_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{120}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z T \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \mid\mid 2_{210}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{120}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z 2_{100} \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{120}\right] \left[\bar{E}2_{100} \mid\mid 2_{210}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{120}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{120}\right]$
C_4 : $\left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \mid\mid 1\right]$
C_7 : $\left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1
χ_3^{co}	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{100} \mid\mid 2_{210}\right]$	$\left[T \mid\mid \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 429.1.948 : ${}^{\bar{1}}\bar{6}^{2}m^{m}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[T \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid 2_{210}\right]$ $\left[2_{001}T \mid\mid 2_{1\bar{1}0}\right]$
$\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{100}\right]$ $\left[2_{001} \mid\mid m_{110}\right]$ $\left[2_{001} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \mid\mid 3^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{100}\right]$	$\left[T \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 429.2.949 : ${}^{\bar{1}}\bar{6}^{2_z}m^{m_z}2^{\infty_x}1 \rightarrow {}^{\bar{1}}\bar{6}^{2_z}m^{m_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1
χ_3^{co}	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 430.1.950 : ${}^{m}\bar{6}^{\bar{1}}m^{2}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	-1
χ_3^{co}	2	-1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{210}\right]$	$\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 430.2.951 : ${}^{m_z}\bar{6}^{\bar{1}}m^{2_z}2^{\infty_x}1 \rightarrow {}^{m_z}\bar{6}^{\bar{1}}m^{2_z}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{210}\right]$	$\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 431.1.952 : ${}^{m}\bar{6}^{2}m^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 431.2.953 : ${}^{m_z}\bar{6}^{2_z}m^{\bar{1}}2^{\infty_x}1 \rightarrow {}^{m_z}\bar{6}^{2_z}m^{\bar{1}}2^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[T \parallel 2_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel 2_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{100}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right]$
C_4 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_5 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_6 : $\left[\bar{E} \parallel 1\right]$
C_7 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	0	0	−2	−2
χ_2^{co}	4	−2	0	0	0	−4	2

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel \bar{6}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 432.1.954 : ${}^{3_z}\bar{6}^{2_x}m^{2_{xy}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$ $\left[3^-_{001} \parallel \bar{6}^-_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right]$ $\left[2_{110} \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[3^-_{001} \parallel \bar{6}^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[3^+_{001} \parallel \bar{6}^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 432.2.955 : ${}^{3_z}\bar{6}^{2_x}m^{2_{xy}}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001} \parallel \bar{6}^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[E \parallel m_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{001}T \parallel 1\right] \left[6^-_{001}T \parallel \bar{6}^+_{001}\right] \left[6^+_{001}T \parallel 3^+_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[6^-_{001}T \parallel 3^-_{001}\right] \left[6^+_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right] \left[2_{1\bar{1}0}T \parallel 2_{210}\right] \left[2_{210}T \parallel 2_{1\bar{1}0}\right] \left[2_{210}T \parallel m_{110}\right] \left[2_{1\bar{1}0}T \parallel m_{010}\right] \left[2_{120}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel \bar{6}^+_{001}\right] \left[3^-_{001} \parallel \bar{6}^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel \bar{6}^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 433.1.956 : ${}^{3_z}\bar{6}^{m_x}m^{m_{xy}}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^+_{001} \mid\mid \bar{6}^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^+_{001}\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[3^+_{001} \mid\mid 3^-_{001}\right]$ $\left[3^-_{001} \mid\mid \bar{6}^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid 2_{210}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{110}T \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid \bar{6}^+_{001}\right]$
C_3 : $\left[3^-_{001} \mid\mid 3^+_{001}\right]$
C_4 : $\left[E \mid\mid m_{001}\right]$
C_5 : $\left[3^+_{001} \mid\mid 3^-_{001}\right]$
C_6 : $\left[3^-_{001} \mid\mid \bar{6}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid \bar{6}^+_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 434.1.957 : ${}^{6_z}\bar{6}^{2_x}m^{2_1}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1	1	1	1	1	1	1
χ_2	1	−1	1	−1	1	−1
χ_3	1	−1	1	−1	−1	1
χ_4	1	1	1	1	−1	−1
χ_5	2	−1	−1	2	0	0
χ_6	2	1	−1	−2	0	0

Irrepresntations of group generators

	$\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$
Γ_1	1	1
Γ_2	−1	1
Γ_3	−1	−1
Γ_4	1	−1
Γ_5	ω_1^*	ω_2
Γ_6	ω_3	ω_2

No 434.2.958 : ${}^{6_z}\bar{6}^{2_x}m^{2_1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$
$\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{120} \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[3^-_{001}T \parallel \bar{6}^+_{001}\right]$ $\left[6^-_{001}T \parallel 3^+_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[6^+_{001}T \parallel 3^-_{001}\right]$ $\left[3^+_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{110}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{110}\right]$ $\left[2_{210}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{210} \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{120} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	−1	1	−1	1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 435.1.959 : ${}^{\bar{3}}{}_{z}\bar{6}^{m_x}m^{2_{xy}}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{110}T \parallel m_{010}\right]$ $\left[2_{100} \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$	$\left[2_{110} \parallel 2_{210}\right]$	$\left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 436.1.960 : ${}^{\bar{3}}{}_{z}\bar{6}^{2}{}_{x}m^{m}{}_{xy}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[3^{+}_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$ $\left[3^{-}_{001}T \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110}T \mid\mid 2_{210}\right]$ $\left[2_{010}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$ $\left[2_{100}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$ $\left[3^{+}_{001} \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{010} \mid\mid m_{110}\right]$ $\left[2_{110} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^{-}_{001} \mid\mid 3^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[3^{+}_{001}T \mid\mid \bar{6}^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 437.1.961 : ${}^{6_z}\bar{6}^{m_x}m^{m_1}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{210}T \parallel 2_{210}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{120}T \parallel 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right]$
C_4 : $\left[2_{001} \parallel m_{001}\right]$
C_5 : $\left[3^-_{001} \parallel 3^-_{001}\right]$
C_6 : $\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	-1	1	-1	1	-1
χ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_5^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel \bar{6}^+_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$
Γ_1^{co}	1	$\mathcal{K}$
Γ_2^{co}	-1	$\mathcal{K}$
Γ_3^{co}	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$e^{\frac{\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-\pi i}{3}}$	$\mathcal{K}$

No 438.1.962 : ${}^{\bar{6}_z}\bar{6}^{m_x}m^{2_1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001}T \mid\mid \bar{6}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001}T \mid\mid \bar{6}^-_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{210} \mid\mid 2_{210}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{110}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{120} \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{210} \mid\mid 2_{210}\right]$ $\left[2_{1\bar{1}0} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{120} \mid\mid 2_{120}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{210} \mid\mid 2_{210}\right]$	$\left[6^+_{001}T \mid\mid \bar{6}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$

No 439.1.963 : ${}^{\bar{6}_z}\bar{6}^{2_x}m^{m_1}2$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001}T \mid\mid \bar{6}^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001}T \mid\mid \bar{6}^-_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{210}T \mid\mid 2_{210}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{120}T \mid\mid 2_{120}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_3 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$

Co-character table

	C_1	C_2	C_3
χ_1^{co}	1	1	1
χ_2^{co}	1	1	−1
χ_3^{co}	2	−1	0

Coirrepresntations of group generators

	$\left[3^+_{001} \mid\mid 3^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[6^+_{001}T \mid\mid \bar{6}^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	σ_x	$\sigma_0\mathcal{K}$

No 440.1.964 : ${}^{1}6/{}^{1}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{210}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[E \parallel 2_{120}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 440.2.965 : ${}^{1}6/{}^{1}m{}^{1}m{}^{1}m{}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[2_{001}T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 440.3.966 : ${}^{1}6/{}^{1}m{}^{1}m{}^{1}m^{\infty_z}1 \rightarrow {}^{1}6/{}^{1}m{}^{1}m{}^{1}m$

No 440.4.967 : ${}^{1}6/{}^{1}m{}^{1}m{}^{1}m{}^{\infty_z m_x}1 \to {}^{1}6/{}^{1}m{}^{1}m{}^{1}m{}^{m_x}1$

No 441.1.968 : ${}^{2}6/{}^{2}m^{1}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 441.2.969 : ${}^{2_z}6/{}^{2_z}m^1m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^-_{001}\right] \left[2_{001} \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^+_{001}\right] \left[E \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[2_{001}T \parallel 3^+_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^-_{001}\right] \left[T \parallel \bar{6}^-_{001}\right] \left[2_{001}T \parallel -1\right] \left[T \parallel \bar{6}^+_{001}\right] \left[2_{001}T \parallel \bar{3}^+_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^+_{001}\right] \left[2_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{3}^-_{001}\right] \left[E \parallel \bar{3}^+_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^-_{001}\right] \left[2_{001} \parallel \bar{6}^+_{001}\right]$
C_8 : $\left[E \parallel -1\right]$
C_9 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_{10} : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 441.3.970 : ${}^{2_z}6/{}^{2_z}m^1m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 441.4.971 : ${}^{2_z}6/{}^{2_z}m^1m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{2_z}m^1m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[E \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E} 2_{100} \parallel 6^-_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^-_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E} \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{210}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{120}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel m_{120}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel 6^+_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^-_{001}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E} 2_{100} \parallel m_{210}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{120}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$

C_4 : $\left[E \parallel \bar{3}^-_{001}\right] \left[E \parallel \bar{3}^+_{001}\right]$

C_5 : $\left[E \parallel -1\right]$

C_6 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$

C_7 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$

C_{10} : $\left[\bar{E} \parallel \bar{3}^-_{001}\right] \left[\bar{E} \parallel \bar{3}^+_{001}\right]$

C_{11} : $\left[\bar{E} \parallel -1\right]$

C_{12} : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

$C_{13}:\ \left[\bar{E}\ ||\ 2_{010}\right]\left[\bar{E}\ ||\ 2_{110}\right]\left[\bar{E}\ ||\ 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	0	2	2	2	2	2	−2	−2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	−2	2	−2	−2	2	2	2	−2
χ_3	2	0	2	2	2	−2	−2	−2	−2	−2	−2	2	2
χ_4	2	0	2	−2	−2	2	−2	−2	−2	2	2	−2	2
χ_5	4	0	−2	−2	4	0	0	−4	2	2	−4	0	0
χ_6	4	0	−2	2	−4	0	0	−4	2	−2	4	0	0

Irrepresntations of group generators

	$\left[2_{100}\ ||\ 6^{+}_{001}\right]$	$\left[2_{100}\ ||\ m_{001}\right]$	$\left[E\ ||\ m_{100}\right]$	$\left[\Sigma_z\ ||\ 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_4	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1^* & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_3^* \\ \omega_3 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_4 & 0 \\ 0 & \omega_4^* \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 442.1.972 : ${}^{m}6/{}^{m}m^{1}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 442.2.973 : ${}^{m_z}6/{}^{m_z}m^1m^{m_z}m^{\infty_x}1 \to {}^{m}6/{}^{m}m^1m^{m}m$

No 443.1.974 : ${}^{\bar{1}}6/{}^{\bar{1}}m^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 443.2.975 : ${}^{\bar{1}}6/{}^{\bar{1}}m^{1}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}6/{}^{\bar{1}}m^{1}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_3 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
C_4 :	$\left[E \parallel -1\right]$	C_{16} :	$\left[\bar{E} \parallel -1\right]$
C_5 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$	C_{17} :	$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$
C_6 :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$	C_{18} :	$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel -1\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel -1\right]$
C_{11} :	$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_{12} :	$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	2	−2	−2	−2	2	0	0	0	0	0	0
χ_3^{co}	2	2	2	2	−2	−2	0	0	0	0	0	0
χ_4^{co}	2	2	−2	−2	2	−2	0	0	0	0	0	0

χ_5^{co}	4	−2	−2	4	0	0	0	0	0	0	0	0
χ_6^{co}	4	−2	2	−4	0	0	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}
χ_1^{co}	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0
χ_2^{co}	−2	−2	2	2	2	−2	0	0	0	0	0	0
χ_3^{co}	−2	−2	−2	−2	2	2	0	0	0	0	0	0
χ_4^{co}	−2	−2	2	2	−2	2	0	0	0	0	0	0
χ_5^{co}	−4	2	2	−4	0	0	0	0	0	0	0	0
χ_6^{co}	−4	2	−2	4	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6^{co}	$\begin{pmatrix} \omega_3^* & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \omega_4 & 0 \\ 0 & \omega_4 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 443.3.976 : ${}^{\bar{1}}6/{}^{\bar{1}}m^{1}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6/{}^{\bar{1}}m^{1}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}\right.$
$\left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{10}\right.$
$\left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right.$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel\right.$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{1}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_7 : $\left[2_{100} \,||\, 6^{+}_{001}\right] \left[2_{100} \,||\, 2_{001}\right] \left[2_{100} \,||\, 6^{-}_{001}\right] \left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[2_{100} \,||\, m_{210}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[2_{100} \,||\, m_{120}\right] \left[2_{100} \,||\, 2_{120}\right] \left[2_{100} \,||\, 2_{1\bar{1}0}\right] \left[2_{100} \,||\, 2_{210}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 3^{+}_{001}\right] \left[\Sigma_z \,||\, 3^{-}_{001}\right] \left[\Sigma_z \,||\, \bar{3}^{-}_{001}\right]$
$\left[\Sigma_z \,||\, -1\right] \left[\Sigma_z \,||\, \bar{3}^{+}_{001}\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z \,||\, m_{110}\right] \left[\Sigma_z \,||\, m_{010}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 2_{110}\right] \left[\Sigma_z \,||\, 2_{100}\right]$
$\left[\Sigma_z 2_{100} \,||\, 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \,||\, 2_{001}\right] \left[\Sigma_z 2_{100} \,||\, 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \,||\, m_{210}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{120}\right] \left[\Sigma_z 2_{100} \,||\, 2_{120}\right] \left[\Sigma_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, 2_{210}\right] \left[\bar{E} 2_{100} \,||\, 6^{+}_{001}\right] \left[\bar{E} 2_{100} \,||\, 2_{001}\right] \left[\bar{E} 2_{100} \,||\, 6^{-}_{001}\right] \left[\bar{E} 2_{100} \,||\, m_{001}\right]$
$\left[\bar{E} 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \,||\, m_{210}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, m_{120}\right] \left[\bar{E} 2_{100} \,||\, 2_{120}\right] \left[\bar{E} 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, 2_{210}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{110}\right]$
$\left[\bar{\Sigma}_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{110}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{210}\right]$

C_8 : $\left[\bar{E} \,||\, 1\right]$

C_9 : $\left[\bar{E} \,||\, 3^{+}_{001}\right] \left[\bar{E} \,||\, 3^{-}_{001}\right]$

C_{10} : $\left[\bar{E} \,||\, \bar{3}^{-}_{001}\right] \left[\bar{E} \,||\, \bar{3}^{+}_{001}\right]$

C_{11} : $\left[\bar{E} \,||\, -1\right]$

C_{12} : $\left[\bar{E} \,||\, m_{100}\right] \left[\bar{E} \,||\, m_{110}\right] \left[\bar{E} \,||\, m_{010}\right]$

C_{13} : $\left[\bar{E} \,||\, 2_{010}\right] \left[\bar{E} \,||\, 2_{110}\right] \left[\bar{E} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	−2	2	0	−2	−2	2	2	2	−2
χ_3^{co}	2	2	2	2	−2	−2	0	−2	−2	−2	−2	2	2
χ_4^{co}	2	2	−2	−2	2	−2	0	−2	−2	2	2	−2	2
χ_5^{co}	4	−2	−2	4	0	0	0	−4	2	2	−4	0	0
χ_6^{co}	4	−2	2	−4	0	0	0	−4	2	−2	4	0	0

Coirrepresntations of group generators

	$\left[E \,||\, \bar{3}^{-}_{001}\right]$	$\left[E \,||\, m_{100}\right]$	$\left[2_{100} \,||\, 6^{+}_{001}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

Γ_6^{co}	$\begin{pmatrix} \omega_3^* & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \omega_4 & 0 \\ 0 & \omega_4^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 444.1.977 : ${}^{2}6/{}^{1}m^{2}m^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[E \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 444.2.978 : ${}^{2_z}6/{}^{1}m^{2_z}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 444.3.979 : ${}^{2_z}6/{}^{1}m^{2_z}m^{1}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 444.4.980 : ${}^{2_z}6/{}^{1}m^{2_z}m^{1}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{1}m^{2_z}m^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$

C_4 : $\left[E \parallel m_{001}\right]$

C_5 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$

C_6 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$

C_7 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

C_{10} : $\left[\bar{E} \parallel m_{001}\right]$

C_{11} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

C_{12} : $\left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{120}\right]$

$C_{13}:\ \left[\bar{E}\ ||\ 2_{010}\right]\left[\bar{E}\ ||\ 2_{110}\right]\left[\bar{E}\ ||\ 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	0	2	2	2	2	2	−2	−2	−2	−2	−2	−2
χ_2	2	0	2	−2	−2	2	−2	−2	−2	2	2	−2	2
χ_3	2	0	2	−2	−2	−2	2	−2	−2	2	2	2	−2
χ_4	2	0	2	2	2	−2	−2	−2	−2	−2	−2	2	2
χ_5	4	0	−2	4	−2	0	0	−4	2	−4	2	0	0
χ_6	4	0	−2	−4	2	0	0	−4	2	4	−2	0	0

Irrepresntations of group generators

	$\left[2_{100}\ ||\ 6^{+}_{001}\right]$	$\left[E\ ||\ m_{001}\right]$	$\left[2_{100}\ ||\ m_{100}\right]$	$\left[\Sigma_z\ ||\ 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	$-\sigma_0$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$\sigma_x\mathcal{K}$	σ_0	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_2 \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_2 \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 445.1.981 : ${}^{m}6/{}^{1}m^{m}m^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[E \parallel m_{210}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 445.2.982 : ${}^{m_z}6/{}^{1}m{}^{m_z}m{}^{1}m{}^{\infty_x}1 \to {}^{m}6/{}^{1}m{}^{m}m{}^{1}m$

No 446.1.983 : ${}^{\bar{1}}6/{}^{1}m{}^{\bar{1}}m{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[E \parallel m_{210}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 446.2.984 : ${}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_3 :	$\left[E \parallel m_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel m_{001}\right]$
C_4 :	$\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$	C_{16} :	$\left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$
C_5 :	$\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$	C_{17} :	$\left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{120}\right]$
C_6 :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$	C_{18} :	$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel m_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel m_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$
C_{11} :	$\left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{120}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
C_{12} :	$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	2	−2	−2	2	−2	0	0	0	0	0	0
χ_3^{co}	2	2	−2	−2	−2	2	0	0	0	0	0	0
χ_4^{co}	2	2	2	2	−2	−2	0	0	0	0	0	0

χ_5^{co}	4	−2	4	−2	0	0	0	0	0	0	0	0
χ_6^{co}	4	−2	−4	2	0	0	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}
χ_1^{co}	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0
χ_2^{co}	−2	−2	2	2	−2	2	0	0	0	0	0	0
χ_3^{co}	−2	−2	2	2	2	−2	0	0	0	0	0	0
χ_4^{co}	−2	−2	−2	−2	2	2	0	0	0	0	0	0
χ_5^{co}	−4	2	−4	2	0	0	0	0	0	0	0	0
χ_6^{co}	−4	2	4	−2	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}_{001}^{-}\right]$	$\left[E \parallel m_{210}\right]$	$\left[T \parallel 6_{001}^{+}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 446.3.985 : ${}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}6/{}^{1}m^{\bar{1}}m^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100} \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \right.$
$\left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{1}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_7 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right]$
$\left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right]$
$\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

C_{10} : $\left[\bar{E} \parallel m_{001}\right]$

C_{11} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

C_{12} : $\left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{120}\right]$

C_{13} : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	2	−2	0	−2	−2	2	2	−2	2
χ_3^{co}	2	2	−2	−2	−2	2	0	−2	−2	2	2	2	−2
χ_4^{co}	2	2	2	2	−2	−2	0	−2	−2	−2	−2	2	2
χ_5^{co}	4	−2	4	−2	0	0	0	−4	2	−4	2	0	0
χ_6^{co}	4	−2	−4	2	0	0	0	−4	2	4	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[E \parallel m_{210}\right]$	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

Γ_6^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1^* \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \omega_1 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 447.1.986 : ${}^{1}6/{}^{1}m^{2}m^{2}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, 6^{+}_{001}\right] \left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[E \,||\, 6^{-}_{001}\right] \left[E \,||\, m_{001}\right] \left[E \,||\, \bar{3}^{-}_{001}\right] \left[E \,||\, \bar{6}^{-}_{001}\right]$
$\left[E \,||\, -1\right] \left[E \,||\, \bar{6}^{+}_{001}\right] \left[E \,||\, \bar{3}^{+}_{001}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{210}\right] \left[2_{001} \,||\, m_{1\bar{1}0}\right] \left[2_{001} \,||\, m_{110}\right] \left[2_{001} \,||\, m_{010}\right] \left[2_{001} \,||\, m_{120}\right]$
$\left[2_{001} \,||\, 2_{120}\right] \left[2_{001} \,||\, 2_{010}\right] \left[2_{001} \,||\, 2_{110}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right] \left[2_{001} \,||\, 2_{210}\right] \left[2_{001} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 6^{+}_{001}\right] \left[E \,||\, 6^{-}_{001}\right]$
C_3 : $\left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, 3^{-}_{001}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$
C_5 : $\left[E \,||\, m_{001}\right]$
C_6 : $\left[E \,||\, \bar{3}^{-}_{001}\right] \left[E \,||\, \bar{3}^{+}_{001}\right]$
C_7 : $\left[E \,||\, \bar{6}^{-}_{001}\right] \left[E \,||\, \bar{6}^{+}_{001}\right]$
C_8 : $\left[E \,||\, -1\right]$
C_9 : $\left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{110}\right] \left[2_{001} \,||\, m_{010}\right]$
C_{10} : $\left[2_{001} \,||\, m_{210}\right] \left[2_{001} \,||\, m_{1\bar{1}0}\right] \left[2_{001} \,||\, m_{120}\right]$
C_{11} : $\left[2_{001} \,||\, 2_{120}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right] \left[2_{001} \,||\, 2_{210}\right]$
C_{12} : $\left[2_{001} \,||\, 2_{010}\right] \left[2_{001} \,||\, 2_{110}\right] \left[2_{001} \,||\, 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \,||\, 6^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{001} \,||\, m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 447.2.987 : ${}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	−1	−1	1	$\mathcal{K}$
Γ_6^{co}	−1	1	−1	$\mathcal{K}$
Γ_7^{co}	−1	−1	−1	$\mathcal{K}$
Γ_8^{co}	−1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 447.3.988 : ${}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 447.4.989 : ${}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}6/{}^{1}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[E \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_{10} : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_{11} : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{12} : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_{13} : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{14} : $\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$

$C_{15}: \left[\Sigma_z \,||\, \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{6}^+_{001}\right]$
$C_{16}: \left[\Sigma_z \,||\, \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{6}^-_{001}\right]$
$C_{17}: \left[\Sigma_z \,||\, \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^-_{001}\right]$
$C_{18}: \left[\bar{E} \,||\, 1\right]$
$C_{19}: \left[\bar{E} \,||\, 6^+_{001}\right] \left[\bar{E} \,||\, 6^-_{001}\right]$
$C_{20}: \left[\bar{E} \,||\, 3^+_{001}\right] \left[\bar{E} \,||\, 3^-_{001}\right]$
$C_{21}: \left[\bar{E} \,||\, 2_{001}\right]$
$C_{22}: \left[\bar{E} \,||\, m_{001}\right]$
$C_{23}: \left[\bar{E} \,||\, \bar{3}^-_{001}\right] \left[\bar{E} \,||\, \bar{3}^+_{001}\right]$
$C_{24}: \left[\bar{E} \,||\, \bar{6}^-_{001}\right] \left[\bar{E} \,||\, \bar{6}^+_{001}\right]$
$C_{25}: \left[\bar{E} \,||\, -1\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}	C_{25}
χ_1	2	2	2	2	2	2	2	2	0	0	0	0	0	0	0	0	0	−2	−2	−2	−2	−2	−2	−2	−2
χ_2	2	2	2	2	−2	−2	−2	−2	0	0	0	0	0	0	0	0	0	−2	−2	−2	−2	2	2	2	2
χ_3	2	−2	2	−2	−2	2	−2	2	0	0	0	0	0	0	0	0	0	−2	2	−2	2	2	−2	2	−2
χ_4	2	−2	2	−2	2	−2	2	−2	0	0	0	0	0	0	0	0	0	−2	2	−2	2	−2	2	−2	2
χ_5	2	−1	−1	2	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2	−2	1	1	−2
χ_6	2	−1	−1	2	−2	1	1	−2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2	2	−1	−1	2
χ_7	2	1	−1	−2	−2	−1	1	2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	−2	−1	1	2	2	1	−1	−2
χ_8	2	1	−1	−2	2	1	−1	−2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2	−2	−1	1	2
χ_9	2	−1	−1	2	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2	−2	1	1	−2
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2	2	−1	−1	2
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2	2	1	−1	−2
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	−2	−1	1	2	−2	−1	1	2

Irrepresntations of group generators

	$\left[E \,||\, 6^+_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

Γ_3	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	ω_1^*	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6	ω_1^*	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_7	ω_3	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_8	ω_3	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_9	ω_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_{10}	ω_1	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_{11}	ω_3^*	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_{12}	ω_3^*	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 448.1.990 : ${}^{1}6/{}^{1}m^{m}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{210}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[2_{001}T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_7 : $\left[E \parallel m_{001}\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ C_8 : $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ C_9 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$ C_{10} : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$ C_{11} : $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$ C_{12} : $\left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{001}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 448.2.991 : ${}^{1}6/{}^{1}m^{m_z}m^{m_z}m^{\infty_x}1 \to {}^{1}6/{}^{1}m^{m}m^{m}m$

No 449.1.992 : ${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{210}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_7 : $\left[E \parallel m_{001}\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ C_8 : $\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ C_9 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$ C_{10} : $\left[E \parallel -1\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$ C_{11} : $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$ C_{12} : $\left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 449.2.993 : ${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_{25} : \left[\bar{E} \parallel 1\right]$
$C_2 : \left[E \parallel 6^{+}_{001}\right]$ $C_{26} : \left[\bar{E} \parallel 6^{+}_{001}\right]$
$C_3 : \left[E \parallel 3^{+}_{001}\right]$ $C_{27} : \left[\bar{E} \parallel 3^{+}_{001}\right]$
$C_4 : \left[E \parallel 2_{001}\right]$ $C_{28} : \left[\bar{E} \parallel 2_{001}\right]$
$C_5 : \left[E \parallel 3^{-}_{001}\right]$ $C_{29} : \left[\bar{E} \parallel 3^{-}_{001}\right]$
$C_6 : \left[E \parallel 6^{-}_{001}\right]$ $C_{30} : \left[\bar{E} \parallel 6^{-}_{001}\right]$
$C_7 : \left[E \parallel m_{001}\right]$ $C_{31} : \left[\bar{E} \parallel m_{001}\right]$
$C_8 : \left[E \parallel \bar{3}^{-}_{001}\right]$ $C_{32} : \left[\bar{E} \parallel \bar{3}^{-}_{001}\right]$
$C_9 : \left[E \parallel \bar{6}^{-}_{001}\right]$ $C_{33} : \left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
$C_{10} : \left[E \parallel -1\right]$ $C_{34} : \left[\bar{E} \parallel -1\right]$
$C_{11} : \left[E \parallel \bar{6}^{+}_{001}\right]$ $C_{35} : \left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$
$C_{12} : \left[E \parallel \bar{3}^{+}_{001}\right]$ $C_{36} : \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
$C_{13} : \left[\Sigma_z \parallel 1\right]$ $C_{37} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_{14} : \left[\Sigma_z \parallel 6^{+}_{001}\right]$ $C_{38} : \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
$C_{15} : \left[\Sigma_z \parallel 3^{+}_{001}\right]$ $C_{39} : \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
$C_{16} : \left[\Sigma_z \parallel 2_{001}\right]$ $C_{40} : \left[\bar{\Sigma}_z \parallel 2_{001}\right]$
$C_{17} : \left[\Sigma_z \parallel 3^{-}_{001}\right]$ $C_{41} : \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
$C_{18} : \left[\Sigma_z \parallel 6^{-}_{001}\right]$ $C_{42} : \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
$C_{19} : \left[\Sigma_z \parallel m_{001}\right]$ $C_{43} : \left[\bar{\Sigma}_z \parallel m_{001}\right]$
$C_{20} : \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right]$ $C_{44} : \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$

$C_{21}: \left[\Sigma_z \,||\, \bar{6}^{-}_{001}\right]$ $C_{45}: \left[\bar{\Sigma}_z \,||\, \bar{6}^{-}_{001}\right]$
$C_{22}: \left[\Sigma_z \,||\, -1\right]$ $C_{46}: \left[\bar{\Sigma}_z \,||\, -1\right]$
$C_{23}: \left[\Sigma_z \,||\, \bar{6}^{+}_{001}\right]$ $C_{47}: \left[\bar{\Sigma}_z \,||\, \bar{6}^{+}_{001}\right]$
$C_{24}: \left[\Sigma_z \,||\, \bar{3}^{+}_{001}\right]$ $C_{48}: \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}
χ_1^{co}	2	2	2	2	2	2	2	2	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	0
χ_2^{co}	2	2	2	2	2	2	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0	0	0	0	0	0	0
χ_3^{co}	2	−2	2	−2	2	−2	−2	2	−2	2	−2	2	0	0	0	0	0	0	0	0	0	0	0	0
χ_4^{co}	2	−2	2	−2	2	−2	2	−2	2	−2	2	−2	0	0	0	0	0	0	0	0	0	0	0	0
χ_5^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_6^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_7^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_8^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_9^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_{10}^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_{11}^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_{12}^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0

	C_{25}	C_{26}	C_{27}	C_{28}	C_{29}	C_{30}	C_{31}	C_{32}	C_{33}	C_{34}	C_{35}	C_{36}	C_{37}	C_{38}	C_{39}	C_{40}	C_{41}	C_{42}	C_{43}	C_{44}	C_{45}	C_{46}	C_{47}	C_{48}
χ_1^{co}	−2	−2	−2	−2	−2	−2	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0	0	0	0	0	0	0
χ_2^{co}	−2	−2	−2	−2	−2	−2	2	2	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	0
χ_3^{co}	−2	2	−2	2	−2	2	2	−2	2	−2	2	−2	0	0	0	0	0	0	0	0	0	0	0	0
χ_4^{co}	−2	2	−2	2	−2	2	−2	2	−2	2	−2	2	0	0	0	0	0	0	0	0	0	0	0	0
χ_5^{co}	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_6^{co}	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_7^{co}	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_8^{co}	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0

χ_9^{co}	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_{10}^{co}	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_{11}^{co}	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0
χ_{12}^{co}	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 6_{001}^{+}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5^*	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_5^*	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_7^{co}	ω_7	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_8^{co}	ω_7	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_9^{co}	ω_5	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	ω_5	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	ω_7^*	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	ω_7^*	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$

No 449.3.994 : ${}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}6/{}^{1}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{10}\right.$
$\left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right.$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[E \parallel -1\right]$

C_9 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{210}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, m_{120}\right] \left[2_{100} \,||\, 2_{120}\right] \left[2_{100} \,||\, 2_{010}\right]$
$\left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, 2_{1\bar{1}0}\right] \left[2_{100} \,||\, 2_{210}\right] \left[2_{100} \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, m_{001}\right] \left[\Sigma_z \,||\, -1\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{210}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{120}\right] \left[\Sigma_z 2_{100} \,||\, 2_{120}\right] \left[\Sigma_z 2_{100} \,||\, 2_{010}\right]$
$\left[\Sigma_z 2_{100} \,||\, 2_{110}\right] \left[\Sigma_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, 2_{210}\right] \left[\Sigma_z 2_{100} \,||\, 2_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{210}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right]$
$\left[\bar{E} 2_{100} \,||\, m_{010}\right] \left[\bar{E} 2_{100} \,||\, m_{120}\right] \left[\bar{E} 2_{100} \,||\, 2_{120}\right] \left[\bar{E} 2_{100} \,||\, 2_{010}\right] \left[\bar{E} 2_{100} \,||\, 2_{110}\right] \left[\bar{E} 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, 2_{210}\right] \left[\bar{E} 2_{100} \,||\, 2_{100}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{001}\right] \left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{100}\right]$

C_{10} : $\left[\Sigma_z \,||\, 6^+_{001}\right] \left[\bar{\Sigma}_z \,||\, 6^-_{001}\right]$

C_{11} : $\left[\Sigma_z \,||\, 3^+_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{001}\right]$

C_{12} : $\left[\Sigma_z \,||\, 3^-_{001}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{001}\right]$

C_{13} : $\left[\Sigma_z \,||\, 6^-_{001}\right] \left[\bar{\Sigma}_z \,||\, 6^+_{001}\right]$

C_{14} : $\left[\Sigma_z \,||\, \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^+_{001}\right]$

C_{15} : $\left[\Sigma_z \,||\, \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{6}^+_{001}\right]$

C_{16} : $\left[\Sigma_z \,||\, \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{6}^-_{001}\right]$

C_{17} : $\left[\Sigma_z \,||\, \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^-_{001}\right]$

C_{18} : $\left[\bar{E} \,||\, 1\right]$

C_{19} : $\left[\bar{E} \,||\, 6^+_{001}\right] \left[\bar{E} \,||\, 6^-_{001}\right]$

C_{20} : $\left[\bar{E} \,||\, 3^+_{001}\right] \left[\bar{E} \,||\, 3^-_{001}\right]$

C_{21} : $\left[\bar{E} \,||\, 2_{001}\right]$

C_{22} : $\left[\bar{E} \,||\, m_{001}\right]$

C_{23} : $\left[\bar{E} \,||\, \bar{3}^-_{001}\right] \left[\bar{E} \,||\, \bar{3}^+_{001}\right]$

C_{24} : $\left[\bar{E} \,||\, \bar{6}^-_{001}\right] \left[\bar{E} \,||\, \bar{6}^+_{001}\right]$

C_{25} : $\left[\bar{E} \,||\, -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}	C_{25}
χ_1^{co}	2	2	2	2	2	2	2	2	0	0	0	0	0	0	0	0	0	−2	−2	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	2	−2	−2	−2	−2	0	0	0	0	0	0	0	0	0	−2	−2	−2	−2	2	2	2	2
χ_3^{co}	2	−2	2	−2	−2	2	−2	2	0	0	0	0	0	0	0	0	0	−2	2	−2	2	2	−2	2	−2
χ_4^{co}	2	−2	2	−2	2	−2	2	−2	0	0	0	0	0	0	0	0	0	−2	2	−2	2	−2	2	−2	2
χ_5^{co}	4	−2	−2	4	4	−2	−2	4	0	0	0	0	0	0	0	0	0	−4	2	2	−4	−4	2	2	−4
χ_6^{co}	4	−2	−2	4	−4	2	2	−4	0	0	0	0	0	0	0	0	0	−4	2	2	−4	4	−2	−2	4
χ_7^{co}	4	2	−2	−4	−4	−2	2	4	0	0	0	0	0	0	0	0	0	−4	−2	2	4	4	2	−2	−4
χ_8^{co}	4	2	−2	−4	4	2	−2	−4	0	0	0	0	0	0	0	0	0	−4	−2	2	4	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_6^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_7^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_8^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$

No 450.1.995 : ${}^{1}6/{}^{2}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_{10} : $\left[E \parallel m_{210}\right]$ $\left[E \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 450.2.996 : ${}^{1}6/{}^{2_z}m^{1}m^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_{10} : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 450.3.997 : ${}^{1}6/{}^{2}{}_{z}m^{1}m^{1}m^{m}{}_{x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_{10} : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 450.4.998 : ${}^{1}6/{}^{2_z}m^{1}m^{1}m^{\infty_x}1 \rightarrow {}^{1}6/{}^{2_z}m^{1}m^{1}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$

C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$

C_4 : $\left[E \parallel 2_{001}\right]$

C_5 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right]$
$\left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

C_6 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$

C_7 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$

C_{10} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

C_{11} : $\left[\bar{E} \parallel 2_{001}\right]$

C_{12} : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

$C_{13}:\ \left[\bar{E}\ ||\ m_{210}\right]\left[\bar{E}\ ||\ m_{1\bar{1}0}\right]\left[\bar{E}\ ||\ m_{120}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	0	2	2	−2	−2	−2	−2	−2	−2
χ_2	2	2	2	2	0	−2	−2	−2	−2	−2	−2	2	2
χ_3	2	−2	2	−2	0	−2	2	−2	2	−2	2	2	−2
χ_4	2	−2	2	−2	0	2	−2	−2	2	−2	2	−2	2
χ_5	4	−2	−2	4	0	0	0	−4	2	2	−4	0	0
χ_6	4	2	−2	−4	0	0	0	−4	−2	2	4	0	0

Irrepresntations of group generators

	$\left[E\ ||\ 6^{+}_{001}\right]$	$\left[2_{100}\ ||\ m_{001}\right]$	$\left[E\ ||\ m_{100}\right]$	$\left[\Sigma_z\ ||\ 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_3	$-\sigma_0$	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 451.1.999 : ${}^{1}6/{}^{m}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 451.2.1000 : ${}^{1}6/{}^{m_z}m{}^{1}m{}^{1}m{}^{\infty_x}1 \rightarrow {}^{1}6/{}^{m}m{}^{1}m{}^{1}m$

No 452.1.1001 : ${}^{1}6/{}^{\bar{1}}m{}^{1}m{}^{1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{3}^{-}_{001}\right]$ $\left[T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[T \mid\mid -1\right]$ $\left[T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{210}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid m_{120}\right]$
$\left[T \mid\mid 2_{120}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{210}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$
C_6 : $\left[E \mid\mid m_{210}\right]$ $\left[E \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 452.2.1002 : ${}^{1}6/^{\bar{1}}m^{1}m^{1}m^{\infty_z}1 \rightarrow {}^{1}6/^{\bar{1}}m^{1}m^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{16} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_5 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$	C_{17} :	$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$
C_6 :	$\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$	C_{18} :	$\left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{120}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_{11} :	$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
C_{12} :	$\left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{120}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	2	2	2	−2	−2	0	0	0	0	0	0
χ_3^{co}	2	−2	2	−2	−2	2	0	0	0	0	0	0
χ_4^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	0

χ_5^{co}	4	−2	−2	4	0	0	0	0	0	0	0	0
χ_6^{co}	4	2	−2	−4	0	0	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}
χ_1^{co}	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0
χ_2^{co}	−2	−2	−2	−2	2	2	0	0	0	0	0	0
χ_3^{co}	−2	2	−2	2	2	−2	0	0	0	0	0	0
χ_4^{co}	−2	2	−2	2	−2	2	0	0	0	0	0	0
χ_5^{co}	−4	2	2	−4	0	0	0	0	0	0	0	0
χ_6^{co}	−4	−2	2	4	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6_{001}^{+}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 452.3.1003 : ${}^{1}6/^{\bar{1}}m^{1}m^{1}m^{\infty_z m_x}1 \rightarrow {}^{1}6/^{\bar{1}}m^{1}m^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[E \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{10}\right.$
$\left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}\right.$
$\left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[E \parallel m_{210}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{120}\right]$

C_7 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z \parallel 6^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{210}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel m_{120}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right]$
$\left[\bar{E} 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{120}\right] \left[\bar{E} 2_{100} \parallel 2_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{210}\right] \left[\bar{E} 2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 6^-_{001}\right]$

C_{10} : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$

C_{11} : $\left[\bar{E} \parallel 2_{001}\right]$

C_{12} : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

C_{13} : $\left[\bar{E} \parallel m_{210}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	2	−2	−2	0	−2	−2	−2	−2	2	2
χ_3^{co}	2	−2	2	−2	−2	2	0	−2	2	−2	2	2	−2
χ_4^{co}	2	−2	2	−2	2	−2	0	−2	2	−2	2	−2	2
χ_5^{co}	4	−2	−2	4	0	0	0	−4	2	2	−4	0	0
χ_6^{co}	4	2	−2	−4	0	0	0	−4	−2	2	4	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

Γ_6^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 453.1.1004 : ${}^{1}6/{}^{2}m^{2}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{210}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[E \parallel 2_{120}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 453.2.1005 : ${}^{1}6/{}^{2_z}m{}^{2_z}m{}^{2_z}m{}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{210}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{210}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{120}\right]$ $\left[2_{001}T \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$
$\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{110}\right]$ $\left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[E \parallel 2_{120}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{10}	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
Γ^{co}_{11}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{12}	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 453.3.1006 : ${}^{1}6/{}^{2_z}m{}^{2_z}m{}^{2_z}m{}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right] \left[2_{100}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 453.4.1007 : ${}^{1}6/^{2_z}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}6/^{2_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{12}\right.$
$\left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$

C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$

C_4 : $\left[E \parallel 2_{001}\right]$

C_5 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right]$
$\left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right]$
$\left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

C_6 : $\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$

C_7 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$

C_{10} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

C_{11} : $\left[\bar{E} \parallel 2_{001}\right]$

C_{12} : $\left[\bar{E} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{210}\right]$

$C_{13}:\ \left[\bar{E}\ ||\ 2_{010}\right]\ \left[\bar{E}\ ||\ 2_{110}\right]\ \left[\bar{E}\ ||\ 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1	2	2	2	2	0	2	2	−2	−2	−2	−2	−2	−2
χ_2	2	2	2	2	0	−2	−2	−2	−2	−2	−2	2	2
χ_3	2	−2	2	−2	0	−2	2	−2	2	−2	2	2	−2
χ_4	2	−2	2	−2	0	2	−2	−2	2	−2	2	−2	2
χ_5	4	−2	−2	4	0	0	0	−4	2	2	−4	0	0
χ_6	4	2	−2	−4	0	0	0	−4	−2	2	4	0	0

Irrepresntations of group generators

	$\left[E\ ||\ 6^{+}_{001}\right]$	$\left[2_{100}\ ||\ m_{001}\right]$	$\left[2_{100}\ ||\ m_{100}\right]$	$\left[\Sigma_z\ ||\ 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	σ_0	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$-\sigma_0$	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2 \\ \omega_2 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_4^* \\ \omega_4^* & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 454.1.1008 : ${}^{1}6/{}^{m}m{}^{m}m{}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{120}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 454.2.1009 : ${}^{1}6/{}^{m_z}m{}^{m_z}m{}^{m_z}m{}^{\infty_x}1 \rightarrow {}^{1}6/{}^{m}m{}^{m}m{}^{m}m$

No 455.1.1010 : ${}^{1}6/{}^{\bar{1}}m{}^{\bar{1}}m{}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{120}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 455.2.1011 : $^{1}6/^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}6/^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_{13} :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_{14} :	$\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_{15} :	$\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{16} :	$\left[\bar{E} \parallel 2_{001}\right]$
C_5 :	$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$	C_{17} :	$\left[\bar{E} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$	C_{18} :	$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$
C_7 :	$\left[\Sigma_z \parallel 1\right]$	C_{19} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_8 :	$\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$	C_{20} :	$\left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_9 :	$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right]$	C_{21} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_{10} :	$\left[\Sigma_z \parallel 2_{001}\right]$	C_{22} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right]$
C_{11} :	$\left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{210}\right]$	C_{23} :	$\left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right]$
C_{12} :	$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$	C_{24} :	$\left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	0
χ_2^{co}	2	2	2	2	−2	−2	0	0	0	0	0	0
χ_3^{co}	2	−2	2	−2	−2	2	0	0	0	0	0	0
χ_4^{co}	2	−2	2	−2	2	−2	0	0	0	0	0	0

χ_5^{co}	4	−2	−2	4	0	0	0	0	0	0	0	0
χ_6^{co}	4	2	−2	−4	0	0	0	0	0	0	0	0

	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}
χ_1^{co}	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0
χ_2^{co}	−2	−2	−2	−2	2	2	0	0	0	0	0	0
χ_3^{co}	−2	2	−2	2	2	−2	0	0	0	0	0	0
χ_4^{co}	−2	2	−2	2	−2	2	0	0	0	0	0	0
χ_5^{co}	−4	2	2	−4	0	0	0	0	0	0	0	0
χ_6^{co}	−4	−2	2	4	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{120}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_6^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \omega_4^* & 0 \\ 0 & \omega_4 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 455.3.1012 : ${}^{1}6/^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}6/^{\bar{1}}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[E \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[2_{100}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right]$
$\left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{12}\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right]$
$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right]$
$\left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{10}\right.$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T\right.$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 2_{120}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{210}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_7 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right]$
$\left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel -1\right]$
$\left[\bar{E} 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{210}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$

C_8 : $\left[\bar{E} \parallel 1\right]$

C_9 : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$

C_{10} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

C_{11} : $\left[\bar{E} \parallel 2_{001}\right]$

C_{12} : $\left[\bar{E} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{210}\right]$

C_{13} : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
χ_1^{co}	2	2	2	2	2	2	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	2	−2	−2	0	−2	−2	−2	−2	2	2
χ_3^{co}	2	−2	2	−2	−2	2	0	−2	2	−2	2	2	−2
χ_4^{co}	2	−2	2	−2	2	−2	0	−2	2	−2	2	−2	2
χ_5^{co}	4	−2	−2	4	0	0	0	−4	2	2	−4	0	0
χ_6^{co}	4	2	−2	−4	0	0	0	−4	−2	2	4	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[E \parallel 2_{120}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{100}T \parallel 1\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$-\sigma_0$	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

Γ_6^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \omega_4^* & 0 \\ 0 & \omega_4^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 456.1.1013 : ${}^{2_z}6/{}^{2_z}m{}^{2_x}m{}^{2_y}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001} \,||\, 6^{+}_{001}\right] \left[E \,||\, 3^{+}_{001}\right] \left[2_{001} \,||\, 2_{001}\right] \left[E \,||\, 3^{-}_{001}\right] \left[2_{001} \,||\, 6^{-}_{001}\right] \left[2_{001} \,||\, m_{001}\right] \left[E \,||\, \bar{3}^{-}_{001}\right] \left[2_{001} \,||\, \bar{6}^{-}_{001}\right]$
$\left[E \,||\, -1\right] \left[2_{001} \,||\, \bar{6}^{+}_{001}\right] \left[E \,||\, \bar{3}^{+}_{001}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{010} \,||\, m_{210}\right] \left[2_{010} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{010} \,||\, m_{120}\right]$
$\left[2_{010} \,||\, 2_{120}\right] \left[2_{100} \,||\, 2_{010}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{010} \,||\, 2_{1\bar{1}0}\right] \left[2_{010} \,||\, 2_{210}\right] \left[2_{100} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, 6^{+}_{001}\right] \left[2_{001} \,||\, 6^{-}_{001}\right]$
C_3 : $\left[E \,||\, 3^{+}_{001}\right] \left[E \,||\, 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$
C_5 : $\left[2_{001} \,||\, m_{001}\right]$
C_6 : $\left[E \,||\, \bar{3}^{-}_{001}\right] \left[E \,||\, \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \,||\, \bar{6}^{-}_{001}\right] \left[2_{001} \,||\, \bar{6}^{+}_{001}\right]$
C_8 : $\left[E \,||\, -1\right]$
C_9 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{010}\right]$
C_{10} : $\left[2_{010} \,||\, m_{210}\right] \left[2_{010} \,||\, m_{1\bar{1}0}\right] \left[2_{010} \,||\, m_{120}\right]$
C_{11} : $\left[2_{010} \,||\, 2_{120}\right] \left[2_{010} \,||\, 2_{1\bar{1}0}\right] \left[2_{010} \,||\, 2_{210}\right]$
C_{12} : $\left[2_{100} \,||\, 2_{010}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \,||\, 6^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 456.2.1014 : ${}^{2_z}6/{}^{2_z}m^{2_x}m^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{100}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right]$
$\left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[E \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_{10} : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$
C_{11} : $\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$
C_{12} : $\left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 456.3.1015 : ${}^{2_z}6/{}^{2_z}m^{2_x}m^{2_y}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[2_{001}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right]$
$\left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$
C_6 :	$\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 457.1.1016 : ${}^{2_z}6/{}^{2_z}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[E \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[E \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 458.1.1017 : ${}^{m_x}6/{}^{m_x}m^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 459.1.1018 : ${}^{2}6/{}^{2}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$ $C_7 : \left[2_{001} \parallel m_{001}\right]$
$C_2 : \left[2_{001} \parallel 6^{+}_{001}\right]$ $C_8 : \left[E \parallel \bar{3}^{-}_{001}\right]$
$C_3 : \left[E \parallel 3^{+}_{001}\right]$ $C_9 : \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$C_4 : \left[2_{001} \parallel 2_{001}\right]$ $C_{10} : \left[E \parallel -1\right]$
$C_5 : \left[E \parallel 3^{-}_{001}\right]$ $C_{11} : \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
$C_6 : \left[2_{001} \parallel 6^{-}_{001}\right]$ $C_{12} : \left[E \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 459.2.1019 : ${}^{2_z}6/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{2_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[E \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[E \parallel \bar{3}^+_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E} \parallel \bar{3}^+_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right]$
$\left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^+_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^+_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right]$
C_4 : $\left[E \parallel 3^-_{001}\right]$
C_5 : $\left[E \parallel \bar{3}^-_{001}\right]$
C_6 : $\left[E \parallel -1\right]$
C_7 : $\left[E \parallel \bar{3}^+_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 3^+_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^-_{001}\right]$
C_{11} : $\left[\bar{E} \parallel \bar{3}^-_{001}\right]$
C_{12} : $\left[\bar{E} \parallel -1\right]$
C_{13} : $\left[\bar{E} \parallel \bar{3}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	0	2	2	2	2	2	-2	-2	-2	-2	-2	-2
χ_2^{co}	2	0	2	2	-2	-2	-2	-2	-2	-2	2	2	2
χ_3^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$
χ_4^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$
χ_5^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$
χ_6^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha^*\mathcal{K}$	$\omega_2^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$\alpha^*\mathcal{K}$	$\omega_4\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\alpha\mathcal{K}$	$\omega_2\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	$\alpha\mathcal{K}$	$\omega_4^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 460.1.1020 : ${}^{\bar{1}}6/{}^{\bar{1}}m^2m^mm$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^-_{001}\right] \left[T \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^+_{001}\right] \left[E \parallel \bar{3}^+_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^-_{001}\right] \left[E \parallel \bar{3}^+_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^-_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 460.2.1021 : ${}^{\bar{1}}6/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/{}^{\bar{1}}m^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_1$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	0	0	0	0	−2	−2	2	2
χ_3^{co}	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2
χ_4^{co}	2	−1	1	−2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	−1	2
χ_5^{co}	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2
χ_6^{co}	2	−1	1	−2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	−1	2

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{3}^{-}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 6^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 461.1.1022 : ${}^{m}6/{}^{m}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 461.2.1023 : ${}^{m_z}6/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}6/{}^{m_z}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[E \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel -1\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	0	0	0	0	−2	−2	2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	−2	2	−4	0	0	0	0	0	−4	2	−2	4

Coirrepresntations of group generators

	$\left[E \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 462.1.1024 : ${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{210}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{010} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel m_{210}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 462.2.1025 : ${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_{10} : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$
C_{11} : $\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right]$
C_{12} : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 462.3.1026 : ${}^{2_z}6/{}^{1}m^{2_x}m^{2_y}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_{10} : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$
C_{11} : $\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right]$
C_{12} : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 463.1.1027 : ${}^{2_z}6/{}^{1}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{210}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 464.1.1028 : ${}^{m_x}6/{}^{1}m^{2_z}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 465.1.1029 : ${}^{2}6/{}^{1}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{210}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[2_{001}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[E \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[E \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 465.2.1030 : ${}^{2_z}6/{}^{1}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{1}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 3^{-}_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[E \parallel \bar{6}^{-}_{001}\right]$
C_7 : $\left[E \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel m_{001}\right]$
C_{12} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	0	2	2	2	2	2	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	0	2	2	−2	−2	−2	−2	−2	−2	2	2	2
χ_3^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_4^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$
χ_5^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_6^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid 6^{+}_{001}\right]$	$\left[E \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\alpha^*\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$\alpha^*\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	$\alpha\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	$\alpha\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 466.1.1031 : ${}^{\bar{1}}6/{}^{1}m^{2}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 466.2.1032 : ${}^{\bar{1}}6/{}^{1}m^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/{}^{1}m^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_1$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
C_6 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel m_{001}\right]$
C_{13} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	0	0	0	0	−2	−2	2	2
χ_3^{co}	2	−1	2	−1	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	−2	1
χ_4^{co}	2	−1	−2	1	0	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	2	−1
χ_5^{co}	2	−1	2	−1	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	−2	1
χ_6^{co}	2	−1	−2	1	0	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	2	−1

Coirrepresntations of group generators

	$\left[E \mid\mid \bar{6}_{001}^{-}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 6_{001}^{+}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 467.1.1033 : ${}^{m}6/{}^{1}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 467.2.1034 : ${}^{m_z}6/{}^{1}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}6/{}^{1}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[E \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[E \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[E \parallel \bar{6}^{-}_{001}\right] \left[E \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
C_6 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{6}^{-}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel m_{001}\right]$
C_{13} : $\left[\bar{E} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E} \parallel \bar{6}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	2	−2	−2	0	0	0	0	0	−2	−2	2	2
χ_3^{co}	4	−2	4	−2	0	0	0	0	0	−4	2	−4	2
χ_4^{co}	4	−2	−4	2	0	0	0	0	0	−4	2	4	−2

Coirrepresntations of group generators

	$\left[E \parallel \bar{6}_{001}^{-}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6_{001}^{+}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3^* & 0 \\ 0 & \omega_3 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 468.1.1035 : ${}^{1}6/{}^{2}{}_{z}m{}^{2}{}_{x}m{}^{2}{}_{x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{210}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{100} \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_{10} : $\left[2_{100} \parallel m_{210}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{010} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{210}\right]$
C_{12} : $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 468.2.1036 : ${}^{1}6/{}^{2_z}m{}^{2_x}m{}^{2_x}m{}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{210}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$ $\left[2_{100} \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 6^{+}_{001}\right]$ $\left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 3^{-}_{001}\right]$ $\left[2_{001}T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$ $\left[2_{100}T \parallel 2_{120}\right]$ $\left[2_{100}T \parallel 2_{010}\right]$ $\left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{100}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel 2_{210}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{100} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{100} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{100} \parallel m_{210}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[2_{010} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{10}	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
Γ^{co}_{11}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{12}	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 468.3.1037 : ${}^{1}6/{}^{2_z}m{}^{2_x}m{}^{2_x}m{}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{010}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[2_{001}T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_{10} : $\left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$
C_{12} : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 468.4.1038 : ${}^{1}6/{}^{2_z}m{}^{2_x}m{}^{2_x}m{}^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{010}T \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[2_{010}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right] \left[2_{010}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{001} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$
C_6 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 469.1.1039 : ${}^{1}6/{}^{2_z}m^{m_x}m^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{210}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{010}T \parallel 2_{110}\right]$ $\left[2_{010}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 470.1.1040 : ${}^{1}6/^{m_x}m^{m_y}m^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$ $\left[2_{100}T \mid\mid m_{001}\right]$ $\left[2_{100}T \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{100}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \mid\mid -1\right]$ $\left[2_{100}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[2_{010}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{210}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{010}T \mid\mid m_{110}\right]$ $\left[2_{010}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid m_{120}\right]$
$\left[2_{001} \mid\mid 2_{120}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{210}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{120}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{210}\right]$
C_6 : $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{120}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 471.1.1041 : ${}^{1}6/{}^{m_x}m^{2_z}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 472.1.1042 : ${}^{1}6/{}^{2}m^{m}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{210}\right]$ $\left[2_{001}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel m_{110}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[2_{001}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$ C_7 : $\left[2_{001} \parallel m_{001}\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ C_8 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ C_9 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$ C_{10} : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$ C_{11} : $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$ C_{12} : $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 472.2.1043 : ${}^{1}6/^{2_z}m^{m_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{1}6/^{2_z}m^{m_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{10}\right.$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel 3^{-}_{001}\right]$
C_6 : $\left[E \parallel 6^{-}_{001}\right]$
C_7 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \parallel 1\right]$
C_9 : $\left[\bar{E} \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 3^{+}_{001}\right]$
C_{11} : $\left[\bar{E} \parallel 2_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	2	2	0	-2	-2	-2	-2	-2	-2
χ_2^{co}	2	-2	2	-2	2	-2	0	-2	2	-2	2	-2	2
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_4^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$
χ_5^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_6^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid 6_{001}^{+}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_7	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_7^*	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$

No 473.1.1044 : ${}^{1}6/{}^{2}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{210}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[E \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[E \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{001} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 473.2.1045 : ${}^{1}6/{}^{2}{}_{z}m^{\bar{1}}m^{\bar{1}}m^{\infty}{}_{x}1 \rightarrow {}^{1}6/{}^{2}{}_{z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma}{}_{x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right] \left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[2_{100} \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[T \mid\mid m_{100}\right] \left[T \mid\mid m_{210}\right] \left[T \mid\mid m_{1\bar{1}0}\right] \left[T \mid\mid m_{110}\right] \left[T \mid\mid m_{010}\right] \left[T \mid\mid m_{120}\right]$
$\left[2_{100}T \mid\mid 2_{120}\right] \left[2_{100}T \mid\mid 2_{010}\right] \left[2_{100}T \mid\mid 2_{110}\right] \left[2_{100}T \mid\mid 2_{1\bar{1}0}\right] \left[2_{100}T \mid\mid 2_{210}\right] \left[2_{100}T \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right] \left[\Sigma_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \mid\mid m_{100}\right] \left[\Sigma_z T \mid\mid m_{210}\right] \left[\Sigma_z T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z T \mid\mid m_{110}\right] \left[\Sigma_z T \mid\mid m_{010}\right] \left[\Sigma_z T \mid\mid m_{120}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{120}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{110}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100}T \mid\mid 2_{100}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E}T \mid\mid m_{100}\right] \left[\bar{E}T \mid\mid m_{210}\right] \left[\bar{E}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\bar{E}T \mid\mid m_{110}\right] \left[\bar{E}T \mid\mid m_{010}\right] \left[\bar{E}T \mid\mid m_{120}\right] \left[\bar{E}2_{100}T \mid\mid 2_{120}\right] \left[\bar{E}2_{100}T \mid\mid 2_{010}\right] \left[\bar{E}2_{100}T \mid\mid 2_{110}\right] \left[\bar{E}2_{100}T \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \mid\mid 2_{210}\right] \left[\bar{E}2_{100}T \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z T \mid\mid m_{210}\right] \left[\bar{\Sigma}_z T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z T \mid\mid m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[E \mid\mid 3^{-}_{001}\right]$
C_6 : $\left[E \mid\mid 6^{-}_{001}\right]$
C_7 : $\left[2_{100} \mid\mid m_{001}\right] \left[2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[2_{100} \mid\mid -1\right] \left[2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid m_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid -1\right]$
$\left[\Sigma_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid m_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid -1\right] \left[\bar{E}2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{3}^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid -1\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{3}^{+}_{001}\right]$
C_8 : $\left[\bar{E} \mid\mid 1\right]$
C_9 : $\left[\bar{E} \mid\mid 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \mid\mid 3^{+}_{001}\right]$
C_{11} : $\left[\bar{E} \mid\mid 2_{001}\right]$
C_{12} : $\left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \mid\mid 6^{-}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	2	2	0	-2	-2	-2	-2	-2	-2
χ_2^{co}	2	-2	2	-2	2	-2	0	-2	2	-2	2	-2	2
χ_3^{co}	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$
χ_4^{co}	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	-2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$
χ_5^{co}	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$
χ_6^{co}	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	-2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[T \mid\mid m_{100}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_7	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_5	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_7^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 474.1.1046 : ${}^{1}6/{}^{\bar{1}}m^{m}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right] \left[T \mid\mid m_{001}\right] \left[T \mid\mid \bar{3}^{-}_{001}\right] \left[T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[T \mid\mid -1\right] \left[T \mid\mid \bar{6}^{+}_{001}\right] \left[T \mid\mid \bar{3}^{+}_{001}\right] \left[2_{001}T \mid\mid m_{100}\right] \left[2_{001}T \mid\mid m_{210}\right] \left[2_{001}T \mid\mid m_{1\bar{1}0}\right] \left[2_{001}T \mid\mid m_{110}\right] \left[2_{001}T \mid\mid m_{010}\right] \left[2_{001}T \mid\mid m_{120}\right]$
$\left[2_{001} \mid\mid 2_{120}\right] \left[2_{001} \mid\mid 2_{010}\right] \left[2_{001} \mid\mid 2_{110}\right] \left[2_{001} \mid\mid 2_{1\bar{1}0}\right] \left[2_{001} \mid\mid 2_{210}\right] \left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{120}\right] \left[2_{001} \mid\mid 2_{1\bar{1}0}\right] \left[2_{001} \mid\mid 2_{210}\right]$
C_6 : $\left[2_{001} \mid\mid 2_{010}\right] \left[2_{001} \mid\mid 2_{110}\right] \left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{120}\right]$	$\left[T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 474.2.1047 : ${}^{1}6/^{\bar{1}}m^{m_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{1}6/^{\bar{1}}m^{m_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right] \left[T \mid\mid m_{001}\right] \left[T \mid\mid \bar{3}^{-}_{001}\right] \left[T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[T \mid\mid -1\right] \left[T \mid\mid \bar{6}^{+}_{001}\right] \left[T \mid\mid \bar{3}^{+}_{001}\right] \left[2_{100}T \mid\mid m_{100}\right] \left[2_{100}T \mid\mid m_{210}\right] \left[2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[2_{100}T \mid\mid m_{110}\right] \left[2_{100}T \mid\mid m_{010}\right] \left[2_{100}T \mid\mid m_{120}\right]$
$\left[2_{100} \mid\mid 2_{120}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{210}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\Sigma_z \mid\mid 3^{+}_{001}\right]$
$\left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\Sigma_z T \mid\mid m_{001}\right] \left[\Sigma_z T \mid\mid \bar{3}^{-}_{001}\right] \left[\Sigma_z T \mid\mid \bar{6}^{-}_{001}\right] \left[\Sigma_z T \mid\mid -1\right] \left[\Sigma_z T \mid\mid \bar{6}^{+}_{001}\right] \left[\Sigma_z T \mid\mid \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \mid\mid m_{100}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{210}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{110}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{010}\right] \left[\Sigma_z 2_{100}T \mid\mid m_{120}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{120}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z$
$\left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right]$
$\left[\bar{E}T \mid\mid m_{001}\right] \left[\bar{E}T \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{E}T \mid\mid \bar{6}^{-}_{001}\right] \left[\bar{E}T \mid\mid -1\right] \left[\bar{E}T \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{E}T \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \mid\mid m_{100}\right] \left[\bar{E}2_{100}T \mid\mid m_{210}\right] \left[\bar{E}2_{100}T \mid\mid m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100}T \mid\mid m_{110}\right] \left[\bar{E}2_{100}T \mid\mid m_{010}\right] \left[\bar{E}2_{100}T \mid\mid m_{120}\right] \left[\bar{E}2_{100} \mid\mid 2_{120}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{210}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid m_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \mid\mid -1\right] \left[\bar{\Sigma}_z T \mid\mid \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \mid\mid \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \mid\mid m_{1}\right.$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[E \mid\mid 2_{001}\right]$
C_5 : $\left[2_{100} \mid\mid 2_{120}\right] \left[2_{100} \mid\mid 2_{010}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 2_{210}\right] \left[2_{100} \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid 2_{001}\right]$
$\left[\Sigma_z 2_{100} \mid\mid 2_{120}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{210}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{100}\right] \left[\bar{E}2_{100} \mid\mid 2_{120}\right] \left[\bar{E}2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{E}2_{100} \mid\mid 2_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 2_{210}\right] \left[\bar{E}2_{100} \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{100}\right]$
C_6 : $\left[\Sigma_z \mid\mid 6^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \mid\mid 3^{+}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \mid\mid 3^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \mid\mid 6^{-}_{001}\right] \left[\bar{\Sigma}_z \mid\mid 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \mid\mid 1\right]$
C_{11} : $\left[\bar{E} \mid\mid 6^{+}_{001}\right] \left[\bar{E} \mid\mid 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \mid\mid 3^{+}_{001}\right] \left[\bar{E} \mid\mid 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \mid\mid 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2
χ_4^{co}	2	1	−1	−2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2
χ_5^{co}	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2
χ_6^{co}	2	1	−1	−2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	−2	−1	1	2

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid 2_{120}\right]$	$\left[T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 475.1.1048 : ${}^{1}6/{}^{\bar{1}}m^{2}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 475.2.1049 : ${}^{1}6/^{\bar{1}}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}6/^{\bar{1}}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{10}\right.$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_6 : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	2	−1	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	1	−2
χ_4^{co}	2	1	−1	−2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−1	1	2
χ_5^{co}	2	−1	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	1	−2
χ_6^{co}	2	1	−1	−2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	−2	−1	1	2

Coirrepresntations of group generators

	$\left[E \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid m_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	ω_3	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	ω_1	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	ω_3^*	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$

No 476.1.1050 : ${}^{1}6/{}^{m}m^{\bar{1}}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[2_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$ $\left[T \parallel m_{100}\right]$ $\left[T \parallel m_{210}\right]$ $\left[T \parallel m_{1\bar{1}0}\right]$ $\left[T \parallel m_{110}\right]$ $\left[T \parallel m_{010}\right]$ $\left[T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right]$ $\left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{120}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 476.2.1051 : ${}^{1}6/{}^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{1}6/{}^{m_z}m^{\bar{1}}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right]$
$\left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	2	−2	−4	0	0	0	0	0	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{120}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 477.1.1052 : ${}^{1}6/{}^{m}m^{2}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 477.2.1053 : ${}^{1}6/{}^{m_z}m^{2_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{1}6/{}^{m_z}m^{2_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[E \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right.$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel\right.$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 6^{+}_{001}\right] \left[E \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_6 : $\left[\Sigma_z \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{-}_{001}\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right]$
C_8 : $\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right]$
C_9 : $\left[\Sigma_z \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 6^{+}_{001}\right]$
C_{10} : $\left[\bar{E} \parallel 1\right]$
C_{11} : $\left[\bar{E} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 6^{-}_{001}\right]$
C_{12} : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_{13} : $\left[\bar{E} \parallel 2_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}

χ_1^{co}	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2
χ_2^{co}	2	−2	2	−2	0	0	0	0	0	−2	2	−2	2
χ_3^{co}	4	−2	−2	4	0	0	0	0	0	−4	2	2	−4
χ_4^{co}	4	2	−2	−4	0	0	0	0	0	−4	−2	2	4

Coirrepresntations of group generators

	$\left[E \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1^* & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} \omega_3 & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$

No 478.1.1054 : ${}^{2_z}6/{}^{2_x}m^1m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{010} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{100} \parallel m_{001}\right]$	C_{11} :	$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$
C_6 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 478.2.1055 : ${}^{2_z}6/{}^{2_x}m^1m^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right]$
C_6 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{010} \parallel -1\right]$
C_9 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_{10} : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_{11} : $\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right]$
C_{12} : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 478.3.1056 : ${}^{2_z}6/{}^{2_x}m^1m^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$ $\left[2_{001} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{100}\right]$ $\left[2_{100}T \parallel 1\right]$ $\left[2_{010}T \parallel 6^{+}_{001}\right]$ $\left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right]$ $\left[2_{100}T \parallel 3^{-}_{001}\right]$ $\left[2_{010}T \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$ $\left[T \parallel 2_{120}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[2_{001}T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right]$ $\left[T \parallel 2_{210}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{010} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{110}\right]$ $\left[E \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right]$ $\left[2_{001} \parallel m_{1\bar{1}0}\right]$ $\left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{100} \parallel m_{001}\right]$	C_{11} :	$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$
C_6 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 478.4.1057 : ${}^{2_z}6/{}^{2_x}m^1m^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right] \left[2_{010}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{010}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{010}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{001}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right]$
$\left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[2_{010} \parallel -1\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$	C_9 :	$\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_5 :	$\left[2_{100} \parallel m_{001}\right]$	C_{11} :	$\left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right]$
C_6 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{010}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 479.1.1058 : ${}^{m_x}6/{}^{m_y}m^1m^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100}T \parallel 6^-_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^-_{001}\right] \left[2_{010}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{010}T \parallel \bar{6}^+_{001}\right] \left[2_{001} \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^-_{001}\right] \left[2_{001} \parallel \bar{3}^+_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^-_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 480.1.1059 : ${}^{m_x}6/{}^{2_z}m^1m^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100}T \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^-_{001}\right] \left[2_{001} \parallel \bar{6}^-_{001}\right]$
$\left[2_{010}T \parallel -1\right] \left[2_{001} \parallel \bar{6}^+_{001}\right] \left[2_{010}T \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{6}^-_{001}\right] \left[2_{001} \parallel \bar{6}^+_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^-_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 481.1.1060 : ${}^{2_z}6/{}^{m_x}m^1m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{001} \parallel 6^-_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^-_{001}\right] \left[2_{100}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^+_{001}\right] \left[2_{010}T \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^+_{001}\right] \left[2_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^+_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 482.1.1061 : ${}^{m}6/{}^{\bar{1}}m^{1}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 6^{-}_{001}\right]$ $\left[T \mid\mid m_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{001} \mid\mid -1\right]$ $\left[T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{210}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid m_{120}\right]$
$\left[T \mid\mid 2_{120}\right]$ $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 2_{210}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid -1\right]$
C_5 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$
C_6 : $\left[2_{001} \mid\mid 2_{010}\right]$ $\left[2_{001} \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{3}^{-}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 482.2.1062 : ${}^{m_z}6/^{\bar{1}}m^1m^{m_z}m^{\infty_x}1 \rightarrow {}^{m_z}6/^{\bar{1}}m^1m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100}T \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[T \parallel \bar{6}^-_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[T \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^-_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z T \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}2_{100}T \parallel 6^-_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}T \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^+_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 483.1.1063 : ${}^{\bar{1}}6/{}^{m}m^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 483.2.1064 : ${}^{\bar{1}}6/^{m_z}m^1m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/^{m_z}m^1m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z T \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^+_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}T \parallel 6^-_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^+_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{010}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{100}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 484.1.1065 : ${}^{m}6/{}^{2}m^{1}m^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001}T \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001}T \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001}T \mid\mid 6^{-}_{001}\right]$ $\left[2_{001} \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$
$\left[T \mid\mid -1\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$ $\left[T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[2_{001}T \mid\mid m_{210}\right]$ $\left[2_{001}T \mid\mid m_{1\bar{1}0}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[2_{001}T \mid\mid m_{120}\right]$
$\left[2_{001} \mid\mid 2_{120}\right]$ $\left[T \mid\mid 2_{010}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{210}\right]$ $\left[T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \mid\mid m_{001}\right]$
C_4 : $\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$ $\left[2_{001} \mid\mid \bar{6}^{+}_{001}\right]$
C_5 : $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{110}\right]$ $\left[E \mid\mid m_{010}\right]$
C_6 : $\left[2_{001} \mid\mid 2_{120}\right]$ $\left[2_{001} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{001} \mid\mid 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid \bar{6}^{-}_{001}\right]$	$\left[E \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 484.2.1066 : ${}^{m_z}6/{}^{2_z}m^1m^{m_z}m^{\infty_x}1 \rightarrow {}^{m_z}6/{}^{2_z}m^1m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100}T \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[T \parallel -1\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z T \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}2_{100}T \parallel 6^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 485.1.1067 : ${}^{2}6/{}^{m}m^{1}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 485.2.1068 : ${}^{2_z}6/{}^{m_z}m^1m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{m_z}m^1m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[T \parallel \bar{3}^-_{001}\right] \left[2_{100}T \parallel \bar{6}^-_{001}\right]$
$\left[T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^+_{001}\right] \left[T \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^-_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^+_{001}\right] \left[\Sigma_z T \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z T \parallel 2_{010}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^-_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^+_{001}\right] \left[\bar{E}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E}T \parallel 2_{010}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z T \parallel 2_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^+_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^+_{001}\right]$
$\left[\Sigma_z \parallel 3^+_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^-_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right]$
$\left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 6^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^-_{001}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_3 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	2	−2	−2	−2	2
χ_3^{co}	4	0	−2	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 486.1.1069 : ${}^{\bar{1}}6/{}^{2}m^{1}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 486.2.1070 : ${}^{\bar{1}}6/^{2_z}m^1m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/^{2_z}m^1m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^+_{001}\right] \left[E \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^-_{001}\right] \left[T \parallel 6^-_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^-_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[2_{100}T \parallel \bar{3}^+_{001}\right] \left[E \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^+_{001}\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z T \parallel 6^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^+_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^+_{001}\right] \left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right] \left[\bar{E}T \parallel 6^-_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^+_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^+_{001}\right] \left[E \parallel 3^-_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^-_{001}\right] \left[2_{100} \parallel \bar{6}^+_{001}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^+_{001}\right]$
$\left[\Sigma_z \parallel 3^-_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^+_{001}\right] \left[\bar{\Sigma}_z \parallel 3^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^+_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^+_{001}\right] \left[\bar{E} \parallel 3^-_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \sigma_x \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 487.1.1071 : ${}^{2}6/^{\bar{1}}m^{1}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 487.2.1072 : ${}^{2_z}6/^{\bar{1}}m^{1}m^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/^{\bar{1}}m^{1}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[E \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel m_{110}\right]$
$\left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel m_{100}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{010}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	2	−2	−2	−2	2
χ_3^{co}	4	0	−2	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_x & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 488.1.1073 : ${}^{2_z}6/{}^{2_x}m{}^{2_y}m{}^{2_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{100} \parallel m_{210}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{100} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$ $\left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right]$
C_6 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{010} \parallel -1\right]$
C_9 : $\left[2_{010} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{100} \parallel m_{210}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{010} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 488.2.1074 : ${}^{2_z}6/{}^{2_x}m{}^{2_y}m{}^{2_x}m{}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[2_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{010}T \parallel \bar{6}^{-}_{001}\right] \left[2_{100}T \parallel -1\right] \left[2_{010}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right] \left[T \parallel 2_{120}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right]$
C_6 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{010} \parallel -1\right]$
C_9 : $\left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{010} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{10}$	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{11}$	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
$\Gamma^{\mathrm{co}}_{12}$	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 488.3.1075 : ${}^{2_z}6/{}^{2_x}m{}^{2_y}m{}^{2_x}m{}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[2_{100}T \parallel 3^{+}_{001}\right]$
$\left[2_{010}T \parallel 2_{001}\right] \left[2_{100}T \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right] \left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right]$
C_6 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{010} \parallel -1\right]$
C_9 : $\left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{010} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	−1	−1	1	$\mathcal{K}$
Γ_6^{co}	−1	1	−1	$\mathcal{K}$
Γ_7^{co}	−1	−1	−1	$\mathcal{K}$
Γ_8^{co}	−1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 488.4.1076 : ${}^{2_z}6/{}^{2_x}m{}^{2_y}m{}^{2_x}m{}^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[2_{010}T \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[2_{010}T \parallel 3^{+}_{001}\right]$
$\left[2_{100}T \parallel 2_{001}\right] \left[2_{010}T \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right]$
$\left[T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right] \left[2_{100}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right]$
$\left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{001}\right]$
C_6 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{010} \parallel -1\right]$
C_9 : $\left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_{12} : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{010} \mid\mid m_{100}\right]$	$\left[2_{010}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	1	$\mathcal{K}$
Γ_5^{co}	−1	−1	1	$\mathcal{K}$
Γ_6^{co}	−1	1	−1	$\mathcal{K}$
Γ_7^{co}	−1	−1	−1	$\mathcal{K}$
Γ_8^{co}	−1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 489.1.1077 : ${}^{m_y}6/{}^{2_z}m^{m_x}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{010}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{210}\right]$	$\left[2_{010}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 490.1.1078 : ${}^{m_y}6/{}^{m_x}m^{2_z}m^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{010}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{010}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{010}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 491.1.1079 : ${}^{2_z}6/{}^{m_x}m^{m_y}m^{m_x}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[2_{001} \mid\mid 6^{+}_{001}\right] \left[E \mid\mid 3^{+}_{001}\right] \left[2_{001} \mid\mid 2_{001}\right] \left[E \mid\mid 3^{-}_{001}\right] \left[2_{001} \mid\mid 6^{-}_{001}\right] \left[2_{100}T \mid\mid m_{001}\right] \left[2_{010}T \mid\mid \bar{3}^{-}_{001}\right] \left[2_{100}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \mid\mid -1\right] \left[2_{100}T \mid\mid \bar{6}^{+}_{001}\right] \left[2_{010}T \mid\mid \bar{3}^{+}_{001}\right] \left[2_{010}T \mid\mid m_{100}\right] \left[2_{100}T \mid\mid m_{210}\right] \left[2_{100}T \mid\mid m_{1\bar{1}0}\right] \left[2_{010}T \mid\mid m_{110}\right] \left[2_{010}T \mid\mid m_{010}\right] \left[2_{100}T \mid\mid m_{120}\right]$
$\left[2_{001} \mid\mid 2_{120}\right] \left[E \mid\mid 2_{010}\right] \left[E \mid\mid 2_{110}\right] \left[2_{001} \mid\mid 2_{1\bar{1}0}\right] \left[2_{001} \mid\mid 2_{210}\right] \left[E \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 6^{+}_{001}\right] \left[2_{001} \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right] \left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[2_{001} \mid\mid 2_{120}\right] \left[2_{001} \mid\mid 2_{1\bar{1}0}\right] \left[2_{001} \mid\mid 2_{210}\right]$
C_6 : $\left[E \mid\mid 2_{010}\right] \left[E \mid\mid 2_{110}\right] \left[E \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid 2_{120}\right]$	$\left[2_{100}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 492.1.1080 : ${}^{m}6/{}^{2}m^{\bar{1}}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{210}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 492.2.1081 : ${}^{m_z}6/{}^{2_z}m^{\bar{1}}m^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}6/{}^{2_z}m^{\bar{1}}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{210}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \omega_2^* & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 493.1.1082 : ${}^{\bar{1}}6/{}^{2}m^{m}m^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{210}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 493.2.1083 : ${}^{\bar{1}}6/^{2_z}m^{m_z}m^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/^{2_z}m^{m_z}m^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100} \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{10}\right.$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{210}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{120}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{210}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{6}^{-}_{001}\right]$	$\left[2_{100} \mid\mid m_{210}\right]$	$\left[T \mid\mid 6^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \omega_2^* & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 494.1.1084 : ${}^{m}6/{}^{\bar{1}}m^{2}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 494.2.1085 : ${}^{m_z}6/{}^{\bar{1}}m^{2_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}6/{}^{\bar{1}}m^{2_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z 2_{100}T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{100}T \parallel 6^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 495.1.1086 : ${}^{2}6/^{\bar{1}}m^{m}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{120}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 495.2.1087 : ${}^{2_z}6/^{\bar{1}}m^{m_z}m^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}6/^{\bar{1}}m^{m_z}m^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{210}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{210}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{210}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$

C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

C_5 : $\left[\bar{E} \parallel 1\right]$

C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$

C_7 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	2	−2	−2	−2	2
χ_3^{co}	4	0	−2	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{120}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 496.1.1088 : ${}^{\bar{1}}6/{}^{m}m^{2}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 496.2.1089 : ${}^{\bar{1}}6/^{m_z}m^{2_z}m^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}6/^{m_z}m^{2_z}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z T \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z T \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z T \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}T \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}T \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}T \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T$
$\left[\bar{\Sigma}_z T \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	2	0	2	−2	−2	−2
χ_2^{co}	2	2	0	−2	−2	−2	2
χ_3^{co}	4	−2	0	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{3}^{-}_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[T \mid\mid 6^{+}_{001}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 497.1.1090 : ${}^{2}6/{}^{m}m^{\bar{1}}m^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[2_{001} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel 2_{120}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{210}\right]$
C_6 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{001} \parallel 2_{120}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 497.2.1091 : ${}^{2_z}6/{}^{m_z}m^{\bar{1}}m^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}6/{}^{m_z}m^{\bar{1}}m^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{100}T \parallel m_{210}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[E \parallel 2_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{210}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel m_{120}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right]$
$\left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{-}_{001}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{001}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{210}\right] \left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel m_{120}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{210}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{120}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 6^{+}_{001}\right] \left[2_{100} \parallel 2_{001}\right] \left[2_{100} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 6^{+}_{001}\right]$
$\left[\Sigma_z \parallel 3^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{001}\right] \left[\Sigma_z \parallel 3^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 6^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{120}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{210}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\bar{E}2_{100} \parallel 6^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 6^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{120}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{210}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 6^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{120}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{210}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{010}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{100}\right]$
C_5 : $\left[\bar{E} \parallel 1\right]$
C_6 : $\left[\bar{E} \parallel 3^{+}_{001}\right] \left[\bar{E} \parallel 3^{-}_{001}\right]$
C_7 : $\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
χ_1^{co}	2	0	2	2	−2	−2	−2
χ_2^{co}	2	0	2	−2	−2	−2	2
χ_3^{co}	4	0	−2	0	−4	2	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{120}\right]$	$\left[2_{100}T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} 0 & \omega_1 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \sigma_x & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$

No 498.1.1092 : ${}^{3_z}6/{}^{1}m^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{210}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{110} \parallel m_{010}\right]$ $\left[2_{100} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{110} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110} \parallel 2_{210}\right]$ $\left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[E \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{010} \parallel m_{110}\right]$ $\left[2_{110} \parallel m_{010}\right]$
C_{10} : $\left[2_{110} \parallel m_{210}\right]$ $\left[2_{010} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{110} \parallel 2_{210}\right]$
C_{12} : $\left[2_{110} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 498.2.1093 : ${}^{3_z}6/{}^{1}m^{2_x}m^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right] \left[3^{-}_{001} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right] \left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[E \parallel -1\right] \left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right] \left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{110} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[6^{-}_{001}T \parallel 6^{+}_{001}\right] \left[6^{+}_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right] \left[6^{-}_{001}T \parallel 3^{-}_{001}\right] \left[6^{+}_{001}T \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[6^{-}_{001}T \parallel \bar{3}^{-}_{001}\right] \left[6^{+}_{001}T \parallel \bar{6}^{-}_{001}\right] \left[2_{001}T \parallel -1\right] \left[6^{-}_{001}T \parallel \bar{6}^{+}_{001}\right] \left[6^{+}_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right] \left[2_{1\bar{1}0}T \parallel m_{210}\right] \left[2_{210}T \parallel m_{1\bar{1}0}\right] \left[2_{210}T \parallel m_{110}\right] \left[2_{1\bar{1}0}T \parallel m_{010}\right] \left[2_{120}T \parallel m_{120}\right] \left[2_{120}T \parallel 2_{120}\right] \left[2_{1\bar{1}0}T \parallel 2_{010}\right] \left[2_{210}T \parallel 2_{110}\right]$
$\left[2_{210}T \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0}T \parallel 2_{210}\right] \left[2_{120}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right] \left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_2 :	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 6^{-}_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right]$	C_9 :	$\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_5 :	$\left[E \parallel m_{001}\right]$	C_{11} :	$\left[2_{100} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{110} \parallel 2_{210}\right]$
C_6 :	$\left[3^{+}_{001} \parallel \bar{3}^{-}_{001}\right] \left[3^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$	C_{12} :	$\left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{10}	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
Γ^{co}_{11}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{12}	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 499.1.1094 : ${}^{3_z}6/{}^{1}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[3^{+}_{001} \,||\, 6^{+}_{001}\right] \left[3^{-}_{001} \,||\, 3^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[3^{+}_{001} \,||\, 3^{-}_{001}\right] \left[3^{-}_{001} \,||\, 6^{-}_{001}\right] \left[E \,||\, m_{001}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{001}\right] \left[3^{-}_{001} \,||\, \bar{6}^{-}_{001}\right]$
$\left[E \,||\, -1\right] \left[3^{+}_{001} \,||\, \bar{6}^{+}_{001}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{001}\right] \left[2_{100}T \,||\, m_{100}\right] \left[2_{110}T \,||\, m_{210}\right] \left[2_{010}T \,||\, m_{1\bar{1}0}\right] \left[2_{010}T \,||\, m_{110}\right] \left[2_{110}T \,||\, m_{010}\right] \left[2_{100}T \,||\, m_{120}\right]$
$\left[2_{100}T \,||\, 2_{120}\right] \left[2_{110}T \,||\, 2_{010}\right] \left[2_{010}T \,||\, 2_{110}\right] \left[2_{010}T \,||\, 2_{1\bar{1}0}\right] \left[2_{110}T \,||\, 2_{210}\right] \left[2_{100}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \,||\, 1\right]$	C_7 :	$\left[E \,||\, m_{001}\right]$
C_2 :	$\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$	C_8 :	$\left[3^{+}_{001} \,||\, \bar{3}^{-}_{001}\right]$
C_3 :	$\left[3^{-}_{001} \,||\, 3^{+}_{001}\right]$	C_9 :	$\left[3^{-}_{001} \,||\, \bar{6}^{-}_{001}\right]$
C_4 :	$\left[E \,||\, 2_{001}\right]$	C_{10} :	$\left[E \,||\, -1\right]$
C_5 :	$\left[3^{+}_{001} \,||\, 3^{-}_{001}\right]$	C_{11} :	$\left[3^{+}_{001} \,||\, \bar{6}^{+}_{001}\right]$
C_6 :	$\left[3^{-}_{001} \,||\, 6^{-}_{001}\right]$	C_{12} :	$\left[3^{-}_{001} \,||\, \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$	$\left[E \,||\, m_{001}\right]$	$\left[2_{100}T \,||\, m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 500.1.1095 : ${}^{m_z}6/{}^{2_z}m{}^{m_x}m{}^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[2_{001} \parallel \bar{6}^{-}_{001}\right] \left[2_{001} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$
C_6 : $\left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{010} \parallel m_{210}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 501.1.1096 : ${}^{2_z}6/{}^{m_z}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[2_{001} \mid\mid 6^{+}_{001}\right]$ $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$ $\left[2_{001} \mid\mid 6^{-}_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[T \mid\mid \bar{3}^{-}_{001}\right]$ $\left[2_{001}T \mid\mid \bar{6}^{-}_{001}\right]$
$\left[T \mid\mid -1\right]$ $\left[2_{001}T \mid\mid \bar{6}^{+}_{001}\right]$ $\left[T \mid\mid \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \mid\mid m_{100}\right]$ $\left[2_{010}T \mid\mid m_{210}\right]$ $\left[2_{010}T \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{100}T \mid\mid m_{110}\right]$ $\left[2_{100}T \mid\mid m_{010}\right]$ $\left[2_{010}T \mid\mid m_{120}\right]$
$\left[2_{010} \mid\mid 2_{120}\right]$ $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[2_{010} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{210}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[2_{001} \mid\mid 6^{+}_{001}\right]$ $\left[2_{001} \mid\mid 6^{-}_{001}\right]$
C_3 : $\left[E \mid\mid 3^{+}_{001}\right]$ $\left[E \mid\mid 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[2_{010} \mid\mid 2_{120}\right]$ $\left[2_{010} \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{010} \mid\mid 2_{210}\right]$
C_6 : $\left[2_{100} \mid\mid 2_{010}\right]$ $\left[2_{100} \mid\mid 2_{110}\right]$ $\left[2_{100} \mid\mid 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{010} \mid\mid 2_{120}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 502.1.1097 : ${}^{2_z}6/{}^{m_z}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[T \parallel \bar{3}^{-}_{001}\right] \left[2_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[2_{001}T \parallel \bar{6}^{+}_{001}\right] \left[T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 503.1.1098 : ${}^{m_z}6/{}^{m_x}m^{2_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{010} \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 504.1.1099 : ${}^{m_z}6/{}^{2_x}m^{m_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[T \parallel 2_{120}\right] \left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{210}\right] \left[2_{001} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right]$
C_4 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$
C_6 : $\left[2_{001} \parallel 2_{010}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{010} \parallel m_{210}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 505.1.1100 : ${}^{2_z}6/{}^{2_x}m^{m_y}m^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[2_{100} \parallel m_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right]$ $\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{010}T \parallel m_{100}\right]$ $\left[2_{100}T \parallel m_{210}\right]$ $\left[2_{100}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right]$ $\left[T \parallel 2_{010}\right]$ $\left[T \parallel 2_{110}\right]$ $\left[2_{001}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001}T \parallel 2_{210}\right]$ $\left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{010}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 506.1.1101 : ${}^{2_z}6/{}^{m_x}m^{2_y}m^{2_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{001}T \parallel 2_{120}\right] \left[T \parallel 2_{010}\right] \left[T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{210}\right] \left[T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{100} \parallel m_{210}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel m_{100}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 507.1.1102 : ${}^{m_z}6/{}^{2_x}m^{\bar{1}}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right] \left[T \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{110}\right] \left[T \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right]$
C_4 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_6 : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{210}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 508.1.1103 : ${}^{m_z}6/{}^{m_x}m^{2_z}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{010} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 509.1.1104 : ${}^{m_z}6/^{\bar{1}}m^{2_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{-}_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 510.1.1105 : ${}^{2_z}6/{}^{m_x}m^{m_z}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{010} \parallel 2_{120}\right] \left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{010} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$
C_6 : $\left[2_{100} \parallel 2_{010}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{010} \parallel 2_{120}\right]$	$\left[2_{100}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 511.1.1106 : ${}^{2_z}6/{}^{2_x}m^{m_z}m^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{010} \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[T \parallel m_{210}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{100} \parallel m_{001}\right]$
C_2 :	$\left[2_{001} \parallel 6^{+}_{001}\right]$	C_8 :	$\left[2_{010} \parallel \bar{3}^{-}_{001}\right]$
C_3 :	$\left[E \parallel 3^{+}_{001}\right]$	C_9 :	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{010} \parallel -1\right]$
C_5 :	$\left[E \parallel 3^{-}_{001}\right]$	C_{11} :	$\left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_6 :	$\left[2_{001} \parallel 6^{-}_{001}\right]$	C_{12} :	$\left[2_{010} \parallel \bar{3}^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{001}\right]$	$\left[2_{001}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 512.1.1107 : ${}^{2_z}6/^{\bar{1}}m^{m_x}m^{m_y}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[E \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$ $\left[T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[T \parallel \bar{6}^{+}_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{100}T \parallel m_{110}\right]$ $\left[2_{100}T \parallel m_{010}\right]$ $\left[2_{010}T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right]$ $\left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right]$ $\left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{100} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{100} \parallel 2_{210}\right]$
C_6 : $\left[2_{010} \parallel 2_{010}\right]$ $\left[2_{010} \parallel 2_{110}\right]$ $\left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel 2_{120}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 513.1.1108 : ${}^{2_z}6/^{\bar{1}}m^{2_x}m^{2_y}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{-}_{001}\right] \left[T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[T \parallel \bar{6}^{+}_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{010} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 2_{210}\right] \left[2_{010}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 6^{+}_{001}\right] \left[2_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 514.1.1109 : ${}^{\bar{1}}6/{}^{m_x}m^{2_z}m^{m_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100} \parallel -1\right] \left[2_{100}T \parallel \bar{6}^{+}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001}T \parallel m_{210}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right] \left[2_{001}T \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{001}\right]$
C_4 : $\left[2_{100} \parallel -1\right]$
C_5 : $\left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{010}\right]$
C_6 : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{3}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{100}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 515.1.1110 : ${}^{\bar{1}}6/{}^{2_x}m^{m_z}m^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 6^{+}_{001}\right] \left[E \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[E \parallel 3^{-}_{001}\right] \left[T \parallel 6^{-}_{001}\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{100}T \parallel -1\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{001}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001} \parallel m_{120}\right]$
$\left[2_{010}T \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{010} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{001}\right] \left[E \parallel 3^{-}_{001}\right]$
C_3 : $\left[2_{100} \parallel m_{001}\right]$
C_4 : $\left[2_{100} \parallel \bar{6}^{-}_{001}\right] \left[2_{100} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{210}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{120}\right]$
C_6 : $\left[2_{010} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{001} \parallel m_{210}\right]$	$\left[T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 516.1.1111 : ${}^{6_z}6/{}^{2_z}m^{2_x}m^{2_1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^+_{001} \parallel 6^+_{001}\right] \left[3^+_{001} \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right] \left[6^-_{001} \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[3^-_{001} \parallel \bar{3}^-_{001}\right] \left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[6^+_{001} \parallel \bar{6}^+_{001}\right] \left[3^+_{001} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{120} \parallel m_{120}\right]$
$\left[2_{120} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^+_{001} \parallel 6^+_{001}\right] \left[6^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel 3^+_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[3^-_{001} \parallel \bar{3}^-_{001}\right] \left[3^+_{001} \parallel \bar{3}^+_{001}\right]$
C_7 : $\left[6^-_{001} \parallel \bar{6}^-_{001}\right] \left[6^+_{001} \parallel \bar{6}^+_{001}\right]$
C_8 : $\left[E \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{120} \parallel m_{120}\right]$
C_{11} : $\left[2_{120} \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 2_{210}\right]$
C_{12} : $\left[2_{010} \parallel 2_{010}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[6^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 516.2.1112 : ${}^{6_z}6/{}^{2_z}m^{2_x}m^{2_1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^+_{001} \parallel 6^+_{001}\right] \left[3^+_{001} \parallel 3^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right] \left[6^-_{001} \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[3^-_{001} \parallel \bar{3}^-_{001}\right] \left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[6^+_{001} \parallel \bar{6}^+_{001}\right] \left[3^+_{001} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{120} \parallel m_{120}\right]$
$\left[2_{120} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[3^-_{001}T \parallel 6^+_{001}\right] \left[6^-_{001}T \parallel 3^+_{001}\right]$
$\left[T \parallel 2_{001}\right] \left[6^+_{001}T \parallel 3^-_{001}\right] \left[3^+_{001}T \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[6^+_{001}T \parallel \bar{3}^-_{001}\right] \left[3^+_{001}T \parallel \bar{6}^-_{001}\right] \left[2_{001}T \parallel -1\right] \left[3^-_{001}T \parallel \bar{6}^+_{001}\right] \left[6^-_{001}T \parallel \bar{3}^+_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{210}\right] \left[2_{110}T \parallel m_{1\bar{1}0}\right] \left[2_{1\bar{1}0}T \parallel m_{110}\right] \left[2_{210}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right] \left[2_{100}T \parallel 2_{120}\right] \left[2_{210}T \parallel 2_{010}\right] \left[2_{1\bar{1}0}T \parallel 2_{110}\right]$
$\left[2_{110}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{120}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[6^-_{001} \parallel \bar{6}^-_{001}\right] \left[6^+_{001} \parallel \bar{6}^+_{001}\right]$
C_2 :	$\left[6^+_{001} \parallel 6^+_{001}\right] \left[6^-_{001} \parallel 6^-_{001}\right]$	C_8 :	$\left[E \parallel -1\right]$
C_3 :	$\left[3^+_{001} \parallel 3^+_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right]$	C_9 :	$\left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{120} \parallel m_{120}\right]$
C_5 :	$\left[2_{001} \parallel m_{001}\right]$	C_{11} :	$\left[2_{120} \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 2_{210}\right]$
C_6 :	$\left[3^-_{001} \parallel \bar{3}^-_{001}\right] \left[3^+_{001} \parallel \bar{3}^+_{001}\right]$	C_{12} :	$\left[2_{010} \parallel 2_{010}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[6^{+}_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 517.1.1113 : ${}^{\bar{3}_z}6/{}^{\bar{1}}m^{2_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001}T \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001}T \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[3^+_{001} \parallel \bar{3}^-_{001}\right] \left[3^-_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[3^+_{001}T \parallel \bar{6}^+_{001}\right] \left[3^-_{001} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[3^+_{001} \parallel \bar{3}^-_{001}\right] \left[3^-_{001} \parallel \bar{3}^+_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel \bar{3}^-_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[3^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 518.1.1114 : ${}^{6_z}6/{}^{2_z}m^{m_x}m^{m_1}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[6^+_{001} \mid\mid 6^+_{001}\right] \left[3^+_{001} \mid\mid 3^+_{001}\right] \left[2_{001} \mid\mid 2_{001}\right] \left[3^-_{001} \mid\mid 3^-_{001}\right] \left[6^-_{001} \mid\mid 6^-_{001}\right] \left[2_{001} \mid\mid m_{001}\right] \left[3^-_{001} \mid\mid \bar{3}^-_{001}\right] \left[6^-_{001} \mid\mid \bar{6}^-_{001}\right]$
$\left[E \mid\mid -1\right] \left[6^+_{001} \mid\mid \bar{6}^+_{001}\right] \left[3^+_{001} \mid\mid \bar{3}^+_{001}\right] \left[2_{100}T \mid\mid m_{100}\right] \left[2_{210}T \mid\mid m_{210}\right] \left[2_{1\bar{1}0}T \mid\mid m_{1\bar{1}0}\right] \left[2_{110}T \mid\mid m_{110}\right] \left[2_{010}T \mid\mid m_{010}\right] \left[2_{120}T \mid\mid m_{120}\right]$
$\left[2_{120}T \mid\mid 2_{120}\right] \left[2_{010}T \mid\mid 2_{010}\right] \left[2_{110}T \mid\mid 2_{110}\right] \left[2_{1\bar{1}0}T \mid\mid 2_{1\bar{1}0}\right] \left[2_{210}T \mid\mid 2_{210}\right] \left[2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \mid\mid 1\right]$ $C_7 : \left[2_{001} \mid\mid m_{001}\right]$
$C_2 : \left[6^+_{001} \mid\mid 6^+_{001}\right]$ $C_8 : \left[3^-_{001} \mid\mid \bar{3}^-_{001}\right]$
$C_3 : \left[3^+_{001} \mid\mid 3^+_{001}\right]$ $C_9 : \left[6^-_{001} \mid\mid \bar{6}^-_{001}\right]$
$C_4 : \left[2_{001} \mid\mid 2_{001}\right]$ $C_{10} : \left[E \mid\mid -1\right]$
$C_5 : \left[3^-_{001} \mid\mid 3^-_{001}\right]$ $C_{11} : \left[6^+_{001} \mid\mid \bar{6}^+_{001}\right]$
$C_6 : \left[6^-_{001} \mid\mid 6^-_{001}\right]$ $C_{12} : \left[3^+_{001} \mid\mid \bar{3}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100}T \mid\mid m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 519.1.1115 : ${}^{\bar{6}_z}6/{}^{m_z}m^{m_x}m^{2_1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^+_{001}T \parallel 6^+_{001}\right] \left[3^+_{001} \parallel 3^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right] \left[6^-_{001}T \parallel 6^-_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[3^-_{001} \parallel \bar{3}^-_{001}\right] \left[6^-_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[E \parallel -1\right] \left[6^+_{001}T \parallel \bar{6}^+_{001}\right] \left[3^+_{001} \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{110}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{120} \parallel m_{120}\right]$
$\left[2_{120} \parallel 2_{120}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{110}T \parallel 2_{110}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel \bar{3}^-_{001}\right] \left[3^+_{001} \parallel \bar{3}^+_{001}\right]$
C_4 : $\left[E \parallel -1\right]$
C_5 : $\left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{120} \parallel m_{120}\right]$
C_6 : $\left[2_{120} \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[3^-_{001} \parallel \bar{3}^-_{001}\right]$	$\left[2_{210} \parallel m_{210}\right]$	$\left[6^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4	$\sigma_0\mathcal{K}$

No 520.1.1116 : ${}^{6_z}6/{}^{1}m^{2_x}m^{2_1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{+}_{001} \parallel 3^{+}_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[3^{-}_{001} \parallel 3^{-}_{001}\right] \left[6^{-}_{001} \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right] \left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{120} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{210} \parallel 2_{010}\right] \left[2_{1\bar{1}0} \parallel 2_{110}\right] \left[2_{110} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{120} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^{+}_{001} \parallel 6^{+}_{001}\right] \left[6^{-}_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right] \left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right] \left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{120} \parallel m_{120}\right]$
C_{11} : $\left[2_{100} \parallel 2_{120}\right] \left[2_{110} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$
C_{12} : $\left[2_{210} \parallel 2_{010}\right] \left[2_{1\bar{1}0} \parallel 2_{110}\right] \left[2_{120} \parallel 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 520.2.1117 : ${}^{6_z}6/{}^{1}m^{2_x}m^{2_1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$ $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{210} \parallel m_{210}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$ $\left[2_{120} \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{210} \parallel 2_{010}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{110}\right]$ $\left[2_{110} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{210}\right]$ $\left[2_{120} \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[3^{-}_{001}T \parallel 6^{+}_{001}\right]$ $\left[6^{-}_{001}T \parallel 3^{+}_{001}\right]$
$\left[T \parallel 2_{001}\right]$ $\left[6^{+}_{001}T \parallel 3^{-}_{001}\right]$ $\left[3^{+}_{001}T \parallel 6^{-}_{001}\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[3^{-}_{001}T \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{6}^{-}_{001}\right]$ $\left[T \parallel -1\right]$ $\left[6^{+}_{001}T \parallel \bar{6}^{+}_{001}\right]$ $\left[3^{+}_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{210}\right]$ $\left[2_{110}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{110}\right]$ $\left[2_{210}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{120}\right]$ $\left[2_{120}T \parallel 2_{120}\right]$ $\left[2_{010}T \parallel 2_{010}\right]$ $\left[2_{110}T \parallel 2_{110}\right]$
$\left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{210}T \parallel 2_{210}\right]$ $\left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$ $\left[6^{-}_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{001}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right]$
C_5 : $\left[E \parallel m_{001}\right]$
C_6 : $\left[6^{+}_{001} \parallel \bar{3}^{-}_{001}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[3^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$ $\left[3^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right]$ $\left[2_{110} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{010}\right]$
C_{10} : $\left[2_{210} \parallel m_{210}\right]$ $\left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$ $\left[2_{120} \parallel m_{120}\right]$
C_{11} : $\left[2_{100} \parallel 2_{120}\right]$ $\left[2_{110} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010} \parallel 2_{210}\right]$
C_{12} : $\left[2_{210} \parallel 2_{010}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{110}\right]$ $\left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[6^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ^{co}_{1}	1	1	1	$\mathcal{K}$
Γ^{co}_{2}	1	-1	-1	$\mathcal{K}$
Γ^{co}_{3}	1	1	-1	$\mathcal{K}$
Γ^{co}_{4}	1	-1	1	$\mathcal{K}$
Γ^{co}_{5}	-1	-1	1	$\mathcal{K}$
Γ^{co}_{6}	-1	1	-1	$\mathcal{K}$
Γ^{co}_{7}	-1	-1	-1	$\mathcal{K}$
Γ^{co}_{8}	-1	1	1	$\mathcal{K}$
Γ^{co}_{9}	ω^{*}_{1}	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{10}	ω^{*}_{1}	$-\sigma_0$	ω^{*}_{4}	$\sigma_0\mathcal{K}$
Γ^{co}_{11}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ^{co}_{12}	ω_3	σ_0	ω^{*}_{4}	$\sigma_0\mathcal{K}$

No 521.1.1118 : ${}^{\bar{3}_z}6/{}^{1}m^{m_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^{+}_{001}T \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[T \parallel 2_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right] \left[3^{-}_{001}T \parallel 6^{-}_{001}\right] \left[E \parallel m_{001}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{001}\right] \left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[T \parallel -1\right] \left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{110}T \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right] \left[3^{+}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_5 : $\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_6 : $\left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel \bar{6}^{-}_{001}\right]$	$\left[2_{110} \parallel m_{210}\right]$	$\left[3^{+}_{001}T \parallel 6^{+}_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 522.1.1119 : ${}^{6_z}6/{}^{1}m^{m_x}m^{m_1}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[6^+_{001} \parallel 6^+_{001}\right]$ $\left[3^+_{001} \parallel 3^+_{001}\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[3^-_{001} \parallel 3^-_{001}\right]$ $\left[6^-_{001} \parallel 6^-_{001}\right]$ $\left[E \parallel m_{001}\right]$ $\left[6^+_{001} \parallel \bar{3}^-_{001}\right]$ $\left[3^+_{001} \parallel \bar{6}^-_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[3^-_{001} \parallel \bar{6}^+_{001}\right]$ $\left[6^-_{001} \parallel \bar{3}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{210}T \parallel m_{210}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[2_{120}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right]$ $\left[2_{210}T \parallel 2_{010}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{110}\right]$ $\left[2_{110}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel 2_{210}\right]$ $\left[2_{120}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[E \parallel m_{001}\right]$
C_2 :	$\left[6^+_{001} \parallel 6^+_{001}\right]$	C_8 :	$\left[6^+_{001} \parallel \bar{3}^-_{001}\right]$
C_3 :	$\left[3^+_{001} \parallel 3^+_{001}\right]$	C_9 :	$\left[3^+_{001} \parallel \bar{6}^-_{001}\right]$
C_4 :	$\left[2_{001} \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[3^-_{001} \parallel 3^-_{001}\right]$	C_{11} :	$\left[3^-_{001} \parallel \bar{6}^+_{001}\right]$
C_6 :	$\left[6^-_{001} \parallel 6^-_{001}\right]$	C_{12} :	$\left[6^-_{001} \parallel \bar{3}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel 6^+_{001}\right]$	$\left[E \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 523.1.1120 : ${}^{\bar{6}_z}6/{}^1m^{2_x}m^{m_1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^+_{001}T \parallel 6^+_{001}\right] \left[3^+_{001} \parallel 3^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right] \left[6^-_{001}T \parallel 6^-_{001}\right] \left[E \parallel m_{001}\right] \left[6^+_{001}T \parallel \bar{3}^-_{001}\right] \left[3^+_{001} \parallel \bar{6}^-_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[3^-_{001} \parallel \bar{6}^+_{001}\right] \left[6^-_{001}T \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{210}T \parallel m_{210}\right] \left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{120}T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{210}T \parallel 2_{010}\right] \left[2_{1\bar{1}0}T \parallel 2_{110}\right] \left[2_{110} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right] \left[2_{120}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[E \parallel m_{001}\right]$
C_4 : $\left[3^+_{001} \parallel \bar{6}^-_{001}\right] \left[3^-_{001} \parallel \bar{6}^+_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{100} \parallel 2_{120}\right] \left[2_{110} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{210}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel \bar{6}^-_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[6^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_2	$\sigma_0\mathcal{K}$

No 524.1.1121 : ${}^{3_z}6/{}^{2_z}m{}^{2_x}m{}^{2_{xy}}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$ $\left[3^{-}_{001} \,||\, 3^{+}_{001}\right]$ $\left[E \,||\, 2_{001}\right]$ $\left[3^{+}_{001} \,||\, 3^{-}_{001}\right]$ $\left[3^{-}_{001} \,||\, 6^{-}_{001}\right]$ $\left[2_{001} \,||\, m_{001}\right]$ $\left[6^{-}_{001} \,||\, \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001} \,||\, \bar{6}^{-}_{001}\right]$
$\left[2_{001} \,||\, -1\right]$ $\left[6^{-}_{001} \,||\, \bar{6}^{+}_{001}\right]$ $\left[6^{+}_{001} \,||\, \bar{3}^{+}_{001}\right]$ $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{110} \,||\, m_{210}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{110} \,||\, m_{010}\right]$ $\left[2_{100} \,||\, m_{120}\right]$
$\left[2_{120} \,||\, 2_{120}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{010}\right]$ $\left[2_{210} \,||\, 2_{110}\right]$ $\left[2_{210} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{210}\right]$ $\left[2_{120} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$ $\left[3^{-}_{001} \,||\, 6^{-}_{001}\right]$
C_3 : $\left[3^{-}_{001} \,||\, 3^{+}_{001}\right]$ $\left[3^{+}_{001} \,||\, 3^{-}_{001}\right]$
C_4 : $\left[E \,||\, 2_{001}\right]$
C_5 : $\left[2_{001} \,||\, m_{001}\right]$
C_6 : $\left[6^{-}_{001} \,||\, \bar{3}^{-}_{001}\right]$ $\left[6^{+}_{001} \,||\, \bar{3}^{+}_{001}\right]$
C_7 : $\left[6^{+}_{001} \,||\, \bar{6}^{-}_{001}\right]$ $\left[6^{-}_{001} \,||\, \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \,||\, -1\right]$
C_9 : $\left[2_{100} \,||\, m_{100}\right]$ $\left[2_{010} \,||\, m_{110}\right]$ $\left[2_{110} \,||\, m_{010}\right]$
C_{10} : $\left[2_{110} \,||\, m_{210}\right]$ $\left[2_{010} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{100} \,||\, m_{120}\right]$
C_{11} : $\left[2_{120} \,||\, 2_{120}\right]$ $\left[2_{210} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{210}\right]$
C_{12} : $\left[2_{1\bar{1}0} \,||\, 2_{010}\right]$ $\left[2_{210} \,||\, 2_{110}\right]$ $\left[2_{120} \,||\, 2_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1	1	1	1	1	1	1	1	1	1	1	1	1
χ_2	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Irrepresntations of group generators

	$\left[3^{+}_{001} \,||\, 6^{+}_{001}\right]$	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$

Γ_1	1	1	1
Γ_2	1	-1	-1
Γ_3	1	1	-1
Γ_4	1	-1	1
Γ_5	-1	-1	1
Γ_6	-1	1	-1
Γ_7	-1	-1	-1
Γ_8	-1	1	1
Γ_9	ω_1^*	σ_0	ω_2
Γ_{10}	ω_1^*	$-\sigma_0$	ω_4^*
Γ_{11}	ω_3	$-\sigma_0$	ω_2
Γ_{12}	ω_3	σ_0	ω_4^*

No 524.2.1122 : ${}^{3_z}6/{}^{2_z}m^{2_x}m^{2_{xy}}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right] \left[3^{-}_{001} \parallel 6^{-}_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{001}\right] \left[6^{+}_{001} \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[6^{-}_{001} \parallel \bar{6}^{+}_{001}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{120} \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{210} \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \parallel 2_{210}\right] \left[2_{120} \parallel 2_{100}\right] \left[2_{001}T \parallel 1\right] \left[6^{-}_{001}T \parallel 6^{+}_{001}\right] \left[6^{+}_{001}T \parallel 3^{+}_{001}\right]$
$\left[2_{001}T \parallel 2_{001}\right] \left[6^{-}_{001}T \parallel 3^{-}_{001}\right] \left[6^{+}_{001}T \parallel 6^{-}_{001}\right] \left[T \parallel m_{001}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{001}\right] \left[3^{-}_{001}T \parallel \bar{6}^{-}_{001}\right] \left[T \parallel -1\right] \left[3^{+}_{001}T \parallel \bar{6}^{+}_{001}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{001}\right]$
$\left[2_{120}T \parallel m_{100}\right] \left[2_{1\bar{1}0}T \parallel m_{210}\right] \left[2_{210}T \parallel m_{1\bar{1}0}\right] \left[2_{210}T \parallel m_{110}\right] \left[2_{1\bar{1}0}T \parallel m_{010}\right] \left[2_{120}T \parallel m_{120}\right] \left[2_{100}T \parallel 2_{120}\right] \left[2_{110}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{001} \parallel m_{001}\right]$
C_6 : $\left[6^{-}_{001} \parallel \bar{3}^{-}_{001}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{001}\right]$
C_7 : $\left[6^{+}_{001} \parallel \bar{6}^{-}_{001}\right] \left[6^{-}_{001} \parallel \bar{6}^{+}_{001}\right]$
C_8 : $\left[2_{001} \parallel -1\right]$
C_9 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_{10} : $\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_{11} : $\left[2_{120} \parallel 2_{120}\right] \left[2_{210} \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \parallel 2_{210}\right]$
C_{12} : $\left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1	−1	−1	−1	−1	1	1
χ_3^{co}	1	1	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	1	1	1	−1	−1	−1	−1	1	1	−1	−1
χ_5^{co}	1	−1	1	−1	−1	1	−1	1	1	−1	−1	1
χ_6^{co}	1	−1	1	−1	1	−1	1	−1	−1	1	−1	1
χ_7^{co}	1	−1	1	−1	−1	1	−1	1	−1	1	1	−1
χ_8^{co}	1	−1	1	−1	1	−1	1	−1	1	−1	1	−1
χ_9^{co}	2	−1	−1	2	2	−1	−1	2	0	0	0	0
χ_{10}^{co}	2	−1	−1	2	−2	1	1	−2	0	0	0	0
χ_{11}^{co}	2	1	−1	−2	−2	−1	1	2	0	0	0	0
χ_{12}^{co}	2	1	−1	−2	2	1	−1	−2	0	0	0	0

Coirrepresntations of group generators

	$\left[3^{+}_{001} \mid\mid 6^{+}_{001}\right]$	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	-1	$\mathcal{K}$
Γ_3^{co}	1	1	-1	$\mathcal{K}$
Γ_4^{co}	1	-1	1	$\mathcal{K}$
Γ_5^{co}	-1	-1	1	$\mathcal{K}$
Γ_6^{co}	-1	1	-1	$\mathcal{K}$
Γ_7^{co}	-1	-1	-1	$\mathcal{K}$
Γ_8^{co}	-1	1	1	$\mathcal{K}$
Γ_9^{co}	ω_1^*	σ_0	ω_2	$\sigma_0\mathcal{K}$
Γ_{10}^{co}	ω_1^*	$-\sigma_0$	ω_4^*	$\sigma_0\mathcal{K}$
Γ_{11}^{co}	ω_3	$-\sigma_0$	ω_2	$\sigma_0\mathcal{K}$
Γ_{12}^{co}	ω_3	σ_0	ω_4^*	$\sigma_0\mathcal{K}$

No 525.1.1123 : ${}^{3_z}6/^{\bar{1}}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[E \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[3^+_{001}T \parallel \bar{3}^-_{001}\right] \left[3^-_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[T \parallel -1\right] \left[3^+_{001}T \parallel \bar{6}^+_{001}\right] \left[3^-_{001}T \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{110}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{110}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{100} \parallel 2_{120}\right] \left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{110} \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel 2_{120}\right] \left[2_{010} \parallel 2_{1\bar{1}0}\right] \left[2_{110} \parallel 2_{210}\right]$
C_6 : $\left[2_{110} \parallel 2_{010}\right] \left[2_{010} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel 2_{120}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 526.1.1124 : ${}^{3_z}6/^{\bar{1}}m^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[E \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[3^+_{001}T \parallel \bar{3}^-_{001}\right] \left[3^-_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[T \parallel -1\right] \left[3^+_{001}T \parallel \bar{6}^+_{001}\right] \left[3^-_{001}T \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{110}T \parallel 2_{010}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 2_{210}\right] \left[2_{100}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 527.1.1125 : ${}^{3_z}6/{}^{2_z}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[3^+_{001} \parallel 6^+_{001}\right]$ $\left[3^-_{001} \parallel 3^+_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^+_{001} \parallel 3^-_{001}\right]$ $\left[3^-_{001} \parallel 6^-_{001}\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[6^-_{001} \parallel \bar{3}^-_{001}\right]$ $\left[6^+_{001} \parallel \bar{6}^-_{001}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[6^-_{001} \parallel \bar{6}^+_{001}\right]$ $\left[6^+_{001} \parallel \bar{3}^+_{001}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{110}T \parallel m_{210}\right]$ $\left[2_{010}T \parallel m_{1\bar{1}0}\right]$ $\left[2_{010}T \parallel m_{110}\right]$ $\left[2_{110}T \parallel m_{010}\right]$ $\left[2_{100}T \parallel m_{120}\right]$
$\left[2_{120}T \parallel 2_{120}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{010}\right]$ $\left[2_{210}T \parallel 2_{110}\right]$ $\left[2_{210}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{210}\right]$ $\left[2_{120}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_7 :	$\left[2_{001} \parallel m_{001}\right]$
C_2 :	$\left[3^+_{001} \parallel 6^+_{001}\right]$	C_8 :	$\left[6^-_{001} \parallel \bar{3}^-_{001}\right]$
C_3 :	$\left[3^-_{001} \parallel 3^+_{001}\right]$	C_9 :	$\left[6^+_{001} \parallel \bar{6}^-_{001}\right]$
C_4 :	$\left[E \parallel 2_{001}\right]$	C_{10} :	$\left[2_{001} \parallel -1\right]$
C_5 :	$\left[3^+_{001} \parallel 3^-_{001}\right]$	C_{11} :	$\left[6^-_{001} \parallel \bar{6}^+_{001}\right]$
C_6 :	$\left[3^-_{001} \parallel 6^-_{001}\right]$	C_{12} :	$\left[6^+_{001} \parallel \bar{3}^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1
χ_3^{co}	1	-1	1	-1	1	-1	-1	1	-1	1	-1	1
χ_4^{co}	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1
χ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_6^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_7^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_8^{co}	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	1	$e^{\frac{\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	-1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$
χ_9^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{10}^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{\pi i}{3}}$
χ_{11}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_{12}^{co}	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	-1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{100}T \parallel m_{100}\right]$

Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	-1	$\mathcal{K}$
Γ_3^{co}	-1	-1	$\mathcal{K}$
Γ_4^{co}	-1	1	$\mathcal{K}$
Γ_5^{co}	$e^{\frac{-2\pi i}{3}}$	1	$\mathcal{K}$
Γ_6^{co}	$e^{\frac{-2\pi i}{3}}$	-1	$\mathcal{K}$
Γ_7^{co}	$e^{\frac{\pi i}{3}}$	-1	$\mathcal{K}$
Γ_8^{co}	$e^{\frac{\pi i}{3}}$	1	$\mathcal{K}$
Γ_9^{co}	$e^{\frac{2\pi i}{3}}$	1	$\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$e^{\frac{2\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{11}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	-1	$\mathcal{K}$
$\Gamma_{12}^{\mathrm{co}}$	$e^{\frac{-\pi i}{3}}$	1	$\mathcal{K}$

No 528.1.1126 : ${}^{3_z}6/{}^{m_z}m^{m_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[E \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[6^-_{001}T \parallel \bar{3}^-_{001}\right] \left[6^+_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[6^-_{001}T \parallel \bar{6}^+_{001}\right] \left[6^+_{001}T \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{110}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{110}T \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{120} \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{210} \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \parallel 2_{210}\right] \left[2_{120} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 6^-_{001}\right]$
C_3 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{120} \parallel 2_{120}\right] \left[2_{210} \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \parallel 2_{210}\right]$
C_6 : $\left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^+_{001} \parallel 6^+_{001}\right]$	$\left[2_{120} \parallel 2_{120}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 529.1.1127 : ${}^{3_z}6/{}^{m_z}m^{2_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right] \left[3^{-}_{001} \parallel 6^{-}_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[6^{-}_{001}T \parallel \bar{3}^{-}_{001}\right] \left[6^{+}_{001}T \parallel \bar{6}^{-}_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[6^{-}_{001}T \parallel \bar{6}^{+}_{001}\right] \left[6^{+}_{001}T \parallel \bar{3}^{+}_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{120}T \parallel 2_{120}\right] \left[2_{1\bar{1}0}T \parallel 2_{010}\right] \left[2_{210}T \parallel 2_{110}\right] \left[2_{210}T \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0}T \parallel 2_{210}\right] \left[2_{120}T \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 6^{+}_{001}\right] \left[3^{-}_{001} \parallel 6^{-}_{001}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{+}_{001}\right] \left[3^{+}_{001} \parallel 3^{-}_{001}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 6^{+}_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 530.1.1128 : ${}^{6_z}6/^{\bar{1}}m^{m_x}m^{m_1}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[6^+_{001} \,||\, 6^+_{001}\right]$ $\left[3^+_{001} \,||\, 3^+_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[3^-_{001} \,||\, 3^-_{001}\right]$ $\left[6^-_{001} \,||\, 6^-_{001}\right]$ $\left[T \,||\, m_{001}\right]$ $\left[6^+_{001}T \,||\, \bar{3}^-_{001}\right]$ $\left[3^+_{001}T \,||\, \bar{6}^-_{001}\right]$
$\left[2_{001}T \,||\, -1\right]$ $\left[3^-_{001}T \,||\, \bar{6}^+_{001}\right]$ $\left[6^-_{001}T \,||\, \bar{3}^+_{001}\right]$ $\left[2_{100}T \,||\, m_{100}\right]$ $\left[2_{210}T \,||\, m_{210}\right]$ $\left[2_{1\bar{1}0}T \,||\, m_{1\bar{1}0}\right]$ $\left[2_{110}T \,||\, m_{110}\right]$ $\left[2_{010}T \,||\, m_{010}\right]$ $\left[2_{120}T \,||\, m_{120}\right]$
$\left[2_{100} \,||\, 2_{120}\right]$ $\left[2_{210} \,||\, 2_{010}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{110}\right]$ $\left[2_{110} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{210}\right]$ $\left[2_{120} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[6^+_{001} \,||\, 6^+_{001}\right]$ $\left[6^-_{001} \,||\, 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \,||\, 3^+_{001}\right]$ $\left[3^-_{001} \,||\, 3^-_{001}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$
C_5 : $\left[2_{100} \,||\, 2_{120}\right]$ $\left[2_{110} \,||\, 2_{1\bar{1}0}\right]$ $\left[2_{010} \,||\, 2_{210}\right]$
C_6 : $\left[2_{210} \,||\, 2_{010}\right]$ $\left[2_{1\bar{1}0} \,||\, 2_{110}\right]$ $\left[2_{120} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \,||\, 6^+_{001}\right]$	$\left[2_{100} \,||\, 2_{120}\right]$	$\left[T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 531.1.1129 : ${}^{6_z}6/{}^{m_z}m^{m_x}m^{m_1}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[6^+_{001} \,||\, 6^+_{001}\right] \left[3^+_{001} \,||\, 3^+_{001}\right] \left[2_{001} \,||\, 2_{001}\right] \left[3^-_{001} \,||\, 3^-_{001}\right] \left[6^-_{001} \,||\, 6^-_{001}\right] \left[2_{001}T \,||\, m_{001}\right] \left[3^-_{001}T \,||\, \bar{3}^-_{001}\right] \left[6^-_{001}T \,||\, \bar{6}^-_{001}\right]$
$\left[T \,||\, -1\right] \left[6^+_{001}T \,||\, \bar{6}^+_{001}\right] \left[3^+_{001}T \,||\, \bar{3}^+_{001}\right] \left[2_{100}T \,||\, m_{100}\right] \left[2_{210}T \,||\, m_{210}\right] \left[2_{1\bar{1}0}T \,||\, m_{1\bar{1}0}\right] \left[2_{110}T \,||\, m_{110}\right] \left[2_{010}T \,||\, m_{010}\right] \left[2_{120}T \,||\, m_{120}\right]$
$\left[2_{120} \,||\, 2_{120}\right] \left[2_{010} \,||\, 2_{010}\right] \left[2_{110} \,||\, 2_{110}\right] \left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right] \left[2_{210} \,||\, 2_{210}\right] \left[2_{100} \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[6^+_{001} \,||\, 6^+_{001}\right] \left[6^-_{001} \,||\, 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \,||\, 3^+_{001}\right] \left[3^-_{001} \,||\, 3^-_{001}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$
C_5 : $\left[2_{120} \,||\, 2_{120}\right] \left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right] \left[2_{210} \,||\, 2_{210}\right]$
C_6 : $\left[2_{010} \,||\, 2_{010}\right] \left[2_{110} \,||\, 2_{110}\right] \left[2_{100} \,||\, 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \,||\, 6^+_{001}\right]$	$\left[2_{120} \,||\, 2_{120}\right]$	$\left[2_{001}T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_4^*	$\sigma_0\mathcal{K}$

No 532.1.1130 : ${}^{6_z}6/^{\bar{1}}m^{2_x}m^{2_1}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[6^+_{001} \,||\, 6^+_{001}\right] \left[3^+_{001} \,||\, 3^+_{001}\right] \left[2_{001} \,||\, 2_{001}\right] \left[3^-_{001} \,||\, 3^-_{001}\right] \left[6^-_{001} \,||\, 6^-_{001}\right] \left[T \,||\, m_{001}\right] \left[6^+_{001}T \,||\, \bar{3}^-_{001}\right] \left[3^+_{001}T \,||\, \bar{6}^-_{001}\right]$
$\left[2_{001}T \,||\, -1\right] \left[3^-_{001}T \,||\, \bar{6}^+_{001}\right] \left[6^-_{001}T \,||\, \bar{3}^+_{001}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{210} \,||\, m_{210}\right] \left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right] \left[2_{110} \,||\, m_{110}\right] \left[2_{010} \,||\, m_{010}\right] \left[2_{120} \,||\, m_{120}\right]$
$\left[2_{100}T \,||\, 2_{120}\right] \left[2_{210}T \,||\, 2_{010}\right] \left[2_{1\bar{1}0}T \,||\, 2_{110}\right] \left[2_{110}T \,||\, 2_{1\bar{1}0}\right] \left[2_{010}T \,||\, 2_{210}\right] \left[2_{120}T \,||\, 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[6^+_{001} \,||\, 6^+_{001}\right] \left[6^-_{001} \,||\, 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \,||\, 3^+_{001}\right] \left[3^-_{001} \,||\, 3^-_{001}\right]$
C_4 : $\left[2_{001} \,||\, 2_{001}\right]$
C_5 : $\left[2_{100} \,||\, m_{100}\right] \left[2_{110} \,||\, m_{110}\right] \left[2_{010} \,||\, m_{010}\right]$
C_6 : $\left[2_{210} \,||\, m_{210}\right] \left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right] \left[2_{120} \,||\, m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \,||\, 6^+_{001}\right]$	$\left[2_{100} \,||\, m_{100}\right]$	$\left[T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 533.1.1131 : ${}^{6_z}6/{}^{m_z}m^{2_x}m^{2_1}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[2_{001} \mid\mid 2_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$ $\left[2_{001}T \mid\mid m_{001}\right]$ $\left[3^-_{001}T \mid\mid \bar{3}^-_{001}\right]$ $\left[6^-_{001}T \mid\mid \bar{6}^-_{001}\right]$
$\left[T \mid\mid -1\right]$ $\left[6^+_{001}T \mid\mid \bar{6}^+_{001}\right]$ $\left[3^+_{001}T \mid\mid \bar{3}^+_{001}\right]$ $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{210} \mid\mid m_{210}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$ $\left[2_{120} \mid\mid m_{120}\right]$
$\left[2_{120}T \mid\mid 2_{120}\right]$ $\left[2_{010}T \mid\mid 2_{010}\right]$ $\left[2_{110}T \mid\mid 2_{110}\right]$ $\left[2_{1\bar{1}0}T \mid\mid 2_{1\bar{1}0}\right]$ $\left[2_{210}T \mid\mid 2_{210}\right]$ $\left[2_{100}T \mid\mid 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[6^+_{001} \mid\mid 6^+_{001}\right]$ $\left[6^-_{001} \mid\mid 6^-_{001}\right]$
C_3 : $\left[3^+_{001} \mid\mid 3^+_{001}\right]$ $\left[3^-_{001} \mid\mid 3^-_{001}\right]$
C_4 : $\left[2_{001} \mid\mid 2_{001}\right]$
C_5 : $\left[2_{100} \mid\mid m_{100}\right]$ $\left[2_{110} \mid\mid m_{110}\right]$ $\left[2_{010} \mid\mid m_{010}\right]$
C_6 : $\left[2_{210} \mid\mid m_{210}\right]$ $\left[2_{1\bar{1}0} \mid\mid m_{1\bar{1}0}\right]$ $\left[2_{120} \mid\mid m_{120}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	1	1	−1	−1
χ_3^{co}	1	−1	1	−1	−1	1
χ_4^{co}	1	−1	1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	1	−1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \mid\mid 6^+_{001}\right]$	$\left[2_{100} \mid\mid m_{100}\right]$	$\left[2_{001}T \mid\mid m_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	1	−1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 534.1.1132 : ${}^{\bar{3}_z}6/{}^{m_z}m^{2_x}m^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001}T \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001}T \parallel 6^-_{001}\right] \left[2_{001}T \parallel m_{001}\right] \left[6^-_{001} \parallel \bar{3}^-_{001}\right] \left[6^+_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[6^-_{001}T \parallel \bar{6}^+_{001}\right] \left[6^+_{001} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{110}T \parallel m_{210}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right] \left[2_{100}T \parallel m_{120}\right]$
$\left[2_{120}T \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{210}T \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0}T \parallel 2_{210}\right] \left[2_{120} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[6^-_{001} \parallel \bar{3}^-_{001}\right] \left[6^+_{001} \parallel \bar{3}^+_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{110}\right] \left[2_{110} \parallel m_{010}\right]$
C_6 : $\left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[6^-_{001} \parallel \bar{3}^-_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[3^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 535.1.1133 : ${}^{\bar{3}_z}6/{}^{2_z}m^{m_x}m^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[3^+_{001}T \parallel 6^+_{001}\right] \left[3^-_{001} \parallel 3^+_{001}\right] \left[T \parallel 2_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right] \left[3^-_{001}T \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[6^-_{001}T \parallel \bar{3}^-_{001}\right] \left[6^+_{001} \parallel \bar{6}^-_{001}\right]$
$\left[2_{001}T \parallel -1\right] \left[6^-_{001} \parallel \bar{6}^+_{001}\right] \left[6^+_{001}T \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{110}T \parallel m_{010}\right] \left[2_{100} \parallel m_{120}\right]$
$\left[2_{120}T \parallel 2_{120}\right] \left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{210}T \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0}T \parallel 2_{210}\right] \left[2_{120} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^-_{001} \parallel 3^+_{001}\right] \left[3^+_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[6^+_{001} \parallel \bar{6}^-_{001}\right] \left[6^-_{001} \parallel \bar{6}^+_{001}\right]$
C_5 : $\left[2_{110} \parallel m_{210}\right] \left[2_{010} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{120}\right]$
C_6 : $\left[2_{1\bar{1}0} \parallel 2_{010}\right] \left[2_{210} \parallel 2_{110}\right] \left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel \bar{6}^-_{001}\right]$	$\left[2_{110} \parallel m_{210}\right]$	$\left[3^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 536.1.1134 : ${}^{\bar{6}_z}6/{}^{\bar{1}}m^{2_x}m^{m_1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^+_{001}T \parallel 6^+_{001}\right] \left[3^+_{001} \parallel 3^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right] \left[6^-_{001}T \parallel 6^-_{001}\right] \left[T \parallel m_{001}\right] \left[6^+_{001} \parallel \bar{3}^-_{001}\right] \left[3^+_{001}T \parallel \bar{6}^-_{001}\right]$
$\left[2_{001} \parallel -1\right] \left[3^-_{001}T \parallel \bar{6}^+_{001}\right] \left[6^-_{001} \parallel \bar{3}^+_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{210}T \parallel m_{210}\right] \left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right] \left[2_{120}T \parallel m_{120}\right]$
$\left[2_{100}T \parallel 2_{120}\right] \left[2_{210} \parallel 2_{010}\right] \left[2_{1\bar{1}0} \parallel 2_{110}\right] \left[2_{110}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{210}\right] \left[2_{120} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[6^+_{001} \parallel \bar{3}^-_{001}\right] \left[6^-_{001} \parallel \bar{3}^+_{001}\right]$
C_4 : $\left[2_{001} \parallel -1\right]$
C_5 : $\left[2_{100} \parallel m_{100}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{010} \parallel m_{010}\right]$
C_6 : $\left[2_{210} \parallel 2_{010}\right] \left[2_{1\bar{1}0} \parallel 2_{110}\right] \left[2_{120} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	−1	1
χ_3^{co}	1	1	1	1	−1	−1
χ_4^{co}	1	1	−1	−1	1	−1
χ_5^{co}	2	−1	−1	2	0	0
χ_6^{co}	2	−1	1	−2	0	0

Coirrepresntations of group generators

	$\left[6^+_{001} \parallel \bar{3}^-_{001}\right]$	$\left[2_{100} \parallel m_{100}\right]$	$\left[6^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	$\mathcal{K}$
Γ_4^{co}	−1	1	$\mathcal{K}$
Γ_5^{co}	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3^*	ω_4	$\sigma_0\mathcal{K}$

No 537.1.1135 : ${}^{\bar{6}_z}6/{}^{2_z}m^{m_x}m^{2_1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[6^+_{001}T \parallel 6^+_{001}\right] \left[3^+_{001} \parallel 3^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right] \left[6^-_{001}T \parallel 6^-_{001}\right] \left[2_{001} \parallel m_{001}\right] \left[3^-_{001}T \parallel \bar{3}^-_{001}\right] \left[6^-_{001} \parallel \bar{6}^-_{001}\right]$
$\left[T \parallel -1\right] \left[6^+_{001} \parallel \bar{6}^+_{001}\right] \left[3^+_{001}T \parallel \bar{3}^+_{001}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{110}T \parallel m_{110}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{120} \parallel m_{120}\right]$
$\left[2_{120}T \parallel 2_{120}\right] \left[2_{010} \parallel 2_{010}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right] \left[2_{210}T \parallel 2_{210}\right] \left[2_{100} \parallel 2_{100}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^+_{001} \parallel 3^+_{001}\right] \left[3^-_{001} \parallel 3^-_{001}\right]$
C_3 : $\left[2_{001} \parallel m_{001}\right]$
C_4 : $\left[6^-_{001} \parallel \bar{6}^-_{001}\right] \left[6^+_{001} \parallel \bar{6}^+_{001}\right]$
C_5 : $\left[2_{210} \parallel m_{210}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[2_{120} \parallel m_{120}\right]$
C_6 : $\left[2_{010} \parallel 2_{010}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6
χ_1^{co}	1	1	1	1	1	1
χ_2^{co}	1	1	−1	−1	1	−1
χ_3^{co}	1	1	−1	−1	−1	1
χ_4^{co}	1	1	1	1	−1	−1
χ_5^{co}	2	−1	2	−1	0	0
χ_6^{co}	2	−1	−2	1	0	0

Coirrepresntations of group generators

	$\left[6^-_{001} \parallel \bar{6}^-_{001}\right]$	$\left[2_{210} \parallel m_{210}\right]$	$\left[6^+_{001}T \parallel 6^+_{001}\right]$
Γ_1^{co}	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	$\mathcal{K}$
Γ_3^{co}	−1	−1	$\mathcal{K}$
Γ_4^{co}	1	−1	$\mathcal{K}$
Γ_5^{co}	ω_1^*	ω_2	$\sigma_0\mathcal{K}$
Γ_6^{co}	ω_3	ω_2	$\sigma_0\mathcal{K}$

No 538.1.1136 : $^{1}2^{1}3$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	3	-1	0	0

Irrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_4	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$

No 538.2.1137 : ${}^{1}2^{1}3^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{\bar{1}11}\right]$
$\left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	2	−1	−1
χ_3^{co}	3	−1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 538.3.1138 : ${}^{1}2^{1}3^{\infty_z}1 \to {}^{1}2^{1}3$

No 538.4.1139 : ${}^{1}2^{1}3^{\infty_z m_x}1 \rightarrow {}^{1}2^{1}3^{m_x}1$

No 539.1.1140 : $^{1}2^{3}3$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	3	-1	0	0

Irrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_4	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$

No 539.2.1141 : ${}^{1}2^{3_z}3^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[6^{-}_{001}T \parallel 3^{+}_{111}\right]$ $\left[6^{-}_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[6^{-}_{001}T \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[6^{+}_{001}T \parallel 3^{-}_{111}\right]$
$\left[6^{+}_{001}T \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[6^{+}_{001}T \parallel 3^{+}_{11\bar{1}}\right]$ $\left[6^{+}_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$ $\left[2_{001}T \parallel 2_{010}\right]$ $\left[6^{-}_{001}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	2	−1	−1
χ_3^{co}	3	−1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 540.1.1142 : ${}^{2_z}2^{3_{xyz}}3$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 2_{001}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$
C_3 : $\left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4
χ_1	1	1	1	1
χ_2	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4	3	-1	0	0

Irrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[3^{+}_{111} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_3	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_4	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$

No 541.1.1143 : ${}^{1}m^{1}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel m_{001}\right]$ $\left[E \parallel \bar{3}^{+}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[E \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[E \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel m_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$

C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right]$ $\left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[E \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[E \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	1	−1	−1	1
χ_3	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7	3	−1	0	0	0	0	3	−1
χ_8	3	1	0	0	0	0	−3	−1

Irrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	1
Γ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_4	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$

Γ_6	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_7	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_8	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$

No 541.2.1144 : ${}^{1}m^{1}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right]$
$\left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right]$
$\left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$

C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	2	2	−1	−1	−1	−1	2	2
χ_4^{co}	2	−2	1	−1	−1	1	−2	2
χ_5^{co}	3	−1	0	0	0	0	3	−1
χ_6^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$
Γ_3^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	ω_3^*	ω_1	$\sigma_x\mathcal{K}$

Γ_5^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_6^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 541.3.1145 : ${}^{1}m^{1}\bar{3}^{\infty_z}1 \rightarrow {}^{1}m^{1}\bar{3}$

No 541.4.1146 : ${}^{1}m^{1}\bar{3}^{\infty_z m_x}1 \rightarrow {}^{1}m^{1}\bar{3}^{m_x}1$

No 542.1.1147 : ${}^{2}m^{2}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_7 : $\left[2_{001} \parallel -1\right]$
C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	1	−1	−1	1
χ_3	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7	3	−1	0	0	0	0	3	−1
χ_8	3	1	0	0	0	0	−3	−1

Irrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	1
Γ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_4	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$

Γ_6	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_7	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_8	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$

No 542.2.1148 : ${}^{2_z}m^{2_z}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right]$
$\left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right]$
$\left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$

C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	2	2	−1	−1	−1	−1	2	2
χ_4^{co}	2	−2	1	−1	−1	1	−2	2
χ_5^{co}	3	−1	0	0	0	0	3	−1
χ_6^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$
Γ_3^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	ω_3^*	ω_1	$\sigma_x\mathcal{K}$

Γ_5^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_6^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 542.3.1149 : ${}^{2_z}m^{2_z}\bar{3}^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{111}\right]$
$\left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right] \left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right]$
$\left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$

C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	2	2	−1	−1	−1	−1	2	2
χ_4^{co}	2	−2	1	−1	−1	1	−2	2
χ_5^{co}	3	−1	0	0	0	0	3	−1
χ_6^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$
Γ_3^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	ω_3^*	ω_1	$\sigma_x\mathcal{K}$

Γ_5^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_6^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 542.4.1150 : ${}^{2_z}m^{2_z}\bar{3}^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}\bar{3}^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right]$
$\left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$

C_3 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_5 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_6 : $\left[\bar{E} \parallel 1\right]$

C_7 : $\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$

C_8 : $\left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_9 : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9

χ_1	2	0	2	2	2	−2	−2	−2	−2
χ_2	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2
χ_3	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2
χ_4	6	0	0	0	−2	−6	0	0	2

Irrepresntations of group generators

	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$\omega_6\mathcal{K}$	ω_5	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	$\omega_6^*\mathcal{K}$	ω_5^*	$\sigma_z\mathcal{K}$
Γ_4	$(defabc)\mathcal{K}$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 543.1.1151 : ${}^{m}m^{m}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[2_{001}T \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[2_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	−1	−1	2
χ_3^{co}	3	0	0	−1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[E \parallel 3^{-}_{111}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	ω_1^*	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	(cab)	$(abc)\mathcal{K}$

No 543.2.1152 : ${}^{m_z}m^{m_z}\bar{3}^{\infty_x}1 \to {}^{m}m^{m}\bar{3}$

No 544.1.1153 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	−1	−1	2
χ_3^{co}	3	0	0	−1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[E \parallel 3^{-}_{111}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	ω_1^*	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	(cab)	$(abc)\mathcal{K}$

No 544.2.1154 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\infty_z}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right]$
$\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 :	$\left[E \parallel 1\right]$	C_9 :	$\left[\bar{E} \parallel 1\right]$
C_2 :	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$	C_{10} :	$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 :	$\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$	C_{11} :	$\left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 :	$\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$	C_{12} :	$\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$
C_5 :	$\left[\Sigma_z \parallel 1\right]$	C_{13} :	$\left[\bar{\Sigma}_z \parallel 1\right]$
C_6 :	$\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$	C_{14} :	$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$
C_7 :	$\left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$	C_{15} :	$\left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_8 :	$\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$	C_{16} :	$\left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	-2	-2	-2	-2	0	0	0	0
χ_2^{co}	2	-1	-1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	0	-2	1	1	-2	0	$-\sqrt{3}i$	$\sqrt{3}i$	0
χ_3^{co}	2	-1	-1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	0	-2	1	1	-2	0	$\sqrt{3}i$	$-\sqrt{3}i$	0
χ_4^{co}	6	0	0	-2	0	0	0	0	-6	0	0	2	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[E \parallel 3^{-}_{111}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	ω_1^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	ω_1^*	ω_1	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{c}\bar{a}\bar{e}\bar{f}d)$	$(cab\bar{f}\bar{d}e)$	$(defabc)\mathcal{K}$	$(abcd\bar{e}\bar{f})\mathcal{K}$

No 544.3.1155 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right]$
$\left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}\right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}\right.$
$\left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{111}\right]$
$\left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel\right.$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_5 : $\left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{3}^{+}_{111}\right] \left[2_{100} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, -1\right] \left[2_{100} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[2_{100} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^{-}_{111}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^{-}_{111}\right]$
$\left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 3^{+}_{111}\right] \left[\Sigma_z \,||\, 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{+}_{111}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, -1\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^{-}_{111}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\bar{E} 2_{100} \,||\, m_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{+}_{111}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{E} 2_{100} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, -1\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^{-}_{111}\right] \left[\bar{E} 2_{100} \,||\, m_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{111}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{111}\right]$
$\left[\bar{\Sigma}_z \,||\, 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right]$

C_6 : $\left[\bar{E} \,||\, 1\right]$

C_7 : $\left[\bar{E} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^{+}_{111}\right] \left[\bar{E} \,||\, 3^{-}_{11\bar{1}}\right]$

C_8 : $\left[\bar{E} \,||\, 3^{-}_{111}\right] \left[\bar{E} \,||\, 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^{+}_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	2	0	−2	−2	−2	−2
χ_2^{co}	4	−2	−2	4	0	−4	2	2	−4
χ_3^{co}	6	0	0	−2	0	−6	0	0	2

Coirrepresntations of group generators

	$\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^{-}_{1\bar{1}1}\right]$	$\left[2_{100} \,||\, \bar{3}^{+}_{111}\right]$	$\left[2_{100} T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x \mathcal{K}$	σ_x	$\sigma_z \mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_5 & 0 \\ 0 & \omega_5^* \end{pmatrix}$	$\begin{pmatrix} \omega_5 & 0 \\ 0 & \omega_5^* \end{pmatrix}$	$\begin{pmatrix} \omega_6 & 0 \\ 0 & \omega_6^* \end{pmatrix} \mathcal{K}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix} \mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{c}\bar{a}\bar{e}\bar{f}d)$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 545.1.1156 : ${}^{1}m^{3}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel m_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$

C_3 : $\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[3^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	1	−1	−1	1
χ_3	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7	3	−1	0	0	0	0	3	−1
χ_8	3	1	0	0	0	0	−3	−1

Irrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	1
Γ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_4	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$

Γ_6	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_7	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_8	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$

No 545.2.1157 : ${}^{1}m^{3_z}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel m_{001}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$ $\left[2_{001}T \parallel 1\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{+}_{111}\right]$
$\left[6^{-}_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[6^{-}_{001}T \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[6^{+}_{001}T \parallel 3^{-}_{111}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{001}T \parallel -1\right]$ $\left[2_{001}T \parallel 2_{001}\right]$ $\left[2_{001}T \parallel 2_{100}\right]$
$\left[2_{001}T \parallel 2_{010}\right]$ $\left[6^{-}_{001}T \parallel 3^{+}_{111}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{-}_{111}\right]$ $\left[6^{+}_{001}T \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[6^{+}_{001}T \parallel 3^{+}_{11\bar{1}}\right]$ $\left[6^{+}_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[6^{-}_{001}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel m_{001}\right]$ $\left[E \parallel m_{100}\right]$ $\left[E \parallel m_{010}\right]$

C_3 : $\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[3^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	2	2	−1	−1	−1	−1	2	2
χ_4^{co}	2	−2	1	−1	−1	1	−2	2
χ_5^{co}	3	−1	0	0	0	0	3	−1
χ_6^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$
Γ_3^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	ω_3^*	ω_1	$\sigma_x\mathcal{K}$

Γ_5^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_6^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 546.1.1158 : ${}^{2}m^{6}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel m_{001}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{111}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{111}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right]$ $\left[2_{001} \parallel m_{100}\right]$ $\left[2_{001} \parallel m_{010}\right]$

C_3 : $\left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[6^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{+}_{001} \parallel 3^{-}_{111}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[6^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$ $\left[6^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	1	−1	−1	1
χ_3	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7	3	−1	0	0	0	0	3	−1
χ_8	3	1	0	0	0	0	−3	−1

Irrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	1
Γ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_4	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$

Γ_6	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_7	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_8	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$

No 546.2.1159 : ${}^{2_z}m^{6_z}\bar{3}^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{-}_{111}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{-}_{001} \parallel 3^{+}_{111}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel m_{001}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{111}\right]$
$\left[6^{+}_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[6^{-}_{001}T \parallel 3^{-}_{111}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right]$
$\left[2_{001}T \parallel 2_{010}\right] \left[6^{+}_{001}T \parallel 3^{+}_{111}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{111}\right] \left[6^{-}_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[6^{-}_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[6^{-}_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[6^{+}_{001}T \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$

C_3 : $\left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{+}_{001} \parallel 3^{-}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[6^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	2	2	−1	−1	−1	−1	2	2
χ_4^{co}	2	−2	1	−1	−1	1	−2	2
χ_5^{co}	3	−1	0	0	0	0	3	−1
χ_6^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$
Γ_3^{co}	σ_0	ω_1	ω_1	$\sigma_x\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	ω_3^*	ω_1	$\sigma_x\mathcal{K}$

Γ_5^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_6^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 547.1.1160 : ${}^{\bar{1}}m^{\bar{3}}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	−1	−1	2
χ_3^{co}	3	0	0	−1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[3^{-}_{001} \parallel 3^{-}_{111}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	ω_1^*	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	(cab)	$(abc)\mathcal{K}$

No 548.1.1161 : ${}^{m}m^{\bar{6}}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[6^{+}_{001}T \parallel \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{-}_{\bar{1}11}\right] \left[6^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{-}_{001} \parallel 3^{+}_{\bar{1}11}\right] \left[6^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[6^{-}_{001}T \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{\bar{1}11}\right] \left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{-}_{\bar{1}11}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	−1	−1	2
χ_3^{co}	3	0	0	−1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[3^{+}_{001} \parallel 3^{-}_{\bar{1}11}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	ω_1^*	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	(cab)	$(abc)\mathcal{K}$

No 549.1.1162 : ${}^{2_z}m^{3_{xyz}}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[3^{+}_{111} \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{111} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$

C_3 : $\left[3^{+}_{111} \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[3^{+}_{1\bar{1}1} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{111} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \parallel -1\right]$

C_8 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	1	−1	−1	1
χ_3	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7	3	−1	0	0	0	0	3	−1
χ_8	3	1	0	0	0	0	−3	−1

Irrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[3^{+}_{111} \parallel \bar{3}^{+}_{111}\right]$	$\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	−1	1
Γ_3	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_4	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
Γ_5	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$

Γ_6	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
Γ_7	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_8	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$

No 550.1.1163 : ${}^{m_z}m^{\bar{3}}{}_{xyz}\bar{3}$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[3^{+}_{111}T \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{111}T \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
C_4 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	2	−1	−1	2
χ_3^{co}	3	0	0	−1

Coirrepresntations of group generators

	$\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[3^{-}_{111} \parallel 3^{-}_{111}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	ω_1^*	$\sigma_x\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	(cab)	$(abc)\mathcal{K}$

No 551.1.1164 : ${}^{1}4^{1}3^{1}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 2_{011}\right]$ $\left[E \parallel 2_{101}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[E \parallel 4^{+}_{100}\right]$ $\left[E \parallel 4^{+}_{010}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 4^{-}_{010}\right]$ $\left[E \parallel 4^{-}_{100}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{01\bar{1}}\right]$ $\left[E \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right]$ $\left[E \parallel 4^{-}_{001}\right]$ $\left[E \parallel 4^{+}_{100}\right]$ $\left[E \parallel 4^{+}_{010}\right]$ $\left[E \parallel 4^{-}_{010}\right]$ $\left[E \parallel 4^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel 2_{011}\right]$ $\left[E \parallel 2_{101}\right]$ $\left[E \parallel 2_{110}\right]$ $\left[E \parallel 2_{1\bar{1}0}\right]$ $\left[E \parallel 2_{01\bar{1}}\right]$ $\left[E \parallel 2_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	1	−1	0	−1
χ_5	3	−1	−1	0	1

Irrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$

No 551.2.1165 : ${}^{1}4^{1}3^{1}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right]$
$\left[2_{001}T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{101}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[2_{001}T \parallel 4^{-}_{010}\right] \left[2_{001}T \parallel 4^{-}_{100}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 4^{+}_{001}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel 2_{011}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	1	−1	0	−1
χ_5^{co}	3	−1	−1	0	1

Coirrepresntations of group generators

	$\left[E \parallel 4^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 551.3.1166 : ${}^{1}4^{1}3^{1}2^{\infty_z}1 \to {}^{1}4^{1}3^{1}2$

No 551.4.1167 : ${}^{1}4^{1}3^{1}2^{\infty}{}_{z}m_{x}1 \rightarrow {}^{1}4^{1}3^{1}2^{m_{x}}1$

No 552.1.1168 : ${}^{2}4^{1}3^{2}2$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[E \parallel 2_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[E \parallel 3^{+}_{111}\right]$ $\left[2_{001} \parallel 2_{011}\right]$ $\left[2_{001} \parallel 2_{101}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right]$ $\left[2_{001} \parallel 4^{+}_{010}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[2_{001} \parallel 4^{-}_{010}\right]$ $\left[2_{001} \parallel 4^{-}_{100}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{001} \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{01\bar{1}}\right]$ $\left[2_{001} \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^{+}_{001}\right]$ $\left[2_{001} \parallel 4^{-}_{001}\right]$ $\left[2_{001} \parallel 4^{+}_{100}\right]$ $\left[2_{001} \parallel 4^{+}_{010}\right]$ $\left[2_{001} \parallel 4^{-}_{010}\right]$ $\left[2_{001} \parallel 4^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{-}_{111}\right]$ $\left[E \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[E \parallel 3^{+}_{11\bar{1}}\right]$ $\left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel 2_{011}\right]$ $\left[2_{001} \parallel 2_{101}\right]$ $\left[2_{001} \parallel 2_{110}\right]$ $\left[2_{001} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{001} \parallel 2_{01\bar{1}}\right]$ $\left[2_{001} \parallel 2_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	1	−1	0	−1
χ_5	3	−1	−1	0	1

Irrepresntations of group generators

	$\left[2_{001} \parallel 4^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$

No 552.2.1169 : ${}^{2_z}4^1 3^{2_z} 2^{m_z} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[E \parallel 3^+_{111}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{101}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right]$
$\left[2_{001} \parallel 4^+_{100}\right] \left[2_{001} \parallel 4^+_{010}\right] \left[E \parallel 3^-_{111}\right] \left[2_{001} \parallel 4^-_{010}\right] \left[2_{001} \parallel 4^-_{100}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^-_{11\bar{1}}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel 4^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right]$
$\left[T \parallel 4^-_{001}\right] \left[2_{001}T \parallel 3^+_{111}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{101}\right] \left[2_{001}T \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^-_{1\bar{1}1}\right] \left[T \parallel 4^+_{100}\right] \left[T \parallel 4^+_{010}\right] \left[2_{001}T \parallel 3^-_{111}\right]$
$\left[T \parallel 4^-_{010}\right] \left[T \parallel 4^-_{100}\right] \left[2_{001}T \parallel 3^+_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^+_{11\bar{1}}\right] \left[2_{001}T \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^-_{11\bar{1}}\right]$
$\left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[2_{001} \parallel 4^+_{100}\right] \left[2_{001} \parallel 4^+_{010}\right] \left[2_{001} \parallel 4^-_{010}\right] \left[2_{001} \parallel 4^-_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^+_{111}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right] \left[E \parallel 3^-_{111}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	1	−1	0	−1
χ_5^{co}	3	−1	−1	0	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel 3^+_{111}\right]$	$\left[E \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 552.3.1170 : ${}^{2_z}4^1 3^{2_z} 2^{m_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[E \parallel 3^+_{111}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{101}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right]$
$\left[2_{001} \parallel 4^+_{100}\right] \left[2_{001} \parallel 4^+_{010}\right] \left[E \parallel 3^-_{111}\right] \left[2_{001} \parallel 4^-_{010}\right] \left[2_{001} \parallel 4^-_{100}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^-_{11\bar{1}}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel 4^+_{001}\right] \left[2_{100}T \parallel 2_{001}\right]$
$\left[2_{010}T \parallel 4^-_{001}\right] \left[2_{100}T \parallel 3^+_{111}\right] \left[2_{010}T \parallel 2_{011}\right] \left[2_{010}T \parallel 2_{101}\right] \left[2_{100}T \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^-_{1\bar{1}1}\right] \left[2_{010}T \parallel 4^+_{100}\right] \left[2_{010}T \parallel 4^+_{010}\right] \left[2_{100}T \parallel 3^-_{111}\right]$
$\left[2_{010}T \parallel 4^-_{010}\right] \left[2_{010}T \parallel 4^-_{100}\right] \left[2_{100}T \parallel 3^+_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^+_{11\bar{1}}\right] \left[2_{100}T \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^-_{11\bar{1}}\right]$
$\left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{01\bar{1}}\right] \left[2_{010}T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 4^+_{001}\right] \left[2_{001} \parallel 4^-_{001}\right] \left[2_{001} \parallel 4^+_{100}\right] \left[2_{001} \parallel 4^+_{010}\right] \left[2_{001} \parallel 4^-_{010}\right] \left[2_{001} \parallel 4^-_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^+_{111}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right] \left[E \parallel 3^-_{111}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	1	−1	0	−1
χ_5^{co}	3	−1	−1	0	1

Coirrepresntations of group generators

	$\left[2_{001} \parallel 4^+_{001}\right]$	$\left[E \parallel 3^+_{111}\right]$	$\left[E \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 552.4.1171 : ${}^{2_z}4^1 3^{2_z} 2^{\infty_x} 1 \rightarrow {}^{2_z}4^1 3^{2_z} 2^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[E \parallel 3^+_{111}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{101}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right]$
$\left[2_{100} \parallel 4^+_{100}\right] \left[2_{100} \parallel 4^+_{010}\right] \left[E \parallel 3^-_{111}\right] \left[2_{100} \parallel 4^-_{010}\right] \left[2_{100} \parallel 4^-_{100}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^-_{11\bar{1}}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^-_{001}\right] \left[\Sigma_z \parallel 3^+_{111}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel 4^+_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^+_{010}\right] \left[\Sigma_z \parallel 3^-_{111}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^-_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{100}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^+_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^-_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} \parallel 3^+_{111}\right] \left[\bar{E} 2_{100} \parallel 2_{011}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{101}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel 4^+_{100}\right] \left[\bar{E} 2_{100} \parallel 4^+_{010}\right] \left[\bar{E} \parallel 3^-_{111}\right] \left[\bar{E} 2_{100} \parallel 4^-_{010}\right] \left[\bar{E} 2_{100} \parallel 4^-_{100}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}1}\right]$
$\left[\bar{E} \parallel 3^+_{11\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^-_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{010}\right] \left[\bar{\Sigma}_z \parallel 3^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{100}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 4^+_{100}\right] \left[2_{100} \parallel 4^+_{010}\right] \left[2_{100} \parallel 4^-_{010}\right] \left[2_{100} \parallel 4^-_{100}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 4^+_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^+_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^-_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 4^+_{001}\right] \left[\bar{E} 2_{100} \parallel 4^-_{001}\right] \left[\bar{E} 2_{100} \parallel 2_{011}\right] \left[\bar{E} 2_{100} \parallel 2_{101}\right]$
$\left[\bar{E} 2_{100} \parallel 4^+_{100}\right] \left[\bar{E} 2_{100} \parallel 4^+_{010}\right] \left[\bar{E} 2_{100} \parallel 4^-_{010}\right] \left[\bar{E} 2_{100} \parallel 4^-_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^+_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right]$

C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_4 : $\left[E \parallel 3^+_{111}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right] \left[E \parallel 3^-_{111}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{11\bar{1}}\right]$

C_5 : $\left[\Sigma_z \parallel 3^+_{111}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \parallel 3^-_{111}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^+_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^+_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \parallel 1\right]$

C_8 : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

C_9 : $\left[\bar{E} \parallel 3^+_{111}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^-_{111}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^+_{11\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{11\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9

χ_1	2	0	2	2	0	0	−2	−2	−2
χ_2	2	0	2	−1	$\sqrt{3}i$	$-\sqrt{3}i$	−2	−2	1
χ_3	2	0	2	−1	$-\sqrt{3}i$	$\sqrt{3}i$	−2	−2	1
χ_4	6	0	−2	0	0	0	−6	2	0

Irrepresntations of group generators

	$\left[2_{100} \mid\mid 4^{+}_{001}\right]$	$\left[E \mid\mid 3^{+}_{111}\right]$	$\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	ω_1	ω_1	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	ω_1^*	ω_1^*	$\sigma_z\mathcal{K}$
Γ_4	$(d\bar{e}\bar{f}abc)\mathcal{K}$	$(bca\bar{f}d\bar{e})$	$(\bar{b}\bar{c}afde)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 553.1.1172 : ${}^{m}4^{1}3^{m}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001}T \parallel 4^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right] \left[E \parallel 3^{-}_{\bar{1}11}\right] \left[2_{001}T \parallel 4^{-}_{010}\right] \left[2_{001}T \parallel 4^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{\bar{1}\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 553.2.1173 : ${}^{m_z}4^1 3^{m_z} 2^{\infty_x} 1 \to {}^{m}4^1 3^{m} 2$

No 554.1.1174 : ${}^{\bar{1}}4^{1}3^{\bar{1}}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel 4^{-}_{010}\right] \left[T \parallel 4^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 554.2.1175 : ${}^{\bar{1}}4^{1}3^{\bar{1}}2^{\infty_z}1 \rightarrow {}^{\bar{1}}4^{1}3^{\bar{1}}2^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel 4^{-}_{010}\right] \left[T \parallel 4^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel 2_{011}\right]$
$\left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right]$
$\left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel 2_{011}\right] \left[\bar{\Sigma}_z T \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$
$C_2 : \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
$C_3 : \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
$C_4 : \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$C_5 : \left[\Sigma_z \parallel 1\right]$
$C_6 : \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$C_7 : \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$C_8 : \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$C_9 : \left[\bar{E} \parallel 1\right]$
$C_{10} : \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$
$C_{11} : \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$C_{12} : \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$C_{13} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_{14} : \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$C_{15} : \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$
$C_{16} : \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	−2	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0
χ_3^{co}	2	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	−2	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0
χ_4^{co}	6	−2	0	0	0	0	0	0	−6	2	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \parallel 4^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	ω_5	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	ω_5^*	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(bca\bar{f}d\bar{e})$	$(\bar{b}\bar{c}afde)$	$(d\bar{e}\bar{f}abc)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 554.3.1176 : ${}^{\bar{1}}4^{1}3^{\bar{1}}2^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}4^{1}3^{\bar{1}}2^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel 4^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel 4^{-}_{010}\right] \left[T \parallel 4^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{10\bar{1}}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right]$
$\left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel 2_{011}\right]$
$\left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right]$
$\left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3\right.$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right]$
$\left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right]$
$\left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}1}\right]$
$\left[\bar{E}2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel 2_{011}\right] \left[\bar{\Sigma}_z T \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_4 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right]$
$\left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right]$
$\left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right]$

C_5 : $\left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \,||\, 3^-_{\bar{1}11}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

C_9 : $\left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{\bar{1}11}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	4	4	−2	0	0	0	−4	−4	2
χ_3^{co}	6	−2	0	0	0	0	−6	2	0

Coirrepresntations of group generators

	$\left[E \,||\, 2_{001}\right]$	$\left[2_{100} \,||\, 4^+_{100}\right]$	$\left[E \,||\, 3^+_{111}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_3^{co}	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(f\bar{d}\bar{e}\bar{b}\bar{c}a)\mathcal{K}$	$(bca\bar{f}d\bar{e})$	$(defa\bar{b}\bar{c})$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 555.1.1177 : ${}^{2_x}4^{3_z}3^{2_x}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{101}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{010} \parallel 4^{+}_{100}\right] \left[2_{110} \parallel 4^{+}_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[2_{110} \parallel 4^{-}_{010}\right] \left[2_{010} \parallel 4^{-}_{100}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{010} \parallel 4^{+}_{100}\right] \left[2_{110} \parallel 4^{+}_{010}\right] \left[2_{110} \parallel 4^{-}_{010}\right] \left[2_{010} \parallel 4^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	1	−1	0	−1
χ_5	3	−1	−1	0	1

Irrepresntations of group generators

	$\left[2_{100} \parallel 4^{+}_{001}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$

No 555.2.1178 : ${}^{2_x}4^{3_z}3^{2_x}2^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel 4^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[3^+_{001} \parallel 3^+_{111}\right] \left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{101}\right] \left[3^+_{001} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^+_{001} \parallel 3^-_{1\bar{1}1}\right]$
$\left[2_{010} \parallel 4^+_{100}\right] \left[2_{110} \parallel 4^+_{010}\right] \left[3^-_{001} \parallel 3^-_{111}\right] \left[2_{110} \parallel 4^-_{010}\right] \left[2_{010} \parallel 4^-_{100}\right] \left[3^-_{001} \parallel 3^+_{1\bar{1}1}\right] \left[3^-_{001} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{001} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^+_{001} \parallel 3^-_{11\bar{1}}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[2_{120}T \parallel 4^+_{001}\right] \left[2_{001}T \parallel 2_{001}\right]$
$\left[2_{120}T \parallel 4^-_{001}\right] \left[6^-_{001}T \parallel 3^+_{111}\right] \left[2_{210}T \parallel 2_{011}\right] \left[2_{1\bar{1}0}T \parallel 2_{101}\right] \left[6^-_{001}T \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[6^-_{001}T \parallel 3^-_{1\bar{1}1}\right] \left[2_{210}T \parallel 4^+_{100}\right] \left[2_{1\bar{1}0}T \parallel 4^+_{010}\right] \left[6^+_{001}T \parallel 3^-_{111}\right]$
$\left[2_{1\bar{1}0}T \parallel 4^-_{010}\right] \left[2_{210}T \parallel 4^-_{100}\right] \left[6^+_{001}T \parallel 3^+_{1\bar{1}1}\right] \left[6^+_{001}T \parallel 3^+_{11\bar{1}}\right] \left[6^+_{001}T \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{120}T \parallel 2_{110}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[6^-_{001}T \parallel 3^-_{11\bar{1}}\right]$
$\left[2_{120}T \parallel 2_{1\bar{1}0}\right] \left[2_{210}T \parallel 2_{01\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel 4^+_{001}\right] \left[2_{100} \parallel 4^-_{001}\right] \left[2_{010} \parallel 4^+_{100}\right] \left[2_{110} \parallel 4^+_{010}\right] \left[2_{110} \parallel 4^-_{010}\right] \left[2_{010} \parallel 4^-_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[3^+_{001} \parallel 3^+_{111}\right] \left[3^+_{001} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^+_{001} \parallel 3^-_{1\bar{1}1}\right] \left[3^-_{001} \parallel 3^-_{111}\right] \left[3^-_{001} \parallel 3^+_{1\bar{1}1}\right] \left[3^-_{001} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{001} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[3^+_{001} \parallel 3^-_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	1	−1	0	−1
χ_5^{co}	3	−1	−1	0	1

Coirrepresntations of group generators

	$\left[2_{100} \parallel 4^+_{001}\right]$	$\left[3^+_{001} \parallel 3^+_{111}\right]$	$\left[3^+_{001} \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 556.1.1179 : ${}^{m_x}4^{3_z}3^{m_x}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel 2_{011}\right] \left[2_{110}T \parallel 2_{101}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{010}T \parallel 4^{+}_{100}\right] \left[2_{110}T \parallel 4^{+}_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[2_{110}T \parallel 4^{-}_{010}\right] \left[2_{010}T \parallel 4^{-}_{100}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 2_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 2_{01\bar{1}}\right] \left[2_{110}T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \parallel 4^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 557.1.1180 : ${}^{4_z}4^{3_{xyz}}3^{2_{xy}}2$

Full symmetry operations

$\left[E \,||\, 1\right] \left[4^+_{001} \,||\, 4^+_{001}\right] \left[2_{001} \,||\, 2_{001}\right] \left[4^-_{001} \,||\, 4^-_{001}\right] \left[3^+_{111} \,||\, 3^+_{111}\right] \left[2_{011} \,||\, 2_{011}\right] \left[2_{101} \,||\, 2_{101}\right] \left[3^+_{1\bar{1}\bar{1}} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \,||\, 3^-_{1\bar{1}1}\right]$
$\left[4^+_{100} \,||\, 4^+_{100}\right] \left[4^+_{010} \,||\, 4^+_{010}\right] \left[3^-_{111} \,||\, 3^-_{111}\right] \left[4^-_{010} \,||\, 4^-_{010}\right] \left[4^-_{100} \,||\, 4^-_{100}\right] \left[3^+_{\bar{1}\bar{1}1} \,||\, 3^+_{\bar{1}\bar{1}1}\right] \left[3^+_{11\bar{1}} \,||\, 3^+_{11\bar{1}}\right] \left[3^-_{\bar{1}\bar{1}\bar{1}} \,||\, 3^-_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{110} \,||\, 2_{110}\right]$
$\left[2_{100} \,||\, 2_{100}\right] \left[2_{010} \,||\, 2_{010}\right] \left[3^-_{11\bar{1}} \,||\, 3^-_{11\bar{1}}\right] \left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right] \left[2_{01\bar{1}} \,||\, 2_{01\bar{1}}\right] \left[2_{10\bar{1}} \,||\, 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[4^+_{001} \,||\, 4^+_{001}\right] \left[4^-_{001} \,||\, 4^-_{001}\right] \left[4^+_{100} \,||\, 4^+_{100}\right] \left[4^+_{010} \,||\, 4^+_{010}\right] \left[4^-_{010} \,||\, 4^-_{010}\right] \left[4^-_{100} \,||\, 4^-_{100}\right]$
C_3 : $\left[2_{001} \,||\, 2_{001}\right] \left[2_{100} \,||\, 2_{100}\right] \left[2_{010} \,||\, 2_{010}\right]$
C_4 : $\left[3^+_{111} \,||\, 3^+_{111}\right] \left[3^+_{1\bar{1}\bar{1}} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \,||\, 3^-_{1\bar{1}1}\right] \left[3^-_{111} \,||\, 3^-_{111}\right] \left[3^+_{\bar{1}\bar{1}1} \,||\, 3^+_{\bar{1}\bar{1}1}\right] \left[3^+_{11\bar{1}} \,||\, 3^+_{11\bar{1}}\right] \left[3^-_{\bar{1}\bar{1}\bar{1}} \,||\, 3^-_{\bar{1}\bar{1}\bar{1}}\right] \left[3^-_{11\bar{1}} \,||\, 3^-_{11\bar{1}}\right]$
C_5 : $\left[2_{011} \,||\, 2_{011}\right] \left[2_{101} \,||\, 2_{101}\right] \left[2_{110} \,||\, 2_{110}\right] \left[2_{1\bar{1}0} \,||\, 2_{1\bar{1}0}\right] \left[2_{01\bar{1}} \,||\, 2_{01\bar{1}}\right] \left[2_{10\bar{1}} \,||\, 2_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	1	−1	0	−1
χ_5	3	−1	−1	0	1

Irrepresntations of group generators

	$\left[4^+_{001} \,||\, 4^+_{001}\right]$	$\left[3^+_{111} \,||\, 3^+_{111}\right]$	$\left[3^+_{1\bar{1}\bar{1}} \,||\, 3^+_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$

No 558.1.1181 : ${}^{\bar{4}_z}4^{3_{xyz}}3^{m_{xy}}2$

Full symmetry operations

$\left[E \parallel 1\right] \left[4^+_{001}T \parallel 4^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[4^-_{001}T \parallel 4^-_{001}\right] \left[3^+_{111} \parallel 3^+_{111}\right] \left[2_{011}T \parallel 2_{011}\right] \left[2_{101}T \parallel 2_{101}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel 3^-_{1\bar{1}1}\right]$
$\left[4^+_{100}T \parallel 4^+_{100}\right] \left[4^+_{010}T \parallel 4^+_{010}\right] \left[3^-_{111} \parallel 3^-_{111}\right] \left[4^-_{010}T \parallel 4^-_{010}\right] \left[4^-_{100}T \parallel 4^-_{100}\right] \left[3^+_{1\bar{1}1} \parallel 3^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{110}T \parallel 2_{110}\right]$
$\left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^-_{11\bar{1}} \parallel 3^-_{11\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right] \left[2_{01\bar{1}}T \parallel 2_{01\bar{1}}\right] \left[2_{10\bar{1}}T \parallel 2_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$
C_3 : $\left[3^+_{111} \parallel 3^+_{111}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel 3^-_{1\bar{1}1}\right] \left[3^-_{11\bar{1}} \parallel 3^-_{11\bar{1}}\right]$
C_4 : $\left[3^-_{111} \parallel 3^-_{111}\right] \left[3^+_{1\bar{1}1} \parallel 3^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel 3^-_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[3^+_{111} \parallel 3^+_{111}\right]$	$\left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[4^+_{001}T \parallel 4^+_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 559.1.1182 : ${}^{1}\bar{4}^{1}3^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel m_{011}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	−1	−1	0	1
χ_5	3	1	−1	0	−1

Irrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$

No 559.2.1183 : ${}^{1}\bar{4}^{1}3^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right]$
$\left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel m_{101}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{4}^{+}_{100}\right] \left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel m_{011}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	−1	−1	0	1
χ_5^{co}	3	1	−1	0	−1

Coirrepresntations of group generators

	$\left[E \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 559.3.1184 : ${}^{1}\bar{4}^{1}3^{1}m^{\infty_z}1 \rightarrow {}^{1}\bar{4}^{1}3^{1}m$

No 559.4.1185 : ${}^{1}\bar{4}^{1}3^{1}m^{\infty_z m_x}1 \rightarrow {}^{1}\bar{4}^{1}3^{1}m^{m_x}1$

No 560.1.1186 : ${}^{2}\bar{4}^{1}3^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{\bar{1}11}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	−1	−1	0	1
χ_5	3	1	−1	0	−1

Irrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$

No 560.2.1187 : ${}^{2_z}\bar{4}^{1}3^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right]$
$\left[T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[T \parallel m_{011}\right] \left[T \parallel m_{101}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	−1	−1	0	1
χ_5^{co}	3	1	−1	0	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 560.3.1188 : ${}^{2_z}\bar{4}^{1}3^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right]$
$\left[2_{010}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel m_{011}\right] \left[2_{010}T \parallel m_{101}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{4}^{+}_{100}\right] \left[2_{010}T \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{010}T \parallel \bar{4}^{-}_{010}\right] \left[2_{010}T \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel m_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{01\bar{1}}\right] \left[2_{010}T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	−1	−1	0	1
χ_5^{co}	3	1	−1	0	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel \bar{4}^{+}_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 560.4.1189 : ${}^{2_z}\bar{4}^1 3^{2_z} m^{\infty_x} 1 \rightarrow {}^{2_z}\bar{4}^1 3^{2_z} m^{\Sigma_x} 1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{4}^+_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel \bar{4}^-_{001}\right] \left[E \parallel 3^+_{111}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel m_{101}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right]$
$\left[2_{100} \parallel \bar{4}^+_{100}\right] \left[2_{100} \parallel \bar{4}^+_{010}\right] \left[E \parallel 3^-_{111}\right] \left[2_{100} \parallel \bar{4}^-_{010}\right] \left[2_{100} \parallel \bar{4}^-_{100}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^-_{11\bar{1}}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^-_{001}\right] \left[\Sigma_z \parallel 3^+_{111}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{010}\right] \left[\Sigma_z \parallel 3^-_{111}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^-_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^-_{100}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^+_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^-_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{E} \parallel 3^+_{111}\right] \left[\bar{E} 2_{100} \parallel m_{011}\right]$
$\left[\bar{E} 2_{100} \parallel m_{101}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{100}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{010}\right] \left[\bar{E} \parallel 3^-_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{100}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}1}\right]$
$\left[\bar{E} \parallel 3^+_{11\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^-_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z \parallel 3^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z \parallel 3^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel \bar{4}^+_{001}\right] \left[2_{100} \parallel \bar{4}^-_{001}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel \bar{4}^+_{100}\right] \left[2_{100} \parallel \bar{4}^+_{010}\right] \left[2_{100} \parallel \bar{4}^-_{010}\right] \left[2_{100} \parallel \bar{4}^-_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^-_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^+_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^-_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^-_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \parallel m_{011}\right] \left[\bar{E} 2_{100} \parallel m_{101}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{4}^+_{100}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^+_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^-_{100}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^+_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right]$

C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_4 : $\left[E \parallel 3^+_{111}\right] \left[E \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}1}\right] \left[E \parallel 3^-_{111}\right] \left[E \parallel 3^+_{1\bar{1}1}\right] \left[E \parallel 3^+_{11\bar{1}}\right] \left[E \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^-_{11\bar{1}}\right]$

C_5 : $\left[\Sigma_z \parallel 3^+_{111}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \parallel 3^-_{111}\right] \left[\Sigma_z \parallel 3^+_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^+_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^+_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \parallel 1\right]$

C_8 : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

C_9 : $\left[\bar{E} \parallel 3^+_{111}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^-_{111}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^+_{11\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{11\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9

χ_1	2	0	2	2	0	0	−2	−2	−2
χ_2	2	0	2	−1	$\sqrt{3}i$	$-\sqrt{3}i$	−2	−2	1
χ_3	2	0	2	−1	$-\sqrt{3}i$	$\sqrt{3}i$	−2	−2	1
χ_4	6	0	−2	0	0	0	−6	2	0

Irrepresntations of group generators

	$\left[2_{100} \mid\mid \bar{4}^{+}_{001}\right]$	$\left[E \mid\mid 3^{+}_{111}\right]$	$\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	ω_1	ω_1	$\sigma_z\mathcal{K}$
Γ_3	$\sigma_x\mathcal{K}$	ω_1^*	ω_1^*	$\sigma_z\mathcal{K}$
Γ_4	$(d\bar{e}\bar{f}abc)\mathcal{K}$	$(bca\bar{f}d\bar{e})$	$(\bar{b}\bar{c}afde)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 561.1.1190 : ${}^{m}\bar{4}^{1}3^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{001}T \parallel \bar{4}^{+}_{100}\right] \left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 561.2.1191 : ${}^{m_z}\bar{4}^{1}3^{m_z}m^{\infty_x}1 \to {}^{m}\bar{4}^{1}3^{m}m$

No 562.1.1192 : ${}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel m_{011}\right] \left[T \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 562.2.1193 : ${}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel m_{011}\right] \left[T \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right] \left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel m_{011}\right]$
$\left[\bar{E}T \parallel m_{101}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right]$
$\left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right] \left[\bar{E}T \parallel m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel m_{011}\right] \left[\bar{\Sigma}_z T \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

$C_1 : \left[E \parallel 1\right]$
$C_2 : \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
$C_3 : \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
$C_4 : \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$C_5 : \left[\Sigma_z \parallel 1\right]$
$C_6 : \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$C_7 : \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$C_8 : \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$C_9 : \left[\bar{E} \parallel 1\right]$
$C_{10} : \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$
$C_{11} : \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$C_{12} : \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$C_{13} : \left[\bar{\Sigma}_z \parallel 1\right]$
$C_{14} : \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$
$C_{15} : \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$
$C_{16} : \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	0	0	0	0	−2	−2	−2	−2	0	0	0	0
χ_2^{co}	2	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	0	0	0	0	−2	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	0	0	0	0
χ_3^{co}	2	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	0	0	0	0	−2	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	0	0	0	0
χ_4^{co}	6	−2	0	0	0	0	0	0	−6	2	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[E \parallel 3^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \parallel \bar{4}^{+}_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$

Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	ω_5	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	ω_5^*	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(bca\bar{f}d\bar{e})$	$(\bar{b}\bar{c}afde)$	$(d\bar{e}\bar{f}abc)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 562.3.1194 : ${}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}\bar{4}^{1}3^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel m_{011}\right] \left[T \parallel m_{101}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel 2_{001}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel m_{011}\right]$
$\left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right]$
$\left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right] \left[\Sigma_z T \parallel m_{10\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel \right.$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel 2_{001}\right]$
$\left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right] \left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100} \parallel m_{011}\right]$
$\left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}1}\right]$
$\left[\bar{E}2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right] \left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{10\right.$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel m_{011}\right] \left[\bar{\Sigma}_z T \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_3 : $\left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_4 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel m_{101}\right]$
$\left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right]$

C_5 : $\left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \,||\, 3^-_{\bar{1}11}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

C_9 : $\left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{\bar{1}11}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	4	4	−2	0	0	0	−4	−4	2
χ_3^{co}	6	−2	0	0	0	0	−6	2	0

Coirrepresntations of group generators

	$\left[E \,||\, 2_{001}\right]$	$\left[2_{100} \,||\, \bar{4}^+_{100}\right]$	$\left[E \,||\, 3^+_{111}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_3^{co}	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(f\bar{d}\bar{e}\bar{b}\bar{c}a)\mathcal{K}$	$(bca\bar{f}d\bar{e})$	$(defa\bar{b}\bar{c})$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 563.1.1195 : ${}^{2_x}\bar{4}^{3_z}3^{2_x}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[E \,||\, 2_{001}\right] \left[2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[3^{+}_{001} \,||\, 3^{+}_{111}\right] \left[2_{010} \,||\, m_{011}\right] \left[2_{110} \,||\, m_{101}\right] \left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}1}\right]$
$\left[2_{010} \,||\, \bar{4}^{+}_{100}\right] \left[2_{110} \,||\, \bar{4}^{+}_{010}\right] \left[3^{-}_{001} \,||\, 3^{-}_{111}\right] \left[2_{110} \,||\, \bar{4}^{-}_{010}\right] \left[2_{010} \,||\, \bar{4}^{-}_{100}\right] \left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, m_{110}\right]$
$\left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[3^{+}_{001} \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{010} \,||\, m_{01\bar{1}}\right] \left[2_{110} \,||\, m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[2_{010} \,||\, \bar{4}^{+}_{100}\right] \left[2_{110} \,||\, \bar{4}^{+}_{010}\right] \left[2_{110} \,||\, \bar{4}^{-}_{010}\right] \left[2_{010} \,||\, \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_4 : $\left[3^{+}_{001} \,||\, 3^{+}_{111}\right] \left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{-}_{111}\right] \left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \,||\, m_{011}\right] \left[2_{110} \,||\, m_{101}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{010} \,||\, m_{01\bar{1}}\right] \left[2_{110} \,||\, m_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	−1	−1	0	1
χ_5	3	1	−1	0	−1

Irrepresntations of group generators

	$\left[2_{100} \,||\, \bar{4}^{+}_{001}\right]$	$\left[3^{+}_{001} \,||\, 3^{+}_{111}\right]$	$\left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$

No 563.2.1196 : ${}^{2_x}\bar{4}^{3_z}3^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[2_{010} \parallel m_{011}\right] \left[2_{110} \parallel m_{101}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{010} \parallel \bar{4}^{+}_{100}\right] \left[2_{110} \parallel \bar{4}^{+}_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[2_{110} \parallel \bar{4}^{-}_{010}\right] \left[2_{010} \parallel \bar{4}^{-}_{100}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{01\bar{1}}\right] \left[2_{110} \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel 1\right] \left[2_{120}T \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel 2_{001}\right]$
$\left[2_{120}T \parallel \bar{4}^{-}_{001}\right] \left[6^{-}_{001}T \parallel 3^{+}_{111}\right] \left[2_{210}T \parallel m_{011}\right] \left[2_{1\bar{1}0}T \parallel m_{101}\right] \left[6^{-}_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[6^{-}_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{210}T \parallel \bar{4}^{+}_{100}\right] \left[2_{1\bar{1}0}T \parallel \bar{4}^{+}_{010}\right] \left[6^{+}_{001}T \parallel 3^{-}_{111}\right]$
$\left[2_{1\bar{1}0}T \parallel \bar{4}^{-}_{010}\right] \left[2_{210}T \parallel \bar{4}^{-}_{100}\right] \left[6^{+}_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[6^{+}_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[6^{+}_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{120}T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[6^{-}_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{120}T \parallel m_{1\bar{1}0}\right] \left[2_{210}T \parallel m_{01\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{010} \parallel \bar{4}^{+}_{100}\right] \left[2_{110} \parallel \bar{4}^{+}_{010}\right] \left[2_{110} \parallel \bar{4}^{-}_{010}\right] \left[2_{010} \parallel \bar{4}^{-}_{100}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel m_{011}\right] \left[2_{110} \parallel m_{101}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{010} \parallel m_{01\bar{1}}\right] \left[2_{110} \parallel m_{10\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	−1	1	1	−1
χ_3^{co}	2	0	2	−1	0
χ_4^{co}	3	−1	−1	0	1
χ_5^{co}	3	1	−1	0	−1

Coirrepresntations of group generators

	$\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	1	$\mathcal{K}$
Γ_3^{co}	σ_x	ω_1	ω_1	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 564.1.1197 : ${}^{m_x}\bar{4}^{3_z}3^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[E \parallel 2_{001}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel m_{011}\right] \left[2_{110}T \parallel m_{101}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$
$\left[2_{010}T \parallel \bar{4}^{+}_{100}\right] \left[2_{110}T \parallel \bar{4}^{+}_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[2_{110}T \parallel \bar{4}^{-}_{010}\right] \left[2_{010}T \parallel \bar{4}^{-}_{100}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{110}\right]$
$\left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel m_{01\bar{1}}\right] \left[2_{110}T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_3 : $\left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_4 : $\left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[E \parallel 2_{001}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{111}\right]$	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 565.1.1198 : ${}^{4_z}\bar{4}^{3_{xyz}}3^{2_{xy}}m$

Full symmetry operations

$\left[E \,||\, 1\right]$ $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$ $\left[2_{001} \,||\, 2_{001}\right]$ $\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[3^+_{111} \,||\, 3^+_{111}\right]$ $\left[2_{011} \,||\, m_{011}\right]$ $\left[2_{101} \,||\, m_{101}\right]$ $\left[3^+_{1\bar{1}\bar{1}} \,||\, 3^+_{1\bar{1}\bar{1}}\right]$ $\left[3^-_{1\bar{1}1} \,||\, 3^-_{1\bar{1}1}\right]$
$\left[4^+_{100} \,||\, \bar{4}^+_{100}\right]$ $\left[4^+_{010} \,||\, \bar{4}^+_{010}\right]$ $\left[3^-_{111} \,||\, 3^-_{111}\right]$ $\left[4^-_{010} \,||\, \bar{4}^-_{010}\right]$ $\left[4^-_{100} \,||\, \bar{4}^-_{100}\right]$ $\left[3^+_{1\bar{1}1} \,||\, 3^+_{1\bar{1}1}\right]$ $\left[3^+_{11\bar{1}} \,||\, 3^+_{11\bar{1}}\right]$ $\left[3^-_{1\bar{1}\bar{1}} \,||\, 3^-_{1\bar{1}\bar{1}}\right]$ $\left[2_{110} \,||\, m_{110}\right]$
$\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$ $\left[3^-_{11\bar{1}} \,||\, 3^-_{11\bar{1}}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{01\bar{1}} \,||\, m_{01\bar{1}}\right]$ $\left[2_{10\bar{1}} \,||\, m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$ $\left[4^-_{001} \,||\, \bar{4}^-_{001}\right]$ $\left[4^+_{100} \,||\, \bar{4}^+_{100}\right]$ $\left[4^+_{010} \,||\, \bar{4}^+_{010}\right]$ $\left[4^-_{010} \,||\, \bar{4}^-_{010}\right]$ $\left[4^-_{100} \,||\, \bar{4}^-_{100}\right]$
C_3 : $\left[2_{001} \,||\, 2_{001}\right]$ $\left[2_{100} \,||\, 2_{100}\right]$ $\left[2_{010} \,||\, 2_{010}\right]$
C_4 : $\left[3^+_{111} \,||\, 3^+_{111}\right]$ $\left[3^+_{1\bar{1}\bar{1}} \,||\, 3^+_{1\bar{1}\bar{1}}\right]$ $\left[3^-_{1\bar{1}1} \,||\, 3^-_{1\bar{1}1}\right]$ $\left[3^-_{111} \,||\, 3^-_{111}\right]$ $\left[3^+_{1\bar{1}1} \,||\, 3^+_{1\bar{1}1}\right]$ $\left[3^+_{11\bar{1}} \,||\, 3^+_{11\bar{1}}\right]$ $\left[3^-_{1\bar{1}\bar{1}} \,||\, 3^-_{1\bar{1}\bar{1}}\right]$ $\left[3^-_{11\bar{1}} \,||\, 3^-_{11\bar{1}}\right]$
C_5 : $\left[2_{011} \,||\, m_{011}\right]$ $\left[2_{101} \,||\, m_{101}\right]$ $\left[2_{110} \,||\, m_{110}\right]$ $\left[2_{1\bar{1}0} \,||\, m_{1\bar{1}0}\right]$ $\left[2_{01\bar{1}} \,||\, m_{01\bar{1}}\right]$ $\left[2_{10\bar{1}} \,||\, m_{10\bar{1}}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5
χ_1	1	1	1	1	1
χ_2	1	−1	1	1	−1
χ_3	2	0	2	−1	0
χ_4	3	−1	−1	0	1
χ_5	3	1	−1	0	−1

Irrepresntations of group generators

	$\left[4^+_{001} \,||\, \bar{4}^+_{001}\right]$	$\left[3^+_{111} \,||\, 3^+_{111}\right]$	$\left[3^+_{1\bar{1}\bar{1}} \,||\, 3^+_{1\bar{1}\bar{1}}\right]$
Γ_1	1	1	1
Γ_2	−1	1	1
Γ_3	σ_x	ω_1	ω_1
Γ_4	$(\bar{a}\bar{c}b)$	(bca)	$(\bar{b}\bar{c}a)$
Γ_5	$(ac\bar{b})$	(bca)	$(\bar{b}\bar{c}a)$

No 566.1.1199 : ${}^{\bar{4}_z}\bar{4}^{3_{xyz}}3^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[4^+_{001}T \parallel \bar{4}^+_{001}\right] \left[2_{001} \parallel 2_{001}\right] \left[4^-_{001}T \parallel \bar{4}^-_{001}\right] \left[3^+_{111} \parallel 3^+_{111}\right] \left[2_{011}T \parallel m_{011}\right] \left[2_{101}T \parallel m_{101}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel 3^-_{1\bar{1}1}\right]$
$\left[4^+_{100}T \parallel \bar{4}^+_{100}\right] \left[4^+_{010}T \parallel \bar{4}^+_{010}\right] \left[3^-_{111} \parallel 3^-_{111}\right] \left[4^-_{010}T \parallel \bar{4}^-_{010}\right] \left[4^-_{100}T \parallel \bar{4}^-_{100}\right] \left[3^+_{1\bar{1}1} \parallel 3^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{110}T \parallel m_{110}\right]$
$\left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^-_{11\bar{1}} \parallel 3^-_{11\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right] \left[2_{01\bar{1}}T \parallel m_{01\bar{1}}\right] \left[2_{10\bar{1}}T \parallel m_{10\bar{1}}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$
C_3 : $\left[3^+_{111} \parallel 3^+_{111}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel 3^-_{1\bar{1}1}\right] \left[3^-_{11\bar{1}} \parallel 3^-_{11\bar{1}}\right]$
C_4 : $\left[3^-_{111} \parallel 3^-_{111}\right] \left[3^+_{1\bar{1}1} \parallel 3^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel 3^-_{1\bar{1}\bar{1}}\right]$

Co-character table

	C_1	C_2	C_3	C_4
χ_1^{co}	1	1	1	1
χ_2^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$
χ_3^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$
χ_4^{co}	3	-1	0	0

Coirrepresntations of group generators

	$\left[2_{001} \parallel 2_{001}\right]$	$\left[3^+_{111} \parallel 3^+_{111}\right]$	$\left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[4^+_{001}T \parallel \bar{4}^+_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_3^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 567.1.1200 : ${}^{1}m^{1}\bar{3}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_9 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$
C_{10} : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[E \parallel 4^{-}_{010}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 567.2.1201 : ${}^{1}m^{1}\bar{3}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{001}T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 4^{-}_{010}\right] \left[2_{001}T \parallel 4^{-}_{100}\right] \left[2_{001}T \parallel m_{101}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right] \left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{101}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right] \left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_9 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$
C_{10} : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[E \parallel 4^{-}_{010}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 567.3.1202 : ${}^{1}m^{1}\bar{3}^{1}m^{\infty_z}1 \to {}^{1}m^{1}\bar{3}^{1}m$

No 567.4.1203 : ${}^{1}m^{1}\bar{3}^{1}m^{\infty_{z}m_{x}}1 \to {}^{1}m^{1}\bar{3}^{1}m^{m_{x}}1$

No 568.1.1204 : ${}^{1}m^{1}\bar{3}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_9 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel 4^{-}_{010}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 568.2.1205 : ${}^{1}m^{1}\bar{3}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right] \left[T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right]$
$\left[T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right]$
$\left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right] \left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_9 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel 4_{010}^{-}\right]$	$\left[E \parallel \bar{3}_{111}^{+}\right]$	$\left[2_{001}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 568.3.1206 : ${}^{1}m^{1}\bar{3}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{010}T \parallel m_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 4^{-}_{010}\right] \left[2_{010}T \parallel 4^{-}_{100}\right] \left[2_{010}T \parallel m_{101}\right] \left[2_{010}T \parallel m_{01\bar{1}}\right] \left[2_{010}T \parallel m_{10\bar{1}}\right] \left[2_{010}T \parallel m_{011}\right] \left[2_{010}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{010}T \parallel \bar{4}^{+}_{010}\right] \left[2_{010}T \parallel 2_{011}\right] \left[2_{010}T \parallel 2_{10\bar{1}}\right] \left[2_{010}T \parallel 2_{01\bar{1}}\right] \left[2_{010}T \parallel 2_{101}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel \bar{4}^{-}_{001}\right] \left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{010}T \parallel \bar{4}^{-}_{010}\right] \left[2_{010}T \parallel \bar{4}^{-}_{100}\right] \left[2_{010}T \parallel 4^{+}_{010}\right] \left[2_{010}T \parallel 4^{+}_{100}\right] \left[2_{010}T \parallel 4^{-}_{001}\right] \left[2_{010}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_9 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[2_{001} \parallel 4_{010}^{-}\right]$	$\left[E \parallel \bar{3}_{111}^{+}\right]$	$\left[2_{100}T \parallel 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 568.4.1207 : ${}^{1}m^{1}\bar{3}^{2_z}m^{\infty_x}1 \rightarrow {}^{1}m^{1}\bar{3}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid m_{001}\right] \left[E \mid\mid \bar{3}^{+}_{111}\right] \left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}1}\right] \left[E \mid\mid 3^{-}_{111}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \mid\mid -1\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 2_{100}\right] \left[E \mid\mid 2_{010}\right] \left[E \mid\mid 3^{+}_{111}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{-}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{111}\right]$
$\left[E \mid\mid 3^{+}_{1\bar{1}1}\right] \left[E \mid\mid 3^{+}_{11\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{010}\right] \left[E \mid\mid 3^{-}_{11\bar{1}}\right] \left[2_{100} \mid\mid m_{110}\right] \left[2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[2_{100} \mid\mid 4^{-}_{010}\right]$
$\left[2_{100} \mid\mid 4^{-}_{100}\right] \left[2_{100} \mid\mid m_{101}\right] \left[2_{100} \mid\mid m_{01\bar{1}}\right] \left[2_{100} \mid\mid m_{10\bar{1}}\right] \left[2_{100} \mid\mid m_{011}\right] \left[2_{100} \mid\mid \bar{4}^{+}_{100}\right] \left[2_{100} \mid\mid \bar{4}^{+}_{010}\right] \left[2_{100} \mid\mid 2_{011}\right] \left[2_{100} \mid\mid 2_{10\bar{1}}\right]$
$\left[2_{100} \mid\mid 2_{01\bar{1}}\right] \left[2_{100} \mid\mid 2_{101}\right] \left[2_{100} \mid\mid 2_{110}\right] \left[2_{100} \mid\mid m_{1\bar{1}0}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{010}\right] \left[2_{100} \mid\mid \bar{4}^{-}_{100}\right] \left[2_{100} \mid\mid 4^{+}_{010}\right]$
$\left[2_{100} \mid\mid 4^{+}_{100}\right] \left[2_{100} \mid\mid 4^{-}_{001}\right] \left[2_{100} \mid\mid 4^{+}_{001}\right] \left[\Sigma_z \mid\mid 1\right] \left[\Sigma_z \mid\mid m_{001}\right] \left[\Sigma_z \mid\mid \bar{3}^{+}_{111}\right] \left[\Sigma_z \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \mid\mid 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \mid\mid 3^{-}_{111}\right]$
$\left[\Sigma_z \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \mid\mid -1\right] \left[\Sigma_z \mid\mid 2_{001}\right] \left[\Sigma_z \mid\mid 2_{100}\right] \left[\Sigma_z \mid\mid 2_{010}\right] \left[\Sigma_z \mid\mid 3^{+}_{111}\right] \left[\Sigma_z \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z \mid\mid \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z \mid\mid \bar{3}^{-}_{111}\right] \left[\Sigma_z \mid\mid 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \mid\mid 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \mid\mid m_{100}\right] \left[\Sigma_z \mid\mid m_{010}\right] \left[\Sigma_z \mid\mid 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \mid\mid m_{110}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \mid\mid m_{101}\right] \left[\Sigma_z 2_{100} \mid\mid m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \mid\mid m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \mid\mid m_{011}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}\right.$
$\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{011}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{101}\right] \left[\Sigma_z 2_{100} \mid\mid 2_{110}\right] \left[\Sigma_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right.$
$\left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \mid\mid \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \mid\mid 4^{+}_{001}\right] \left[\bar{E} \mid\mid 1\right] \left[\bar{E} \mid\mid m_{001}\right] \left[\bar{E} \mid\mid \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \mid\mid 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \mid\mid 3^{-}_{111}\right] \left[\bar{E} \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \mid\mid -1\right] \left[\bar{E} \mid\mid 2_{001}\right] \left[\bar{E} \mid\mid 2_{100}\right]$
$\left[\bar{E} \mid\mid 2_{010}\right] \left[\bar{E} \mid\mid 3^{+}_{111}\right] \left[\bar{E} \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} \mid\mid \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} \mid\mid \bar{3}^{-}_{111}\right] \left[\bar{E} \mid\mid 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \mid\mid 3^{+}_{11\bar{1}}\right] \left[\bar{E} \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} \mid\mid m_{100}\right] \left[\bar{E} \mid\mid m_{010}\right] \left[\bar{E} \mid\mid 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \mid\mid m_{110}\right] \left[\bar{E}2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{010}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{100}\right] \left[\bar{E}2_{100} \mid\mid m_{101}\right] \left[\bar{E}2_{100} \mid\mid m_{01\bar{1}}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{10\bar{1}}\right] \left[\bar{E}2_{100} \mid\mid m_{011}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \mid\mid 2_{011}\right] \left[\bar{E}2_{100} \mid\mid 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \mid\mid 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \mid\mid 2_{101}\right] \left[\bar{E}2_{100} \mid\mid 2_{110}\right]$
$\left[\bar{E}2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \mid\mid \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{010}\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{100}\right] \left[\bar{E}2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{E}2_{100} \mid\mid 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \mid\mid 1\right] \left[\bar{\Sigma}_z \mid\mid m_{001}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{111}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z \mid\mid -1\right] \left[\bar{\Sigma}_z \mid\mid 2_{001}\right] \left[\bar{\Sigma}_z \mid\mid 2_{100}\right] \left[\bar{\Sigma}_z \mid\mid 2_{010}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{111}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \mid\mid \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \mid\mid 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \mid\mid 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \mid\mid m_{100}\right] \left[\bar{\Sigma}_z \mid\mid m_{010}\right] \left[\bar{\Sigma}_z \mid\mid 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{1}\right.$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{010}\right.$
$\left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \mid\mid 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$
C_2 : $\left[E \mid\mid m_{001}\right] \left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{010}\right]$
C_3 : $\left[E \mid\mid \bar{3}^{+}_{111}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{-}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}1}\right] \left[E \mid\mid 3^{-}_{111}\right] \left[E \mid\mid 3^{+}_{111}\right] \left[E \mid\mid 3^{+}_{1\bar{1}1}\right] \left[E \mid\mid 3^{+}_{11\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \mid\mid -1\right]$
C_6 : $\left[E \mid\mid 2_{001}\right] \left[E \mid\mid 2_{100}\right] \left[E \mid\mid 2_{010}\right]$

C_7 : $\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right]$
$\left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E} 2_{100} \parallel m_{101}\right] \left[\bar{E} 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{011}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{011}\right] \left[\bar{E} 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{101}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

C_8 : $\left[\Sigma_z \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{111}\right]$

C_9 : $\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_{10} : $\left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$

C_{11} : $\left[\Sigma_z \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_{12} : $\left[\bar{E} \parallel 1\right]$

C_{13} : $\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right]$

C_{14} : $\left[\bar{E} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{111}\right]$

C_{15} : $\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$

C_{16} : $\left[\bar{E} \parallel -1\right]$

C_{17} : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}
χ_1	2	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	−2
χ_2	2	−2	−2	2	−2	2	0	0	0	0	0	−2	2	2	−2	2	−2
χ_3	2	2	−1	−1	2	2	0	$\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$-\sqrt{3}i$	−2	−2	1	1	−2	−2
χ_4	2	−2	1	−1	−2	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	−2	2	−1	1	2	−2
χ_5	2	2	−1	−1	2	2	0	$-\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$\sqrt{3}i$	−2	−2	1	1	−2	−2
χ_6	2	−2	1	−1	−2	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	$\sqrt{3}i$	$-\sqrt{3}i$	−2	2	−1	1	2	−2

χ_7	6	−2	0	0	6	−2	0	0	0	0	0	−6	2	0	0	−6	2
χ_8	6	2	0	0	−6	−2	0	0	0	0	0	−6	−2	0	0	6	2

Irrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[2_{100} \parallel 4^{-}_{010}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2	$-\sigma_0$	$-\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_z\mathcal{K}$
Γ_3	σ_0	$\omega_2\mathcal{K}$	ω_1	$\sigma_z\mathcal{K}$
Γ_4	$-\sigma_0$	$\omega_4^*\mathcal{K}$	ω_3^*	$\sigma_z\mathcal{K}$
Γ_5	σ_0	$\omega_2^*\mathcal{K}$	ω_1^*	$\sigma_z\mathcal{K}$
Γ_6	$-\sigma_0$	$\omega_4\mathcal{K}$	ω_3	$\sigma_z\mathcal{K}$
Γ_7	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(efd\bar{c}\bar{a}b)\mathcal{K}$	$(bca\bar{f}\bar{d}e)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_8	$(\bar{a}bc\bar{d}ef)$	$(\bar{e}\bar{f}\bar{d}ca\bar{b})\mathcal{K}$	$(\bar{b}\bar{c}\bar{a}fd\bar{e})$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 569.1.1208 : ${}^{1}m^{1}\bar{3}^{m}m$

Full symmetry operations

$\left[E \mid\mid 1\right] \left[E \mid\mid m_{001}\right] \left[E \mid\mid \bar{3}^{+}_{111}\right] \left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}1}\right] \left[E \mid\mid 3^{-}_{111}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \mid\mid -1\right] \left[E \mid\mid 2_{001}\right] \left[E \mid\mid 2_{100}\right] \left[E \mid\mid 2_{010}\right] \left[E \mid\mid 3^{+}_{111}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{-}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{111}\right]$
$\left[E \mid\mid 3^{+}_{1\bar{1}1}\right] \left[E \mid\mid 3^{+}_{11\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{010}\right] \left[E \mid\mid 3^{-}_{11\bar{1}}\right] \left[2_{001}T \mid\mid m_{110}\right] \left[2_{001}T \mid\mid 2_{1\bar{1}0}\right] \left[2_{001}T \mid\mid 4^{-}_{010}\right]$
$\left[2_{001}T \mid\mid 4^{-}_{100}\right] \left[2_{001}T \mid\mid m_{101}\right] \left[2_{001}T \mid\mid m_{01\bar{1}}\right] \left[2_{001}T \mid\mid m_{10\bar{1}}\right] \left[2_{001}T \mid\mid m_{011}\right] \left[2_{001}T \mid\mid \bar{4}^{+}_{100}\right] \left[2_{001}T \mid\mid \bar{4}^{+}_{010}\right] \left[2_{001}T \mid\mid 2_{011}\right] \left[2_{001}T \mid\mid 2_{10\bar{1}}\right]$
$\left[2_{001}T \mid\mid 2_{01\bar{1}}\right] \left[2_{001}T \mid\mid 2_{101}\right] \left[2_{001}T \mid\mid 2_{110}\right] \left[2_{001}T \mid\mid m_{1\bar{1}0}\right] \left[2_{001}T \mid\mid \bar{4}^{-}_{001}\right] \left[2_{001}T \mid\mid \bar{4}^{+}_{001}\right] \left[2_{001}T \mid\mid \bar{4}^{-}_{010}\right] \left[2_{001}T \mid\mid \bar{4}^{-}_{100}\right] \left[2_{001}T \mid\mid 4^{+}_{010}\right]$
$\left[2_{001}T \mid\mid 4^{+}_{100}\right] \left[2_{001}T \mid\mid 4^{-}_{001}\right] \left[2_{001}T \mid\mid 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$

C_2 : $\left[E \mid\mid m_{001}\right] \left[E \mid\mid m_{100}\right] \left[E \mid\mid m_{010}\right]$

C_3 : $\left[E \mid\mid \bar{3}^{+}_{111}\right] \left[E \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}1}\right] \left[E \mid\mid 3^{+}_{111}\right] \left[E \mid\mid 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \mid\mid 3^{-}_{111}\right] \left[E \mid\mid 3^{+}_{1\bar{1}1}\right] \left[E \mid\mid 3^{+}_{11\bar{1}}\right] \left[E \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[E \mid\mid \bar{3}^{+}_{1\bar{1}1}\right] \left[E \mid\mid \bar{3}^{+}_{11\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \mid\mid \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \mid\mid -1\right]$

C_8 : $\left[E \mid\mid 2_{001}\right] \left[E \mid\mid 2_{100}\right] \left[E \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	-1	-1	1	1	-1	-1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	1
χ_7^{co}	3	-1	0	0	0	0	3	-1
χ_8^{co}	3	1	0	0	0	0	-3	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{001}\right]$	$\left[E \mid\mid \bar{3}^{+}_{111}\right]$	$\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \mid\mid m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	-1	-1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 569.2.1209 : ${}^{1}m^{1}\bar{3}^{m_z}m^{\infty_x}1 \to {}^{1}m^{1}\bar{3}^{m}m$

No 570.1.1210 : ${}^{1}m^{1}\bar{3}^{\bar{1}}m$

Full symmetry operations

$\left[E \mid\mid 1\right]$ $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid \bar{3}^{+}_{111}\right]$ $\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \mid\mid 3^{-}_{1\bar{1}1}\right]$ $\left[E \mid\mid 3^{-}_{111}\right]$ $\left[E \mid\mid \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[E \mid\mid \bar{3}^{+}_{11\bar{1}}\right]$ $\left[E \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \mid\mid -1\right]$ $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid 2_{010}\right]$ $\left[E \mid\mid 3^{+}_{111}\right]$ $\left[E \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \mid\mid \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[E \mid\mid \bar{3}^{-}_{11\bar{1}}\right]$ $\left[E \mid\mid \bar{3}^{-}_{111}\right]$
$\left[E \mid\mid 3^{+}_{1\bar{1}1}\right]$ $\left[E \mid\mid 3^{+}_{11\bar{1}}\right]$ $\left[E \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$ $\left[E \mid\mid 3^{-}_{11\bar{1}}\right]$ $\left[T \mid\mid m_{110}\right]$ $\left[T \mid\mid 2_{1\bar{1}0}\right]$ $\left[T \mid\mid 4^{-}_{010}\right]$
$\left[T \mid\mid 4^{-}_{100}\right]$ $\left[T \mid\mid m_{101}\right]$ $\left[T \mid\mid m_{01\bar{1}}\right]$ $\left[T \mid\mid m_{10\bar{1}}\right]$ $\left[T \mid\mid m_{011}\right]$ $\left[T \mid\mid \bar{4}^{+}_{100}\right]$ $\left[T \mid\mid \bar{4}^{+}_{010}\right]$ $\left[T \mid\mid 2_{011}\right]$ $\left[T \mid\mid 2_{10\bar{1}}\right]$
$\left[T \mid\mid 2_{01\bar{1}}\right]$ $\left[T \mid\mid 2_{101}\right]$ $\left[T \mid\mid 2_{110}\right]$ $\left[T \mid\mid m_{1\bar{1}0}\right]$ $\left[T \mid\mid \bar{4}^{-}_{001}\right]$ $\left[T \mid\mid \bar{4}^{+}_{001}\right]$ $\left[T \mid\mid \bar{4}^{-}_{010}\right]$ $\left[T \mid\mid \bar{4}^{-}_{100}\right]$ $\left[T \mid\mid 4^{+}_{010}\right]$
$\left[T \mid\mid 4^{+}_{100}\right]$ $\left[T \mid\mid 4^{-}_{001}\right]$ $\left[T \mid\mid 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \mid\mid 1\right]$

C_2 : $\left[E \mid\mid m_{001}\right]$ $\left[E \mid\mid m_{100}\right]$ $\left[E \mid\mid m_{010}\right]$

C_3 : $\left[E \mid\mid \bar{3}^{+}_{111}\right]$ $\left[E \mid\mid \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \mid\mid \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[E \mid\mid \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[E \mid\mid 3^{-}_{1\bar{1}1}\right]$ $\left[E \mid\mid 3^{+}_{111}\right]$ $\left[E \mid\mid 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \mid\mid 3^{-}_{111}\right]$ $\left[E \mid\mid 3^{+}_{1\bar{1}1}\right]$ $\left[E \mid\mid 3^{+}_{11\bar{1}}\right]$ $\left[E \mid\mid 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[E \mid\mid \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[E \mid\mid \bar{3}^{+}_{11\bar{1}}\right]$ $\left[E \mid\mid \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$ $\left[E \mid\mid \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \mid\mid -1\right]$

C_8 : $\left[E \mid\mid 2_{001}\right]$ $\left[E \mid\mid 2_{100}\right]$ $\left[E \mid\mid 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	-1	-1	1	1	-1	-1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	1
χ_7^{co}	3	-1	0	0	0	0	3	-1
χ_8^{co}	3	1	0	0	0	0	-3	-1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{001}\right]$	$\left[E \mid\mid \bar{3}^{+}_{111}\right]$	$\left[E \mid\mid 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \mid\mid m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	-1	-1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 570.2.1211 : ${}^{1}m^{1}\bar{3}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{1}m^{1}\bar{3}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right] \left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E}T \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{100}\right] \left[\bar{\Sigma}_z \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{101}\right] \left[\bar{\Sigma}_z T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{011}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{011}\right] \left[\bar{\Sigma}_z T \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{101}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$

C_3 : $\left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[E \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_9 : $\left[\Sigma_z \parallel 1\right]$

C_{17} : $\left[\bar{E} \parallel 1\right]$

C_{18} : $\left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right]$

C_{19} : $\left[\bar{E} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_{20} : $\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$

C_{21} : $\left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_{22} : $\left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{111}\right]$

C_{23} : $\left[\bar{E} \parallel -1\right]$

C_{24} : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

C_{25} : $\left[\bar{\Sigma}_z \parallel 1\right]$

$C_{10}: \left[\Sigma_z \,||\, m_{001}\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z \,||\, m_{010}\right]$

$C_{11}: \left[\Sigma_z \,||\, \bar{3}^{+}_{111}\right] \left[\Sigma_z \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z \,||\, \bar{3}^{-}_{11\bar{1}}\right]$

$C_{12}: \left[\Sigma_z \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^{+}_{111}\right] \left[\Sigma_z \,||\, 3^{-}_{11\bar{1}}\right]$

$C_{13}: \left[\Sigma_z \,||\, 3^{-}_{111}\right] \left[\Sigma_z \,||\, 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^{-}_{1\bar{1}\bar{1}}\right]$

$C_{14}: \left[\Sigma_z \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, \bar{3}^{-}_{111}\right]$

$C_{15}: \left[\Sigma_z \,||\, -1\right]$

$C_{16}: \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 2_{010}\right]$

$C_{26}: \left[\bar{\Sigma}_z \,||\, m_{001}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right]$

$C_{27}: \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{11\bar{1}}\right]$

$C_{28}: \left[\bar{\Sigma}_z \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{11\bar{1}}\right]$

$C_{29}: \left[\bar{\Sigma}_z \,||\, 3^{-}_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^{-}_{1\bar{1}\bar{1}}\right]$

$C_{30}: \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^{-}_{111}\right]$

$C_{31}: \left[\bar{\Sigma}_z \,||\, -1\right]$

$C_{32}: \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}
χ_1^{co}	2	2	2	2	2	2	2	2	0	0	0	0	0	0	0	0
χ_2^{co}	2	−2	−2	2	2	−2	−2	2	0	0	0	0	0	0	0	0
χ_3^{co}	2	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	2	0	0	0	0	0	0	0	0
χ_4^{co}	2	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	2	0	0	0	0	0	0	0	0
χ_5^{co}	2	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	2	0	0	0	0	0	0	0	0
χ_6^{co}	2	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	2	0	0	0	0	0	0	0	0
χ_7^{co}	6	−2	0	0	0	0	6	−2	0	0	0	0	0	0	0	0
χ_8^{co}	6	2	0	0	0	0	−6	−2	0	0	0	0	0	0	0	0

	C_{17}	C_{18}	C_{19}	C_{20}	C_{21}	C_{22}	C_{23}	C_{24}	C_{25}	C_{26}	C_{27}	C_{28}	C_{29}	C_{30}	C_{31}	C_{32}
χ_1^{co}	−2	−2	−2	−2	−2	−2	−2	−2	0	0	0	0	0	0	0	0
χ_2^{co}	−2	2	2	−2	−2	2	2	−2	0	0	0	0	0	0	0	0
χ_3^{co}	−2	−2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	−2	−2	0	0	0	0	0	0	0	0
χ_4^{co}	−2	2	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	−2	0	0	0	0	0	0	0	0
χ_5^{co}	−2	−2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	−2	−2	0	0	0	0	0	0	0	0
χ_6^{co}	−2	2	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	−2	0	0	0	0	0	0	0	0
χ_7^{co}	−6	2	0	0	0	0	−6	2	0	0	0	0	0	0	0	0
χ_8^{co}	−6	−2	0	0	0	0	6	2	0	0	0	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel m_{001}\right]$	$\left[E \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \parallel m_{110}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_0$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	σ_0	ω_5	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$-\sigma_0$	ω_7^*	ω_5	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_5^{co}	σ_0	ω_5^*	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_7	ω_5^*	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(bca\bar{f}\bar{d}e)$	$(\bar{b}\bar{c}afd\bar{d}\bar{e})$	$(defabc)\mathcal{K}$	$(abcd\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_8^{co}	$(\bar{a}bcd\bar{d}ef)$	$(\bar{b}\bar{c}\bar{a}fd\bar{e})$	$(\bar{b}\bar{c}afd\bar{d}\bar{e})$	$(defabc)\mathcal{K}$	$(abcd\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 570.3.1212 : ${}^{1}m^{1}\bar{3}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{1}m^{1}\bar{3}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[E \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[E \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[E \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel m_{001}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{111}\right]$
$\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z \parallel m_{100}\right] \left[\Sigma_z \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{10}\right.$
$\left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2\right.$
$\left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2\right.$
$\left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{1}\right.$
$\left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right.$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} \parallel m_{001}\right] \left[\bar{E} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel m_{100}\right] \left[\bar{E} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right] \left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right]$
$\left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 2\right.$
$\left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel\right.$
$\left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right] \left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel m_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel \bar{3}^{-}_{111}\right]$

$\left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{110}\right] \left[\bar{\Sigma}_z T \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, 4^-_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{101}\right] \left[\bar{\Sigma}_z T \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{011}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z T \,||\, 2_{011}\right] \left[\bar{\Sigma}_z T \,||\, 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, 2_{101}\right] \left[\bar{\Sigma}_z T \,||\, 2_{110}\right] \left[\bar{\Sigma}_z T \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z T \,||\, 4^+_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z T \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z T \,||\, 4^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{111}\right]$
$\left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z 2\right.$
$\left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{3}^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{010}\right] \left[\bar{\Sigma}\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{001}\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$

C_2 : $\left[E \,||\, m_{001}\right] \left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right]$

C_3 : $\left[E \,||\, \bar{3}^+_{111}\right] \left[E \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[E \,||\, \bar{3}^+_{11\bar{1}}\right] \left[E \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[E \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[E \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[E \,||\, \bar{3}^-_{11\bar{1}}\right] \left[E \,||\, \bar{3}^-_{111}\right]$

C_4 : $\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}1}\right] \left[E \,||\, 3^-_{111}\right] \left[E \,||\, 3^+_{111}\right] \left[E \,||\, 3^+_{1\bar{1}1}\right] \left[E \,||\, 3^+_{11\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{11\bar{1}}\right]$

C_5 : $\left[E \,||\, -1\right]$

C_6 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

C_7 : $\left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, 2_{1\bar{1}0}\right] \left[2_{100} \,||\, 4^-_{010}\right] \left[2_{100} \,||\, 4^-_{100}\right] \left[2_{100} \,||\, m_{101}\right] \left[2_{100} \,||\, m_{01\bar{1}}\right] \left[2_{100} \,||\, m_{10\bar{1}}\right] \left[2_{100} \,||\, m_{011}\right]$
$\left[2_{100} \,||\, \bar{4}^+_{100}\right] \left[2_{100} \,||\, \bar{4}^+_{010}\right] \left[2_{100} \,||\, 2_{011}\right] \left[2_{100} \,||\, 2_{10\bar{1}}\right] \left[2_{100} \,||\, 2_{01\bar{1}}\right] \left[2_{100} \,||\, 2_{101}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[2_{100} \,||\, \bar{4}^-_{001}\right] \left[2_{100} \,||\, \bar{4}^+_{001}\right] \left[2_{100} \,||\, \bar{4}^-_{010}\right] \left[2_{100} \,||\, \bar{4}^-_{100}\right] \left[2_{100} \,||\, 4^+_{010}\right] \left[2_{100} \,||\, 4^+_{100}\right] \left[2_{100} \,||\, 4^-_{001}\right] \left[2_{100} \,||\, 4^+_{001}\right]$
$\left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, m_{001}\right] \left[\Sigma_z \,||\, -1\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, m_{100}\right] \left[\Sigma_z \,||\, m_{010}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{010}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{101}\right] \left[\Sigma_z 2_{100} \,||\, m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, m_{011}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{100}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{011}\right] \left[\Sigma_z 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, 2_{101}\right] \left[\Sigma_z 2_{100} \,||\, 2_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{100}\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{010}\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{100}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{001}\right]$
$\left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, 4^-_{010}\right] \left[\bar{E} 2_{100} \,||\, 4^-_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{101}\right] \left[\bar{E} 2_{100} \,||\, m_{01\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, m_{10\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, m_{011}\right]$
$\left[\bar{E} 2_{100} \,||\, \bar{4}^+_{100}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\bar{E} 2_{100} \,||\, 2_{011}\right] \left[\bar{E} 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, 2_{101}\right] \left[\bar{E} 2_{100} \,||\, 2_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\bar{E} 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{100}\right] \left[\bar{E} 2_{100} \,||\, 4^+_{010}\right] \left[\bar{E} 2_{100} \,||\, 4^+_{100}\right] \left[\bar{E} 2_{100} \,||\, 4^-_{001}\right] \left[\bar{E} 2_{100} \,||\, 4^+_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, m_{001}\right] \left[\bar{\Sigma}_z \,||\, -1\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, m_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{011}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{001}\right]$

C_8 : $\left[\Sigma_z \,||\, \bar{3}^+_{111}\right] \left[\Sigma_z \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^-_{111}\right]$

C_9 : $\left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

C_{10} : $\left[\Sigma_z \,||\, 3^-_{111}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_{11} : $\left[\Sigma_z \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, \bar{3}^-_{111}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^+_{111}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, \bar{3}^-_{11\bar{1}}\right]$

$C_{12}: \left[\bar{E} \,||\, 1\right]$

$C_{13}: \left[\bar{E} \,||\, m_{001}\right] \left[\bar{E} \,||\, m_{100}\right] \left[\bar{E} \,||\, m_{010}\right]$

$C_{14}: \left[\bar{E} \,||\, \bar{3}^{+}_{111}\right] \left[\bar{E} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} \,||\, \bar{3}^{-}_{111}\right]$

$C_{15}: \left[\bar{E} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^{-}_{111}\right] \left[\bar{E} \,||\, 3^{+}_{111}\right] \left[\bar{E} \,||\, 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^{+}_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^{-}_{11\bar{1}}\right]$

$C_{16}: \left[\bar{E} \,||\, -1\right]$

$C_{17}: \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}
χ_1^{co}	2	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	−2
χ_2^{co}	2	−2	−2	2	−2	2	0	0	0	0	0	−2	2	2	−2	2	−2
χ_3^{co}	4	4	−2	−2	4	4	0	0	0	0	0	−4	−4	2	2	−4	−4
χ_4^{co}	4	−4	2	−2	−4	4	0	0	0	0	0	−4	4	−2	2	4	−4
χ_5^{co}	6	−2	0	0	6	−2	0	0	0	0	0	−6	2	0	0	−6	2
χ_6^{co}	6	2	0	0	−6	−2	0	0	0	0	0	−6	−2	0	0	6	2

Coirrepresntations of group generators

	$\left[E \,||\, m_{001}\right]$	$\left[2_{100} \,||\, 4^{-}_{010}\right]$	$\left[E \,||\, \bar{3}^{+}_{111}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	$\sigma_x\mathcal{K}$	σ_0	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	$-\sigma_0$	$-\sigma_x\mathcal{K}$	$-\sigma_0$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_0 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$\begin{pmatrix} -\sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}$	$\begin{pmatrix} \omega_4^* & 0 \\ 0 & \omega_4^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_3^* & 0 \\ 0 & \omega_3^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & -\sigma_z \end{pmatrix}\mathcal{K}$
Γ_5^{co}	$(a\bar{b}\bar{c}d\bar{e}\bar{f})$	$(efd\bar{c}\bar{a}b)\mathcal{K}$	$(bca\bar{f}\bar{d}e)$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_6^{co}	$(\bar{a}bc\bar{d}ef)$	$(\bar{e}\bar{f}\bar{d}ca\bar{b})\mathcal{K}$	$(\bar{b}\bar{c}\bar{a}fd\bar{e})$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 571.1.1213 : ${}^{2}m^{2}\bar{3}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_9 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid 4^{-}_{010}\right]$	$\left[2_{001} \mid\mid \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 571.2.1214 : ${}^{2_z}m^{2_z}\bar{3}^{1}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right]$
$\left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{001}T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right] \left[T \parallel 4^{-}_{100}\right] \left[2_{001}T \parallel m_{101}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right] \left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{\bar{1}\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_9 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid 4^{-}_{010}\right]$	$\left[2_{001} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 571.3.1215 : ${}^{2_z}m^{2_z}\bar{3}^{1}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010}T \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100}T \parallel m_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 4^{-}_{010}\right] \left[2_{010}T \parallel 4^{-}_{100}\right] \left[2_{100}T \parallel m_{101}\right] \left[2_{100}T \parallel m_{01\bar{1}}\right] \left[2_{100}T \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel m_{011}\right] \left[2_{100}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[2_{010}T \parallel 2_{011}\right] \left[2_{010}T \parallel 2_{10\bar{1}}\right] \left[2_{010}T \parallel 2_{01\bar{1}}\right] \left[2_{010}T \parallel 2_{101}\right] \left[2_{010}T \parallel 2_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[2_{010}T \parallel 4^{+}_{010}\right] \left[2_{010}T \parallel 4^{+}_{100}\right] \left[2_{010}T \parallel 4^{-}_{001}\right] \left[2_{010}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_9 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{001} \mid\mid 4^{-}_{010}\right]$	$\left[2_{001} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 571.4.1216 : ${}^{2_z}m^{2_z}\bar{3}^1m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}\bar{3}^1m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z \parallel m_{101}\right] \left[\Sigma_z \parallel m_{01\bar{1}}\right] \left[\Sigma_z \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel m_{011}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E} \parallel m_{101}\right] \left[\bar{E} \parallel m_{01\bar{1}}\right]$
$\left[\bar{E} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel m_{011}\right] \left[\bar{E} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{011}\right] \left[\bar{E} 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{101}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right]$
$\left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{\bar{1}\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{011}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 2_{011}\right]$
$\left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z \parallel m_{101}\right] \left[\Sigma_z \parallel m_{01\bar{1}}\right] \left[\Sigma_z \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel m_{011}\right]$
$\left[\Sigma_z \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E} 2_{100} \parallel 2_{011}\right]$
$\left[\bar{E} 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel 2_{101}\right] \left[\bar{E} 2_{100} \parallel 2_{110}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E} 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{011}\right]$
$\left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

C_3 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_5 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$

C_6 : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

C_7 : $\left[\bar{E} \parallel 1\right]$

C_8 : $\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

C_{10} : $\left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{101}\right] \left[\bar{E} \parallel m_{01\bar{1}}\right] \left[\bar{E} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel m_{011}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$

C_{11} : $\left[\bar{E} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1	2	0	2	2	2	2	−2	−2	−2	−2	−2
χ_2	2	0	2	2	−2	−2	−2	−2	−2	2	2
χ_3	4	0	−2	4	0	0	−4	2	−4	0	0
χ_4	6	0	0	−2	2	−2	−6	0	2	−2	2

χ_5	6	0	0	−2	−2	2	−6	0	2	2	−2

Irrepresntations of group generators

	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid 4^{-}_{010}\right]$	$\left[2_{100} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_2^* \\ \omega_2^* & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_4	$(defabc)\mathcal{K}$	$(edf\bar{b}\bar{a}c)\mathcal{K}$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(abcd\bar{e}\bar{f})\mathcal{K}$
Γ_5	$(defabc)\mathcal{K}$	$(\bar{e}\bar{d}\bar{f}ba\bar{c})\mathcal{K}$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(abcd\bar{e}\bar{f})\mathcal{K}$

No 572.1.1217 : ${}^{m}m^{m}\bar{3}^{1}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001}T \,||\, m_{001}\right] \left[2_{001}T \,||\, \bar{3}^{+}_{111}\right] \left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}1}\right] \left[E \,||\, 3^{-}_{111}\right] \left[2_{001}T \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[E \,||\, 3^{+}_{111}\right] \left[2_{001}T \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \,||\, \bar{3}^{-}_{111}\right]$
$\left[E \,||\, 3^{+}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{11\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \,||\, m_{100}\right] \left[2_{001}T \,||\, m_{010}\right] \left[E \,||\, 3^{-}_{11\bar{1}}\right] \left[E \,||\, m_{110}\right] \left[2_{001}T \,||\, 2_{1\bar{1}0}\right] \left[2_{001}T \,||\, 4^{-}_{010}\right]$
$\left[2_{001}T \,||\, 4^{-}_{100}\right] \left[E \,||\, m_{101}\right] \left[E \,||\, m_{01\bar{1}}\right] \left[E \,||\, m_{10\bar{1}}\right] \left[E \,||\, m_{011}\right] \left[E \,||\, \bar{4}^{+}_{100}\right] \left[E \,||\, \bar{4}^{+}_{010}\right] \left[2_{001}T \,||\, 2_{011}\right] \left[2_{001}T \,||\, 2_{10\bar{1}}\right]$
$\left[2_{001}T \,||\, 2_{01\bar{1}}\right] \left[2_{001}T \,||\, 2_{101}\right] \left[2_{001}T \,||\, 2_{110}\right] \left[E \,||\, m_{1\bar{1}0}\right] \left[E \,||\, \bar{4}^{-}_{001}\right] \left[E \,||\, \bar{4}^{+}_{001}\right] \left[E \,||\, \bar{4}^{-}_{010}\right] \left[E \,||\, \bar{4}^{-}_{100}\right] \left[2_{001}T \,||\, 4^{+}_{010}\right]$
$\left[2_{001}T \,||\, 4^{+}_{100}\right] \left[2_{001}T \,||\, 4^{-}_{001}\right] \left[2_{001}T \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}1}\right] \left[E \,||\, 3^{-}_{111}\right] \left[E \,||\, 3^{+}_{111}\right] \left[E \,||\, 3^{+}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{11\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_4 : $\left[E \,||\, m_{110}\right] \left[E \,||\, m_{101}\right] \left[E \,||\, m_{01\bar{1}}\right] \left[E \,||\, m_{10\bar{1}}\right] \left[E \,||\, m_{011}\right] \left[E \,||\, m_{1\bar{1}0}\right]$
C_5 : $\left[E \,||\, \bar{4}^{+}_{100}\right] \left[E \,||\, \bar{4}^{+}_{010}\right] \left[E \,||\, \bar{4}^{-}_{001}\right] \left[E \,||\, \bar{4}^{+}_{001}\right] \left[E \,||\, \bar{4}^{-}_{010}\right] \left[E \,||\, \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^{-}_{1\bar{1}1}\right]$	$\left[E \,||\, \bar{4}^{+}_{100}\right]$	$\left[2_{001}T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 572.2.1218 : ${}^{m_z}m^{m_z}\bar{3}^1m^{\infty_x}1 \to {}^{m}m^{m}\bar{3}^1m$

No 573.1.1219 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[E \parallel \bar{4}^{+}_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 573.2.1220 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\infty_z}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z \parallel m_{101}\right] \left[\Sigma_z \parallel m_{01\bar{1}}\right] \left[\Sigma_z \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel m_{011}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E} \parallel m_{101}\right] \left[\bar{E} \parallel m_{01\bar{1}}\right]$
$\left[\bar{E} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel m_{011}\right] \left[\bar{E} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E}T \parallel 2_{110}\right]$
$\left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{011}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{011}\right] \left[\bar{\Sigma}_z T \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{101}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel m_{110}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right]$
C_6 : $\left[\Sigma_z \parallel 1\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
C_8 : $\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
C_9 : $\left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z \parallel m_{101}\right] \left[\Sigma_z \parallel m_{01\bar{1}}\right] \left[\Sigma_z \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel m_{011}\right] \left[\Sigma_z \parallel m_{1\bar{1}0}\right]$

$C_{10}: \left[\Sigma_z \parallel \bar{4}^+_{100}\right] \left[\Sigma_z \parallel \bar{4}^+_{010}\right] \left[\Sigma_z \parallel \bar{4}^-_{001}\right] \left[\Sigma_z \parallel \bar{4}^+_{001}\right] \left[\Sigma_z \parallel \bar{4}^-_{010}\right] \left[\Sigma_z \parallel \bar{4}^-_{100}\right]$

$C_{11}: \left[\bar{E} \parallel 1\right]$

$C_{12}: \left[\bar{E} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^-_{111}\right] \left[\bar{E} \parallel 3^+_{111}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^+_{11\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{11\bar{1}}\right]$

$C_{13}: \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

$C_{14}: \left[\bar{E} \parallel m_{110}\right] \left[\bar{E} \parallel m_{101}\right] \left[\bar{E} \parallel m_{01\bar{1}}\right] \left[\bar{E} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel m_{011}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right]$

$C_{15}: \left[\bar{E} \parallel \bar{4}^+_{100}\right] \left[\bar{E} \parallel \bar{4}^+_{010}\right] \left[\bar{E} \parallel \bar{4}^-_{001}\right] \left[\bar{E} \parallel \bar{4}^+_{001}\right] \left[\bar{E} \parallel \bar{4}^-_{010}\right] \left[\bar{E} \parallel \bar{4}^-_{100}\right]$

$C_{16}: \left[\bar{\Sigma}_z \parallel 1\right]$

$C_{17}: \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^-_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{11\bar{1}}\right]$

$C_{18}: \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$

$C_{19}: \left[\bar{\Sigma}_z \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel m_{101}\right] \left[\bar{\Sigma}_z \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z \parallel m_{011}\right] \left[\bar{\Sigma}_z \parallel m_{1\bar{1}0}\right]$

$C_{20}: \left[\bar{\Sigma}_z \parallel \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z \parallel \bar{4}^-_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}
χ_1^{co}	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	0	0	0	0	0
χ_2^{co}	2	2	2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	0	0	0	0	0
χ_3^{co}	4	−2	4	0	0	0	0	0	0	0	−4	2	−4	0	0	0	0	0	0	0
χ_4^{co}	6	0	−2	2	−2	0	0	0	0	0	−6	0	2	−2	2	0	0	0	0	0
χ_5^{co}	6	0	−2	−2	2	0	0	0	0	0	−6	0	2	2	−2	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^-_{1\bar{1}1}\right]$	$\left[E \parallel \bar{4}^+_{100}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{c}\bar{a}\bar{e}\bar{f}d)$	$(c\bar{b}\bar{a}\bar{f}\bar{e}d)$	$(defabc)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{c}\bar{a}\bar{e}\bar{f}d)$	$(\bar{c}bafe\bar{d})$	$(defabc)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 573.3.1221 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{1}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[E \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[E \parallel m_{101}\right] \left[E \parallel m_{01\bar{1}}\right] \left[E \parallel m_{10\bar{1}}\right] \left[E \parallel m_{011}\right] \left[E \parallel \bar{4}^{+}_{100}\right] \left[E \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[E \parallel m_{1\bar{1}0}\right] \left[E \parallel \bar{4}^{-}_{001}\right] \left[E \parallel \bar{4}^{+}_{001}\right] \left[E \parallel \bar{4}^{-}_{010}\right] \left[E \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100}T \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100}T \parallel m_{101}\right] \left[2_{100}T \parallel m_{01\bar{1}}\right] \left[2_{100}T \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel m_{011}\right] \left[2_{100}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right]$
$\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z \parallel m_{101}\right] \left[\Sigma_z \parallel m_{01\bar{1}}\right]$
$\left[\Sigma_z \parallel m_{10\bar{1}}\right] \left[\Sigma_z \parallel m_{011}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z T \parallel 2_{110}\right]$
$\left[\Sigma_z \parallel m_{1\bar{1}0}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}\right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}\right.$
$\left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{10}\right.$
$\left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{101}\right] \left[\Sigma_z 2_{100}T \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{011}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z\right.$
$\left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{E} \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E} \parallel m_{101}\right] \left[\bar{E} \parallel m_{01\bar{1}}\right] \left[\bar{E} \parallel m_{10\bar{1}}\right] \left[\bar{E} \parallel m_{011}\right] \left[\bar{E} \parallel \bar{4}^{+}_{100}\right]$
$\left[\bar{E} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E}T \parallel 2_{110}\right] \left[\bar{E} \parallel m_{1\bar{1}0}\right] \left[\bar{E} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel m_{101}\right] \left[\bar{E}2_{100}T \parallel m_{01\bar{1}}\right.$
$\left[\bar{E}2_{100}T \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel m_{011}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right] \left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$

$\left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, m_{110}\right] \left[\bar{\Sigma}_z T \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, 4^-_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z \,||\, m_{101}\right] \left[\bar{\Sigma}_z \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, m_{011}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z T \,||\, 2_{011}\right] \left[\bar{\Sigma}_z T \,||\, 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, 2_{101}\right] \left[\bar{\Sigma}_z T \,||\, 2_{110}\right] \left[\bar{\Sigma}_z \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z T \,||\, 4^+_{010}\right]$
$\left[\bar{\Sigma}_z T \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z T \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z T \,||\, 4^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{111}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} T\right.$
$\left[\bar{\Sigma}_z 2_{100} T \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{101}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{011}\right] \left[\bar{\Sigma}_z 2\right.$
$\left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} T\right.$
$\left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$

C_2 : $\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}1}\right] \left[E \,||\, 3^-_{111}\right] \left[E \,||\, 3^+_{111}\right] \left[E \,||\, 3^+_{1\bar{1}1}\right] \left[E \,||\, 3^+_{11\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{11\bar{1}}\right]$

C_3 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

C_4 : $\left[E \,||\, m_{110}\right] \left[E \,||\, m_{101}\right] \left[E \,||\, m_{01\bar{1}}\right] \left[E \,||\, m_{10\bar{1}}\right] \left[E \,||\, m_{011}\right] \left[E \,||\, m_{1\bar{1}0}\right]$

C_5 : $\left[E \,||\, \bar{4}^+_{100}\right] \left[E \,||\, \bar{4}^+_{010}\right] \left[E \,||\, \bar{4}^-_{001}\right] \left[E \,||\, \bar{4}^+_{001}\right] \left[E \,||\, \bar{4}^-_{010}\right] \left[E \,||\, \bar{4}^-_{100}\right]$

C_6 : $\left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{3}^+_{111}\right] \left[2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, -1\right] \left[2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^-_{111}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, 2_{1\bar{1}0}\right] \left[2_{100} \,||\, 4^-_{010}\right] \left[2_{100} \,||\, 4^-_{100}\right] \left[2_{100} \,||\, 2_{011}\right]$
$\left[2_{100} \,||\, 2_{10\bar{1}}\right] \left[2_{100} \,||\, 2_{01\bar{1}}\right] \left[2_{100} \,||\, 2_{101}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, 4^+_{010}\right] \left[2_{100} \,||\, 4^+_{100}\right] \left[2_{100} \,||\, 4^-_{001}\right] \left[2_{100} \,||\, 4^+_{001}\right]$
$\left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^-_{111}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 3^+_{111}\right]$
$\left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\Sigma_z \,||\, m_{110}\right] \left[\Sigma_z \,||\, m_{101}\right] \left[\Sigma_z \,||\, m_{01\bar{1}}\right] \left[\Sigma_z \,||\, m_{10\bar{1}}\right]$
$\left[\Sigma_z \,||\, m_{011}\right] \left[\Sigma_z \,||\, \bar{4}^+_{100}\right] \left[\Sigma_z \,||\, \bar{4}^+_{010}\right] \left[\Sigma_z \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z \,||\, \bar{4}^-_{001}\right] \left[\Sigma_z \,||\, \bar{4}^+_{001}\right] \left[\Sigma_z \,||\, \bar{4}^-_{010}\right] \left[\Sigma_z \,||\, \bar{4}^-_{100}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, -1\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{010}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{100}\right] \left[\Sigma_z 2_{100} \,||\, 2_{011}\right]$
$\left[\Sigma_z 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, 2_{101}\right] \left[\Sigma_z 2_{100} \,||\, 2_{110}\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{010}\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{100}\right] \left[\Sigma_z 2_{100} \,||\, 4^-_{001}\right] \left[\Sigma_z 2_{100} \,||\, 4^+_{001}\right]$
$\left[\bar{E} 2_{100} \,||\, m_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, -1\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[\bar{E} 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\bar{E} 2_{100} \,||\, m_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{010}\right] \left[\bar{E} 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, 4^-_{010}\right] \left[\bar{E} 2_{100} \,||\, 4^-_{100}\right] \left[\bar{E} 2_{100} \,||\, 2_{011}\right]$
$\left[\bar{E} 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, 2_{101}\right] \left[\bar{E} 2_{100} \,||\, 2_{110}\right] \left[\bar{E} 2_{100} \,||\, 4^+_{010}\right] \left[\bar{E} 2_{100} \,||\, 4^+_{100}\right] \left[\bar{E} 2_{100} \,||\, 4^-_{001}\right] \left[\bar{E} 2_{100} \,||\, 4^+_{001}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right]$
$\left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, m_{110}\right] \left[\bar{\Sigma}_z \,||\, m_{101}\right] \left[\bar{\Sigma}_z \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \,||\, m_{011}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z \,||\, \bar{4}^-_{100}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{011}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, 4^+_{001}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

C_{10} : $\left[\bar{E} \,||\, m_{110}\right] \left[\bar{E} \,||\, m_{101}\right] \left[\bar{E} \,||\, m_{01\bar{1}}\right] \left[\bar{E} \,||\, m_{10\bar{1}}\right] \left[\bar{E} \,||\, m_{011}\right] \left[\bar{E} \,||\, m_{1\bar{1}0}\right]$

C_{11} : $\left[\bar{E} \,||\, \bar{4}^{+}_{100}\right] \left[\bar{E} \,||\, \bar{4}^{+}_{010}\right] \left[\bar{E} \,||\, \bar{4}^{-}_{001}\right] \left[\bar{E} \,||\, \bar{4}^{+}_{001}\right] \left[\bar{E} \,||\, \bar{4}^{-}_{010}\right] \left[\bar{E} \,||\, \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1^{co}	2	2	2	2	2	0	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	−2	−2	0	−2	−2	−2	2	2
χ_3^{co}	4	−2	4	0	0	0	−4	2	−4	0	0
χ_4^{co}	6	0	−2	2	−2	0	−6	0	2	−2	2
χ_5^{co}	6	0	−2	−2	2	0	−6	0	2	2	−2

Coirrepresntations of group generators

	$\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, \bar{4}^{+}_{100}\right]$	$\left[2_{100} \,||\, 4^{-}_{010}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$-\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_2^* \\ \omega_2^* & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(c\bar{b}\bar{a}\bar{f}\bar{e}d)$	$(edf\bar{b}\bar{a}c)\mathcal{K}$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(\bar{c}bafe\bar{d})$	$(\bar{e}\bar{d}\bar{f}ba\bar{c})\mathcal{K}$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 574.1.1222 : ${}^{2}m^{2}\bar{3}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_9 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[E \mid\mid 4^{-}_{010}\right]$	$\left[2_{001} \mid\mid \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 574.2.1223 : ${}^{2_z}m^{2_z}\bar{3}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 4^{-}_{010}\right] \left[2_{001}T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right]$
$\left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right] \left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{101}\right] \left[2_{001}T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right]$
$\left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right] \left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_9 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[E \mid\mid 4^{-}_{010}\right]$	$\left[2_{001} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 574.3.1224 : ${}^{2_z}m^{2_z}\bar{3}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{010}T \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 4^{-}_{010}\right] \left[2_{100}T \parallel 4^{-}_{100}\right] \left[2_{010}T \parallel m_{101}\right] \left[2_{010}T \parallel m_{01\bar{1}}\right] \left[2_{010}T \parallel m_{10\bar{1}}\right] \left[2_{010}T \parallel m_{011}\right] \left[2_{010}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{010}T \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 2_{011}\right] \left[2_{100}T \parallel 2_{10\bar{1}}\right] \left[2_{100}T \parallel 2_{01\bar{1}}\right] \left[2_{100}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel \bar{4}^{-}_{001}\right] \left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{010}T \parallel \bar{4}^{-}_{010}\right] \left[2_{010}T \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 4^{+}_{010}\right] \left[2_{100}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_9 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[E \mid\mid 4^{-}_{010}\right]$	$\left[2_{001} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 574.4.1225 : ${}^{2_z}m^{2_z}\bar{3}^{2_z}m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}\bar{3}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 4^{-}_{010}\right] \left[\Sigma_z \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z \parallel 2_{011}\right] \left[\Sigma_z \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 2_{01\bar{1}}\right] \left[\Sigma_z \parallel 2_{101}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 4^{+}_{010}\right] \left[\Sigma_z \parallel 4^{+}_{100}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 4^{-}_{010}\right] \left[\bar{E} \parallel 4^{-}_{100}\right] \left[\bar{E} 2_{100} \parallel m_{101}\right] \left[\bar{E} 2_{100} \parallel m_{01\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{011}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} \parallel 2_{011}\right] \left[\bar{E} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 2_{01\bar{1}}\right] \left[\bar{E} \parallel 2_{101}\right] \left[\bar{E} \parallel 2_{110}\right]$
$\left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} \parallel 4^{+}_{010}\right] \left[\bar{E} \parallel 4^{+}_{100}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{011}\right] \left[\bar{\Sigma}_z \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

$C_1 : \quad \left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right]$
$\left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right]$
$\left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 4^{-}_{010}\right] \left[\Sigma_z \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z \parallel 2_{011}\right] \left[\Sigma_z \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 2_{01\bar{1}}\right] \left[\Sigma_z \parallel 2_{101}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 4^{+}_{010}\right] \left[\Sigma_z \parallel 4^{+}_{100}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} 2_{100} \parallel m_{110}\right] \left[\bar{E} 2_{100} \parallel m_{101}\right] \left[\bar{E} 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel m_{011}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} 2_{100} \parallel \bar{4}^{-}_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{011}\right] \left[\bar{\Sigma}_z \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$

C_3 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_5 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$

C_6 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

C_7 : $\left[\bar{E} \parallel 1\right]$

C_8 : $\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

C_{10} : $\left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{011}\right] \left[\bar{E} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 2_{01\bar{1}}\right] \left[\bar{E} \parallel 2_{101}\right] \left[\bar{E} \parallel 2_{110}\right]$

C_{11} : $\left[\bar{E} \parallel 4^{-}_{010}\right] \left[\bar{E} \parallel 4^{-}_{100}\right] \left[\bar{E} \parallel 4^{+}_{010}\right] \left[\bar{E} \parallel 4^{+}_{100}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 4^{+}_{001}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1	2	0	2	2	2	2	−2	−2	−2	−2	−2
χ_2	2	0	2	2	−2	−2	−2	−2	−2	2	2
χ_3	4	0	−2	4	0	0	−4	2	−4	0	0
χ_4	6	0	0	−2	−2	2	−6	0	2	2	−2

χ_5	6	0	0	−2	2	−2	−6	0	2	−2	2

Irrepresntations of group generators

	$\left[2_{100} \mid\mid m_{001}\right]$	$\left[E \mid\mid 4^{-}_{010}\right]$	$\left[2_{100} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[\Sigma_z \mid\mid 1\right]$
Γ_1	$\sigma_x\mathcal{K}$	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2	$\sigma_x\mathcal{K}$	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_1 \\ \omega_1 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_4	$(defabc)\mathcal{K}$	$(\bar{b}aced\bar{d}f)$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(abcd\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_5	$(defabc)\mathcal{K}$	$(b\bar{a}\bar{c}\bar{e}d\bar{f})$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(abcd\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 575.1.1226 : ${}^{m}m^{m}\bar{3}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[2_{001}T \parallel m_{101}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel \bar{4}^{+}_{100}\right] \left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_5 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[E \parallel 4^{-}_{010}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 575.2.1227 : ${}^{m_z}m^{m_z}\bar{3}^{m_z}m^{\infty_x}1 \to {}^{m}m^{m}\bar{3}^{m}m$

No 576.1.1228 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_5 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[E \parallel 4^{-}_{010}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 576.2.1229 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\infty_z}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\Sigma_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 4^{-}_{010}\right] \left[\Sigma_z \parallel 4^{-}_{100}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right] \left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z \parallel 2_{011}\right] \left[\Sigma_z \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 2_{01\bar{1}}\right] \left[\Sigma_z \parallel 2_{101}\right] \left[\Sigma_z \parallel 2_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 4^{+}_{010}\right] \left[\Sigma_z \parallel 4^{+}_{100}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 4^{-}_{010}\right] \left[\bar{E} \parallel 4^{-}_{100}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} \parallel 2_{011}\right] \left[\bar{E} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 2_{01\bar{1}}\right] \left[\bar{E} \parallel 2_{101}\right] \left[\bar{E} \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} \parallel 4^{+}_{010}\right] \left[\bar{E} \parallel 4^{+}_{100}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{101}\right] \left[\bar{\Sigma}_z T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{011}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z \parallel 2_{011}\right] \left[\bar{\Sigma}_z \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right] \left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right]$
C_5 : $\left[E \parallel 4^{-}_{010}\right] \left[E \parallel 4^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right] \left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right]$
C_6 : $\left[\Sigma_z \parallel 1\right]$
C_7 : $\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
C_8 : $\left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
C_9 : $\left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 2_{011}\right] \left[\Sigma_z \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 2_{01\bar{1}}\right] \left[\Sigma_z \parallel 2_{101}\right] \left[\Sigma_z \parallel 2_{110}\right]$

$C_{10}: \left[\Sigma_z \parallel 4^-_{010}\right] \left[\Sigma_z \parallel 4^-_{\bar{1}00}\right] \left[\Sigma_z \parallel 4^+_{010}\right] \left[\Sigma_z \parallel 4^+_{100}\right] \left[\Sigma_z \parallel 4^-_{001}\right] \left[\Sigma_z \parallel 4^+_{001}\right]$

$C_{11}: \left[\bar{E} \parallel 1\right]$

$C_{12}: \left[\bar{E} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^-_{111}\right] \left[\bar{E} \parallel 3^+_{111}\right] \left[\bar{E} \parallel 3^+_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^+_{11\bar{1}}\right] \left[\bar{E} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^-_{11\bar{1}}\right]$

$C_{13}: \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right]$

$C_{14}: \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 2_{011}\right] \left[\bar{E} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 2_{01\bar{1}}\right] \left[\bar{E} \parallel 2_{101}\right] \left[\bar{E} \parallel 2_{110}\right]$

$C_{15}: \left[\bar{E} \parallel 4^-_{010}\right] \left[\bar{E} \parallel 4^-_{100}\right] \left[\bar{E} \parallel 4^+_{010}\right] \left[\bar{E} \parallel 4^+_{100}\right] \left[\bar{E} \parallel 4^-_{001}\right] \left[\bar{E} \parallel 4^+_{001}\right]$

$C_{16}: \left[\bar{\Sigma}_z \parallel 1\right]$

$C_{17}: \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^-_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{111}\right] \left[\bar{\Sigma}_z \parallel 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^-_{11\bar{1}}\right]$

$C_{18}: \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right]$

$C_{19}: \left[\bar{\Sigma}_z \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \parallel 2_{011}\right] \left[\bar{\Sigma}_z \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 2_{101}\right] \left[\bar{\Sigma}_z \parallel 2_{110}\right]$

$C_{20}: \left[\bar{\Sigma}_z \parallel 4^-_{010}\right] \left[\bar{\Sigma}_z \parallel 4^-_{100}\right] \left[\bar{\Sigma}_z \parallel 4^+_{010}\right] \left[\bar{\Sigma}_z \parallel 4^+_{100}\right] \left[\bar{\Sigma}_z \parallel 4^-_{001}\right] \left[\bar{\Sigma}_z \parallel 4^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}	C_{20}
χ_1^{co}	2	2	2	2	2	0	0	0	0	0	−2	−2	−2	−2	−2	0	0	0	0	0
χ_2^{co}	2	2	2	−2	−2	0	0	0	0	0	−2	−2	−2	2	2	0	0	0	0	0
χ_3^{co}	4	−2	4	0	0	0	0	0	0	0	−4	2	−4	0	0	0	0	0	0	0
χ_4^{co}	6	0	−2	−2	2	0	0	0	0	0	−6	0	2	2	−2	0	0	0	0	0
χ_5^{co}	6	0	−2	2	−2	0	0	0	0	0	−6	0	2	−2	2	0	0	0	0	0

Coirrepresntations of group generators

	$\left[E \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^-_{1\bar{1}1}\right]$	$\left[E \parallel 4^-_{010}\right]$	$\left[T \parallel m_{001}\right]$	$\left[\Sigma_z \parallel 1\right]$
Γ_1^{co}	σ_0	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	σ_0	$-\sigma_0$	$\sigma_x\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2 \end{pmatrix}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{c}\bar{a}\bar{e}\bar{f}d)$	$(\bar{b}ac\bar{e}\bar{d}f)$	$(defabc)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{c}\bar{a}\bar{e}\bar{f}d)$	$(b\bar{a}\bar{c}\bar{e}d\bar{f})$	$(defabc)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 576.3.1230 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\infty_z m_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{\bar{1}}m^{\Sigma_z m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[E \parallel 2_{1\bar{1}0}\right] \left[E \parallel 4^{-}_{010}\right]$
$\left[E \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[E \parallel 2_{011}\right] \left[E \parallel 2_{10\bar{1}}\right]$
$\left[E \parallel 2_{01\bar{1}}\right] \left[E \parallel 2_{101}\right] \left[E \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[E \parallel 4^{+}_{010}\right]$
$\left[E \parallel 4^{+}_{100}\right] \left[E \parallel 4^{-}_{001}\right] \left[E \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 4^{-}_{010}\right] \left[2_{100}T \parallel 4^{-}_{100}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 2_{011}\right] \left[2_{100}T \parallel 2_{10\bar{1}}\right] \left[2_{100}T \parallel 2_{01\bar{1}}\right] \left[2_{100}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 4^{+}_{010}\right] \left[2_{100}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right]$
$\left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z \parallel 4^{-}_{010}\right] \left[\Sigma_z \parallel 4^{-}_{100}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z \parallel 2_{011}\right] \left[\Sigma_z \parallel 2_{10\bar{1}}\right] \left[\Sigma_z \parallel 2_{01\bar{1}}\right] \left[\Sigma_z \parallel 2_{101}\right] \left[\Sigma_z \parallel 2_{110}\right]$
$\left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 4^{+}_{010}\right] \left[\Sigma_z \parallel 4^{+}_{100}\right] \left[\Sigma_z \parallel 4^{-}_{001}\right] \left[\Sigma_z \parallel 4^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}\right.$
$\left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z 2_{100}T \parallel 2_{001}\right] \left[\Sigma_z 2_{100}T \parallel 2_{100}\right] \left[\Sigma_z 2_{100}T \parallel 2_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}\right.$
$\left[\Sigma_z 2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z 2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{10}\right.$
$\left[\Sigma_z 2_{100}T \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{011}\right] \left[\Sigma_z 2_{100}T\right.$
$\left[\Sigma_z 2_{100}T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 2_{101}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100}T\right.$
$\left[\Sigma_z 2_{100}T \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right]$
$\left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right] \left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{110}\right] \left[\bar{E} \parallel 2_{1\bar{1}0}\right] \left[\bar{E} \parallel 4^{-}_{010}\right] \left[\bar{E} \parallel 4^{-}_{100}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right] \left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right]$
$\left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E} \parallel 2_{011}\right] \left[\bar{E} \parallel 2_{10\bar{1}}\right] \left[\bar{E} \parallel 2_{01\bar{1}}\right] \left[\bar{E} \parallel 2_{101}\right] \left[\bar{E} \parallel 2_{110}\right] \left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E} \parallel 4^{+}_{010}\right] \left[\bar{E} \parallel 4^{+}_{100}\right] \left[\bar{E} \parallel 4^{-}_{001}\right] \left[\bar{E} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel 1\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100}T \parallel 2_{001}\right] \left[\bar{E}2_{100}T \parallel 2_{100}\right]$
$\left[\bar{E}2_{100}T \parallel 2_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E}2_{100}T \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right.$
$\left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{011}\right] \left[\bar{E}2_{100}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 2_{101}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$

$\left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{010}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{110}\right] \left[\bar{\Sigma}_z \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, 4^-_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z T \,||\, m_{101}\right] \left[\bar{\Sigma}_z T \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z T \,||\, m_{011}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z \,||\, 2_{011}\right] \left[\bar{\Sigma}_z \,||\, 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 2_{101}\right] \left[\bar{\Sigma}_z \,||\, 2_{110}\right] \left[\bar{\Sigma}_z T \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z T \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z \,||\, 4^+_{010}\right]$
$\left[\bar{\Sigma}_z \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{111}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} T\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 4^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{011}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 4^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} T \,||\, 4^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$

C_2 : $\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}1}\right] \left[E \,||\, 3^-_{111}\right] \left[E \,||\, 3^+_{111}\right] \left[E \,||\, 3^+_{1\bar{1}1}\right] \left[E \,||\, 3^+_{11\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{11\bar{1}}\right]$

C_3 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

C_4 : $\left[E \,||\, 2_{1\bar{1}0}\right] \left[E \,||\, 2_{011}\right] \left[E \,||\, 2_{10\bar{1}}\right] \left[E \,||\, 2_{01\bar{1}}\right] \left[E \,||\, 2_{101}\right] \left[E \,||\, 2_{110}\right]$

C_5 : $\left[E \,||\, 4^-_{010}\right] \left[E \,||\, 4^-_{100}\right] \left[E \,||\, 4^+_{010}\right] \left[E \,||\, 4^+_{100}\right] \left[E \,||\, 4^-_{001}\right] \left[E \,||\, 4^+_{001}\right]$

C_6 : $\left[2_{100} \,||\, m_{001}\right] \left[2_{100} \,||\, \bar{3}^+_{111}\right] \left[2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, -1\right] \left[2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[2_{100} \,||\, \bar{3}^-_{111}\right] \left[2_{100} \,||\, m_{100}\right] \left[2_{100} \,||\, m_{010}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, m_{101}\right] \left[2_{100} \,||\, m_{01\bar{1}}\right] \left[2_{100} \,||\, m_{10\bar{1}}\right]$
$\left[2_{100} \,||\, m_{011}\right] \left[2_{100} \,||\, \bar{4}^+_{100}\right] \left[2_{100} \,||\, \bar{4}^+_{010}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, \bar{4}^-_{001}\right] \left[2_{100} \,||\, \bar{4}^+_{001}\right] \left[2_{100} \,||\, \bar{4}^-_{010}\right] \left[2_{100} \,||\, \bar{4}^-_{100}\right]$
$\left[\Sigma_z \,||\, 1\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^-_{111}\right] \left[\Sigma_z \,||\, 2_{001}\right] \left[\Sigma_z \,||\, 2_{100}\right] \left[\Sigma_z \,||\, 2_{010}\right] \left[\Sigma_z \,||\, 3^+_{111}\right]$
$\left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\Sigma_z \,||\, 2_{1\bar{1}0}\right] \left[\Sigma_z \,||\, 4^-_{010}\right] \left[\Sigma_z \,||\, 4^-_{100}\right] \left[\Sigma_z \,||\, 2_{011}\right]$
$\left[\Sigma_z \,||\, 2_{10\bar{1}}\right] \left[\Sigma_z \,||\, 2_{01\bar{1}}\right] \left[\Sigma_z \,||\, 2_{101}\right] \left[\Sigma_z \,||\, 2_{110}\right] \left[\Sigma_z \,||\, 4^+_{010}\right] \left[\Sigma_z \,||\, 4^+_{100}\right] \left[\Sigma_z \,||\, 4^-_{001}\right] \left[\Sigma_z \,||\, 4^+_{001}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, -1\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\Sigma_z 2_{100} \,||\, m_{100}\right] \left[\Sigma_z 2_{100} \,||\, m_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{110}\right] \left[\Sigma_z 2_{100} \,||\, m_{101}\right] \left[\Sigma_z 2_{100} \,||\, m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \,||\, m_{10\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \,||\, m_{011}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{100}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\Sigma_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\Sigma_z 2_{100} \,||\, \bar{4}^-_{100}\right]$
$\left[\bar{E} 2_{100} \,||\, m_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, -1\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[\bar{E} 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\bar{E} 2_{100} \,||\, m_{100}\right] \left[\bar{E} 2_{100} \,||\, m_{010}\right] \left[\bar{E} 2_{100} \,||\, m_{110}\right] \left[\bar{E} 2_{100} \,||\, m_{101}\right] \left[\bar{E} 2_{100} \,||\, m_{01\bar{1}}\right] \left[\bar{E} 2_{100} \,||\, m_{10\bar{1}}\right]$
$\left[\bar{E} 2_{100} \,||\, m_{011}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{100}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\bar{E} 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\bar{E} 2_{100} \,||\, \bar{4}^-_{100}\right]$
$\left[\bar{\Sigma}_z \,||\, 1\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 2_{001}\right] \left[\bar{\Sigma}_z \,||\, 2_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{010}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right]$
$\left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z \,||\, 4^-_{010}\right] \left[\bar{\Sigma}_z \,||\, 4^-_{100}\right] \left[\bar{\Sigma}_z \,||\, 2_{011}\right]$
$\left[\bar{\Sigma}_z \,||\, 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 2_{101}\right] \left[\bar{\Sigma}_z \,||\, 2_{110}\right] \left[\bar{\Sigma}_z \,||\, 4^+_{010}\right] \left[\bar{\Sigma}_z \,||\, 4^+_{100}\right] \left[\bar{\Sigma}_z \,||\, 4^-_{001}\right] \left[\bar{\Sigma}_z \,||\, 4^+_{001}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, -1\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{3}^-_{111}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \,||\, m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{100}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^+_{001}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{010}\right] \left[\bar{\Sigma}_z 2_{100} \,||\, \bar{4}^-_{100}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

C_{10} : $\left[\bar{E} \,||\, 2_{1\bar{1}0}\right] \left[\bar{E} \,||\, 2_{011}\right] \left[\bar{E} \,||\, 2_{10\bar{1}}\right] \left[\bar{E} \,||\, 2_{01\bar{1}}\right] \left[\bar{E} \,||\, 2_{101}\right] \left[\bar{E} \,||\, 2_{110}\right]$

C_{11} : $\left[\bar{E} \,||\, 4^-_{010}\right] \left[\bar{E} \,||\, 4^-_{100}\right] \left[\bar{E} \,||\, 4^+_{010}\right] \left[\bar{E} \,||\, 4^+_{100}\right] \left[\bar{E} \,||\, 4^-_{001}\right] \left[\bar{E} \,||\, 4^+_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
χ_1^{co}	2	2	2	2	2	0	−2	−2	−2	−2	−2
χ_2^{co}	2	2	2	−2	−2	0	−2	−2	−2	2	2
χ_3^{co}	4	−2	4	0	0	0	−4	2	−4	0	0
χ_4^{co}	6	0	−2	−2	2	0	−6	0	2	2	−2
χ_5^{co}	6	0	−2	2	−2	0	−6	0	2	−2	2

Coirrepresntations of group generators

	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 4^-_{010}\right]$	$\left[2_{100} \,||\, \bar{4}^+_{100}\right]$	$\left[2_{100}T \,||\, 1\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_2^{co}	σ_0	$-\sigma_0$	$-\sigma_x\mathcal{K}$	σ_x	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1 \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2^* \end{pmatrix}$	$\begin{pmatrix} 0 & \omega_2 \\ \omega_2 & 0 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(\bar{b}ace\bar{d}f)$	$(\bar{f}\bar{e}\bar{d}c\bar{b}a)\mathcal{K}$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(b\bar{a}\bar{c}\bar{e}d\bar{f})$	$(fed\bar{c}b\bar{a})\mathcal{K}$	$(defabc)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 577.1.1231 : ${}^{2_y}m^{2_y}\bar{3}^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_3 : $\left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
C_9 : $\left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[2_{010} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid 4^{-}_{010}\right]$	$\left[2_{010} \mid\mid \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 577.2.1232 : ${}^{2_y}m^{2_y}\bar{3}^{2_z}m^{m_y}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{010}T \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[2_{010}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{010}T \parallel 3^{-}_{111}\right]$
$\left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel -1\right] \left[2_{010}T \parallel 2_{001}\right] \left[2_{010}T \parallel 2_{100}\right] \left[2_{010}T \parallel 2_{010}\right] \left[2_{010}T \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right] \left[2_{010}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{010}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{010}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[2_{010}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{100}T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 4^{-}_{010}\right] \left[2_{001}T \parallel 4^{-}_{100}\right] \left[2_{100}T \parallel m_{101}\right] \left[2_{100}T \parallel m_{01\bar{1}}\right] \left[2_{100}T \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel m_{011}\right] \left[2_{100}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right] \left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{101}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right] \left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_3 : $\left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
C_9 : $\left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid 4^{-}_{010}\right]$	$\left[2_{010} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{010}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 577.3.1233 : ${}^{2_y}m^{2_y}\bar{3}^{2_z}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{-}_{111}\right]$
$\left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[2_{001}T \parallel 3^{+}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right]$
$\left[2_{100}T \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[2_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{010}\right] \left[2_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[T \parallel m_{110}\right] \left[2_{010}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 4^{-}_{010}\right] \left[2_{010}T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right]$
$\left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{010}T \parallel 2_{011}\right] \left[2_{010}T \parallel 2_{10\bar{1}}\right] \left[2_{010}T \parallel 2_{01\bar{1}}\right] \left[2_{010}T \parallel 2_{101}\right] \left[2_{010}T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right]$
$\left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{010}T \parallel 4^{+}_{010}\right] \left[2_{010}T \parallel 4^{+}_{100}\right] \left[2_{010}T \parallel 4^{-}_{001}\right] \left[2_{010}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_3 : $\left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{\bar{1}\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{\bar{1}\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
C_9 : $\left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid 4^{-}_{010}\right]$	$\left[2_{010} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 577.4.1234 : ${}^{2_y}m^{2_y}\bar{3}^{2_z}m^{m_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{100}T \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{-}_{111}\right]$
$\left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel -1\right] \left[2_{100}T \parallel 2_{001}\right] \left[2_{100}T \parallel 2_{100}\right] \left[2_{100}T \parallel 2_{010}\right] \left[2_{100}T \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right] \left[2_{100}T \parallel 3^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel 3^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[2_{100}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{010}T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right] \left[T \parallel 4^{-}_{100}\right] \left[2_{010}T \parallel m_{101}\right] \left[2_{010}T \parallel m_{01\bar{1}}\right] \left[2_{010}T \parallel m_{10\bar{1}}\right] \left[2_{010}T \parallel m_{011}\right] \left[2_{010}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{010}T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right] \left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[2_{010}T \parallel m_{1\bar{1}0}\right] \left[2_{010}T \parallel \bar{4}^{-}_{001}\right] \left[2_{010}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{010}T \parallel \bar{4}^{-}_{010}\right] \left[2_{010}T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right] \left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{010} \parallel m_{001}\right] \left[2_{010} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_3 : $\left[2_{010} \parallel \bar{3}^{+}_{111}\right] \left[2_{010} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{010} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{\bar{1}\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{\bar{1}\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{010} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
C_9 : $\left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{010} \mid\mid m_{001}\right]$	$\left[2_{100} \mid\mid 4^{-}_{010}\right]$	$\left[2_{010} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{100}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 578.1.1235 : ${}^{m_x}m^{m_x}\bar{3}^{m_y}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{100}T \,||\, m_{001}\right] \left[2_{100}T \,||\, \bar{3}^{+}_{111}\right] \left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}1}\right] \left[E \,||\, 3^{-}_{111}\right] \left[2_{100}T \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100}T \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100}T \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100}T \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[E \,||\, 3^{+}_{111}\right] \left[2_{100}T \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100}T \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100}T \,||\, \bar{3}^{-}_{111}\right]$
$\left[E \,||\, 3^{+}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{11\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \,||\, m_{100}\right] \left[2_{100}T \,||\, m_{010}\right] \left[E \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{010}T \,||\, m_{110}\right] \left[2_{001} \,||\, 2_{1\bar{1}0}\right] \left[2_{001} \,||\, 4^{-}_{010}\right]$
$\left[2_{001} \,||\, 4^{-}_{100}\right] \left[2_{010}T \,||\, m_{101}\right] \left[2_{010}T \,||\, m_{01\bar{1}}\right] \left[2_{010}T \,||\, m_{10\bar{1}}\right] \left[2_{010}T \,||\, m_{011}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{100}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{010}\right] \left[2_{001} \,||\, 2_{011}\right] \left[2_{001} \,||\, 2_{10\bar{1}}\right]$
$\left[2_{001} \,||\, 2_{01\bar{1}}\right] \left[2_{001} \,||\, 2_{101}\right] \left[2_{001} \,||\, 2_{110}\right] \left[2_{010}T \,||\, m_{1\bar{1}0}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{001}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{001}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{010}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{100}\right] \left[2_{001} \,||\, 4^{+}_{010}\right]$
$\left[2_{001} \,||\, 4^{+}_{100}\right] \left[2_{001} \,||\, 4^{-}_{001}\right] \left[2_{001} \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}1}\right] \left[E \,||\, 3^{-}_{111}\right] \left[E \,||\, 3^{+}_{111}\right] \left[E \,||\, 3^{+}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{11\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_4 : $\left[2_{001} \,||\, 2_{1\bar{1}0}\right] \left[2_{001} \,||\, 2_{011}\right] \left[2_{001} \,||\, 2_{10\bar{1}}\right] \left[2_{001} \,||\, 2_{01\bar{1}}\right] \left[2_{001} \,||\, 2_{101}\right] \left[2_{001} \,||\, 2_{110}\right]$
C_5 : $\left[2_{001} \,||\, 4^{-}_{010}\right] \left[2_{001} \,||\, 4^{-}_{100}\right] \left[2_{001} \,||\, 4^{+}_{010}\right] \left[2_{001} \,||\, 4^{+}_{100}\right] \left[2_{001} \,||\, 4^{-}_{001}\right] \left[2_{001} \,||\, 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^{-}_{1\bar{1}1}\right]$	$\left[2_{001} \,||\, 4^{-}_{010}\right]$	$\left[2_{100}T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 579.1.1236 : ${}^{2_z}m^{2_z}\bar{3}^{m_y}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001} \,||\, m_{001}\right] \left[2_{001} \,||\, \bar{3}^{+}_{111}\right] \left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}1}\right] \left[E \,||\, 3^{-}_{111}\right] \left[2_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[E \,||\, 3^{+}_{111}\right] \left[2_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \,||\, \bar{3}^{-}_{111}\right]$
$\left[E \,||\, 3^{+}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{11\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{010}\right] \left[E \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{010}T \,||\, m_{110}\right] \left[2_{100}T \,||\, 2_{1\bar{1}0}\right] \left[2_{100}T \,||\, 4^{-}_{010}\right]$
$\left[2_{100}T \,||\, 4^{-}_{100}\right] \left[2_{010}T \,||\, m_{101}\right] \left[2_{010}T \,||\, m_{01\bar{1}}\right] \left[2_{010}T \,||\, m_{10\bar{1}}\right] \left[2_{010}T \,||\, m_{011}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{100}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{010}\right] \left[2_{100}T \,||\, 2_{011}\right] \left[2_{100}T \,||\, 2_{10\bar{1}}\right]$
$\left[2_{100}T \,||\, 2_{01\bar{1}}\right] \left[2_{100}T \,||\, 2_{101}\right] \left[2_{100}T \,||\, 2_{110}\right] \left[2_{010}T \,||\, m_{1\bar{1}0}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{001}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{001}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{010}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{100}\right] \left[2_{100}T \,||\, 4^{+}_{010}\right]$
$\left[2_{100}T \,||\, 4^{+}_{100}\right] \left[2_{100}T \,||\, 4^{-}_{001}\right] \left[2_{100}T \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$

C_2 : $\left[2_{001} \,||\, m_{001}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{010}\right]$

C_3 : $\left[2_{001} \,||\, \bar{3}^{+}_{111}\right] \left[2_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{111}\right] \left[E \,||\, 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \,||\, 3^{-}_{111}\right] \left[E \,||\, 3^{+}_{1\bar{1}1}\right] \left[E \,||\, 3^{+}_{11\bar{1}}\right] \left[E \,||\, 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[2_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \,||\, \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \,||\, -1\right]$

C_8 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7^{co}	3	−1	0	0	0	0	3	−1
χ_8^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{001}\right]$	$\left[2_{001} \,||\, \bar{3}^{+}_{111}\right]$	$\left[E \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{010}T \,||\, m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 580.1.1237 : ${}^{m_y}m^{m_y}\bar{3}^{2_z}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{010}T \parallel m_{001}\right] \left[2_{010}T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{010}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{010}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{010}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{010}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{010}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 4^{-}_{010}\right]$
$\left[2_{100}T \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 2_{011}\right] \left[2_{100}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{100}T \parallel 2_{01\bar{1}}\right] \left[2_{100}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 4^{+}_{010}\right]$
$\left[2_{100}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{001} \parallel \bar{4}^{+}_{100}\right]$	$\left[2_{010}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 581.1.1238 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001}T \parallel m_{101}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel \bar{4}^{+}_{100}\right] \left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_5 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{001} \parallel 4^{-}_{010}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 581.2.1239 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{m_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100}T \parallel m_{101}\right] \left[2_{100}T \parallel m_{01\bar{1}}\right] \left[2_{100}T \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel m_{011}\right] \left[2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{101}\right] \left[\Sigma_z 2_{100}T \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{011}\right] \left[\Sigma_z 2$
$\left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \right.$
$\left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel m_{101}\right] \left[\bar{E}2_{100}T \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}2_{100}T \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel m_{011}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right] \left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}\right.$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_4 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right]$
$\left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

C_5 : $\left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \,||\, 3^-_{111}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	−2
χ_3^{co}	2	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	−2
χ_4^{co}	6	0	−2	0	0	0	−6	0	2

Coirrepresntations of group generators

	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^-_{1\bar{1}1}\right]$	$\left[2_{100} \,||\, 4^-_{010}\right]$	$\left[T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	$\omega_2\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	ω_1^*	$\omega_2^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a\bar{f}d\bar{e})$	$(b\bar{c}\bar{a}f\bar{d}\bar{e})$	$(e\bar{f}\bar{d}c\bar{a}\bar{b})\mathcal{K}$	$(abcdef)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 582.1.1240 : ${}^{m}m^{m}\bar{3}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 4^{-}_{010}\right]$
$\left[2_{001} \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right]$
$\left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right]$
$\left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{001} \parallel 2_{1\bar{1}0}\right] \left[2_{001} \parallel 2_{011}\right] \left[2_{001} \parallel 2_{10\bar{1}}\right] \left[2_{001} \parallel 2_{01\bar{1}}\right] \left[2_{001} \parallel 2_{101}\right] \left[2_{001} \parallel 2_{110}\right]$
C_5 : $\left[2_{001} \parallel 4^{-}_{010}\right] \left[2_{001} \parallel 4^{-}_{100}\right] \left[2_{001} \parallel 4^{+}_{010}\right] \left[2_{001} \parallel 4^{+}_{100}\right] \left[2_{001} \parallel 4^{-}_{001}\right] \left[2_{001} \parallel 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{001} \parallel 4^{-}_{010}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 582.2.1241 : ${}^{m_z}m^{m_z}\bar{3}^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{m_z}m^{m_z}\bar{3}^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right]$
$\left[2_{100} \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{100} \parallel 4^{+}_{010}\right]$
$\left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right] \left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right] \left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{101}\right] \left[\bar{\Sigma}_z T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{011}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_4 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{100} \parallel 4^{-}_{010}\right] \left[2_{100} \parallel 4^{-}_{100}\right] \left[2_{100} \parallel 2_{011}\right] \left[2_{100} \parallel 2_{10\bar{1}}\right] \left[2_{100} \parallel 2_{01\bar{1}}\right] \left[2_{100} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
$\left[2_{100} \parallel 4^{+}_{010}\right] \left[2_{100} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel 2_{011}\right] \left[\Sigma_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel 2_{101}\right] \left[\Sigma_z 2_{100} \parallel 2_{110}\right]$
$\left[\Sigma_z 2_{100} \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel 4^{+}_{001}\right] \left[\bar{E}2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel 2_{011}\right]$
$\left[\bar{E}2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel 2_{101}\right] \left[\bar{E}2_{100} \parallel 2_{110}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100} \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100} \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{011}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel 2_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel 4^{+}_{001}\right]$

C_5 : $\left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{\bar{1}\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{\bar{1}\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \,||\, 3^-_{\bar{1}11}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{\bar{1}\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	4	−2	4	0	0	0	−4	2	−4
χ_3^{co}	6	0	−2	0	0	0	−6	0	2

Coirrepresntations of group generators

	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^-_{1\bar{1}1}\right]$	$\left[2_{100} \,||\, 4^-_{010}\right]$	$\left[2_{100}T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2 & 0 \\ 0 & \omega_2^* \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}a\bar{f}d\bar{e})$	$(b\bar{c}\bar{a}f\bar{d}\bar{e})$	$(e\bar{f}\bar{d}c\bar{a}\bar{b})\mathcal{K}$	$(abcdef)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 583.1.1242 : ${}^{2}m^{2}\bar{3}^{m}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[2_{001}T \parallel m_{101}\right] \left[2_{001}T \parallel m_{01\bar{1}}\right] \left[2_{001}T \parallel m_{10\bar{1}}\right] \left[2_{001}T \parallel m_{011}\right] \left[2_{001}T \parallel \bar{4}^{+}_{100}\right] \left[2_{001}T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[2_{001}T \parallel m_{1\bar{1}0}\right] \left[2_{001}T \parallel \bar{4}^{-}_{001}\right] \left[2_{001}T \parallel \bar{4}^{+}_{001}\right] \left[2_{001}T \parallel \bar{4}^{-}_{010}\right] \left[2_{001}T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$

C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7^{co}	3	−1	0	0	0	0	3	−1
χ_8^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{001}T \parallel m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 583.2.1243 : ${}^{2_z}m^{2_z}\bar{3}^{m_z}m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}\bar{3}^{m_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[2_{100}T \parallel m_{101}\right] \left[2_{100}T \parallel m_{01\bar{1}}\right] \left[2_{100}T \parallel m_{10\bar{1}}\right] \left[2_{100}T \parallel m_{011}\right] \left[2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{101}\right] \left[\Sigma_z 2_{100}T \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{011}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}\right.$
$\left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100}T \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E} 2_{100}T \parallel m_{101}\right] \left[\bar{E} 2_{100}T \parallel m_{01\bar{1}}\right]$
$\left[\bar{E} 2_{100}T \parallel m_{10\bar{1}}\right] \left[\bar{E} 2_{100}T \parallel m_{011}\right] \left[\bar{E} 2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E} 2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E}T \parallel 2_{110}\right]$
$\left[\bar{E} 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{E} 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E} 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E} 2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E} 2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{011}\right] \left[\bar{\Sigma}_z T \parallel 2\right.$
$\left[\bar{\Sigma}_z T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{101}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right]$
$\left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel m_{001}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel -1\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E} 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} 2_{100} \parallel m_{100}\right] \left[\bar{E} 2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$

$C_3: \quad \left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}1}\right] \left[E \,||\, 3^+_{111}\right] \left[E \,||\, 3^-_{11\bar{1}}\right]$

$C_4: \quad \left[E \,||\, 3^-_{111}\right] \left[E \,||\, 3^+_{1\bar{1}1}\right] \left[E \,||\, 3^+_{11\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

$C_5: \quad \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

$C_6: \quad \left[\bar{E} \,||\, 1\right]$

$C_7: \quad \left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

$C_8: \quad \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

$C_9: \quad \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	2	-2	-2	-2	-2
χ_2^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2
χ_3^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2
χ_4^{co}	6	0	0	0	-2	-6	0	0	2

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, \bar{3}^+_{111}\right]$	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \,||\, m_{110}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$\omega_6\mathcal{K}$	ω_5	σ_0	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\sigma_x\mathcal{K}$	$\omega_6^*\mathcal{K}$	ω_5^*	σ_0	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(defabc)\mathcal{K}$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(abcdef)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 584.1.1244 : ${}^{m}m^{m}\bar{3}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[2_{001}T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{001} \parallel \bar{4}^{+}_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 584.2.1245 : ${}^{m_z}m^{m_z}\bar{3}^{2_z}m^{\infty_x}1 \rightarrow {}^{m_z}m^{m_z}\bar{3}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100}T \parallel m_{001}\right] \left[2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100}T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{100}T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[T \parallel 2_{1\bar{1}0}\right] \left[T \parallel 4^{-}_{010}\right]$
$\left[T \parallel 4^{-}_{100}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[T \parallel 2_{011}\right] \left[T \parallel 2_{10\bar{1}}\right]$
$\left[T \parallel 2_{01\bar{1}}\right] \left[T \parallel 2_{101}\right] \left[T \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[T \parallel 4^{+}_{010}\right]$
$\left[T \parallel 4^{+}_{100}\right] \left[T \parallel 4^{-}_{001}\right] \left[T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100}T \parallel m_{001}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel m_{100}\right] \left[\Sigma_z 2_{100}T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z T \parallel 4^{-}_{010}\right] \left[\Sigma_z T \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z T \parallel 2_{011}\right] \left[\Sigma_z T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z T \parallel 2_{101}\right] \left[\Sigma_z T \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z T \parallel 4^{+}_{010}\right] \left[\Sigma_z T \parallel 4^{+}_{100}\right] \left[\Sigma_z T \parallel 4^{-}_{001}\right] \left[\Sigma_z T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100}T \parallel m_{001}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}2_{100}T \parallel m_{100}\right] \left[\bar{E}2_{100}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}T \parallel 4^{-}_{010}\right] \left[\bar{E}T \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}T \parallel 2_{011}\right] \left[\bar{E}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}T \parallel 2_{101}\right] \left[\bar{E}T \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}T \parallel 4^{+}_{010}\right] \left[\bar{E}T \parallel 4^{+}_{100}\right] \left[\bar{E}T \parallel 4^{-}_{001}\right] \left[\bar{E}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z T \parallel 2_{011}\right] \left[\bar{\Sigma}_z T \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel 2_{101}\right] \left[\bar{\Sigma}_z T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_4 : $\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right]$

C_5 : $\left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{\bar{1}\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{\bar{1}\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \,||\, 3^-_{\bar{1}11}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{\bar{1}\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{\bar{1}\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{\bar{1}11}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	4	−2	4	0	0	0	−4	2	−4
χ_3^{co}	6	0	−2	0	0	0	−6	0	2

Coirrepresntations of group generators

	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^-_{1\bar{1}1}\right]$	$\left[2_{100} \,||\, \bar{4}^+_{100}\right]$	$\left[2_{100}T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	σ_0	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_1 & 0 \\ 0 & \omega_1^* \end{pmatrix}$	$\begin{pmatrix} \omega_2^* & 0 \\ 0 & \omega_2 \end{pmatrix}\mathcal{K}$	$\begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix}$	$\begin{pmatrix} \sigma_z & 0 \\ 0 & \sigma_z \end{pmatrix}\mathcal{K}$
Γ_3^{co}	$(\bar{b}\bar{c}af\bar{d}\bar{e})$	$(b\bar{c}\bar{a}\bar{f}d\bar{e})$	$(fde\bar{b}c\bar{a})\mathcal{K}$	$(abcdef)$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 585.1.1246 : ${}^{2}m^{2}\bar{3}^{\bar{1}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 4^{-}_{010}\right]$
$\left[2_{001}T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{101}\right] \left[2_{001}T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right]$
$\left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$

C_3 : $\left[2_{001} \parallel \bar{3}^{+}_{111}\right] \left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[2_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel \bar{3}^{-}_{111}\right]$

C_7 : $\left[2_{001} \parallel -1\right]$

C_8 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7^{co}	3	−1	0	0	0	0	3	−1
χ_8^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[2_{001} \parallel \bar{3}^{+}_{111}\right]$	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[T \parallel m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 585.2.1247 : ${}^{2_z}m^{2_z}\bar{3}^{\bar{1}}m^{\infty_x}1 \rightarrow {}^{2_z}m^{2_z}\bar{3}^{\bar{1}}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{100} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[T \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 4^{-}_{010}\right]$
$\left[2_{100}T \parallel 4^{-}_{100}\right] \left[T \parallel m_{101}\right] \left[T \parallel m_{01\bar{1}}\right] \left[T \parallel m_{10\bar{1}}\right] \left[T \parallel m_{011}\right] \left[T \parallel \bar{4}^{+}_{100}\right] \left[T \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 2_{011}\right] \left[2_{100}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{100}T \parallel 2_{01\bar{1}}\right] \left[2_{100}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[T \parallel m_{1\bar{1}0}\right] \left[T \parallel \bar{4}^{-}_{001}\right] \left[T \parallel \bar{4}^{+}_{001}\right] \left[T \parallel \bar{4}^{-}_{010}\right] \left[T \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 4^{+}_{010}\right]$
$\left[2_{100}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z T \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{100}\right] \left[\Sigma_z T \parallel m_{101}\right] \left[\Sigma_z T \parallel m_{01\bar{1}}\right] \left[\Sigma_z T \parallel m_{10\bar{1}}\right] \left[\Sigma_z T \parallel m_{011}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{100}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{011}\right] \left[\Sigma_z 2_{100}T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 2_{101}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z T \parallel m_{1\bar{1}0}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z T \parallel \bar{4}^{+}_{001}\right]$
$\left[\Sigma_z T \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z T \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{100}\right] \left[\bar{E}T \parallel m_{101}\right] \left[\bar{E}T \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{10\bar{1}}\right] \left[\bar{E}T \parallel m_{011}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{011}\right] \left[\bar{E}2_{100}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 2_{101}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right]$
$\left[\bar{E}T \parallel m_{1\bar{1}0}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}T \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{101}\right] \left[\bar{\Sigma}_z T \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{011}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z T \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z T \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[2_{100} \parallel m_{001}\right] \left[2_{100} \parallel \bar{3}^{+}_{111}\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel -1\right] \left[2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[2_{100} \parallel \bar{3}^{-}_{111}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z 2_{100} \parallel m_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right]$
$\left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right]$
$\left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{001}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel -1\right] \left[\bar{E}2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{E}2_{100} \parallel m_{100}\right] \left[\bar{E}2_{100} \parallel m_{010}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{11\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{1\bar{1}1}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{3}^{-}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right]$

$C_3: \quad \left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}1}\right] \left[E \,||\, 3^+_{111}\right] \left[E \,||\, 3^-_{11\bar{1}}\right]$

$C_4: \quad \left[E \,||\, 3^-_{111}\right] \left[E \,||\, 3^+_{1\bar{1}1}\right] \left[E \,||\, 3^+_{11\bar{1}}\right] \left[E \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

$C_5: \quad \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

$C_6: \quad \left[\bar{E} \,||\, 1\right]$

$C_7: \quad \left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

$C_8: \quad \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

$C_9: \quad \left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	0	2	2	2	-2	-2	-2	-2
χ_2^{co}	2	0	$2e^{\frac{2\pi i}{3}}$	$2e^{\frac{-2\pi i}{3}}$	2	-2	$2e^{\frac{-\pi i}{3}}$	$2e^{\frac{\pi i}{3}}$	-2
χ_3^{co}	2	0	$2e^{\frac{-2\pi i}{3}}$	$2e^{\frac{2\pi i}{3}}$	2	-2	$2e^{\frac{\pi i}{3}}$	$2e^{\frac{-\pi i}{3}}$	-2
χ_4^{co}	6	0	0	0	-2	-6	0	0	2

Coirrepresntations of group generators

	$\left[2_{100} \,||\, m_{001}\right]$	$\left[2_{100} \,||\, \bar{3}^+_{111}\right]$	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[T \,||\, m_{110}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	$\sigma_x\mathcal{K}$	$\sigma_x\mathcal{K}$	σ_0	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	$\sigma_x\mathcal{K}$	$\omega_6\mathcal{K}$	ω_5	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	$\sigma_x\mathcal{K}$	$\omega_6^*\mathcal{K}$	ω_5^*	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(defabc)\mathcal{K}$	$(e\bar{f}\bar{d}\bar{b}\bar{c}a)\mathcal{K}$	$(\bar{b}\bar{c}ae\bar{f}\bar{d})$	$(abcdef)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 586.1.1248 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{2}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{001} \parallel m_{110}\right] \left[2_{001}T \parallel 2_{1\bar{1}0}\right] \left[2_{001}T \parallel 4^{-}_{010}\right]$
$\left[2_{001}T \parallel 4^{-}_{100}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001}T \parallel 2_{011}\right] \left[2_{001}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{001}T \parallel 2_{01\bar{1}}\right] \left[2_{001}T \parallel 2_{101}\right] \left[2_{001}T \parallel 2_{110}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right] \left[2_{001}T \parallel 4^{+}_{010}\right]$
$\left[2_{001}T \parallel 4^{+}_{100}\right] \left[2_{001}T \parallel 4^{-}_{001}\right] \left[2_{001}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{001} \parallel m_{110}\right] \left[2_{001} \parallel m_{101}\right] \left[2_{001} \parallel m_{01\bar{1}}\right] \left[2_{001} \parallel m_{10\bar{1}}\right] \left[2_{001} \parallel m_{011}\right] \left[2_{001} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{001} \parallel \bar{4}^{+}_{100}\right] \left[2_{001} \parallel \bar{4}^{+}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{001}\right] \left[2_{001} \parallel \bar{4}^{+}_{001}\right] \left[2_{001} \parallel \bar{4}^{-}_{010}\right] \left[2_{001} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[E \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{001} \parallel \bar{4}^{+}_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 586.2.1249 : ${}^{\bar{1}}m^{\bar{1}}\bar{3}^{2_z}m^{\infty_x}1 \rightarrow {}^{\bar{1}}m^{\bar{1}}\bar{3}^{2_z}m^{\Sigma_x}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[T \parallel \bar{3}^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[E \parallel 3^{+}_{111}\right] \left[T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[T \parallel \bar{3}^{-}_{111}\right]$
$\left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{100}T \parallel 4^{-}_{010}\right]$
$\left[2_{100}T \parallel 4^{-}_{100}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100}T \parallel 2_{011}\right] \left[2_{100}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{100}T \parallel 2_{01\bar{1}}\right] \left[2_{100}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[2_{100}T \parallel 4^{+}_{010}\right]$
$\left[2_{100}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z T \parallel m_{001}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{-}_{111}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel -1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right] \left[\Sigma_z \parallel 3^{+}_{111}\right] \left[\Sigma_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[\Sigma_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\Sigma_z T \parallel \bar{3}^{-}_{111}\right] \left[\Sigma_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\Sigma_z \parallel 3^{+}_{11\bar{1}}\right] \left[\Sigma_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\Sigma_z T \parallel m_{100}\right] \left[\Sigma_z T \parallel m_{010}\right] \left[\Sigma_z \parallel 3^{-}_{11\bar{1}}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{100}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 2_{011}\right] \left[\Sigma_z 2_{100}T \parallel 2_{10\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 2_{01\bar{1}}\right] \left[\Sigma_z 2_{100}T \parallel 2_{101}\right] \left[\Sigma_z 2_{100}T \parallel 2_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100}\right.$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{010}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{100}\right] \left[\Sigma_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\Sigma_z 2_{100}T \parallel 4^{+}_{001}\right] \left[\bar{E} \parallel 1\right] \left[\bar{E}T \parallel m_{001}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{111}\right]$
$\left[\bar{E} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{-}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel -1\right] \left[\bar{E} \parallel 2_{001}\right] \left[\bar{E} \parallel 2_{100}\right]$
$\left[\bar{E} \parallel 2_{010}\right] \left[\bar{E} \parallel 3^{+}_{111}\right] \left[\bar{E}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{E}T \parallel \bar{3}^{-}_{111}\right] \left[\bar{E} \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{E} \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{E} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{E}T \parallel m_{100}\right] \left[\bar{E}T \parallel m_{010}\right] \left[\bar{E} \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{010}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 2_{011}\right] \left[\bar{E}2_{100}T \parallel 2_{10\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 2_{01\bar{1}}\right] \left[\bar{E}2_{100}T \parallel 2_{101}\right] \left[\bar{E}2_{100}T \parallel 2_{110}\right]$
$\left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{010}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{100}\right] \left[\bar{E}2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{E}2_{100}T \parallel 4^{+}_{001}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z T \parallel m_{001}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{111}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[\bar{\Sigma}_z T \parallel -1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{111}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel \bar{3}^{-}_{111}\right]$
$\left[\bar{\Sigma}_z \parallel 3^{+}_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \parallel 3^{+}_{11\bar{1}}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z T \parallel m_{100}\right] \left[\bar{\Sigma}_z T \parallel m_{010}\right] \left[\bar{\Sigma}_z \parallel 3^{-}_{11\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{010}\right]$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{011}\right] \left[\bar{\Sigma}_z 2_{100}T\right.$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 2_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{101}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 2_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T\right.$
$\left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$

C_2 : $\left[E \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}1}\right] \left[E \parallel 3^{-}_{111}\right] \left[E \parallel 3^{+}_{111}\right] \left[E \parallel 3^{+}_{1\bar{1}1}\right] \left[E \parallel 3^{+}_{11\bar{1}}\right] \left[E \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel 3^{-}_{11\bar{1}}\right]$

C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$

C_4 : $\left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel m_{101}\right] \left[2_{100} \parallel m_{01\bar{1}}\right] \left[2_{100} \parallel m_{10\bar{1}}\right] \left[2_{100} \parallel m_{011}\right] \left[2_{100} \parallel \bar{4}^{+}_{100}\right] \left[2_{100} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{100} \parallel \bar{4}^{-}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\Sigma_z \parallel 1\right] \left[\Sigma_z \parallel 2_{001}\right] \left[\Sigma_z \parallel 2_{100}\right] \left[\Sigma_z \parallel 2_{010}\right]$
$\left[\Sigma_z 2_{100} \parallel m_{110}\right] \left[\Sigma_z 2_{100} \parallel m_{101}\right] \left[\Sigma_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{10\bar{1}}\right] \left[\Sigma_z 2_{100} \parallel m_{011}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\Sigma_z 2_{100} \parallel m_{1\bar{1}0}\right]$
$\left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\Sigma_z 2_{100} \parallel \bar{4}^{-}_{100}\right] \left[\bar{E}2_{100} \parallel m_{110}\right] \left[\bar{E}2_{100} \parallel m_{101}\right] \left[\bar{E}2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{E}2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{E}2_{100} \parallel m_{011}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{E}2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{E}2_{100} \parallel \bar{4}^{-}_{100}\right]$
$\left[\bar{\Sigma}_z \parallel 1\right] \left[\bar{\Sigma}_z \parallel 2_{001}\right] \left[\bar{\Sigma}_z \parallel 2_{100}\right] \left[\bar{\Sigma}_z \parallel 2_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{110}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{101}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{01\bar{1}}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{10\bar{1}}\right]$
$\left[\bar{\Sigma}_z 2_{100} \parallel m_{011}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{100}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel m_{1\bar{1}0}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{+}_{001}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{010}\right] \left[\bar{\Sigma}_z 2_{100} \parallel \bar{4}^{-}_{100}\right]$

C_5 : $\left[\Sigma_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{111}\right] \left[\Sigma_z \,||\, 3^-_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{11\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[\Sigma_z \,||\, 3^-_{111}\right] \left[\Sigma_z \,||\, 3^+_{1\bar{1}1}\right] \left[\Sigma_z \,||\, 3^+_{11\bar{1}}\right] \left[\Sigma_z \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{\Sigma}_z \,||\, 3^+_{111}\right] \left[\bar{\Sigma}_z \,||\, 3^-_{11\bar{1}}\right]$

C_7 : $\left[\bar{E} \,||\, 1\right]$

C_8 : $\left[\bar{E} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^-_{111}\right] \left[\bar{E} \,||\, 3^+_{111}\right] \left[\bar{E} \,||\, 3^+_{1\bar{1}1}\right] \left[\bar{E} \,||\, 3^+_{11\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[\bar{E} \,||\, 3^-_{11\bar{1}}\right]$

C_9 : $\left[\bar{E} \,||\, 2_{001}\right] \left[\bar{E} \,||\, 2_{100}\right] \left[\bar{E} \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9
χ_1^{co}	2	2	2	0	0	0	−2	−2	−2
χ_2^{co}	2	−1	2	0	$\sqrt{3}i$	$-\sqrt{3}i$	−2	1	−2
χ_3^{co}	2	−1	2	0	$-\sqrt{3}i$	$\sqrt{3}i$	−2	1	−2
χ_4^{co}	6	0	−2	0	0	0	−6	0	2

Coirrepresntations of group generators

	$\left[E \,||\, 3^+_{1\bar{1}\bar{1}}\right]$	$\left[E \,||\, 3^-_{1\bar{1}1}\right]$	$\left[2_{100} \,||\, \bar{4}^+_{100}\right]$	$\left[T \,||\, m_{001}\right]$	$\left[\Sigma_z \,||\, 1\right]$
Γ_1^{co}	σ_0	σ_0	$\sigma_x\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_2^{co}	ω_1	ω_1	$\omega_2^*\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_3^{co}	ω_1^*	ω_1^*	$\omega_2\mathcal{K}$	$\sigma_0\mathcal{K}$	$\sigma_z\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}af\bar{d}\bar{e})$	$(b\bar{c}\bar{a}\bar{f}d\bar{e})$	$(fde\bar{b}c\bar{a})\mathcal{K}$	$(abcdef)\mathcal{K}$	$(abc\bar{d}\bar{e}\bar{f})\mathcal{K}$

No 587.1.1250 : ${}^{1}m^{3_z}\bar{3}^{2_x}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, m_{001}\right] \left[3^{+}_{001} \,||\, \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{-}_{111}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[3^{+}_{001} \,||\, 3^{+}_{111}\right] \left[3^{+}_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right] \left[3^{+}_{001} \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{100} \,||\, 2_{1\bar{1}0}\right] \left[2_{110} \,||\, 4^{-}_{010}\right]$
$\left[2_{010} \,||\, 4^{-}_{100}\right] \left[2_{110} \,||\, m_{101}\right] \left[2_{010} \,||\, m_{01\bar{1}}\right] \left[2_{110} \,||\, m_{10\bar{1}}\right] \left[2_{010} \,||\, m_{011}\right] \left[2_{010} \,||\, \bar{4}^{+}_{100}\right] \left[2_{110} \,||\, \bar{4}^{+}_{010}\right] \left[2_{010} \,||\, 2_{011}\right] \left[2_{110} \,||\, 2_{10\bar{1}}\right]$
$\left[2_{010} \,||\, 2_{01\bar{1}}\right] \left[2_{110} \,||\, 2_{101}\right] \left[2_{100} \,||\, 2_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[2_{110} \,||\, \bar{4}^{-}_{010}\right] \left[2_{010} \,||\, \bar{4}^{-}_{100}\right] \left[2_{110} \,||\, 4^{+}_{010}\right]$
$\left[2_{010} \,||\, 4^{+}_{100}\right] \left[2_{100} \,||\, 4^{-}_{001}\right] \left[2_{100} \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[E \,||\, m_{001}\right] \left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right]$
C_3 : $\left[3^{+}_{001} \,||\, \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{111}\right]$
C_4 : $\left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{-}_{111}\right] \left[3^{+}_{001} \,||\, 3^{+}_{111}\right] \left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \,||\, -1\right]$
C_6 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_7 : $\left[2_{100} \,||\, m_{110}\right] \left[2_{110} \,||\, m_{101}\right] \left[2_{010} \,||\, m_{01\bar{1}}\right] \left[2_{110} \,||\, m_{10\bar{1}}\right] \left[2_{010} \,||\, m_{011}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right]$
C_8 : $\left[2_{100} \,||\, 2_{1\bar{1}0}\right] \left[2_{010} \,||\, 2_{011}\right] \left[2_{110} \,||\, 2_{10\bar{1}}\right] \left[2_{010} \,||\, 2_{01\bar{1}}\right] \left[2_{110} \,||\, 2_{101}\right] \left[2_{100} \,||\, 2_{110}\right]$
C_9 : $\left[2_{110} \,||\, 4^{-}_{010}\right] \left[2_{010} \,||\, 4^{-}_{100}\right] \left[2_{110} \,||\, 4^{+}_{010}\right] \left[2_{010} \,||\, 4^{+}_{100}\right] \left[2_{100} \,||\, 4^{-}_{001}\right] \left[2_{100} \,||\, 4^{+}_{001}\right]$
C_{10} : $\left[2_{010} \,||\, \bar{4}^{+}_{100}\right] \left[2_{110} \,||\, \bar{4}^{+}_{010}\right] \left[2_{100} \,||\, \bar{4}^{-}_{001}\right] \left[2_{100} \,||\, \bar{4}^{+}_{001}\right] \left[2_{110} \,||\, \bar{4}^{-}_{010}\right] \left[2_{010} \,||\, \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[E \mid\mid m_{001}\right]$	$\left[2_{110} \mid\mid 4^{-}_{010}\right]$	$\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 587.2.1251 : ${}^{1}m^{3_z}\bar{3}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[E \parallel m_{001}\right] \left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{110} \parallel 4^{-}_{010}\right]$
$\left[2_{010} \parallel 4^{-}_{100}\right] \left[2_{110} \parallel m_{101}\right] \left[2_{010} \parallel m_{01\bar{1}}\right] \left[2_{110} \parallel m_{10\bar{1}}\right] \left[2_{010} \parallel m_{011}\right] \left[2_{010} \parallel \bar{4}^{+}_{100}\right] \left[2_{110} \parallel \bar{4}^{+}_{010}\right] \left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right]$
$\left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{110} \parallel \bar{4}^{-}_{010}\right] \left[2_{010} \parallel \bar{4}^{-}_{100}\right] \left[2_{110} \parallel 4^{+}_{010}\right]$
$\left[2_{010} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[6^{-}_{001}T \parallel \bar{3}^{+}_{111}\right] \left[6^{-}_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[6^{-}_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[6^{+}_{001}T \parallel 3^{-}_{111}\right]$
$\left[6^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{+}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[6^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[6^{-}_{001}T \parallel 3^{+}_{111}\right] \left[6^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[6^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{-}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[6^{+}_{001}T \parallel \bar{3}^{-}_{111}\right] \left[6^{+}_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[6^{+}_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[6^{+}_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \parallel m_{100}\right] \left[2_{001}T \parallel m_{010}\right] \left[6^{-}_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{120}T \parallel m_{110}\right] \left[2_{120}T \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0}T \parallel 4^{-}_{010}\right] \left[2_{210}T \parallel 4^{-}_{100}\right] \left[2_{1\bar{1}0}T \parallel m_{101}\right] \left[2_{210}T \parallel m_{01\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel m_{10\bar{1}}\right] \left[2_{210}T \parallel m_{011}\right] \left[2_{210}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{1\bar{1}0}T \parallel \bar{4}^{+}_{010}\right] \left[2_{210}T \parallel 2_{011}\right] \left[2_{1\bar{1}0}T \parallel 2_{10\bar{1}}\right] \left[2_{210}T \parallel 2_{01\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel 2_{101}\right] \left[2_{120}T \parallel 2_{110}\right] \left[2_{120}T \parallel m_{1\bar{1}0}\right] \left[2_{120}T \parallel \bar{4}^{-}_{001}\right] \left[2_{120}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{1\bar{1}0}T \parallel \bar{4}^{-}_{010}\right] \left[2_{210}T \parallel \bar{4}^{-}_{100}\right] \left[2_{1\bar{1}0}T \parallel 4^{+}_{010}\right] \left[2_{210}T \parallel 4^{+}_{100}\right] \left[2_{120}T \parallel 4^{-}_{001}\right] \left[2_{120}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[E \parallel m_{001}\right] \left[E \parallel m_{100}\right] \left[E \parallel m_{010}\right]$
C_3 : $\left[3^{+}_{001} \parallel \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{100} \parallel m_{110}\right] \left[2_{110} \parallel m_{101}\right] \left[2_{010} \parallel m_{01\bar{1}}\right] \left[2_{110} \parallel m_{10\bar{1}}\right] \left[2_{010} \parallel m_{011}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right] \left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
C_9 : $\left[2_{110} \parallel 4^{-}_{010}\right] \left[2_{010} \parallel 4^{-}_{100}\right] \left[2_{110} \parallel 4^{+}_{010}\right] \left[2_{010} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{010} \parallel \bar{4}^{+}_{100}\right] \left[2_{110} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{110} \parallel \bar{4}^{-}_{010}\right] \left[2_{010} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[E \mid\mid m_{001}\right]$	$\left[2_{110} \mid\mid 4^{-}_{010}\right]$	$\left[3^{+}_{001} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 588.1.1252 : ${}^{1}m^{3_z}\bar{3}^{m_x}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[E \,||\, m_{001}\right] \left[3^{+}_{001} \,||\, \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{-}_{\bar{1}11}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[3^{+}_{001} \,||\, 3^{+}_{111}\right] \left[3^{+}_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{\bar{1}11}\right]$
$\left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right] \left[3^{+}_{001} \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{100}T \,||\, m_{110}\right] \left[2_{100}T \,||\, 2_{1\bar{1}0}\right] \left[2_{110}T \,||\, 4^{-}_{010}\right]$
$\left[2_{010}T \,||\, 4^{-}_{100}\right] \left[2_{110}T \,||\, m_{101}\right] \left[2_{010}T \,||\, m_{01\bar{1}}\right] \left[2_{110}T \,||\, m_{10\bar{1}}\right] \left[2_{010}T \,||\, m_{011}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{100}\right] \left[2_{110}T \,||\, \bar{4}^{+}_{010}\right] \left[2_{010}T \,||\, 2_{011}\right] \left[2_{110}T \,||\, 2_{10\bar{1}}\right]$
$\left[2_{010}T \,||\, 2_{01\bar{1}}\right] \left[2_{110}T \,||\, 2_{101}\right] \left[2_{100}T \,||\, 2_{110}\right] \left[2_{100}T \,||\, m_{1\bar{1}0}\right] \left[2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[2_{100}T \,||\, \bar{4}^{+}_{001}\right] \left[2_{110}T \,||\, \bar{4}^{-}_{010}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{100}\right] \left[2_{110}T \,||\, 4^{+}_{010}\right]$
$\left[2_{010}T \,||\, 4^{+}_{100}\right] \left[2_{100}T \,||\, 4^{-}_{001}\right] \left[2_{100}T \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$

C_2 : $\left[E \,||\, m_{001}\right] \left[E \,||\, m_{100}\right] \left[E \,||\, m_{010}\right]$

C_3 : $\left[3^{+}_{001} \,||\, \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{+}_{111}\right] \left[3^{+}_{001} \,||\, 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{-}_{001} \,||\, 3^{-}_{\bar{1}11}\right] \left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[3^{-}_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \,||\, \bar{3}^{-}_{\bar{1}11}\right]$

C_7 : $\left[E \,||\, -1\right]$

C_8 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7^{co}	3	−1	0	0	0	0	3	−1
χ_8^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[E \,||\, m_{001}\right]$	$\left[3^{+}_{001} \,||\, \bar{3}^{+}_{111}\right]$	$\left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \,||\, m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 589.1.1253 : ${}^{2_z}m^{6_z}\bar{3}^{2_x}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001} \,||\, m_{001}\right] \left[6^+_{001} \,||\, \bar{3}^+_{111}\right] \left[3^-_{001} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{001} \,||\, 3^-_{1\bar{1}1}\right] \left[3^+_{001} \,||\, 3^-_{111}\right] \left[6^-_{001} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[6^-_{001} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[6^-_{001} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[3^-_{001} \,||\, 3^+_{111}\right] \left[6^+_{001} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[6^+_{001} \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[6^+_{001} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[6^-_{001} \,||\, \bar{3}^-_{111}\right]$
$\left[3^+_{001} \,||\, 3^+_{1\bar{1}1}\right] \left[3^+_{001} \,||\, 3^+_{11\bar{1}}\right] \left[3^+_{001} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{010}\right] \left[3^-_{001} \,||\, 3^-_{11\bar{1}}\right] \left[2_{100} \,||\, m_{110}\right] \left[2_{120} \,||\, 2_{1\bar{1}0}\right] \left[2_{210} \,||\, 4^-_{010}\right]$
$\left[2_{1\bar{1}0} \,||\, 4^-_{100}\right] \left[2_{010} \,||\, m_{101}\right] \left[2_{110} \,||\, m_{01\bar{1}}\right] \left[2_{010} \,||\, m_{10\bar{1}}\right] \left[2_{110} \,||\, m_{011}\right] \left[2_{110} \,||\, \bar{4}^+_{100}\right] \left[2_{010} \,||\, \bar{4}^+_{010}\right] \left[2_{1\bar{1}0} \,||\, 2_{011}\right] \left[2_{210} \,||\, 2_{10\bar{1}}\right]$
$\left[2_{1\bar{1}0} \,||\, 2_{01\bar{1}}\right] \left[2_{210} \,||\, 2_{101}\right] \left[2_{120} \,||\, 2_{110}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right] \left[2_{100} \,||\, \bar{4}^-_{001}\right] \left[2_{100} \,||\, \bar{4}^+_{001}\right] \left[2_{010} \,||\, \bar{4}^-_{010}\right] \left[2_{110} \,||\, \bar{4}^-_{100}\right] \left[2_{210} \,||\, 4^+_{010}\right]$
$\left[2_{1\bar{1}0} \,||\, 4^+_{100}\right] \left[2_{120} \,||\, 4^-_{001}\right] \left[2_{120} \,||\, 4^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[2_{001} \,||\, m_{001}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{010}\right]$
C_3 : $\left[6^+_{001} \,||\, \bar{3}^+_{111}\right] \left[6^-_{001} \,||\, \bar{3}^+_{1\bar{1}1}\right] \left[6^-_{001} \,||\, \bar{3}^+_{11\bar{1}}\right] \left[6^-_{001} \,||\, \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[6^+_{001} \,||\, \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[6^+_{001} \,||\, \bar{3}^-_{1\bar{1}1}\right] \left[6^+_{001} \,||\, \bar{3}^-_{11\bar{1}}\right] \left[6^-_{001} \,||\, \bar{3}^-_{111}\right]$
C_4 : $\left[3^-_{001} \,||\, 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{001} \,||\, 3^-_{1\bar{1}1}\right] \left[3^+_{001} \,||\, 3^-_{111}\right] \left[3^-_{001} \,||\, 3^+_{111}\right] \left[3^+_{001} \,||\, 3^+_{1\bar{1}1}\right] \left[3^+_{001} \,||\, 3^+_{11\bar{1}}\right] \left[3^+_{001} \,||\, 3^-_{1\bar{1}\bar{1}}\right] \left[3^-_{001} \,||\, 3^-_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \,||\, -1\right]$
C_6 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_7 : $\left[2_{100} \,||\, m_{110}\right] \left[2_{010} \,||\, m_{101}\right] \left[2_{110} \,||\, m_{01\bar{1}}\right] \left[2_{010} \,||\, m_{10\bar{1}}\right] \left[2_{110} \,||\, m_{011}\right] \left[2_{100} \,||\, m_{1\bar{1}0}\right]$
C_8 : $\left[2_{120} \,||\, 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \,||\, 2_{011}\right] \left[2_{210} \,||\, 2_{10\bar{1}}\right] \left[2_{1\bar{1}0} \,||\, 2_{01\bar{1}}\right] \left[2_{210} \,||\, 2_{101}\right] \left[2_{120} \,||\, 2_{110}\right]$
C_9 : $\left[2_{210} \,||\, 4^-_{010}\right] \left[2_{1\bar{1}0} \,||\, 4^-_{100}\right] \left[2_{210} \,||\, 4^+_{010}\right] \left[2_{1\bar{1}0} \,||\, 4^+_{100}\right] \left[2_{120} \,||\, 4^-_{001}\right] \left[2_{120} \,||\, 4^+_{001}\right]$
C_{10} : $\left[2_{110} \,||\, \bar{4}^+_{100}\right] \left[2_{010} \,||\, \bar{4}^+_{010}\right] \left[2_{100} \,||\, \bar{4}^-_{001}\right] \left[2_{100} \,||\, \bar{4}^+_{001}\right] \left[2_{010} \,||\, \bar{4}^-_{010}\right] \left[2_{110} \,||\, \bar{4}^-_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{210} \mid\mid 4^{-}_{010}\right]$	$\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 589.2.1254 : ${}^{2_z}m^{6_z}\bar{3}^{2_x}m^{m_z}1$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{-}_{111}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{-}_{001} \parallel 3^{+}_{111}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right] \left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{120} \parallel 2_{1\bar{1}0}\right] \left[2_{210} \parallel 4^{-}_{010}\right]$
$\left[2_{1\bar{1}0} \parallel 4^{-}_{100}\right] \left[2_{010} \parallel m_{101}\right] \left[2_{110} \parallel m_{01\bar{1}}\right] \left[2_{010} \parallel m_{10\bar{1}}\right] \left[2_{110} \parallel m_{011}\right] \left[2_{110} \parallel \bar{4}^{+}_{100}\right] \left[2_{010} \parallel \bar{4}^{+}_{010}\right] \left[2_{1\bar{1}0} \parallel 2_{011}\right] \left[2_{210} \parallel 2_{10\bar{1}}\right]$
$\left[2_{1\bar{1}0} \parallel 2_{01\bar{1}}\right] \left[2_{210} \parallel 2_{101}\right] \left[2_{120} \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{010} \parallel \bar{4}^{-}_{010}\right] \left[2_{110} \parallel \bar{4}^{-}_{100}\right] \left[2_{210} \parallel 4^{+}_{010}\right]$
$\left[2_{1\bar{1}0} \parallel 4^{+}_{100}\right] \left[2_{120} \parallel 4^{-}_{001}\right] \left[2_{120} \parallel 4^{+}_{001}\right] \left[2_{001}T \parallel 1\right] \left[T \parallel m_{001}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{111}\right] \left[6^{+}_{001}T \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001}T \parallel 3^{-}_{1\bar{1}1}\right] \left[6^{-}_{001}T \parallel 3^{-}_{111}\right]$
$\left[3^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel -1\right] \left[2_{001}T \parallel 2_{001}\right] \left[2_{001}T \parallel 2_{100}\right] \left[2_{001}T \parallel 2_{010}\right] \left[6^{+}_{001}T \parallel 3^{+}_{111}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$
$\left[3^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{111}\right] \left[6^{-}_{001}T \parallel 3^{+}_{1\bar{1}1}\right] \left[6^{-}_{001}T \parallel 3^{+}_{11\bar{1}}\right] \left[6^{-}_{001}T \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[6^{+}_{001}T \parallel 3^{-}_{11\bar{1}}\right]$
$\left[2_{120}T \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{010}T \parallel 4^{-}_{010}\right] \left[2_{110}T \parallel 4^{-}_{100}\right] \left[2_{210}T \parallel m_{101}\right] \left[2_{1\bar{1}0}T \parallel m_{01\bar{1}}\right] \left[2_{210}T \parallel m_{10\bar{1}}\right] \left[2_{1\bar{1}0}T \parallel m_{011}\right] \left[2_{1\bar{1}0}T \parallel \bar{4}^{+}_{100}\right]$
$\left[2_{210}T \parallel \bar{4}^{+}_{010}\right] \left[2_{110}T \parallel 2_{011}\right] \left[2_{010}T \parallel 2_{10\bar{1}}\right] \left[2_{110}T \parallel 2_{01\bar{1}}\right] \left[2_{010}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{120}T \parallel m_{1\bar{1}0}\right] \left[2_{120}T \parallel \bar{4}^{-}_{001}\right] \left[2_{120}T \parallel \bar{4}^{+}_{001}\right]$
$\left[2_{210}T \parallel \bar{4}^{-}_{010}\right] \left[2_{1\bar{1}0}T \parallel \bar{4}^{-}_{100}\right] \left[2_{010}T \parallel 4^{+}_{010}\right] \left[2_{110}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{001} \parallel m_{100}\right] \left[2_{001} \parallel m_{010}\right]$
C_3 : $\left[6^{+}_{001} \parallel \bar{3}^{+}_{111}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[6^{-}_{001} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[2_{001} \parallel -1\right]$
C_6 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_7 : $\left[2_{100} \parallel m_{110}\right] \left[2_{010} \parallel m_{101}\right] \left[2_{110} \parallel m_{01\bar{1}}\right] \left[2_{010} \parallel m_{10\bar{1}}\right] \left[2_{110} \parallel m_{011}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{120} \parallel 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \parallel 2_{011}\right] \left[2_{210} \parallel 2_{10\bar{1}}\right] \left[2_{1\bar{1}0} \parallel 2_{01\bar{1}}\right] \left[2_{210} \parallel 2_{101}\right] \left[2_{120} \parallel 2_{110}\right]$
C_9 : $\left[2_{210} \parallel 4^{-}_{010}\right] \left[2_{1\bar{1}0} \parallel 4^{-}_{100}\right] \left[2_{210} \parallel 4^{+}_{010}\right] \left[2_{1\bar{1}0} \parallel 4^{+}_{100}\right] \left[2_{120} \parallel 4^{-}_{001}\right] \left[2_{120} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[2_{110} \parallel \bar{4}^{+}_{100}\right] \left[2_{010} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{010} \parallel \bar{4}^{-}_{010}\right] \left[2_{110} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1^{co}	1	1	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3^{co}	1	1	1	1	1	1	−1	−1	−1	−1
χ_4^{co}	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5^{co}	2	2	−1	−1	2	2	0	0	0	0

χ_6^{co}	2	−2	1	−1	−2	2	0	0	0	0
χ_7^{co}	3	−1	0	0	3	−1	−1	−1	1	1
χ_8^{co}	3	1	0	0	−3	−1	1	−1	1	−1
χ_9^{co}	3	−1	0	0	3	−1	1	1	−1	−1
χ_{10}^{co}	3	1	0	0	−3	−1	−1	1	−1	1

Coirrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[2_{210} \mid\mid 4^{-}_{010}\right]$	$\left[6^{+}_{001} \mid\mid \bar{3}^{+}_{111}\right]$	$\left[2_{001}T \mid\mid 1\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	1	−1	$\mathcal{K}$
Γ_3^{co}	1	−1	1	$\mathcal{K}$
Γ_4^{co}	−1	−1	−1	$\mathcal{K}$
Γ_5^{co}	σ_0	ω_2^*	ω_1	$\sigma_0\mathcal{K}$
Γ_6^{co}	$-\sigma_0$	ω_2^*	ω_3^*	$\sigma_0\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$
Γ_9^{co}	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)	$(abc)\mathcal{K}$
$\Gamma_{10}^{\mathrm{co}}$	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$	$(abc)\mathcal{K}$

No 590.1.1255 : ${}^{\bar{1}}m^{\bar{3}_z}\bar{3}^{m_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100}T \parallel m_{110}\right] \left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{110} \parallel 4^{-}_{010}\right]$
$\left[2_{010} \parallel 4^{-}_{100}\right] \left[2_{110}T \parallel m_{101}\right] \left[2_{010}T \parallel m_{01\bar{1}}\right] \left[2_{110}T \parallel m_{10\bar{1}}\right] \left[2_{010}T \parallel m_{011}\right] \left[2_{010}T \parallel \bar{4}^{+}_{100}\right] \left[2_{110}T \parallel \bar{4}^{+}_{010}\right] \left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right]$
$\left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right] \left[2_{100}T \parallel m_{1\bar{1}0}\right] \left[2_{100}T \parallel \bar{4}^{-}_{001}\right] \left[2_{100}T \parallel \bar{4}^{+}_{001}\right] \left[2_{110}T \parallel \bar{4}^{-}_{010}\right] \left[2_{010}T \parallel \bar{4}^{-}_{100}\right] \left[2_{110} \parallel 4^{+}_{010}\right]$
$\left[2_{010} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel 2_{1\bar{1}0}\right] \left[2_{010} \parallel 2_{011}\right] \left[2_{110} \parallel 2_{10\bar{1}}\right] \left[2_{010} \parallel 2_{01\bar{1}}\right] \left[2_{110} \parallel 2_{101}\right] \left[2_{100} \parallel 2_{110}\right]$
C_5 : $\left[2_{110} \parallel 4^{-}_{010}\right] \left[2_{010} \parallel 4^{-}_{100}\right] \left[2_{110} \parallel 4^{+}_{010}\right] \left[2_{010} \parallel 4^{+}_{100}\right] \left[2_{100} \parallel 4^{-}_{001}\right] \left[2_{100} \parallel 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{110} \parallel 4^{-}_{010}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 591.1.1256 : ${}^{\bar{1}}m^{\bar{3}_z}\bar{3}^{2_x}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[T \parallel m_{001}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{-}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{+}_{001}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{+}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{001}T \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[T \parallel m_{100}\right] \left[T \parallel m_{010}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{100} \parallel m_{110}\right] \left[2_{100}T \parallel 2_{1\bar{1}0}\right] \left[2_{110}T \parallel 4^{-}_{010}\right]$
$\left[2_{010}T \parallel 4^{-}_{100}\right] \left[2_{110} \parallel m_{101}\right] \left[2_{010} \parallel m_{01\bar{1}}\right] \left[2_{110} \parallel m_{10\bar{1}}\right] \left[2_{010} \parallel m_{011}\right] \left[2_{010} \parallel \bar{4}^{+}_{100}\right] \left[2_{110} \parallel \bar{4}^{+}_{010}\right] \left[2_{010}T \parallel 2_{011}\right] \left[2_{110}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{010}T \parallel 2_{01\bar{1}}\right] \left[2_{110}T \parallel 2_{101}\right] \left[2_{100}T \parallel 2_{110}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{110} \parallel \bar{4}^{-}_{010}\right] \left[2_{010} \parallel \bar{4}^{-}_{100}\right] \left[2_{110}T \parallel 4^{+}_{010}\right]$
$\left[2_{010}T \parallel 4^{+}_{100}\right] \left[2_{100}T \parallel 4^{-}_{001}\right] \left[2_{100}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{-}_{111}\right] \left[3^{+}_{001} \parallel 3^{+}_{111}\right] \left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{-}_{001} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right] \left[E \parallel 2_{100}\right] \left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel m_{110}\right] \left[2_{110} \parallel m_{101}\right] \left[2_{010} \parallel m_{01\bar{1}}\right] \left[2_{110} \parallel m_{10\bar{1}}\right] \left[2_{010} \parallel m_{011}\right] \left[2_{100} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{010} \parallel \bar{4}^{+}_{100}\right] \left[2_{110} \parallel \bar{4}^{+}_{010}\right] \left[2_{100} \parallel \bar{4}^{-}_{001}\right] \left[2_{100} \parallel \bar{4}^{+}_{001}\right] \left[2_{110} \parallel \bar{4}^{-}_{010}\right] \left[2_{010} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{010} \parallel \bar{4}^{+}_{100}\right]$	$\left[T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 592.1.1257 : ${}^{2_z}m^{6_z}\bar{3}^{m_x}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001} \,||\, m_{001}\right] \left[6^{+}_{001} \,||\, \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{-}_{\bar{1}11}\right] \left[6^{-}_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001} \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[3^{-}_{001} \,||\, 3^{+}_{\bar{1}11}\right] \left[6^{+}_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[6^{-}_{001} \,||\, \bar{3}^{-}_{\bar{1}11}\right]$
$\left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{010}\right] \left[3^{-}_{001} \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{100}T \,||\, m_{110}\right] \left[2_{120}T \,||\, 2_{1\bar{1}0}\right] \left[2_{210}T \,||\, 4^{-}_{010}\right]$
$\left[2_{1\bar{1}0}T \,||\, 4^{-}_{100}\right] \left[2_{010}T \,||\, m_{101}\right] \left[2_{110}T \,||\, m_{01\bar{1}}\right] \left[2_{010}T \,||\, m_{10\bar{1}}\right] \left[2_{110}T \,||\, m_{011}\right] \left[2_{110}T \,||\, \bar{4}^{+}_{100}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{010}\right] \left[2_{1\bar{1}0}T \,||\, 2_{011}\right] \left[2_{210}T \,||\, 2_{10\bar{1}}\right]$
$\left[2_{1\bar{1}0}T \,||\, 2_{01\bar{1}}\right] \left[2_{210}T \,||\, 2_{101}\right] \left[2_{120}T \,||\, 2_{110}\right] \left[2_{100}T \,||\, m_{1\bar{1}0}\right] \left[2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[2_{100}T \,||\, \bar{4}^{+}_{001}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{010}\right] \left[2_{110}T \,||\, \bar{4}^{-}_{100}\right] \left[2_{210}T \,||\, 4^{+}_{010}\right]$
$\left[2_{1\bar{1}0}T \,||\, 4^{+}_{100}\right] \left[2_{120}T \,||\, 4^{-}_{001}\right] \left[2_{120}T \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$

C_2 : $\left[2_{001} \,||\, m_{001}\right] \left[2_{001} \,||\, m_{100}\right] \left[2_{001} \,||\, m_{010}\right]$

C_3 : $\left[6^{+}_{001} \,||\, \bar{3}^{+}_{111}\right] \left[6^{+}_{001} \,||\, \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[6^{+}_{001} \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001} \,||\, \bar{3}^{-}_{11\bar{1}}\right]$

C_4 : $\left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{001} \,||\, 3^{+}_{\bar{1}11}\right] \left[3^{-}_{001} \,||\, 3^{-}_{11\bar{1}}\right]$

C_5 : $\left[3^{+}_{001} \,||\, 3^{-}_{\bar{1}11}\right] \left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right]$

C_6 : $\left[6^{-}_{001} \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001} \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001} \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[6^{-}_{001} \,||\, \bar{3}^{-}_{\bar{1}11}\right]$

C_7 : $\left[2_{001} \,||\, -1\right]$

C_8 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	-1	-1	1	1	-1	-1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	-1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	-1	1
χ_7^{co}	3	-1	0	0	0	0	3	-1
χ_8^{co}	3	1	0	0	0	0	-3	-1

Coirrepresntations of group generators

	$\left[2_{001} \,||\, m_{001}\right]$	$\left[6^{+}_{001} \,||\, \bar{3}^{+}_{111}\right]$	$\left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[2_{100}T \,||\, m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	-1	-1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 593.1.1258 : ${}^{m_z}m^{\bar{6}_z}\bar{3}^{2_x}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{\bar{1}11}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{-}_{\bar{1}\bar{1}\bar{1}}\right]$
$\left[2_{001}T \parallel -1\right]$ $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{\bar{1}11}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[6^{+}_{001}T \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[6^{-}_{001}T \parallel \bar{3}^{-}_{\bar{1}11}\right]$
$\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{001}T \parallel m_{100}\right]$ $\left[2_{001}T \parallel m_{010}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$ $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{120}T \parallel 2_{1\bar{1}0}\right]$ $\left[2_{210}T \parallel 4^{-}_{010}\right]$
$\left[2_{1\bar{1}0}T \parallel 4^{-}_{100}\right]$ $\left[2_{010} \parallel m_{101}\right]$ $\left[2_{110} \parallel m_{01\bar{1}}\right]$ $\left[2_{010} \parallel m_{10\bar{1}}\right]$ $\left[2_{110} \parallel m_{011}\right]$ $\left[2_{110} \parallel \bar{4}^{+}_{100}\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{010}\right]$ $\left[2_{1\bar{1}0}T \parallel 2_{011}\right]$ $\left[2_{210}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{1\bar{1}0}T \parallel 2_{01\bar{1}}\right]$ $\left[2_{210}T \parallel 2_{101}\right]$ $\left[2_{120}T \parallel 2_{110}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{-}_{010}\right]$ $\left[2_{110} \parallel \bar{4}^{-}_{100}\right]$ $\left[2_{210}T \parallel 4^{+}_{010}\right]$
$\left[2_{1\bar{1}0}T \parallel 4^{+}_{100}\right]$ $\left[2_{120}T \parallel 4^{-}_{001}\right]$ $\left[2_{120}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{\bar{1}11}\right]$ $\left[3^{-}_{001} \parallel 3^{+}_{\bar{1}11}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{001} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{+}_{001} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{001} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \parallel 2_{001}\right]$ $\left[E \parallel 2_{100}\right]$ $\left[E \parallel 2_{010}\right]$
C_4 : $\left[2_{100} \parallel m_{110}\right]$ $\left[2_{010} \parallel m_{101}\right]$ $\left[2_{110} \parallel m_{01\bar{1}}\right]$ $\left[2_{010} \parallel m_{10\bar{1}}\right]$ $\left[2_{110} \parallel m_{011}\right]$ $\left[2_{100} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[2_{110} \parallel \bar{4}^{+}_{100}\right]$ $\left[2_{010} \parallel \bar{4}^{+}_{010}\right]$ $\left[2_{100} \parallel \bar{4}^{-}_{001}\right]$ $\left[2_{100} \parallel \bar{4}^{+}_{001}\right]$ $\left[2_{010} \parallel \bar{4}^{-}_{010}\right]$ $\left[2_{110} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{-}_{001} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[2_{110} \parallel \bar{4}^{+}_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$

No 594.1.1259 : ${}^{m_z}m^{\bar{6}_z}\bar{3}^{m_x}m$

Full symmetry operations

$\left[E \,||\, 1\right] \left[2_{001}T \,||\, m_{001}\right] \left[6^{+}_{001}T \,||\, \bar{3}^{+}_{111}\right] \left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{-}_{\bar{1}11}\right] \left[6^{-}_{001}T \,||\, \bar{3}^{+}_{1\bar{1}1}\right] \left[6^{-}_{001}T \,||\, \bar{3}^{+}_{11\bar{1}}\right] \left[6^{-}_{001}T \,||\, \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[2_{001}T \,||\, -1\right] \left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right] \left[3^{-}_{001} \,||\, 3^{+}_{\bar{1}11}\right] \left[6^{+}_{001}T \,||\, \bar{3}^{+}_{\bar{1}\bar{1}\bar{1}}\right] \left[6^{+}_{001}T \,||\, \bar{3}^{-}_{1\bar{1}1}\right] \left[6^{+}_{001}T \,||\, \bar{3}^{-}_{11\bar{1}}\right] \left[6^{-}_{001}T \,||\, \bar{3}^{-}_{\bar{1}11}\right]$
$\left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{001}T \,||\, m_{100}\right] \left[2_{001}T \,||\, m_{010}\right] \left[3^{-}_{001} \,||\, 3^{-}_{11\bar{1}}\right] \left[2_{100}T \,||\, m_{110}\right] \left[2_{120} \,||\, 2_{1\bar{1}0}\right] \left[2_{210} \,||\, 4^{-}_{010}\right]$
$\left[2_{1\bar{1}0} \,||\, 4^{-}_{100}\right] \left[2_{010}T \,||\, m_{101}\right] \left[2_{110}T \,||\, m_{01\bar{1}}\right] \left[2_{010}T \,||\, m_{10\bar{1}}\right] \left[2_{110}T \,||\, m_{011}\right] \left[2_{110}T \,||\, \bar{4}^{+}_{100}\right] \left[2_{010}T \,||\, \bar{4}^{+}_{010}\right] \left[2_{1\bar{1}0} \,||\, 2_{011}\right] \left[2_{210} \,||\, 2_{10\bar{1}}\right]$
$\left[2_{1\bar{1}0} \,||\, 2_{01\bar{1}}\right] \left[2_{210} \,||\, 2_{101}\right] \left[2_{120} \,||\, 2_{110}\right] \left[2_{100}T \,||\, m_{1\bar{1}0}\right] \left[2_{100}T \,||\, \bar{4}^{-}_{001}\right] \left[2_{100}T \,||\, \bar{4}^{+}_{001}\right] \left[2_{010}T \,||\, \bar{4}^{-}_{010}\right] \left[2_{110}T \,||\, \bar{4}^{-}_{100}\right] \left[2_{210} \,||\, 4^{+}_{010}\right]$
$\left[2_{1\bar{1}0} \,||\, 4^{+}_{100}\right] \left[2_{120} \,||\, 4^{-}_{001}\right] \left[2_{120} \,||\, 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \,||\, 1\right]$
C_2 : $\left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{-}_{\bar{1}11}\right] \left[3^{-}_{001} \,||\, 3^{+}_{\bar{1}11}\right] \left[3^{+}_{001} \,||\, 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{001} \,||\, 3^{+}_{11\bar{1}}\right] \left[3^{+}_{001} \,||\, 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{001} \,||\, 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[E \,||\, 2_{001}\right] \left[E \,||\, 2_{100}\right] \left[E \,||\, 2_{010}\right]$
C_4 : $\left[2_{120} \,||\, 2_{1\bar{1}0}\right] \left[2_{1\bar{1}0} \,||\, 2_{011}\right] \left[2_{210} \,||\, 2_{10\bar{1}}\right] \left[2_{1\bar{1}0} \,||\, 2_{01\bar{1}}\right] \left[2_{210} \,||\, 2_{101}\right] \left[2_{120} \,||\, 2_{110}\right]$
C_5 : $\left[2_{210} \,||\, 4^{-}_{010}\right] \left[2_{1\bar{1}0} \,||\, 4^{-}_{100}\right] \left[2_{210} \,||\, 4^{+}_{010}\right] \left[2_{1\bar{1}0} \,||\, 4^{+}_{100}\right] \left[2_{120} \,||\, 4^{-}_{001}\right] \left[2_{120} \,||\, 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[3^{-}_{001} \,||\, 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{-}_{001} \,||\, 3^{-}_{1\bar{1}1}\right]$	$\left[2_{210} \,||\, 4^{-}_{010}\right]$	$\left[2_{001}T \,||\, m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 595.1.1260 : ${}^{2_z}m^{3_{xyz}}\bar{3}^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[3^{+}_{111} \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{111} \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[4^{-}_{010} \parallel 4^{-}_{010}\right]$
$\left[4^{-}_{100} \parallel 4^{-}_{100}\right] \left[2_{101} \parallel m_{101}\right] \left[2_{01\bar{1}} \parallel m_{01\bar{1}}\right] \left[2_{10\bar{1}} \parallel m_{10\bar{1}}\right] \left[2_{011} \parallel m_{011}\right] \left[4^{+}_{100} \parallel \bar{4}^{+}_{100}\right] \left[4^{+}_{010} \parallel \bar{4}^{+}_{010}\right] \left[2_{011} \parallel 2_{011}\right] \left[2_{10\bar{1}} \parallel 2_{10\bar{1}}\right]$
$\left[2_{01\bar{1}} \parallel 2_{01\bar{1}}\right] \left[2_{101} \parallel 2_{101}\right] \left[2_{110} \parallel 2_{110}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[4^{-}_{001} \parallel \bar{4}^{-}_{001}\right] \left[4^{+}_{001} \parallel \bar{4}^{+}_{001}\right] \left[4^{-}_{010} \parallel \bar{4}^{-}_{010}\right] \left[4^{-}_{100} \parallel \bar{4}^{-}_{100}\right] \left[4^{+}_{010} \parallel 4^{+}_{010}\right]$
$\left[4^{+}_{100} \parallel 4^{+}_{100}\right] \left[4^{-}_{001} \parallel 4^{-}_{001}\right] \left[4^{+}_{001} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_3 : $\left[3^{+}_{111} \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}} \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{111} \parallel \bar{3}^{-}_{111}\right]$
C_4 : $\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$
C_5 : $\left[E \parallel -1\right]$
C_6 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$
C_7 : $\left[2_{110} \parallel m_{110}\right] \left[2_{101} \parallel m_{101}\right] \left[2_{01\bar{1}} \parallel m_{01\bar{1}}\right] \left[2_{10\bar{1}} \parallel m_{10\bar{1}}\right] \left[2_{011} \parallel m_{011}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
C_8 : $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right] \left[2_{011} \parallel 2_{011}\right] \left[2_{10\bar{1}} \parallel 2_{10\bar{1}}\right] \left[2_{01\bar{1}} \parallel 2_{01\bar{1}}\right] \left[2_{101} \parallel 2_{101}\right] \left[2_{110} \parallel 2_{110}\right]$
C_9 : $\left[4^{-}_{010} \parallel 4^{-}_{010}\right] \left[4^{-}_{100} \parallel 4^{-}_{100}\right] \left[4^{+}_{010} \parallel 4^{+}_{010}\right] \left[4^{+}_{100} \parallel 4^{+}_{100}\right] \left[4^{-}_{001} \parallel 4^{-}_{001}\right] \left[4^{+}_{001} \parallel 4^{+}_{001}\right]$
C_{10} : $\left[4^{+}_{100} \parallel \bar{4}^{+}_{100}\right] \left[4^{+}_{010} \parallel \bar{4}^{+}_{010}\right] \left[4^{-}_{001} \parallel \bar{4}^{-}_{001}\right] \left[4^{+}_{001} \parallel \bar{4}^{+}_{001}\right] \left[4^{-}_{010} \parallel \bar{4}^{-}_{010}\right] \left[4^{-}_{100} \parallel \bar{4}^{-}_{100}\right]$

Character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}
χ_1	1	1	1	1	1	1	1	1	1	1
χ_2	1	−1	−1	1	−1	1	−1	1	1	−1
χ_3	1	1	1	1	1	1	−1	−1	−1	−1
χ_4	1	−1	−1	1	−1	1	1	−1	−1	1
χ_5	2	2	−1	−1	2	2	0	0	0	0
χ_6	2	−2	1	−1	−2	2	0	0	0	0
χ_7	3	−1	0	0	3	−1	−1	−1	1	1
χ_8	3	1	0	0	−3	−1	1	−1	1	−1
χ_9	3	−1	0	0	3	−1	1	1	−1	−1

χ_{10}	3	1	0	0	−3	−1	−1	1	−1	1

Irrepresntations of group generators

	$\left[2_{001} \mid\mid m_{001}\right]$	$\left[4^{-}_{010} \mid\mid 4^{-}_{010}\right]$	$\left[3^{+}_{111} \mid\mid \bar{3}^{+}_{111}\right]$
Γ_1	1	1	1
Γ_2	−1	1	−1
Γ_3	1	−1	1
Γ_4	−1	−1	−1
Γ_5	σ_0	ω_2^*	ω_1
Γ_6	$-\sigma_0$	ω_2^*	ω_3^*
Γ_7	$(a\bar{b}\bar{c})$	$(\bar{b}ac)$	(bca)
Γ_8	$(\bar{a}bc)$	$(\bar{b}ac)$	$(\bar{b}\bar{c}\bar{a})$
Γ_9	$(a\bar{b}\bar{c})$	$(b\bar{a}\bar{c})$	(bca)
Γ_{10}	$(\bar{a}bc)$	$(b\bar{a}\bar{c})$	$(\bar{b}\bar{c}\bar{a})$

No 596.1.1261 : ${}^{2_z}m^{3_{xyz}}\bar{3}^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001} \parallel m_{001}\right] \left[3^+_{111} \parallel \bar{3}^+_{111}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel 3^-_{1\bar{1}1}\right] \left[3^-_{111} \parallel 3^-_{111}\right] \left[3^+_{1\bar{1}1} \parallel \bar{3}^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel \bar{3}^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel \bar{3}^-_{1\bar{1}\bar{1}}\right]$
$\left[E \parallel -1\right] \left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^+_{111} \parallel 3^+_{111}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel \bar{3}^-_{1\bar{1}1}\right] \left[3^-_{11\bar{1}} \parallel \bar{3}^-_{11\bar{1}}\right] \left[3^-_{111} \parallel \bar{3}^-_{111}\right]$
$\left[3^+_{1\bar{1}1} \parallel 3^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel 3^-_{1\bar{1}\bar{1}}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right] \left[3^-_{11\bar{1}} \parallel 3^-_{11\bar{1}}\right] \left[2_{110}T \parallel m_{110}\right] \left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right] \left[4^-_{010}T \parallel 4^-_{010}\right]$
$\left[4^-_{100}T \parallel 4^-_{100}\right] \left[2_{101}T \parallel m_{101}\right] \left[2_{01\bar{1}}T \parallel m_{01\bar{1}}\right] \left[2_{10\bar{1}}T \parallel m_{10\bar{1}}\right] \left[2_{011}T \parallel m_{011}\right] \left[4^+_{100}T \parallel \bar{4}^+_{100}\right] \left[4^+_{010}T \parallel \bar{4}^+_{010}\right] \left[2_{011}T \parallel 2_{011}\right] \left[2_{10\bar{1}}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{01\bar{1}}T \parallel 2_{01\bar{1}}\right] \left[2_{101}T \parallel 2_{101}\right] \left[2_{110}T \parallel 2_{110}\right] \left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right] \left[4^-_{001}T \parallel \bar{4}^-_{001}\right] \left[4^+_{001}T \parallel \bar{4}^+_{001}\right] \left[4^-_{010}T \parallel \bar{4}^-_{010}\right] \left[4^-_{100}T \parallel \bar{4}^-_{100}\right] \left[4^+_{010}T \parallel 4^+_{010}\right]$
$\left[4^+_{100}T \parallel 4^+_{100}\right] \left[4^-_{001}T \parallel 4^-_{001}\right] \left[4^+_{001}T \parallel 4^+_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[2_{001} \parallel m_{001}\right] \left[2_{100} \parallel m_{100}\right] \left[2_{010} \parallel m_{010}\right]$
C_3 : $\left[3^+_{111} \parallel \bar{3}^+_{111}\right] \left[3^+_{1\bar{1}\bar{1}} \parallel \bar{3}^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel \bar{3}^-_{1\bar{1}1}\right] \left[3^-_{11\bar{1}} \parallel \bar{3}^-_{11\bar{1}}\right]$
C_4 : $\left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right] \left[3^-_{1\bar{1}1} \parallel 3^-_{1\bar{1}1}\right] \left[3^+_{111} \parallel 3^+_{111}\right] \left[3^-_{11\bar{1}} \parallel 3^-_{11\bar{1}}\right]$
C_5 : $\left[3^-_{111} \parallel 3^-_{111}\right] \left[3^+_{1\bar{1}1} \parallel 3^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel 3^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel 3^-_{1\bar{1}\bar{1}}\right]$
C_6 : $\left[3^+_{1\bar{1}1} \parallel \bar{3}^+_{1\bar{1}1}\right] \left[3^+_{11\bar{1}} \parallel \bar{3}^+_{11\bar{1}}\right] \left[3^-_{1\bar{1}\bar{1}} \parallel \bar{3}^-_{1\bar{1}\bar{1}}\right] \left[3^-_{111} \parallel \bar{3}^-_{111}\right]$
C_7 : $\left[E \parallel -1\right]$
C_8 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
χ_1^{co}	1	1	1	1	1	1	1	1
χ_2^{co}	1	−1	−1	1	1	−1	−1	1
χ_3^{co}	1	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	1	1
χ_4^{co}	1	−1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{\pi i}{3}}$	−1	1
χ_5^{co}	1	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	1	1
χ_6^{co}	1	−1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$e^{\frac{-\pi i}{3}}$	−1	1
χ_7^{co}	3	−1	0	0	0	0	3	−1
χ_8^{co}	3	1	0	0	0	0	−3	−1

Coirrepresntations of group generators

	$\left[2_{001} \parallel m_{001}\right]$	$\left[3^+_{111} \parallel \bar{3}^+_{111}\right]$	$\left[3^+_{1\bar{1}\bar{1}} \parallel 3^+_{1\bar{1}\bar{1}}\right]$	$\left[2_{110}T \parallel m_{110}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	−1	−1	1	$\mathcal{K}$

Γ_3^{co}	1	$e^{\frac{2\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_4^{co}	-1	$e^{\frac{-\pi i}{3}}$	$e^{\frac{2\pi i}{3}}$	$\mathcal{K}$
Γ_5^{co}	1	$e^{\frac{-2\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_6^{co}	-1	$e^{\frac{\pi i}{3}}$	$e^{\frac{-2\pi i}{3}}$	$\mathcal{K}$
Γ_7^{co}	$(a\bar{b}\bar{c})$	(bca)	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$
Γ_8^{co}	$(\bar{a}bc)$	$(\bar{b}\bar{c}\bar{a})$	$(\bar{b}\bar{c}a)$	$(abc)\mathcal{K}$

No 597.1.1262 : ${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}^{m_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right]$ $\left[2_{001}T \parallel m_{001}\right]$ $\left[3^{+}_{111}T \parallel \bar{3}^{+}_{111}\right]$ $\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{111} \parallel 3^{-}_{111}\right]$ $\left[3^{+}_{1\bar{1}1}T \parallel \bar{3}^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{11\bar{1}}T \parallel \bar{3}^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{1\bar{1}\bar{1}}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right]$ $\left[2_{001} \parallel 2_{001}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{010}\right]$ $\left[3^{+}_{111} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{1\bar{1}\bar{1}}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{1\bar{1}1}T \parallel \bar{3}^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{11\bar{1}}T \parallel \bar{3}^{-}_{11\bar{1}}\right]$ $\left[3^{-}_{111}T \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[2_{100}T \parallel m_{100}\right]$ $\left[2_{010}T \parallel m_{010}\right]$ $\left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$ $\left[2_{110}T \parallel m_{110}\right]$ $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[4^{-}_{010} \parallel 4^{-}_{010}\right]$
$\left[4^{-}_{100} \parallel 4^{-}_{100}\right]$ $\left[2_{101}T \parallel m_{101}\right]$ $\left[2_{01\bar{1}}T \parallel m_{01\bar{1}}\right]$ $\left[2_{10\bar{1}}T \parallel m_{10\bar{1}}\right]$ $\left[2_{011}T \parallel m_{011}\right]$ $\left[4^{+}_{100}T \parallel \bar{4}^{+}_{100}\right]$ $\left[4^{+}_{010}T \parallel \bar{4}^{+}_{010}\right]$ $\left[2_{011} \parallel 2_{011}\right]$ $\left[2_{10\bar{1}} \parallel 2_{10\bar{1}}\right]$
$\left[2_{01\bar{1}} \parallel 2_{01\bar{1}}\right]$ $\left[2_{101} \parallel 2_{101}\right]$ $\left[2_{110} \parallel 2_{110}\right]$ $\left[2_{1\bar{1}0}T \parallel m_{1\bar{1}0}\right]$ $\left[4^{-}_{001}T \parallel \bar{4}^{-}_{001}\right]$ $\left[4^{+}_{001}T \parallel \bar{4}^{+}_{001}\right]$ $\left[4^{-}_{010}T \parallel \bar{4}^{-}_{010}\right]$ $\left[4^{-}_{100}T \parallel \bar{4}^{-}_{100}\right]$ $\left[4^{+}_{010} \parallel 4^{+}_{010}\right]$
$\left[4^{+}_{100} \parallel 4^{+}_{100}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$ $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right]$ $\left[3^{-}_{111} \parallel 3^{-}_{111}\right]$ $\left[3^{+}_{111} \parallel 3^{+}_{111}\right]$ $\left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right]$ $\left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right]$ $\left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right]$ $\left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right]$ $\left[2_{100} \parallel 2_{100}\right]$ $\left[2_{010} \parallel 2_{010}\right]$
C_4 : $\left[2_{1\bar{1}0} \parallel 2_{1\bar{1}0}\right]$ $\left[2_{011} \parallel 2_{011}\right]$ $\left[2_{10\bar{1}} \parallel 2_{10\bar{1}}\right]$ $\left[2_{01\bar{1}} \parallel 2_{01\bar{1}}\right]$ $\left[2_{101} \parallel 2_{101}\right]$ $\left[2_{110} \parallel 2_{110}\right]$
C_5 : $\left[4^{-}_{010} \parallel 4^{-}_{010}\right]$ $\left[4^{-}_{100} \parallel 4^{-}_{100}\right]$ $\left[4^{+}_{010} \parallel 4^{+}_{010}\right]$ $\left[4^{+}_{100} \parallel 4^{+}_{100}\right]$ $\left[4^{-}_{001} \parallel 4^{-}_{001}\right]$ $\left[4^{+}_{001} \parallel 4^{+}_{001}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	−1	1
χ_5^{co}	3	0	−1	1	−1

Coirrepresntations of group generators

	$\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[4^{-}_{010} \parallel 4^{-}_{010}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2^*	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{b}ac)$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(b\bar{a}\bar{c})$	$(abc)\mathcal{K}$

No 598.1.1263 : ${}^{m_z}m^{\bar{3}_{xyz}}\bar{3}^{2_{xy}}m$

Full symmetry operations

$\left[E \parallel 1\right] \left[2_{001}T \parallel m_{001}\right] \left[3^{+}_{111}T \parallel \bar{3}^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{1\bar{1}1}T \parallel \bar{3}^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}}T \parallel \bar{3}^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}}T \parallel \bar{3}^{-}_{1\bar{1}\bar{1}}\right]$
$\left[T \parallel -1\right] \left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}\bar{1}}T \parallel \bar{3}^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1}T \parallel \bar{3}^{-}_{1\bar{1}1}\right] \left[3^{-}_{11\bar{1}}T \parallel \bar{3}^{-}_{11\bar{1}}\right] \left[3^{-}_{111}T \parallel \bar{3}^{-}_{111}\right]$
$\left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[2_{100}T \parallel m_{100}\right] \left[2_{010}T \parallel m_{010}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right] \left[2_{110} \parallel m_{110}\right] \left[2_{1\bar{1}0}T \parallel 2_{1\bar{1}0}\right] \left[4^{-}_{010}T \parallel 4^{-}_{010}\right]$
$\left[4^{-}_{100}T \parallel 4^{-}_{100}\right] \left[2_{101} \parallel m_{101}\right] \left[2_{01\bar{1}} \parallel m_{01\bar{1}}\right] \left[2_{10\bar{1}} \parallel m_{10\bar{1}}\right] \left[2_{011} \parallel m_{011}\right] \left[4^{+}_{100} \parallel \bar{4}^{+}_{100}\right] \left[4^{+}_{010} \parallel \bar{4}^{+}_{010}\right] \left[2_{011}T \parallel 2_{011}\right] \left[2_{10\bar{1}}T \parallel 2_{10\bar{1}}\right]$
$\left[2_{01\bar{1}}T \parallel 2_{01\bar{1}}\right] \left[2_{101}T \parallel 2_{101}\right] \left[2_{110}T \parallel 2_{110}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right] \left[4^{-}_{001} \parallel \bar{4}^{-}_{001}\right] \left[4^{+}_{001} \parallel \bar{4}^{+}_{001}\right] \left[4^{-}_{010} \parallel \bar{4}^{-}_{010}\right] \left[4^{-}_{100} \parallel \bar{4}^{-}_{100}\right] \left[4^{+}_{010}T \parallel 4^{+}_{010}\right]$
$\left[4^{+}_{100}T \parallel 4^{+}_{100}\right] \left[4^{-}_{001}T \parallel 4^{-}_{001}\right] \left[4^{+}_{001}T \parallel 4^{+}_{001}\right]$

Classes of maximal unitary subgroup

C_1 : $\left[E \parallel 1\right]$
C_2 : $\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right] \left[3^{-}_{111} \parallel 3^{-}_{111}\right] \left[3^{+}_{111} \parallel 3^{+}_{111}\right] \left[3^{+}_{1\bar{1}1} \parallel 3^{+}_{1\bar{1}1}\right] \left[3^{+}_{11\bar{1}} \parallel 3^{+}_{11\bar{1}}\right] \left[3^{-}_{1\bar{1}\bar{1}} \parallel 3^{-}_{1\bar{1}\bar{1}}\right] \left[3^{-}_{11\bar{1}} \parallel 3^{-}_{11\bar{1}}\right]$
C_3 : $\left[2_{001} \parallel 2_{001}\right] \left[2_{100} \parallel 2_{100}\right] \left[2_{010} \parallel 2_{010}\right]$
C_4 : $\left[2_{110} \parallel m_{110}\right] \left[2_{101} \parallel m_{101}\right] \left[2_{01\bar{1}} \parallel m_{01\bar{1}}\right] \left[2_{10\bar{1}} \parallel m_{10\bar{1}}\right] \left[2_{011} \parallel m_{011}\right] \left[2_{1\bar{1}0} \parallel m_{1\bar{1}0}\right]$
C_5 : $\left[4^{+}_{100} \parallel \bar{4}^{+}_{100}\right] \left[4^{+}_{010} \parallel \bar{4}^{+}_{010}\right] \left[4^{-}_{001} \parallel \bar{4}^{-}_{001}\right] \left[4^{+}_{001} \parallel \bar{4}^{+}_{001}\right] \left[4^{-}_{010} \parallel \bar{4}^{-}_{010}\right] \left[4^{-}_{100} \parallel \bar{4}^{-}_{100}\right]$

Co-character table

	C_1	C_2	C_3	C_4	C_5
χ_1^{co}	1	1	1	1	1
χ_2^{co}	1	1	1	−1	−1
χ_3^{co}	2	−1	2	0	0
χ_4^{co}	3	0	−1	1	−1
χ_5^{co}	3	0	−1	−1	1

Coirrepresntations of group generators

	$\left[3^{+}_{1\bar{1}\bar{1}} \parallel 3^{+}_{1\bar{1}\bar{1}}\right]$	$\left[3^{-}_{1\bar{1}1} \parallel 3^{-}_{1\bar{1}1}\right]$	$\left[4^{+}_{100} \parallel \bar{4}^{+}_{100}\right]$	$\left[2_{001}T \parallel m_{001}\right]$
Γ_1^{co}	1	1	1	$\mathcal{K}$
Γ_2^{co}	1	1	−1	$\mathcal{K}$
Γ_3^{co}	ω_1	ω_1	ω_2	$\sigma_0\mathcal{K}$
Γ_4^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(c\bar{b}\bar{a})$	$(abc)\mathcal{K}$
Γ_5^{co}	$(\bar{b}\bar{c}a)$	$(b\bar{c}\bar{a})$	$(\bar{c}ba)$	$(abc)\mathcal{K}$